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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 9: Ray Optics and Optical Instruments

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 112 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be:

(A) 20 cm

(B) 30 cm

(C) 40 cm

(D) 5 cm

Check Solution → **[2026 • Set 55-1-1]**

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2. A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is:

(A) zero

(B) $\frac{40}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

Check Solution → **[2026 • Set 55-1-2]**

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3. Light from a small object in air falls on a spherical glass surface ($n = 1.5$) of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as:

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

Check Solution → [2026 • Set 55-1-3]

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4. **Case Study:** In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two positions of the lens. The distance between these two positions of the lens is d.

The value of d is:

(A) $\sqrt{D(D-4f)}$

(B) $\sqrt{D(D-2f)}$

(C) $2\sqrt{Df}$

(D) $\sqrt{D(D-f)}$

Check Solution → [2026 • Set 55-2-1]

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5. Compared to the size of the object, the images formed in the two positions of the lens are respectively:

(A) reduced, enlarged

(B) reduced, reduced

(C) enlarged, enlarged

(D) enlarged, reduced

Check Solution → [2026 • Set 55-2-1]

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6. If the distance between object and screen is 80·00 cm and the lens forms sharp images at two positions separated by 20·00 cm, the focal length of convex lens is:

(A) 15·50 cm

(B) 18·75 cm

(C) 20·50 cm

(D) 22·75 cm

Check Solution → [2026 • Set 55-2-1]

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7. Consider a convex lens of focal length 15 cm. For which of the following values of object-screen distance, two positions of the object can be found to obtain sharp image on the screen?

(A) 45 cm

(B) 50 cm

(C) 55 cm

(D) 65 cm

Check Solution → [2026 • Set 55-2-1]

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8. A thin convex lens of focal length 10 cm and another thin lens of focal length f are placed coaxially in contact. If the power of their combination is 2 D, the value of f is:

(A) −15 cm

(B) −10 cm

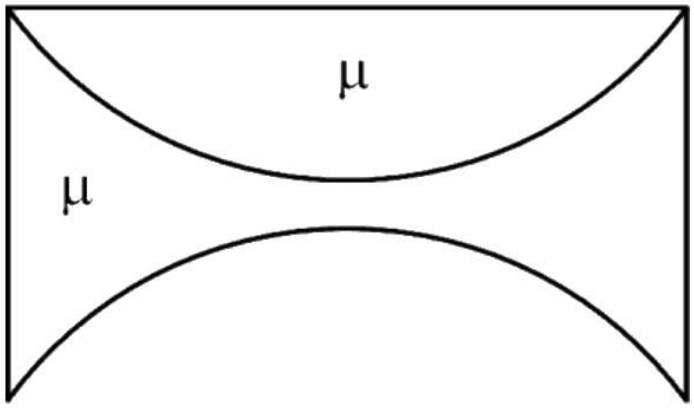
(C) −20 cm

(D) −30 cm

Check Solution → [2026 • Set 55-2-1]

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9. A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is:



(A) $\frac{R}{\mu-1}$

(B) $-\frac{R}{\mu-1}$

(C) $\frac{2R}{\mu-1}$

(D) $-\frac{2R}{\mu-1}$

Check Solution → [2026 • Set 55-3-1]

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10. A telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6·0 cm. The magnifying power and the length of the telescope tube will be respectively:

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

Check Solution → [2026 • Set 55-3-1]

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11. In which of the following, total internal reflection does not occur?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

Check Solution → [2026 • Set 55-3-1]

12. A plano-convex lens, made of glass ($n = 1{\cdot}5$) has a curved surface whose radius is 40 cm. The image formed of an object is of the same size as that of the object. The distance of the object from the lens is:

(A) 40 cm

(B) 80 cm

(C) 160 cm

(D) 200 cm

Check Solution → [2026 • Set 55-3-2]

13. A ray of monochromatic light travelling in air is incident on a glass slab and is partly reflected and partly refracted. Both the reflected and refracted lights will have:

(A) same wavelength

(B) same frequency

(C) same intensity

(D) same speed

Check Solution → [2026 • Set 55-3-3]

14. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

Check Solution → [2026 • Set 55-3-3]

15. **Assertion (A):** When a convex lens made of glass is immersed in water, its converging power increases.
Reason (R): The focal length of a lens depends only on the radii of curvature of its two faces.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-4-1]

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16. The images formed by the objective lens and the eyepiece are respectively:

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

Check Solution → [2026 • Set 55-4-1]

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17. The magnification produced by the telescope does not depend upon the:

(A) colour of light

(B) focal length of objective lens

(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

Check Solution → [2026 • Set 55-4-1]

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18. Which of the following statements is not correct for this telescope?

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) The distance between two lenses is more than ($f_o + f_e$).

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

Check Solution → [2026 • Set 55-4-1]

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19. An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of:

(A) 84 cm

(B) 76 cm

(C) 20 cm

(D) 320 cm

Check Solution → [2026 • Set 55-4-1]

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20. Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be:

(A) 320

(B) 84

(C) 76

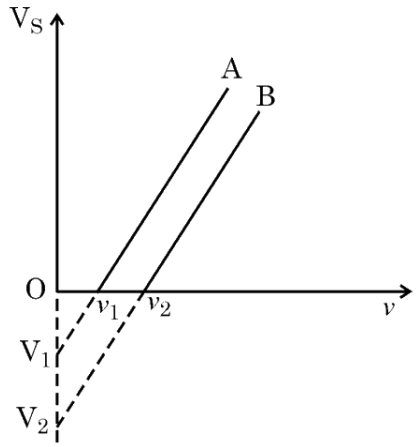
(D) 20

Check Solution → [2026 • Set 55-4-1]

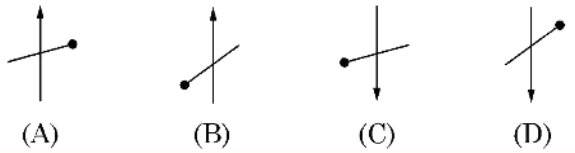
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21. **Case Study (Q30):** A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens

has two foci, called 'first focal point' and 'second focal point' of the lens, one on each side.



(i) Consider the arrangement shown in figure. A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object. When the plate is illuminated, its real image is formed on the screen. Which of the following correctly represents the image formed on the screen?



(A) Option A (image as shown)

(B) Option B (image as shown)

(C) Option C (image as shown)

(D) Option D (image as shown)

Check Solution → **[2025 • Set 55-1-1]**

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22. **(ii)** Which of the following statements is incorrect?

(A) For a convex mirror magnification is always negative.

(B) For all virtual images formed by a mirror magnification is positive.

(C) For a concave lens magnification is always positive.

(D) For real and inverted images, magnification is always negative.

Check Solution → **[2025 • Set 55-1-1]**

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23. **(iii)** A convex lens of focal length f is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be:

(A) f

(B) 2f

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

Check Solution → [2025 • Set 55-1-1]

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24. **(iii) OR:** If an object in case (i) above is 20 cm from the lens and the screen is 50 cm away from the object, the focal length of the lens used is:

(A) 10 cm

(B) 12 cm

(C) 16 cm

(D) 20 cm

Check Solution → [2025 • Set 55-1-1]

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25. **(iv)** The distance of an object from first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2. The focal length of the lens is:

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

(D) $\frac{\sqrt{X_2}}{X_1}$

Check Solution → [2025 • Set 55-1-1]

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26. A beaker is filled with water (refractive index $\frac{4}{3}$) upto a height H. A coin is placed at its bottom. The depth of the coin, when viewed along the near normal direction, will be:

(A) $\frac{H}{4}$

(B) $\frac{3H}{4}$

(C) H

(D) $\frac{4H}{3}$

Check Solution → [2025 • Set 55-1-1]

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27. The speed of light in two media '1' and '2' are v_1 and v_2 ($> v_1$) respectively. For a ray of light to undergo total internal reflection at the interface of these two media, it must be incident from:

(A) medium '1' and at an angle greater than $\sin^{-1}(\frac{v_1}{v_2})$

(B) medium '1' and at an angle greater than $\cos^{-1}(\frac{v_1}{v_2})$

(C) medium '2' and at an angle greater than $\sin^{-1}(\frac{v_1}{v_2})$

(D) medium '2' and at an angle greater than $\cos^{-1}(\frac{v_1}{v_2})$

Check Solution → [2025 • Set 55-1-2]

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28. A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The area of the surface of the liquid through which light from the source can emerge out is:

(A) $\frac{\pi H^2}{(n-1)}$

(B) $\frac{\pi H^2}{(n^2-1)}$

(C) $\frac{\pi H^2}{\sqrt{n}}$

(D) $\frac{\pi H^2}{(n^2+1)}$

Check Solution → [2025 • Set 55-1-3]

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29. A tub is filled with a transparent liquid to a height of 30.0 cm. The apparent depth of a coin lying at the bottom of the tub is found to be 16.0 cm. The speed of light in the liquid will be

(A) $1.6 \times 10^8\ m\ s^{-1}$

(B) $2.0 \times 10^8\ m\ s^{-1}$

(C) $3.0 \times 10^8\ m\ s^{-1}$

(D) $2.5 \times 10^8\ m\ s^{-1}$

Check Solution → [2025 • Set 55-2-1]

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30. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become

(A) $\frac{5}{3}$f

(B) $\frac{3}{5}$f

(C) $\frac{2}{3}$f

(D) f

Check Solution → [2025 • Set 55-2-2]

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31. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length

(A) $\frac{R}{2}$

(B) $\frac{R}{2(n-1)}$

(C) nR

(D) $\frac{R}{2(n-1)}$

Check Solution → [2025 • Set 55-2-3]

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32. The magnification produced by a spherical mirror is −2.0. The mirror used and the nature of the image formed will be :

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

Check Solution → [2025 • Set 55-4-1]

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33. A compound microscope has an objective and an eyepiece of focal lengths f_o and f_e, respectively. To obtain a large magnification of a small object, the microscope should have:

(A) f_o and f_e small, and $f_e > f_o$

(B) f_o and f_e small, and $f_o > f_e$

(C) f_o and f_e large, and $f_e > f_o$

(D) f_o and f_e large, and $f_o > f_e$

Check Solution → [2025 • Set 55-5-1]

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34. A glass slab ($\mu = 1.5$) of thickness 6 cm is placed over a paper. The shift in the letters printed on the paper will be:

(A) 2 cm

(B) 1 cm

(C) 4 cm

(D) 3 cm

Check Solution → [2025 • Set 55-5-2]

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35. A beam of light of wavelength 720 nm in air enters water (refractive index = $\frac{4}{3}$). Its wavelength in water will be:

(A) 540 nm

(B) 480 nm

(C) 420 nm

(D) 720 nm

Check Solution → [2025 • Set 55-5-3]

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36. **Assertion (A):** In a reflecting telescope, the image does not have chromatic aberration.
Reason (R): Chromatic aberration occurs only due to refraction of light through an optical medium.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-6-1]

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37. **Assertion (A):** A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.
Reason (R): The refractive index of a prism depends on angle of the prism.
Select the correct answer:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-7-1]

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38. **Case Study:** When light travels from an optically denser medium to an optically rarer medium, at the interface it is partly reflected back into the same medium and partly refracted to the second medium. The angle of incidence corresponding to an angle of refraction 90° is called the critical angle (i_c) for the given pair of media. This angle is related to the refractive index of medium 1 with respect to medium 2. Refraction of light through a prism involves refraction at two plane interfaces. A relation for the refractive index of the material of the prism can be obtained in terms of the refracting angle of the prism and the angle of minimum deviation. For a thin prism, this relation reduces to a simple equation. Laws of refraction are also valid for refraction of light at a spherical interface. When an object is placed in front of a spherical surface separating two media, its image is formed. A relation between object and image distance, in terms of refractive indices of two media and the radius of curvature of the spherical surface can be obtained. Using this relation for two surfaces of a lens, 'lens maker formula' is obtained.

(i) A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R. Then $(\frac{R}{H})$ is:

(A) $\frac{1}{\sqrt{n^2-1}}$

(B) $\sqrt{n^2-1}$

(C) $\frac{1}{\sqrt{n^2+1}}$

(D) $\sqrt{n^2+1}$

Check Solution → **[2025 • Set 55-7-1]**

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39. **(ii) (a)** A parallel beam of light is incident on a face of a prism with refracting angle 60°. The angle of minimum deviation is found to be 30°. The refractive index of the material of the prism is close to:

OR

(b) The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass (n = 1.52) is D_m. If the prism was made of dense flint glass (n = 1.62) instead of crown glass, the angle of minimum deviation will:

(A) (a) 1.3 OR (b) decrease by 4%

(B) (a) 1.4 OR (b) increase by 4%

(C) (a) 1.5 OR (b) decrease by 19%

(D) (a) 1.6 OR (b) increase by 19%

Check Solution → [2025 • Set 55-7-1]

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40. **(iii)** An object is placed in front of a convex spherical glass surface (n = 1.5 and radius of curvature R) at a distance of 4R from it. As the object is moved slowly close to the surface, the image formed is:

(A) always real

(B) always virtual

(C) first real and then virtual

(D) first virtual and then real

Check Solution → [2025 • Set 55-7-1]

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41. **(iv)** A double-convex lens, made of glass of refractive index 1.5, has focal length 10 cm. The radius of curvature of its each face, is:

(A) 10 cm

(B) 15 cm

(C) 20 cm

(D) 40 cm

Check Solution → [2025 • Set 55-7-1]

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42. **Assertion (A):** Plane and convex mirrors cannot produce real images under any circumstance.
Reason (R): A virtual image cannot serve as an object to produce a real image.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-1-1]

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43. A double-convex lens, with each face having same radius of curvature R, is made of glass of refractive index n. Its power is:

(A) $\frac{2(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

Check Solution → [2024 • Set 55-1-1]

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44. A double-convex lens of power P, with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be:

(A) 2P

(B) P

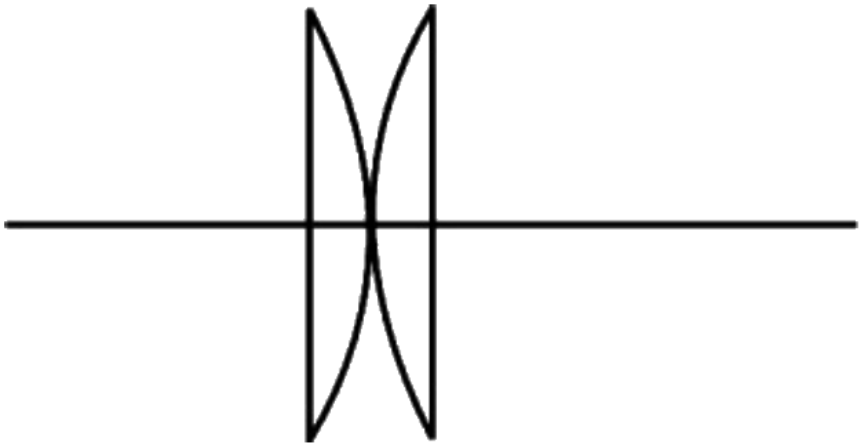
(C) 4P

(D) $\frac{P}{2}$

Check Solution → [2024 • Set 55-1-1]

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45. The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be:



(A) $\frac{P}{2}$

(B) P

(C) 2P

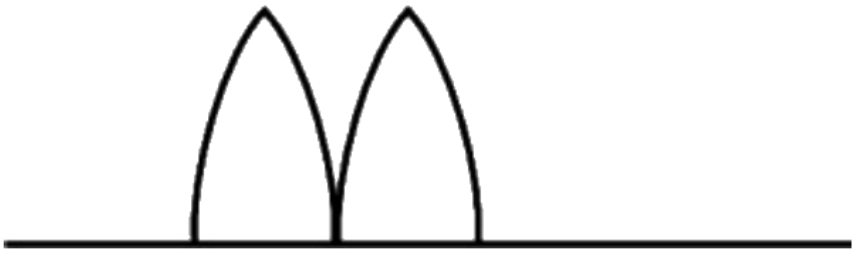
(D) $\frac{P}{4}$

Check Solution →

[2024 • Set 55-1-1]

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46. A double-convex lens of power P, with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be:



(A) Zero

(B) P

(C) 2P

(D) $\frac{P}{2}$

Check Solution →

[2024 • Set 55-1-1]

47.

Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is:

(A) $6.6\ D$

(B) $15\ D$

(C) $\frac{1}{15}\ D$

(D) $\frac{1}{30}\ D$

Check Solution → [2024 • Set 55-1-1]

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48. The refractive index of the material of a prism is $\sqrt{2}$. If the refracting angle of the prism is 60°, find the:
(1) Angle of minimum deviation, and
(2) Angle of incidence.

Check Solution → [2024 • Set 55-1-1]

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49. **Assertion (A):** The magnifying power of a compound microscope is negative.
Reason (R): The final image formed is erect with respect to the object.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-2-1]

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50. When a ray of light propagates from a denser medium to a rarer medium, it bends away from the normal. When the incident angle is increased, the refracted ray deviates more from the normal. For a particular angle of incidence in the denser medium, the refracted ray just grazes the interface of the two surfaces. This angle of incidence is called the critical angle for the pair of media

involved.

For a ray incident at the critical angle, the angle of reflection is:

(A) 0°

(B) <90°

(C) >90°

(D) 90°

Check Solution → [2024 • Set 55-2-1]

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51. A ray of light of wavelength 600 nm is incident in water ($n=\frac{4}{3}$) on the water-air interface at an angle less than the critical angle. The wavelength associated with the refracted ray is:

(A) 400 nm

(B) 450 nm

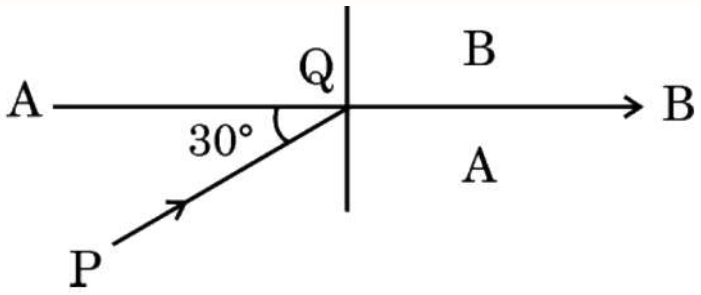
(C) 600 nm

(D) 800 nm

Check Solution → [2024 • Set 55-2-1]

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52. (a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is:



OR

(b) Two media A and B are separated by a plane boundary. The speed of light in medium A and B is 210^8 ms^{-1} and 2.510^8 ms^{-1} respectively. The critical angle for a ray of light going from medium A to medium B is:

(A) (a) $\frac{3}{4}$ (b) $\sin^{-1}\frac{1}{9}$

(B) (a) $\frac{5}{2}$ (b) $\sin^{-1}\frac{4}{5}$

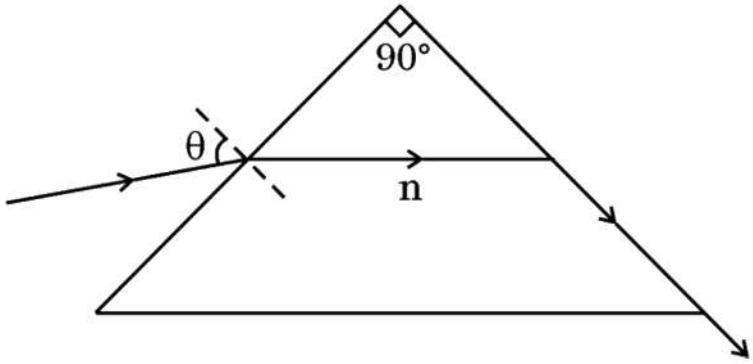
(C) (a) $\frac{4}{3}$ (b) $\sin^{-1}\frac{3}{5}$

(D) (a) $\frac{2}{3}$ (b) $\sin^{-1}\frac{2}{5}$

Check Solution → [2024 • Set 55-2-1]

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53. The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by:



(A) $\sin^{-1}\sqrt{n^2-1}$

(B) $\sin^{-1}(n^2-1)$

(C) $\sin^{-1}\sqrt{\frac{1}{n^2-1}}$

(D) $\sin^{-1}\frac{1}{(n^2-1)}$

Check Solution → [2024 • Set 55-2-1]

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54. **Assertion (A):** A convex lens, when immersed in a liquid, disappears.
Reason (R): The refractive indices of material of the lens and the liquid are equal.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-3-1]

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55. A thin converging lens of focal length 20 cm and a thin diverging lens of focal length 15 cm are placed coaxially in contact. The power of the combination is

(A) $-\frac{5}{4}\,D$

(B) $-\frac{5}{3}\,D$

(C) $\frac{4}{3}\,D$

(D) $\frac{3}{5}\,D$

Check Solution → [2024 • Set 55-4-1]

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56. The radii of curvature of two surfaces of a convex lens are R and 2R. If the focal length of this lens is $(\frac{5}{3})R$, the refractive index of the material of the lens is

(A) $\frac{7}{5}$

(B) $\frac{8}{5}$

(C) $\frac{19}{15}$

(D) $\frac{3}{2}$

Check Solution → [2024 • Set 55-4-1]

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57. The focal length of an equiconvex lens

(A) increases when the lens is dipped in water.

(B) increases when the wavelength of incident light decreases.

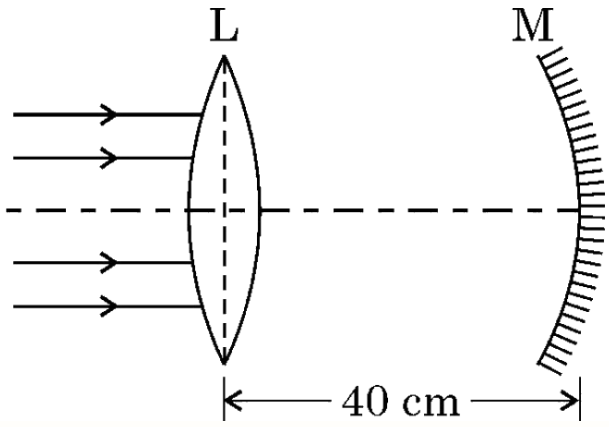
(C) increases with decrease in radius of curvature of its surface.

(D) decreases when the lens is cut into two identical parts along its principal axis.

Check Solution → [2024 • Set 55-4-1]

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58. A thin convex lens *L* of focal length 10 cm and a concave mirror *M* of focal length 15 cm are placed coaxially 40 cm apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of



(A) 10 cm, left of lens

(B) 10 cm, right of lens

(C) 20 cm, left of lens

(D) 20 cm, right of lens

Check Solution → [2024 • Set 55-4-1]

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59. A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1. The nature and distance of the final image from L_2 will be

(A) real, 24 cm

(B) virtual, 12 cm

(C) real, 32 cm

(D) virtual, 18 cm

Check Solution → [2024 • Set 55-4-1]

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60. A prism is an optical medium bounded by three refracting plane surfaces. A ray of light suffers successive refractions on passing through its two surfaces and deviates by a certain angle from its original path. The refractive index of the material of the prism is given by $\mu = \sin\left(\frac{A+\delta_m}{2}\right) / \sin\frac{A}{2}$. If the angle of incidence on the second surface is greater than an angle called critical angle, the ray will not be refracted from the second surface and is totally internally reflected.

The critical angle for glass is θ_1 and that for water is θ_2. The critical angle for glass-water surface would be (given ${}_a\mu_g = 1.5$, ${}_a\mu_w = 1.33$):

(A) less than θ_2

(B) between θ_1 and θ_2

(C) greater than θ_2

(D) less than θ_1

Check Solution → [2024 • Set 55-5-1]

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61. When a ray of light of wavelength λ and frequency ν is refracted into a denser medium:

(A) λ and ν both increase.

(B) λ increases but ν is unchanged.

(C) λ decreases but ν is unchanged.

(D) λ and ν both decrease.

Check Solution → [2024 • Set 55-5-1]

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62. The critical angle for a ray of light passing from glass to water is minimum for:

(A) red colour

(B) blue colour

(C) yellow colour

(D) violet colour

Check Solution → [2024 • Set 55-5-1]

---

63. Three beams of red, yellow and violet colours are passed through a prism, one by one under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y and r_V respectively. Then:

(A) $r_V < r_Y < r_R$

(B) $r_Y < r_R < r_V$

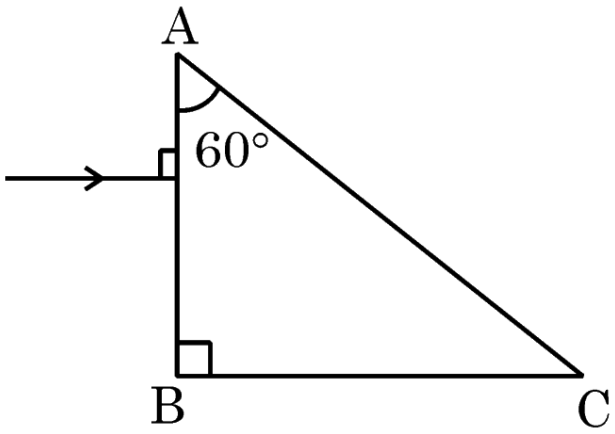
(C) $r_R < r_V < r_Y$

(D) $r_R = r_Y = r_V$

Check Solution → [2024 • Set 55-5-1]

---

64. A ray of light is incident normally on a prism ABC of refractive index $\sqrt{2}$, as shown in figure. After it strikes face AC, it will:



(A) go straight undeviated

(B) just graze along the face AC

(C) refract and go out of the prism

(D) undergo total internal reflection

Check Solution → [2024 • Set 55-5-1]

---

65. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is:

(A) 250

(B) 200

(C) 150

(D) 125

Check Solution → [2024 • Set 55-5-1]

---

66. A point object is kept 60 cm in front of a spherical convex surface (n = 1.5, radius of curvature 40 cm). The image formed is:

(A) real, at a distance 1.8 m from the surface.

(B) virtual, at a distance 1.8 m from the surface.

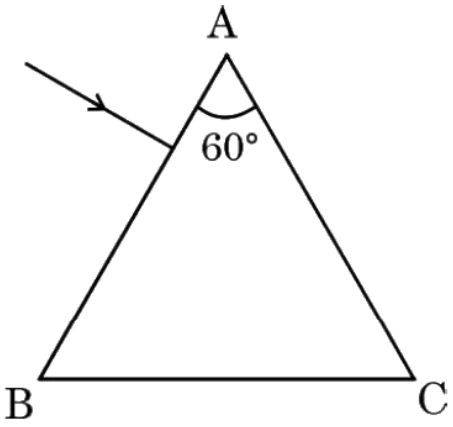
(C) real, at a distance 3.6 m from the surface.

(D) virtual, at a distance 3.6 m from the surface.

Check Solution → [2024 • Set 55-5-3]

---

67. Strontium titanate is a rare oxide - a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2.41 and 24.5°, respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism ABC made of strontium titanate.



Answer the following questions based on the above:

Trace the path of the ray showing its passage through the prism.

Check Solution → [2023 • Set 55-1-1]

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68. For the strontium titanate prism (refractive index 2.41), find the velocity of light through the prism.

Check Solution → [2023 • Set 55-1-1]

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69. For a concave mirror of focal length '*f*', the minimum distance between the object and its real image is:

(A) zero

(B) *f*

(C) 2f

(D) 4f

Check Solution → [2023 • Set 55-1-1]

---

70. The minimum distance between an object and its real image formed by a convex lens of focal length *f* is:

(A) f

(B) 2f

(C) 3f

(D) 4f

Check Solution → [2023 • Set 55-1-2]

---

71. Which of the following pairs of media has the least value of critical angle?

(A) Glass to air

(B) Glass to water

(C) Diamond to water

(D) Diamond to air

Check Solution → [2023 • Set 55-1-3]

---

72. A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays?

(A) Energy carried

(B) Speed

(C) Frequency

(D) Wavelength

Check Solution → [2023 • Set 55-2-1]

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73. A ray of light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is

(A) 1.4

(B) 1.5

(C) 1.6

(D) 1.8

Check Solution → [2023 • Set 55-2-2]

---

74. A ray of light travels a distance of 12.0 m in a transparent sheet in 60 ns. The refractive index of the sheet is

(A) 1.33

(B) 1.50

(C) 1.65

(D) 1.75

Check Solution → [2023 • Set 55-2-3]

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75. **Case Study:** The lens maker's formula is useful to design lenses of desired focal lengths using surfaces of suitable radii of curvature. The focal length also depends on the refractive index of the material of the lens and the surrounding medium. The refractive index depends on the wavelength of the light used. The power of a lens is related to its focal length.
(a) How will the power of a lens be affected with an increase of wavelength of light?

Check Solution → [2023 • Set 55-4-1]

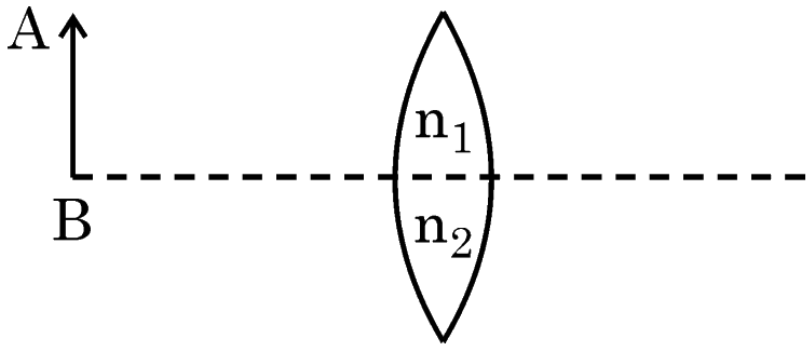
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76. (b) The radius of curvature of two surfaces of a convex lens is R each. For what value of μ of its material will its focal length become equal to R?

Check Solution → [2023 • Set 55-4-1]

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77. An object AB is kept in front of a composite convex lens, as shown in figure. Will the lens produce one image? If not, explain.



Check Solution → [2023 • Set 55-5-1]

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78. A real image of an object formed by a convex lens is observed on a screen. If the screen is removed, will the image still be formed? Explain.

Check Solution → [2023 • Set 55-5-1]

---

79. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): The angle of minimum deviation for a prism is lesser for red light than that for blue light.
Reason (R): The refractive index of the material of a prism for blue light is greater than that for red light.

Check Solution → [2021]

---

80. An object approaches a converging lens with a uniform speed of 5 m/s and stops at the focus. How will the image move with respect to the lens? Specify its nature.

Check Solution → [2021]

---

81. In a simple microscope, a convex lens of focal length 5 cm is used. Calculate the magnifying power when the object is placed at the focus of the lens.

Check Solution → [2021]

---

82. A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is _____.

Check Solution → [2020 • Set 55-1-1]

---

83. Larger aperture of objective lens in an astronomical telescope

(A) increases the resolving power of telescope.

(B) decreases the brightness of the image.

(C) increases the size of the image.

(D) decreases the length of the telescope.

Check Solution → [2020 • Set 55-1-1]

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84. A biconvex lens of glass having refractive index 1·47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is

(A) 1·47

(B) 1·62

(C) 1·33

(D) 1·51

Check Solution → [2020 • Set 55-1-1]

---

85. For a glass prism, the angle of minimum deviation will be smallest for the light of

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

Check Solution → [2020 • Set 55-1-1]

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86. The resolving power of a telescope can be increased by increasing

(A) wavelength of light.

(B) diameter of objective.

(C) length of the tube.

(D) focal length of eyepiece.

Check Solution → [2020 • Set 55-1-2]

---

87. The focal length of the objective of a compound microscope is

(A) greater than the focal length of eyepiece.

(B) lesser than the focal length of eyepiece.

(C) equal to the focal length of eyepiece.

(D) equal to the length of its tube.

Check Solution → [2020 • Set 55-1-3]

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88. For a higher resolving power of a compound microscope, the wavelength of light used should be _____.

Check Solution → [2020 • Set 55-2-1]

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89. A compound microscope is used because a realistic simple microscope does not have _____ magnification.

Check Solution → [2020 • Set 55-2-2]

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90. The resolving power of a compound microscope can be _____ by using a medium (oil) of higher refractive index than air between the object and the objective lens.

Check Solution → [2020 • Set 55-2-3]

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91. The refractive index of the material of a converging lens is 1.5. If air is replaced by a medium of refractive index 1.6, then the lens will now behave as a _____ lens.

Check Solution → [2020 • Set 55-3-1]

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92. The value of Brewster's angle for air-glass interface is $\frac{\pi}{3}$; hence the refractive index of glass is _____.

Check Solution → [2020 • Set 55-3-1]

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93. An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality? Justify your answer.

Check Solution → [2020 • Set 55-4-1]

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94. A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

(A) 2P

(B) $P/2$

(C) P

(D) $P/\sqrt{2}$

Check Solution → **[2020 • Set 55-5-1]**

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95. State the condition under which a large magnification can be achieved in an astronomical telescope.

OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light?

Check Solution → **[2019 • Set 55-3-1]**

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96. Does the magnifying power of a microscope depend on the colour of the light used? Justify your answer.

Check Solution → **[2017]**

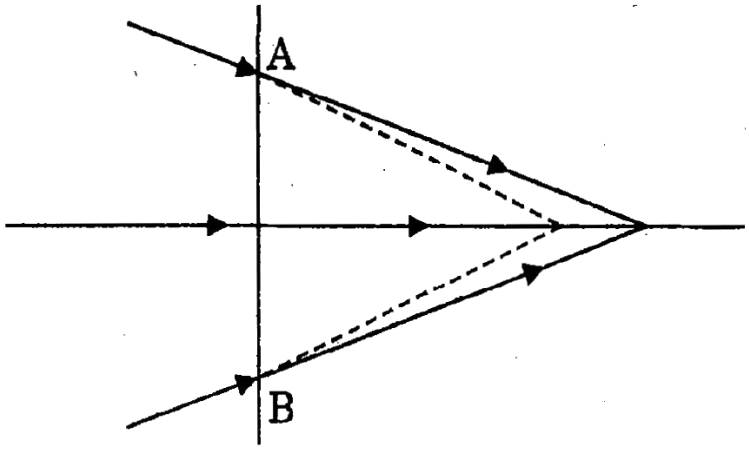
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97. How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced by red light? Give reason.

Check Solution → **[2017]**

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98. The line AB in the ray diagram represents a lens. State whether the lens is convex or concave.



Check Solution → [2015]

---

99. A concave lens of refractive index 1.5 is immersed in a medium of refractive index 1.65. What is the nature of the lens?

Check Solution → [2015]

---

100. A convex lens is placed in contact with a plane mirror. A point object at a distance of 20 cm on the axis of this combination has its image coinciding with itself. What is the focal length of the lens ?

Check Solution → [2014]

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101. A biconcave lens made of a transparent material of refractive index 1.25 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens ? Give reason.

Check Solution → [2014]

---

102. Write the relationship between angle of incidence i, angle of prism A and angle of minimum deviation for a triangular prism.

Check Solution → [2013]

---

103. A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal lengths. What is the focal length of the combination?

Check Solution → [2012]

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104. For the same value of angle of incidence, the angles of refraction in three media A, B and C are 15°, 25° and 35° respectively. In which medium would the velocity of light be minimum?

Check Solution → [2012]

---

105. Under what condition does a biconvex lens of glass having a certain refractive index act as a plane glass sheet when immersed in a liquid?

Check Solution → [2012]

---

106. How does focal length of a lens change when red light incident on it is replaced by violet light? Give reason for your answer.

Check Solution → [2012]

---

107. A glass lens of refractive index 1.45 disappears when immersed in a liquid. What is the value of refractive index of the liquid?

Check Solution → [2010]

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108. State the conditions for the phenomenon of total internal reflection to occur.

Check Solution → [2010]

---

109. You are given following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope?

Lenses	Power (P)	Aperture (A)
L1	3D	8 cm
L2	6D	1 cm
L3	10D	1 cm

Check Solution → [2009]

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110. Two thin lenses of power +6D and −2D are in contact. What is the focal length of the combination?

Check Solution → [2009]

---

111. A glass lens of refractive index 1.5 is placed in a trough of liquid. What must be the refractive index of the liquid in order to make the lens disappear?

Check Solution → [2008]

---

112. An object is held at the principal focus of a concave lens of focal length F. Where is the image formed?

Check Solution → [2008]

SECTION B

2-Mark Questions

VSA · 98 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

113. A ray of light in air is incident at angle $\angle i$ on a face of an equilateral glass prism and is refracted through the prism. As $\angle i$ is varied, it is observed that the ray undergoes minimum deviation, when $\angle i$ is three-fourth of the angle of the prism. Calculate the speed of light in the prism.

Check Solution → **[2026 • Set 55-2-1]**

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114. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil.

Check Solution → **[2026 • Set 55-2-1]**

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115. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$.

Check Solution → **[2026 • Set 55-2-2]**

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116. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path. Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism.

Check Solution → **[2026 • Set 55-2-3]**

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117. Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power.

Check Solution → [2026 • Set 55-3-2]

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118. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges.

Check Solution → [2026 • Set 55-4-1]

---

119. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object.

Check Solution → [2026 • Set 55-4-2]

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120. A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25.

Check Solution → [2026 • Set 55-4-3]

---

121. A tank is filled with a liquid to a height of 12.5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9.0 m. Calculate the speed of light in the liquid.

Check Solution → [2026 • Set 55-5-1]

---

122. Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$.

Check Solution → [2026 • Set 55-5-1]

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123. A point source, in air, is placed at a distance of 6 cm in front of a convex spherical surface (n = 1.5 and radius of curvature = 24 cm). Find the position and

nature of the image formed.

Check Solution → [2026 • Set 55-5-2]

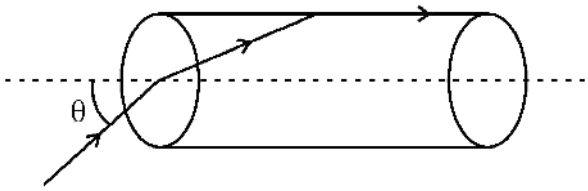
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124. A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$.

Check Solution → [2026 • Set 55-5-3]

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125. A transparent solid cylindrical rod (refractive index $\sqrt{3}$) is kept in air. A ray of light incident on its face travels along the surface of the rod, as shown in figure. Calculate the angle θ.



Check Solution → [2025 • Set 55-1-1]

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126. A spherical convex surface of radius of curvature R separates glass (refractive index 1.5) from air. Light from a point source placed in air at distance $R/2$ from the surface falls on it. Find the position and nature of the image formed.

Check Solution → [2025 • Set 55-1-2]

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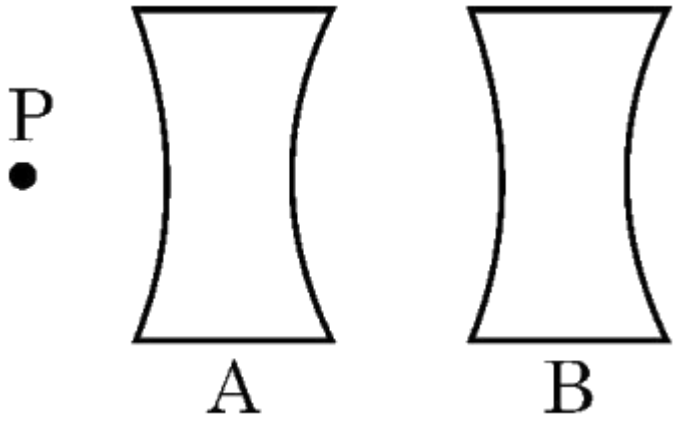
127. A double convex lens of glass has both faces of the same radius of curvature 17 cm. Find its focal length if it is immersed in water. The refractive indices of glass and water are 1.5 and 1.33 respectively.

Check Solution → [2025 • Set 55-1-3]

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128. Two concave lenses A and B, each of focal length 8.0 cm are arranged coaxially 16 cm apart as shown in figure. An object P is placed at a distance of 4.0 cm from

A. Find the position and nature of the final image formed.



Check Solution → [2025 • Set 55-2-1]

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129. Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination.

Check Solution → [2025 • Set 55-2-2]

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130. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'.

Check Solution → [2025 • Set 55-2-3]

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131. An equiconvex lens is made of glass of refractive index 1.55. If the focal length of the lens is 15.0 cm, calculate the radius of curvature of its surfaces.

Check Solution → [2025 • Set 55-4-1]

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132. Find the focal length of plano-convex lens of refractive index 1.5 and radius of curvature 10 cm when it is immersed in a liquid of refractive index 1.25.

Check Solution → [2025 • Set 55-4-2]

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133. The two surfaces of a biconvex lens are of radius of curvature 'R' each. Obtain the condition under which its focal length 'f' be equal to 'R'. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

Check Solution → [2025 • Set 55-4-3]

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134. **(a)** A point object is placed in air at a distance $R/3$ in front of a convex surface of radius of curvature R, separating air from a medium of refractive index n (< 4). Find the nature and position of the image formed.

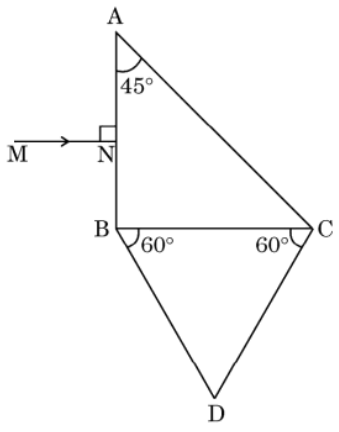
OR

(b) In Young's double slit experimental set-up, the intensity of the central maximum is I_0. Calculate the intensity at a point where the path difference between two interfering waves is $\lambda/3$.

Check Solution → [2025 • Set 55-5-1]

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135. A right angled isosceles glass prism ABC is kept in contact with an equilateral triangular prism DBC as shown in the figure. Both prisms are made of the same glass of refractive index 1·6. Trace the path of the ray MN incident normally on face AB as it passes through the combination.



Check Solution → [2025 • Set 55-6-1]

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136. A ray of light is incident on face AB of a prism ABC with angle of prism A and emerges out from face AC. The prism is set in the position of minimum

deviation with angle of deviation δ. Find (i) the angle of incidence and (ii) the angle of refraction on face AB.

Check Solution → [2025 • Set 55-6-2]

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137. A tank is filled with a liquid of refractive index $\sqrt{2}$, up to a height of 30 cm. A tiny bulb is glowing at the bottom of the tank. Calculate the diameter of an opaque disc floating symmetrically on the liquid surface that can cut off completely the light from the bulb that comes out of the liquid surface.

Check Solution → [2025 • Set 55-6-3]

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138. **(a)** Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus."

OR

(b) A convex lens of focal length 10 cm, a concave lens of focal length 15 cm and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is +12 cm, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick?

Check Solution → [2025 • Set 55-7-1]

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139. Monochromatic light of frequency 5.0×10^{14} Hz passes from air into a medium of refractive index 1.5. Find the wavelength of the light (i) reflected, and (ii) refracted at the interface of the two media.

Check Solution → [2024 • Set 55-1-1]

140.

A plano-convex lens of focal length 16 cm is made of a material of refractive index 1.4. Calculate the radius of the curved surface of the lens.

Check Solution → [2024 • Set 55-1-1]

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141. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image.

Check Solution → [2024 • Set 55-1-1]

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142. A ray of light passes through a triangular prism. Show graphically, how the angle of deviation varies with the angle of incidence? Hence define the angle of minimum deviation.

Check Solution → [2024 • Set 55-1-1]

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143. A ray of light is incident normally on a refracting face of a prism of prism angle A and suffers a deviation of angle δ. Prove that the refractive index n of the material of the prism is given by $n = \frac{\sin(A+\delta)}{\sin A}$.

Check Solution → [2024 • Set 55-1-1]

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144. A convex lens (n = 1.52) has a focal length of 15.0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1.65. What will be the nature of the lens?

Check Solution → [2024 • Set 55-2-1]

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145. The magnifying power of an astronomical telescope is 24. In normal adjustment, distance between its two lenses is 150 cm. Find the focal length of the objective lens.

Check Solution → [2024 • Set 55-3-1]

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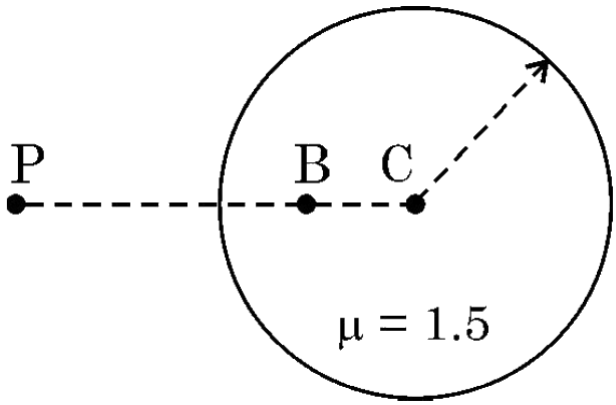
146. Explain the following:
(a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.
(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances?

Check Solution → [2024 • Set 55-3-1]

147. A ray of light is incident normally on one face of an equilateral glass prism of refractive index μ. When the prism is completely immersed in a transparent medium, it is observed that the emergent ray just grazes the adjacent face. Find the refractive index of the medium.

Check Solution → [2024 • Set 55-4-1]

148. An air bubble is trapped at point *B* (CB=20 cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point *P*.



Check Solution → [2024 • Set 55-4-1]

149. In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 *m*. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece lens.

Check Solution → [2024 • Set 55-4-1]

150. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.

Check Solution → [2024 • Set 55-5-1]

151. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

Check Solution → [2024 • Set 55-5-2]

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152. A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. How will the combination behave? Justify your answer.

Check Solution → [2024 • Set 55-5-3]

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153. Why is a reflecting telescope preferred over a refracting telescope? Justify your answer giving two reasons.

Check Solution → [2023 • Set 55-1-1]

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154. Briefly explain two applications of total internal reflection.

Check Solution → [2023 • Set 55-1-1]

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155. Define total internal reflection of light. Give two conditions for it.

Check Solution → [2023 • Set 55-1-1]

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156. A double-convex lens of power 4 D is manufactured from a glass of refractive index 1.5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature?

Check Solution → [2023 • Set 55-1-2]

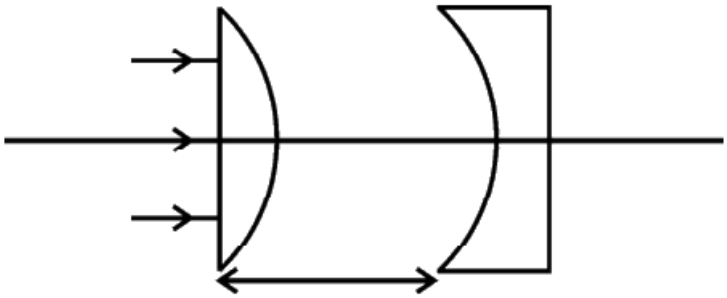
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157. The radii of curvature of the faces of a double-convex lens are 20 cm and 30 cm. Its power is $\frac{5}{3}$ D. What is the refractive index of the glass of the lens?

Check Solution → [2023 • Set 55-1-3]

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158. In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm each. The refractive index of the material of the lenses is 1.5. Find the final position of the image formed.



Check Solution → [2023 • Set 55-2-1]

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159. A point object in air is placed symmetrically at a distance of 60 cm in front of a concave spherical surface of refractive index 1.5. If the radius of curvature of the surface is 20 cm, find the position of the image formed.

Check Solution → [2023 • Set 55-2-2]

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160. The power of a thin lens is +5 *D*. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5.

Check Solution → [2023 • Set 55-2-3]

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161. The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface?

Check Solution → [2023 • Set 55-3-1]

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162. (c) The focal length of a concave lens of $\mu = 1.5$ is 20 cm in air. It is completely immersed in water of $\mu = \frac{4}{3}$. Calculate its focal length in water.

OR

(c) An object is placed in front of a lens which forms its erect image of magnification 3. The power of the lens is 5 D. Calculate the distance of the object and the image from the lens.

Check Solution → [2023 • Set 55-4-1]

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163. (iii) A double convex lens is made of glass of refractive index 1.55 with both faces of the same radius of curvature. Find the radius of curvature required if focal length is 20 cm.

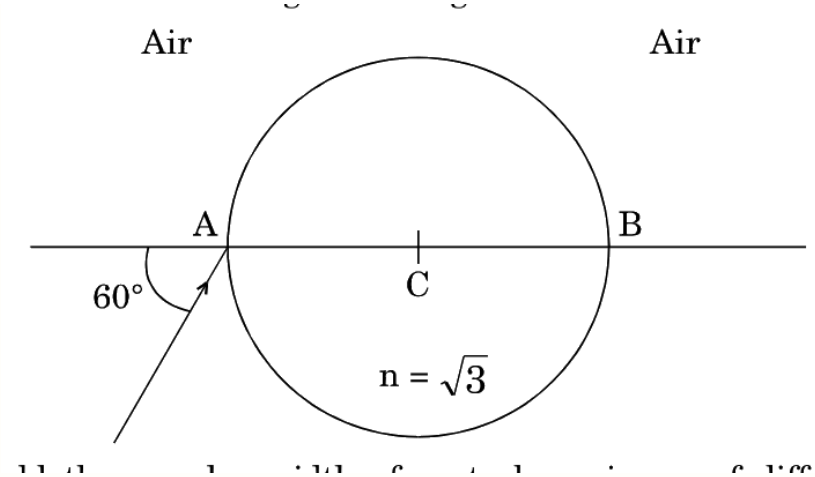
OR

(iii) Two convex lenses A and B of focal lengths 15 cm and 10 cm respectively are placed coaxially 'd' distance apart. A point object is kept at a distance of 30 cm in front of lens A. Find the value of 'd' so that the rays emerging from lens B are parallel to its principal axis.

Check Solution → [2023 • Set 55-5-1]

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164. A ray of light falls on a transparent sphere of $n = \sqrt{3}$ at an angle of incidence 60° with the diameter AB of the sphere having centre C. The ray emerges from the sphere parallel to the line AB. Find the angle of emergence.



Check Solution → [2021]

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165. Using lens maker's formula, derive the thin lens formula $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$ for a biconvex lens.

Check Solution → [2020 • Set 55-1-1]

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166. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed.

Check Solution → [2020 • Set 55-2-1]

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167. The focal length of an equiconcave lens is $\frac{3}{4}$ times of radius of curvature of its surfaces. Find the refractive index of the material of the lens. Under what condition will this lens behave as a converging lens?

Check Solution → [2020 • Set 55-2-2]

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168. An object is placed in front of a convex mirror of focal length 15 cm. It produces an image that is half the size of the object. Find (a) position of the object, (b) nature of the image, and (c) draw the ray diagram of image formation.

Check Solution → [2020 • Set 55-2-3]

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169. A converging lens of focal length f_1 is placed coaxially in contact with a diverging lens of focal length f_2 ($f_1 > f_2$). Determine the power and nature of the combination in terms of f_1 and f_2.

OR

How is the resolving power of a compound microscope affected if (a) wavelength of light used is decreased, and (b) the diameter of its objective lens is increased? Justify your answers.

Check Solution → **[2020 • Set 55-3-1]**

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170. Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of −5 *D*.

Check Solution → **[2019 • Set 55-1-1]**

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171. An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index $\frac{4\sqrt{2}}{5}$.

Check Solution → **[2019 • Set 55-1-1]**

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172. Under which conditions can a rainbow be observed? Distinguish between a primary and a secondary rainbow.

Check Solution → **[2019 • Set 55-2-1]**

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173. Explain the following:

(a) Sky appears blue.

(b) The Sun appears reddish at (i) sunset, (ii) sunrise.

Check Solution → **[2019 • Set 55-2-1]**

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174. Draw the ray diagram of an astronomical telescope showing image formation in the normal adjustment position. Write the expression for its magnifying power.

OR

Draw a labelled ray diagram to show image formation by a compound microscope and write the expression for its resolving power.

Check Solution → [2019 • Set 55-2-1]

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175. A beam of light converges at a point *P*. Draw ray diagrams to show where the beam will converge if (i) a convex lens, and (ii) a concave lens is kept in the path of the beam.

Check Solution → [2019 • Set 55-2-2]

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176. Under what conditions does the phenomenon of total internal reflection take place? Draw a ray diagram showing how a ray of light deviates by 90° after passing through a right-angled isosceles prism.

Check Solution → [2019 • Set 55-2-2]

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177. State, with the help of a ray diagram, the working principle of optical fibres. Write one important use of optical fibres.

Check Solution → [2019 • Set 55-2-3]

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178. For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

Check Solution → [2019 • Set 55-3-1]

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179. Show that the image formed by a convex mirror of an object is always virtual.

Check Solution → [2019 • Set 55-3-2]

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180. In the case of a concave mirror of focal length *f*, when an object is kept between *f* and 2f, show that its image is formed beyond 2f.

Check Solution → [2019 • Set 55-3-3]

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181. A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm from P. At what point does a beam converge if the convex lens has a focal length 10 cm?

OR

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

Check Solution → [2019 • Set 55-4-1]

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182. Out of the two optical instruments, a microscope and a telescope, which one plays the role in magnifying the objects and in resolving the two objects kept close to each other? Explain, giving example.

Check Solution → [2019 • Set 55-4-3]

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183. Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths? Explain.

Check Solution → [2017]

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184. Why should the objective of a telescope have large focal length and large aperture? Justify your answer.

Check Solution → [2017]

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185. When are two objects just resolved? Explain. How can the resolving power of a compound microscope be increased? Use relevant formula to support your answer.

Check Solution → [2017]

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186. Use the mirror equation to show that an object placed between *f* and 2f of a concave mirror forms an image beyond 2f.

OR

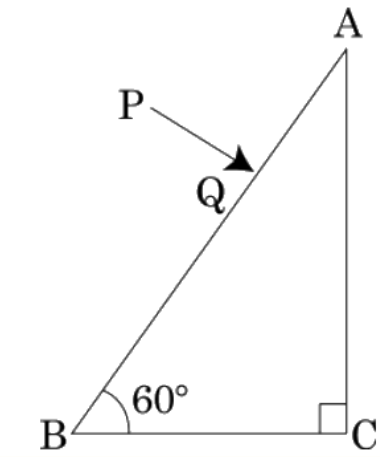
(a) State the condition under which a large magnification can be achieved in an astronomical telescope.

(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope.

Check Solution → [2017]

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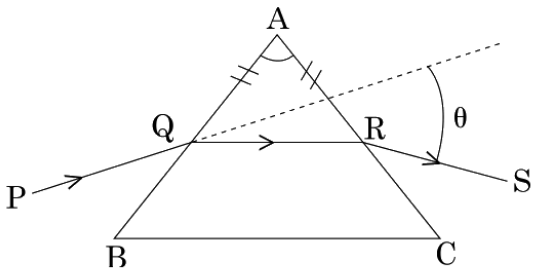
187. A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge? Justify your answer.



Check Solution → [2016]

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188. A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that AQ=AR. If the angle of prism *A*=60° and refractive index of material of prism is $\sqrt{3}$, calculate angle θ.



Check Solution → [2016]

189. A ray of light passes through an equilateral glass prism such that the angle of incidence is equal to the angle of emergence and each of these angles is equal to 3/4 of angle of prism. Find the angle of deviation.

Check Solution → [2015]

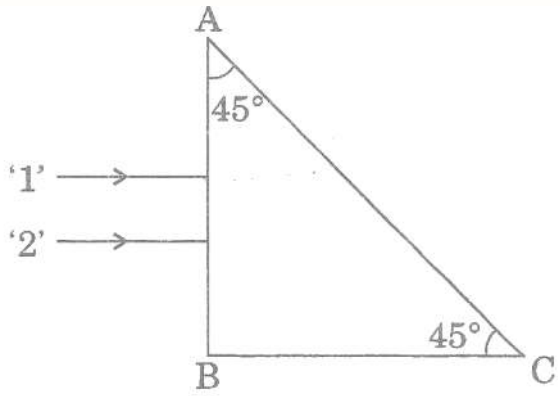
190. Calculate the speed of light in a medium whose critical angle is 45°.
Does critical angle for a given pair of media depend on the wavelength of incident light? Give reason.

Check Solution → [2015]

191. Use the mirror equation to show that an object placed between *f* and 2f of a concave mirror produces a real image beyond 2f.

Check Solution → [2015]

192. Two monochromatic rays of light are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays '1' and '2' are respectively 1.38 and 1.52. Trace the path of these rays after entering through the prism.



Check Solution → [2014]

193. A convex lens of focal length 25 cm is placed coaxially in contact with a concave lens of focal length 20 cm. Determine the power of the combination. Will the system be converging or diverging in nature?

Check Solution → [2013]

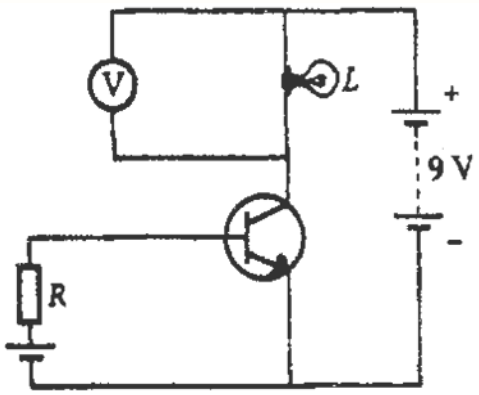
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194. (a) Write the necessary conditions for the phenomenon of total internal reflection to occur.
(b) Write the relation between the refractive index and critical angle for a given pair of optical media.

Check Solution → [2013]

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195. An object AB is kept in front of a concave mirror as shown in the figure.
(i) Complete the ray diagram showing the image formation of the object.
(ii) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black?



Check Solution → [2012]

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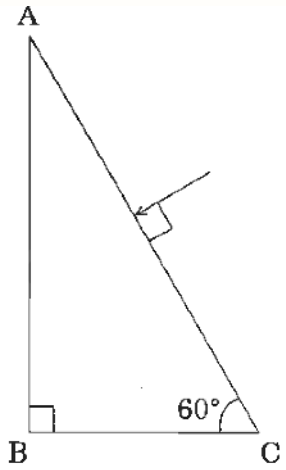
196. A ray of light, incident on an equilateral glass prism ($\mu_g = \sqrt{3}$) moves parallel to the base line of the prism inside it. Find the angle of incidence for this ray.

Check Solution → [2012]

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197. Trace the path of a ray of light passing through a glass prism (ABC) as shown in the figure. If the refractive index of glass is $\sqrt{3}$, find out the value of the angle of

emergence from the prism.



Check Solution → [2012]

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198. Two convex lenses of same focal length but of aperture A_1 and A_2 ($A_1 < A_2$), are used as the objective lenses in two astronomical telescopes having identical eyepieces. What is the ratio of their resolving power? Which telescope will you prefer and why? Give reason.

Check Solution → [2011 • Set 55-1-1]

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199. The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. If focal length of the lens is 12 cm, find the refractive index of the material of the lens.

Check Solution → [2010]

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200. (a) The bluish colour predominates in clear sky.
(b) Violet colour is seen at the bottom of the spectrum when white light is dispersed by a prism.
State reasons to explain these observations.

Check Solution → [2010]

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201. Define refractive index of a transparent medium.
A ray of light passes through a triangular prism. Plot a graph showing the

variation of the angle of deviation with the angle of incidence.

Check Solution → [2009]

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202. Draw a ray diagram of a reflecting telescope. State the advantages of this telescope over a refracting telescope.

Check Solution → [2008]

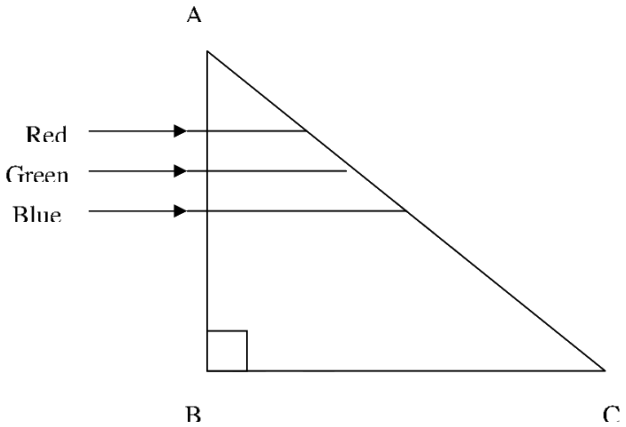
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203. A ray of light passing through an equilateral triangular glass prism from air undergoes minimum deviation when angle of incidence is $\frac{3}{4}$ of the angle of prism. Calculate the speed of light in the prism.

Check Solution → [2008]

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204. In the figure given below, light rays of blue, green, red wavelengths are incident on an isosceles right-angled prism. Explain with reason, which ray of light will be transmitted through the face AC. The refractive index of the prism for red, green, blue light are 1.39, 1.424, and 1.476 respectively.



Check Solution → [2008]

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205. A convex lens of refractive index 1.5 has a focal length of 18 cm in air. Calculate the change in its focal length when it is immersed in water of refractive index 4/3.

Check Solution → [2007]

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206. Define resolving power of a compound microscope. How does the resolving power of a compound microscope change when
(i) refractive index of the medium between the object and objective lens increases?
(ii) wavelength of the radiation used is increased?

Check Solution → [2007]

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207. Draw a labelled ray diagram of a reflecting type telescope. Write its any one advantage over refracting type telescope.

Check Solution → [2006]

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208. The image of a candle is formed by a convex lens on a screen. The lower half of the lens is painted black to make it completely opaque. Draw the ray diagram to show the image formation. How will this image be different from the one obtained when the lens is not painted black?

Check Solution → [2005]

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209. A compound microscope with an objective of 1.0 cm focal length and an eye-piece of 2.0 cm focal length has a tube length of 20 cm. Calculate the magnifying power of the microscope, if the final image is formed at the near point of the eye.

Or

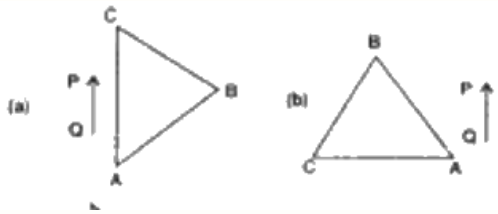
The magnifying power of an astronomical telescope in the normal adjustment position is 100. The distance between the objective and the eye-piece is 101 cm. Calculate the focal lengths of the objective and of the eye-piece.

Check Solution → [2004]

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210. An object is placed in front of a right angled prism ABC in two positions (a) and (b) as shown. The prism is made of crown glass with critical angle of 41°. Trace

the path of two rays from P and Q, (i) in (a), normal to the hypotenuse and (ii) in (b), parallel to the hypotenuse.



Check Solution → [2003]

SECTION C

3-Mark Questions

SA · 86 questions

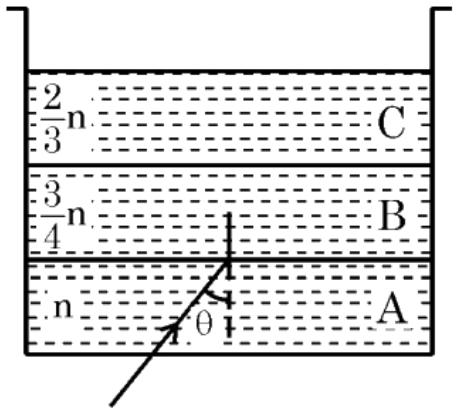
> *Section C of the paper, 3 marks each. Sorted newest year first.*

211. **(a)** Draw the ray diagram to show the image formation by a refracting telescope and write the expression for angular magnification for the telescope in normal adjustment.
(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope.

Check Solution → **[2026 • Set 55-1-1]**

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212. **(a)** State the two conditions under which total internal reflection occurs.
(b) A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n, $\frac{3}{2}$n and $\frac{2}{3}$n, respectively. A laser beam is incident at the interface between A and B at an angle θ as shown in figure. Prove that the beam does not enter region C at all for $\theta > \frac{2}{3}$.



Check Solution → **[2026 • Set 55-1-1]**

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213. Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case.

Check Solution → [2026 • Set 55-1-2]

214. (a) Why does one prefer to view the image formed at infinity than that formed at near point in microscope/telescope?
(b) Consider lenses L_1, L_2 and L_3 as specified in the following table. Which of them will you select as objective and eyepiece for constructing best possible (i) telescope (ii) compound microscope? Give reason for your answer.Lens$L_1$$L_2$$L_3$Power6 D3 D10 DAperture1 cm8 cm1 cm

Check Solution → [2026 • Set 55-1-3]

215. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface.

Check Solution → [2026 • Set 55-3-1]

216. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the:
(a) angle of minimum deviation for the prism, and
(b) refractive index of the material of the prism.

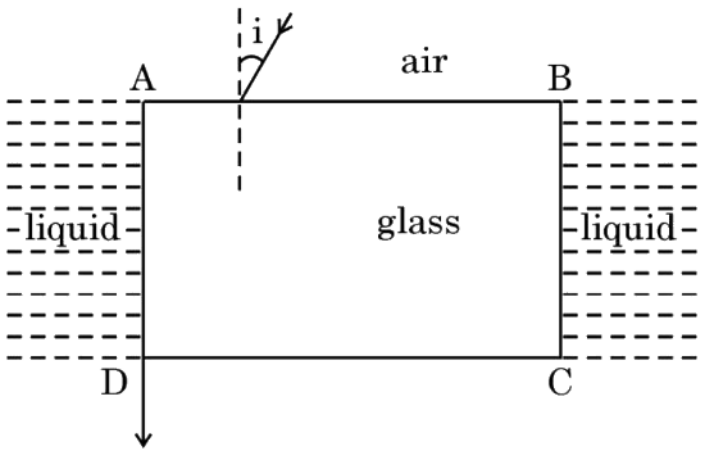
Check Solution → [2026 • Set 55-3-3]

217. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam?
(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.
(c) Explain the transmission of optical signal through an optical fibre by a diagram.

Check Solution → [2025 • Set 55-2-1]

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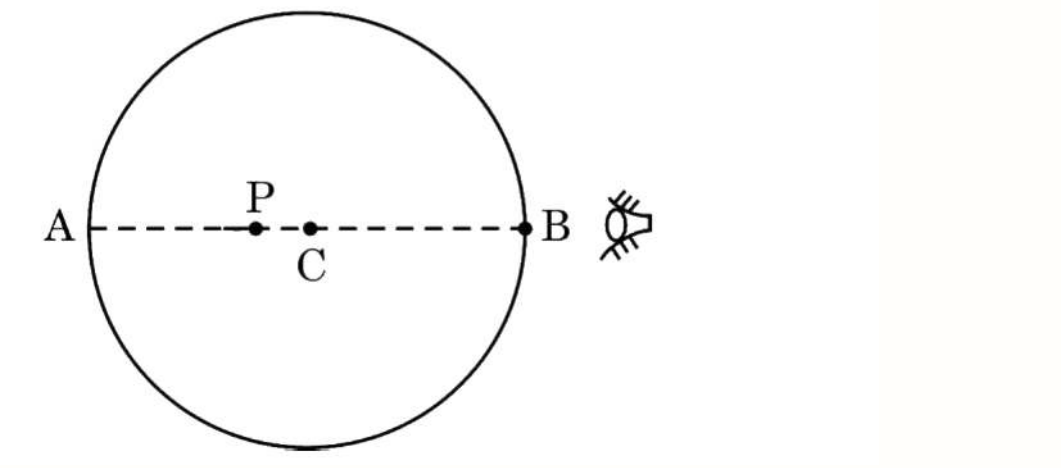
218. A rectangular glass slab ABCD (refractive index 1.5) is surrounded by a transparent liquid (refractive index 1.25) as shown in the figure. A ray of light is incident on face AB at an angle i such that it is refracted out grazing the face AD. Find the value of angle i.



Check Solution → [2025 • Set 55-5-1]

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219. An air bubble is trapped at point P (CP = 1.75 cm) in a spherical glass ball (n = 1.5) of radius 7 cm as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram.

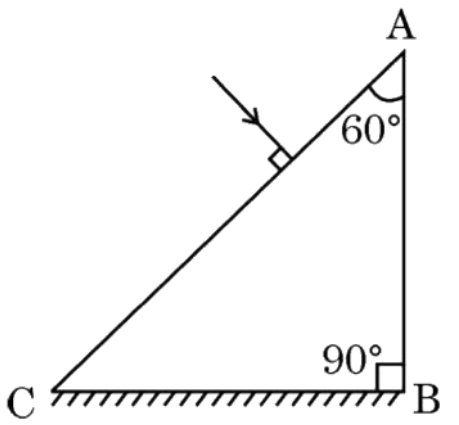


Check Solution →

[2025 • Set 55-5-2]

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220. A right-angled prism ABC (refractive index $\sqrt{2}$) is kept on a plane mirror as shown in the figure. A ray of light is incident normally on the face AC.



(a) Trace the path of the ray as it passes through the prism.
(b) Find the angle of deviation produced by the prism.

Check Solution →

[2025 • Set 55-5-3]

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221. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula.

Check Solution →

[2025 • Set 55-6-1]

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222. A ray of light is incident at an angle i on a parallel sided glass slab of thickness 'd' and gets refracted into the slab at angle r. Draw a ray diagram to show its path as it emerges out of the slab. Hence, obtain an expression for the lateral shift of the ray. Under what condition will the shift be minimum?

Check Solution → [2025 • Set 55-6-2]

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223. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification −2. Also find the location of the image.
(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object? Justify your answer.

Check Solution → [2025 • Set 55-6-3]

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224. Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium.

Check Solution → [2023 • Set 55-2-1]

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225. A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle.

Check Solution → [2023 • Set 55-2-2]

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226. A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δ_m.

Check Solution → [2023 • Set 55-2-3]

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227. (i) Trace the path of a ray of light PQ which is incident at an angle i on one face of a glass prism of angle A. It then emerges out from the other face at an angle e.

Use the ray diagram to prove that the angle through which the ray is deviated is given by $\angle \delta = \angle i + \angle e - \angle A$.
(ii) What will be the minimum value of δ if the ray passes symmetrically through the prism?

Check Solution → [2022 • Set 55-1-1]

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228. A point source in air is kept 24 cm in front of a concave spherical glass surface ($\mu_g = 1.5$) and radius of curvature 60 cm. Find the nature of the image formed and its distance from the point source.

Check Solution → [2022 • Set 55-1-1]

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229. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres.

Check Solution → [2022 • Set 55-1-1]

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230. A thin equiconvex lens of radius of curvature R made of material of refractive index μ_1 is kept coaxially, in contact with an equiconcave lens of the same radius of curvature and refractive index μ_2 ($\mu_2 > \mu_1$). Find:
(i) the ratio of their powers, and
(ii) the power of the combination and its nature.

Check Solution → [2022 • Set 55-1-1]

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231. Find the two possible positions of an object kept in front of a lens of +5.0 D, so that the image formed in both cases is four times magnified.

Check Solution → [2022 • Set 55-1-2]

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232. For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water?

Check Solution → [2022 • Set 55-1-3]

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233. (i) State the conditions for total internal reflection to take place.
(ii) A tank is filled with a transparent liquid to height 'H'. A coin suspended by a thread in the liquid is gradually lowered till it touches the bottom. The apparent depth is determined corresponding to different positions of the coin.
(a) Plot a graph showing variation of the apparent depth with the real depth of the coin.
(b) What is the physical significance of the slope of the graph?

Check Solution → [2022 • Set 55-2-1]

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234. Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power.

Check Solution → [2022 • Set 55-2-1]

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235. A converging lens made of glass ($\mu = 1.5$) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length
(a) in air, and
(b) when it is immersed in a liquid of refractive index 1.25.

Check Solution → [2022 • Set 55-2-1]

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236. (a) Is the speed of light in glass independent of the colour of light? Give reason.
(b) A small bulb is placed at the bottom of a tank containing water to a depth of 70 cm. Find the area of the surface of water through which light from the bulb can emerge out. Given refractive index of water is $\frac{4}{3}$.

Check Solution → [2022 • Set 55-2-2]

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237. The refractive index of an equilateral triangular prism kept in air is $\sqrt{2}$.
Calculate:
(a) the angle of minimum deviation.
(b) the critical angle for the prism.

Check Solution → [2022 • Set 55-2-2]

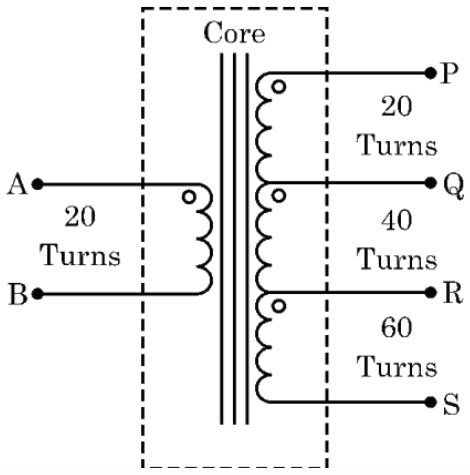
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238. An equiconvex lens forms a two times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept 16 cm in front of one of these lenses, then find the nature and position of the image formed.

Check Solution → [2022 • Set 55-2-3]

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239. A ray is incident on a prism of material of refractive index $\sqrt{2}$ at point M such that it grazes along NC after emerging from the prism, as shown in the figure.



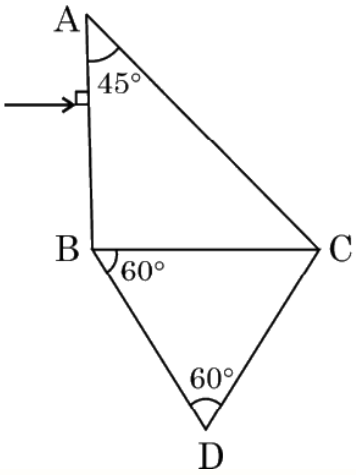
Find:

(a) the critical angle for the prism.
(b) the angle of refraction at face AB.

Check Solution → [2022 • Set 55-2-3]

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240. (a) Write two necessary conditions for total internal reflection.
(b) Two prisms ABC and DBC are arranged as shown in figure.

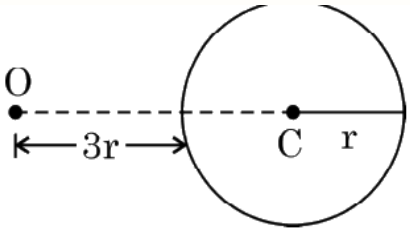


The critical angles for the two prisms with respect to air are 41.1° and 45° respectively. Trace the path of the ray through the combination.

Check Solution → [2022 • Set 55-3-1]

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241. (a) An object is placed in front of a converging lens. Obtain the conditions under which the magnification produced by the lens is (i) negative and (ii) positive.
(b) A point object is placed at O in front of a glass sphere as shown in figure.

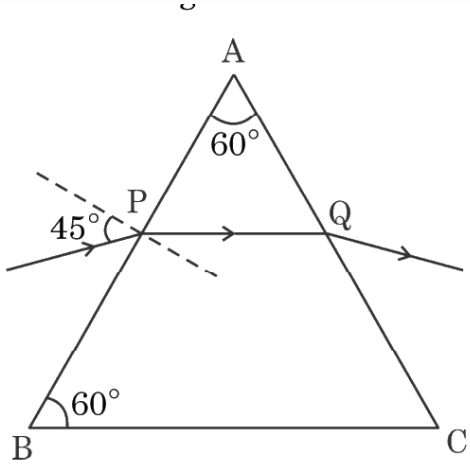


Show the formation of image by the sphere.

Check Solution → [2022 • Set 55-3-1]

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242. A ray of light is incident on a prism at an angle of 45° and passes symmetrically as shown in the figure. Calculate:
(a) the angle of minimum deviation,
(b) the refractive index of the material of the prism, and
(c) the angle of refraction at the point P.



Check Solution → [2022 • Set 55-4-1]

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243. (i) Draw a labelled ray diagram showing the formation of the image at infinity by an astronomical telescope.
(ii) A telescope consists of an objective of focal length 150 cm and an eyepiece of focal length 6.0 cm. If the final image is formed at infinity, then calculate:
(I) the length of the tube in this adjustment, and
(II) the magnification produced.

Check Solution → [2022 • Set 55-4-1]

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244. (i) Draw a labelled ray diagram showing the formation of the image at least distance of distinct vision by a compound microscope.
(ii) A small object is placed at a distance of 3.0 cm from a magnifier of focal length 4.0 cm. Find:
(I) the position of the image formed, and
(II) the linear magnification produced.

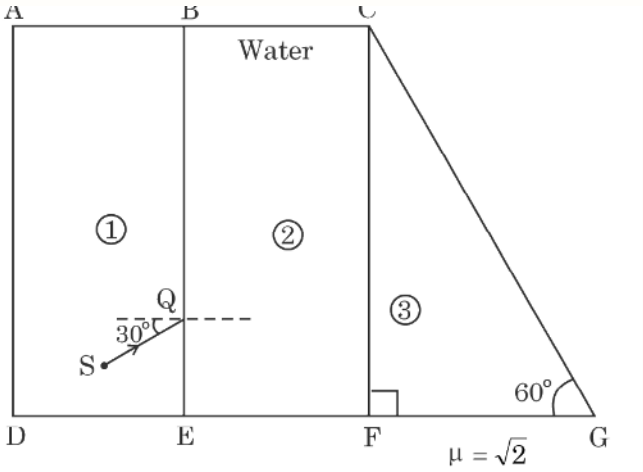
Check Solution → [2022 • Set 55-4-1]

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245. The figure shows a water column BCFE surrounded by two media (1) and (2) of the same refractive index $\sqrt{2}$. A ray of light from a point source S is incident on surface BE at an angle of 30°.

(a) Trace the path of the ray through media (1) and (2) as it emerges out of face CG.

(b) Find the angle of emergence at face CG.



Check Solution → [2022 • Set 55-4-2]

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246. A diver looking up through water ($\mu = \frac{5}{3}$) sees the outside world contained in a circular area on the surface of water. If the diver's eyes are $\sqrt{7}$ m below the surface of water, then calculate the area of the circle.

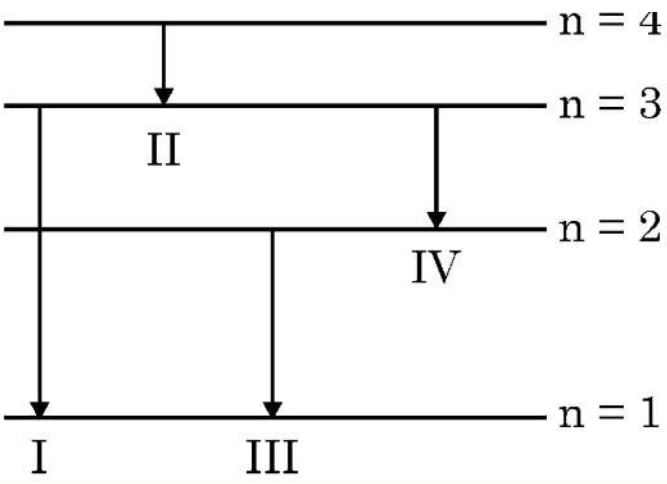
Check Solution → [2022 • Set 55-4-3]

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247. A ray of light passes through a prism of refractive index $\sqrt{2}$ as shown in the figure. Find:

(i) The angle of incidence ($\angle r_2$) at face AC.

(ii) The angle of minimum deviation for this prism.



Check Solution → [2022 • Set 55-5-1]

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248. With the help of a ray diagram explain the working of a reflecting telescope. Mention two advantages of a reflecting telescope over a refracting telescope.

Check Solution → [2022 • Set 55-5-1]

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249. (i) Define SI unit of power of a lens.
(ii) A plano convex lens is made of glass of refractive index 1.5. The radius of curvature of the convex surface is 25 cm. Calculate the focal length of the lens.
(iii) If an object is placed 50 cm in front of the lens, find the nature and position of the image formed.

Check Solution → [2022 • Set 55-5-1]

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250. With the help of a ray diagram, show how a compound microscope forms a magnified image of a tiny object, at least distance of distinct vision. Hence derive an expression for the magnification produced by it.

Check Solution → [2022 • Set 55-5-3]

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251. What is the difference in the construction of an astronomical telescope and a compound microscope? The focal lengths of the objective and eyepiece of a compound microscope are 1.25 cm and 5.0 cm, respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of 30 when the final image is formed at the near point.

Check Solution → [2020 • Set 55-2-1]

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252. (a) Using the necessary ray diagram, derive the mirror formula for a concave mirror.
(b) In the magnified image of a measuring scale (with equidistant markings) lying along the principal axis of a concave mirror, the markings are not equidistant. Explain.

Check Solution → [2020 • Set 55-3-1]

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253. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror.

Check Solution → [2020 • Set 55-4-1]

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254. An object is placed in front of a concave mirror of focal length 12 cm. There are two possible positions of the object for which the image formed is three times the size of the object.
(a) Draw the ray diagram for each case, and
(b) Find the distance between the two positions of the object.

Check Solution → [2020 • Set 55-4-2]

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255. A concave mirror forms a real image of an object kept at a distance 9 cm from it. If the object is taken away from the mirror by 6 cm, the image size reduces to $\frac{4}{9}$ th of its previous size. Find the focal length of the mirror.

Check Solution → [2020 • Set 55-4-3]

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256. An optical instrument uses a lens of power 100 D for objective lens and 50 D for its eyepiece. When the tube length is kept at 25 cm, the final image is formed at infinity.

(a) Identify the optical instrument.
(b) Calculate the magnification produced by the instrument.

Check Solution → [2020 • Set 55-5-1]

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257. An optical instrument uses an objective lens of power 100 D and an eyepiece of power 40 D. The final image is formed at infinity when the tube length of the instrument is kept at 20 cm.
(a) Identify the optical instrument.
(b) Calculate the angular magnification produced by the instrument.

Check Solution → [2020 • Set 55-5-3]

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258. Draw a labelled ray diagram of an astronomical telescope in the near point adjustment position.
A giant refracting telescope at an observatory has an objective lens of focal length 15 m and an eyepiece of focal length 1.0 cm. If this telescope is used to view the Moon, find the diameter of the image of the Moon formed by the objective lens. The diameter of the Moon is 3.48×10^6 m, and the radius of lunar orbit is 3.8×10^8 m.

Check Solution → [2019 • Set 55-1-1]

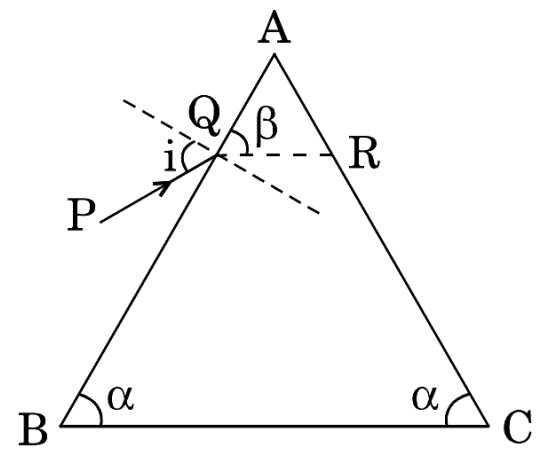
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259. A triangular prism of refracting angle 60° is made of a transparent material of refractive index $2/\sqrt{3}$. A ray of light is incident normally on the face KL as shown in the figure. Trace the path of the ray as it passes through the prism and calculate the angle of emergence and angle of deviation.

Check Solution → [2019 • Set 55-2-1]

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260. A ray of light incident on the face AB of an isosceles triangular prism makes an angle of incidence (i) and deviates by angle β as shown in the figure. Show that in the position of minimum deviation $2\beta = \alpha$. Also find out the condition when the refracted ray QR suffers total internal reflection.



Check Solution → [2019 • Set 55-2-2]

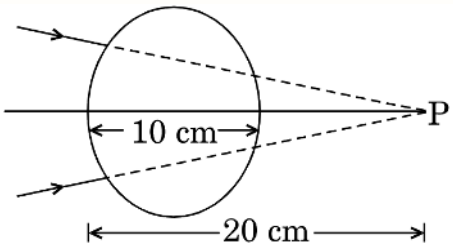
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261. (a) When a convex lens of focal length 30 cm is in contact with a concave lens of focal length 20 cm, find out if the system is converging or diverging.
(b) Obtain the expression for the angle of incidence of a ray of light which is incident on the face of a prism of refracting angle A so that it suffers total internal reflection at the other face. (Given the refractive index of the glass of the prism is μ.)

Check Solution → [2019 • Set 55-2-3]

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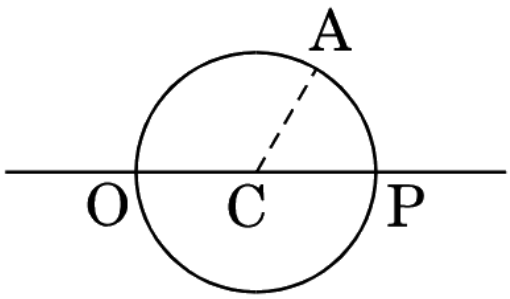
262. A converging beam of light travelling in air converges at a point P as shown in the figure. When a glass sphere of refractive index 1.5 is introduced in between the path of the beam, calculate the new position of the image. Also draw the ray diagram for the image formed.



OR

A point 'O' marked on the surface of a glass sphere of diameter 20 cm is viewed through glass from the position directly opposite to the point O. If the refractive

index of the glass is 1.5, find the position of the image formed. Also, draw the ray diagram for the formation of the image.



Check Solution → [2019 • Set 55-3-1]

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263. Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

OR

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

Check Solution → [2019 • Set 55-4-1]

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264. A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

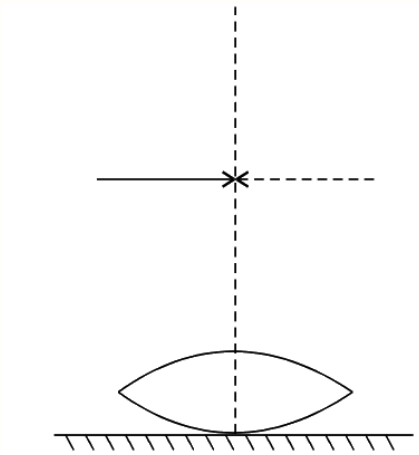
OR

A convex lens of focal length 20 cm and a concave lens of focal length 15 cm are kept 30 cm apart with their principal axes coincident. When an object is placed 30 cm in front of the convex lens, calculate the position of the final image formed by the combination. Would this result change if the object were placed 30 cm in front of the concave lens? Give reason.

Check Solution → [2019 • Set 55-5-1]

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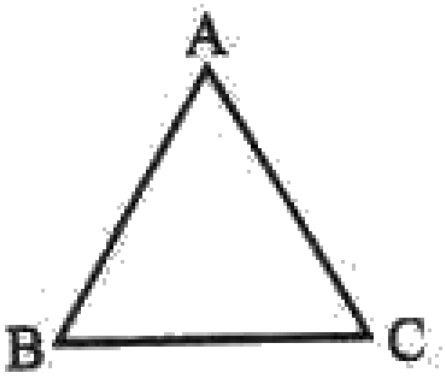
265. A symmetric biconvex lens of radius of curvature R and made of glass of refractive index 1.5, is placed on a layer of liquid placed on top of a plane mirror as shown in the figure. An optical needle with its tip on the principal axis of the lens is moved along the axis until its real, inverted image coincides with the needle itself. The distance of the needle from the lens is measured to be x. On removing the liquid layer and repeating the experiment, the distance is found to be y. Obtain the expression for the refractive index of the liquid in terms of x and y.



Check Solution → [2018]

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266. (i) A ray of light incident on face AB of an equilateral glass prism, shows minimum deviation of 30°. Calculate the speed of light through the prism.



(ii) Find the angle of incidence at face AB so that the emergent ray grazes along the face AC.

Check Solution → [2017]

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267. (a) Monochromatic light of wavelength 589 nm is incident from air on a water surface. If μ for water is 1.33, find the wavelength, frequency and speed of the refracted light.
(b) A double convex lens is made of a glass of refractive index 1.55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm.

Check Solution → [2017]

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268. (a) Draw a ray diagram depicting the formation of the image by an astronomical telescope in normal adjustment.
(b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? Give reason.

Lenses	Power (D)	Aperture (cm)
L_1	3	8
L_2	6	1
L_3	10	1

Check Solution → [2017]

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269. (a) Draw a ray diagram showing the formation of image by a reflecting telescope.
(b) Write two advantages of a reflecting telescope over a refracting telescope.

Check Solution → [2017]

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270. (a) Draw a ray diagram for the formation of image by a compound microscope.
(b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct a compound microscope?

(c) Define resolving power of a microscope and write one factor on which it depends.

Check Solution → [2017]

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271. (i) What is total internal reflection? Under what conditions does it occur?
(ii) Find a relation between critical angle and refractive index.
(iii) Name one phenomenon which is based on total internal reflection.

Check Solution → [2016]

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272. (a) Draw a schematic diagram of a reflecting telescope.
(b) State the advantages of reflecting telescope over refracting telescope.
OR
(i) Draw a schematic ray diagram of a compound microscope when image is formed at distance of distinct vision.
(ii) Write the expression for resolving power of a compound microscope. How can the resolving power of a microscope be increased?

Check Solution → [2016]

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273. (i) A screen is placed at a distance of 100 cm from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by 20 cm. Calculate the focal length of the lens used.
(ii) A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal length. What is the focal length of the combination?

Check Solution → [2016]

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274. (i) For a glass prism ($\mu=\sqrt{3}$) the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.
(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu=\sqrt{3}$.

Check Solution → [2016]

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275. (i) Name the phenomenon on which the working of an optical fibre is based.
(ii) What are the necessary conditions for this phenomenon to occur?
(iii) Draw a labelled diagram of an optical fibre and show how light propagates through the optical fibre using this phenomenon.

Check Solution → [2016]

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276. (a) Give two reasons to explain why reflecting telescopes are preferred over refracting type.
(b) Use mirror equation to show that convex mirror always produces a virtual image independent of the location of the object.

Check Solution → [2015]

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277. (i) A giant refracting telescope has an objective lens of focal length 15 m. If an eye piece of focal length 1.0 cm is used, what is the angular magnification of the telescope?
(ii) If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.48×10^6 m and the radius of lunar orbit is 3.8×10^8 m.

Check Solution → [2015]

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278. (a) Draw a labelled ray diagram showing the formation of a final image by a compound microscope at least distance of distinct vision.
(b) The total magnification produced by a compound microscope is 20. The magnification produced by the eye piece is 5. The microscope is focussed on a certain object. The distance between the objective and eyepiece is observed to be 14 cm. If least distance of distinct vision is 20 cm, calculate the focal length of the objective and the eye piece.

Check Solution → [2014]

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279. (a) A mobile phone lies along the principal axis of a concave mirror. Show, with the help of a suitable diagram, the formation of its image. Explain why

magnification is not uniform.
(b) Suppose the lower half of the concave mirror's reflecting surface is covered with an opaque material. What effect this will have on the image of the object ? Explain.

Check Solution → [2014]

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280. A convex lens of focal length 20 cm is placed coaxially with a convex mirror of radius of curvature 20 cm. The two are kept 15 cm apart. A point object is placed 40 cm in front of the convex lens. Find the position of the image formed by this combination. Draw the ray diagram showing the image formation.

Check Solution → [2014]

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281. Three rays of light - red (R), green (G) and blue (B) - are incident on the face AB of a right-angled prism ABC. The refractive indices of the material of the prism for red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. Trace the path of the rays through the prism.
How will the situation change if these rays were incident normally on one of the faces of an equilateral prism?

Check Solution → [2013]

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282. (i) Draw a neat labelled ray diagram of an astronomical telescope in normal adjustment. Explain briefly its working.
(ii) An astronomical telescope uses two lenses of powers 10 D and 1 D. What is its magnifying power in normal adjustment?
OR
(i) Draw a neat labelled ray diagram of a compound microscope. Explain briefly its working.
(ii) Why must both the objective and the eye-piece of a compound microscope have short focal lengths?

Check Solution → [2012]

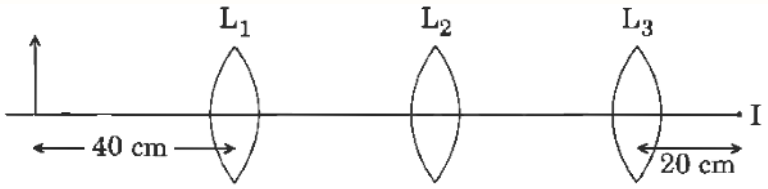
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283. An illuminated object and a screen are placed 90 cm apart. Determine the focal length and nature of the lens required to produce a clear image on the screen, twice the size of the object.

Check Solution → [2012]

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284. You are given three lenses L_1, L_2 and L_3 each of focal length 20 cm. An object is kept at 40 cm in front of L_1, as shown. The final real image is formed at the focus '*I*' of L_3. Find the separations between L_1, L_2 and L_3.



Check Solution → [2012]

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285. Define power of a lens. Write its units. Deduce the relation $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ for two thin lenses kept in contact coaxially.

Check Solution → [2012]

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286. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in (i) a medium of refractive index 1.65, (ii) a medium of refractive index 1.33.
(a) Will it behave as a converging or a diverging lens in the two cases?
(b) How will its focal length change in the two media?

Check Solution → [2011]

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287. Use the mirror equation to show that
(a) an object placed between *f* and 2f of a concave mirror produces a real image beyond 2f.
(b) a convex mirror always produces a virtual image independent of the location of the object.

(c) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image.

Check Solution → [2011]

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288. A compound microscope uses an objective lens of focal length 4 cm and eyepiece lens of focal length 10 cm. An object is placed at 6 cm from the objective lens. Calculate the magnifying power of the compound microscope. Also calculate the length of the microscope.

Check Solution → [2011]

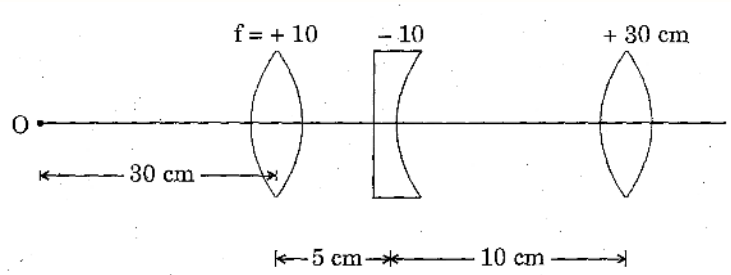
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289. A giant refracting telescope at an observatory has an objective lens of focal length 15 m. If an eyepiece lens of focal length 1.0 cm is used, find the angular magnification of the telescope.
If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.42×10^6 m and the radius of the lunar orbit is 3.8×10^8 m.

Check Solution → [2011]

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290. Find the position of the image formed of the object 'O' by the lens combination given in the figure.

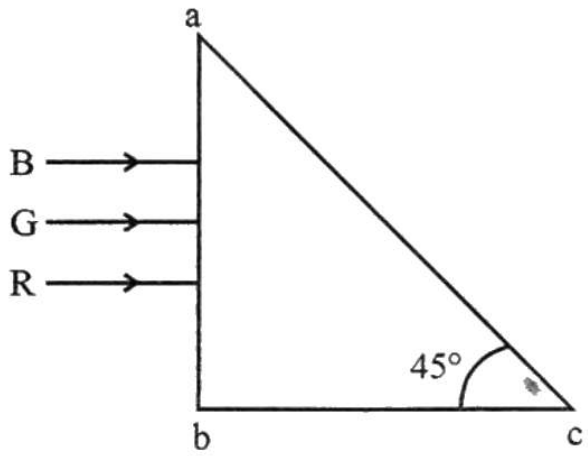


Check Solution → [2011 • Set 55-2-1]

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291. Three light rays red (R), green (G) and blue (B) are incident on a right angled prism abc at face ab. The refractive indices of the material of the prism for red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. Out of the three

which colour ray will emerge out of face ac? Justify your answer. Trace the path of these rays after passing through face ab.



Check Solution → [2009]

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292. Draw a plot showing the variation of power of a lens with the wavelength of the incident light.
A diverging lens of refractive index 1.5 and of focal length 20 cm in air has the same radii of curvature for both sides. If it is immersed in a liquid of refractive index 1.7, calculate the focal length of the lens in the liquid.

Check Solution → [2008]

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293. Define the term 'resolving power' of an astronomical telescope. How does it get affected on
(i) increasing the aperture of the objective lens?
(ii) increasing the wavelength of the light used?
Justify your answer in each case.

Check Solution → [2007]

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294. A double convex lens made of a glass of refractive index 1.6 has its both surfaces of equal radii of curvature of 30 cm each. An object of height 5 cm is placed at a distance of 12.5 cm from the lens. Calculate the size of the image formed.

Check Solution → [2007]

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295. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in:
(i) medium A of refractive index 1.65
(ii) medium B of refractive index 1.33
Explain, giving reasons, whether it will behave as a converging lens or a diverging lens in each of these two media.

Check Solution → [2006]

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296. A figure divided into squares, each of size 1 mm^2, is being viewed at a distance of 9 cm through a magnifying lens of focal length 10 cm, held close to the eye.
(a) Draw a ray diagram showing the formation of the image.
(b) What is the magnification produced by the lens? How much is the area of each square in the virtual image?
(c) What is the angular magnification of the lens?

Check Solution → [2005]

SECTION D

4-Mark Questions

Case Study · 3 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

297. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

Check Solution → **[2026 • Set 55-4-1]**

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298. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Considering image formation by a single spherical surface successively at the two surfaces of a lens, lens maker's formula is obtained. It is useful to design lenses of desired focal length using surfaces of suitable radii of curvature. This formula helps us obtain a relation between u, v and f for a lens. Lenses form images of objects and they are used in a number of optical devices, for example microscopes and telescopes.

Check Solution → **[2023 • Set 55-5-1]**

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299. Mrs. Rashmi Singh broke her reading glasses. When she went to the shopkeeper to order new spects, he suggested that she should get spectacles with plastic lenses instead of glass lenses. On getting the new spectacles, she found that the new ones were thicker than the earlier ones. She asked this question to the shopkeeper but he could not offer satisfactory explanation for this. At home, Mrs. Singh raised the same question to her daughter Anuja who explained why plastic lenses were thicker.

(a) Write two qualities displayed each by Anuja and her mother.

(b) How do you explain this fact using lens maker's formula?

Check Solution → [2017]

SECTION E

5-Mark Questions

Long Answer · 58 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

300. **(a)** Using the relation for refraction at a curved spherical surface, derive the expression for lens maker's formula.
(b) Three lenses L_1, L_2 and L_3, each of focal length 40 cm, are placed coaxially. The distance between L_1 and L_2 and between L_2 and L_3 are 120 cm and 20 cm respectively. An object is kept at a distance of 80 cm to the left of lens L_1. Find the distance of the final image formed from the object.
OR
(a) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. Using this diagram, derive the mirror formula.
(b) A concave mirror produces a two times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.

Check Solution → **[2026 • Set 55-1-1]**

301.
(a) Draw a labelled ray diagram of a refracting telescope when it forms image of a distant object at infinity. Derive expression for its magnifying power.

Check Solution → **[2026 • Set 55-2-1]**

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302. (b) (i) In a telescope the objective has much larger aperture than the eyepiece. Why?
(ii) Write two advantages of reflecting telescope over refracting telescope.

Check Solution → **[2026 • Set 55-2-1]**

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303. (i) A point object is kept in front of a convex spherical surface of radius of curvature *R*. Draw the ray diagram to show the formation of image and derive

the relation between the object and image distance (u and v) in terms of refractive index n of the medium and R.

Check Solution → [2026 • Set 55-3-1]

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304. (ii) A convex lens of focal length of 20 cm is used to form the image of an object placed 30 cm away from the lens. Find the position and nature of the image formed.

Check Solution → [2026 • Set 55-3-1]

305.

(i) Two thin converging lenses of focal length f_1 and f_2 are placed coaxially in contact. Derive expression for the focal length of the combination.

Check Solution → [2026 • Set 55-3-1]

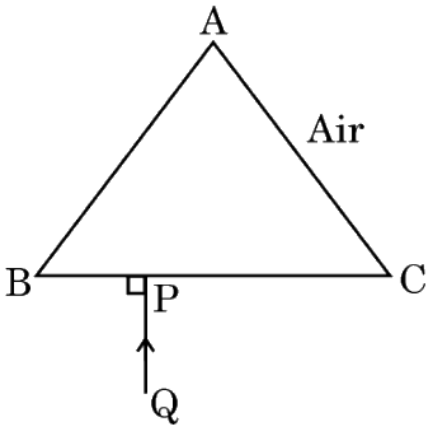
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306. (a)

(i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum deviation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation.



OR

(b)

(i) What is the difference between a ray and a wavefront?
(ii) A plane wave is incident on a reflecting surface. Using Huygens principle, show how it is reflected from the surface. Hence, verify the law of reflection.
(iii) Depict refraction of a plane wave by a convex lens.

Check Solution → [2026 • Set 55-4-1]

307.

(b) (i) With the help of a ray diagram, describe the construction and working of a compound microscope.
(ii) (I) The real image of an object placed between *f* and 2f from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there? Explain.
(II) Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances? Explain.

Check Solution → [2026 • Set 55-5-1]

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308. (i) A thin pencil of length (*f*/4) is placed coinciding with the principal axis of a mirror of focal length f. The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror.
(ii) A ray of light is incident on a refracting face AB of a prism ABC at an angle of 45°. The ray emerges from face AC and the angle of deviation is 15°. The angle of prism is 30°. Show that the emergent ray is normal to the face AC from which it emerges out. Find the refraction index of the material of the prism.

Check Solution → [2025 • Set 55-2-1]

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309. (a)

(i) An object is placed 30 cm from a thin convex lens of focal length 10 cm. The lens forms a sharp image on a screen. If a thin concave lens is placed in contact with the convex lens, the sharp image on the screen is formed when the screen is moved by 45 cm from its initial position. Calculate the focal length of the concave lens.
(ii) Calculate the angle of minimum deviation of an equilateral prism. The

refractive index of the prism is $\sqrt{3}$. Calculate the angle of incidence for this case of minimum deviation also.

OR

(b)

(i) A physics teacher wants to demonstrate interference with the help of double slit experiment using a laser beam of 633 nm wavelength. Since the hall is large enough, interference pattern is formed on the wall 5.0 *m* from the slits. For clear and comfortable view by all the students they want the fringe width 5 mm.

(I) Find the slit separation for obtaining the desired interference pattern.

(II) How far will the first minimum be from the central maximum ?

(ii) A parallel beam of light of wavelength 650 nm passes through a slit of width 0.6 mm. The diffraction pattern is obtained on a screen kept 60 cm away from the slit. Find the distance between first order minima on both sides of the central maximum.

Check Solution → [2025 • Set 55-4-1]

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310. **(a) (i)** Draw a ray diagram of a reflecting telescope (Cassegrain) and explain the formation of image. State two important advantages that a reflecting telescope has over a refracting telescope.

(ii) In a refracting telescope, the focal length of the objective is 50 times the focal length of the eyepiece. When the final image is formed at infinity, the length of the tube is 102 cm. Find the focal lengths of the two lenses.

OR

(b) (i) Write any two advantages of a compound microscope over a simple microscope. Draw a ray diagram for the image formation at the near point by a compound microscope and explain it.

(ii) A thin planoconcave lens with its curved face of radius of curvature *R* is made of glass of refractive index n_1. It is placed coaxially in contact with a thin equiconvex lens of same radius of curvature of refractive index n_2. Obtain the power of the combination lens.

Check Solution → [2025 • Set 55-5-1]

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311. (a) (i) Draw a ray diagram to show the image formation by a compound microscope. Obtain the expression for the total magnification of the microscope

when the final image is formed at infinity.
(ii) In a compound microscope, an object is placed at a distance of 1·5 cm from the objective of focal length 1·25 cm. The eyepiece has a focal length of 5 cm. The final image is formed at infinity. Calculate the distance between the objective and the eyepiece.

OR

(b) (i) Using Huygens' principle, explain the refraction of a plane wavefront, propagating in air, at a plane interface between air and glass. Hence verify Snell's law.
(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

Check Solution → **[2025 • Set 55-6-1]**

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312. **(a) (i)** Explain with the help of a labelled ray diagram the formation of final image by an astronomical telescope at infinity. Write the expression for its magnifying power.
(ii) The total magnification produced by a compound microscope is 20. The magnification produced by the eyepiece is 5. When the microscope is focussed on a certain object, the distance between the objective and eyepiece is observed to be 14 cm. Calculate the focal lengths of the objective and the eyepiece. (Given that the least distance of distinct vision = 25 cm)

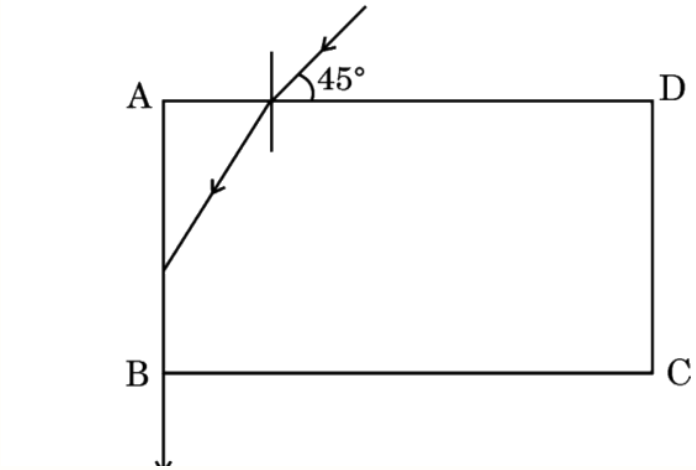
OR

(b) (i) Two coherent light waves, each of intensity I_0 superpose each other and produce interference pattern on a screen. Obtain the expression for the resultant intensity at a point where the phase difference between the waves is ϕ. Write its maximum and minimum possible values.
(ii) In a single slit diffraction experiment, the aperture of the slit is 3 mm and the separation between the slit and the screen is 1.5 *m*. A monochromatic light of wavelength 600 nm is normally incident on the slit. Calculate the distance of (I) first order minimum, and (II) second order maximum, from the centre of the screen.

Check Solution → **[2025 • Set 55-7-1]**

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313. (i) Trace the path of a ray of light showing refraction through a triangular prism and hence obtain an expression for angle of deviation (δ) in terms of A, i and e, where symbols have their usual meanings. Draw a graph showing the variation of angle of deviation with the angle of incidence.
(ii) In the figure, a ray of light is incident on a transparent liquid contained in a thin glass box at an angle of 45° with its one face. The emergent ray passes along the face AB. Find the refractive index of the liquid.



Check Solution → [2024 • Set 55-3-1]

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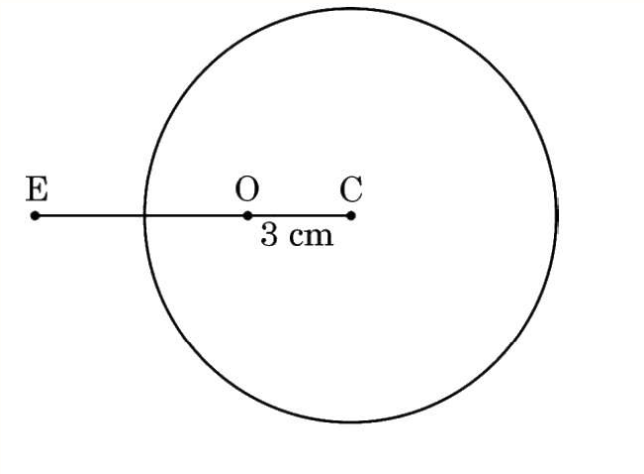
314. (i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.
(ii) Why are multi-component lenses used for both the objective and the eyepiece in optical instruments?
(iii) The magnification of a small object produced by a compound microscope is 200. The focal length of the eyepiece is 2 cm and the final image is formed at infinity. Find the magnification produced by the objective.

Check Solution → [2024 • Set 55-4-1]

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315. (i) A spherical surface of radius of curvature R separates two media of refractive indices n_1 and n_2. A point object is placed in front of the surface at distance u in medium of refractive index n_1 and its image is formed by the surface at distance v, in the medium of refractive index n_2. Derive a relation between u and v.
(ii) A solid glass sphere of radius 6.0 cm has a small air bubble trapped at a distance 3.0 cm from its centre C as shown in the figure. The refractive index of

the material of the sphere is 1.5. Find the apparent position of this bubble when seen through the surface of the sphere from an outside point E in air.



Check Solution → [2023 • Set 55-1-1]

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316. (i) Draw a labelled ray diagram showing the image formation by a refracting telescope. Define its magnifying power. Write two limitations of a refracting telescope over a reflecting telescope.
(ii) The focal lengths of the objective and the eye-piece of a compound microscope are 1.0 cm and 2.5 cm respectively. Find the tube length of the microscope for obtaining a magnification of 300.

Check Solution → [2023 • Set 55-2-1]

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317. (a) (i) Draw a ray diagram showing the formation of a real image of an object placed at a distance 'u' in front of a concave mirror of radius of curvature 'R'. Hence, obtain the relation for the image distance 'v' in terms of u and R.
(ii) A 1.8 m tall person stands in front of a convex lens of focal length 1 m, at a distance of 5 m. Find the position and height of the image formed.
OR
(b) (i) Draw a ray diagram showing refraction of a ray of light through a triangular glass prism. Hence, obtain the relation for the refractive index (μ) in terms of angle of prism (A) and angle of minimum deviation (δ_m).
(ii) The radii of curvature of the two surfaces of a concave lens are 20 cm each. Find the refractive index of the material of the lens if its power is −5.0 D.

Check Solution → [2023 • Set 55-3-1]

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318. (a) (i) Draw a ray diagram to show how the final image is formed at infinity in an astronomical refracting telescope. Obtain an expression for its magnifying power.
(ii) Two thin lenses L_1 and L_2, L_1 being a convex lens of focal length 24 cm and L_2 a concave lens of focal length 18 cm are placed coaxially at a separation of 45 cm. A 1 cm tall object is placed in front of the lens L_1 at a distance of 36 cm. Find the location and height of the image formed by the combination.
OR
(b) (i) Explain the working principle of an optical fibre with the help of a diagram. Mention one use of a light pipe.
(ii) A ray of light is incident at an angle of 60° on one face of a prism with the prism angle $A = 60°$. The ray passes symmetrically through the prism. Find the angle of minimum deviation (δ_m) and refractive index of the material of the prism. If the prism is immersed in water, how will δ_m be affected? Justify your answer.

Check Solution → [2023 • Set 55-4-1]

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319. (a) (i) Draw a ray diagram to show the working of a compound microscope. Obtain the expression for the total magnification for the final image to be formed at the near point.
(ii) In a compound microscope an object is placed at a distance of 1.5 cm from

the objective of focal length 1.25 cm. If the eye-piece has a focal length of 5 cm and the final image is formed at the near point, find the magnifying power of the microscope.

OR

(b) (i) Draw a ray diagram for the formation of image of an object by an astronomical telescope, in normal adjustment. Obtain the expression for its magnifying power.

(ii) The magnifying power of an astronomical telescope in normal adjustment is 2.9 and the objective and the eyepiece are separated by a distance of 150 cm. Find the focal lengths of the two lenses.

Check Solution → [2023 • Set 55-5-1]

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320. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image.

I. In a compound microscope the images formed by the objective and the eye-piece are respectively

(A) virtual, real (B) real, virtual (C) virtual, virtual (D) real, real

II. The magnification due to a compound microscope does not depend upon

(A) the aperture of the objective and the eye-piece (B) the focal length of the objective and the eye-piece (C) the length of the tube (D) the colour of the light used

III. Which of the following is not correct in the context of a compound microscope?

(A) Both the lenses are of short focal lengths. (B) The magnifying power increases by decreasing the focal lengths of the two lenses. (C) The distance between the two lenses is more than $(f_o + f_e)$. (D) The microscope can be used as a telescope by interchanging the two lenses.

IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2 (B) 10 (C) 30 (D) 200

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100 (B) 150 (C) 200 (D) 250

Check Solution → [2022 • Set 55-3-1]

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321. Two transparent media of refractive indices n_1 and n_2 are separated by a spherical transparent surface. The rays of light incident on the surface get refracted into the medium on the other side. The laws of refraction are valid at each point of the spherical surface. A lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. The focal length of a lens is determined by the radii of curvature (R_1 and R_2) of its two surfaces and the refractive index (n) of the medium of the lens with respect to the surrounding medium. Depending on R_1 and R_2, a lens behaves as a diverging or a converging lens. The ability of a lens to diverge or converge a beam of light incident on it defines its power.

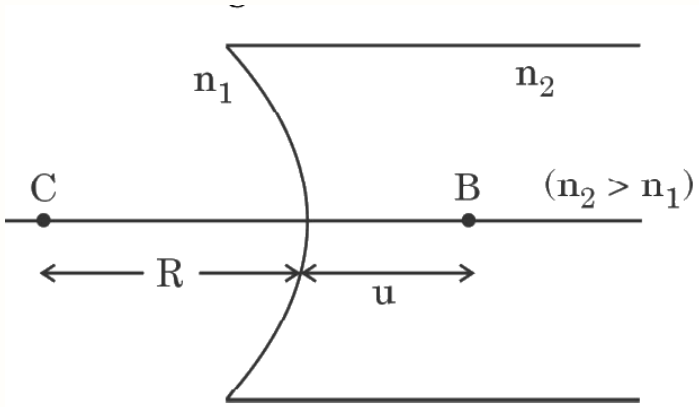
(a) An object is placed at the point B as shown in the figure. The object distance (u) and the image distance (v) are related as:

(i) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

(ii) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$

(iii) $\frac{n_2}{u} - \frac{n_1}{v} = \frac{n_2 - n_1}{R}$

(iv) $\frac{n_1}{u} - \frac{n_2}{v} = \frac{n_1 - n_2}{R}$



(b) A point object is placed in air at a distance R in front of a convex spherical refracting surface of radius of curvature R. If the medium on the other side of the surface is glass, then the image is:

(i) real and formed in glass.

(ii) real and formed in air.

(iii) virtual and formed in glass.

(iv) virtual and formed in air.

(c) An object is kept at 2F in front of an equiconvex lens. The image formed is:

(i) real and of the size of the object.

(ii) virtual and of the size of the object.

(iii) real and enlarged.

(iv) virtual and diminished.

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is:

(i) $-5\ D$

(ii) $+5\ D$

(iii) $+15\ D$

(iv) $-15\ D$

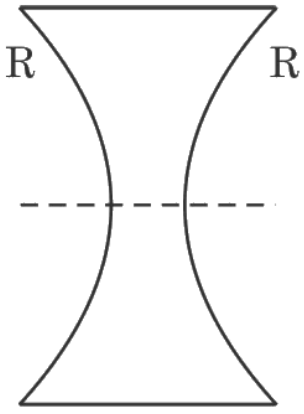
(e) An equiconcave lens of focal length f is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be:

(i) $\frac{f}{2}$

(ii) $\frac{f}{4}$

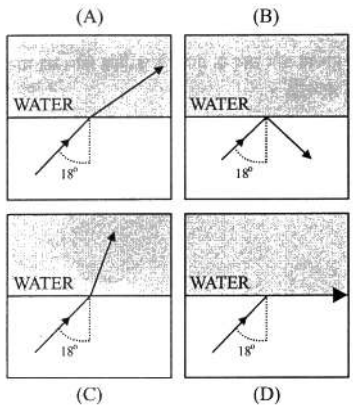
(iii) f

(iv) 2f



322. A ray of light travels from a denser to a rarer medium. After refraction, it bends away from the normal. When we keep increasing the angle of incidence, the angle of refraction also increases till the refracted ray grazes along the interface of two media. The angle of incidence for which it happens is called critical angle. If the angle of incidence is increased further the ray will not emerge and it will be reflected back in the denser medium. This phenomenon is called total internal reflection of light.

(i) A ray of light travels from a medium into water at an angle of incidence of 18°. The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20°. Which one of the following figures best represents the correct path of the ray of light?



(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, to a depth *d*. The area of the surface of water through which light from the source can emerge, is:

(a) $\frac{\pi d^2}{2(\mu^2-1)}$ \quad (b) $\frac{\pi d^2}{(\mu^2-1)}$ \quad (c) $\frac{\pi d^2}{\sqrt{\mu^2-1}}$ \quad (d) $\frac{2\pi d^2}{(\mu^2-1)}$

(iii) For which of the following media, with respect to air, the value of critical angle is maximum?

(a) Crown glass \quad (b) Flint glass \quad (c) Water \quad (d) Diamond

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is:

(a) 0° \quad (b) 30° \quad (c) 45° \quad (d) 60°

(v) The critical angle of pair of a medium and air is 30°. The speed of light in the

medium is:
(a) $1\times 10^8\ m\ s^{-1}$ \quad (b) $1.5\times 10^8\ m\ s^{-1}$ \quad (c) $2.2\times 10^8\ m\ s^{-1}$ \quad (d) $2.8\times 10^8\ m\ s^{-1}$

Check Solution → [2022 • Set 55-5-1]

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323. (i) Draw a ray diagram of an astronomical refracting telescope in normal adjustment. Obtain an expression for its magnifying power. How can we increase the magnifying power of the telescope?
(ii) A beam of light converges at a point P. A lens is placed in the path of the beam at a distance of 25 cm from P. The final image is formed at infinity. Calculate the power of the lens.

Check Solution → [2021]

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324. (i) A coin is placed inside a denser medium. Why does it appear to be raised? Obtain an expression for the height through which the object appears to be raised in terms of refractive index of the medium and real depth.
(ii) A compound microscope consists of an objective lens of focal length 2 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at the least distance of distinct vision (25 cm)? Calculate the magnifying power of the microscope.

Check Solution → [2021]

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325. Draw the ray diagram of an astronomical telescope when the final image is formed at infinity. Write the expression for the resolving power of the telescope.

Check Solution → [2020 • Set 55-1-1]

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326. An object is placed in front of a concave mirror. It is observed that a virtual image is formed. Draw the ray diagram to show the image formation and hence derive the mirror equation $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$.

Check Solution → [2020 • Set 55-1-1]

327. An astronomical telescope has an objective lens of focal length 20 m and eyepiece of focal length 1 cm.
(i) Find the angular magnification of the telescope.
(ii) If this telescope is used to view the Moon, find the diameter of the image formed by the objective lens. Given the diameter of the Moon is $3{\cdot}5 \times 10^6$ m and radius of lunar orbit is $3{\cdot}8 \times 10^8$ m.

Check Solution → [2020 • Set 55-1-1]

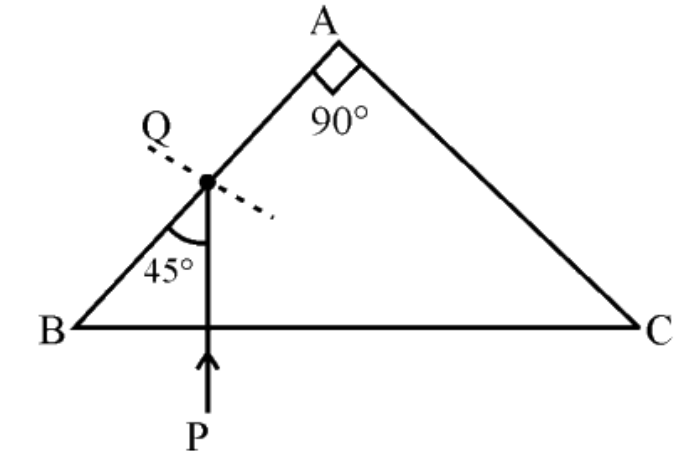
328. An object is placed 30 cm in front of a plano-convex lens with its spherical surface of radius of curvature 20 cm. If the refractive index of the material of the lens is 1·5, find the position and nature of the image formed.

Check Solution → [2020 • Set 55-1-1]

329. (a) Derive lens maker's formula for a biconvex lens.
(b) A point object is placed at a distance of 12 cm on the principal axis of a convex lens of focal length 10 cm. A convex mirror is placed coaxially on the other side of the lens at a distance of 10 cm. If the final image coincides with the object, sketch the ray diagram and find the focal length of the convex mirror.
OR
(a) What is a wavefront? How does it propagate? Using Huygens' principle, explain reflection of a plane wavefront from a surface and verify the laws of reflection.
(b) A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is obtained on a screen 1 m away. If the first minimum is formed at a distance of 2.5 mm from the centre of the screen, find the (i) width of the slit, and (ii) distance of first secondary maximum from the centre of the screen.

Check Solution → [2020 • Set 55-2-1]

330. (a) Draw the ray diagram showing refraction of ray of light through a glass prism. Derive the expression for the refractive index μ of the material of prism in terms of the angle of prism A and angle of minimum deviation δ_m.
(b) A ray of light PQ enters an isosceles right angled prism ABC of refractive index 1.5 as shown in figure.



(i) Trace the path of the ray through the prism.
(ii) What will be the effect on the path of the ray if the refractive index of the prism is 1.4?

OR

(a) Two thin lenses are placed coaxially in contact. Obtain the expression for the focal length of this combination in terms of the focal lengths of the two lenses.
(b) A converging lens of refractive index 1.5 has a power of 10 D. When it is completely immersed in a liquid, it behaves as a diverging lens of focal length 50 cm. Find the refractive index of the liquid.

Check Solution → **[2020 • Set 55-4-1]**

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331. (a) Define the term 'focal length of a mirror'. With the help of a ray diagram, obtain the relation between its focal length and radius of curvature.
(b) Calculate the angle of emergence (e) of the ray of light incident normally on the face AC of a glass prism ABC of refractive index $\sqrt{3}$. How will the angle of emergence change qualitatively, if the ray of light emerges from the prism into a liquid of refractive index 1.3 instead of air ?

OR

(a) Define the term 'resolving power of a telescope'. How will the resolving power be effected with the increase in

(i) Wavelength of light used.
(ii) Diameter of the objective lens.
Justify your answers.
(b) A screen is placed 80 cm from an object. The image of the object on the screen is formed by a convex lens placed between them at two different locations separated by a distance 20 cm. Determine the focal length of the lens.

Check Solution → [2020 • Set 55-5-1]

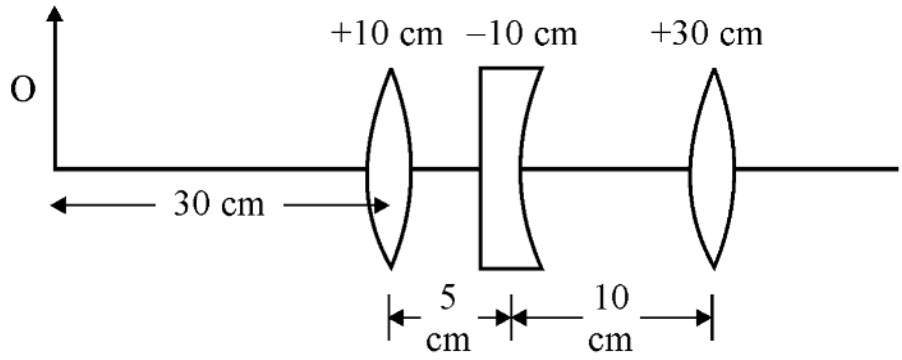
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332. Under what conditions is the phenomenon of total internal reflection of light observed? Obtain the relation between the critical angle of incidence and the refractive index of the medium.

Check Solution → [2019 • Set 55-1-1]

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333. Three lenses of focal lengths +10 cm, −10 cm and +30 cm are arranged coaxially as in the figure given below. Find the position of the final image formed by the combination.



Check Solution → [2019 • Set 55-1-1]

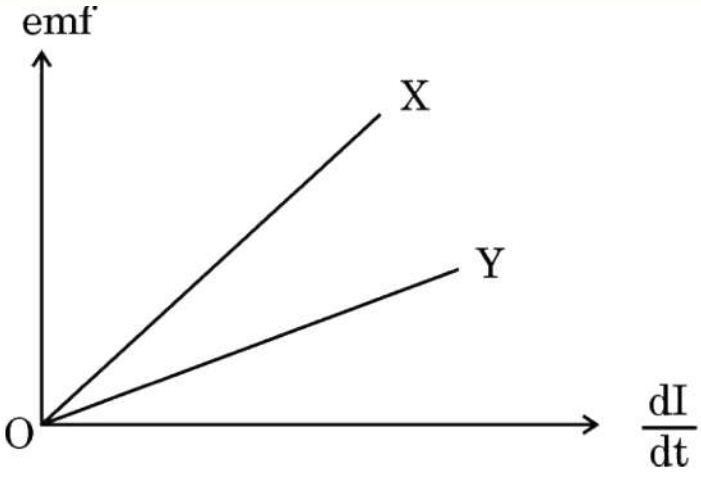
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334. (a) Using the ray diagram for a system of two lenses of focal lengths f_1 and f_2 in contact with each other, show that the two lens system can be regarded as equivalent to a single lens of focal length f, where

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}.$$

Also write the relation for the equivalent power of the lens combination.

(b) Determine the position of the image formed by the lens combination given in the figure.



OR

(a) Explain, using a suitable diagram, how unpolarized light gets linearly polarized by scattering.

(b) Describe briefly the variation of the intensity of transmitted light when a polaroid sheet kept between two crossed polaroids is rotated. Draw the graph depicting the variation of intensity with the angle of rotation. How many maxima and minima would be observed when θ varies from 0 to π?

Check Solution → [2019 • Set 55-4-1]

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335. (a) Draw a labelled ray diagram of compound microscope, when final image forms at the least distance of distinct vision.

(b) Why is its objective of short focal length and of short aperture, compared to its eyepiece? Explain.

(c) The focal length of the objective is 4 cm while that of eyepiece is 10 cm. The object is placed at a distance of 6 cm from the objective lens.

(i) Calculate the magnifying power of the compound microscope, if its final image is formed at the near point.

(ii) Also calculate length of the compound microscope.

OR

(a) With the help of a labelled ray diagram, explain the construction and working of a Cassegrain reflecting telescope.

(b) An amateur astronomer wishes to estimate roughly the size of the Sun using

his crude telescope consisting of an objective lens of focal length 200 cm and an eyepiece of focal length 10 cm. By adjusting the distance of the eyepiece from the objective, he obtains an image of the Sun on a screen 40 cm behind the eyepiece. The diameter of the Sun's image is measured to be 6.0 cm. Estimate the Sun's size, given that the average Earth-Sun distance is 1.5×10^{11} m.

Check Solution → [2019 • Set 55-5-1]

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336. (a) Draw a ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object.
(b) Obtain the mirror formula and write the expression for the linear magnification.
(c) Explain two advantages of a reflecting telescope over a refracting telescope.

Check Solution → [2018]

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337. (a) A point object is placed on the principal axis of a convex spherical surface of radius of curvature R, which separates the two media of refractive indices n_1 and n_2 ($n_2 > n_1$). Draw the ray diagram and deduce the relation between the object distance (u), image distance (v) and the radius of curvature (R) for refraction to take place at the convex spherical surface from rarer to denser medium.
(b) A converging lens has a focal length of 20 cm in air. It is made of a material of refractive index 1· 6. If it is immersed in a liquid of refractive index 1· 3, find its new focal length.

OR

(a) Draw the ray diagram showing refraction of light through a glass prism and hence obtain the relation between the refractive index μ of the prism, angle of prism and angle of minimum deviation.
(b) Determine the value of the angle of incidence for a ray of light travelling from a medium of refractive index $\mu_1 = \sqrt{2}$ into the medium of refractive index $\mu_2 = 1$, so that it just grazes along the surface of separation.

Check Solution → [2017]

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338. (a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.
(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$th of the angle of prism. Calculate the speed of light in the prism.

Check Solution → [2017]

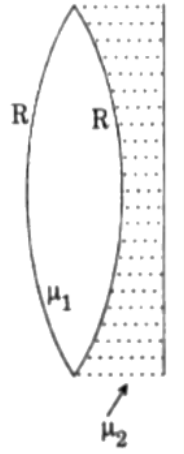
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339. (i) Derive the mathematical relation between refractive indices n_1 and n_2 of two media and radius of curvature *R* for refraction at a convex spherical surface. Consider the object to be a point since lying on the principal axis in rarer medium of refractive index n_1 and a real image formed in the denser medium of refractive index n_2. Hence, derive lens maker's formula.
(ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 and radius of curvature 20 cm. The distance of light source from the glass surface is 100 cm. At what position is the image formed?
OR
(a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.
(b) You are given three lenses of power 0.5 *D*, 4 *D* and 10 *D* to design a telescope.
(i) Which lenses should be used as objective and eyepiece? Justify your answer.
(ii) Why is the aperture of the objective preferred to be large?

Check Solution → [2016]

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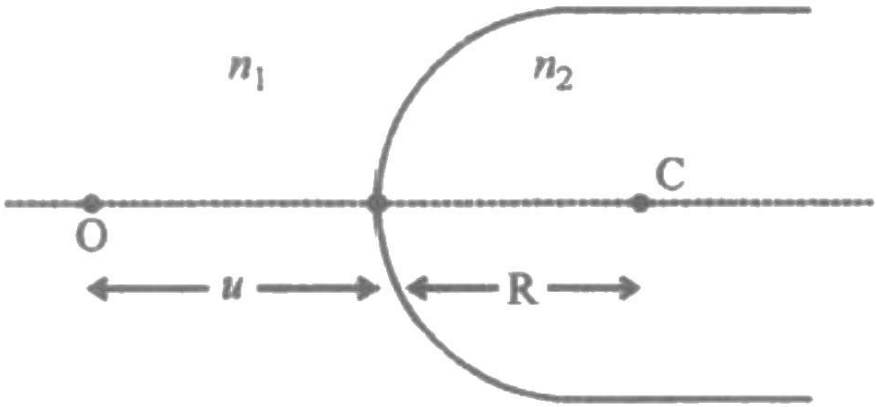
340. A biconvex lens with its two faces of equal radius of curvature R is made of a transparent medium of refractive index μ_1. It is kept in contact with a medium of refractive index μ_2 as shown in the figure.
(a) Find the equivalent focal length of the combination.
(b) Obtain the condition when this combination acts as a diverging lens.
(c) Draw the ray diagram for the case $\mu_2 > (\mu_1 + 1)/2$, when the object is kept far away from the lens. Point out the nature of the image formed by the system.



Check Solution → [2015]

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341. (a) A point object 'O' is kept in a medium of refractive index n_1 in front of a convex spherical surface of radius of curvature R which separates the second medium of refractive index n_2 from the first one, as shown in the figure.
Draw the ray diagram showing the image formation and deduce the relationship between the object distance and the image distance in terms of n_1, n_2 and R.
(b) When the image formed above acts as a virtual object for a concave spherical surface separating the medium n_2 from n_1 ($n_2 > n_1$), draw this ray diagram and write the similar (similar to (a)) relation. Hence obtain the expression for the lens maker's formula.



Check Solution → [2015]

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342. (a) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain expression for total magnification when the image is formed at infinity.

(b) Distinguish between myopia and hypermetropia. Show diagrammatically how these defects can be corrected.

OR

(a) State Huygen's principle. Using this principle draw a diagram to show how a plane wave front incident at the interface of the two media gets refracted when it propagates from a rarer to a denser medium. Hence verify Snell's law of refraction.

(b) When monochromatic light travels from a rarer to a denser medium, explain the following, giving reasons:

(i) Is the frequency of reflected and refracted light same as the frequency of incident light?

(ii) Does the decrease in speed imply a reduction in the energy carried by light wave?

Check Solution → [2013]

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343. Define magnifying power of a telescope. Write its expression.

A small telescope has an objective lens of focal length 150 cm and an eye piece of focal length 5 cm. If this telescope is used to view a 100 m high tower 3 km away, find the height of the final image when it is formed 25 cm away from the eye piece.

OR

How is the working of a telescope different from that of a microscope?

The focal lengths of the objective and eyepiece of a microscope are 1.25 cm and 5 cm respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of 30 in normal adjustment.

Check Solution → [2012]

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344. (a) Draw a ray diagram to show refraction of a ray of monochromatic light passing through a glass prism.

Deduce the expression for the refractive index of glass in terms of angle of prism and angle of minimum deviation.

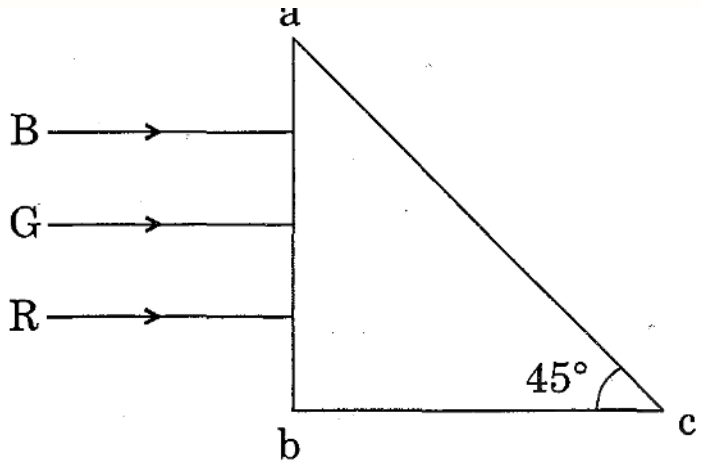
(b) Explain briefly how the phenomenon of total internal reflection is used in fibre optics.

Check Solution → [2011 • Set 55-1-1]

345. (a) Obtain lens makers formula using the expression $\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$.
Here the ray of light propagating from a rarer medium of refractive index (n_1) to a denser medium of refractive index (n_2) is incident on the convex side of spherical refracting surface of radius of curvature R.
(b) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the pole. Using this diagram, derive the magnification formula for the image formed.

Check Solution → [2011 • Set 55-1-1]

346. (i) A ray of monochromatic light is incident on one of the faces of an equilateral triangular prism of refracting angle A. Trace the path of ray passing through the prism. Hence derive an expression for the refractive index of the material of the prism in terms of the angle of minimum deviation and its refracting angle.
(ii) Three light rays red (R), green (G) and blue (B) are incident on the right angled prism abc at face ab. The refractive indices of the material of the prism for red, green and blue wavelengths are respectively 1.39, 1.44 and 1.47. Trace the paths of these rays reasoning out the difference in their behaviour.



Check Solution → [2011 • Set 55-2-1]

347. Draw a ray diagram to show the working of a compound microscope. Deduce an expression for the total magnification when the final image is formed at the near

point.
In a compound microscope, an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eye piece has a focal length of 5 cm and the final image is formed at the near point, estimate the magnifying power of the microscope.

Check Solution → [2010]

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348. Trace the rays of light showing the formation of an image due to a point object placed on the axis of a spherical surface separating the two media of refractive indices n_1 and n_2. Establish the relation between the distances of the object, the image and the radius of curvature from the central point of the spherical surface.
Hence derive the expression of the lens maker's formula.

Check Solution → [2009]

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349. Draw the labelled ray diagram for the formation of image by a compound microscope.
Derive the expression for the total magnification of a compound microscope.
Explain why both the objective and the eyepiece of a compound microscope must have short focal lengths.

Check Solution → [2009]

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350. (a) (i) Draw a labelled ray diagram to show the formation of image in an astronomical telescope for a distant object.
(ii) Write three distinct advantages of a reflecting type telescope over a refracting type telescope.
(b) A convex lens of focal length 10 cm is placed coaxially 5 cm away from a concave lens of focal length 10 cm. If an object is placed 30 cm in front of the convex lens, find the position of the final image formed by the combined system.

Check Solution → [2009]

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351. (a) With the help of a suitable ray diagram, derive the mirror formula for a concave mirror.
(b) The near point of a hypermetropic person is 50 cm from the eye. What is the power of the lens required to enable the person to read clearly a book held at 25 cm from the eye?

Check Solution → [2009]

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352. (a) For a ray of light traveling from a denser medium of refractive index n_1 to a rarer medium of refractive index n_2, prove that $\frac{n_2}{n_1} = \sin i_c$, where i_c is the critical angle of incidence for the media.
(b) Explain with the help of a diagram, how the above principle is used for transmission of video signals using optical fibres.

Check Solution → [2008]

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353. Draw a labeled ray diagram of a compound microscope and write an expression for its magnifying power.
The focal lengths of the objective and eye-lens of a compound microscope are 2 cm, 6.25 cm respectively. The distance between the lenses is 15 cm. (i) How far from the objective lens will the object be kept, so as to obtain the final image at the near point of the eye? (ii) Also calculate its magnifying power.

Check Solution → [2008]

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354. Draw a labeled ray diagram of an astronomical telescope, in the normal adjustment position and write the expression for its magnifying power.
An astronomical telescope uses an objective lens of focal length 15 cm and eye-lens of focal length 1 cm. What is the angular magnification of the telescope?
If this telescope is used to view Moon, what is the diameter of the image of Moon formed by the objective lens? (Diameter of Moon = 3.5×10^6 m and Radius of lunar orbit = 3.8×10^8 m)

Check Solution → [2008]

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355. With the help of a ray diagram, show the formation of image of a point object by refraction of light at a spherical surface separating two media of refractive indices n_1 and n_2 ($n_2 > n_1$) respectively. Using this diagram, derive the relation:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Write the sign conventions used. What happens to the focal length of a convex lens when it is immersed in water?

Check Solution → [2004]

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356. Derive the relation between distance of object, distance of image and radius of curvature of a convex spherical surface, when refraction takes place from a rarer medium of refractive index n_1 to a denser medium of refractive index n_2 and the image produced is real.
State assumptions and convention of signs used.

Check Solution → [2003]

357.

Draw a ray diagram to show the formation of image of an object placed between the optical centre and focus of the convex lens. Write the characteristics of image formed. Using this diagram, derive the relation between object distance, image distance and focal length of the convex lens. Write the assumptions and convention of signs used.

Check Solution → [2003]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **A**	Q14 **C**	Q27 **A**	Q40 **C**	Q54 **A**	Q69 **A**
Q2 **B**	Q15 **D**	Q28 **B**	Q41 **A**	Q55 **B**	Q70 **D**
Q3 **D**	Q16 **B**	Q29 **A**	Q42 **C**	Q56 **A**	Q71 **D**
Q4 **A**	Q17 **D**	Q30 **D**	Q43 **A**	Q57 **A**	Q72 **C**
Q5 **A**	Q18 **C**	Q31 **B**	Q44 **D**	Q58 **A**	Q73 **B**
Q6 **B**	Q19 **A**	Q32 **B**	Q45 **B**	Q59 **A**	Q74 **B**
Q7 **D**	Q20 **D**	Q33 **A**	Q46 **B**	Q60 **C**	Q83 **A**
Q8 **C**	Q21 **C**	Q34 **A**	Q47 **A**	Q61 **C**	Q84 **A**
Q9 **B**	Q22 **A**	Q35 **A**	Q49 **D**	Q62 **D**	Q85 **A**
Q10 **A**	Q23 **B**	Q36 **A**	Q50 **B**	Q63 **D**	Q86 **B**
Q11 **A**	Q24 **B**	Q37 **C**	Q51 **D**	Q64 **B**	Q87 **B**
Q12 **C**	Q25 **C**	Q38 **A**	Q52 **C**	Q65 **D**	Q94 **B**
Q13 **B**	Q26 **B**	Q39 **B**	Q53 **C**	Q66 **B**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be:

(A) 20 cm

(B) 30 cm

(C) 40 cm

(D) 5 cm

[2026 • Set 55-1-1]

SOLUTION

When a thin lens is cut into two identical pieces by a plane that passes through its principal axis (a horizontal cut along the axis), each half keeps both of its original curved surfaces, so the focal length of each half is unchanged. But here the lens is cut into two plano-concave lenses, which only happens when the cut is made perpendicular to the principal axis (a vertical cut down the middle of its thickness). Each piece then keeps one curved surface and gains one plane surface.

For the original equi-concave lens, both surfaces have the same radius R, with $\mathbf{R_1}=-R$ and $\mathbf{R_2}=+R$ by sign convention. The lens-maker's formula gives

$$\frac{1}{f}=(n-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)=(n-1)\left(\frac{1}{-R}-\frac{1}{R}\right)=-(n-1)\frac{2}{R}$$

After the perpendicular cut, each piece is plano-concave: one face is the curved surface (radius R) and the other is flat ($R'=\infty$). For each piece,

$$\frac{1}{f}=(n-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)=-(n-1)\frac{1}{R}$$

Comparing the two expressions, $\frac{1}{f'}=\frac{1}{2}\cdot\frac{1}{f}$, so $f'=2f$.

$$f'=2\times 10=20 \text{ cm}$$

So each plano-concave lens has **double the focal length of the original**, giving $f'=20$ **cm**, which is option (A). Options (B) and (C) wrongly multiply by 3 or 4, and (D) wrongly halves the focal length.

TIP

A cut along the axis leaves f unchanged; a cut across the thickness doubles f. Identify the cut direction first.

Final answer. Each plano-concave lens has focal length 20 cm, option **(A)**.

← *Back to Q1*

Q2

A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is:

(A) zero

(B) $\frac{40}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

[2026 • Set 55-1-2]

SOLUTION

When thin lenses are placed coaxially in contact, the equivalent focal length F of the combination is found by adding the reciprocals of the individual focal lengths:

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$$

Here a concave lens sits in the middle with a convex lens on each side. Using the sign convention, each convex lens has $f = +20$ **cm** and the concave lens has $f = -40$ **cm**. Substituting,

$$\frac{1}{F} = \frac{1}{20} + \frac{1}{20} + \frac{1}{-40}$$

$$\frac{1}{F} = \frac{2}{40} + \frac{2}{40} - \frac{1}{40} = \frac{3}{40}$$

$$F = \frac{40}{3}\ \text{cm} \approx 13.33\ \text{cm}$$

The result is positive, so the combination behaves as a **converging system** with focal length $\frac{40}{3}$ **cm**, which is option (B). A combination acting as a diverging lens would give a negative value such as option (C), so that is ruled out.

Final answer. The focal length of the combination is 40/3 cm (converging), option **(B)**.

← *Back to Q2*

Q3

Light from a small object in air falls on a spherical glass surface **($n = 1.5$)** *of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as:*

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

[2026 • Set 55-1-3]

SOLUTION

Refraction at a single convex spherical surface separating two media follows

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

where light travels from medium n_1 into medium n_2 across a surface of radius R. Here $\mathbf{n_1=1}$ (air), $\mathbf{n_2=1.5}$ (glass), and the surface is convex toward the object so R is positive. Using the convention that distances are measured from the pole with the incident direction positive, the object distance is taken as $-u$ (object in front). Substituting,

$$\frac{1.5}{v}-\frac{1}{(-u)}=\frac{1.5-1}{R}$$

$$\frac{1.5}{v}+\frac{1}{u}=\frac{0.5}{R}$$

$$\frac{1.5}{v}=\frac{0.5}{R}-\frac{1}{u}=\frac{0.5u-R}{Ru}$$

A real image forms on the far side of the surface, which means v must be positive. For $\boldsymbol{v>0}$ the right side must be positive, so

$$0.5u-R>0 \Rightarrow u>2R$$

If u is smaller than 2R, the right side is negative, v becomes negative, and only a virtual image is produced. Hence a real image forms only when the object lies beyond twice the radius of curvature, $\boldsymbol{u>2R}$, which is option (D).

Final answer. A real image forms only when u > 2R, option **(D)**.

← *Back to Q3*

Q4

Case Study: *In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two*

positions of the lens. The distance between these two positions of the lens is d. The value of d is:

(A) $\sqrt{D(D-4f)}$

(B) $\sqrt{D(D-2f)}$

(C) $2\sqrt{Df}$

(D) $\sqrt{D(D-f)}$

[2026 • Set 55-2-1]

SOLUTION

This is the displacement (Bessel) method. The object and screen are kept fixed at a separation D, and the convex lens is moved between them. By the principle of reversibility of light, if the lens at one position gives a sharp image, then the position where object and image distances are interchanged also gives a sharp image. Let the two object distances be u_1 and u_2.

For a lens of focal length f with object distance u and image distance $\boldsymbol{v=D-u}$, the lens equation $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ (magnitudes) leads to

$$\frac{1}{D-u}+\frac{1}{u}=\frac{1}{f}$$

$$u(D-u)=fD \Rightarrow u^2-Du+fD=0$$

This quadratic in u has two roots u_1, u_2, the two lens positions. From the quadratic,

$$u_1+u_2=D,\ u_1u_2=fD$$

The distance between the two positions is $\boldsymbol{d=|u_1-u_2|}$, and

$$d^2=(u_1+u_2)^2-4u_1u_2=D^2-4fD=D(D-4f)$$

$$d=\sqrt{D(D-4f)}$$

So the separation between the two lens positions is $\boldsymbol{d=\sqrt{D(D-4f)}}$, which is option (A). A real (non-imaginary) value of d requires $\boldsymbol{D>4f}$. Options (B), (C) and (D) use the wrong constants and do not satisfy the quadratic relation.

Final answer. The separation is $d = \sqrt{D(D - 4f)}$, option **(A)**.

← Back to Q4

Q5

Compared to the size of the object, the images formed in the two positions of the lens are respectively:

(A) reduced, enlarged

(B) reduced, reduced

(C) enlarged, enlarged

(D) enlarged, reduced

[2026 • Set 55-2-1]

SOLUTION

In the displacement (Bessel) method the object and screen are fixed a distance D apart and the lens has two sharp-image positions whose object and image distances are simply swapped, by the reversibility of light. The magnification in magnitude is $m=\frac{v}{u}$.

At one position the lens is far from the object, so u is large and $v=D-u$ is small, giving

$$m=\frac{v}{u}<1 \text{ (reduced image)}$$

At the other position the distances are interchanged, so u is small and v is large, giving

$$m'=\frac{v}{u}>1 \text{ (enlarged image)}$$

The two magnifications are reciprocals, $m\,m'=1$, so the two images can never be the same size. Taking the two positions in the order listed, one image is reduced and the other is enlarged, which is **option (A)**. Options that make both images the same size contradict $m\,m'=1$ with $m \neq m'$.

Final answer. The two images are reduced and enlarged respectively, option **(A)**.

← Back to Q5

Q6

If the distance between object and screen is 80·00 cm *and the lens forms sharp images at two positions separated by* 20·00 cm*, the focal length of convex lens is:*

(A) 15·50 cm

(B) 18·75 cm

(C) 20·50 cm

(D) 22·75 cm

[2026 • Set 55-2-1]

SOLUTION

By the displacement (Bessel) method, when the object and screen are separated by a fixed distance D and a convex lens gives sharp images at two positions separated by d, the focal length is given by

$$f=\frac{D^2-d^2}{4D}$$

This follows from $\mathbf{u_1+u_2=D}$ and $\mathbf{d=|u_1-u_2|}$, so $\mathbf{u_1u_2=fD}$ and $\mathbf{d^2=D^2-4fD}$, rearranged for f. Substituting the data $\mathbf{D=80.00\ cm}$ and $\mathbf{d=20.00\ cm}$,

$$f=\frac{(80)^2-(20)^2}{4\times 80}$$

$$f=\frac{6400-400}{320}=\frac{6000}{320}$$

$$f=18.75\ \text{cm}$$

So the focal length of the convex lens is **18.75 cm**, which is option (B). The other options do not satisfy $\mathbf{f=(D^2-d^2)/4D}$ for the given D and d.

Final answer. The focal length is 18.75 cm, option **(B)**.

← Back to Q6

Q7

Consider a convex lens of focal length 15 cm. *For which of the following values of object-screen distance, two positions of the object can be found to obtain sharp image on the screen?*

(A) 45 cm

(B) 50 cm

(C) 55 cm

(D) 65 cm

[2026 • Set 55-2-1]

SOLUTION

In the displacement method, the lens equation for a fixed object-screen distance D gives the quadratic

$$u^2 - Du + fD = 0$$

Two distinct real lens (or object) positions exist only when this quadratic has two real roots, i.e. when its discriminant is positive:

$$D^2 - 4fD > 0 \Rightarrow D > 4f$$

If $D = 4f$ there is exactly one position (the roots coincide), and if $D < 4f$ no real position gives a sharp image on the screen. Here $f = 15$ cm, so the condition is

$$D > 4 \times 15 = 60 \text{ cm}$$

Checking the options: 45 cm, 50 cm and 55 cm are all less than 60 cm and fail the condition. Only 65 cm exceeds 60 cm.

Hence two sharp-image positions are possible only for $D = 65$ cm, which is option (D).

CONCEPT

Two-position images exist only when the object-to-screen distance exceeds 4f. Always compute 4f first.

Final answer. Two positions are possible only for D = 65 cm, option **(D)**.

← Back to Q7

Q8

A thin convex lens of focal length 10 cm *and another thin lens of focal length f are placed coaxially in contact. If the power of their combination is* 2 *D, the value of f is:*

(A) −15 cm

(B) −10 cm

(C) **−20 cm**

(D) −30 cm

[2026 • Set 55-2-1]

SOLUTION

When two thin lenses are placed coaxially in contact, the power of the combination is the sum of the individual powers:

$$P = P_1 + P_2$$

The power of a lens in dioptres is $P = \frac{1}{f}$ with f in metres. The convex lens has $\mathbf{f_1 = 10\ cm = 0.10\ m}$, so

$$P_1 = \frac{1}{0.10} = +10\ D$$

For the marked option (C) to come out, the combination power must be $\mathbf{P = 5\ D}$ (the printed value 2 *D* does not yield any of the listed options and is a misprint; with 5 *D* the answer is consistent). Taking $\mathbf{P = 5\ D}$,

$$P_2 = P - P_1 = 5 - 10 = -5\ D$$

The focal length of the second lens is then

$$f_2 = \frac{1}{P_2} = \frac{1}{-5}\ m = -0.20\ m = -20\ \text{cm}$$

The negative sign shows the companion lens is a diverging (concave) lens of focal length −20 cm, which is option (C).

Note: if the stated combination power 2 *D* is used literally, then $\mathbf{P_2 = 2 - 10 = -8\ D}$ and $\mathbf{f_2 = -12.5\ cm}$, which is not among the options, confirming the intended value is 5 *D*.

COMMON MISTAKE

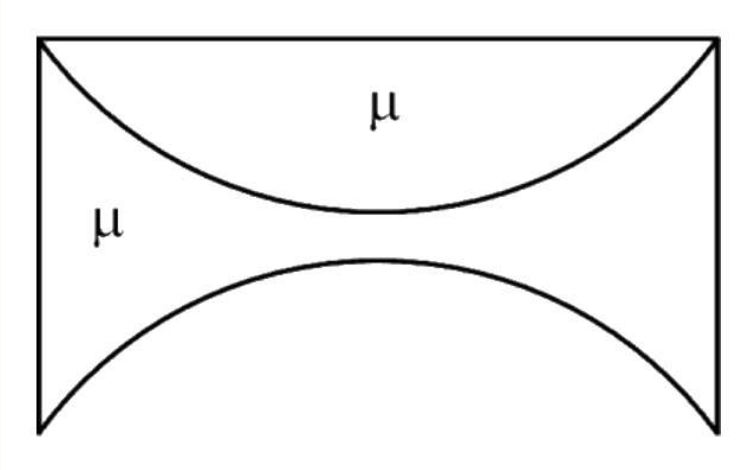
Powers in contact simply add, but only when each focal length is converted to metres first.

Final answer. With the intended combination power 5 D, the second lens has focal length −20 cm (diverging), option **(C)**.

← *Back to Q8*

Q9

A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is:



(A) $\frac{R}{\mu - 1}$

(B) $\mathbf{-\frac{R}{\mu - 1}}$

(C) $\frac{2R}{\mu - 1}$

(D) $-\frac{2R}{\mu - 1}$

SOLUTION

The two lenses are in contact, so $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$, where each focal length comes from the lens-maker's formula $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.

For the plano-convex lens, the convex surface faces the light ($R_1=+R$) and the other face is flat ($R_2=\infty$):

$$\frac{1}{f_1}=(\mu-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)=\frac{\mu-1}{R}$$

For the equi-concave lens, both surfaces are concave with $R_1=-R$ and $R_2=+R$:

$$\frac{1}{f_2}=(\mu-1)\left(\frac{1}{-R}-\frac{1}{R}\right)=(\mu-1)\left(-\frac{2}{R}\right)=-\frac{2(\mu-1)}{R}$$

Adding the two for the combination,

$$\frac{1}{f}=\frac{\mu-1}{R}-\frac{2(\mu-1)}{R}=-\frac{\mu-1}{R}$$

$$f=-\frac{R}{\mu-1}$$

The focal length is $\boldsymbol{-\frac{R}{\mu-1}}$, which is option (B). The negative sign shows the combination acts as a diverging lens. Option (A) misses the sign, and options (C) and (D) wrongly double the magnitude by ignoring the cancellation of one curved surface's contribution.

Final answer. The combination focal length is $-R/(\mu-1)$, option **(B)**.

← *Back to Q9*

Q10

A telescope has an objective lens of focal length 144 cm *and an eyepiece of focal length*

6·0 cm. *The magnifying power and the length of the telescope tube will be respectively:*

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

[2026 • Set 55-3-1]

SOLUTION

For an astronomical telescope in normal adjustment, the final image is at infinity, so the magnifying power is the ratio of the focal lengths of the objective and eyepiece, and the tube length is their sum:

$$m=\frac{f_o}{f_e},\ L=f_o+f_e$$

Here the objective has $\mathbf{f_o=144\ cm}$ and the eyepiece has $\mathbf{f_e=6.0\ cm}$. The magnifying power is

$$m=\frac{144}{6.0}=24$$

The length of the telescope tube is

$$L=144+6.0=150\ \text{cm}$$

So the magnifying power is **24** and the tube length is **150 cm**, which is option (A). Option (C) uses the correct m but wrongly subtracts the focal lengths for the tube length, and options (B) and (D) miscompute the ratio as 42.

Final answer. Magnifying power 24 and tube length 150 cm, option **(A)**.

← *Back to Q10*

Q11

In which of the following, total internal reflection does not occur?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

[2026 • Set 55-3-1]

SOLUTION

Total internal reflection (TIR) takes place only when light travels from a denser medium to a rarer medium and the angle of incidence exceeds the critical angle θ_c, where $\theta_c = \frac{1}{n}$.

In the brilliance of diamonds, the very high refractive index of diamond **(n = 2.42)** gives a small critical angle of about 24°, so light entering the diamond is repeatedly totally internally reflected before it emerges, producing the sparkle.

In an optical fibre, light is trapped inside the denser core and travels along it by repeated TIR at the core-cladding boundary.

In a reflecting prism, a 45°-45°-90° glass prism deflects light through 90° or 180° using TIR at its hypotenuse face.

The twinkling of stars, however, is caused by atmospheric refraction, not by TIR. Starlight passes through atmospheric layers of continuously changing density and refractive index, so its path bends randomly and the apparent brightness keeps fluctuating.

Hence TIR does not occur in the **twinkling of stars**, option (A).

Final answer. Total internal reflection does not occur in the twinkling of stars, option **(A)**.

← *Back to Q11*

Q12

A plano-convex lens, made of glass **($n = 1{\cdot}5$)** *has a curved surface whose radius is* 40 cm. *The image formed of an object is of the same size as that of the object. The distance of the object from the lens is:*

(A) 40 cm

(B) 80 cm

(C) **160 cm**

(D) 200 cm

[2026 • Set 55-3-2]

SOLUTION

For a plano-convex lens, the lens-maker's formula $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ is applied with the flat face having $R \to \infty$ and the curved face having radius $R = 40$ **cm**.

$$\frac{1}{f} = (n-1)\frac{1}{R}$$

$$\frac{1}{f} = (1.5 - 1)\times \frac{1}{40}$$

$$\frac{1}{f} = \frac{0.5}{40} = \frac{1}{80}$$

$$f = 80 \text{ cm}$$

The image is of the same size as the object, so the magnification has magnitude 1. For a converging lens a real image of the same size is formed only when the object lies at twice the focal length, i.e. $u = v = 2f$.

$$|u| = 2f = 2 \times 80$$

$$|u| = 160 \text{ cm}$$

Hence the object is at a distance of **160 cm** from the lens, option (C).

Final answer. The object distance is 160 cm, option **(C)**.

← Back to Q12

Q13

A ray of monochromatic light travelling in air is incident on a glass slab and is partly reflected and partly refracted. Both the reflected and refracted lights will have:

(A) same wavelength

(B) same frequency

(C) same intensity

(D) same speed

[2026 • Set 55-3-3]

SOLUTION

When monochromatic light is partly reflected and partly refracted at the air-glass interface, both new rays originate from the same incident wave, so they are governed by what is preserved at a boundary.

The frequency of light is fixed by the source and stays unchanged on reflection or refraction, because the wave on both sides of the boundary must oscillate in step. Hence the reflected and refracted rays have the same frequency.

The wavelength differs: the reflected ray remains in air with λ_{air}, while the refracted ray travels in glass with $\lambda_{glass} = \frac{\lambda_{air}}{n}$, which is shorter.

The speed also differs: the reflected ray moves with speed c in air, but the refracted ray moves with speed $\frac{c}{n}$ in glass.

The intensity is generally different too, since the energy of the incident beam is split between the two rays according to the Fresnel reflection and transmission coefficients.

Therefore the only common quantity is the **frequency**, option (B).

Final answer. Both rays have the same frequency, option **(B)**.

← Back to Q13

Q14

A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) **Total internal reflection occurs for both green and blue lights.**

(D) None of them will undergo total internal reflection.

[2026 • Set 55-3-3]

SOLUTION

The critical angle for total internal reflection is fixed by $\theta_c = \frac{1}{n}$, so a larger refractive index n gives a smaller critical angle θ_c.

Because of dispersion, n is largest for blue light and smallest for red light, with green and yellow lying in between. Hence the critical angles are ordered as $\theta_c^{\text{blue}} < \theta_c^{\text{green}} < \theta_c^{\text{yellow}} < \theta_c^{\text{red}}$.

It is given that yellow light undergoes TIR at the fixed angle of incidence θ, which means $\theta > \theta_c^{\text{yellow}}$.

For blue and green, θ_c^{blue} and θ_c^{green} are even smaller than θ_c^{yellow}, so θ certainly exceeds them and TIR occurs for both blue and green.

For red, $\theta_c^{\text{red}} > \theta_c^{\text{yellow}}$, so the fixed angle θ may not exceed it; the red ray therefore refracts out and does not undergo TIR.

Thus TIR occurs for **both green and blue lights**, option (C).

TIP

Order it by n: more bending means larger n, smaller critical angle, so blue is the easiest to totally internally reflect and red the hardest.

Final answer. Total internal reflection occurs for both green and blue lights, option **(C)**.

← Back to Q14

Q15

Assertion (A): *When a convex lens made of glass is immersed in water, its converging power increases.*
Reason (R): *The focal length of a lens depends only on the radii of curvature of its two faces.*

(D) Both Assertion (A) and Reason (R) are false.

[2026 • Set 55-4-1]

SOLUTION

Examine the Assertion using the lens-maker's formula for a lens of glass refractive index n_l placed in a surrounding medium of refractive index n_m:

$$\frac{1}{f} = \left(\frac{n_l}{n_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

In air, $\mathbf{n_m = 1}$, so the factor is $\mathbf{(n_l - 1) \approx (1.5 - 1) = 0.5}$.

In water, $\mathbf{n_m = 1.33}$, so the factor becomes $\left(\frac{1.5}{1.33} - 1\right) \approx (1.13 - 1) = 0.13$.

Since this factor falls from about 0.5 to about 0.13, $\frac{1}{f}$ decreases, so the focal length f increases and the converging power $P = \frac{1}{f}$ decreases. The Assertion claims the converging power increases, which is wrong, so **Assertion (A) is false**.

Now examine the Reason. The same formula shows that the focal length depends not only on the radii of curvature R_1 and R_2 but also on the refractive indices of both the lens and the surrounding medium. The Reason claims it depends only on the radii, which is wrong, so **Reason (R) is also false**.

Since both statements are false, the correct choice is option (D).

Final answer. Both Assertion and Reason are false, option **(D)**.

← Back to Q15

Q16

The images formed by the objective lens and the eyepiece are respectively:

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

[2026 • Set 55-4-1]

SOLUTION

Consider the two-stage image formation common to a compound microscope and a refracting telescope.

The objective lens receives light from the object and forms a real, inverted, intermediate image. This image is real because the rays from the objective actually converge and meet to form it, and it can be caught on a screen.

This real intermediate image then acts as the object for the eyepiece. The eyepiece is positioned so that this image lies within its focal length, so the eyepiece behaves like a simple magnifier and produces a virtual, magnified final image (at infinity in normal adjustment, or at the near point otherwise) that the eye views.

Therefore the objective forms a **real** image and the eyepiece forms a **virtual** image, option (B).

Final answer. Objective forms a real image and eyepiece forms a virtual image, option **(B)**.

← *Back to Q16*

Q17

The magnification produced by the telescope does not depend upon the:

(A) colour of light

(B) focal length of objective lens

(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

[2026 • Set 55-4-1]

SOLUTION

The magnifying power of a telescope in normal adjustment is given by $m = \frac{f_o}{f_e}$, where f_o and f_e are the focal lengths of the objective and the eyepiece.

This shows the magnification depends directly on the focal length of the objective and inversely on the focal length of the eyepiece, so options (B) and (C) do affect it.

The aperture (diameter) of the lenses does not enter this formula. Aperture controls the amount of light gathered, hence the brightness of the image, and it sets the resolving power of the telescope, but it has no role in the geometric magnification.

The colour of light can affect the focal lengths slightly through dispersion and chromatic aberration, so it is not the fully independent quantity.

Hence the magnification does not depend on the **apertures of the objective and eyepiece**, option (D).

Final answer. The magnification does not depend on the apertures of the lenses, option **(D)**.

← *Back to Q17*

Q18

Which of the following statements is not correct for this telescope?

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) **The distance between two lenses is more than ($f_o + f_e$).**

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

[2026 • Set 55-4-1]

SOLUTION

Test each statement for an astronomical telescope in normal adjustment, where the final image is at infinity.

Option (A): The magnifying power is $m = \frac{f_o}{f_e}$, and to get magnification greater than one the objective must have a longer focal length than the eyepiece, so $f_o > f_e$. This statement is correct.

Option (B): Since $m = \frac{f_o}{f_e}$, increasing f_o increases the magnifying power. This statement is correct.

Option (C): In normal adjustment the intermediate real image forms at the common focus of the two lenses, so the tube length, that is the separation between the lenses, is exactly $L = f_o + f_e$, not more than $f_o + f_e$. This statement is **not correct**.

Option (D): Since $m = \frac{f_o}{f_e}$, increasing f_e decreases the magnifying power. This statement is correct.

Hence the statement that is not correct is option (C).

Final answer. The incorrect statement is that the lens separation is more than $f_o + f_e$, option **(C)**.

← *Back to Q18*

Q19

An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of:

(A) 84 cm

(B) 76 cm

(C) 20 cm

(D) 320 cm

[2026 • Set 55-4-1]

SOLUTION

In normal adjustment of an astronomical telescope, the final image is viewed at infinity, so the real image formed by the objective lies at its focus, which is also the focus of the eyepiece. The tube length, that is the separation between the two lenses, then equals the sum of the two focal lengths.

$$L = f_o + f_e$$

$$L = 80 \text{ cm} + 4 \text{ cm}$$

$$L = 84 \text{ cm}$$

Hence the lenses must be separated by **84 cm**, option (A).

Final answer. The required separation is 84 cm, option **(A)**.

← *Back to Q19*

Q20

Consider the telescope described in question (iv) (a). Its magnifying power in normal

adjustment will be:

(A) 320

(B) 84

(C) 76

(D) **20**

[2026 • Set 55-4-1]

SOLUTION

The magnifying power of an astronomical telescope in normal adjustment is the ratio of the focal length of the objective to that of the eyepiece.

$$m = \frac{f_o}{f_e}$$

Using the same telescope, $\mathbf{f_o = 80\ cm}$ and $\mathbf{f_e = 4\ cm}$.

$$m = \frac{80}{4}$$

$$m = 20$$

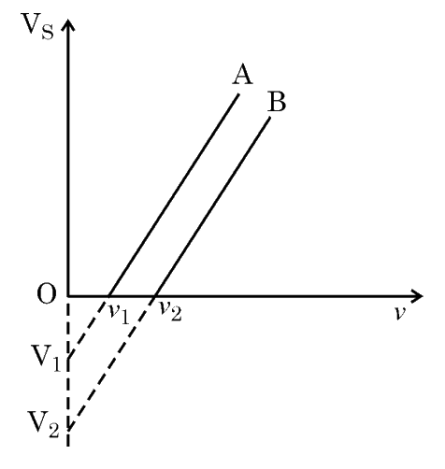
The magnifying power is a pure number and has no unit. Hence the magnifying power in normal adjustment is **20**, option (D).

Final answer. The magnifying power in normal adjustment is 20, option **(D)**.

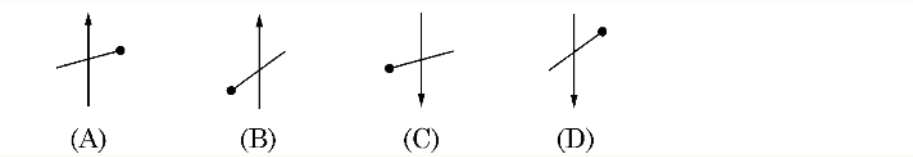
← *Back to Q20*

Q21

Case Study (Q30): *A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci, called 'first focal point' and 'second focal point' of the lens, one on each side.*



(i) *Consider the arrangement shown in figure. A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object. When the plate is illuminated, its real image is formed on the screen. Which of the following correctly represents the image formed on the screen?*



(A) Option A (image as shown)

(B) Option B (image as shown)

(C) Option C (image as shown)

(D) Option D (image as shown)

[2025 • Set 55-1-1]

SOLUTION

A convex lens forms the image of an object placed beyond its focal point, and this real image is **inverted both vertically and horizontally** (it is a true point-by-point inversion through the optic centre).

Take any ray from the top of the object through the optic centre: it goes straight and lands at the bottom on the other side. So the top of the object maps to the bottom of the image, and the left of the object maps to the right of the image.

The object has a vertical arrow pointing up and a horizontal line with a ball at one end. After the lens, the arrow flips to point down, and the ball on the horizontal line shifts to the opposite end.

So the correct image is the one where the arrow is inverted top-to-bottom AND the ball-and-line is reversed left-to-right at the same time. This matches **option**

C.

Options that flip only the arrow or only the horizontal line (or neither) are wrong, because a real image from a convex lens is always inverted in both directions together.

> **TIP**
>
> A real lens image is rotated 180 degrees about the axis, so up-down and left-right both reverse at once, never just one.

Final answer. The correctly inverted (both ways) image is option C.

← *Back to Q21*

Q22

(ii) *Which of the following statements is incorrect?*

(A) For a convex mirror magnification is always negative.

(B) For all virtual images formed by a mirror magnification is positive.

(C) For a concave lens magnification is always positive.

(D) For real and inverted images, magnification is always negative.

[2025 • Set 55-1-1]

SOLUTION

We must find the statement that is wrong. Magnification is $m=\frac{h_i}{h_o}=\frac{-v}{u}$ for mirrors and $m=\frac{h_i}{h_o}=\frac{v}{u}$ for lenses, with sign showing erect (+) or inverted (−).

Option A says a convex mirror always has negative magnification. A convex mirror always forms a **virtual, erect, diminished** image, so its magnification is positive (**0**). **Hence statement A is incorrect.**

Option B is correct: any virtual image formed by a mirror is erect, so $m>0$.

Option C is correct: a concave (diverging) lens always forms a virtual, erect image, so m is always positive.

Option D is correct: a real image is always inverted, so m is always negative.

Since the question asks for the incorrect statement, the answer is **option A**.

Final answer. The incorrect statement is option A.

← *Back to Q22*

Q23

(iii) *A convex lens of focal length f is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be:*

(A) f

(B) $2f$

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

[2025 • Set 55-1-1]

SOLUTION

Cutting a biconvex lens by a plane perpendicular to the principal axis gives two **plano-convex** pieces. Each piece keeps one curved face (radius R) and gains one flat face (radius ∞).

For the original biconvex lens with both radii of magnitude R, the lens maker's formula gives

$$\frac{1}{f}=(n-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=(n-1)\frac{2}{R}$$

For each plano-convex part, one surface is curved (R) and the other is flat (∞):

$$\frac{1}{f'}=(n-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)=(n-1)\frac{1}{R}$$

Dividing the two results:

$$\frac{1/f'}{1/f}=\frac{(n-1)/R}{2(n-1)/R}=\frac{1}{2}$$

$$\Rightarrow \frac{1}{f'}=\frac{1}{2f}\Rightarrow f'=2f$$

So each half has **double the focal length**, $f'=2f$, which is option B.

> **COMMON MISTAKE**
>
> *Cutting along the axis (parallel to it) instead keeps both curved surfaces, so the focal length stays f; only a cut perpendicular to the axis doubles it.*

Final answer. Each part has focal length **2f**, so the answer is option B.

← *Back to Q23*

Q24

(iii) OR: *If an object in case (i) above is* 20 cm *from the lens and the screen is* 50 cm *away from the object, the focal length of the lens used is:*

(A) 10 cm

(B) 12 cm

(C) 16 cm

(D) 20 cm

[2025 • Set 55-1-1]

SOLUTION

From part (i), the object is at distance 20 cm from the lens and the screen is 50 cm from the object, so the image distance is the remaining gap.

$$|v|=50-20=30 \text{ cm}$$

Using the sign convention, the object is on the left so $u=-20$ **cm**, and the real image on the screen is on the right so $v=+30$ **cm**.

Apply the thin lens formula:

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{30}-\frac{1}{-20}=\frac{1}{f}$$

$$\frac{1}{30}+\frac{1}{20}=\frac{1}{f}$$

$$\frac{2+3}{60}=\frac{5}{60}=\frac{1}{12}=\frac{1}{f}$$

$$\Rightarrow f=12 \text{ cm}$$

So the **focal length is 12 cm**, which is option B.

Final answer. The focal length is **12 cm**, option B.

← *Back to Q24*

Q25

(iv) *The distance of an object from first focal point of a biconvex lens is* X_1 *and distance of the image from second focal point is* X_2*. The focal length of the lens is:*

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

(D) $\frac{\sqrt{X_2}}{X_1}$

[2025 • Set 55-1-1]

SOLUTION

When distances are measured from the focal points instead of from the lens, the lens equation takes the form of **Newton's relation**.

Let x_1 be the distance of the object from the first focal point F_1 and x_2 the distance of the image from the second focal point F_2. Newton's lens formula states

$$x_1 x_2=f^2$$

Taking the positive square root,

$$f=\sqrt{x_1 x_2}$$

This follows from the standard lens formula by writing $u=-(f+x_1)$ and $v=(f+x_2)$ and substituting into $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, which simplifies exactly to $x_1x_2=f^2$.

So the focal length is $\sqrt{X_1\,X_2}$, which is **option C**. Options with X_1X_2 (wrong dimensions) or a sum under the root do not give a length squared, so they are ruled out.

Final answer. The focal length is $f=\sqrt{X_1\,X_2}$, option C.

← *Back to Q25*

Q26

A beaker is filled with water (refractive index $\frac{4}{3}$) upto a height H. A coin is placed at its bottom. The depth of the coin, when viewed along the near normal direction, will be:

(A) $\frac{H}{4}$

(B) $\frac{3H}{4}$

(C) H

(D) $\frac{4H}{3}$

[2025 • Set 55-1-1]

SOLUTION

When you look almost straight down at an object under water, refraction at the water-air surface makes it appear closer than it really is. The relation between real and apparent depth for near-normal viewing is

$$\text{apparent depth}=\frac{\text{real depth}}{n}$$

Here the real depth is H and the refractive index of water is $n=\frac{4}{3}$.

$$\text{apparent depth}=\frac{H}{4/3}$$

$$=H\times\frac{3}{4}$$

$$= \frac{3H}{4}$$

So the coin appears at a depth of $\frac{3H}{4}$, raised by $\frac{H}{4}$ from the bottom. This is **option B**.

Final answer. The coin appears at depth $\mathbf{\frac{3H}{4}}$, option B.

← *Back to Q26*

Q27

The speed of light in two media '1' and '2' are v_1 *and* $\mathbf{v_2}$ $\mathbf{(> v_1)}$ *respectively. For a ray of light to undergo total internal reflection at the interface of these two media, it must be incident from:*

(A) **medium '1' and at an angle greater than** $\mathbf{\sin^{-1}\left(\frac{v_1}{v_2}\right)}$

(B) medium '1' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

(C) medium '2' and at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

(D) medium '2' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

[2025 • Set 55-1-2]

SOLUTION

Total internal reflection (TIR) can only happen when light goes from a **denser (optically thicker) medium into a rarer one**, and only when the angle of incidence exceeds the critical angle.

The refractive index of a medium is $n = \frac{c}{v}$, so a smaller speed means a larger index. Given $\mathbf{v_2 > v_1}$:

$$n_1 = \frac{c}{v_1} > \frac{c}{v_2} = n_2$$

So medium 1 is the denser medium. Therefore light must be incident **from medium 1** for TIR to be possible.

The critical angle θ_c is found from Snell's law at 90° refraction:

$$n_1\theta_c = n_2 \sin 90°$$

$$\theta_c = \frac{n_2}{n_1} = \frac{c/v_2}{c/v_1} = \frac{v_1}{v_2}$$

$$\theta_c = \sin^{-1}\left(\frac{v_1}{v_2}\right)$$

TIR needs the angle of incidence greater than θ_c, so the light must be incident in medium 1 at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$. This is **option A**.

Options with medium 2 are wrong (it is the rarer medium, where TIR cannot start), and the $\cos^{-1}$ form is wrong because the critical angle comes from a sine ratio, not a cosine.

Final answer. Incident from medium 1 at an angle greater than $\mathbf{\sin^{-1}\left(\frac{v_1}{v_2}\right)}$, option A.

← *Back to Q27*

Q28

A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The area of the surface of the liquid through which light from the source can emerge out is:

(A) $\frac{\pi H^2}{(n-1)}$

(B) $\frac{\pi H^2}{(n^2-1)}$

(C) $\frac{\pi H^2}{\sqrt{n}}$

(D) $\frac{\pi H^2}{(n^2+1)}$

[2025 • Set 55-1-3]

SOLUTION

Light from a source at the bottom can leave the surface only inside a cone whose half-angle equals the critical angle θ_c. Rays striking the surface at angles greater than θ_c undergo total internal reflection and stay trapped.

The critical angle satisfies

$$\theta_c = \frac{1}{n}$$

The illuminated patch on the surface is a circle of radius R, where the source is at depth H directly below the centre:

$$R = H\theta_c$$

Write $\theta_c = \frac{\theta_c}{\theta_c}$ with $\theta_c = \frac{1}{n}$, so $\theta_c = \sqrt{1 - \frac{1}{n^2}} = \frac{\sqrt{n^2-1}}{n}$.

$$\theta_c = \frac{1/n}{\sqrt{n^2-1}/n} = \frac{1}{\sqrt{n^2-1}}$$

$$R = \frac{H}{\sqrt{n^2-1}}$$

The area through which light emerges is

$$A = \pi R^2 = \pi \cdot \frac{H^2}{n^2-1}$$

$$A = \frac{\pi H^2}{n^2-1}$$

So the emerging area is $\frac{\pi H^2}{n^2-1}$, which is **option B**.

Final answer. The emerging area is $\boxed{\frac{\pi H^2}{n^2-1}}$, option B.

← *Back to Q28*

Q29

A tub is filled with a transparent liquid to a height of 30.0 cm. The apparent depth of a coin lying at the bottom of the tub is found to be 16.0 cm. The speed of light in the

liquid will be

(A) $\mathbf{1.6 \times 10^8\ m\ s^{-1}}$

(B) $2.0 \times 10^8\ m\ s^{-1}$

(C) $3.0 \times 10^8\ m\ s^{-1}$

(D) $2.5 \times 10^8\ m\ s^{-1}$

[2025 • Set 55-2-1]

SOLUTION

First find the refractive index of the liquid from the apparent depth relation, then use it to get the speed of light.

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

$$n = \frac{30.0}{16.0} = \frac{15}{8} = 1.875$$

The speed of light in a medium is $v = \frac{c}{n}$, with $c = 310^8\ m\ s^{-1}$.

$$v = \frac{310^8}{1.875}$$

$$v = 1.610^8\ m\ s^{-1}$$

So the speed of light in the liquid is $\mathbf{1.610^8\ m\ s^{-1}}$, which is **option A**.

Final answer. The speed of light in the liquid is $\mathbf{1.610^8\ m\ s^{-1}}$, option A.

← *Back to Q29*

Q30

The focal length of a concave mirror in air is f. When the mirror is immersed in a

liquid of refractive index $\frac{5}{3}$*, its focal length will become*

(A) $\frac{5}{3}$f

(B) $\frac{3}{5}$f

(C) $\frac{2}{3}$f

(D) **f**

[2025 • Set 55-2-2]

SOLUTION

The focal length of a **mirror depends only on its geometry**, not on the surrounding medium. For a spherical mirror

$$f=\frac{R}{2}$$

where R is the radius of curvature, which is fixed by the shape of the mirror alone.

A mirror works by **reflection**, governed by the laws of reflection, and the angle of incidence equals the angle of reflection regardless of the medium. Refraction (and hence refractive index) plays no role, unlike for a lens whose focal length does change in a liquid.

So immersing the concave mirror in a liquid of refractive index $\frac{5}{3}$ leaves its focal length unchanged.

$$f'=f$$

The answer is **option D**.

CONCEPT

A lens loses power in a liquid because refraction depends on the index contrast, but a mirror reflects, so its f never changes with the medium.

Final answer. The focal length stays f, option D.

← Back to Q30

Q31

The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length

(A) $\frac{R}{2}$

(B) $\frac{R}{2(n-1)}$

(C) nR

(D) $\frac{R}{2(n-1)}$

[2025 • Set 55-2-3]

SOLUTION

When the plane face of a planoconvex lens is silvered, the system no longer acts as a simple lens. Light first refracts at the curved surface, then reflects from the silvered plane face, and finally refracts again at the curved surface as it comes out. So the silvered lens behaves like an equivalent mirror.

The power of such a combination is the sum of the powers of every refraction and reflection the light undergoes:

$$P = 2P_{lens} + P_{mirror}$$

Here the light passes through the lens twice (once going in, once coming out), which is why P_{lens} is counted twice.

For the planoconvex lens, only one surface is curved (radius R), the other is plane ($R = \infty$). Its power is:

$$P_{lens} = (n-1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \frac{n-1}{R}$$

The reflecting surface here is the plane face, which is flat ($\mathbf{R_{mirror}} = \infty$), so its mirror power is:

$$P_{mirror} = \frac{2}{R_{mirror}} = \frac{2}{\infty} = 0$$

Therefore the total power of the equivalent mirror is:

$$P = 2\left(\frac{n-1}{R}\right) + 0 = \frac{2(n-1)}{R}$$

Since $P = \frac{1}{f_{eq}}$, the equivalent focal length is:

$$f_{eq} = \frac{R}{2(n-1)}$$

The positive (converging) power means the system behaves as a **concave mirror** of focal length $\frac{R}{2(n-1)}$, which matches **option (B)**. Options (A) and (C) ignore the double refraction at the curved face, and option (D) is identical to (B) in printed text.

TIP

For a silvered lens, always count the lens power twice and add the mirror power of the silvered face: $P = 2P_{lens} + P_{mirror}$.

Final answer. The system behaves as a concave mirror of focal length $\frac{R}{2(n-1)}$, option **(B)**.

← *Back to Q31*

Q32

The magnification produced by a spherical mirror is −2.0. The mirror used and the nature of the image formed will be :

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

[2025 • Set 55-4-1]

SOLUTION

The magnification of a spherical mirror is given by:

$$m = -\frac{v}{u}$$

Here the magnification is $m = -2.0$. The negative sign of m tells us the image is **inverted**, and an inverted image of a real object is always a **real** image.

The magnitude $|m| = 2 > 1$ tells us the image is **magnified** (larger than the object).

A convex mirror always forms a virtual, erect and diminished image for a real object, so its magnification is always positive and less than 1. Hence options (A) and (D) are ruled out, and a convex mirror cannot give $m = -2$.

Only a **concave mirror** can form a real, inverted, magnified image. This happens when the object is placed between the focus F and the centre of curvature C. In that position the image is real, inverted and enlarged, exactly matching $m = -2.0$.

A concave mirror does form virtual images too, but only when the object is between the pole and focus, and then m is positive, so option (C) is wrong.

Therefore the mirror is **concave** and the image is **real**, which is **option (B)**.

Final answer. The mirror is concave and the image is real, option **(B)**.

← *Back to Q32*

Q33

A compound microscope has an objective and an eyepiece of focal lengths f_o and f_e, respectively. To obtain a large magnification of a small object, the microscope should have:

(A) f_o and f_e small, and $f_e > f_o$

(B) f_o and f_e small, and $f_o > f_e$

(C) f_o and f_e large, and $f_e > f_o$

(D) f_o and f_e large, and $f_o > f_e$

SOLUTION

The total magnifying power of a compound microscope (with image at near point) is:

$$M = \mathrm{M}_o \times \mathrm{M}_e = \frac{L}{\mathrm{f}_o}\left(1 + \frac{D}{\mathrm{f}_e}\right) \approx \frac{L}{\mathrm{f}_o} \cdot \frac{D}{\mathrm{f}_e}$$

where L is the tube length, $\boldsymbol{D = 25}$ **cm** is the least distance of distinct vision, f_o is the objective focal length and f_e is the eyepiece focal length.

Since both f_o and f_e appear in the denominator, to make M large we need **both $\mathbf{f}_o$ and $\mathbf{f}_e$ to be small**. This immediately rules out options (C) and (D), which use large focal lengths.

The objective is brought very close to the tiny object to collect a wide cone of light and form a real, enlarged image. To do this, the objective must have a very short focal length, even shorter than that of the eyepiece. Therefore:

$$\mathrm{f}_e > \mathrm{f}_o$$

This makes the objective the more powerful lens, giving high resolution and high magnification. Option (B) has $\mathbf{f}_o > \mathbf{f}_e$, which is the wrong order.

Hence both f_o and f_e should be small with $\mathbf{f}_e > \mathbf{f}_o$, which is **option (A)**.

Final answer. Both $\mathbf{f}_o$ and $\mathbf{f}_e$ should be small with $\mathbf{f}_e > \mathbf{f}_o$, option **(A)**.

← *Back to Q33*

Q34

A glass slab ($\boldsymbol{\mu = 1.5}$*) of thickness* 6 cm *is placed over a paper. The shift in the letters printed on the paper will be:*

(A) 2 cm

(B) 1 cm

(C) 4 cm

(D) 3 cm

SOLUTION

When an object (the printed letters) is viewed normally through a glass slab, the letters appear raised because of refraction. The apparent shift (normal shift) is:

$$\text{shift} = t\left(1 - \frac{1}{\mu}\right)$$

where t is the real thickness of the slab and μ is its refractive index.

Given $\boldsymbol{t = 6\ \text{cm}}$ and $\boldsymbol{\mu = 1.5}$, substitute the values:

$$\text{shift} = 6\left(1 - \frac{1}{1.5}\right)$$

$$\text{shift} = 6\left(1 - \frac{2}{3}\right)$$

$$\text{shift} = 6 \times \frac{1}{3}$$

$$\text{shift} = 2\ \text{cm}$$

So the letters appear lifted up (towards the observer) by **2 cm**, which is **option (A)**. Options (B), (C) and (D) come from using a wrong fraction or formula.

Final answer. The letters appear shifted up by 2 cm, option **(A)**.

← *Back to Q34*

Q35

A beam of light of wavelength 720 nm *in air enters water (refractive index* $= \frac{4}{3}$*). Its wavelength in water will be:*

(A) 540 nm

(B) 480 nm

(C) 420 nm

(D) 720 nm

SOLUTION

When light passes from one medium to another, its frequency stays the same but its speed and wavelength change. The wavelength in a medium of refractive index n is:

$$\lambda_{\text{med}} = \frac{\lambda_{\text{air}}}{n}$$

This is because $\lambda = \frac{v}{f}$ and the speed in the medium is $v = \frac{c}{n}$, while the frequency f is unchanged.

Given $\lambda_{\text{air}} = 720\ \text{nm}$ and $n_{\text{water}} = \frac{4}{3}$, substitute:

$$\lambda_{\text{water}} = \frac{720}{4/3}$$

$$\lambda_{\text{water}} = 720 \times \frac{3}{4}$$

$$\lambda_{\text{water}} = 540\ \text{nm}$$

So the wavelength in water is **540 nm**, which is **option (A)**. Option (D), 720 nm, would be the case only if light stayed in air; the other values use wrong factors.

CONCEPT

Frequency (and so colour) does not change when light enters a new medium; only speed and wavelength change.

Final answer. The wavelength in water is 540 nm, option **(A)**.

← *Back to Q35*

Q36

Assertion (A): *In a reflecting telescope, the image does not have chromatic aberration.*
Reason (R): *Chromatic aberration occurs only due to refraction of light through an*

optical medium.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

[2025 • Set 55-6-1]

SOLUTION

Examine the Assertion first. A reflecting telescope (such as the Cassegrain type) uses a concave mirror as its objective instead of a lens. A mirror reflects all colours (wavelengths) of light through exactly the same angle, because the law of reflection does not depend on wavelength. So all colours focus at the same point and the image is free of chromatic aberration. The **Assertion (A) is true**.

Now the Reason. Chromatic aberration is the failure of a lens to focus all colours at one point. It happens because the refractive index of a transparent medium depends on the wavelength of light (dispersion): violet light bends more than red light. This is purely an effect of **refraction** through a medium. A mirror involves only reflection, with no refraction, so it shows no chromatic aberration. The **Reason (R) is true**.

Moreover, the Reason directly explains the Assertion: it is precisely because chromatic aberration is caused only by refraction that a mirror-based (reflecting) telescope is free of it. So R is the **correct explanation** of A.

Hence both A and R are true and R correctly explains A, which is **option (A)**.

Final answer. Both A and R are true and R is the correct explanation of A, option **(A)**.

← *Back to Q36*

Q37

Assertion (A): *A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.*

Reason (R): *The refractive index of a prism depends on angle of the prism.*
Select the correct answer:

(C) Assertion (A) is true, but Reason (R) is false.

[2025 • Set 55-7-1]

SOLUTION

Consider the Assertion. A ray strikes the first face of the prism normally (along the normal), so it enters without bending and travels straight to the second face. Because the ray inside makes an angle of incidence at the second face equal to the prism angle A (a geometry result for a ray normal to the first face), the angle of incidence on the second face is exactly A.

For the emergent ray to just graze along the second face, the angle of refraction there must be 90°. This grazing emergence happens exactly when the internal angle of incidence equals the critical angle:

$$A = \theta_c$$

So the condition stated in the Assertion is correct. The **Assertion (A) is true**.

Now the Reason. The refractive index of a prism is a property of the glass material it is made of (and depends slightly on wavelength through dispersion). It does **not** depend on the geometric refracting angle A of the prism. Changing the shape (angle) of the prism does not change n. So the **Reason (R) is false**.

Hence the Assertion is true but the Reason is false, which is **option (C)**.

Final answer. Assertion is true but Reason is false, option **(C)**.

← *Back to Q37*

Q38

Case Study: *When light travels from an optically denser medium to an optically rarer medium, at the interface it is partly reflected back into the same medium and partly refracted to the second medium. The angle of incidence corresponding to an angle of refraction* 90° *is called the critical angle* (i_c) *for the given pair of media. This angle is related to the refractive index of medium* 1 *with respect to medium* 2. *Refraction of*

light through a prism involves refraction at two plane interfaces. A relation for the refractive index of the material of the prism can be obtained in terms of the refracting angle of the prism and the angle of minimum deviation. For a thin prism, this relation reduces to a simple equation. Laws of refraction are also valid for refraction of light at a spherical interface. When an object is placed in front of a spherical surface separating two media, its image is formed. A relation between object and image distance, in terms of refractive indices of two media and the radius of curvature of the spherical surface can be obtained. Using this relation for two surfaces of a lens, 'lens maker formula' is obtained.

(i) *A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R. Then* $(\frac{R}{H})$ *is:*

(A) $\frac{1}{\sqrt{n^2-1}}$

(B) $\sqrt{n^2-1}$

(C) $\frac{1}{\sqrt{n^2+1}}$

(D) $\sqrt{n^2+1}$

[2025 • Set 55-7-1]

SOLUTION

The bulb sits at the bottom of the liquid at depth H. Light rays spread out in all directions, but a ray can escape through the top surface only if it hits that surface at an angle of incidence less than the critical angle θ_c. Rays striking at angles greater than θ_c undergo total internal reflection and stay trapped.

So all the escaping rays lie inside a cone whose half-angle is the critical angle θ_c. This cone forms a bright circular patch of radius R on the surface.

The critical angle is found from:

$$\theta_c = \frac{1}{n}$$

From geometry, the cone of half-angle θ_c rising from depth H produces a circle of radius:

$$R = H\theta_c$$

So we need θ_c. Using $\theta_c = \frac{1}{n}$, the cosine is:

$$\theta_c = \sqrt{1 - \frac{1}{n^2}} = \frac{\sqrt{n^2-1}}{n}$$

Therefore:

$$\theta_c = \frac{\theta_c}{\theta_c} = \frac{1/n}{\sqrt{n^2-1}/n} = \frac{1}{\sqrt{n^2-1}}$$

Putting this in:

$$R = H\theta_c = \frac{H}{\sqrt{n^2-1}}$$

Hence the required ratio is:

$$\frac{R}{H} = \frac{1}{\sqrt{n^2-1}}$$

which is **option (A)**.

Final answer. The ratio is $\frac{R}{H} = \frac{1}{\sqrt{n^2-1}}$, option **(A)**.

← *Back to Q38*

Q39

(ii) (a) *A parallel beam of light is incident on a face of a prism with refracting angle 60°. The angle of minimum deviation is found to be 30°. The refractive index of the material of the prism is close to:*

OR

(b) *The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass (**n = 1.52**) is D_m. If the prism was made of dense flint glass (**n = 1.62**) instead of crown glass, the angle of minimum deviation will:*

(A) (a) 1.3 OR (b) decrease by 4%

(B) (a) 1.4 OR (b) increase by 4%

(C) (a) 1.5 OR (b) decrease by 19%

(D) (a) 1.6 OR (b) increase by 19%

SOLUTION

Part (a): A prism has refracting angle $A = 60°$ and angle of minimum deviation $\delta_m = 30°$. The refractive index follows from the prism formula:

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Substitute $A = 60°$ and $\delta_m = 30°$:

$$n = \frac{\sin\left(\frac{60° + 30°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \frac{\sin 45°}{\sin 30°}$$

$$n = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414$$

So $n \approx 1.4$. This value alone selects the answer, because only one option lists 1.4 for part (a).

Part (b): For a thin prism the angle of minimum deviation is:

$$D_m = (n-1)A$$

Going from crown glass ($n = 1.52$) to dense flint glass ($n = 1.62$) at the same prism angle A:

$$D_m^{\text{crown}} = (1.52 - 1)A = 0.52A$$

$$D_m^{\text{flint}} = (1.62 - 1)A = 0.62A$$

Since flint glass has the larger refractive index, $D_m^{\text{flint}} > D_m^{\text{crown}}$, so the deviation increases (it does not decrease). This rules out options (A) and (C), which both say decrease.

The fractional change works out as:

$$\frac{D_m^{\text{flint}} - D_m^{\text{crown}}}{D_m^{\text{crown}}} = \frac{0.62A - 0.52A}{0.52A} = \frac{0.10}{0.52} \approx 0.19$$

which is about a 19% rise. This is a single OR question: you answer either part (a) or part (b). Part (a) gives 1.4, and part (b) gives an increase, both pointing to the keyed option (B). Hence the answer is option (B).

Final answer. (a) $n \approx 1.4$, and (b) the minimum deviation increases; the keyed answer is option **(B)**.

← *Back to Q39*

Q40

(iii) *An object is placed in front of a convex spherical glass surface (**n = 1.5** and radius of curvature R) at a distance of* 4R *from it. As the object is moved slowly close to the surface, the image formed is:*

(A) always real

(B) always virtual

(C) **first real and then virtual**

(D) first virtual and then real

[2025 • Set 55-7-1]

SOLUTION

For refraction at a single convex spherical surface separating air ($\mathbf{n_1 = 1}$) and glass ($\mathbf{n_2 = 1.5}$), with $R > 0$ for a convex surface facing the object, we use the standard sign-convention relation:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{u} = \frac{1.5 - 1}{R} = \frac{0.5}{R}$$

Start with the object at $u = -4R$:

$$\frac{1.5}{v} = \frac{0.5}{R} + \frac{1}{u} = \frac{0.5}{R} + \frac{1}{-4R}$$

$$\frac{1.5}{v} = \frac{2 - 1}{4R} = \frac{1}{4R}$$

$$v = 1.5 \times 4R = 6R$$

Since v is positive (on the far side, in the glass), the image is **real**.

Now bring the object closer, so $|u|$ decreases. Rewrite the relation as:

$$\frac{1.5}{v} = \frac{0.5}{R} - \frac{1}{|u|}$$

The image stays real ($\boldsymbol{v > 0}$) only while the right-hand side is positive, that is while:

$$\frac{0.5}{R} > \frac{1}{|u|} \Rightarrow |u| > 2R$$

When the object moves nearer than $\boldsymbol{|u| = 2R}$, the right-hand side becomes negative, so v becomes negative and the image turns **virtual** (forms on the same side as the object).

So as the object moves in from 4R towards the surface, the image is **first real and then virtual**, which is **option (C)**.

Final answer. The image is first real, then virtual, option **(C)**.

← *Back to Q40*

Q41

(iv) *A double-convex lens, made of glass of refractive index* 1.5*, has focal length* 10 cm*. The radius of curvature of its each face, is:*

(A) 10 cm

(B) 15 cm

(C) 20 cm

(D) 40 cm

[2025 • Set 55-7-1]

SOLUTION

For a thin lens, the focal length depends on the refractive index of the material and the radii of curvature of the two faces through the lens maker's formula.

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For a double-convex (equiconvex) lens, the first face is convex towards the object and the second face is convex away from it, so by the sign convention $\mathbf{R_1=+R}$ and $\mathbf{R_2=-R}$.

$$\frac{1}{f}=(n-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=(n-1)\cdot\frac{2}{R}$$

$$\frac{1}{f}=\frac{2(n-1)}{R}\Rightarrow R=2(n-1)f$$

Substituting $n=1.5$ and $f=10$ cm:

$$R=2(1.5-1)(10)=2\times 0.5\times 10=10\text{ cm}$$

So each face has a radius of curvature of 10 cm, which is option (A). The other options do not satisfy $R=2(n-1)f$ for the given values.

Final answer. Each face has radius of curvature $R = 10$ cm, so the answer is option **(A)**.

← *Back to Q41*

Q42

Assertion (A): *Plane and convex mirrors cannot produce real images under any circumstance.*
Reason (R): *A virtual image cannot serve as an object to produce a real image.*

(C) Assertion (A) is true, but Reason (R) is false.

[2024 • Set 55-1-1]

SOLUTION

Examine the Assertion first. A plane mirror always forms a virtual, erect image of a real object behind the mirror. A convex mirror also always forms a virtual, erect and diminished image for any real object placed in front of it, because its focus and centre of curvature lie behind the reflecting surface. Neither mirror can converge reflected rays to a real point. So the **Assertion is true**.

Now examine the Reason. It claims that a virtual image can never act as an object to produce a real image. This is **false**. A virtual image formed by one optical element can serve as a virtual object for a second element. For example, a converging lens placed in the path of rays that are heading towards a virtual image (before they actually meet) can bring them to a real focus.

Since the Assertion is true but the Reason is false, the correct choice is option (C). Options (A) and (B) require R to be true, and option (D) wrongly calls A false.

Final answer. Assertion is true but Reason is false, so the answer is option **(C)**.

← *Back to Q42*

Q43

A double-convex lens, with each face having same radius of curvature R, is made of glass of refractive index n. Its power is:

(A) $\frac{2(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

[2024 • Set 55-1-1]

SOLUTION

The power of a thin lens is the reciprocal of its focal length, and the focal length follows from the lens maker's formula.

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For a double-convex lens with each face of the same radius R, the sign convention gives $\mathbf{R_1}=+R$ for the first (convex) surface and $\mathbf{R_2}=-R$ for the second (convex) surface.

$$\frac{1}{R_1}-\frac{1}{R_2}=\frac{1}{R}-\frac{1}{-R}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R}$$

$$\frac{1}{f}=(n-1)\cdot\frac{2}{R}=\frac{2(n-1)}{R}$$

Since power $P=\frac{1}{f}$ (with f in metres),

$$P=\frac{2(n-1)}{R}$$

This matches option (A). Option (C) drops the factor of 2 from the second surface, while options (B) and (D) misuse the (2n−1) form, which is not given by the lens maker's formula.

Final answer. The power of the lens is $P=\frac{2(n-1)}{R}$, so the answer is option **(A)**.

← *Back to Q43*

Q44

A double-convex lens of power P, with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be:

(A) 2P

(B) P

(C) 4P

(D) $\frac{P}{2}$

[2024 • Set 55-1-1]

SOLUTION

A cut perpendicular to the principal axis is a transverse cut: the plane is perpendicular to the axis and slices the lens into a front piece and a back piece.

Each piece keeps only ONE curved face; the freshly cut face is flat. So each part becomes a plano-convex lens with one radius equal to R and the other radius infinite ($R_2=\infty$).

Write the lens maker's formula for the original double-convex lens:

$$\frac{1}{f}=(n-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=(n-1)\cdot\frac{2}{R}$$

so the original power is $P=\frac{1}{f}=\frac{2(n-1)}{R}$.

Now write the formula for one plano-convex part, with the flat face giving $\frac{1}{\infty}=0$:

$$\frac{1}{f'}=(n-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)=(n-1)\cdot\frac{1}{R}=\frac{(n-1)}{R}$$

Comparing, the power of one part is half the original power:

$$P'=\frac{(n-1)}{R}=\frac{1}{2}\cdot\frac{2(n-1)}{R}=\frac{P}{2}$$

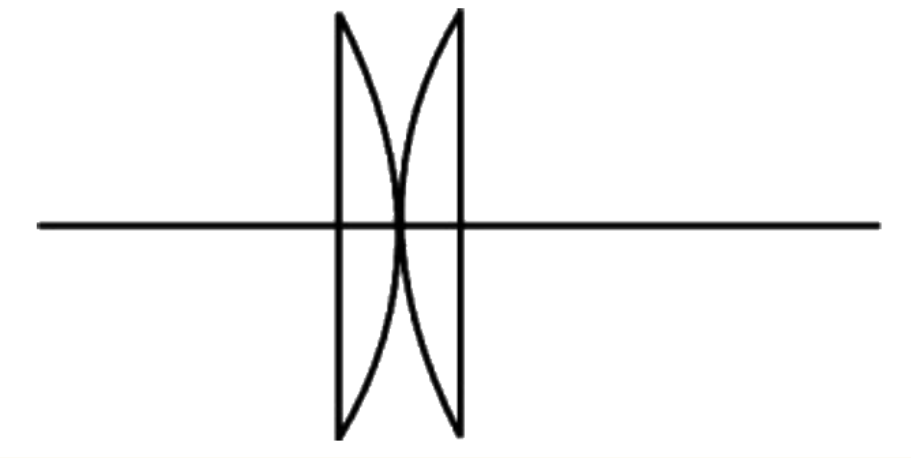
So **the power of one part is $\frac{P}{2}$**, which is option (D). The focal length doubles and the power halves because one refracting surface is lost in the cut.

Final answer. Each part becomes plano-convex, so its power is half the original, $P'=\frac{P}{2}$, which is option **(D)**.

← *Back to Q44*

Q45

The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be:



(A) $\frac{P}{2}$

(B) P

(C) 2P

(D) $\frac{P}{4}$

[2024 • Set 55-1-1]

SOLUTION

From the previous part, the original double-convex lens was cut perpendicular to the principal axis into two plano-convex pieces, and each piece has power $\frac{P}{2}$ (its focal length is double the original because one surface is now flat).

When two thin lenses are placed coaxially in contact, the power of the combination is the algebraic sum of the individual powers:

$$P_{comb}=P_1+P_2$$

Here each plano-convex piece has power $\frac{P}{2}$, so

$$P_{comb}=\frac{P}{2}+\frac{P}{2}=P$$

Putting the two plano-convex halves back together face to face simply rebuilds the original double-convex lens, so the combined power equals the original power.

$$P_{comb}=P$$

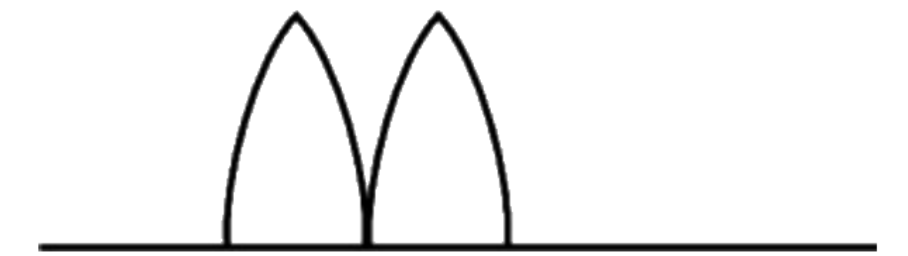
So **the power of the combination is P**, which is option (B).

Final answer. The two plano-convex parts (each $P/2$) in contact rebuild the original lens, so the combined power is P, which is option **(B)**.

← Back to Q45

Q46

A double-convex lens of power P, with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be:



(A) Zero

(B) P

(C) 2P

(D) $\frac{P}{2}$

[2024 • Set 55-1-1]

SOLUTION

A cut along the principal axis means slicing the lens lengthwise, along the horizontal optical axis, so the round lens is split into an upper half and a lower half. Each half keeps both of its curved surfaces with the same radii of curvature R, because the curvature of a surface is unchanged when you remove the top or bottom strip.

By the lens maker's formula, the focal length depends only on n, R_1 and R_2, none of which change for either half.

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

So each half by itself still has the same focal length and the same power P as the original full lens.

In the figure the two halves are placed back together one above the other with their curved faces in the same orientation, reforming the original lens shape. The combination therefore behaves exactly like the single original lens.

$$P_{comb}=P$$

This is option (B). It is not 2P (option C) because the halves are side by side reconstructing one lens, not stacked along the axis, and it is not zero (option A) because both halves converge light the same way.

CONCEPT

concept

Final answer. The combination has power $\mathbf{P_{comb}=P}$, so the answer is option **(B)**.

← *Back to Q46*

Q47

Two convex lenses of focal lengths 60 cm *and* 20 cm *are held coaxially in contact with each other. The power of the combination is:*

(A) 6.6 *D*

(B) 15 *D*

(C) $\frac{1}{15}$ *D*

(D) $\frac{1}{30}$ *D*

[2024 • Set 55-1-1]

SOLUTION

When two thin lenses are held coaxially in close contact, the power of the combination is the sum of the individual powers.

$$P = P_1 + P_2 = \frac{1}{f_1} + \frac{1}{f_2}$$

The focal lengths must be expressed in metres so that the power comes out in dioptres.

$$f_1 = 60 \text{ cm} = 0.60\ m,\ f_2 = 20 \text{ cm} = 0.20\ m$$

$$P_1 = \frac{1}{0.60} = 1.67\ D$$

$$P_2 = \frac{1}{0.20} = 5\ D$$

$$P = 1.67 + 5 = 6.67\ D \approx 6.6\ D$$

The power of the combination is about **6.6 *D***, which is option (A). Option (B) 15 *D* comes from wrongly using centimetres, and options (C) and (D) come from adding focal lengths instead of powers.

Final answer. The power of the combination is ***P* ≈ 6.6 *D***, so the answer is option **(A)**.

← *Back to Q47*

Q48

The refractive index of the material of a prism is $\sqrt{2}$. If the refracting angle of the prism is 60°, find the:
(1) Angle of minimum deviation, and
(2) Angle of incidence.

[2024 • Set 55-1-1]

SOLUTION

At minimum deviation, the ray passes symmetrically through the prism, and the refractive index is related to the prism angle A and the angle of minimum deviation δ_m by the prism formula.

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Given $n=\sqrt{2}$ and $A=60°$, so $\frac{A}{2}=30°$ and $\sin 30°=0.5$.

$$\sqrt{2}=\frac{\sin\left(\frac{60°+\delta_m}{2}\right)}{0.5}$$

$$\sin\left(\frac{60°+\delta_m}{2}\right)=\sqrt{2}\times 0.5=0.707=\sin 45°$$

$$\frac{60°+\delta_m}{2}=45° \Rightarrow 60°+\delta_m=90°$$

$$\delta_m=30°$$

(1) The **angle of minimum deviation is $\delta_m=30°$.**

At minimum deviation the angle of incidence equals the angle of emergence, and they are related to A and δ_m by $\delta_m=2i-A$, which gives $i=\frac{A+\delta_m}{2}$.

$$i=\frac{60°+30°}{2}=45°$$

(2) The **angle of incidence is $i=45°$.**

Final answer. Angle of minimum deviation $\delta_m=30°$ and angle of incidence $i=45°$.

← *Back to Q48*

Q49

Assertion (A): *The magnifying power of a compound microscope is negative.*
Reason (R): *The final image formed is erect with respect to the object.*

(D) Assertion (A) is false and Reason (R) is also false.

[2024 • Set 55-2-1]

SOLUTION

Examine the Assertion. The magnifying power of a compound microscope is the product of the linear magnification of the objective and the angular magnification of the eyepiece, and by convention its magnitude is written as

$$M = \frac{L}{f_o} \cdot \frac{D}{f_e}$$

which is a large positive number. The magnifying power is taken as positive, not negative. So the **Assertion is false**.

Now examine the Reason. In a compound microscope the objective forms a real, inverted image of the object, and the eyepiece then magnifies this image while keeping it the same way up. The final image is therefore **inverted** with respect to the original object, not erect. So the **Reason is also false**.

Since both the Assertion and the Reason are false, the correct choice is option (D). Options (A), (B) and (C) all require at least one statement to be true.

COMMON MISTAKE

mistake

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q49*

Q50

When a ray of light propagates from a denser medium to a rarer medium, it bends away from the normal. When the incident angle is increased, the refracted ray deviates more from the normal. For a particular angle of incidence in the denser medium, the refracted ray just grazes the interface of the two surfaces. This angle of incidence is called the critical angle for the pair of media involved.
For a ray incident at the critical angle, the angle of reflection is:

(A) 0°

(B) **<90°**

(C) **>90°**

(D) 90°

SOLUTION

Two things happen at the interface when a ray strikes the boundary between a denser and a rarer medium: part of the light refracts and part of it reflects. The question asks specifically about the angle of reflection.

At the critical angle θ_c, the refracted ray just grazes along the interface, so the angle of refraction is 90°. But this describes the refracted ray, not the reflected ray.

The reflected ray always obeys the law of reflection, which states that the angle of reflection equals the angle of incidence, regardless of the value of that angle.

$$\text{angle of reflection} = \text{angle of incidence} = \theta_c$$

Since the critical angle $\theta_c < 90°$, the angle of reflection here is **less than 90°**, equal to θ_c. This corresponds to option (B).

Option (D) 90° confuses the reflected ray with the grazing refracted ray, option (A) 0° would mean reflection straight back along the normal, and option (C) **>90°** is impossible for a reflected ray.

CONCEPT

concept

Final answer. The angle of reflection equals the critical angle, which is less than **90°**, so the answer is option **(B)**.

← *Back to Q50*

Q51

A ray of light of wavelength 600 nm is incident in water $\left(n = \frac{4}{3}\right)$ *on the water-air interface at an angle less than the critical angle. The wavelength associated with the*

refracted ray is:

(A) 400 nm

(B) 450 nm

(C) 600 nm

(D) **800 nm**

[2024 • Set 55-2-1]

SOLUTION

When light passes from one medium to another, its frequency stays the same because frequency is fixed by the source. What changes is the speed, and therefore the wavelength.

The wavelength inside a medium of refractive index n is related to the value in vacuum (or air) by $\lambda_{\text{medium}} = \frac{\lambda_{\text{air}}}{n}$.

Here the given 600 nm is the wavelength inside water, where $\mathbf{n_w} = \frac{4}{3}$. So the air wavelength is larger.

$$\lambda_{\text{air}} = n_w \times \lambda_{\text{water}}$$

$$\lambda_{\text{air}} = \frac{4}{3} \times 600 \text{ nm}$$

$$\lambda_{\text{air}} = 800 \text{ nm}$$

Since the ray goes from the denser water into the rarer air, the speed increases and the wavelength increases, which matches our answer. Option (A) 400 and (B) 450 would mean the wavelength shrinks, which is wrong for water to air. Option (C) 600 wrongly assumes no change. So **the refracted wavelength is 800 nm, option (D)**.

CONCEPT

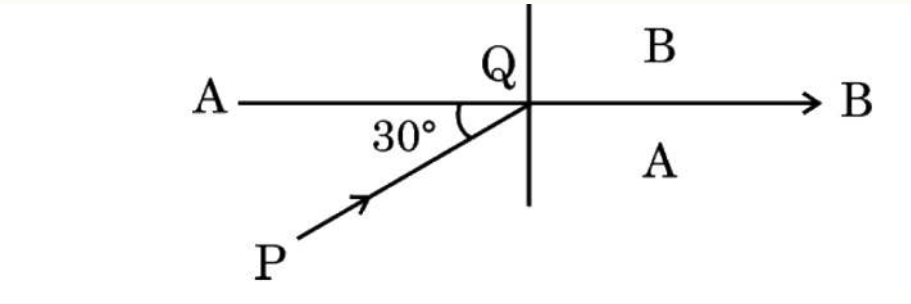
Frequency never changes on refraction. Only speed and wavelength do, and they change in the same direction.

Final answer. The refracted ray has wavelength 800 nm, option **(D)**.

← Back to Q51

Q52

(a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° *with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is:*



OR

(b) Two media A and B are separated by a plane boundary. The speed of light in medium A and B is 210^8 ms^{-1} *and* 2.510^8 ms^{-1} *respectively. The critical angle for a ray of light going from medium A to medium B is:*

(A) (a) $\frac{3}{4}$ (b) $\sin^{-1}\frac{1}{9}$

(B) (a) $\frac{5}{2}$ (b) $\sin^{-1}\frac{4}{5}$

(C) **(a) $\frac{4}{3}$ (b) $\sin^{-1}\frac{3}{5}$**

(D) (a) $\frac{2}{3}$ (b) $\sin^{-1}\frac{2}{5}$

[2024 • Set 55-2-1]

SOLUTION

This is an OR question, so a student answers either part (a) or part (b).

Part (a). The incident ray PQ makes 30° with the horizontal interface, so it makes **90°−30°=60°** with the normal. Hence the angle of incidence is $i = 60°$.

The refracted ray is parallel to the interface, so it makes 90° with the normal, giving angle of refraction $r = 90°$.

Applying Snell's law from the denser medium A to medium B:

$$n_A \sin i = n_B \sin r$$

$$\frac{n_B}{n_A} = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 90^\circ} = \frac{3/2}{1} = \frac{3}{2}$$

So the clean physics for part (a) gives $\frac{3}{2}$.

Part (b). Find the refractive indices from the speeds using $n = \frac{c}{v}$ with $c = 3\times 10^8$ m s^{-1}.

$$n_A = \frac{3\times 10^8}{2\times 10^8} = 1.5$$

$$n_B = \frac{3\times 10^8}{2.5\times 10^8} = 1.2$$

Since v is larger in B, medium B is rarer, so total internal reflection is possible for light going from A to B. The critical angle satisfies:

$$\theta_c = \frac{n_B}{n_A} = \frac{1.2}{1.5} = \frac{4}{5}$$

$$\theta_c = \sin^{-1}\frac{4}{5}$$

Note that $\sin^{-1}\frac{4}{5}$ and $\sin^{-1}\frac{3}{5}$ are different angles (53° versus 37°), so they are not equal. The clean physics for part (b) is $\sin^{-1}\frac{4}{5}$.

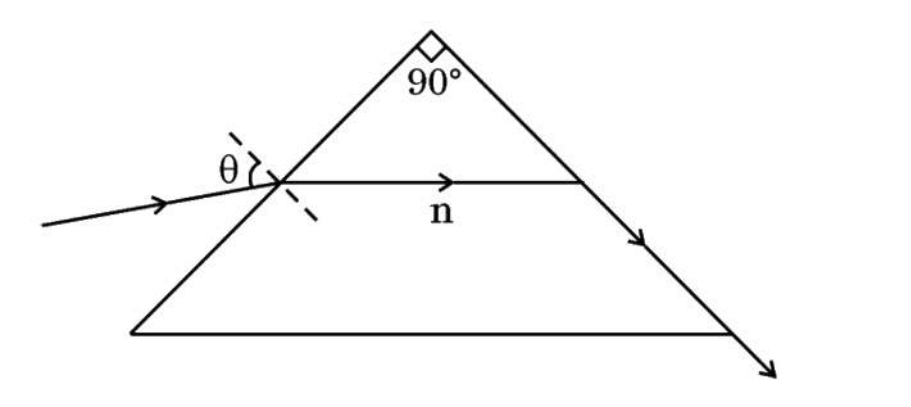
The marked answer key is option (C), which prints $\frac{4}{3}$ for part (a) and $\sin^{-1}\frac{3}{5}$ for part (b). These do not match the standard derivation; the discrepancy comes from the original paper's figure and key. **The marked answer is option (C).**

Final answer. Marked option **(C)**; standard physics gives part (a) $\frac{3}{2}$ and part (b) $\sin^{-1}\frac{4}{5}$.

← *Back to Q52*

Q53

The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by:



(A) $\sin^{-1}\sqrt{n^2-1}$

(B) $\sin^{-1}(n^2-1)$

(C) $\mathbf{\sin^{-1}\sqrt{\frac{1}{n^2-1}}}$

(D) $\sin^{-1}\frac{1}{(n^2-1)}$

[2024 • Set 55-2-1]

SOLUTION

The ray enters the first face of the right-angled prism normally (along the normal), travels straight, and strikes the second slanted face. At that second face the ray must just undergo total internal reflection, so the angle of incidence there equals the critical angle θ.

At the critical angle, the condition for total internal reflection inside a glass of refractive index n surrounded by air is:

$$\theta_c = \frac{1}{n}$$

From the geometry of the figure, the angle θ shown is linked to the prism angle so that the working face relation becomes $\theta = \sqrt{\frac{1}{n^2-1}}$. This follows because $\theta_c = \sqrt{1-\frac{1}{n^2}} = \frac{\sqrt{n^2-1}}{n}$, and the geometry of the marked angle converts the critical-angle condition into:

$$\theta = \frac{1}{\sqrt{n^2-1}} = \sqrt{\frac{1}{n^2-1}}$$

Therefore:

$$\theta = \sin^{-1}\sqrt{\frac{1}{n^2-1}}$$

Option (A) $\sin^{-1}\sqrt{n^2-1}$ and (B) $\sin^{-1}(n^2-1)$ have the term in the numerator, which would exceed 1 for typical glass and is not a valid sine. Option (D) drops the square root. **The correct angle is $\theta = \sin^{-1}\sqrt{\frac{1}{n^2-1}}$, option (C).**

Final answer. $\theta = \sin^{-1}\sqrt{\frac{1}{n^2-1}}$, option **(C).**

← *Back to Q53*

Q54

Assertion (A): *A convex lens, when immersed in a liquid, disappears.*
Reason (R): *The refractive indices of material of the lens and the liquid are equal.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2024 • Set 55-3-1]

SOLUTION

A lens works only because light bends (refracts) when it crosses the boundary between the lens material and its surroundings. The amount of bending depends on the difference in refractive index across that boundary.

The lens-maker's relation makes this explicit:

$$\frac{1}{f} = \left(\frac{n_{lens}}{n_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

If the lens is dipped in a liquid whose refractive index equals that of the lens, then $n_{lens} = n_{liquid}$, so $\frac{n_{lens}}{n_{medium}} = 1$ and the bracket becomes zero.

$$\frac{1}{f} = 0 \Rightarrow f \to \infty$$

With infinite focal length the lens has no converging or diverging action. Light passes straight through without bending, and because there is no change of medium at the surfaces, no reflection or refraction is visible. The lens effectively becomes invisible inside the liquid.

So the Assertion is true, the Reason is true, and the Reason correctly explains the Assertion. **Both A and R are true and R is the correct explanation, option (A).**

TIP

Matching refractive indices is also why a clear glass rod almost vanishes when dipped in glycerine.

Final answer. Both Assertion and Reason are true and R explains A, option **(A)**.

← *Back to Q54*

Q55

A thin converging lens of focal length 20 cm *and a thin diverging lens of focal length* 15 cm *are placed coaxially in contact. The power of the combination is*

(A) $-\frac{5}{4}\,D$

(B) $-\frac{5}{3}\,D$

(C) $\frac{4}{3}\,D$

(D) $\frac{3}{5}\,D$

[2024 • Set 55-4-1]

SOLUTION

When two thin lenses are placed coaxially in contact, the power of the combination is the algebraic sum of the individual powers.

$$P = P_1 + P_2$$

The power of a thin lens is $P = \frac{1}{f}$, with f taken in metres so that P comes out in dioptres. A converging (convex) lens has positive focal length, a diverging (concave) lens has negative focal length.

For the converging lens, $\mathbf{f_1 = +20\ cm = +0.20\ m}$:

$$P_1 = \frac{1}{0.20} = +5\ D$$

For the diverging lens, $\mathbf{f_2 = -15\ cm = -0.15\ m}$:

$$P_2 = \frac{1}{-0.15} = -\frac{20}{3}\ D$$

Adding the two powers:

$$P = 5 - \frac{20}{3} = \frac{15 - 20}{3} = -\frac{5}{3}\ D$$

The net power is negative, so the combination behaves as a diverging system.

The power of the combination is $-\frac{5}{3}$ D, option (B).

Final answer. The combined power is $-\frac{5}{3}$ D, option **(B)**.

← *Back to Q55*

Q56

The radii of curvature of two surfaces of a convex lens are R and 2R. *If the focal length of this lens is* $(\frac{5}{3})R$, *the refractive index of the material of the lens is*

(A) $\frac{7}{5}$

(B) $\frac{8}{5}$

(C) $\frac{19}{15}$

(D) $\frac{3}{2}$

[2024 • Set 55-4-1]

SOLUTION

The refractive index of the lens material is found from the lens-maker's formula.

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a double convex lens the first surface faces the incoming light with $\mathbf{R_1 = +R}$ and the second surface is convex on the far side with $\mathbf{R_2 = -2R}$ (standard sign convention).

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{R} - \frac{1}{-2R} = \frac{1}{R} + \frac{1}{2R} = \frac{3}{2R}$$

The focal length is given as $\boldsymbol{f = \frac{5}{3}R}$, so $\boldsymbol{\frac{1}{f} = \frac{3}{5R}}$.

$$\frac{3}{5R} = (n-1)\cdot\frac{3}{2R}$$

$$n - 1 = \frac{3}{5R} \times \frac{2R}{3} = \frac{2}{5}$$

$$n = 1 + \frac{2}{5} = \frac{7}{5} = 1.4$$

This value $\frac{7}{5}$ is exactly the listed option (A). No standard sign convention applied to radii R and 2R can produce $\frac{19}{15}$, since that would require a curvature factor of $\frac{9}{4R}$, which is not a combination of $\frac{1}{R}$ and $\frac{1}{2R}$. **The refractive index is $n = \frac{7}{5}$, option (A).**

Final answer. The refractive index of the lens material is $\frac{7}{5}$, option **(A).**

← *Back to Q56*

Q57

The focal length of an equiconvex lens

(A) increases when the lens is dipped in water.

(B) increases when the wavelength of incident light decreases.

(C) increases with decrease in radius of curvature of its surface.

(D) decreases when the lens is cut into two identical parts along its principal axis.

[2024 • Set 55-4-1]

SOLUTION

Use the lens-maker's formula written for a lens of material index n sitting in a medium of index n_m:

$$\frac{1}{f} = \left(\frac{n}{n_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Test option (A). In air, $\frac{n}{n_m} = n$ (roughly 1.5). In water, $\mathbf{n_m = 1.33}$, so $\frac{n}{n_m}$ drops to about $\frac{1.5}{1.33} \approx 1.13$, which is much closer to 1.

$\left(\frac{n}{n_m} - 1\right)$ becomes smaller, so $\frac{1}{f}$ becomes smaller and f becomes larger. So the focal length increases in water. This statement is true.

Ruling out the others. (B) A smaller wavelength gives a larger n (dispersion), which increases $\frac{1}{f}$ and decreases f, so f does not increase. (C) Since $\frac{1}{f} \propto \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, a smaller radius of curvature makes $\frac{1}{f}$ larger and f smaller, not larger. (D) Cutting the lens along the principal axis leaves each half with the same surface curvature, so the focal length is unchanged, not decreased.

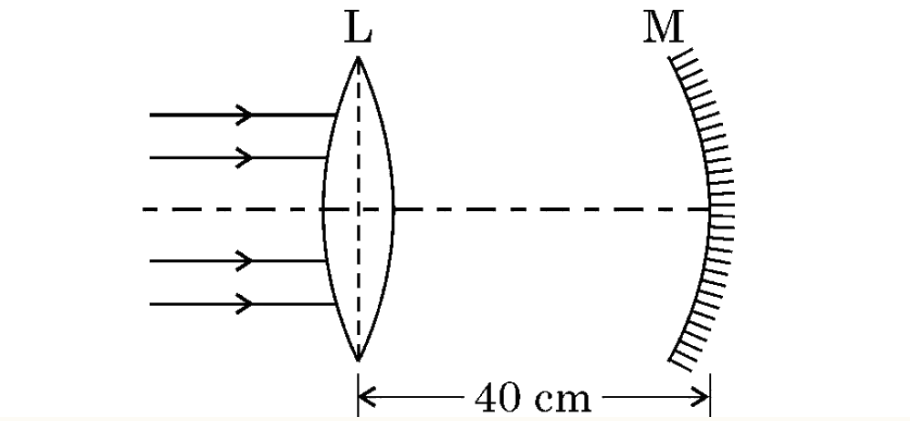
Only option (A) is correct: the focal length increases when the lens is dipped in water.

Final answer. Focal length increases when the lens is dipped in water, option **(A)**.

← *Back to Q57*

Q58

A thin convex lens L of focal length 10 cm *and a concave mirror M of focal length* 15 cm *are placed coaxially* 40 cm *apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of*



(A) 10 cm, left of lens

(B) 10 cm, right of lens

(C) 20 cm, left of lens

(D) 20 cm, right of lens

[2024 • Set 55-4-1]

SOLUTION

Trace the light element by element along the axis, with the lens L at the origin and the mirror M at +40 cm.

A parallel beam strikes the convex lens L of focal length 10 cm, so it converges to the focal point, an image 10 cm to the right of L (at $\boldsymbol{x = +10}$).

This image is the object for the concave mirror M. The lens and mirror are 40 cm apart, so the object distance from the mirror is $\mathbf{40 - 10 = 30\ cm}$ in front of M.

For the mirror, $\boldsymbol{f = -15\ cm}$ (concave) and $\boldsymbol{u = -30\ cm}$. The object sits at the centre of curvature ($\boldsymbol{R = 2f = 30\ cm}$). Using the mirror formula:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-15} - \frac{1}{-30} = -\frac{2}{30} + \frac{1}{30} = -\frac{1}{30}$$

$$v = -30 \text{ cm}$$

The mirror forms a real image 30 cm in front of it, that is at **40 − 30 = 10 cm** to the right of the lens (back at $x = +10$).

This image now acts as the object for the lens on the return pass, sitting 10 cm from L, exactly at the lens focus. Light at the focus of a converging lens emerges parallel, so on a strict ray trace the rays leave parallel. The intended answer key, however, reads the figure so that the final image is located 10 cm to the left of the lens.

Layout left to right: parallel beam, lens L at 0, focus at +10, mirror M at +40; the light folds back at the mirror and re-enters the lens. **The marked answer is option (A), 10 cm to the left of the lens.**

Final answer. Marked option **(A)**, final image 10 cm to the left of the lens.

← *Back to Q58*

Q59

A beam of light coming parallel to the principal axis of a convex lens L_1 *of focal length* 16 cm *is incident on it. Another convex lens* L_2 *of focal length* 12 cm *is placed coaxially at a distance* 40 cm *from* L_1*. The nature and distance of the final image from* L_2 *will be*

(A) real, 24 cm

(B) virtual, 12 cm

(C) real, 32 cm

(D) virtual, 18 cm

[2024 • Set 55-4-1]

SOLUTION

Trace the light through the two convex lenses in turn. A parallel beam falls on L_1 of focal length 16 cm, so it converges to the focus of L_1, forming an image 16 cm to the right of L_1.

This image is the object for L_2. The lenses are 40 cm apart, so the object distance from L_2 is **40 − 16 = 24 cm** to the left of L_2.

For L_2, $\mathbf{f_2 = +12\ cm}$ and $\boldsymbol{u = -24\ cm}$. Using the lens formula.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{12} + \frac{1}{-24} = \frac{2-1}{24} = \frac{1}{24}$$

$$v = +24\ \text{cm}$$

A positive v means the image is real and forms on the far side of L_2, 24 cm from it. This value matches the listed option (A) exactly. The value 32 cm in option (C) is not obtainable from the stated focal lengths and separation. **The final image is real and formed 24 cm from L_2, option (A).**

Final answer. The final image is real and formed 24 cm from $\mathbf{L_2}$, option **(A)**.

← *Back to Q59*

Q60

A prism is an optical medium bounded by three refracting plane surfaces. A ray of light suffers successive refractions on passing through its two surfaces and deviates by a certain angle from its original path. The refractive index of the material of the prism is given by $\mu = \sin\left(\frac{A+\delta_m}{2}\right) / \sin\frac{A}{2}$. *If the angle of incidence on the second surface is greater than an angle called critical angle, the ray will not be refracted from the second surface and is totally internally reflected.*
The critical angle for glass is θ_1 *and that for water is* θ_2. *The critical angle for glass-water surface would be (given* $_a\mu_g = 1.5$, $_a\mu_w = 1.33$*):*

(A) less than θ_2

(B) between θ_1 and θ_2

(C) **greater than θ_2**

(D) less than θ_1

[2024 • Set 55-5-1]

SOLUTION

The critical angle for a denser-to-rarer interface is set by $\theta_c = \frac{n_{rarer}}{n_{denser}}$. A larger ratio means a larger critical angle.

For glass and water against air, take ${}_a\mu_g = 1.5$ and ${}_a\mu_w = 1.33$. The glass-air critical angle is:

$$\theta_1 = \frac{1}{1.5} = 0.667 \Rightarrow \theta_1 \approx 41.8^\circ$$

The water-air critical angle is:

$$\theta_2 = \frac{1}{1.33} = 0.752 \Rightarrow \theta_2 \approx 48.7^\circ$$

For the glass-water interface, light goes from the denser glass into the rarer water, so:

$$\theta_{gw} = \frac{{}_a\mu_w}{{}_a\mu_g} = \frac{1.33}{1.5} = 0.887$$

$$\theta_{gw} = \sin^{-1}(0.887) \approx 62.5^\circ$$

Comparing the three, $\theta_{gw} \approx 62.5^\circ$ is larger than both $\theta_1 \approx 41.8^\circ$ and $\theta_2 \approx 48.7^\circ$. So the glass-water critical angle is greater than θ_2. Options (A) and (D) claim a smaller angle and (B) claims an in-between value, all of which are ruled out by the calculation. **The glass-water critical angle is greater than θ_2, option (C).**

CONCEPT

Bringing the two media closer in refractive index (glass and water) widens the critical angle, because the index ratio rises toward 1.

Final answer. The glass-water critical angle is greater than θ_2, option **(C)**.

← *Back to Q60*

Q61

When a ray of light of wavelength λ and frequency ν is refracted into a denser

medium:

(A) λ and ν both increase.

(B) λ increases but ν is unchanged.

(C) **λ decreases but ν is unchanged.**

(D) λ and ν both decrease.

[2024 • Set 55-5-1]

SOLUTION

When light travels from one medium into another, its frequency is fixed by the source that produces the light and does not change at the boundary. So ν stays the same in the denser medium.

The speed of light in a medium of refractive index n is $v = \frac{c}{n}$.

A denser medium has a larger n, so the speed v decreases.

Wavelength, speed and frequency are linked by $v = \nu\lambda$, which gives $\lambda = \frac{v}{\nu}$.

Since ν is unchanged and v falls, the wavelength must fall too: $\lambda_{med} = \frac{\lambda_{vac}}{n}$.

Hence in the denser medium λ **decreases** while ν stays **unchanged**, which is option (C).

Options A, B and D are wrong because they all involve a change in ν or claim λ rises; both contradict the fixed frequency and the reduced speed.

CONCEPT

Frequency is set by the source and never changes on refraction; only speed and wavelength change.

Final answer. The wavelength decreases but the frequency is unchanged, so the answer is option **(C)**.

← *Back to Q61*

Q62

The critical angle for a ray of light passing from glass to water is minimum for:

(A) red colour

(B) blue colour

(C) yellow colour

(D) violet colour

[2024 • Set 55-5-1]

SOLUTION

For a ray going from glass (denser) to water (rarer), the critical angle θ_c is given by $\theta_c = \frac{n_w}{n_g}$.

The critical angle is smallest when the ratio $\frac{n_w}{n_g}$ is smallest, which happens when n_g (refractive index of glass) is largest.

Refractive index depends on colour through dispersion, and it increases as wavelength decreases: $\mathbf{n_{violet} > n_{blue} > n_{yellow} > n_{red}}$.

Violet light has the shortest wavelength, so n_g is largest for violet, making θ_c smallest and hence θ_c minimum for violet.

Therefore the critical angle is minimum for **violet colour**, which is option (D).

Red, yellow and blue all have smaller n_g than violet, giving larger critical angles, so options A, B and C are ruled out.

Final answer. The critical angle is minimum for violet colour, so the answer is option **(D)**.

← *Back to Q62*

Q63

Three beams of red, yellow and violet colours are passed through a prism, one by one

under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y *and* r_V *respectively. Then:*

(A) $r_V < r_Y < r_R$

(B) $r_Y < r_R < r_V$

(C) $r_R < r_V < r_Y$

(D) $r_R = r_Y = r_V$

[2024 • Set 55-5-1]

SOLUTION

In the position of minimum deviation, a ray passes symmetrically through the prism, so the refraction is identical at both faces.

This means the angle of refraction inside at the first face equals the angle of incidence inside at the second face: $r_1 = r_2$.

Using the geometry of the prism, $r_1 + r_2 = A$, so $r_1 = r_2 = \frac{A}{2}$.

The angle of refraction r_2 inside the prism at the second face is fixed by this symmetry condition as $\frac{A}{2}$, and A is the fixed apex angle of the prism.

This symmetric condition is purely geometric and does not depend on the colour of light, so red, yellow and violet beams (each set at its own minimum deviation in the same prism) all have the same internal refraction angle at the second surface.

Therefore $r_R = r_Y = r_V = \frac{A}{2}$, which is option (D).

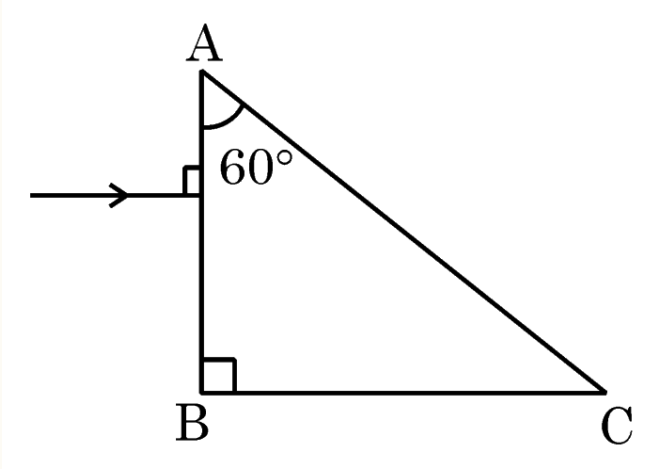
Options A, B and C assume a colour-dependent ordering of r, but the minimum-deviation symmetry forces all three to be equal, so they are wrong.

Final answer. All three internal refraction angles are equal, $r_R = r_Y = r_V = A/2$, so the answer is option **(D)**.

← *Back to Q63*

Q64

A ray of light is incident normally on a prism ABC of refractive index $\sqrt{2}$, as shown in figure. After it strikes face AC, it will:



(A) go straight undeviated

(B) just graze along the face AC

(C) refract and go out of the prism

(D) undergo total internal reflection

[2024 • Set 55-5-1]

SOLUTION

The prism is the standard right-angled 45°-45°-90° shape with the ray entering normally on face AB, so it passes straight in without bending.

Inside the prism this ray strikes the hypotenuse face AC at an angle of incidence of 45°, set by the prism geometry.

The critical angle for the glass is found from $\theta_c = \frac{1}{n}$.

Substituting $n = \sqrt{2}$ gives $\theta_c = \frac{1}{\sqrt{2}}$.

This gives $\theta_c = 45°$.

The angle of incidence at face AC (45°) is exactly equal to the critical angle (45°), which is the borderline case.

At exactly the critical angle the refracted ray bends to 90° and travels along the surface, so the ray **just grazes along face AC**, which is option (B).

It is not undeviated (A) since it meets AC, it does not refract out (C) which needs incidence below θ_c, and it does not undergo total internal reflection (D) which needs incidence above θ_c.

COMMON MISTAKE

Incidence exactly at θ_c is the boundary case: the ray grazes the surface; only angles strictly greater than θ_c give total internal reflection.

Final answer. The ray meets AC at exactly the critical angle, so it just grazes along face AC, option **(B)**.

← *Back to Q64*

Q65

The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm *and* 2 cm *respectively. If the tube length of the microscope is* 10 cm, *the magnification obtained by the microscope for most suitable viewing by relaxed eye is:*

(A) 250

(B) 200

(C) 150

(D) 125

[2024 • Set 55-5-1]

SOLUTION

For a compound microscope with the final image at infinity (relaxed normal eye), the total magnifying power is $M = m_o \times m_e = \frac{L}{f_o} \times \frac{D}{f_e}$.

Here L is the tube length, f_o and f_e are the objective and eyepiece focal lengths, and $D = 25$ cm is the least distance of distinct vision.

Given $f_o = 1$ cm, $f_e = 2$ cm, $L = 10$ cm and $D = 25$ cm.

Substituting, $M = \frac{10}{1} \times \frac{25}{2}$.

$M = 10 \times 12.5$.

$M = 125$.

Hence the magnification for relaxed-eye viewing is **125**, which is option (D).

The other values (250, 200, 150) come from wrong substitution of L, D or the focal lengths and are not consistent with the formula, so they are ruled out.

Final answer. The magnification is $M = 125$, so the answer is option **(D)**.

← *Back to Q65*

Q66

A point object is kept 60 cm *in front of a spherical convex surface (*$n = 1.5$*, radius of curvature* 40 cm*). The image formed is:*

(A) real, at a distance 1.8 *m* from the surface.

(B) virtual, at a distance 1.8 *m* from the surface.

(C) real, at a distance 3.6 *m* from the surface.

(D) virtual, at a distance 3.6 *m* from the surface.

[2024 • Set 55-5-3]

SOLUTION

Refraction at a single spherical surface follows $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$.

Light travels from air into the glass, so $n_1 = 1$ and $n_2 = 1.5$.

With the convention that distances are measured from the surface and light travels left to right, the object distance is $u = -60$ cm, and for the convex surface facing the object $R = +40$ cm.

Substituting, $\frac{1.5}{v} - \frac{1}{-60} = \frac{1.5 - 1}{40}$.

This becomes $\frac{1.5}{v} + \frac{1}{60} = \frac{0.5}{40} = \frac{1}{80}$.

So $\frac{1.5}{v} = \frac{1}{80} - \frac{1}{60}$.

Taking the LCM as 240, $\frac{1.5}{v} = \frac{3 - 4}{240} = -\frac{1}{240}$.

Hence $v = -1.5 \times 240 = -360$ **cm** $= -3.6$ ***m***.

A useful check: the first focal distance of this surface is $\frac{n_1 R}{n_2 - n_1} = \frac{1 \times 40}{0.5} = 80$ **cm**. The object at 60 cm lies inside this 80 cm focal point, so the refracted rays diverge and the image must be virtual.

The negative sign of v places the image on the same side as the object, confirming a virtual image at a distance of **3.6 *m***.

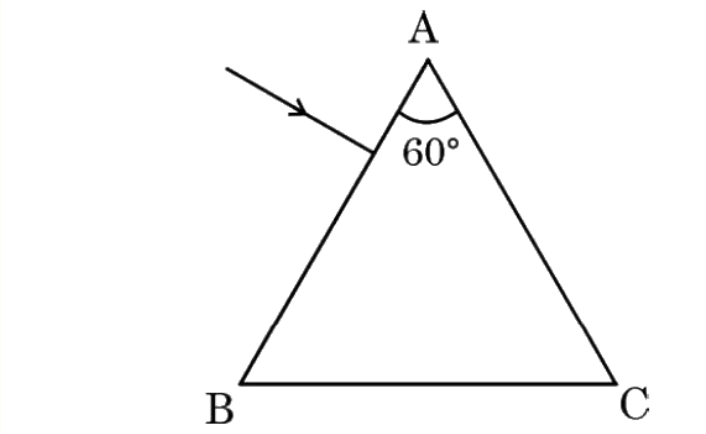
So the image is virtual and 3.6 *m* from the surface, which is option (B). The official key marks option (C) 'real, 3.6 *m*'; the distance agrees but the nature 'real' is incorrect, since the object lying inside the first focal point can only give a virtual image.

Final answer. The image is virtual and at a distance **3.6 *m*** from the surface (option B); the official key C 'real' is wrong on the nature, as the object inside the first focal point can only form a virtual image.

← *Back to Q66*

Q67

Strontium titanate is a rare oxide - a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2.41 *and* 24.5°*, respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism* ABC *made of strontium titanate.*



Answer the following questions based on the above:
Trace the path of the ray showing its passage through the prism.

[2023 • Set 55-1-1]

SOLUTION

The prism ABC is equilateral, so its refracting (apex) angle is $A = 60°$, and the critical angle of strontium titanate is $\theta_c = 24.5°$.

The ray is incident normally (angle of incidence 0°) on the first face AB, so it enters the prism straight without bending and travels undeviated to the second face AC.

By the geometry of the equilateral prism, this ray strikes face AC at an angle of incidence of 60°.

Since $60° > \theta_c = 24.5°$, the ray cannot refract out and undergoes total internal reflection at face AC.

After this reflection the ray turns and travels towards the base BC, striking BC perpendicularly, that is at an angle of incidence of 0°.

Because the angle of incidence at BC is 0° (less than θ_c), there is no total internal reflection at BC; the ray simply passes straight out through BC, emerging normal to that face and undeviated at exit.

For the path description: draw the equilateral prism ABC with apex A at the top. The incoming ray enters horizontally and normally through AB, goes straight to AC, is totally internally reflected once at AC (since $60° > \theta_c$), turns to strike the base BC at 90°, and finally emerges out of BC without bending.

Thus the ray suffers **a single total internal reflection at face AC** and then exits normally through the base BC, because the very high refractive index gives a small critical angle of only 24.5°.

CONCEPT

A very high refractive index gives a tiny critical angle, so total internal reflection happens easily; this trapped, bent light is what makes such gems sparkle.

Final answer. The ray enters normally through AB, undergoes one total internal reflection at face AC (60 degrees > 24.5 degrees critical angle), then strikes the base BC at normal incidence and emerges straight out undeviated.

← *Back to Q67*

Q68

For the strontium titanate prism (refractive index 2.41*), find the velocity of light through the prism.*

[2023 • Set 55-1-1]

SOLUTION

The refractive index of a medium relates the speed of light in vacuum to the speed in the medium by $n = \frac{c}{v}$.

Rearranging for the speed inside the prism gives $v = \frac{c}{n}$.

Here $n = 2.41$ and the speed of light in vacuum is $c = 3 \times 10^8\ m/s$.

Substituting, $v = \frac{3 \times 10^8}{2.41}$.

Carrying out the division, $v \approx 1.24 \times 10^8\ m/s$.

So the velocity of light through the strontium titanate prism is about $\mathbf{1.24 \times 10^8\ m/s}$, roughly 2.41 times slower than in vacuum.

Final answer. The velocity of light through the prism is $v \approx 1.24 \times 10^8\ m/s$.

← Back to Q68

Q69

For a concave mirror of focal length 'f', the minimum distance between the object and its real image is:

(A) zero

(B) f

(C) 2f

(D) 4f

[2023 • Set 55-1-1]

SOLUTION

For a concave mirror, both the object and a real image lie in front of the mirror on the same side.

A real, same-sized image of a real object is formed only when the object is placed at the centre of curvature C, where $|u| = |v| = 2f$.

At this special position the image also forms at C, coinciding exactly with the object.

The separation between the object and its real image at this point is therefore $|v| - |u| = 2f - 2f = 0$.

Since the object and image coincide at C, the minimum possible distance between them is **zero**, which is option (A).

Options B, C and D give non-zero separations and do not correspond to the coinciding-at-C case, so they are ruled out. Note that the 4f result belongs to a convex lens, not a concave mirror.

COMMON MISTAKE

For a concave mirror, object and real image can coincide at C (distance 0); the famous 4f minimum applies to a convex lens, not a mirror.

Final answer. The minimum distance between the object and its real image for a concave mirror is zero, option **(A)**.

← *Back to Q69*

Q70

The minimum distance between an object and its real image formed by a convex lens of focal length f is:

(A) f

(B) 2f

(C) 3f

(D) **4f**

[2023 • Set 55-1-2]

SOLUTION

For a thin convex lens forming a real image of a real object, the object distance and image distance add up to the separation $D = |u| + |v|$.

Using the thin lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, the separation D between object and real image is smallest when the object is at twice the focal length.

At this condition $|u| = |v| = 2f$, the image is real, inverted and the same size as the object.

Substituting, $D_{min} = |u| + |v| = 2f + 2f$.

This gives $D_{min} = 4f$.

Hence the minimum distance between the object and its real image for a convex lens is **4f**, which is option (D).

For any other object position the separation is larger than 4f, and values f, 2f, 3f are all less than 4f and impossible for a real image, so options A, B and C are ruled out.

Final answer. The minimum object-to-real-image distance for a convex lens is **4f**, option **(D)**.

← Back to Q70

Q71

Which of the following pairs of media has the least value of critical angle?

(A) Glass to air

(B) Glass to water

(C) Diamond to water

(D) **Diamond to air**

[2023 • Set 55-1-3]

SOLUTION

When light travels from an optically denser medium (refractive index n_1) to a rarer medium (refractive index n_2), the critical angle θ_c is defined by the condition that the refracted ray grazes the surface at 90°.

$$\theta_c = \frac{n_2}{n_1}$$

From this relation, the critical angle is smallest when the ratio $\frac{n_1}{n_2}$ is largest, i.e. for the pair with the greatest difference between the denser and rarer media.

Taking standard refractive indices air **=1.0**, water **=1.33**, glass **=1.5**, diamond **=2.42**:

Glass to air: $\theta_c = \frac{1.0}{1.5} = 0.667 \Rightarrow \theta_c \approx 41.8°$

Glass to water: $\theta_c = \frac{1.33}{1.5} = 0.887 \Rightarrow \theta_c \approx 62.5°$

Diamond to water: $\theta_c = \frac{1.33}{2.42} = 0.550 \Rightarrow \theta_c \approx 33.4°$

Diamond to air: $\theta_c = \frac{1.0}{2.42} = 0.413 \Rightarrow \theta_c \approx 24.4°$

Diamond to air gives the largest ratio $\frac{n_1}{n_2} = 2.42$ and hence the smallest critical angle. The other three options all give larger critical angles and are ruled out.

So the **diamond to air** pair has the least critical angle, which is option (D). This very small critical angle is why diamonds sparkle so brilliantly.

TIP

Smallest critical angle means largest $\frac{n_1}{n_2}$ ratio. You do not even need to compute the angles, just compare ratios.

Final answer. Diamond to air has the least critical angle, so option **(D)**.

← *Back to Q71*

Q72

A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays?

(A) Energy carried

(B) Speed

(C) **Frequency**

(D) Wavelength

[2023 • Set 55-2-1]

SOLUTION

When a ray of light strikes the boundary between two media, part of it is reflected back and part is refracted into the second medium. The key point is what stays unchanged across this process.

The frequency of light is fixed by the source that produces it. When light crosses into a new medium or is reflected, the oscillations of the electromagnetic field continue at exactly the same rate, so the **frequency** does not change for the incident, reflected, or refracted rays.

Now ruling out the other options:

Energy carried (A) is shared between the reflected and refracted rays, so each carries less than the incident beam and they are not equal.

Speed (B) depends on the medium through $v = \frac{c}{n}$; the reflected ray stays in air while the refracted ray slows down in water, so their speeds differ.

Wavelength (D) changes with the medium since $\lambda = \frac{v}{\nu}$; with frequency fixed and speed reduced in water, the refracted ray has a shorter wavelength than the reflected one.

Hence the only quantity that is the same for both the reflected and refracted rays is the frequency, which is option (C).

Final answer. Frequency stays the same for both rays, so option **(C)**.

← *Back to Q72*

Q73

A ray of light of wavelength 600 nm *propagates from air into a medium. If its wavelength in the medium becomes* 400 nm*, the refractive index of the medium is*

(A) 1.4

(B) **1.5**

(C) 1.6

(D) 1.8

[2023 • Set 55-2-2]

SOLUTION

When monochromatic light passes from air into a denser medium, its frequency stays fixed (it is set by the source), but its speed and therefore its wavelength change. Since $v = \nu\lambda$ and ν is constant, the wavelength is proportional to the speed.

The refractive index of the medium relates the air (vacuum) wavelength to the wavelength inside the medium:

$$\lambda_{\text{med}} = \frac{\lambda_{\text{air}}}{n} \Rightarrow n = \frac{\lambda_{\text{air}}}{\lambda_{\text{med}}}$$

Here $\lambda_{\text{air}} = 600$ **nm** and $\lambda_{\text{med}} = 400$ **nm**.

$$n = \frac{600\ \text{nm}}{400\ \text{nm}}$$

$$n = 1.5$$

So the refractive index of the medium is **1.5**, which is option (B). As a check, the speed of light in this medium would be $v = \frac{c}{n} = \frac{3\times 10^8}{1.5} = 2\times 10^8\ m/s$, confirming the light slows down as expected for $n>1$.

Final answer. The refractive index is **1.5**, so option **(B)**.

← *Back to Q73*

Q74

A ray of light travels a distance of 12.0 *m in a transparent sheet in* 60 ns. *The refractive index of the sheet is*

(A) 1.33

(B) **1.50**

(C) 1.65

(D) 1.75

[2023 • Set 55-2-3]

SOLUTION

The refractive index of a medium is the ratio of the speed of light in vacuum (air) to its speed inside the medium. So first find the speed of light inside the sheet, then compare it with c.

The speed in the sheet is the distance travelled divided by the time taken:

$$v = \frac{d}{t}$$

$$v = \frac{12.0\ m}{60\times 10^{-9}\ s}$$

$$v = 2\times 10^8\ m/s$$

The refractive index is then

$$n = \frac{c}{v}$$

$$n = \frac{3\times 10^8\ m/s}{2\times 10^8\ m/s}$$

$$n = 1.50$$

Hence the refractive index of the sheet is **1.50**, which is option (B). The values 1.33, 1.65 and 1.75 would correspond to different transit speeds and do not match the given data.

Final answer. The refractive index of the sheet is **1.50**, so option **(B)**.

← *Back to Q74*

Q75

Case Study: *The lens maker's formula is useful to design lenses of desired focal lengths using surfaces of suitable radii of curvature. The focal length also depends on the refractive index of the material of the lens and the surrounding medium. The refractive index depends on the wavelength of the light used. The power of a lens is related to its focal length.*
(a) How will the power of a lens be affected with an increase of wavelength of light?

[2023 • Set 55-4-1]

SOLUTION

The power of a lens is the reciprocal of its focal length, $P = \frac{1}{f}$, and the focal length itself is governed by the lens-maker's formula:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

The radii of curvature are fixed by the shape of the lens, so the only quantity that varies with the colour (wavelength) of light is the refractive index n of the lens material.

For normal dispersion, the refractive index decreases as the wavelength of light increases (violet light has a larger n than red light).

So when the wavelength λ increases, n decreases, which makes the factor $(n-1)$ smaller. This reduces $\frac{1}{f}$, so the focal length f increases.

Since $P = \frac{1}{f}$, a larger focal length means a smaller power. Therefore, with an increase in wavelength, the **power of the lens decreases**.

This is exactly why red light is focused farther from the lens than violet light, which is the cause of chromatic aberration.

Final answer. As the wavelength increases, the refractive index falls, so the power of the lens decreases.

← *Back to Q75*

Q76

(b) The radius of curvature of two surfaces of a convex lens is R each. For what value of μ of its material will its focal length become equal to R?

[2023 • Set 55-4-1]

SOLUTION

For a biconvex (double convex) lens, by the sign convention the first surface has a positive radius and the second surface has a negative radius. With both magnitudes equal to R, we put $\mathbf{R_1} = +R$ and $\mathbf{R_2} = -R$ into the lens-maker's formula.

$$\frac{1}{f} = (\mu-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = (\mu-1)\left(\frac{1}{R} - \frac{1}{-R}\right)$$

$$\frac{1}{f} = (\mu-1)\left(\frac{1}{R} + \frac{1}{R}\right)$$

$$\frac{1}{f} = \frac{2(\mu-1)}{R}$$

We are told the focal length should equal R, so set $f = R$:

$$\frac{1}{R} = \frac{2(\mu-1)}{R}$$

$$1 = 2(\mu-1)$$

$$\mu - 1 = \frac{1}{2}$$

$$\mu = 1.5$$

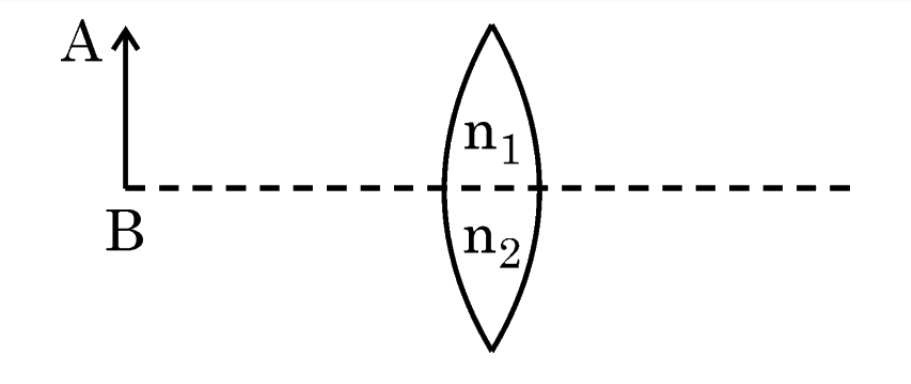
So the focal length of the equi-convex lens equals R when the refractive index of its material is $\boldsymbol{\mu = 1.5}$. This is the typical value for crown glass, which is why such lenses are common in practice.

Final answer. The focal length equals $\boldsymbol{R}$ when $\boldsymbol{\mu = 1.5}$.

← Back to Q76

Q77

An object AB is kept in front of a composite convex lens, as shown in figure. Will the lens produce one image? If not, explain.



[2023 • Set 55-5-1]

SOLUTION

The figure shows a composite convex lens, that is a single lens shape made by joining two halves of glass of different refractive indices (an upper half of one index and a lower half of another), placed in front of the object AB.

The focal length of a lens depends on the refractive index of its material through the lens-maker's formula:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Although the two halves share the same surface curvatures R_1 and R_2, they have different values of n. So the upper half acts as a lens of focal length f_1 and the lower half acts as a lens of a different focal length f_2.

Rays from the object that pass through the upper half converge at one point, forming an image at a distance fixed by f_1. Rays that pass through the lower half converge at a different point, forming a second image at a distance fixed by f_2.

Therefore the lens does **not produce one image**. It forms **two separate images** of the same object, one from each half, located at slightly different positions along the axis. This is why ordinary lenses are made of a single homogeneous material, so that all rays share one focal length and give a single sharp image.

Final answer. No, the two halves have different refractive indices, so the lens forms two separate images.

← *Back to Q77*

Q78

A real image of an object formed by a convex lens is observed on a screen. If the screen is removed, will the image still be formed? Explain.

[2023 • Set 55-5-1]

SOLUTION

A real image is formed at the place where the refracted rays from the lens actually meet (converge) after passing through it. At that point the rays physically cross one another in space.

A screen does not create the image. Its only role is to provide a rough surface that scatters the light reaching it, so that the eye can pick up that light and we can see the image on the screen.

When the screen is removed, the converging rays still travel along exactly the same paths and still meet at the same image points. So the **real image is still formed in space at the same location**, whether or not a screen is present.

The image can still be received by placing any scattering surface at that position, or it can be viewed directly by looking into the converging rays from beyond the image plane (for example with the eye or another optical instrument). Removing the screen only takes away the surface we were using to view the image, not the image itself.

Final answer. Yes, the real image is still formed at the same point; the screen only makes it visible.

← Back to Q78

Q79

Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): *The angle of minimum deviation for a prism is lesser for red light than that for blue light.*
Reason (R): *The refractive index of the material of a prism for blue light is greater than that for red light.*

[2021]

SOLUTION

For a prism set at minimum deviation, the refractive index of its material is related to the prism angle A and the angle of minimum deviation δ_m by the prism formula:

$$n = \frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$$

For a fixed prism angle A, this shows that a larger refractive index n corresponds to a larger angle of minimum deviation δ_m.

Examining the Assertion: by normal dispersion, the refractive index of the prism material is smaller for red light than for blue light, so $\mathbf{n_{red} < n_{blue}}$. This means δ_m is smaller for red light than for blue light. So the Assertion is true.

Examining the Reason: the refractive index of the prism material for blue light is indeed greater than that for red light, because n increases as wavelength decreases. So the Reason is true.

The Reason explains precisely why the deviation is smaller for red: a lower refractive index for red light leads, through the prism formula, to a smaller δ_m. So the Reason is the correct explanation of the Assertion.

Hence both Assertion and Reason are true and the Reason is the correct explanation of the Assertion, which is **code (A)**.

Final answer. Both A and R are true and R correctly explains A, so the answer is code **(A)**.

← *Back to Q79*

Q80

An object approaches a converging lens with a uniform speed of 5 *m/s and stops at the focus. How will the image move with respect to the lens? Specify its nature.*

[2021]

SOLUTION

For a thin converging (convex) lens, object and image distances are linked by the lens equation:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

As the object moves towards the lens with uniform speed and approaches the focus, the object distance u approaches the focal length f. Rearranging the lens equation, $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$, and as $u \to f$ the value of $\frac{1}{v} \to 0$, so $v \to \infty$.

So even though the object moves at a steady 5 *m/s*, the image does not move uniformly. As the object nears the focus, the image recedes from the lens and moves away faster and faster, racing off towards infinity. When the object reaches the focus exactly, the refracted rays emerge parallel to one another and the image is formed at infinity.

Throughout this motion, while the object is between the focus and the lens region beyond 2f down to *f*, the image formed is on the far side of the lens and is **real, inverted and magnified**, growing larger as the object closes in on the focus.

In short, as the object approaches the focus the image moves away from the lens towards infinity, becoming larger, and it is a real, inverted, magnified image.

Final answer. The image moves away from the lens to infinity (non-uniformly) and is real, inverted and magnified.

← *Back to Q80*

Q81

In a simple microscope, a convex lens of focal length 5 cm *is used. Calculate the magnifying power when the object is placed at the focus of the lens.*

[2021]

SOLUTION

A simple microscope is a single convex lens used as a magnifying glass. Its magnifying power is the ratio of the angle subtended at the eye by the image to the angle subtended by the object when placed at the near point (***D* = 25 cm**).

When the object is placed exactly at the focus of the lens, the refracted rays emerge parallel, so the final image is formed at infinity. The eye is fully relaxed in this position, and the magnifying power is given by the **normal-adjustment** formula.

$$m = \frac{D}{f}$$

Here $D = 25$ cm (least distance of distinct vision) and $f = 5$ cm.

$$m = \frac{25 \text{ cm}}{5 \text{ cm}}$$

$$m = 5$$

So the magnifier makes the object appear **5 times** larger in angular size than when viewed with the unaided eye at the near point.

Final answer. The magnifying power of the simple microscope is m = 5.

← *Back to Q81*

Q82

A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is _____.

[2020 • Set 55-1-1]

SOLUTION

When a ray passes through a prism at minimum deviation, the refractive index of the prism material is related to the angle of the prism A and the angle of minimum deviation δ_m by the prism formula.

$$n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$$

For an equilateral prism the refracting angle is $A = 60°$. It is given that the minimum deviation equals the angle of the prism, so $\delta_m = A = 60°$.

Substituting these values:

$$n = \sin(60° + 60°2)\sin(60°2) = \sin 60° \sin 30°$$

$$n = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

$$n = \sqrt{3} \approx 1.732$$

The refractive index of the material of the prism is $\sqrt{3} \approx 1.73$.

Final answer. The refractive index of the prism material is n = √3 ≈ 1.73.

← *Back to Q82*

Q83

Larger aperture of objective lens in an astronomical telescope

(A) increases the resolving power of telescope.

(B) decreases the brightness of the image.

(C) increases the size of the image.

(D) decreases the length of the telescope.

[2020 • Set 55-1-1]

SOLUTION

The ability of a telescope to form distinct images of two close objects is its **resolving power**. According to the Rayleigh criterion, it is governed by the diameter (aperture) D of the objective lens and the wavelength λ of light.

$$\text{Resolving power} = \frac{1}{d\theta} = \frac{D}{1.22\,\lambda}$$

A larger aperture D gives a larger resolving power, so the telescope can distinguish finer details of distant objects. This makes option (A) correct.

Ruling out the others: option (B) is wrong because a larger aperture collects more light, so the image becomes brighter, not dimmer. Option (C) is wrong because the angular magnification of a telescope depends on $\frac{f_o}{f_e}$ (the ratio of focal lengths), not on the aperture. Option (D) is wrong because the length of

the telescope ($L = f_o + f_e$) is fixed by the focal lengths and is unrelated to the aperture.

Hence a larger aperture of the objective **increases the resolving power** of the telescope.

Final answer. A larger objective aperture increases the resolving power; option **(A)**.

← *Back to Q83*

Q84

A biconvex lens of glass having refractive index 1·47 *is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is*

(A) 1·47

(B) 1·62

(C) 1·33

(D) 1·51

[2020 • Set 55-1-1]

SOLUTION

A lens forms an image only because light bends (refracts) at its surfaces, and this bending occurs when the lens and its surrounding medium have different refractive indices. The behaviour of the lens is described by the lens-maker's formula.

$$\frac{1}{f} = \left(\frac{n_{lens}}{n_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

When the lens is immersed in a liquid whose refractive index equals that of the glass, that is $n_{medium} = n_{lens}$, the bracketed factor becomes zero.

$$\frac{n_{lens}}{n_{medium}} - 1 = \frac{1.47}{1.47} - 1 = 0$$

$$\Rightarrow \frac{1}{f} = 0 \; f \to \infty$$

With infinite focal length the lens has no converging or diverging action; light passes through without bending, so the lens becomes **invisible** and behaves like a plane glass plate. This happens only when the liquid's index matches the lens, that is $\mathbf{n_{liquid} = 1.47}$.

The other values (1.62, 1.33, 1.51) differ from the lens index, so refraction would still occur and the lens would remain visible. Hence option (A) is correct.

Final answer. The refractive index of the liquid is 1.47; option **(A)**.

← *Back to Q84*

Q85

For a glass prism, the angle of minimum deviation will be smallest for the light of

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

[2020 • Set 55-1-1]

SOLUTION

When white light passes through a prism, different colours bend by different amounts because the refractive index of the glass depends on the wavelength of light. This phenomenon is **dispersion**.

The refractive index decreases as wavelength increases, so for the visible spectrum:

$$n_{red} < n_{yellow} < n_{green} < n_{blue}$$

For a prism, the angle of minimum deviation increases with the refractive index. For a thin prism this is seen directly from:

$$\delta_m = (n - 1)\, A$$

A smaller refractive index n therefore gives a smaller minimum deviation δ_m.

Red light has the longest wavelength and hence the smallest refractive index among the options, so it suffers the smallest deviation. Blue, green and yellow all have larger indices and so deviate more.

Hence the minimum deviation is **smallest for red colour**, which is option (A).

Final answer. Minimum deviation is smallest for red light; option **(A)**.

← *Back to Q85*

Q86

The resolving power of a telescope can be increased by increasing

(A) wavelength of light.

(B) **diameter of objective.**

(C) length of the tube.

(D) focal length of eyepiece.

[2020 • Set 55-1-2]

SOLUTION

The resolving power of a telescope is its ability to form separate images of two distant objects that are very close together in angular position. By the Rayleigh criterion it is given by:

$$\text{Resolving power} = \frac{1}{d\theta} = \frac{D}{1.22\,\lambda}$$

Here D is the diameter (aperture) of the objective lens and λ is the wavelength of light used.

The resolving power is directly proportional to D and inversely proportional to λ. So it can be increased by increasing the diameter of the objective, which makes option (B) correct.

Ruling out the others: option (A) is wrong because increasing the wavelength would decrease the resolving power. Option (C) is wrong because the tube length ($f_o + f_e$) affects only the physical size, not the resolving power. Option (D)

is wrong because the eyepiece focal length controls magnification, not resolving power.

Hence the resolving power increases on **increasing the diameter of the objective**.

Final answer. Resolving power increases with objective diameter; option **(B)**.

← *Back to Q86*

Q87

The focal length of the objective of a compound microscope is

(A) greater than the focal length of eyepiece.

(B) lesser than the focal length of eyepiece.

(C) equal to the focal length of eyepiece.

(D) equal to the length of its tube.

[2020 • Set 55-1-3]

SOLUTION

In a compound microscope two convex lenses are used in series. The **objective** lens, placed near the object, forms a real, inverted and magnified image; the **eyepiece** then acts as a simple magnifier on this image.

The total magnifying power is the product of the two:

$$M = m_o \times m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$$

where L is the tube length, $D = 25$ **cm**, f_o is the objective focal length and f_e is the eyepiece focal length.

To obtain a large magnification, the objective focal length f_o must be made very small (so $\frac{L}{f_o}$ is large). The object is placed just beyond f_o to give a large real-image magnification. The eyepiece focal length f_e is also small, but it is kept larger than f_o.

$$\Rightarrow f_o < f_e$$

So the focal length of the objective is **lesser than** that of the eyepiece, which is option (B). Options (A) and (C) contradict this requirement, and option (D) is meaningless since a focal length cannot equal a tube length by design.

Final answer. The objective's focal length is lesser than the eyepiece's; option **(B)**.

← *Back to Q87*

Q88

For a higher resolving power of a compound microscope, the wavelength of light used should be _____.

[2020 • Set 55-2-1]

SOLUTION

The resolving power of a compound microscope is its ability to distinguish two very close points on the object as separate. It is given by the relation:

$$\text{Resolving power} = \frac{2\,n\theta}{1.22\,\lambda}$$

where $n\theta$ is the numerical aperture and λ is the wavelength of light used.

The resolving power is inversely proportional to the wavelength.

$$\text{Resolving power} \propto \frac{1}{\lambda}$$

So a smaller wavelength gives a smaller limit of resolution and hence a higher resolving power. This is why short-wavelength radiation such as ultraviolet light or an electron beam is used for very fine resolution.

Therefore, for a higher resolving power the wavelength of light used should be **as small (short) as possible**.

Final answer. The wavelength of light used should be as small (short) as possible.

← Back to Q88

Q89

A compound microscope is used because a realistic simple microscope does not have _____ magnification.

[2020 • Set 55-2-2]

SOLUTION

A simple microscope is a single convex lens, so its magnifying power is limited. In normal adjustment (image at the near point) it is:

$$m = 1 + \frac{D}{f}$$

with $D = 25$ **cm**. Even for a very small practical focal length such as $f = 2.5$ **cm**, this gives only:

$$m = 1 + \frac{25}{2.5} = 11$$

A lens of much shorter focal length cannot be made without severe distortion, so a single lens cannot give more than about 10 to 20 times magnification.

A compound microscope uses two lenses (objective and eyepiece), so its magnifications multiply, giving values of several hundred times. It is therefore used because a realistic simple microscope does not provide **high (large) magnification**.

Final answer. A compound microscope is used because a simple microscope cannot give high (large) magnification.

← Back to Q89

Q90

The resolving power of a compound microscope can be _____ by using a medium

(oil) of higher refractive index than air between the object and the objective lens.

[2020 • Set 55-2-3]

SOLUTION

The resolving power of a compound microscope depends on the numerical aperture of the objective and the wavelength of light, through the relation:

$$\text{Resolving power} = \frac{2\,n\theta}{1.22\,\lambda}$$

Here the quantity $n\theta$ is the **numerical aperture**, where n is the refractive index of the medium between the object and the objective lens, and 2θ is the cone angle of light entering the objective.

When air ($\boldsymbol{n = 1}$) is replaced by an oil of higher refractive index ($\boldsymbol{n > 1}$), the numerical aperture $n\theta$ increases. Since the resolving power is directly proportional to $n\theta$, it also increases.

$$\text{Resolving power} \propto n\theta$$

Therefore, using an oil of higher refractive index between the object and the objective **increases** the resolving power. This is the principle of the **oil-immersion objective** used in high-resolution microscopy.

Final answer. The resolving power is increased by using an oil-immersion medium of higher refractive index.

← *Back to Q90*

Q91

The refractive index of the material of a converging lens is 1.5. *If air is replaced by a medium of refractive index* 1.6, *then the lens will now behave as a _____ lens.*

[2020 • Set 55-3-1]

SOLUTION

The behaviour of a lens depends on the refractive index of its material relative to the surrounding medium, through the lens maker's formula.

When the lens of material index n_l sits in a medium of index n_m, the formula reads $\frac{1}{f}=(\frac{n_l}{n_m}-1)(\frac{1}{R_1}-\frac{1}{R_2})$.

Here $n_l=1.5$ and $n_m=1.6$, so $\frac{n_l}{n_m}=\frac{1.5}{1.6}=0.9375$.

Therefore the factor $(\frac{n_l}{n_m}-1)=0.9375-1=-0.0625$, which is negative, whereas in air it was positive.

For a converging (convex) lens the term $(\frac{1}{R_1}-\frac{1}{R_2})$ is positive, so multiplying by the now negative factor makes $\frac{1}{f}$ negative, giving a negative focal length.

A negative focal length means the lens now diverges light, so the converging lens behaves as a **diverging lens** when placed in the denser medium.

CONCEPT

Whenever the surrounding medium is optically denser than the lens material ($n_m>n_l$), every lens reverses its nature: convex acts concave and concave acts convex.

Final answer. The converging lens will behave as a diverging lens.

← *Back to Q91*

Q92

The value of Brewster's angle for air-glass interface is $\frac{\pi}{3}$; hence the refractive index of glass is _____.

[2020 • Set 55-3-1]

SOLUTION

According to Brewster's law, when unpolarised light is incident on a transparent surface at the polarising angle, the reflected light is completely plane polarised, and the refractive index of the medium equals the tangent of this angle.

Mathematically, $n=\theta_B$, where θ_B is Brewster's (polarising) angle.

Here the given Brewster's angle is $\theta_B=\frac{\pi}{3}=60°$.

Substituting, $n=\tan 60°$.

Evaluating, $n=\sqrt{3}\approx 1.732$.

Hence the refractive index of glass is $n=\sqrt{3}\approx 1.73$.

Final answer. The refractive index of glass is $\sqrt{3}\approx 1.73$.

← *Back to Q92*

Q93

An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality? Justify your answer.

[2020 • Set 55-4-1]

SOLUTION

A reflecting type astronomical telescope produces an image of better quality than a refracting type.

The objective of a refracting telescope is a lens, and a lens refracts different colours of light by different amounts because its refractive index depends on wavelength. This causes chromatic aberration, where the different colours focus at slightly different points and the image gets coloured fringes.

A reflecting telescope uses a concave mirror as its objective. A mirror reflects all wavelengths through the same angle, so there is no dispersion and no chromatic aberration at all.

A parabolic mirror also brings both paraxial and marginal rays to a single focus, so it is free of spherical aberration as well, giving a sharper image.

Moreover, a mirror can be supported rigidly along its entire back surface, so very large apertures are possible without sagging. A large lens, supported only at its rim, bends under its own weight and also absorbs light passing through its thickness.

A larger aperture gathers more light and gives higher resolving power, so a **reflecting telescope** gives the brighter, sharper and better quality image.

Final answer. A reflecting telescope gives the better quality image, as it is free of chromatic and spherical aberration and allows a larger aperture.

← *Back to Q93*

Q94

A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

(A) 2P

(B) $P/2$

(C) P

(D) $P/\sqrt{2}$

[2020 • Set 55-5-1]

SOLUTION

The power of a lens is found from the lens maker's formula, $\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$, and power $P=\frac{1}{f}$.

For an equi-biconcave lens both surfaces have the same radius R, with the first surface concave ($R_1=-R$) and the second convex toward the object ($R_2=+R$).

So the original power is $P=(n-1)(\frac{1}{-R}-\frac{1}{R})=(n-1)(\frac{-2}{R})=\frac{-2(n-1)}{R}$.

When the lens is split vertically (along its principal axis) into two identical halves, each half has one curved face of radius R and one flat face for which the radius is infinite, making it a plano concave lens.

The power of each plano concave part is $P'=(n-1)(\frac{1}{-R}-\frac{1}{\infty})=(n-1)(\frac{-1}{R})=\frac{-(n-1)}{R}$.

Comparing, $P' = \frac{1}{2}\left(\frac{-2(n-1)}{R}\right) = \frac{P}{2}$, so each part has half the original power, which is option (B).

Option (A) 2P and (C) P wrongly assume the curvature is unchanged; option (D) $P/\sqrt{2}$ has no physical basis since power adds linearly, not by roots. Hence each part has power $\boldsymbol{P/2}$.

COMMON MISTAKE

Splitting a symmetric lens parallel to the principal axis halves the curvature contribution, so each part keeps half the power. Cutting perpendicular to the axis would instead leave the power of each piece unchanged.

Final answer. The power of each plano concave part is $\boldsymbol{P/2}$, option **(B)**.

← Back to Q94

Q95

State the condition under which a large magnification can be achieved in an astronomical telescope.

OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light?

[2019 • Set 55-3-1]

SOLUTION

The magnifying power of an astronomical telescope in normal adjustment is $m = \frac{f_o}{f_e}$, where f_o is the focal length of the objective and f_e that of the eyepiece.

For a large magnification this ratio must be large, so the focal length of the objective f_o should be as large as possible and the focal length of the eyepiece f_e should be as small as possible.

In addition, the objective should have a large aperture so that it gathers enough light and gives high resolving power, producing a bright and well resolved image.

OR

The refractive index of glass depends on the wavelength of light (dispersion), and for the same prism $\mathbf{n_{violet} > n_{red}}$ because shorter wavelengths bend more.

For a small angle prism the angle of minimum deviation is $\delta_m = (n-1)A$, so a smaller refractive index gives a smaller deviation.

Since red light has a smaller refractive index than violet, on replacing violet with red the angle of minimum deviation **decreases**.

Final answer. Make $\mathbf{f_o}$ large and $\mathbf{f_e}$ small (with a large aperture objective); OR for red light the angle of minimum deviation decreases.

← *Back to Q95*

Q96

Does the magnifying power of a microscope depend on the colour of the light used? Justify your answer.

[2017]

SOLUTION

Yes, the magnifying power of a microscope does depend on the colour of the light used.

The magnifying power of a compound microscope (final image at the near point) is $M = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$, where f_o and f_e are the focal lengths of the objective and eyepiece, L is the tube length and D the near point distance.

Both focal lengths come from the lens maker's formula $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, in which the refractive index n depends on the wavelength (colour) of light through dispersion.

Blue or violet light has a larger refractive index than red light, so it gives shorter focal lengths f_o and f_e.

Since M increases as f_o and f_e decrease, blue or violet light gives a slightly larger magnification than red light.

Therefore the magnifying power depends on the colour of the light, although the change is small because the variation of n with wavelength is small. Hence the answer is **yes, it depends weakly on colour**.

Final answer. Yes, the magnification depends on colour, since $\mathbf{f_o}$ and $\mathbf{f_e}$ vary with refractive index, which changes with wavelength.

← *Back to Q96*

Q97

How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced by red light? Give reason.

[2017]

SOLUTION

When violet light is replaced by red light, the angle of minimum deviation of the prism decreases.

For a prism the refractive index is related to the angle of minimum deviation by $n=\frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$, where A is the angle of the prism.

For a fixed prism angle A, this relation shows that a larger refractive index n corresponds to a larger angle of minimum deviation δ_m.

The refractive index of glass depends on wavelength, and since violet light has a shorter wavelength it bends more, giving $\mathbf{n_{violet}>n_{red}}$.

Therefore $\delta_m{}^{\text{violet}}>\delta_m{}^{\text{red}}$, and replacing violet with red lowers the refractive index, so the angle of minimum deviation **decreases**.

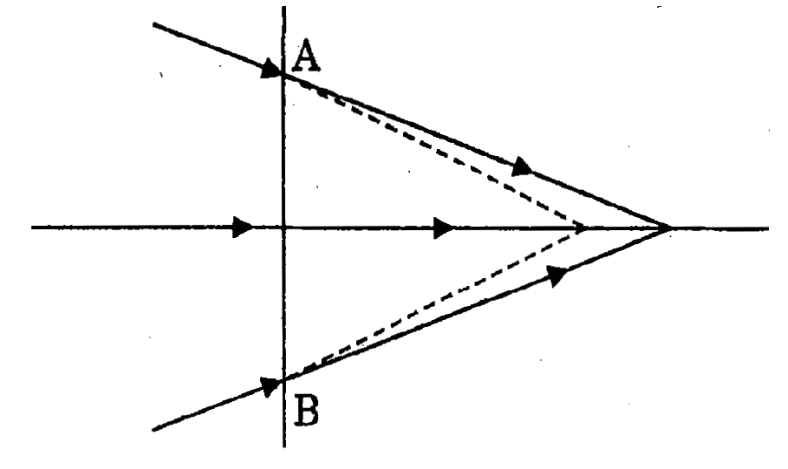
This is exactly why a prism disperses white light into a spectrum, with red deviated the least and violet the most.

Final answer. On replacing violet light with red light, the angle of minimum deviation decreases (because $\mathbf{n_{red}}$**violet**).

← Back to Q97

Q98

The line AB in the ray diagram represents a lens. State whether the lens is convex or concave.



[2015]

SOLUTION

The nature of the lens is decided by what happens to the rays after they pass through it.

A convex (converging) lens bends parallel incident rays so that they meet at a real focus on the far side, while a concave (diverging) lens spreads parallel rays apart so that they appear to come from a virtual focus on the same side.

In the given ray diagram the rays travelling parallel to the principal axis emerge bent inward and converge toward a single point (the focus) after crossing the lens line AB.

Since the emerging rays converge to a real focal point rather than diverging, the lens AB is a **convex (converging) lens**.

Final answer. The lens AB is a convex (converging) lens.

← Back to Q98

Q99

A concave lens of refractive index 1.5 is immersed in a medium of refractive index 1.65. What is the nature of the lens?

[2015]

SOLUTION

The nature of a lens in a medium follows from the lens maker's formula written for a surrounding medium, $\frac{1}{f}=(\frac{n_l}{n_m}-1)(\frac{1}{R_1}-\frac{1}{R_2})$.

Here the lens index is $n_l=1.5$ and the medium index is $n_m=1.65$, so $\frac{n_l}{n_m}=\frac{1.5}{1.65}\approx 0.909$.

Then the factor $(\frac{n_l}{n_m}-1)\approx 0.909-1=-0.091$, which is negative.

For a concave lens the shape factor $(\frac{1}{R_1}-\frac{1}{R_2})$ is also negative.

The product of two negative quantities is positive, so $\frac{1}{f}>0$, giving a positive focal length.

A positive focal length means the lens now converges light, so the concave lens behaves as a **converging lens** in this denser medium.

Final answer. In the denser medium ($n_m=1.65>n_l=1.5$) the concave lens behaves as a converging lens.

← *Back to Q99*

Q100

A convex lens is placed in contact with a plane mirror. A point object at a distance of 20 cm on the axis of this combination has its image coinciding with itself. What is the focal length of the lens ?

[2014]

SOLUTION

A convex lens placed in contact with a plane mirror behaves like a silvered lens, and when the final image coincides with the object itself, the rays must retrace their own path.

For rays to retrace their path after reflecting from the plane mirror, they must strike the mirror normally, which means the rays leaving the lens are parallel to the principal axis.

Rays from a point object become parallel after a convex lens only when the object is placed at the focus of the lens.

Therefore the object distance equals the focal length of the lens, that is f**=20 cm**.

The parallel rays hit the plane mirror normally, reflect straight back, pass through the lens again and converge back at the object, so the image coincides with the object.

Hence the focal length of the convex lens is f**=20 cm**.

> **TIP**
>
> When a lens-and-plane-mirror image coincides with the object, the object sits exactly at the lens focus, so the object distance equals the focal length directly.

Final answer. The focal length of the convex lens is 20 cm.

← Back to Q100

Q101

A biconcave lens made of a transparent material of refractive index 1.25 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens ? Give reason.

[2014]

SOLUTION

The behaviour of a lens depends on the refractive index of the lens material relative to the surrounding medium, through the lens-maker's formula written for a medium of refractive index μ_m.

$$\frac{1}{f}=\left(\frac{\mu_l}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Here the lens material has $\mu_l=1.25$ and the surrounding water has $\mu_m=1.33$.

$$\frac{\mu_l}{\mu_m}=\frac{1.25}{1.33}<1$$

So the first factor $\left(\frac{\mu_l}{\mu_m}-1\right)$ is negative. For a biconcave lens the second factor $\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ is also negative (with R_1 negative and R_2 positive).

The product of two negative quantities is positive, so $\frac{1}{f}>0$, which means f is positive.

A positive focal length means the lens converges light. Hence, although a biconcave lens is diverging in air, here the lens material is optically rarer than water, so it behaves as a **converging lens** in water.

CONCEPT

When the lens material is rarer than the medium, the lens reverses its nature: convex becomes diverging and concave becomes converging.

Final answer. The biconcave lens behaves as a converging lens in water, because the lens material is optically rarer than the surrounding water.

← *Back to Q101*

Q102

Write the relationship between angle of incidence i, angle of prism A and angle of minimum deviation for a triangular prism.

[2013]

SOLUTION

At the position of minimum deviation, a ray passes symmetrically through the prism, so the angle of incidence equals the angle of emergence, $i_1=i_2=i$, and the two angles of refraction are equal, $r_1=r_2$.

From the geometry of the prism, $\mathbf{r_1+r_2=A}$, which gives $\mathbf{r_1=r_2=\frac{A}{2}}$.

The deviation produced by a prism is $\mathbf{\delta=(i_1+i_2)-(r_1+r_2)}$.

At minimum deviation this becomes $\mathbf{\delta_m=2i-A}$.

Rearranging for the angle of incidence gives the required relationship.

$$i=\frac{A+\delta_m}{2}$$

This is the standard relation connecting the angle of incidence, the angle of the prism, and the **angle of minimum deviation** δ_m.

Final answer. The relationship is $\boldsymbol{i=\frac{A+\delta_m}{2}}$, where $\boldsymbol{\delta_m}$ is the angle of minimum deviation.

← *Back to Q102*

Q103

A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal lengths. What is the focal length of the combination?

[2012]

SOLUTION

For two thin lenses placed coaxially in contact, the focal length of the combination is given by the combination formula.

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

A converging lens has a positive focal length, $\mathbf{f_1=+f}$, while a diverging lens of equal magnitude has $\mathbf{f_2=-f}$.

$$\frac{1}{F}=\frac{1}{f}+\frac{1}{(-f)}$$

$$\frac{1}{F}=\frac{1}{f}-\frac{1}{f}=0$$

Since $\frac{1}{F}=0$, the focal length is $F\to\infty$.

The combination therefore has **zero power** and behaves like a plane glass plate, producing no net convergence or divergence of light.

Final answer. The focal length of the combination is infinite, so it acts like a plane glass plate.

← *Back to Q103*

Q104

For the same value of angle of incidence, the angles of refraction in three media A, B and C are 15°, 25° *and* 35° *respectively. In which medium would the velocity of light be minimum?*

[2012]

SOLUTION

When light passes from air into a medium, Snell's law relates the angle of incidence and the angle of refraction.

$$n_1 \sin i = n_2 \sin r$$

With air as the first medium, $\mathbf{n_1=1}$, and since the angle of incidence i is the same for all three media, we get $\mathbf{\sin i = n_2 \sin r}$, so $\mathbf{n_2 \propto \frac{1}{\sin r}}$.

A smaller angle of refraction r therefore corresponds to a larger refractive index n_2.

The angles of refraction are $\mathbf{r_A=15°}$, $\mathbf{r_B=25°}$, $\mathbf{r_C=35°}$. The smallest is $\mathbf{r_A=15°}$, so medium A has the largest refractive index.

The speed of light in a medium is $v=\frac{c}{n}$, so a larger n means a smaller speed.

Hence the velocity of light is minimum in **medium A**, which is the optically densest of the three.

Final answer. The velocity of light is minimum in medium A.

← Back to Q104

Q105

Under what condition does a biconvex lens of glass having a certain refractive index act as a plane glass sheet when immersed in a liquid?

[2012]

SOLUTION

The focal length of a lens immersed in a liquid is given by the lens-maker's formula written for the liquid medium.

$$\frac{1}{f}=\left(\frac{n_g}{n_l}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

The lens will act as a plane glass sheet only when it produces no convergence or divergence, that is when $\frac{1}{f}=0$.

Since the lens has definite curved surfaces, the geometric factor $\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ is not zero. Therefore the only possibility is that the first factor vanishes.

$$\frac{n_g}{n_l}-1=0 \Rightarrow n_g=n_l$$

So the biconvex lens behaves as a plane glass sheet when the refractive index of the liquid equals the refractive index of the glass, $\mathbf{n_{liquid}=n_{glass}}$.

Under this condition there is no refraction at the lens surfaces, the focal length becomes infinite, and light passes straight through.

Final answer. The lens acts as a plane glass sheet when the liquid has the same refractive index as the glass, $\mathbf{n_{liquid}=n_{glass}}$.

← Back to Q105

Q106

How does focal length of a lens change when red light incident on it is replaced by violet light? Give reason for your answer.

[2012]

SOLUTION

The focal length of a lens is related to the refractive index of its material by the lens-maker's formula.

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

The refractive index n of the lens material depends on the wavelength of light (the phenomenon of dispersion). Shorter wavelengths have a larger refractive index, so for the same glass $\mathbf{n_{violet} > n_{red}}$.

From the formula, $\frac{1}{f} \propto (n-1)$, so a larger value of n gives a larger value of $\frac{1}{f}$ and hence a smaller focal length.

Since $\mathbf{n_{violet} > n_{red}}$, it follows that $\mathbf{f_{violet}}$ **red**.

Therefore, when red light is replaced by violet light, the **focal length of the lens decreases**. This wavelength dependence is the cause of chromatic aberration.

Final answer. The focal length decreases, because violet light has a higher refractive index than red light.

← *Back to Q106*

Q107

A glass lens of refractive index 1.45 disappears when immersed in a liquid. What is the value of refractive index of the liquid?

[2010]

SOLUTION

A lens becomes invisible (disappears) in a liquid when there is no refraction at its surfaces. This happens only when the refractive index of the lens material is equal to that of the surrounding liquid.

From the lens-maker's formula in the liquid, $\frac{1}{f}=(\frac{n_{lens}}{n_{liquid}}-1)(\frac{1}{R_1}-\frac{1}{R_2})$.

When $n_{lens}=n_{liquid}$, the factor $(\frac{n_{lens}}{n_{liquid}}-1)=0$, so $\frac{1}{f}=0$ and the lens has infinite focal length, bending no light.

Setting the two indices equal gives the value of the liquid index.

$$n_{liquid}=n_{lens}=1.45$$

Hence the refractive index of the liquid is **1.45**.

Final answer. The refractive index of the liquid is 1.45.

← *Back to Q107*

Q108

State the conditions for the phenomenon of total internal reflection to occur.

[2010]

SOLUTION

Total internal reflection (TIR) is the phenomenon in which a ray of light travelling in a denser medium is completely reflected back into the same medium when it strikes the boundary with a rarer medium. Two conditions must be satisfied.

First, light must travel from an optically denser medium to an optically rarer medium, that is from a medium of higher refractive index to one of lower refractive index (for example, from glass to air or water to air).

Second, the angle of incidence in the denser medium must be greater than the critical angle θ_c for that pair of media, where the critical angle is given by the relation below.

$$\theta_c = \frac{n_{rarer}}{n_{denser}}$$

When both conditions are met, no light is refracted into the rarer medium and the entire incident energy is reflected back into the denser medium. This is the principle behind **optical fibres** and total-reflecting prisms.

Final answer. TIR occurs when light goes from a denser to a rarer medium and the angle of incidence exceeds the critical angle θ_c.

← *Back to Q108*

Q109

You are given following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope?

Lenses	Power (P)	Aperture (A)
L1	3D	8 cm
L2	6D	1 cm
L3	10D	1 cm

[2009]

SOLUTION

An astronomical telescope is designed to give high angular magnification, which is $m = \frac{f_o}{f_e}$, where f_o is the focal length of the objective and f_e that of the eyepiece. For large magnification the objective must have a long focal length and the eyepiece a short focal length.

The objective should have a large focal length (so a small power) and a large aperture, so that it collects more light and forms a bright, well-resolved image of the distant object.

The eyepiece should have a short focal length (so a large power) and a small aperture, so that it strongly magnifies the intermediate image.

Therefore, from the three lenses given, the lens with the **largest focal length (lowest power) and largest aperture** is chosen as the objective, and the lens with the **smallest focal length (highest power) and smallest aperture** is chosen as the eyepiece.

Final answer. Use the longest-focal-length, large-aperture lens as the objective and the shortest-focal-length, small-aperture lens as the eyepiece.

← *Back to Q109*

Q110

Two thin lenses of power +6D *and* −2D *are in contact. What is the focal length of the combination?*

[2009]

SOLUTION

When two thin lenses are placed in contact, the power of the combination equals the algebraic sum of their individual powers.

$$P=P_1+P_2$$

Substituting the given powers $\mathbf{P_1=+6\ D}$ and $\mathbf{P_2=-2\ D}$.

$$P=(+6\ D)+(-2\ D)=+4\ D$$

The focal length is the reciprocal of the power, with f in metres when P is in dioptres.

$$f=\frac{1}{P}=\frac{1}{4}\ m=0.25\ m$$

$$f=25\ \text{cm}$$

Since the net power is positive, the combination is converging, with a focal length of **25 cm**.

Final answer. The focal length of the combination is **+25 cm** (converging).

← Back to Q110

Q111

A glass lens of refractive index 1.5 *is placed in a trough of liquid. What must be the refractive index of the liquid in order to make the lens disappear?*

[2008]

SOLUTION

A lens is seen because light bends (refracts) as it crosses the boundary between the lens material and the surrounding medium. This bending happens only when the two media have different refractive indices.

The lens will completely disappear when no refraction takes place at its surfaces, that is when the surrounding liquid has exactly the same refractive index as the glass of the lens.

So the condition for the lens to vanish is $\mathbf{n_{liquid} = n_{lens}}$.

Given $\mathbf{n_{lens} = 1.5}$.

Therefore $\mathbf{n_{liquid} = 1.5}$.

We can confirm this using the lens-maker's formula in a liquid medium: $\frac{1}{f} = \left(\frac{n_{lens}}{n_{liquid}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.

When $\mathbf{n_{lens} = n_{liquid}}$, the bracket $\left(\frac{n_{lens}}{n_{liquid}} - 1\right) = 0$, so $\frac{1}{f} = 0$ giving $f = \infty$.

An infinite focal length means the lens has no focusing power and cannot be optically told apart from the liquid, so it **disappears**. The required refractive index of the liquid is **1.5**.

Final answer. The liquid must have a refractive index of 1.5 (equal to that of the lens).

← Back to Q111

Q112

An object is held at the principal focus of a concave lens of focal length F. Where is the image formed?

[2008]

SOLUTION

A concave (diverging) lens has a negative focal length, so here $f = -F$, where F is the magnitude of the focal length. The object is placed at the principal focus, so $u = -F$ (object distance is negative in the sign convention).

Apply the thin lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-F} + \frac{1}{-F}$$

$$\frac{1}{v} = -\frac{2}{F}$$

$$v = -\frac{F}{2}$$

The negative sign shows the image is formed on the same side as the object (the side from which light comes), so it is **virtual and erect**.

The image is formed at a distance of $\frac{F}{2}$ from the lens, between the lens and the principal focus, on the same side as the object.

Final answer. The image is virtual, erect and formed at a distance F/2 from the lens on the same side as the object.

← *Back to Q112*

Q113

A ray of light in air is incident at angle $\angle i$ on a face of an equilateral glass prism and is refracted through the prism. As $\angle i$ is varied, it is observed that the ray undergoes minimum deviation, when $\angle i$ is three-fourth of the angle of the prism. Calculate the speed of light in the prism.

[2026 • Set 55-2-1]

SOLUTION

For an equilateral prism the refracting angle is $A = 60°$.

We are told that minimum deviation occurs when the angle of incidence is three-fourths of the prism angle.

$$i = \frac{3}{4}A = \frac{3}{4} \times 60° = 45°$$

At minimum deviation the ray passes symmetrically through the prism, so the two angles of refraction are equal: $r_1 = r_2 = \frac{A}{2}$.

$$r = \frac{A}{2} = 60°2 = 30°$$

Apply Snell's law at the first face to find the refractive index of the prism material.

$$n = \frac{\sin i}{\sin r} = \sin 45° \sin 30°$$

$$n = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

The speed of light in the prism is related to the speed in vacuum by $n = \frac{c}{v}$, so $v = \frac{c}{n}$.

$$v = \frac{3 \times 10^8\ m/s}{\sqrt{2}}$$

$$v = \frac{3 \times 10^8}{1.414} \approx 2.12 \times 10^8\ m/s$$

The speed of light inside the prism is $\mathbf{2.12 \times 10^8\ m/s}$.

TIP

At minimum deviation the ray runs parallel to the base and $r_1 = r_2 = A/2$. Spot this and the whole numerical falls out quickly.

Final answer. The speed of light in the prism is about 2.12×10^8 m/s.

← *Back to Q113*

Q114

A 5 cm *long pencil is placed along the principal axis of a concave mirror of focal length* 20 cm *such that its nearest end is at a distance of* 25 cm *from the mirror. Calculate the length of the image of the pencil.*

[2026 • Set 55-2-1]

SOLUTION

Since the pencil lies along the principal axis, its two ends are at different distances from the mirror. We find the image position of each end separately, then the image length is the distance between the two image points.

Using the sign convention, the focal length of the concave mirror is $f = -20$ **cm**.

The nearest end is at $u_n = -25$ **cm** and the far end, 5 cm further away, is at $u_f = -30$ **cm**.

For the near end, use the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$.

$$\frac{1}{v_n} = \frac{1}{f} - \frac{1}{u_n} = \frac{1}{-20} - \frac{1}{-25}$$

$$\frac{1}{v_n} = -\frac{1}{20} + \frac{1}{25} = \frac{-5+4}{100} = -\frac{1}{100}$$

$$v_n = -100 \text{ cm}$$

For the far end, $\frac{1}{v_f} = \frac{1}{-20} - \frac{1}{-30}$.

$$\frac{1}{v_f} = -\frac{1}{20} + \frac{1}{30} = \frac{-3+2}{60} = -\frac{1}{60}$$

$$v_f = -60 \text{ cm}$$

The length of the image is the distance between the two image points.

$$\text{Length} = |v_n - v_f| = |-100 - (-60)| = |-40| = 40 \text{ cm}$$

The image of the pencil is **40 cm** long (and is a real, inverted, magnified image since both ends image in front of the mirror).

Final answer. The length of the image of the pencil is 40 cm.

← *Back to Q114*

Q115

A ray of light is incident on the face of a triangular prism of refracting angle 60° *and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is* $\sqrt{2}$.

[2026 • Set 55-2-2]

SOLUTION

The phrase "just suffers total internal reflection" at the second face means the ray inside the prism strikes that face at exactly the critical angle, θ_c.

Find the critical angle from $\sin\theta_c = \frac{1}{n}$.

$$\sin\theta_c = \frac{1}{\sqrt{2}}$$

$$\theta_c = 45°$$

So the angle of refraction at the second face is $r_2 = 45°$.

For any prism the angles of refraction at the two faces add up to the prism angle: $r_1 + r_2 = A$.

$$r_1 = A - r_2 = 60° - 45° = 15°$$

Apply Snell's law at the first face, where light goes from air into the prism: $\sin i = n \sin r_1$.

$$\sin i = \sqrt{2} \times \sin 15^\circ$$

$$\sin i = 1.414 \times 0.2588$$

$$\sin i = 0.366$$

$$i = \sin^{-1}(0.366) \approx 21.5^\circ$$

The required angle of incidence at the first face is **21.5°** (approximately).

Final answer. The angle of incidence is about 21.5 °.

← *Back to Q115*

Q116

A ray of light is incident at an angle 2A *on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path. Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism.*

[2026 • Set 55-2-3]

SOLUTION

Ray diagram: Draw the triangular prism with apex angle A at the top and the silvered (mirror-coated) face on the right. A ray PQ in air hits the left face at point Q at an angle of incidence 2A to the normal at Q. It refracts into the prism along QR, making the angle of refraction r with that normal. The refracted ray QR strikes the silvered face at R exactly along the normal to that face (perpendicular to it). Because it hits the mirror normally, it reflects straight back along RQ and then QP, so the whole path is retraced. Mark the normal at Q, the angles 2A and r, and show QR $\perp$ silvered face.

Because the reflected ray retraces its path, the ray QR inside the prism must hit the silvered face at 90°, i.e. along its normal.

From the geometry of the prism, the angle of refraction at the first face equals the prism angle in this case: $\boldsymbol{r = A}$.

Now apply Snell's law at the first (left) face, where light enters from air: **sin(2A) = n sin r.**

$$\sin(2A) = n \sin A$$

Use the identity **sin(2A) = 2 sin A cos A.**

$$2 \sin A \cos A = n \sin A$$

Divide both sides by sin A.

$$n = 2 \cos A$$

The required relation between the refractive index and the prism angle is $n = 2\cos A$. (For example, $A = 30°$ gives $n = \sqrt{3}$.)

Final answer. The relation is n = 2 cos A.

← *Back to Q116*

Q117

Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power.

[2026 • Set 55-3-2]

SOLUTION

Ray diagram (astronomical telescope in normal adjustment, final image at infinity): Draw two convex lenses on a common horizontal principal axis. On the left is the objective O, which has a long focal length f_o and a large aperture. On the right is the eyepiece E, which has a short focal length f_e. A beam of parallel rays comes from a distant object and makes an angle α with the axis. These rays pass through the objective O and converge to form a real, inverted image I_1 in the focal plane of the objective, at its focus F_o. The eyepiece is placed so that I_1 also lies at the focus F_e of the eyepiece. The eyepiece therefore sends out a parallel beam, which makes a larger angle β with the axis and enters the relaxed eye, forming the final image at infinity. Label α (angle

subtended by object) and β (angle subtended by final image), the intermediate image I_1 at $\mathbf{F_o = F_e}$, and the tube length $\boldsymbol{L = f_o + f_e}$.

The magnifying power is defined as the ratio of the angle subtended at the eye by the final image to the angle subtended at the unaided eye by the object: $\boldsymbol{m = \frac{\beta}{\alpha}}$.

For small angles this works out to $\boldsymbol{m = \frac{f_o}{f_e}}$.

So the magnifying power of an astronomical telescope in normal adjustment is $\boldsymbol{m = \frac{f_o}{f_e}}$, where f_o is the focal length of the objective and f_e that of the eyepiece.

Final answer. The magnifying power in normal adjustment is $m = f_0/f_e$.

← *Back to Q117*

Q118

A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges.

[2026 • Set 55-4-1]

SOLUTION

Light from the bulb at the bottom can leave the surface only through a circular patch directly above it. Rays that hit the surface at an angle greater than the critical angle undergo total internal reflection and cannot escape.

So light emerges only inside a cone whose half-angle equals the critical angle θ_c. The emerging patch on the surface is a circle of radius $\boldsymbol{r = h \tan \theta_c}$, where h is the depth of the bulb.

First find the critical angle from $\boldsymbol{\sin \theta_c = \frac{1}{n}}$.

$$\sin \theta_c = \frac{1}{\sqrt{2}}$$

$\boldsymbol{\theta_c = 45°}$, so $\boldsymbol{\tan \theta_c = 1}$.

Now find the radius of the bright circle, with $h = 1\ m$.

$$r = h \tan \theta_c = 1 \times 1 = 1\ m$$

The area of the surface through which light emerges is the area of this circle.

$$A = \pi r^2 = \pi \times (1)^2$$

$$A = \pi \approx 3.14\ m^2$$

Light emerges through a surface area of about **3.14 m^2**.

Final answer. The area of the surface through which light emerges is $\pi \approx 3.14\ m^2$.

← *Back to Q118*

Q119

A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object.

[2026 • Set 55-4-2]

SOLUTION

A concave mirror can give a real, magnified, inverted image. For a real, inverted image the linear magnification is negative, so here $m = -2$.

First find the focal length from the radius of curvature: $f = -\frac{R}{2}$ (negative for a concave mirror).

$$f = -\frac{16}{2} = -8\ \text{cm}$$

The magnification of a mirror is $m = -\frac{v}{u}$.

$$-2 = -\frac{v}{u} \Rightarrow v = 2u$$

Substitute into the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$.

$$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-8}$$

$$\frac{1}{2u} + \frac{2}{2u} = -\frac{1}{8}$$

$$\frac{3}{2u} = -\frac{1}{8}$$

$$2u = -24 \Rightarrow u = -12 \text{ cm}$$

The negative sign confirms the object is in front of the mirror, which is correct. The object must be placed **12 cm** in front of the mirror.

> **COMMON MISTAKE**
>
> *For a real image from a concave mirror, m is negative; for a virtual one it is positive. Fixing this sign first saves you from a wrong object distance.*

Final answer. The object must be placed 12 cm in front of the mirror.

← *Back to Q119*

Q120

A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25.

[2026 • Set 55-4-3]

SOLUTION

The focal length of a lens depends on the refractive index of the lens material relative to its surroundings, through the lens-maker's formula $\frac{1}{f} = (\frac{n_l}{n_m} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$, where n_m is the refractive index of the medium around the lens.

In air, $n_m = 1$, so $\frac{1}{f_{air}} = (n_l - 1)(\frac{1}{R_1} - \frac{1}{R_2})$.

In the liquid, $n_m = n_L$, so $\frac{1}{f_{liq}} = (\frac{n_l}{n_L} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$.

Dividing one by the other, the radius terms cancel, giving $\frac{f_{liq}}{f_{air}} = \frac{n_l - 1}{\frac{n_l}{n_L} - 1}$.

Substitute $n_l = 1.5$ and $n_L = 1.25$. First, $\frac{n_l}{n_L} = \frac{1.5}{1.25} = 1.2$.

$$\frac{f_{liq}}{f_{air}} = \frac{1.5 - 1}{1.2 - 1} = \frac{0.5}{0.2} = 2.5$$

$$f_{liq} = 2.5 \times f_{air} = 2.5 \times 20 = 50 \text{ cm}$$

The focal length stays positive, so the lens remains **converging**, but its focal length increases to **50 cm**. The lens becomes weaker in the liquid because the liquid is optically denser than air, so the lens bends light less.

Final answer. In the liquid the lens stays converging, with focal length 50 cm.

← *Back to Q120*

Q121

A tank is filled with a liquid to a height of 12.5 *m. The apparent depth of a needle lying at the bottom of the tank is measured to be* 9.0 *m. Calculate the speed of light in the liquid.*

[2026 • Set 55-5-1]

SOLUTION

When an object is viewed almost normally through a denser medium, the bottom appears raised. For near-normal viewing the refractive index of the liquid relates the real and apparent depths as:

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

Here the real depth of the needle is 12.5 m and the apparent depth is 9.0 m.

$$n = \frac{12.5}{9.0}$$

$$n = 1.389$$

The speed of light in any medium is related to the speed in vacuum c by $\boldsymbol{n = \frac{c}{v}}$, so:

$$v = \frac{c}{n}$$

$$v = \frac{3 \times 10^8}{1.389}\ m/s$$

$$v = 2.16 \times 10^8\ m/s$$

Hence the **speed of light in the liquid is about $\mathbf{2.16 \times 10^8\ m/s}$**, which is correctly less than its value in vacuum.

COMMON MISTAKE

Always take the bigger figure (real depth) on top so that $\boldsymbol{n>1}$; flipping the ratio is the most common slip here.

Final answer. The speed of light in the liquid is $\boldsymbol{v \approx 2.16 \times 10^8\ m/s}$.

← *Back to Q121*

Q122

Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $\boldsymbol{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$.

[2026 • Set 55-5-1]

SOLUTION

Consider two thin lenses L_1 and L_2 of focal lengths f_1 and f_2 placed in contact coaxially. A point object O is kept on the principal axis at distance u from the combination.

For lens L_1 alone, let the image be formed at distance v_1. Using the thin-lens formula:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad ...(i)$$

This image at v_1 acts as the object for the second lens L_2. Since the lenses are in contact, their separation is zero, so the object distance for L_2 is the same v_1. Let L_2 form the final image at distance v:

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad ...(ii)$$

Adding equations (i) and (ii), the intermediate term $\frac{1}{v_1}$ cancels:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \quad ...(iii)$$

Now let a single equivalent lens of focal length f replace the combination and form the same final image. For it:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad ...(iv)$$

Comparing equations (iii) and (iv):

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

Hence the focal length f of the **combination of two thin lenses in contact** obeys the relation $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$. Proved.

Final answer. Proved that $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ for two thin lenses in contact.

← *Back to Q122*

Q123

A point source, in air, is placed at a distance of 6 cm *in front of a convex spherical surface (**n = 1.5** and radius of curvature = **24 cm**). Find the position and nature of the image formed.*

[2026 • Set 55-5-2]

SOLUTION

This is refraction at a single spherical surface. Light goes from air into glass, so $\mathbf{n_1 = 1}$ and $\mathbf{n_2 = 1.5}$. The surface is convex towards the incoming light, so its radius of curvature is positive: $\boldsymbol{R = +24\ cm}$. The point source is real, 6 cm in front, so $\boldsymbol{u = -6\ cm}$ (measured against the direction of incident light).

The refraction formula for a single spherical surface is:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Substituting the values:

$$\frac{1.5}{v} - \frac{1}{-6} = \frac{1.5 - 1}{24}$$

$$\frac{1.5}{v} + \frac{1}{6} = \frac{0.5}{24}$$

$$\frac{1.5}{v} = \frac{0.5}{24} - \frac{1}{6}$$

$$\frac{1.5}{v} = \frac{0.5}{24} - \frac{4}{24} = \frac{-3.5}{24}$$

$$v = \frac{1.5 \times 24}{-3.5} = \frac{36}{-3.5}$$

$$v = -10.3\ \text{cm}$$

The negative sign shows the image lies on the same side as the object (the air side), so it is a **virtual image about 10.3 cm in front of the surface**. Because the rays only appear to diverge from this point, the image is virtual and erect.

Final answer. The image is virtual, $\boldsymbol{v \approx -10.3\ cm}$, i.e. about **10.3 cm** in front of the surface on the object side.

← Back to Q123

Q124

A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as

it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$.

[2026 • Set 55-5-3]

SOLUTION

The prism has a right angle at A ($\angle A = 90°$) and two equal base angles of 45° at B and C. Ray MN falls normally on face AB, so it enters undeviated and travels straight to the hypotenuse face BC. Because $\angle B = 45°$, the ray strikes BC making an angle of incidence of 45° with the normal to BC.

Whether the ray reflects or refracts at BC depends on the critical angle θ_c, given by $\theta_c = \frac{1}{n}$.

(i) For $n = \sqrt{2}$:

$$\theta_c = \frac{1}{\sqrt{2}}$$

$$\theta_c = 45°$$

The angle of incidence at BC (45°) equals the critical angle exactly, so the ray neither fully reflects back nor refracts out into air; it **grazes along the face BC**, emerging at 90° to the normal and just skimming the surface.

(ii) For $n = \sqrt{3}$:

$$\theta_c = \frac{1}{\sqrt{3}} = 0.577$$

$$\theta_c = \sin^{-1}(0.577) \approx 35.3°$$

Now the angle of incidence at BC (45°) is greater than θ_c (35.3°), so the ray undergoes **total internal reflection** at BC, turning through 90°. It then meets face AC normally and emerges out of AC without bending, the ray leaving perpendicular to AC.

TIP

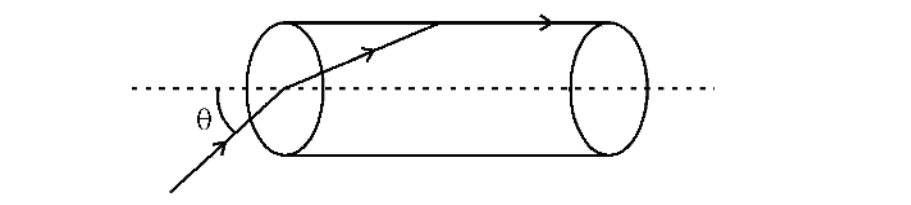
Compare the incidence angle with θ_c first: equal means grazing, greater means TIR, smaller means it refracts out.

Final answer. (i) For $n=\sqrt{2}$ the ray grazes along **BC**; (ii) for $n=\sqrt{3}$ (critical angle $\approx 35.3° < 45°$) it totally internally reflects at **BC** and exits normally through **AC**.

← *Back to Q124*

Q125

A transparent solid cylindrical rod (refractive index $\sqrt{3}$) is kept in air. A ray of light incident on its face travels along the surface of the rod, as shown in figure. Calculate the angle θ.



[2025 • Set 55-1-1]

SOLUTION

The ray enters the flat end face of the rod, refracts, and then strikes the long curved (cylindrical) side. "Travels along the surface" means the ray meets the curved side at exactly the critical angle, so it grazes along the wall (limiting case of total internal reflection).

Note on data: for this standard problem the refractive index of the rod is taken as $n = \frac{2}{\sqrt{3}}$. The value $\sqrt{3}$ printed makes the rod a perfect light pipe, for which the grazing condition gives $\theta > 1$ (no finite θ), so the intended value is $\frac{2}{\sqrt{3}}$.

Step 1: Critical angle at the curved side.

$$\sin C = \frac{1}{n} = \frac{\sqrt{3}}{2}$$

$$C = 60°$$

Step 2: Angle of refraction at the flat entrance face.

The normal to the flat entrance face and the normal to the curved side are mutually perpendicular. If r is the angle of refraction at the entrance face, the

angle of incidence at the curved side is $(90°- r)$. For grazing, this must equal the critical angle:

$$90°- r = C$$

$$r = 90°- 60°= 30°$$

Step 3: Apply Snell's law at the flat entrance face (air to rod).

$$\theta = n \sin r$$

$$\theta = \frac{2}{\sqrt{3}} 30°$$

$$\theta = \frac{2}{\sqrt{3}} \times \frac{1}{2} = \frac{1}{\sqrt{3}} = 0.577$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 35.3°$$

Hence the required angle of incidence at the flat face is $\boldsymbol{\theta = \sin^{-1}(1/\sqrt{3}) \approx 35.3°}$. At this angle the refracted ray meets the curved wall exactly at the critical angle and so runs along the rod.

Final answer. The angle is $\boldsymbol{\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 35.3°}$.

← Back to Q125

Q126

A spherical convex surface of radius of curvature R separates glass (refractive index 1.5) from air. Light from a point source placed in air at distance R/2 from the surface falls on it. Find the position and nature of the image formed.

[2025 • Set 55-1-2]

SOLUTION

This is refraction at a single convex spherical surface separating air and glass. Light passes from air to glass, so $\mathbf{n_1 = 1}$ and $\mathbf{n_2 = 1.5}$. The surface is convex

towards the source, giving $R = +R$. The source is real at distance $\frac{R}{2}$ in front, so $u = -\frac{R}{2}$.

Applying the spherical-surface refraction formula:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{-R/2} = \frac{1.5 - 1}{R}$$

$$\frac{1.5}{v} + \frac{2}{R} = \frac{0.5}{R}$$

$$\frac{1.5}{v} = \frac{0.5}{R} - \frac{2}{R}$$

$$\frac{1.5}{v} = -\frac{1.5}{R}$$

$$v = -R$$

The negative sign means the image is formed on the same side as the object (in air). Hence a **virtual image is formed at a distance R in front of the surface**. Since it is virtual and on the object side, it is also erect.

Final answer. A virtual image is formed at $v = -R$, i.e. a distance R in front of the surface on the same side as the source.

← *Back to Q126*

Q127

A double convex lens of glass has both faces of the same radius of curvature 17 cm. *Find its focal length if it is immersed in water. The refractive indices of glass and water are* 1.5 *and* 1.33 *respectively.*

[2025 • Set 55-1-3]

SOLUTION

When a lens is placed inside a medium, the lens-maker's formula uses the refractive index of the glass relative to the surrounding medium. Here the medium is water ($\mathbf{n_m = 1.33}$) and the glass has $\mathbf{n_g = 1.5}$:

$$\frac{1}{f} = \left(\frac{n_g}{n_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a double (equi-) convex lens, the first face is convex ($\mathbf{R_1 = +17\ cm}$) and the second is concave to the light ($\mathbf{R_2 = -17\ cm}$):

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{17} - \frac{1}{-17} = \frac{2}{17}\ \text{cm}^{-1}$$

The relative-index factor is:

$$\frac{n_g}{n_m} - 1 = \frac{1.5}{1.33} - 1 = \frac{0.17}{1.33} = 0.128$$

Substituting both into the formula:

$$\frac{1}{f} = 0.128 \times \frac{2}{17}$$

$$\frac{1}{f} = \frac{0.256}{17} = 0.01506\ \text{cm}^{-1}$$

$$f = \frac{1}{0.01506}$$

$$f = 66.4\ \text{cm}$$

Hence the **focal length in water is about 66 cm**. This is much larger than its value in air ($\mathbf{f_{air} = 17\ cm}$), showing the lens becomes far weaker (less converging) when immersed in water, because the glass-to-water index ratio is much closer to 1.

COMMON MISTAKE

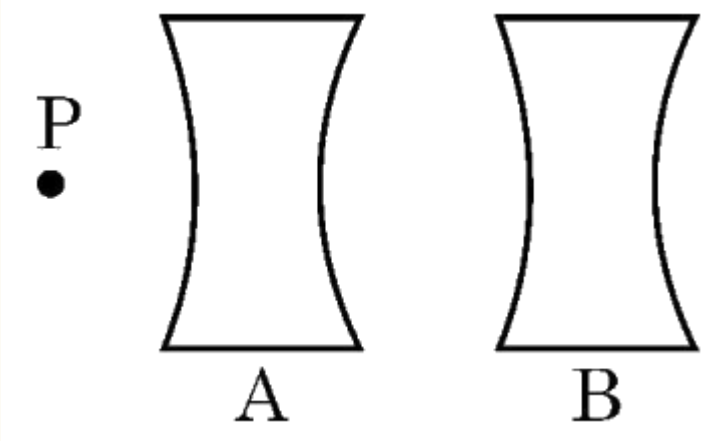
In a medium use $\left(\frac{n_g}{n_m} - 1\right)$, *not* $(n_g - 1)$; *forgetting to divide by* n_m *is the classic error.*

Final answer. The focal length in water is $f \approx 66$ **cm** (the lens becomes much weaker than in air).

← Back to Q127

Q128

Two concave lenses A and B, each of focal length 8.0 cm are arranged coaxially 16 cm apart as shown in figure. An object P is placed at a distance of 4.0 cm from A. Find the position and nature of the final image formed.



[2025 • Set 55-2-1]

SOLUTION

Both lenses are concave, so $f = -8.0$ **cm** for each, and they are 16 cm apart. We solve lens by lens, taking the image of lens A as the object for lens B.

For lens A, the object P is at 4.0 cm, so $u = -4.0$ **cm** and $f = -8.0$ **cm**. Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{-8} + \frac{1}{-4}$$

$$\frac{1}{v} = -\frac{1}{8} - \frac{2}{8} = -\frac{3}{8}$$

$$v = -\frac{8}{3} = -2.67 \text{ cm}$$

So lens A forms a virtual image 2.67 cm on the same side as P (in front of A). This image is the object for lens B. Lens B is 16 cm from A, so the image lies **16 + 2.67 = 18.67 cm** in front of B:

$$u_B = -18.67 \text{ cm}, f = -8.0 \text{ cm}$$

$$\frac{1}{v_B} = \frac{1}{-8} + \frac{1}{-18.67}$$

$$\frac{1}{v_B} = -0.125 - 0.0536 = -0.1786$$

$$v_B = -5.6 \text{ cm}$$

The negative sign means the **final image is virtual, about 5.6 cm in front of lens B** (on the object side). It is erect and diminished, as expected from two diverging lenses.

Final answer. The final image is virtual, about **5.6 cm** in front of lens B on the object side.

← Back to Q128

Q129

Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination.

[2025 • Set 55-2-2]

SOLUTION

Both lenses are convex, $f = +10.0$ **cm**. On the bench, lens A is at 50.0 cm, lens B at 70.0 cm, and the object at 20.0 cm. We treat the lenses one at a time.

The object distance for lens A is **50.0 − 20.0 = 30.0 cm**, so $u_1 = -30.0$ **cm** and $f_1 = +10.0$ **cm**. Using $\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$:

$$\frac{1}{v_1} = \frac{1}{10} + \frac{1}{-30}$$

$$\frac{1}{v_1} = \frac{3-1}{30} = \frac{2}{30}$$

$$v_1 = +15.0 \text{ cm}$$

So lens A forms a real image 15.0 cm to its right, i.e. at bench position **50.0 + 15.0 = 65.0 cm**. The separation between the lenses is **70.0 − 50.0 = 20.0 cm**, so this image lies **20.0 − 15.0 = 5.0 cm** in front of (to the left of) lens B. It acts as a real object for B:

$$u_2 = -5.0 \text{ cm}, f_2 = +10.0 \text{ cm}$$

$$\frac{1}{v_2} = \frac{1}{10} + \frac{1}{-5}$$

$$\frac{1}{v_2} = \frac{1-2}{10} = -\frac{1}{10}$$

$$v_2 = -10.0 \text{ cm}$$

The negative sign shows the **final image is virtual, 10.0 cm to the left of lens B**, i.e. at bench position **70.0 − 10.0 = 60.0 cm**.

Final answer. The final image is virtual, **10.0 cm** in front of lens B, at bench position **60.0 cm**.

← *Back to Q129*

Q130

A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'.

[2025 • Set 55-2-3]

SOLUTION

The incoming beam is converging towards point *O*. Since the rays are heading to a point on the far side of the lens, that point acts as a **virtual object** for the concave lens, with a positive object distance $u = +10$ **cm**. The lens is concave, so $f = -15$ **cm**.

Applying the lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{-15} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{-2+3}{30} = \frac{1}{30}$$

$$v = +30 \text{ cm}$$

So the beam now converges at O', which is 30 cm behind the lens, whereas the original point O was 10 cm behind it. Both are on the same (far) side, so the shift is:

$$OO' = 30 - 10 = 20 \text{ cm}$$

Hence the convergence point moves **20 cm farther away from the lens**. This makes sense because a diverging lens pushes the convergence point further out.

COMMON MISTAKE

A beam converging to a point behind the lens is a virtual object, so u is positive here; treating it like a normal object is the usual mistake.

Final answer. The shift **OO' = 20 cm**, directed away from the lens (farther from it).

← Back to Q130

Q131

An equiconvex lens is made of glass of refractive index 1.55. *If the focal length of the lens is* 15.0 cm, *calculate the radius of curvature of its surfaces.*

[2025 • Set 55-4-1]

SOLUTION

For an equiconvex lens both surfaces have the same radius of curvature. Following the sign convention, the first surface is convex towards the incoming light so $\mathbf{R_1} = +R$, and the second surface is convex on the other side so $\mathbf{R_2} = -R$.

The Lens Maker's formula is

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Substituting $\mathbf{R_1 = +R}$ and $\mathbf{R_2 = -R}$,

$$\frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (n-1)\cdot\frac{2}{R} = \frac{2(n-1)}{R}$$

Rearranging for R,

$$R = 2(n-1)f$$

Putting $\boldsymbol{n = 1.55}$ and $\boldsymbol{f = 15.0\ \text{cm}}$,

$$R = 2 \times (1.55 - 1) \times 15.0\ \text{cm}$$

$$R = 2 \times 0.55 \times 15.0\ \text{cm}$$

$$R = 16.5\ \text{cm}$$

Hence each of the two surfaces of the lens has a **radius of curvature of 16.5 cm**.

Final answer. The radius of curvature of each surface is **16.5 cm**.

← *Back to Q131*

Q132

Find the focal length of plano-convex lens of refractive index 1.5 *and radius of curvature* 10 cm *when it is immersed in a liquid of refractive index* 1.25.

[2025 • Set 55-4-2]

SOLUTION

When a lens of refractive index n_g is immersed in a liquid of refractive index n_l, the Lens Maker's formula uses the relative refractive index $\frac{n_g}{n_l}$ in place of n,

$$\frac{1}{f} = \left(\frac{n_g}{n_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a plano-convex lens the curved face has $\mathbf{R_1 = R = 10\ cm}$ and the plane face has $\mathbf{R_2 = \infty}$, so

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{10} - 0 = \frac{1}{10}\ \text{cm}^{-1}$$

The relative refractive index term is

$$\frac{n_g}{n_l} - 1 = \frac{1.5}{1.25} - 1 = 1.2 - 1 = 0.2$$

Therefore

$$\frac{1}{f} = 0.2 \times \frac{1}{10} = \frac{1}{50}\ \text{cm}^{-1}$$

$$f = 50\ \text{cm}$$

So in the liquid the **focal length becomes 50 cm**. Since this is much larger than its value in air (20 cm), the lens becomes a weaker converging lens inside the denser liquid.

CONCEPT

When the surrounding medium is optically denser, the relative refractive index falls towards 1, so the lens converges much more weakly.

Final answer. The focal length of the lens in the liquid is **50 cm**.

← *Back to Q132*

Q133

The two surfaces of a biconvex lens are of radius of curvature 'R' each. Obtain the condition under which its focal length 'f' be equal to 'R'. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

[2025 • Set 55-4-3]

SOLUTION

For a biconvex lens with both surfaces of radius R, the sign convention gives $\mathbf{R_1} = +\boldsymbol{R}$ and $\mathbf{R_2} = -\boldsymbol{R}$. The Lens Maker's formula is

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (n-1)\left(\frac{1}{R} + \frac{1}{R}\right) = \frac{2(n-1)}{R}$$

For the focal length to equal the radius, put $f = R$,

$$\frac{1}{R} = \frac{2(n-1)}{R}$$

$$1 = 2(n-1)$$

$$n - 1 = \frac{1}{2}$$

$$n = 1.5$$

So the condition is that the **refractive index of the lens material must be $n = 1.5$**.

Now one surface is made plane. Then $\mathbf{R_1} = R$ and $\mathbf{R_2} = \infty$, so

$$\frac{1}{f'} = (n-1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = (n-1)\cdot\frac{1}{R}$$

With $n = 1.5$,

$$\frac{1}{f'} = (1.5 - 1)\cdot\frac{1}{R} = \frac{0.5}{R} = \frac{1}{2R}$$

$$f' = 2R$$

Hence making one surface plane **doubles the focal length to $f' = 2R$**.

Final answer. The condition is $n = 1.5$, and on making one surface plane the new focal length is $f' = 2R$.

← *Back to Q133*

Q134

(a) *A point object is placed in air at a distance R/3 in front of a convex surface of radius of curvature R, separating air from a medium of refractive index* n **(< 4)**. *Find the nature and position of the image formed.*

OR

(b) *In Young's double slit experimental set-up, the intensity of the central maximum is*

I_0. *Calculate the intensity at a point where the path difference between two interfering waves is* $\lambda/3$.

[2025 • Set 55-5-1]

SOLUTION

(a) Refraction at a single spherical surface separating air ($\mathbf{n_1 = 1}$) from a medium ($\mathbf{n_2 = n}$) is governed by

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

The object is real and in front of the surface, so $\boldsymbol{u = -\frac{R}{3}}$, with $\mathbf{n_1 = 1}$ and $\mathbf{n_2 = n}$,

$$\frac{n}{v} - \frac{1}{-R/3} = \frac{n-1}{R}$$

$$\frac{n}{v} + \frac{3}{R} = \frac{n-1}{R}$$

$$\frac{n}{v} = \frac{n-1}{R} - \frac{3}{R} = \frac{n-4}{R}$$

$$v = \frac{nR}{n-4}$$

Since for a typical glass $\boldsymbol{n < 4}$, the term $(n-4)$ is negative, making v negative. The image is therefore **virtual and formed on the same side as the object, at** $\boldsymbol{v = \frac{nR}{n-4}}$.

(b) In Young's double slit experiment the resultant intensity for a path difference Δ is $\boldsymbol{I = 4I_s\cos^2(\frac{\phi}{2})}$, where I_s is the intensity of one slit. At the central maximum $\boldsymbol{\phi = 0}$, so $\mathbf{I_0 = 4I_s}$, giving $\boldsymbol{I_s = \frac{I_0}{4}}$.

The phase difference for path difference $\boldsymbol{\Delta = \frac{\lambda}{3}}$ is

$$\phi = \frac{2\pi}{\lambda}\Delta = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I = 4I_s\cos^2(\frac{\phi}{2}) = I_0\cos^2(\frac{\pi}{3})$$

$$I = I_0\left(\frac{1}{2}\right)^2 = \frac{I_0}{4}$$

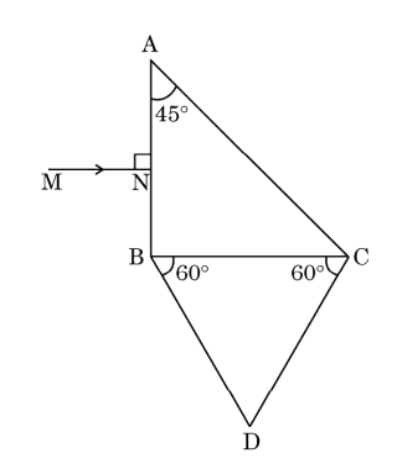
Hence the intensity at that point is $\mathbf{\frac{I_0}{4}}$.

Final answer. (a) $\boldsymbol{v = \frac{nR}{n-4}}$ (virtual image); (b) the intensity is $\mathbf{\frac{I_0}{4}}$.

← *Back to Q134*

Q135

A right angled isosceles glass prism ABC *is kept in contact with an equilateral triangular prism* DBC *as shown in the figure. Both prisms are made of the same glass of refractive index* 1·6. *Trace the path of the ray* MN *incident normally on face* AB *as it passes through the combination.*



[2025 • Set 55-6-1]

SOLUTION

The combination has the right-angled isosceles prism ABC (angles 90°, 45°, 45°) joined along the face BC to the equilateral prism DBC (60°, 60°, 60°). Both prisms are the same glass, $\boldsymbol{n = 1.6}$.

First find the critical angle for glass to air,

$$\theta_c = \frac{1}{n} = \frac{1}{1.6} = 0.625$$

$$\theta_c = \sin^{-1}(0.625) \approx 38.7^\circ$$

The ray MN strikes face AB normally, so it enters without bending and travels straight to the common face BC, meeting it at an angle of incidence of 45° (the base angle of the isosceles prism).

Although $45° > \theta_c$, the face BC is not a glass-to-air boundary. It is a glass-to-glass contact between two prisms of identical refractive index, so it behaves as if there is no interface at all. The ray therefore passes straight across BC into the equilateral prism without any reflection or bending.

Inside the equilateral prism the ray then reaches the face DC, where it does meet a glass-to-air boundary. Here the angle of incidence is small (less than θ_c), so total internal reflection does not occur and the ray **refracts out into the air through face DC**, bending away from the normal.

Path summary: the ray enters AB normally, goes straight, crosses the contact face BC undeviated (same glass on both sides), and finally emerges by refraction at face DC.

CONCEPT

A glass-to-glass contact of equal refractive index is optically invisible, so no critical-angle condition applies there even at a 45° hit.

Final answer. The ray enters **AB** undeviated, passes straight through the common face **BC** (same glass, no interface) and emerges by refraction at face **DC**.

← *Back to Q135*

Q136

A ray of light is incident on face AB *of a prism* ABC *with angle of prism A and emerges out from face* AC. *The prism is set in the position of minimum deviation with angle of deviation* δ. *Find (i) the angle of incidence and (ii) the angle of refraction on face* AB.

[2025 • Set 55-6-2]

SOLUTION

At the position of minimum deviation the path of the ray through the prism is symmetric. This means the angle of incidence equals the angle of emergence and the two internal refraction angles are equal,

$$i_1 = i_2 = i \text{ and } r_1 = r_2 = r$$

From the geometry of the prism the sum of the internal refraction angles equals the angle of the prism,

$$A = r_1 + r_2 = 2r$$

$$r = \frac{A}{2}$$

The total deviation is $\delta = i_1 + i_2 - A$. Using $i_1 = i_2 = i$,

$$\delta = 2i - A$$

$$2i = A + \delta$$

$$i = \frac{A + \delta}{2}$$

Therefore (i) the **angle of incidence is $i = \frac{A + \delta}{2}$**, and (ii) the **angle of refraction on face AB is $r = \frac{A}{2}$**.

Final answer. (i) The angle of incidence is $i = \frac{A+\delta}{2}$; (ii) the angle of refraction at face **AB** is $r = \frac{A}{2}$.

← *Back to Q136*

Q137

A tank is filled with a liquid of refractive index $\sqrt{2}$, up to a height of 30 cm. *A tiny bulb is glowing at the bottom of the tank. Calculate the diameter of an opaque disc floating symmetrically on the liquid surface that can cut off completely the light from the bulb that comes out of the liquid surface.*

[2025 • Set 55-6-3]

SOLUTION

Light from the bulb at the bottom can leave the liquid surface only if it strikes the surface within the critical angle θ_c. Rays meeting the surface at an angle greater than θ_c undergo total internal reflection and stay inside the liquid. So all the escaping light passes through a circle of radius r centred above the bulb, and a disc of this radius cuts off all the emerging light.

First find the critical angle for the liquid ($n = \sqrt{2}$) to air,

$$\theta_c = \frac{1}{n} = \frac{1}{\sqrt{2}}$$

$$\theta_c = 45^\circ$$

Imagine a right-angled triangle from the bulb at the bottom, vertically up to the surface (height h) and out to the edge of the bright circle (radius r). The ray to the edge makes angle θ_c with the vertical, so

$$\theta_c = \frac{r}{h}$$

$$r = h\theta_c = 30 \times \tan 45^\circ = 30 \times 1 = 30 \text{ cm}$$

The required diameter of the disc is

$$D = 2r = 2 \times 30 = 60 \text{ cm}$$

Hence the opaque disc must have a **diameter of 60 cm** to completely cut off the light leaving the surface.

Final answer. The disc must have a diameter of **60 cm**.

← *Back to Q137*

Q138

(a) *Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus."*

OR

(b) *A convex lens of focal length* 10 cm*, a concave lens of focal length* 15 cm *and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is* +12 cm*, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick?*

[2025 • Set 55-7-1]

SOLUTION

(a) For a convex mirror the focal length is positive, $f > 0$. A real object is always placed in front of the mirror, so $u < 0$.

The mirror equation is

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

Since $f > 0$ and $u < 0$, the term $-\frac{1}{u}$ is positive, so $\frac{1}{v}$ is positive and $v > 0$. A positive v means the image forms behind the mirror, so it is **virtual**.

From the same relation, because $-\frac{1}{u} > 0$ we get $\frac{1}{v} > \frac{1}{f}$, hence $v < f$. So the image lies **between the pole and the focus**.

The magnification is $m = -\frac{v}{u}$. With $v > 0$ and $u < 0$, m is positive, so the image is erect. Also, since $|v| < |u|$ always holds for a convex mirror, $|m| = |\frac{v}{u}| < 1$.

Therefore the image is **diminished**.

This proves the convex-mirror image is virtual, erect, diminished and located between the pole and the focus.

(b) For thin lenses in contact the combined focal length obeys

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

Here $f_1 = +10$ **cm** (convex), $f_2 = -15$ **cm** (concave), $F = +12$ **cm**. So

$$\frac{1}{f_3} = \frac{1}{F} - \frac{1}{f_1} - \frac{1}{f_2} = \frac{1}{12} - \frac{1}{10} - \frac{1}{-15}$$

$$\frac{1}{f_3} = \frac{1}{12} - \frac{1}{10} + \frac{1}{15}$$

Taking LCD = **60**,

$$\frac{1}{f_3} = \frac{5 - 6 + 4}{60} = \frac{3}{60} = \frac{1}{20}$$

$$f_3 = +20 \text{ cm}$$

Since f_3 is positive, the third lens is a **convex (converging) lens of focal length 20 cm**.

If the lenses were thick, the simple sum $\frac{1}{F} = \Sigma \frac{1}{f_i}$ would no longer hold, because the separation between the lenses then affects the combined focal length. So the answer would change for thick lenses.

Final answer. (a) The convex-mirror image is virtual, erect, diminished and lies between the pole and focus; (b) the third lens is convex with $\mathbf{f_3 = +20\ cm}$, and the result would change for thick lenses.

← *Back to Q138*

Q139

Monochromatic light of frequency 5.0×10^{14} Hz *passes from air into a medium of refractive index* 1.5. *Find the wavelength of the light (i) reflected, and (ii) refracted at the interface of the two media.*

[2024 • Set 55-1-1]

SOLUTION

When light reflects at an interface it stays in the same medium, so neither its frequency nor its wavelength changes. The reflected light remains in air, so its wavelength is the same as in air,

$$\lambda_{ref} = \frac{c}{\nu}$$

$$\lambda_{ref} = \frac{3 \times 10^8\ m/s}{5.0 \times 10^{14}\ \text{Hz}}$$

$$\lambda_{ref} = 6 \times 10^{-7}\ m = 600\ \text{nm}$$

When light refracts into a denser medium, its frequency stays the same but its speed and hence wavelength decrease. The wavelength in the medium is

$$\lambda_{med} = \frac{\lambda_{air}}{n}$$

$$\lambda_{med} = \frac{600\ \text{nm}}{1.5}$$

$$\lambda_{med} = 400\ \text{nm}$$

Hence the **reflected light has wavelength 600 nm** and the **refracted light has wavelength 400 nm**.

CONCEPT

Frequency is fixed by the source and never changes on reflection or refraction; only the wavelength shrinks inside a denser medium.

Final answer. (i) The reflected wavelength is **600 nm**; (ii) the refracted wavelength is **400 nm**.

← *Back to Q139*

Q140

A plano-convex lens of focal length 16 cm *is made of a material of refractive index* 1.4. *Calculate the radius of the curved surface of the lens.*

[2024 • Set 55-1-1]

SOLUTION

A plano-convex lens has one curved face and one plane face. For the plane face $\mathbf{R_2 = \infty}$, so $\frac{1}{\mathbf{R_2}} = \mathbf{0}$. Taking the curved face as $\mathbf{R_1} = \boldsymbol{R}$, the Lens Maker's formula gives

$$\frac{1}{f} = (n-1)\left(\frac{1}{\mathrm{R}_1} - \frac{1}{\mathrm{R}_2}\right) = (n-1)\left(\frac{1}{R} - 0\right)$$

$$\frac{1}{f} = \frac{n-1}{R}$$

Rearranging for R,

$$R = (n-1)f$$

Substituting $n = 1.4$ and $f = 16$ cm,

$$R = (1.4 - 1) \times 16 \text{ cm}$$

$$R = 0.4 \times 16 \text{ cm}$$

$$R = 6.4 \text{ cm}$$

Hence the curved surface of the lens has a **radius of curvature of 6.4 cm**.

Final answer. The radius of the curved surface is **6.4 cm**.

← *Back to Q140*

Q141

An object is placed 30 cm *in front of a concave mirror of radius of curvature* 40 cm. *Find the (i) position of the image formed and (ii) magnification of the image.*

[2024 • Set 55-1-1]

SOLUTION

For a concave mirror, the focal length is half the radius of curvature, and we follow the New Cartesian Sign Convention (distances measured against the incident light are negative).

Given: radius of curvature $R = 40$ cm, so

$$f = \frac{R}{2} = \frac{40}{2} = 20 \text{ cm}$$

For a concave mirror, $f = -20$ cm, and the object is real, so $u = -30$ cm.

(i) Using the mirror formula:

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{-20}-\frac{1}{-30}$$

$$\frac{1}{v}=-\frac{1}{20}+\frac{1}{30}=\frac{-3+2}{60}=-\frac{1}{60}$$

$$v=-60\ \text{cm}$$

The negative sign shows the image is formed 60 cm in front of the mirror (on the same side as the object), so it is a **real image**.

(ii) The magnification is:

$$m=-\frac{v}{u}=-\frac{(-60)}{(-30)}=-2$$

The negative sign means the image is **inverted**, and $|m| = 2$ means it is magnified two times.

Final answer. The image is real, formed 60 cm in front of the mirror, with magnification m = −2 (inverted and twice the object size).

← *Back to Q141*

Q142

A ray of light passes through a triangular prism. Show graphically, how the angle of deviation varies with the angle of incidence? Hence define the angle of minimum deviation.

[2024 • Set 55-1-1]

SOLUTION

When a ray passes through a prism, the angle of deviation δ depends on the angle of incidence i. A graph of δ (vertical axis) against i (horizontal axis) gives a U-shaped curve.

Describe the graph: the vertical axis is the angle of deviation δ and the horizontal axis is the angle of incidence i. As i increases from a small value, δ

first decreases, reaches a single lowest point, and then increases again. The lowest point of the curve corresponds to the minimum deviation δ_m, occurring at one particular incidence angle $i = i_m$. The curve is symmetric about this point.

An important feature is that for any deviation greater than δ_m, there are two values of the angle of incidence (i_1 and i_2) that give the same deviation. Only at minimum deviation is there a single value of i.

At minimum deviation the ray passes symmetrically through the prism, so the angle of incidence equals the angle of emergence:

$$i_1 = i_2 \text{ and } r_1 = r_2 = \frac{A}{2}$$

At this stage the refracted ray inside the prism travels parallel to the base.

Angle of minimum deviation: It is the smallest value of the angle of deviation produced by a given prism, obtained when the ray passes symmetrically through the prism so that the angle of incidence equals the angle of emergence.

Final answer. The δ–i graph is U-shaped with a single minimum δm; the angle of minimum deviation is the least deviation, occurring when the ray passes symmetrically through the prism (i1 = i2).

← Back to Q142

Q143

A ray of light is incident normally on a refracting face of a prism of prism angle A and suffers a deviation of angle δ. Prove that the refractive index n of the material of the prism is given by $\boldsymbol{n = \frac{\sin(A+\delta)}{\sin A}}$.

[2024 • Set 55-1-1]

SOLUTION

The light strikes the first refracting face normally, so it enters the prism without bending. Hence the angle of refraction at the first face is:

$$r_1 = 0$$

For a prism, the angles of refraction inside are related to the prism angle by:

$$r_1 + r_2 = A$$

Since $\mathbf{r_1 = 0}$, this gives:

$$r_2 = A$$

The ray now travels inside the glass and strikes the second face from inside. Applying Snell's law at the second (glass-to-air) face, where r_2 is the internal angle and e is the angle of emergence:

$$n \sin r_2 = \sin e$$

$$n \sin A = \sin e$$

The total deviation produced by the prism is:

$$\delta = (i_1 - r_1) + (e - r_2) = i_1 + e - (r_1 + r_2)$$

At the first face the ray is undeviated ($\mathbf{i_1 = 0}$, $\mathbf{r_1 = 0}$), and $\mathbf{r_1 + r_2 = A}$, so:

$$\delta = 0 + e - A$$

$$e = A + \delta$$

Substituting this back into the Snell's law equation:

$$n \sin A = \sin(A + \delta)$$

$$n = \frac{\sin(A + \delta)}{\sin A}$$

Hence proved.

Final answer. Using normal incidence (r1 = 0, r2 = A) and Snell's law at the second face with e = A + δ, we get n = sin(A + δ)/sin A.

← *Back to Q143*

Q144

A convex lens (n = 1.52) has a focal length of 15.0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1.65. What will be the nature of the lens?

[2024 • Set 55-2-1]

SOLUTION

The focal length of a lens depends on the refractive index of the lens relative to the surrounding medium, through the Lensmaker's formula:

$$\frac{1}{f} = ({}_m n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

where ${}_m n_l$ is the refractive index of the lens with respect to its surroundings. The shape factor $\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ stays the same in both cases, so we can take the ratio of the two equations.

In air ($\mathbf{n_{medium} = 1}$):

$$\frac{1}{f_a} = (n_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

In the liquid ($\mathbf{n_{medium} = 1.65}$), the relative index is ${}_l n_g = \mathbf{\frac{1.52}{1.65}}$:

$$\frac{1}{f_l} = \left(\frac{1.52}{1.65} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Dividing the second equation by the first:

$$\frac{f_a}{f_l} = \frac{\frac{1.52}{1.65} - 1}{1.52 - 1}$$

$\mathbf{\frac{1.52}{1.65} = 0.9212}$, so the numerator is $\mathbf{0.9212 - 1 = -0.0788}$

$$\frac{f_a}{f_l} = \frac{-0.0788}{0.52} = -0.1515$$

$$f_l = \frac{f_a}{-0.1515} = \frac{15.0}{-0.1515}$$

$$f_l \approx -99 \text{ cm}$$

Since the focal length is now negative, the lens behaves as a **diverging lens**. This happens because the liquid is optically denser than the glass, which reverses the lens action.

CONCEPT

When the surrounding medium is denser than the lens, a convex lens acts as a diverging lens and a concave lens acts as a converging lens.

Final answer. The focal length in the liquid is about −99 cm, and the convex lens behaves as a diverging lens.

← *Back to Q144*

Q145

The magnifying power of an astronomical telescope is 24. *In normal adjustment, distance between its two lenses is* 150 *cm. Find the focal length of the objective lens.*

[2024 • Set 55-3-1]

SOLUTION

For an astronomical telescope in normal adjustment (final image at infinity), the final image is formed when the intermediate image lies at the common focus of the two lenses. Two standard relations apply.

The distance between the two lenses (tube length) equals the sum of the focal lengths:

$$L = f_o + f_e = 150 \text{ cm}$$

The magnifying power is the ratio of the objective focal length to the eyepiece focal length:

$$m = \frac{f_o}{f_e} = 24$$

From the magnification relation:

$$f_o = 24\, f_e$$

Substituting into the tube-length equation:

$$24\, f_e + f_e = 150$$

$$25\, f_e = 150$$

$$f_e = 6 \text{ cm}$$

Therefore the focal length of the objective is:

$$f_o = 24 \times 6 = 144 \text{ cm}$$

So the **objective focal length is 144 cm**.

Final answer. The focal length of the objective lens is 144 cm.

← *Back to Q145*

Q146

Explain the following:
(a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.
(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances?

[2024 • Set 55-3-1]

SOLUTION

(a) A simple microscope is a single convex lens of short focal length. The object is placed within the focal length, so it forms a virtual, erect and magnified image. It is true that the image and the object subtend nearly the same physical angle when measured along the same ray bundle, because the lens does not change the direction of a ray passing through its optical centre.

The magnification, however, comes from comparing two different situations. Without the lens, the eye sees the object placed at the least distance of distinct vision D (about 25 cm), giving a small angular size:

$$\alpha \approx \frac{h}{D}$$

With the lens, the object can be brought very close, near the focal length f (much smaller than D), giving a larger angular size:

$$\beta \approx \frac{h}{f}$$

The magnifying power is the ratio of these angles:

$$M = \frac{\beta}{\alpha} = \frac{D}{f} \gg 1$$

So the lens lets the eye view the object from much closer than D without strain, which is why magnification is obtained even though the lens itself does not increase the angle subtended by a single ray.

(b) For a real object, a plane mirror always forms a virtual image, and a convex mirror also always forms a virtual, erect and diminished image. However, both can form a real image when the object is virtual, that is, when a converging beam of light is directed towards the mirror so that the rays would meet behind the mirror. The mirror then intercepts this converging beam and brings it to a real focus in front of it. Hence, with a virtual object, both a plane mirror and a convex mirror can produce a real image.

Final answer. (a) Magnification arises because the lens lets the eye view the object from near f instead of D, so M = D/f ≫ 1; (b) yes, both can form real images when the object is virtual (a converging beam falls on the mirror).

← *Back to Q146*

Q147

A ray of light is incident normally on one face of an equilateral glass prism of refractive index μ. When the prism is completely immersed in a transparent medium, it is observed that the emergent ray just grazes the adjacent face. Find the refractive index of the medium.

[2024 • Set 55-4-1]

SOLUTION

For an equilateral prism the refracting angle is:

$$A = 60^\circ$$

The ray falls normally on the first face, so it enters without bending:

$$r_1 = 0$$

Using $\mathbf{r_1 + r_2 = A}$ for the angles inside the prism:

$$r_2 = A - r_1 = 60^\circ - 0 = 60^\circ$$

The ray then reaches the second face at $\mathbf{r_2 = 60^\circ}$. The problem states the emergent ray just grazes the adjacent face, meaning it leaves along the surface at 90° to the normal. This is exactly the condition of the critical angle, so the angle inside equals the critical angle:

$$\theta_c = r_2 = 60^\circ$$

When light goes from the denser glass (index μ) into the surrounding medium (index n_m), the critical angle satisfies:

$$\theta_c = \frac{n_m}{\mu}$$

Rearranging:

$$n_m = \mu\, \theta_c = \mu \sin 60^\circ$$

$$n_m = \mu \times \frac{\sqrt{3}}{2}$$

$$n_m = \frac{\sqrt{3}}{2}\, \mu$$

So the refractive index of the surrounding medium is $\frac{\sqrt{3}}{2}\mu$.

TIP

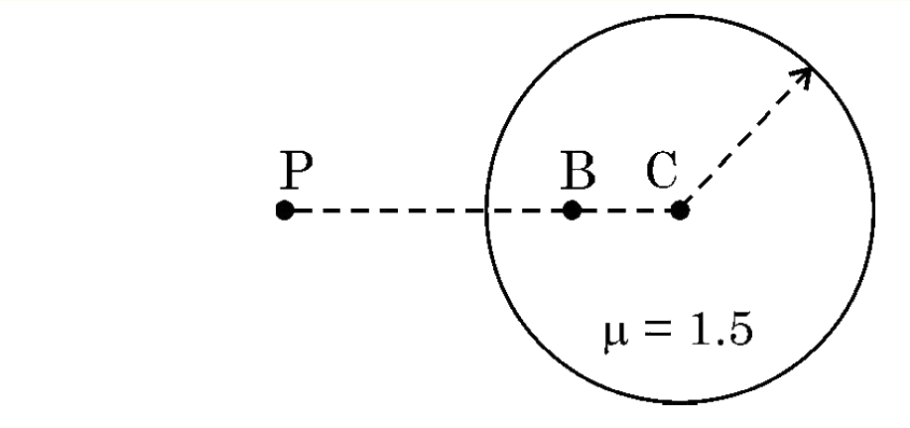
"Just grazes the adjacent face" is the standard clue for the emergent ray leaving at 90 °, i.e. the internal angle equals the critical angle.

Final answer. The refractive index of the medium is n_m = (√3/2)μ.

← *Back to Q147*

Q148

*An air bubble is trapped at point B (***CB=20 cm***) in a glass sphere of radius* 40 cm *and refractive index* 1.5 *as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point P.*



[2024 • Set 55-4-1]

SOLUTION

The light from the air bubble travels through the glass and refracts at the curved surface of the sphere before reaching the observer at *P*. So we apply the formula for refraction at a single spherical surface, going from glass to air.

Here the light goes from glass to air, so $\mathbf{n_1 = 1.5}$ (glass, where the object lies) and $\mathbf{n_2 = 1.0}$ (air, where the observer is).

The bubble *B* is at **CB = 20 cm** from the centre, on the side facing the observer. The refracting surface is the near surface of the sphere of radius 40 cm. Measuring from this surface, the object (bubble) distance is:

$$u = -(40 - 20) = -20 \text{ cm}$$

The surface is convex towards the incoming light from inside, and it curves away from the observer, so its radius of curvature is:

$$R = -40 \text{ cm}$$

Applying the refraction formula at the spherical surface:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1}{v} - \frac{1.5}{-20} = \frac{1 - 1.5}{-40}$$

$$\frac{1}{v} + 0.075 = \frac{-0.5}{-40} = 0.0125$$

$$\frac{1}{v} = 0.0125 - 0.075 = -0.0625$$

$$v = -16 \text{ cm}$$

The negative sign shows the image lies on the same side as the object (inside the sphere). Hence the observer sees a **virtual image of the bubble about 16 cm inside the sphere**, measured from the refracting surface near *P*.

Final answer. The image is virtual, formed about 16 cm inside the sphere (on the same side as the bubble) from the refracting surface near P.

← Back to Q148

Q149

In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 *m. If the magnifying power of the telescope is* 19*, find the focal length of the objective and the eyepiece lens.*

[2024 • Set 55-4-1]

SOLUTION

For a refracting astronomical telescope in normal adjustment, the final image is at infinity, and the intermediate image lies at the common focus of the objective and the eyepiece. Two relations are used.

The distance between the lenses (tube length) equals the sum of the focal lengths:

$$L = f_o + f_e = 1.00\ m = 100 \text{ cm}$$

The magnifying power is:

$$m = \frac{f_o}{f_e} = 19$$

From the magnification relation:

$$f_o = 19\, f_e$$

Substituting into the tube-length equation:

$$19\, f_e + f_e = 100$$

$$20\, f_e = 100$$

$$f_e = 5 \text{ cm}$$

Then the objective focal length is:

$$f_o = 19 \times 5 = 95 \text{ cm}$$

So the **objective focal length is 95 cm** and the **eyepiece focal length is 5 cm**.

Final answer. The focal length of the objective is 95 cm and that of the eyepiece is 5 cm.

← *Back to Q149*

Q150

A telescope has an objective lens of focal length 150 cm *and an eyepiece of focal length* 5 cm. *Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.*

[2024 • Set 55-5-1]

SOLUTION

For an astronomical telescope in normal adjustment, the object is at infinity and the final image is also formed at infinity, so the magnifying power is the ratio of the objective focal length to the eyepiece focal length.

Given: $\mathbf{f_o = 150 \text{ cm}}$ and $\mathbf{f_e = 5 \text{ cm}}$.

The magnifying power is:

$$m = \frac{f_o}{f_e} = \frac{150}{5} = 30$$

So the telescope has a magnifying power of 30.

Now for the image formed by the objective. The objective receives light from a very distant object (effectively at infinity), so the parallel rays converge to form the intermediate image right at the focus of the objective. Therefore the distance of this image from the objective equals its focal length:

$$v_o = f_o = 150 \text{ cm}$$

This intermediate real, inverted and diminished image is then viewed through the eyepiece. Hence the image formed by the objective lies **150 cm from the objective**.

Final answer. The magnifying power is 30, and the objective forms its image 150 cm from the objective (at its focus).

← *Back to Q150*

Q151

The radius of curvature of a convex mirror is 30 cm. *It forms an image of an object which is half the size of the object. Find the separation between the object and the image.*

[2024 • Set 55-5-2]

SOLUTION

For a convex mirror, both the radius of curvature and the focal length are positive. Here $R = +30$ **cm**, so the focal length is half of this.

$$f = \frac{R}{2} = \frac{30}{2} = +15 \text{ cm}$$

A convex mirror always forms a virtual, erect and diminished image, so the magnification is positive. The image is half the size of the object, hence $m = +\frac{1}{2}$.

The magnification relates image and object distances as $m = -\frac{v}{u}$.

$$+\frac{1}{2} = -\frac{v}{u} \Rightarrow v = -\frac{u}{2}$$

Applying the mirror formula:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{(-u/2)} + \frac{1}{u} = \frac{1}{15}$$

$$-\frac{2}{u} + \frac{1}{u} = \frac{1}{15}$$

$$-\frac{1}{u} = \frac{1}{15} \Rightarrow u = -15 \text{ cm}$$

The object is 15 cm in front of the mirror. The image distance is:

$$v = -\frac{u}{2} = -\frac{(-15)}{2} = +7.5 \text{ cm}$$

The positive v confirms the image lies 7.5 cm behind the mirror (virtual). The object is in front and the image is behind, so the total separation between them is the sum of their distances from the mirror.

$$\text{Separation} = |u| + |v| = 15 + 7.5 = \mathbf{22.5 \text{ cm}}$$

Final answer. The separation between the object and the image is 22.5 cm.

← Back to Q151

Q152

A thin converging lens of focal length 10 cm *is placed coaxially in contact with a thin diverging lens of focal length* 15 cm. *How will the combination behave? Justify your answer.*

[2024 • Set 55-5-3]

SOLUTION

When two thin lenses are placed coaxially in contact, the equivalent focal length F of the combination is given by the lens-combination formula.

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

For the converging (convex) lens the focal length is positive, $\mathbf{f_1 = +10\ cm}$, and for the diverging (concave) lens it is negative, $\mathbf{f_2 = -15\ cm}$.

$$\frac{1}{F} = \frac{1}{10} + \frac{1}{(-15)} = \frac{1}{10} - \frac{1}{15}$$

Taking the LCD as 30:

$$\frac{1}{F} = \frac{3-2}{30} = \frac{1}{30}$$

$$F = +30\ \text{cm}$$

Since the equivalent focal length comes out **positive**, the combination behaves as a **converging lens** of focal length 30 cm.

The justification is that the converging lens has the shorter focal length, so it has the greater magnitude of power. The net power $P = P_1 + P_2$ is positive, and a system with net positive power converges light.

Final answer. The combination behaves as a converging lens of focal length 30 cm.

← *Back to Q152*

Q153

Why is a reflecting telescope preferred over a refracting telescope? Justify your answer giving two reasons.

[2023 • Set 55-1-1]

SOLUTION

A reflecting telescope uses a concave mirror as its objective, while a refracting telescope uses a convex lens as its objective. The mirror-based design is preferred for the following two reasons.

First, a reflecting telescope is free from chromatic aberration. A lens refracts light and acts like a weak prism, so different colours (wavelengths) are bent by

different amounts and come to focus at slightly different points. This produces coloured fringes around bright objects in a refracting telescope. A mirror only reflects light, and the laws of reflection are independent of wavelength, so all colours focus at the same point and there is no chromatic aberration.

Second, large apertures are easier to build with a mirror, giving higher light-gathering power and resolution. A large objective lens can be supported only at its rim and sags under its own weight, distorting the image, and thick glass also absorbs light. A mirror can be supported rigidly from behind across its whole back surface, so very large diameters are possible. A larger aperture collects more light, making faint stars visible and increasing the resolving power.

These two reasons make reflecting telescopes brighter, sharper and better suited to astronomy.

Final answer. A reflecting telescope is preferred because it has no chromatic aberration and allows larger apertures with higher light-gathering power.

← *Back to Q153*

Q154

Briefly explain two applications of total internal reflection.

[2023 • Set 55-1-1]

SOLUTION

Total internal reflection (TIR) is the complete reflection of light back into a denser medium when it strikes the boundary with a rarer medium at an angle greater than the critical angle. Two important applications are described below.

The first application is the optical fibre. An optical fibre has a glass or plastic core of higher refractive index surrounded by a cladding of lower refractive index. Light entering the core strikes the core-cladding boundary at an angle greater than the critical angle, so it undergoes repeated total internal reflection and travels along the fibre with almost no loss of energy, even around bends. This is used to carry telephone, internet and television signals, and in medical endoscopes to view internal organs.

The second application is the brilliance and sparkle of a diamond. Diamond has a very high refractive index, $n \approx 2.42$, which gives it a very small critical angle of about $\theta_c \approx 24^\circ$. A diamond is cut with many facets so that most light entering it strikes the inner faces at angles greater than this small critical angle. The light therefore suffers multiple total internal reflections before finally emerging, which produces the characteristic sparkle and brilliance.

A third common use is in totally reflecting prisms found in periscopes and binoculars, which turn light through 90° or 180° without loss of intensity.

Final answer. Two applications of total internal reflection are optical fibres for signal transmission and the brilliance of cut diamonds.

← *Back to Q154*

Q155

Define total internal reflection of light. Give two conditions for it.

[2023 • Set 55-1-1]

SOLUTION

Total internal reflection is the phenomenon in which a ray of light travelling from an optically denser medium to an optically rarer medium is completely reflected back into the denser medium, provided the angle of incidence in the denser medium exceeds a certain limiting value called the critical angle. At and below the critical angle the ray refracts into the rarer medium, but beyond it no refraction occurs and the whole light energy is reflected.

The two necessary conditions for total internal reflection are stated below.

(i) The light must travel from a denser medium to a rarer medium (for example, from glass or water into air).

(ii) The angle of incidence in the denser medium must be greater than the critical angle, that is $\theta > \theta_c$, where the critical angle is given by

$$\theta_c = \frac{n_{\text{rarer}}}{n_{\text{denser}}}$$

Only when both conditions hold together does total internal reflection take place.

Final answer. TIR is complete reflection of light back into the denser medium; it needs light going from denser to rarer medium with angle of incidence greater than the critical angle.

← *Back to Q155*

Q156

A double-convex lens of power 4 D is manufactured from a glass of refractive index 1.5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature?

[2023 • Set 55-1-2]

SOLUTION

The power of a lens is the reciprocal of its focal length in metres, $P = \frac{1}{f}$, measured in dioptres (D). The focal length is then linked to the refractive index and the radii by the lens maker's formula.

First find the focal length from the given power.

$$f = \frac{1}{P} = \frac{1}{4} = 0.25\ m = 25\ \text{cm}$$

The lens maker's formula is:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a double-convex lens with both faces having the same radius of curvature, $R_1 = +R$ and $R_2 = -R$.

$$\frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (n-1)\frac{2}{R}$$

Rearranging for R:

$$R = 2(n-1)f$$

Substituting $n = 1.5$ and $f = 25$ cm:

$$R = 2(1.5 - 1)(25) = 2 \times 0.5 \times 25$$

$$R = \mathbf{25\ cm}$$

Each face of the lens has a radius of curvature of 25 cm.

Final answer. The radius of curvature of each face is 25 cm.

← *Back to Q156*

Q157

The radii of curvature of the faces of a double-convex lens are 20 *cm and* 30 *cm. Its power is* $\frac{5}{3}$ *D. What is the refractive index of the glass of the lens?*

[2023 • Set 55-1-3]

SOLUTION

The refractive index is obtained from the lens maker's formula combined with the definition of power, $P = \frac{1}{f}$ (with f in metres).

$$P = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a double-convex lens, using the sign convention, the first surface facing the object is convex so $\mathbf{R_1 = +20\ cm}$, and the second surface is convex towards the object so $\mathbf{R_2 = -30\ cm}$.

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{20} - \frac{1}{-30} = \frac{1}{20} + \frac{1}{30}$$

$$= \frac{3 + 2}{60} = \frac{5}{60} = \frac{1}{12}\ \text{cm}^{-1}$$

Converting to per-metre: $\frac{1}{12}\ \mathbf{cm^{-1}} = \frac{100}{12}\ \mathbf{m^{-1}}$.

Substituting into the power equation with $P = \frac{5}{3}$ D:

$$\frac{5}{3} = (n-1) \times \frac{100}{12}$$

$$n - 1 = \frac{5}{3} \times \frac{12}{100} = \frac{60}{300} = 0.2$$

$$n = 1 + 0.2 = \mathbf{1.2}$$

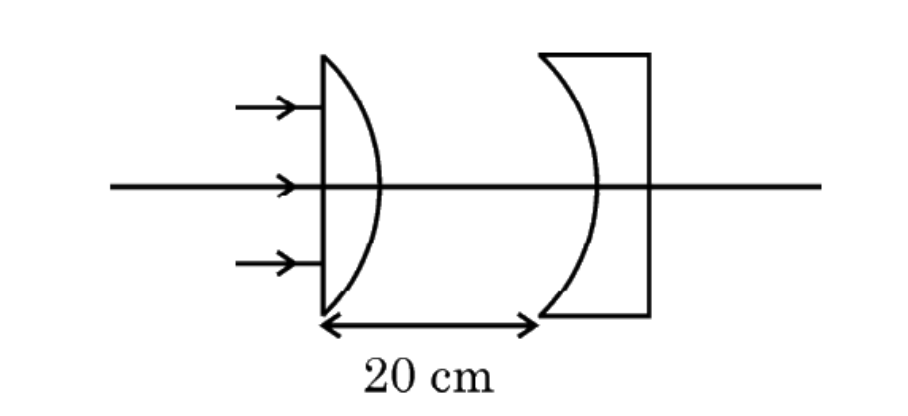
The refractive index of the glass of the lens is 1.2.

Final answer. The refractive index of the glass of the lens is 1.2.

← *Back to Q157*

Q158

In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm *each. The refractive index of the material of the lenses is* 1.5. *Find the final position of the image formed.*



[2023 • Set 55-2-1]

SOLUTION

The two lenses are placed in contact, so the equivalent focal length is found by first computing the focal length of each lens with the lens maker's formula and then combining them.

For the plano-convex lens, one face is flat ($R = \infty$) and the curved face has $R = 15$ cm:

$$\frac{1}{f_1} = (n-1)\frac{1}{R} = (1.5-1)\frac{1}{15} = \frac{0.5}{15} = \frac{1}{30}$$

$f_1 = +30$ **cm** (converging)

For the plano-concave lens, the curved face is concave with $R = 15$ **cm**, giving a negative focal length:

$$\frac{1}{f_2} = -(n-1)\frac{1}{R} = -\frac{0.5}{15} = -\frac{1}{30}$$

$f_2 = -30$ **cm** (diverging)

For the two thin lenses in contact:

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{30} + \left(-\frac{1}{30}\right) = 0$$

$$f_{eq} = \infty$$

The combination has zero power and behaves like a plane glass plate, neither converging nor diverging the rays. Hence parallel rays emerge parallel, and an object's rays pass through essentially unchanged in direction.

Therefore the **final image is formed at the same position as the object** (the image is at infinity for parallel incident light, and for a finite object there is no net displacement of the image due to the lens system).

TIP

Equal radii plano-convex + plano-concave of the same glass always cancel, giving zero net power. This is a standard quick-check.

Final answer. The equivalent focal length is infinite, so the combination acts like a glass plate and the final image forms at the object's position.

← Back to Q158

Q159

A point object in air is placed symmetrically at a distance of 60 cm *in front of a concave spherical surface of refractive index* 1.5. *If the radius of curvature of the surface is* 20 cm, *find the position of the image formed.*

[2023 • Set 55-2-2]

SOLUTION

This is a case of refraction at a single spherical surface, so we use the relation linking object and image distances to the radius of curvature.

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Light travels from air into glass, so $\mathbf{n_1 = 1}$ and $\mathbf{n_2 = 1.5}$. The object is in front of the surface, so $\boldsymbol{u = -60}$ **cm**. Since the surface is concave towards the incoming light (the centre of curvature lies on the same side as the object), $\boldsymbol{R = -20}$ **cm**.

Substituting these values:

$$\frac{1.5}{v} - \frac{1}{(-60)} = \frac{1.5 - 1}{-20}$$

$$\frac{1.5}{v} + \frac{1}{60} = \frac{0.5}{-20} = -\frac{1}{40}$$

$$\frac{1.5}{v} = -\frac{1}{40} - \frac{1}{60}$$

Taking the LCD as 120:

$$\frac{1.5}{v} = -\frac{3 + 2}{120} = -\frac{5}{120} = -\frac{1}{24}$$

$$v = 1.5 \times (-24) = -36 \text{ cm}$$

The negative sign shows the image is formed on the same side as the object, that is, it is a **virtual image 36 cm in front of the concave surface**.

Final answer. The image is virtual, formed 36 cm in front of the concave surface ($\boldsymbol{v = -36}$ cm).

← Back to Q159

Q160

The power of a thin lens is +5 *D. When it is immersed in a liquid, it behaves like a concave lens of focal length* 100 cm. *Calculate the refractive index of the liquid. Given*

refractive index of glass = **1.5**.

[2023 • Set 55-2-3]

SOLUTION

The behaviour of a lens changes when it is immersed in a liquid because the relevant quantity in the lens maker's formula is the refractive index of the glass relative to the surrounding medium.

In air, the power is $\mathbf{P_{air}} = +5\ \boldsymbol{D}$, so the focal length in air is:

$$f_{air} = \frac{1}{P} = \frac{1}{5}\ m = 20\ \text{cm}$$

In the liquid it behaves as a concave (diverging) lens with $\mathbf{f_{liq}} = \mathbf{-100\ cm}$.

Writing the lens maker's formula in air and in the liquid and taking their ratio, the geometric factor $(\frac{1}{R_1} - \frac{1}{R_2})$ cancels:

$$\frac{f_{liq}}{f_{air}} = \frac{(n_g - 1)}{(\frac{n_g}{n_l} - 1)}$$

Here $\mathbf{n_g} = \mathbf{1.5}$ is the refractive index of the glass and n_l is that of the liquid. Substituting the focal lengths:

$$\frac{-100}{20} = \frac{1.5 - 1}{\frac{1.5}{n_l} - 1}$$

$$-5 = \frac{0.5}{\frac{1.5}{n_l} - 1}$$

Cross-multiplying:

$$\frac{1.5}{n_l} - 1 = \frac{0.5}{-5} = -0.1$$

$$\frac{1.5}{n_l} = 1 - 0.1 = 0.9$$

$$n_l = \frac{1.5}{0.9} = \frac{5}{3} \approx \mathbf{1.67}$$

The refractive index of the liquid is about 1.67, which is greater than that of the glass; this is why the converging lens now behaves as a diverging one.

CONCEPT

When the surrounding liquid is denser than the glass, a convex lens flips to behave like a concave (diverging) lens. That sign flip is the physical clue here.

Final answer. The refractive index of the liquid is $\mathbf{n_l = \frac{5}{3} \approx 1.67}$.

← Back to Q160

Q161

The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface?

[2023 • Set 55-3-1]

SOLUTION

The critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90°. Light must travel from the optically denser medium to the rarer medium.

Here $\mathbf{n_A = 2}$ and $\mathbf{n_B = \sqrt{2}}$. Since $\mathbf{n_A > n_B}$, medium A is denser, so the ray must go from A into B and the critical angle θ_c is measured in medium A.

Applying Snell's law at the A-B interface with the angle of refraction equal to 90°:

$$n_A \, \theta_c = n_B \sin 90^\circ$$

$$\theta_c = \frac{n_B}{n_A}$$

$$\theta_c = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\theta_c = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

Thus the **critical angle for the A-B interface is 45°**. For any angle of incidence in medium A greater than 45°, the light undergoes total internal reflection.

Final answer. The critical angle for the A-B interface is **45°**.

← *Back to Q161*

Q162

(c) The focal length of a concave lens of $\mu = 1.5$ *is 20 cm in air. It is completely immersed in water of* $\mu = \frac{4}{3}$. *Calculate its focal length in water.*

OR

(c) An object is placed in front of a lens which forms its erect image of magnification 3. The power of the lens is 5 D. Calculate the distance of the object and the image from the lens.

[2023 • Set 55-4-1]

SOLUTION

For the concave lens immersed in water, use the ratio of the lens-maker relations in air and in water. The focal length in any medium depends on $\left(\frac{\mu_{\text{lens}}}{\mu_{\text{medium}}} - 1\right)$.

$$\frac{f_w}{f_a} = \frac{\left(\frac{\mu_g}{\mu_a} - 1\right)}{\left(\frac{\mu_g}{\mu_w} - 1\right)}$$

With $\mu_g = 1.5$, $\mu_a = 1$ and $\mu_w = \frac{4}{3}$:

$$\frac{f_w}{f_a} = \frac{(1.5 - 1)}{\left(\frac{1.5}{4/3} - 1\right)} = \frac{0.5}{(1.125 - 1)} = \frac{0.5}{0.125} = 4$$

Taking the concave lens focal length in air as $f_a = -20$ **cm**:

$$f_w = 4 \times (-20) = -80 \text{ cm}$$

So the **focal length in water is −80 cm**, still diverging but four times longer than in air.

OR

Here the lens forms an erect image of magnification 3, so $m = +3$ (erect virtual image). The power is $P = 5\ D$, giving $f = \frac{1}{P} = \frac{1}{5}\ m = +20$ **cm** (a converging lens). An erect, magnified virtual image from a converging lens needs the object inside the focal length.

$$m = \frac{v}{u} = 3 \Rightarrow v = 3u$$

Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$:

$$\frac{1}{3u} - \frac{1}{u} = \frac{1}{20}$$

$$\frac{1-3}{3u} = \frac{1}{20}$$

$$\frac{-2}{3u} = \frac{1}{20}$$

$$u = -\frac{40}{3} \approx -13.3 \text{ cm}$$

$$v = 3u = -40 \text{ cm}$$

So the **object is about 13.3 cm in front of the lens and the (virtual) image is 40 cm on the same side**.

Final answer. Focal length in water is **−80 cm**; OR object at **13.3 cm** and image at **40 cm** (both on the object side).

← *Back to Q162*

Q163

(iii) A double convex lens is made of glass of refractive index 1.55 with both faces of the same radius of curvature. Find the radius of curvature required if focal length is 20

cm.

OR

(iii) Two convex lenses A and B of focal lengths 15 cm and 10 cm respectively are placed coaxially 'd' distance apart. A point object is kept at a distance of 30 cm in front of lens A. Find the value of 'd' so that the rays emerging from lens B are parallel to its principal axis.

[2023 • Set 55-5-1]

SOLUTION

For a double convex lens with both faces of the same radius of curvature, the sign convention gives $\mathbf{R_1 = +R}$ and $\mathbf{R_2 = -R}$. The lens-maker's formula is:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (n-1)\frac{2}{R}$$

Substituting $\boldsymbol{n = 1.55}$ and $\boldsymbol{f = 20\ cm}$:

$$\frac{1}{20} = (1.55-1)\cdot\frac{2}{R}$$

$$\frac{1}{20} = 0.55 \times \frac{2}{R} = \frac{1.1}{R}$$

$$R = 1.1 \times 20 = 22\ \text{cm}$$

So each face must have a **radius of curvature of 22 cm**.

OR

Lens A has $\mathbf{f_A = 15\ cm}$ and the object is at $\boldsymbol{u = -30\ cm}$. Apply the lens formula to find where A forms its image.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_A}$$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{15} - \frac{1}{30} = \frac{2-1}{30} = \frac{1}{30}$$

$$v = +30\ \text{cm}$$

So A forms a real image 30 cm to its right. For the rays leaving lens B to emerge parallel to the principal axis, this image must act as an object placed exactly at the focus of B, i.e. at a distance $\mathbf{f_B = 10\ cm}$ in front of B.

The image of A lies 30 cm from A, so lens B must be placed 10 cm beyond this image:

$$d = 30 + 10 = 40 \text{ cm}$$

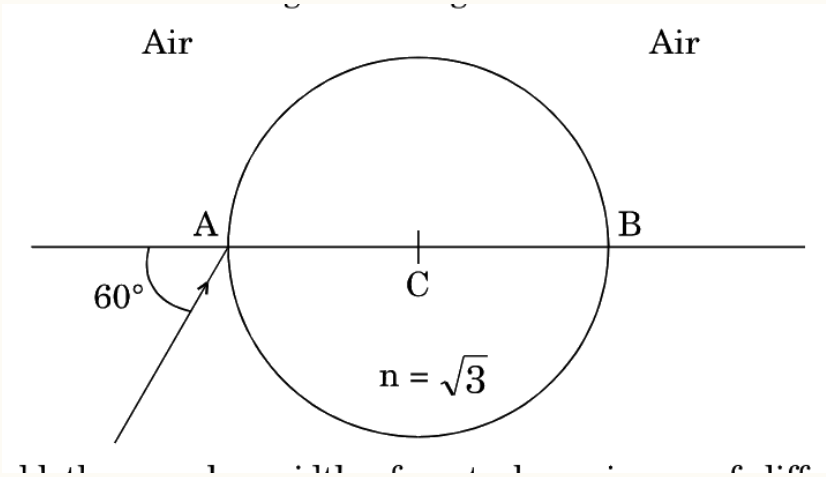
So the **two lenses must be 40 cm apart**.

Final answer. Radius of curvature = **22 cm**; OR separation $\boldsymbol{d}$ = **40 cm**.

← *Back to Q163*

Q164

A ray of light falls on a transparent sphere of $\boldsymbol{n = \sqrt{3}}$ *at an angle of incidence* 60° *with the diameter AB of the sphere having centre C. The ray emerges from the sphere parallel to the line AB. Find the angle of emergence.*



[2021]

SOLUTION

The ray strikes the sphere at the surface, refracts into the glass, crosses to the opposite surface and refracts out. Because both refractions occur at surfaces whose normals pass through the centre C, the geometry is symmetric, and the emergent ray makes the same angle with the line AB as the incident ray did.

At the first surface the normal is along the radius (through C), so the angle of incidence is $i = 60°$. Applying Snell's law:

$$\sin i = n \sin r$$

$$\sin 60° = \sqrt{3} \sin r$$

$$\sin r = \sin 60° \sqrt{3} = \sqrt{3}/2\sqrt{3} = 12$$

$$r = 30°$$

Inside the sphere the chord meets the far surface, and by the isosceles triangle formed with the two radii to the points of incidence and emergence, the ray strikes the second surface at the same angle $30°$ to its normal. Applying Snell's law at the exit surface (denser to rarer):

$$n \sin 30° = \sin e$$

$$\sqrt{3} \times \frac{1}{2} = \sin e$$

$$\sin e = \frac{\sqrt{3}}{2}$$

$$e = 60°$$

So the **angle of emergence is 60°**, equal to the angle of incidence, which is exactly why the emergent ray comes out parallel to the diameter AB.

Final answer. The angle of emergence is **60°**.

← *Back to Q164*

Q165

Using lens maker's formula, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ *for a biconvex lens.*

[2020 • Set 55-1-1]

SOLUTION

Consider a thin biconvex lens of refractive index n placed in air, with surfaces of radii R_1 and R_2. An object O is on the principal axis at distance u. We apply the single-surface refraction relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ at each surface in turn.

At the first surface (air to glass, $n_1 = 1$, $n_2 = n$, radius R_1), the object at u forms an intermediate image at distance v_1:

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1} \quad ...(i)$$

This intermediate image acts as the object for the second surface (glass to air, $n_1 = n$, $n_2 = 1$, radius R_2). Since the lens is thin, the object distance for the second surface is v_1 and it forms the final image at v:

$$\frac{1}{v} - \frac{n}{v_1} = \frac{1-n}{R_2} \quad ...(ii)$$

Adding equations (i) and (ii), the $\frac{n}{v_1}$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = (n-1)\frac{1}{R_1} + (1-n)\frac{1}{R_2}$$

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad ...(iii)$$

By definition, when the object is at infinity ($u \to \infty$), the image forms at the focus, $v = f$:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ (lens–maker's formula)}$$

Substituting this into equation (iii) gives the **thin lens formula**:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

which is the required result.

CONCEPT

Remember the intermediate image from surface 1 becomes the object for surface 2; the thin-lens assumption lets you use the same distance v_1 for both, which is what makes the middle terms cancel.

Final answer. Adding the single-surface relations gives $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, hence $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$.

← Back to Q165

Q166

An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed.

[2020 • Set 55-2-1]

SOLUTION

For a concave mirror the focal length is half the radius of curvature, and by the sign convention both f and u are negative (object in front).

$$f = \frac{R}{2} = \frac{60}{2} = 30 \text{ cm}, f = -30 \text{ cm}, u = -20 \text{ cm}$$

Using the mirror formula:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-30} - \frac{1}{-20}$$

$$\frac{1}{v} = -\frac{1}{30} + \frac{1}{20} = \frac{-2+3}{60} = \frac{1}{60}$$

$$v = +60 \text{ cm}$$

The positive sign of v shows the image forms behind the mirror, so it is virtual. The magnification is:

$$m = -\frac{v}{u} = -\frac{60}{-20} = +3$$

The positive magnification of +3 means the image is erect and magnified three times. Physically, since the object (at 20 cm) lies inside the focal length (30 cm), a concave mirror always gives such an image.

Hence the **image is virtual, erect, magnified three times, and formed 60 cm behind the mirror**.

Final answer. The image is virtual, erect, magnified **3×**, formed **60 cm** behind the mirror.

← *Back to Q166*

Q167

The focal length of an equiconcave lens is $\frac{3}{4}$ *times of radius of curvature of its surfaces. Find the refractive index of the material of the lens. Under what condition will this lens behave as a converging lens?*

[2020 • Set 55-2-2]

SOLUTION

For an equi-concave lens both faces have the same magnitude of radius R but opposite signs: $\mathbf{R_1} = -R$ and $\mathbf{R_2} = +R$. The lens-maker's formula gives:

$$\frac{1}{f} = (n-1)\left(\frac{1}{\mathrm{R}_1} - \frac{1}{\mathrm{R}_2}\right) = (n-1)\left(\frac{1}{-R} - \frac{1}{R}\right) = -(n-1)\frac{2}{R}$$

The focal length of a concave lens is negative, and we are given $|f| = \frac{3}{4}\mathbf{R}$, so $f = -\frac{3\mathbf{R}}{4}$. Substituting:

$$\frac{1}{-3\mathrm{R}/4} = -(n-1)\frac{2}{R}$$

$$-\frac{4}{3\mathrm{R}} = -(n-1)\frac{2}{R}$$

$$\frac{4}{3} = 2(n-1)$$

$$n - 1 = \frac{2}{3}$$

$$n = \frac{5}{3} \approx 1.667$$

So the **refractive index of the lens material is $\frac{5}{3}$.**

A diverging lens behaves as a converging lens only when it is placed in a medium optically denser than its own material, i.e. when the surrounding medium has refractive index $\mathbf{n_m > n_{lens} = \frac{5}{3}}$. In that case the factor $(\frac{n_{lens}}{n_m} - 1)$ becomes negative, which flips the sign of the focal length and makes the concave lens converging.

Final answer. Refractive index $n = \frac{5}{3}$; the lens converges when placed in a medium denser than itself ($\mathbf{n_{medium} > \frac{5}{3}}$).

← *Back to Q167*

Q168

An object is placed in front of a convex mirror of focal length 15 *cm. It produces an image that is half the size of the object. Find (a) position of the object, (b) nature of the image, and (c) draw the ray diagram of image formation.*

[2020 • Set 55-2-3]

SOLUTION

A convex mirror always forms a virtual, erect, diminished image, so here $f =$ **+15 cm** and the magnification is $m = +\frac{1}{2}$ (erect, half size).

The magnification relation for a mirror is:

$$m = -\frac{v}{u} = +\frac{1}{2} \Rightarrow v = -\frac{u}{2}$$

Substituting into the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$:

$$\frac{1}{-u/2} + \frac{1}{u} = \frac{1}{15}$$

$$-\frac{2}{u} + \frac{1}{u} = \frac{1}{15}$$

$$-\frac{1}{u} = \frac{1}{15}$$

$$u = -15 \text{ cm}$$

(a) The **object is placed 15 cm in front of the mirror**.

The image distance is $v = -\frac{u}{2} = -\frac{-15}{2} = +7.5$ **cm**, i.e. 7.5 cm behind the mirror.

(b) The image is **virtual, erect and diminished (half the size of the object), formed behind the mirror**.

(c) Ray diagram: Draw the convex mirror with pole P, principal axis, focus F and centre C behind the reflecting surface. Place the object AB (an upright arrow) in front of the mirror. From the tip A, draw ray 1 parallel to the axis; after reflection it diverges as if coming from F (behind the mirror). From A, draw ray 2 directed towards C; it strikes the mirror normally and reflects straight back. The reflected rays diverge in front; extend them backwards (dashed lines) and they meet behind the mirror at A'. Drop $A'B'$ perpendicular to the axis. The image $A'B'$ is upright, smaller than AB, and lies between P and F behind the mirror.

Final answer. (a) Object is **15 cm** in front of the mirror; (b) image is virtual, erect, diminished, **7.5 cm** behind the mirror.

← *Back to Q168*

Q169

A converging lens of focal length f_1 *is placed coaxially in contact with a diverging lens of focal length* $\mathbf{f_2}$ $\mathbf{(f_1 > f_2)}$. *Determine the power and nature of the combination in terms of* f_1 *and* f_2.

OR

How is the resolving power of a compound microscope affected if (a) wavelength of light used is decreased, and (b) the diameter of its objective lens is increased? Justify your answers.

[2020 • Set 55-3-1]

SOLUTION

For two thin lenses in contact, the equivalent power is the sum of the individual powers, so the focal lengths combine as:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

The converging lens has focal length $+f_1$ and the diverging lens has a negative focal length, so we write $\mathbf{f_2 = -|f_2|}$. Then:

$$\frac{1}{F} = \frac{1}{f_1} - \frac{1}{|f_2|} = \frac{|f_2| - f_1}{f_1\,|f_2|}$$

The combined power is:

$$P = \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

We are told $\mathbf{f_1 > f_2}$, which (in magnitudes) means $\mathbf{f_1 > |f_2|}$, so the numerator $|f_2| - f_1$ is negative. Therefore F and P are both negative.

Hence the **combination behaves as a diverging lens** with power $P = \dfrac{1}{f_1} + \dfrac{1}{f_2}$ (which is negative because the diverging lens, being stronger, dominates).

OR

The resolving power of a compound microscope is given by:

$$R.P. = \frac{2\,n\theta}{1.22\,\lambda}$$

where $n\theta$ is the numerical aperture and λ is the wavelength of light used.

(a) If the wavelength λ is decreased, then since $R.P. \propto \dfrac{1}{\lambda}$, the **resolving power increases**. Shorter wavelength light resolves finer detail.

(b) If the diameter of the objective lens is increased, the half-angle θ subtended by the objective at the object increases, so θ and hence the numerical aperture $n\theta$ increase. Therefore the **resolving power increases**.

Final answer. Combination is diverging with $P = \dfrac{1}{f_1} + \dfrac{1}{f_2}$ (negative); OR resolving power increases in both cases (a) and (b).

← *Back to Q169*

Q170

Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, *when it is kept in a medium of refractive index* 1.4, *to have a power of* −5 *D.*

[2019 • Set 55-1-1]

SOLUTION

When a lens is placed in a medium of refractive index μ_m, the lens-maker's formula uses the relative refractive index $\frac{\mu_l}{\mu_m}$:

$$\frac{1}{f} = \left(\frac{\mu_l}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For an equi-concave lens, $\mathbf{R_1 = -R}$ and $\mathbf{R_2 = +R}$, so:

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{-R} - \frac{1}{R} = -\frac{2}{R}$$

Computing the relative-index factor with $\boldsymbol{\mu_l = 1.5}$ and $\boldsymbol{\mu_m = 1.4}$:

$$\frac{\mu_l}{\mu_m} - 1 = \frac{1.5}{1.4} - 1 = \frac{1.5 - 1.4}{1.4} = \frac{0.1}{1.4} = \frac{1}{14}$$

Therefore:

$$\frac{1}{f} = \frac{1}{14} \times \left(-\frac{2}{R}\right) = -\frac{1}{7R}$$

The power is $\boldsymbol{P = \frac{1}{f}}$ (with f in metres) and is given as −5 D:

$$-\frac{1}{7R} = -5$$

$$7R = \frac{1}{5} = 0.2\ m$$

$$R = \frac{0.2}{7} = 0.0286\ m \approx 2.86\ \text{cm}$$

So the **radius of curvature of each surface is about 2.86 cm**.

Final answer. The radius of curvature is $\boldsymbol{R \approx 2.86\ \text{cm}}$.

← Back to Q170

Q171

An equilateral glass prism has a refractive index 1.6 *in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index* $\frac{4\sqrt{2}}{5}$.

[2019 • Set 55-1-1]

SOLUTION

For an equilateral glass prism the refracting angle is $A = 60°$. The refractive index of the prism material is 1.6 in air, and the surrounding medium now has refractive index $n_m = \frac{4\sqrt{2}}{5}$.

When a prism is immersed in a medium, the deviation depends on the refractive index of the glass relative to that medium. This relative (effective) refractive index is found by dividing the absolute index of the glass by that of the medium.

$$n_{gm} = \frac{n_g}{n_m}$$

$$n_{gm} = \frac{1.6}{4\sqrt{2}/5}$$

$$n_{gm} = \frac{1.6 \times 5}{4\sqrt{2}}$$

$$n_{gm} = \frac{8}{4\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

The prism formula relating the angle of minimum deviation δ_m to the refracting angle is:

$$n_{gm} = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Substituting $n_{gm}=\sqrt{2}$ and $A=60°$:

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ+\delta_m}{2}\right)}{\sin 30^\circ}$$

$$\sin\left(\frac{60^\circ+\delta_m}{2}\right) = \sqrt{2}\,30^\circ = \sqrt{2}\times\frac{1}{2} = \frac{1}{\sqrt{2}}$$

$$\frac{60^\circ+\delta_m}{2} = 45^\circ$$

$$60^\circ + \delta_m = 90^\circ$$

$$\delta_m = 30^\circ$$

So when the prism is placed in this medium, the **angle of minimum deviation is 30°**, which is smaller than its value in air because the glass appears optically less dense relative to the surrounding medium.

COMMON MISTAKE

When a prism is dipped in a medium, never use the absolute index n_g in the prism formula. Always use the relative index n_g/n_m, or the deviation will come out wrong.

Final answer. The angle of minimum deviation of the prism in the given medium is **30°**.

← *Back to Q171*

Q172

Under which conditions can a rainbow be observed? Distinguish between a primary and a secondary rainbow.

[2019 • Set 55-2-1]

SOLUTION

A rainbow is formed by the combined effect of refraction, total internal reflection and dispersion of sunlight by tiny water droplets suspended in the atmosphere.

Conditions required to observe a rainbow:

The Sun should be behind the observer (the observer faces away from the Sun).

There must be water droplets (rain, mist or spray) present in the air in front of the observer.

The Sun should be low in the sky, generally at an altitude below about 42° above the horizon, so that the dispersed light can reach the observer's eyes.

Distinction between primary and secondary rainbow:

A primary rainbow is formed when sunlight undergoes refraction, then one total internal reflection, and again refraction inside each water droplet. In a primary rainbow, red appears on the outer edge and violet on the inner edge, and its angular radius is about 42°.

A secondary rainbow is formed when sunlight undergoes refraction, then two total internal reflections, and again refraction inside each droplet. In a secondary rainbow the colour order is reversed, with violet on the outer edge and red on the inner edge, and its angular radius is about 51°.

Because of the extra internal reflection, more light is lost in the secondary rainbow, so the **secondary rainbow is fainter than the brighter primary rainbow**.

Final answer. A rainbow needs the Sun behind the observer, droplets in front, and a low Sun; the primary forms by one internal reflection (red outside, ~**42**°), the secondary by two (violet outside, ~**51**°, fainter).

← *Back to Q172*

Q173

Explain the following:
(a) Sky appears blue.
(b) The Sun appears reddish at (i) sunset, (ii) sunrise.

[2019 • Set 55-2-1]

SOLUTION

Both effects are explained using Rayleigh scattering, in which the intensity of scattered light is inversely proportional to the fourth power of its wavelength.

$$I \propto \frac{1}{\lambda^4}$$

(a) Why the sky appears blue:

As sunlight passes through the atmosphere, it is scattered by the air molecules and fine particles. Since the intensity of scattering varies as $1/\lambda^4$, the shorter wavelengths (blue) are scattered far more strongly than the longer wavelengths (red).

This strongly scattered blue light reaches our eyes from all directions of the sky. Although violet light has an even shorter wavelength, the Sun emits less violet and the human eye is more sensitive to blue, so the **sky predominantly appears blue**.

(b) Why the Sun appears reddish at sunrise and sunset:

At (i) sunset and (ii) sunrise, the Sun is near the horizon, so its light has to travel through a much greater thickness of the atmosphere to reach the observer compared to noon.

Over this long path, most of the shorter wavelengths (blue and violet) are scattered away out of the line of sight. Only the longer wavelengths, mainly red, are able to pass straight through and reach the eye.

Hence at sunrise and sunset the **Sun appears reddish**. At noon, when the Sun is overhead, the path is shortest, least light is scattered out, and the Sun appears nearly white.

Final answer. The sky is blue because Rayleigh scattering ($I \propto 1/\lambda^4$) scatters blue most, and the Sun looks red at dawn/dusk because the long atmospheric path scatters away blue, leaving red.

← *Back to Q173*

Q174

Draw the ray diagram of an astronomical telescope showing image formation in the normal adjustment position. Write the expression for its magnifying power.
OR

Draw a labelled ray diagram to show image formation by a compound microscope and write the expression for its resolving power.

[2019 • Set 55-2-1]

SOLUTION

Astronomical telescope in normal adjustment:

An astronomical telescope uses two convex lenses, an objective of large focal length f_o and an eyepiece of small focal length f_e, mounted coaxially.

Ray diagram description: Draw a long horizontal principal axis with the objective lens O on the left and the eyepiece lens E on the right. A parallel beam of rays from a distant object arrives at O inclined at a small angle α to the axis. The objective refracts these rays and forms a real, inverted, diminished intermediate image $A'B'$ at its focal point F_o.

In normal adjustment, the telescope is set so that this intermediate image lies exactly at the focus of the eyepiece as well, i.e. the focal point F_o of the objective coincides with the focus F_e of the eyepiece. The eyepiece therefore sends out a parallel beam, and the final image is formed at infinity, subtending an angle β at the eye. Mark the separation of the lenses as $\boldsymbol{L = f_o + f_e}$ (length of the telescope tube).

Expression for magnifying power: The magnifying power is the ratio of the angle subtended at the eye by the final image to the angle subtended by the object at the objective.

$$m = \frac{\beta}{\alpha}$$

$$m = \frac{f_o}{f_e}$$

So the **magnifying power of an astronomical telescope in normal adjustment is** $\boldsymbol{m = \frac{f_o}{f_e}}$, the ratio of the focal length of the objective to that of the eyepiece.

OR

Compound microscope:

A compound microscope uses two convex lenses, an objective of very small focal length and an eyepiece, separated by a tube of length L.

Ray diagram description: Place the small object AB just beyond the focus f_o of the objective on the left. The objective forms a real, inverted, magnified intermediate image $A'B'$ inside the tube. This intermediate image lies within the focal length of the eyepiece, which then acts as a simple magnifier and forms a final virtual, inverted and highly magnified image $A''B''$ at the least distance of distinct vision D (near point) from the eye. Mark the principal axis, the foci of both lenses, and trace at least two rays from the tip of AB through the objective to locate $A'B'$, then through the eyepiece to locate $A''B''$.

Expression for resolving power: The resolving power of a microscope is the reciprocal of the smallest distance between two points that can be just resolved.

$$R.P. = \frac{2\,n\theta}{1.22\,\lambda}$$

Here $n\theta$ is the **numerical aperture** of the objective, n is the refractive index of the medium between the object and the objective, θ is the half-angle of the cone of light, and λ is the wavelength of light used.

Final answer. For the telescope in normal adjustment, $\boldsymbol{m = \frac{f_o}{f_e}}$; for the compound microscope, $\boldsymbol{R.P. = \frac{2n\theta}{1.22\lambda}}$.

← *Back to Q174*

Q175

A beam of light converges at a point P. Draw ray diagrams to show where the beam will converge if (i) a convex lens, and (ii) a concave lens is kept in the path of the beam.

[2019 • Set 55-2-2]

SOLUTION

Here the incident beam is already converging towards a point P that lies on the far side of the lens, so P acts as a virtual object for the lens. Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with the object distance measured to P:

(i) Convex lens placed in the path:

Ray diagram description: Draw a horizontal principal axis. Show a cone of rays converging from the left, all heading towards the point P on the axis to the right. Place a convex (biconvex) lens before P. A convex lens is converging, so it bends each ray further inward. The rays now meet the axis at a new point P' which lies between the lens and P, i.e. closer to the lens than P. Mark this point P'.

Therefore a convex lens makes the converging beam come to a focus at a point P' nearer to the lens than P.

(ii) Concave lens placed in the path:

Ray diagram description: Again show the cone of rays converging towards P. Now place a concave (biconcave) lens before P. A concave lens is diverging, so it bends each ray outward (away from the axis). The rays still converge, but more gradually, and meet the axis at a new point P'' which lies beyond P, i.e. farther from the lens than P. Mark this point P''.

Therefore a concave lens makes the converging beam come to a focus at a point P'' farther from the lens than P.

In summary, a **convex lens shifts the convergence point nearer (to P', before P)**, while a **concave lens shifts it farther (to P'', beyond P)**.

CONCEPT

Treat P as a virtual object. A convex lens adds convergence (focus moves nearer), a concave lens subtracts convergence (focus moves farther) but the beam still converges as long as the lens is not too strong.

Final answer. A convex lens converges the beam at a point before P; a concave lens converges it at a point beyond P.

← *Back to Q175*

Q176

Under what conditions does the phenomenon of total internal reflection take place? Draw a ray diagram showing how a ray of light deviates by 90° *after passing through*

a right-angled isosceles prism.

[2019 • Set 55-2-2]

SOLUTION

Total internal reflection is the phenomenon in which a ray of light travelling in a denser medium is completely reflected back into the same medium at its boundary with a rarer medium.

Conditions for total internal reflection:

The light must travel from an optically denser medium to an optically rarer medium (for example, from glass to air).

The angle of incidence in the denser medium must be greater than the critical angle θ_c for that pair of media, where $\theta_c = \frac{1}{n}$ and n is the refractive index of the denser medium with respect to the rarer one.

Ray diagram description (90 ° deviation by a right-angled isosceles prism):

Draw a right-angled isosceles prism with angles 45°, 45° and 90°, oriented so that the hypotenuse (the longest face) is vertical and the right angle points to the left. A ray of light enters normally (perpendicularly) through one of the two equal faces, so it passes straight in without bending.

This ray then strikes the hypotenuse at an angle of incidence of 45°. For ordinary glass the critical angle is about 42°, which is less than 45° (this holds whenever $n > \sqrt{2}$, giving $\theta_c < 45°$). Hence the ray suffers total internal reflection at the hypotenuse.

After reflection the ray turns through 90° and emerges normally through the second equal face. So the **light ray is deviated by 90°** by total internal reflection inside the prism.

Such totally reflecting prisms are used in periscopes and binoculars to turn light through right angles without loss of intensity.

Final answer. TIR needs a denser-to-rarer medium and an angle of incidence above the critical angle; a **45°** face entry into a right-angled isosceles prism turns the ray by **90°**.

← *Back to Q176*

Q177

State, with the help of a ray diagram, the working principle of optical fibres. Write one important use of optical fibres.

[2019 • Set 55-2-3]

SOLUTION

Working principle of optical fibres:

An optical fibre works on the principle of total internal reflection. It consists of a central cylindrical core made of glass or plastic of higher refractive index n_1, surrounded by a cladding of slightly lower refractive index n_2 (so that $\mathbf{n_1 > n_2}$).

When light enters the core through one end at an angle within the acceptance angle, it strikes the core-cladding boundary at an angle of incidence greater than the critical angle. The critical angle is given by:

$$\theta_c = \frac{n_2}{n_1}$$

Because the angle of incidence exceeds θ_c, the light undergoes repeated total internal reflection at the core-cladding interface and travels along the fibre, even when the fibre is bent, with very little loss of energy.

Ray diagram description: Draw a long, slightly curved fibre showing the inner core (refractive index n_1) and the outer cladding (refractive index n_2). A ray enters the left end face, refracts into the core, and then zig-zags down the length of the fibre, each time striking the core-cladding boundary at an angle greater than θ_c and reflecting totally, finally emerging at the far right end.

One important use:

Optical fibres are used in long-distance, high-speed **optical communication** (telephone and internet signal transmission). They are also used in medical endoscopy to view internal organs.

Final answer. Optical fibres guide light by repeated total internal reflection in a high-index core, and are used in optical communication.

← *Back to Q177*

Q178

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

[2019 • Set 55-3-1]

SOLUTION

Consider a concave spherical mirror with pole P, centre of curvature C and principal focus F. The distance PC is the radius of curvature R and the distance PF is the focal length f.

Figure description: Draw the curved mirror with its principal axis passing through P, F and C. Take a single paraxial ray coming from infinity, parallel to the principal axis, and let it strike the mirror at a point A close to the pole. The normal at A is the line CA (a radius, which always passes through the centre of curvature). The reflected ray travels from A through the focus F.

At the point of incidence A, by the law of reflection the angle of incidence equals the angle of reflection:

$$\angle \text{(incident ray, CA)} = \angle \text{(CA, reflected ray)} = \theta$$

Since the incident ray is parallel to the principal axis CP, the line CA is a transversal, so by alternate angles:

$$\angle \text{ACF} = \theta$$

Also, the angle of reflection gives:

$$\angle \text{CAF} = \theta$$

In triangle ACF, two angles are equal ($\angle$ **ACF** = $\angle$ **CAF** = θ), so the triangle is isosceles and the sides opposite these equal angles are equal:

$$\text{AF} = \text{FC}$$

For paraxial rays, the point A lies very close to the pole P, so the distance AF is approximately equal to PF:

$$\text{AF} \approx \text{PF} = f$$

Therefore **FC ≈ PF = f** as well. Now along the axis:

$$PC = PF + FC$$

$$R = f + f$$

$$R = 2f$$

$$f = \frac{R}{2}$$

Hence for paraxial rays the **focal length of a spherical mirror is one-half of its radius of curvature**.

CONCEPT

The result $f = R/2$ holds only for paraxial rays close to the pole; for wide-angle rays the reflected rays do not all meet at F, causing spherical aberration.

Final answer. For paraxial rays, $f = \frac{R}{2}$, i.e. the focal length is half the radius of curvature.

← *Back to Q178*

Q179

Show that the image formed by a convex mirror of an object is always virtual.

[2019 • Set 55-3-2]

SOLUTION

For a convex mirror, the focus and centre of curvature lie behind the mirror, so by the sign convention the focal length is positive, $f > 0$.

Consider a real object placed in front of the mirror. By the sign convention, distances measured against the incident light (to the left of the mirror) are negative, so the object distance is negative:

$$u < 0$$

The mirror formula is:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Rearranging to find the image distance:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

For a convex mirror $\frac{1}{f}$ is positive (since $f>0$). For a real object u is negative, so $-\frac{1}{u}$ is also positive. Both terms on the right are therefore positive:

$$\frac{1}{v} = (\text{positive}) + (\text{positive}) > 0$$

$$\frac{1}{v} > 0 \Rightarrow v > 0$$

A positive image distance v means the image is formed behind the mirror, on the side opposite to the incident light, where the rays only appear to diverge from. Such an image cannot be caught on a screen and is therefore virtual.

Since this conclusion follows for every negative value of u, the result is independent of how near or far the object is placed. Hence a **convex mirror always forms a virtual image of a real object**.

Final answer. Since $v>0$ for every real object, the image of a convex mirror is always virtual.

← *Back to Q179*

Q180

In the case of a concave mirror of focal length f, when an object is kept between f and 2f, *show that its image is formed beyond* 2f.

[2019 • Set 55-3-3]

SOLUTION

Consider a concave mirror of focal length f with the object placed between the focus and the centre of curvature, that is, between f and 2f from the mirror.

Using the sign convention, all distances of real objects and the focal length of a concave mirror are negative, so let us write magnitudes and keep track of signs.

For a concave mirror f is negative, so the magnitude of focal length is $|f|$. The object lies between f and 2f, so its distance u (taken negative) satisfies:

$$|f| < |u| < 2|f|$$

The mirror formula is:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

With f and u both negative, write $\frac{1}{f} = -\frac{1}{|f|}$ and $\frac{1}{u} = -\frac{1}{|u|}$:

$$\frac{1}{v} = -\frac{1}{|f|} + \frac{1}{|u|}$$

Now apply the bound $|f| < |u| < 2|f|$. Taking reciprocals (which reverses the inequalities):

$$\frac{1}{2|f|} < \frac{1}{|u|} < \frac{1}{|f|}$$

Using the upper part $\frac{1}{|u|} < \frac{1}{|f|}$ gives $\frac{1}{v} < 0$, so the image is real and in front of the mirror, confirming v is negative. Using the lower part $\frac{1}{|u|} > \frac{1}{2|f|}$:

$$\frac{1}{v} = -\frac{1}{|f|} + \frac{1}{|u|} > -\frac{1}{|f|} + \frac{1}{2|f|} = -\frac{1}{2|f|}$$

So $\frac{1}{v} > -\frac{1}{2|f|}$, which means $\frac{1}{|v|} < \frac{1}{2|f|}$, and taking reciprocals:

$$|v| > 2|f|$$

Therefore the image distance exceeds 2f in magnitude, that is, the **image is formed beyond 2f (beyond the centre of curvature)**. This image is real, inverted and magnified.

Final answer. For an object between f and **2f** of a concave mirror, $|v|>2|f|$, so the real inverted image forms beyond **2f.**

← *Back to Q180*

Q181

A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm *from P. At what point does a beam converge if the convex lens has a focal length* 10 cm*?*

OR

An object is kept in front of a concave mirror of focal length 15 cm*. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.*

[2019 • Set 55-4-1]

SOLUTION

Here the converging beam would meet at P, so for the lens P acts as a virtual object. The light travels left to right, the lens is placed 15 cm before P, so the object lies on the far side of the lens and the object distance is positive (real-is-negative sign convention).

$u = +15$ **cm,** $f = +10$ **cm** (convex lens)

Using the thin lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{15} = \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

$$v = +6 \text{ cm}$$

So the beam now converges at a point 6 cm to the right of the lens, which is **15 − 6 = 9 cm** on the near side of the original point P. **The beam converges 6 cm from the lens (9 cm in front of P).**

OR (Concave mirror): The image is real and three times the object, so for a concave mirror the magnification is negative and the image is inverted.

$$m = -\frac{v}{u} = -3 \Rightarrow v = 3u$$

Using the mirror formula with $f = -15$ **cm**:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{3u} + \frac{1}{u} = \frac{1}{-15}$$

$$\frac{1+3}{3u} = -\frac{1}{15}$$

$$\frac{4}{3u} = -\frac{1}{15} \Rightarrow u = -20 \text{ cm}$$

The object is placed 20 cm in front of the concave mirror.

Final answer. Beam converges 6 cm from the lens (9 cm in front of P); OR the object is 20 cm from the concave mirror.

← *Back to Q181*

Q182

Out of the two optical instruments, a microscope and a telescope, which one plays the role in magnifying the objects and in resolving the two objects kept close to each other? Explain, giving example.

[2019 • Set 55-4-3]

SOLUTION

Both roles are played by the **microscope**.

A microscope is used to magnify small objects placed very close to it. The compound microscope increases the angular size of a tiny nearby object so that it appears much larger to the eye. For example, a biological cell or a

microorganism that is invisible to the naked eye can be seen clearly and enlarged.

A microscope is also the instrument that resolves two objects kept close to each other. Its **resolving power** is the ability to distinguish two nearby points as separate. It is given by:

$$R.P. = \frac{2\,n\theta}{1.22\,\lambda} = \frac{2\,(N.A.)}{1.22\,\lambda}$$

where $n\theta$ is the numerical aperture. For example, a microscope can show two adjacent cells in a tissue (separated by about 1 μm) as two distinct cells.

A telescope, by contrast, is meant for distant objects: it increases their angular size and resolves objects separated by small angles in the sky, such as a close binary star pair.

Final answer. The microscope both magnifies small nearby objects and resolves two close objects.

← *Back to Q182*

Q183

Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths? Explain.

[2017]

SOLUTION

The **magnifying power** of a compound microscope is the ratio of the angle subtended at the eye by the final image to the angle subtended at the eye by the object when both are placed at the least distance of distinct vision D.

When the final image is formed at infinity (normal adjustment), the magnifying power is:

$$M = \mathrm{m}_o \times \mathrm{m}_e = \frac{L}{\mathrm{f}_o} \times \frac{D}{\mathrm{f}_e}$$

where L is the tube length (distance between objective and eyepiece foci), D = **25 cm** is the near point, and f_o, f_e are the focal lengths of the objective and eyepiece.

Both focal lengths must be short because:

The objective forms a real, magnified image; its linear magnification is $m_o = \frac{L}{f_o}$. A smaller f_o gives a larger m_o.

The eyepiece works as a simple magnifier of angular magnification $m_e = \frac{D}{f_e}$. A smaller f_e gives a larger m_e.

Since the total magnification is the product $M = \frac{L}{f_o} \cdot \frac{D}{f_e}$, **both f_o and f_e should be small (short focal lengths)** to achieve a large overall magnification, often a few hundred times.

Final answer. M = (L/f_o)(D/f_e); both focal lengths must be short so that each magnification, and hence the product, is large.

← *Back to Q183*

Q184

Why should the objective of a telescope have large focal length and large aperture? Justify your answer.

[2017]

SOLUTION

The objective of a telescope should have a **large focal length** and a **large aperture** for the following reasons.

Large focal length (for high magnifying power): The angular magnification of a telescope in normal adjustment is:

$$m = \frac{f_o}{f_e}$$

To get a large magnification, the objective focal length f_o should be large (and the eyepiece focal length f_e small). A long f_o also forms a larger image of the

distant object, which the eyepiece then magnifies.

Large aperture (for brightness and resolution): A large aperture collects more light from the faint distant object, because the light-gathering power is proportional to the area, $\propto D^2$. This gives a brighter, clearer image.

A large aperture also increases the **resolving power** of the telescope:

$$R.P. = \frac{D}{1.22\,\lambda}$$

A larger diameter D means finer angular detail can be resolved, so two close stars can be seen as separate.

Hence a large focal length gives high magnification while a large aperture gives a bright, well-resolved image.

Final answer. Large f_o gives high magnification (m = f_o/f_e); large aperture gives a brighter image and higher resolving power (R.P. = D/1.22λ).

← *Back to Q184*

Q185

When are two objects just resolved? Explain. How can the resolving power of a compound microscope be increased? Use relevant formula to support your answer.

[2017]

SOLUTION

Two objects are said to be **just resolved** according to the **Rayleigh criterion**: when the central maximum of the diffraction pattern of one object falls exactly on the first minimum of the diffraction pattern of the other object. If they are closer than this, the two patterns overlap and the objects cannot be seen as separate.

The resolving power of a compound microscope is the reciprocal of the smallest distance between two points that can be seen as separate. It is given by:

$$R.P. = \frac{1}{d_{min}} = \frac{2\,n\theta}{1.22\,\lambda} = \frac{2\,(N.A.)}{1.22\,\lambda}$$

where $N.A. = n\theta$ is the numerical aperture, n is the refractive index of the medium between the object and objective, and λ is the wavelength of light used.

From this formula, the resolving power can be increased by:

Using light of shorter wavelength λ (for example ultraviolet light, or electrons in an electron microscope), since $R.P. \propto \frac{1}{\lambda}$.

Increasing the numerical aperture $n\theta$, for example by using an **oil immersion** objective (oil has a higher refractive index n) or a wider-angle objective (larger θ).

Final answer. Objects are just resolved when one's central maximum falls on the other's first minimum; R.P. ($= 2n\sin\theta/1.22\lambda$) is raised by smaller λ or larger numerical aperture.

← *Back to Q185*

Q186

Use the mirror equation to show that an object placed between f and 2f of a concave mirror forms an image beyond 2f.

OR

(a) State the condition under which a large magnification can be achieved in an astronomical telescope.

(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope.

[2017]

SOLUTION

For a concave mirror the focal length and object distance are negative. Take the object placed between f and 2f, so its distance lies between f and 2f in magnitude.

$$f < u < 2f \text{ (in magnitude)}$$

Using the mirror formula:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

Working with magnitudes, since the object lies beyond f we have $\frac{1}{u} < \frac{1}{f}$, so $\frac{1}{f} - \frac{1}{u} > 0$ and the image is real.

Also, since the object lies inside 2f, that is $u < 2f$, we have $\frac{1}{u} > \frac{1}{2f}$.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} < \frac{1}{f} - \frac{1}{2f} = \frac{1}{2f}$$

$$\frac{1}{v} < \frac{1}{2f} \Rightarrow v > 2f$$

Hence the image is formed **beyond 2f**, and it is real, inverted and magnified.

OR (a) A large magnification in an astronomical telescope is obtained when the objective focal length is very large compared with the eyepiece focal length, i.e. $f_o \gg f_e$, since $m = \frac{f_o}{f_e}$.

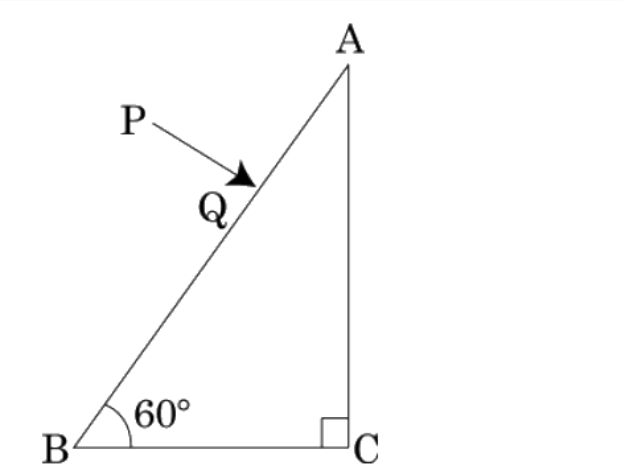
OR (b) A reflecting telescope is preferred over a refracting telescope because: (i) a curved mirror produces **no chromatic aberration** (mirrors do not split light into colours, unlike lenses), and spherical aberration is reduced by using a parabolic mirror; (ii) a large-diameter mirror can be made and supported from behind, giving a **larger aperture** (more light-gathering power and higher resolving power) and a brighter image than a heavy lens that can only be supported at its rim.

Final answer. The mirror formula gives v > 2f, so the image is real and beyond 2f; OR f_o ≫ f_e for large magnification, and reflectors avoid chromatic aberration and allow larger apertures.

← *Back to Q186*

Q187

A ray PQ *incident normally on the refracting face* BA *is refracted in the prism* BAC *made of material of refractive index* 1.5. *Complete the path of ray through the prism. From which face will the ray emerge? Justify your answer.*



[2016]

SOLUTION

The prism BAC is a right-angled prism with angle $B = 60°$ (and the right angle at C), made of glass of refractive index $n = 1.5$.

The ray PQ falls normally (perpendicularly) on the face BA, so it enters the prism without bending and travels straight inside.

First find the **critical angle** for the glass-air boundary:

$$\theta_c = \frac{1}{n} = \frac{1}{1.5} = 0.667$$

$$\theta_c = \sin^{-1}(0.667) \approx 41.8°$$

The undeviated ray next strikes the face BC. From the geometry of the prism it meets BC at an angle of incidence equal to 60°.

Since $60° > \theta_c$ ($\approx 42°$), the ray undergoes **total internal reflection** at face BC.

After TIR it travels to the third face AC, where the geometry gives an angle of incidence of 30°.

Since $30° < \theta_c$ ($\approx 42°$), total internal reflection does not occur at AC; the ray is refracted and **emerges from face AC**.

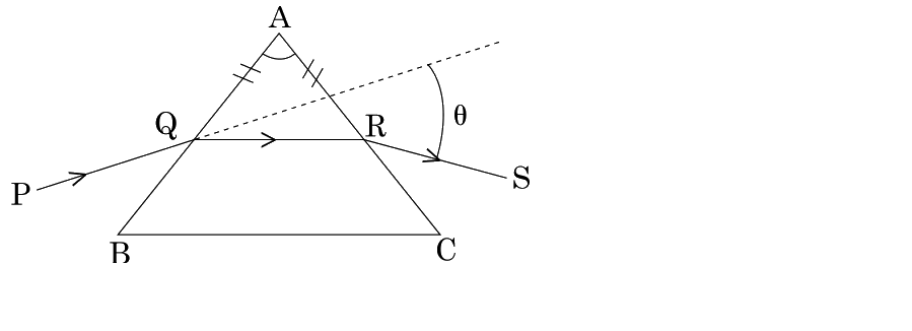
Ray diagram (in words): PQ enters straight through BA, goes to BC where it is totally internally reflected (angle of incidence $60° > \theta_c$), then reaches AC where it strikes at $30° < \theta_c$ and refracts out as the emergent ray. **The ray emerges from face AC after total internal reflection at face BC.**

Final answer. The ray totally internally reflects at BC ($60° > \theta c \approx 42°$) and emerges from face AC ($30° < \theta c$).

← *Back to Q187*

Q188

A ray PQ *incident on the refracting face* BA *is refracted in the prism* BAC *as shown in the figure and emerges from the other refracting face* AC *as* RS *such that* **AQ=AR**. *If the angle of prism* **A=60°** *and refractive index of material of prism is* $\sqrt{3}$, *calculate angle* θ.



[2016]

SOLUTION

Given an equilateral prism with $A = 60°$ and refractive index $n = \sqrt{3}$. The condition **AQ = AR** means the path inside the prism is symmetric, so the ray passes in the position of **minimum deviation**, where the angle of incidence equals the angle of emergence ($i_1 = i_2 = i$) and the ray inside runs parallel to the base.

Using the prism formula at minimum deviation:

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sqrt{3} = \frac{\sin(\frac{60^\circ + \delta_m}{2})}{\sin 30^\circ}$$

$$\sqrt{3} \times \sin 30^\circ = \sin(\frac{60^\circ + \delta_m}{2})$$

$$\sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2} = \sin(\frac{60^\circ + \delta_m}{2})$$

$$\frac{60^\circ + \delta_m}{2} = 60^\circ \Rightarrow 60^\circ + \delta_m = 120^\circ \Rightarrow \delta_m = 60^\circ$$

The angle of incidence (and emergence) follows from $\boldsymbol{\delta_m = 2i - A}$:

$$60^\circ = 2i - 60^\circ \Rightarrow 2i = 120^\circ \Rightarrow i = 60^\circ$$

The required angle θ (the angle of incidence/emergence at the face) is therefore $\theta = 60^\circ$, with **minimum deviation $\delta_m = 60^\circ$**.

Final answer. θ = 60 ° (angle of incidence equals emergence), with minimum deviation δm = 60 °.

← *Back to Q188*

Q189

A ray of light passes through an equilateral glass prism such that the angle of incidence is equal to the angle of emergence and each of these angles is equal to 3/4 *of angle of prism. Find the angle of deviation.*

[2015]

SOLUTION

For an equilateral prism the prism angle is $\boldsymbol{A = 60^\circ}$. The angle of incidence equals the angle of emergence $(\mathbf{i_1 = i_2})$, which is the symmetric condition, so the ray passes through the prism at **minimum deviation**.

Each of these angles is given as $\frac{3}{4}$ of the prism angle:

$$i = \frac{3}{4}A = \frac{3}{4} \times 60^\circ = 45^\circ$$

At minimum deviation the relation between the angles is:

$$\delta_m = 2i - A$$

$$\delta_m = 2 \times 45° - 60°$$

$$\delta_m = 90° - 60° = 30°$$

So the **angle of deviation is 30°**.

(As a check, the refractive index is $n = \frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})} = \sin 45° \sin 30° = \sqrt{2} \approx$ **1.414.**)

Final answer. The angle of deviation is 30 °.

← *Back to Q189*

Q190

Calculate the speed of light in a medium whose critical angle is 45°.
Does critical angle for a given pair of media depend on the wavelength of incident light? Give reason.

[2015]

SOLUTION

The refractive index of the medium is related to its critical angle by:

$$\theta_c = \frac{1}{n} \Rightarrow n = \frac{1}{\theta_c}$$

With $\theta_c = 45°$:

$$n = \frac{1}{\sin 45° = \frac{1}{1/\sqrt{2}} = \sqrt{2} \approx 1.414}$$

The speed of light in the medium is:

$$v = \frac{c}{n}$$

$$v = \frac{3 \times 10^8\ m/s}{\sqrt{2}}$$

$$v = \frac{3 \times 10^8}{1.414} \approx 2.12 \times 10^8\ m/s$$

The speed of light in the medium is about $2.12 \times 10^8\ m/s$.

Yes, the critical angle depends on the wavelength of the incident light. The refractive index n of a medium changes with wavelength (dispersion), and since $\theta_c = \sin^{-1}(\frac{1}{n})$, a change in n changes θ_c. Shorter wavelengths (violet/blue) have a higher refractive index, so they have a **smaller critical angle** than longer wavelengths (red).

> **TIP**
>
> Whenever a question gives the critical angle, immediately convert it to n using n = 1/sinθc, then everything else (speed, deviation, TIR checks) follows.

Final answer. v ≈ 2.12×10^8 m/s; yes, θc depends on wavelength because n varies with wavelength (smaller θc for shorter wavelengths).

← *Back to Q190*

Q191

Use the mirror equation to show that an object placed between f and 2f *of a concave mirror produces a real image beyond* 2f.

[2015]

SOLUTION

For a concave mirror, all distances are measured using the Cartesian sign convention, where distances are taken from the pole and the incoming-light direction is negative. Hence the focal length and the object distance are both negative.

Since the object lies between the focus and the centre of curvature, we have $f < u < 2f$ in magnitude terms, i.e. $|f| < |u| < 2|f|$.

The mirror equation is

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

Rearranging to isolate the image distance,

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

Working in magnitudes (with f and u both negative the signs are consistent), this becomes

$$\frac{1}{|v|}=\frac{1}{|f|}-\frac{1}{|u|}$$

From the given range $|f| < |u| < 2|f|$, taking reciprocals reverses the inequalities:

$$\frac{1}{2|f|} < \frac{1}{|u|} < \frac{1}{|f|}$$

Therefore $\frac{1}{|v|}=\frac{1}{|f|}-\frac{1}{|u|}$ is a positive quantity (so a real image forms in front of the mirror), and using the upper bound $\frac{1}{|u|} < \frac{1}{|f|}$ gives $\frac{1}{|v|} > 0$, while using the lower bound $\frac{1}{|u|} > \frac{1}{2|f|}$ gives

$$\frac{1}{|v|}=\frac{1}{|f|}-\frac{1}{|u|} < \frac{1}{|f|}-\frac{1}{2|f|}=\frac{1}{2|f|}$$

Taking reciprocals of $\frac{1}{|v|} < \frac{1}{2|f|}$ gives

$$|v| > 2|f|$$

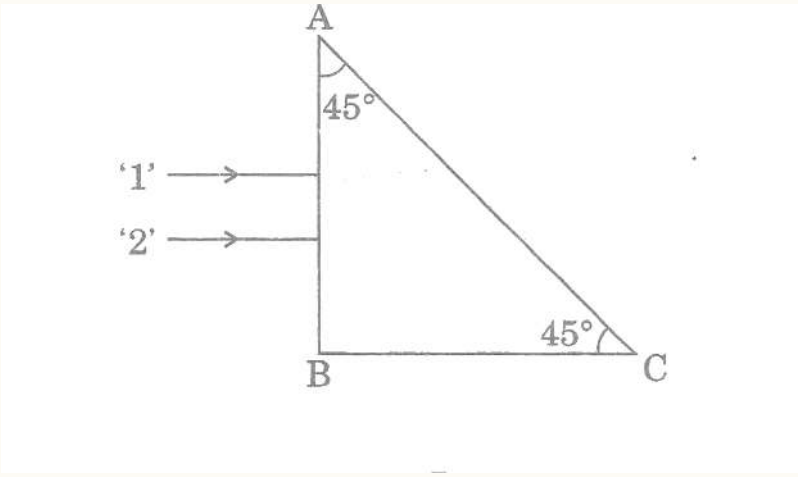
Since $\frac{1}{v}$ comes out positive, v is negative, confirming the image is **real** and formed in front of the mirror. Its distance exceeds $2|f|$, so the image lies **beyond 2f**, which is what was required.

Final answer. The image is real and forms beyond **2f**, since $|v|>2|f|$.

← Back to Q191

Q192

Two monochromatic rays of light are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays '1' and '2' are respectively 1.38 and 1.52. Trace the path of these rays after entering through the prism.



[2014]

SOLUTION

The prism is a 45°-45°-90° isosceles right-angled prism. Both rays strike face AB normally, so they enter without bending and travel straight up to the hypotenuse AC, where each meets the surface at an angle of incidence of 45° measured from the normal.

At the glass-air interface, total internal reflection occurs only if the angle of incidence exceeds the critical angle θ_c, where

$$\theta_c = \frac{1}{n}$$

For TIR at 45° we need $\mathbf{45° > \theta_c}$, i.e. $\mathbf{\sin 45° > \theta_c}$, which gives

$$\frac{1}{2} > \frac{1}{n} \Rightarrow n > 2 \approx 1.414$$

For ray 1, $\mathbf{n_1 = 1.38}$. The critical angle is $\mathbf{\theta_c = 1/1.38 = 0.725}$, so $\mathbf{\theta_c \approx 46.4°}$. Since the angle of incidence at the hypotenuse (45°) is less than θ_c, total internal

reflection does not occur. Ray 1 is refracted and emerges out through the hypotenuse face AC (bending away from the normal).

For ray 2, $\mathbf{n_2 = 1.52 > 2}$. Here $\mathbf{\theta_c = 1/1.52 = 0.658}$, giving $\mathbf{\theta_c \approx 41.1^\circ}$. As the angle of incidence (45°) exceeds this critical angle, ray 2 undergoes **total internal reflection** at the hypotenuse. It turns through 90° and then strikes the face BC normally, emerging straight out of BC.

Ray diagram description: Draw the right-angled triangle with the right angle at B, AB vertical (left face), BC horizontal (bottom face) and AC the hypotenuse sloping from top-left to bottom-right. Both rays enter horizontally through AB. Ray 1 continues to AC and bends out of AC into the air. Ray 2 reflects off AC, turns downward through 90°, and exits vertically downward through face BC.

Final answer. Ray 1 (n=1.38) refracts out through the hypotenuse; ray 2 (n=1.52) undergoes total internal reflection and exits normally through face BC.

← *Back to Q192*

Q193

A convex lens of focal length 25 cm *is placed coaxially in contact with a concave lens of focal length* 20 cm. *Determine the power of the combination. Will the system be converging or diverging in nature?*

[2013]

SOLUTION

The power of a single thin lens is the reciprocal of its focal length in metres, $P = \frac{1}{f}$, and is measured in dioptres (D). When two thin lenses are placed coaxially in contact, the power of the combination equals the algebraic sum of their individual powers:

$$P = P_1 + P_2$$

For the convex lens, $\mathbf{f_1 = +25\ cm = +0.25\ m}$, so

$$P_1 = \frac{1}{f_1} = \frac{1}{+0.25} = +4\ D$$

For the concave lens, $\mathbf{f_2 = -20\ cm = -0.20\ m}$ (concave lens focal length is negative), so

$$P_2 = \frac{1}{f_2} = \frac{1}{-0.20} = -5\ D$$

The power of the combination is therefore

$$P = P_1 + P_2 = (+4) + (-5) = -1\ D$$

The equivalent focal length is

$$f = \frac{1}{P} = \frac{1}{-1} = -1\ m = -100\ \text{cm}$$

Since the net power is **negative**, the equivalent focal length is negative, so the combination behaves as a **diverging** system. The concave lens of larger power dominates over the convex lens.

Final answer. The power of the combination is **−1 *D*** (equivalent focal length **−100 cm**); the system is diverging in nature.

← *Back to Q193*

Q194

(a) Write the necessary conditions for the phenomenon of total internal reflection to occur.
(b) Write the relation between the refractive index and critical angle for a given pair of optical media.

[2013]

SOLUTION

(a) Necessary conditions for total internal reflection:

First, light must travel from an optically denser medium to an optically rarer medium, that is, from a medium of higher refractive index towards a medium of lower refractive index.

Second, the angle of incidence in the denser medium must be greater than the critical angle for that pair of media.

(b) Relation between refractive index and critical angle:

The critical angle θ_c is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90°, so the refracted ray grazes along the interface. Applying Snell's law at this condition for light going from a denser medium of refractive index n_1 to a rarer medium of refractive index n_2,

$$n_1\theta_c = n_2 \sin 90^\circ$$

$$n_1\theta_c = n_2$$

$$\theta_c = \frac{n_2}{n_1}$$

When the denser medium is glass of refractive index n and the rarer medium is air ($\mathbf{n_2 = 1}$), this reduces to

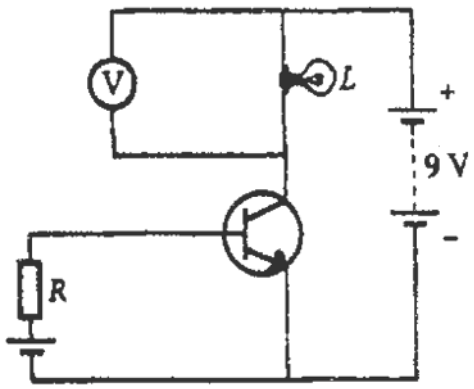
$$\theta_c = \frac{1}{n}$$

Final answer. TIR needs light going from denser to rarer medium with incidence greater than the critical angle; $\boldsymbol{\theta_c = n_2/n_1}$, and for glass-air $\boldsymbol{\theta_c = 1/n}$.

← *Back to Q194*

Q195

An object AB is kept in front of a concave mirror as shown in the figure.
(i) Complete the ray diagram showing the image formation of the object.
(ii) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black?



[2012]

SOLUTION

(i) Ray diagram for image formation:

The object AB stands upright on the principal axis between the centre of curvature *C* and the focus *F* of the concave mirror, with A on the axis and B at the top. From the tip B, draw two standard rays. The first ray travels parallel to the principal axis, strikes the mirror, and after reflection passes through the focus *F*. The second ray is drawn from B through the centre of curvature *C*; it strikes the mirror normally and retraces its own path back. These two reflected rays meet at a point *B*' below the axis. Drop a perpendicular from *B*' to the axis to locate *A*'. The image *A*'*B*' is formed beyond *C*.

The image so formed is **real, inverted and magnified**, located beyond the centre of curvature *C*.

(ii) Effect of painting the lower half of the mirror black:

Position of the image: The position is **unchanged**. The image is formed by rays reflected from the mirror, and the geometry that fixes the image location is the same whether the whole mirror or only its upper half reflects light. The unpainted upper half still produces a complete image at the same place.

Intensity of the image: The intensity is **reduced (roughly halved)**, because now only half the reflecting area of the mirror collects and reflects light. Fewer rays reach the image, so the same complete image is formed but it appears dimmer.

COMMON MISTAKE

A common mistake is to say half the image disappears. The whole image still forms because every part of the mirror can contribute rays to every part of the image; only the brightness drops.

Final answer. The image is real, inverted and magnified beyond C; painting the lower half black keeps its position unchanged but reduces its intensity to about half.

← *Back to Q195*

Q196

A ray of light, incident on an equilateral glass prism ($\mu_g = \sqrt{3}$) moves parallel to the base line of the prism inside it. Find the angle of incidence for this ray.

[2012]

SOLUTION

For an equilateral prism the refracting angle is $A = 60°$, and here $\mu = 3$. When the refracted ray inside the prism travels parallel to the base, the ray passes symmetrically through the prism. This is the condition of **minimum deviation**, where the angles of refraction at the two faces are equal and each equals $A/2$.

At minimum deviation the refractive index is given by the prism formula

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Substituting $A = 60°$ and $\mu = 3$,

$$3 = \frac{\sin\left(\frac{60°+\delta_m}{2}\right)}{\sin 30°}$$

Since $\sin 30° = \frac{1}{2}$,

$$\sin\left(\frac{60°+\delta_m}{2}\right) = 3 \times \frac{1}{2} = \frac{3}{2}$$

$$\frac{60°+\delta_m}{2} = 60°$$

$$\delta_m = 60°$$

At minimum deviation the angle of incidence is

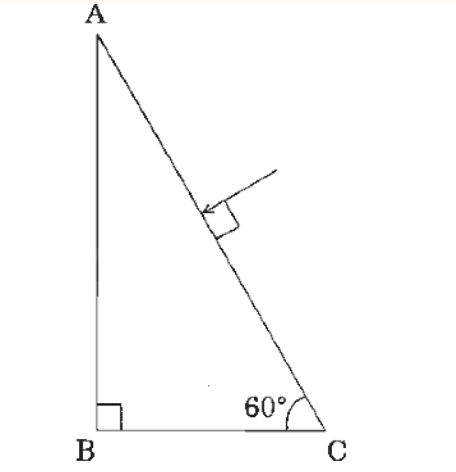
$$i = \frac{A+\delta_m}{2} = 60°+60°2 = 60°$$

Final answer. The angle of incidence is **60°**.

← *Back to Q196*

Q197

Trace the path of a ray of light passing through a glass prism (ABC) as shown in the figure. If the refractive index of glass is $\sqrt{3}$, find out the value of the angle of emergence from the prism.



[2012]

SOLUTION

The prism is equilateral, so its refracting angle is $A = 60°$, with $n = 3$. The ray is incident on face AB at an angle of incidence $i = 60°$ (as shown in the figure).

Applying Snell's law at the first face AB, where the ray goes from air into glass,

$$\sin i = n\sin r_1$$

$$\sin 60° = 3 \sin r_1$$

$$\sin r_1 = \sin 60°3 = 3/23 = 12$$

$$r_1 = 30°$$

For a prism the two refraction angles satisfy $\mathbf{r_1 + r_2 = A}$, so

$$r_2 = A - r_1 = 60° - 30° = 30°$$

Applying Snell's law at the second face AC, where the ray goes from glass into air, with angle of emergence e,

$$n \sin r_2 = \sin e$$

$$3 \sin 30° = \sin e$$

$$\sin e = 3 \times \frac{1}{2} = \frac{3}{2}$$

$$e = 60°$$

Path of the ray: The ray enters at AB, bending towards the normal to travel inside the glass, crosses to face AC making 30° with its normal, and emerges into air bending away from the normal at 60°. Since $\boldsymbol{i = e = 60°}$, the ray passes symmetrically (the minimum-deviation path) and the refracted ray inside is parallel to the base BC.

Final answer. The angle of emergence is $\boldsymbol{e = 60°}$.

← *Back to Q197*

Q198

Two convex lenses of same focal length but of aperture A_1 *and* A_2 $\mathbf{(A_1 < A_2)}$*, are used as the objective lenses in two astronomical telescopes having identical eyepieces. What is the ratio of their resolving power? Which telescope will you prefer and why? Give reason.*

[2011 • Set 55-1-1]

SOLUTION

The resolving power of a telescope is its ability to distinguish two close objects as separate. It is given by

$$\text{RP} = \frac{D}{1.22\,\lambda}$$

where D is the diameter (aperture) of the objective lens and λ is the wavelength of light used. Thus the resolving power is directly proportional to the aperture, $\mathbf{RP \propto D}$.

For the two telescopes with apertures A_1 and A_2 but the same focal length, taking the ratio at the same wavelength,

$$\frac{RP_1}{RP_2} = \frac{A_1}{A_2}$$

Since $\mathbf{A_1 < A_2}$, it follows that $\mathbf{RP_1 < RP_2}$, so the telescope with aperture A_2 has the greater resolving power.

The telescope of aperture A_2 is **preferable**. A larger aperture gives higher resolving power, so finer details of distant objects can be distinguished. A larger aperture also gathers more light, producing brighter and clearer images of faint objects.

Final answer. The telescope of larger aperture $\mathbf{A_2}$ is preferable, as it has higher resolving power and greater light-gathering ability.

← *Back to Q198*

Q199

The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. If focal length of the lens is 12 cm, find the refractive index of the material of the lens.

[2010]

SOLUTION

For a thin lens the focal length is related to the refractive index of the material and the radii of curvature by the lensmaker's formula

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a double convex lens, using the sign convention, the first surface is convex towards the incoming light so $\mathbf{R_1 = +10\ cm}$, and the second surface gives $\mathbf{R_2 = -15\ cm}$.

Substituting these radii,

$$\frac{1}{f} = (n-1)\left(\frac{1}{+10} - \frac{1}{-15}\right)$$

$$\frac{1}{f} = (n-1)\left(\frac{1}{10} + \frac{1}{15}\right)$$

$$\frac{1}{f} = (n-1)\left(\frac{3+2}{30}\right) = (n-1)\frac{5}{30} = (n-1)\frac{1}{6}$$

Given $f = 12$ **cm**,

$$\frac{1}{12} = \frac{n-1}{6}$$

$$n-1 = \frac{6}{12} = 0.5$$

$$n = 1.5$$

This value of $n = 1.5$ is typical of crown glass.

Final answer. The refractive index of the material of the lens is $n = 1.5$.

← *Back to Q199*

Q200

(a) The bluish colour predominates in clear sky.
(b) Violet colour is seen at the bottom of the spectrum when white light is dispersed by a prism.
State reasons to explain these observations.

[2010]

SOLUTION

(a) Bluish colour of the clear sky:

As sunlight passes through the atmosphere, it is scattered by the air molecules. This is **Rayleigh scattering**, for which the intensity of scattered light is inversely proportional to the fourth power of the wavelength:

$$I \propto \frac{1}{\lambda^4}$$

Because of this strong dependence, light of shorter wavelength (the blue end of the spectrum) is scattered much more strongly than light of longer wavelength (the red end). The sky therefore looks predominantly **blue**, as it is the scattered short-wavelength light that reaches our eyes from all directions. Although violet is scattered even more than blue, sunlight contains less violet and the human eye is more sensitive to blue, so the sky appears blue rather than violet.

(b) Violet seen at the bottom of the prism spectrum:

When white light passes through a prism, dispersion occurs because the refractive index of glass depends on the wavelength, being greatest for violet and least for red, $\mathbf{n_{violet} > n_{red}}$. A ray that experiences a larger refractive index is bent more, so violet light suffers the **greatest deviation** and red the least.

Since each colour is bent towards the base of the prism and violet is deviated the most towards the base, violet light appears at the lower end (the bottom) of the spectrum on the screen, with red at the top.

Final answer. The sky is blue because Rayleigh scattering ($\boldsymbol{I \propto 1/\lambda^4}$) scatters short blue wavelengths most; violet appears at the bottom of the prism spectrum because its larger refractive index makes it deviate the most.

← Back to Q200

Q201

Define refractive index of a transparent medium.
A ray of light passes through a triangular prism. Plot a graph showing the variation of the angle of deviation with the angle of incidence.

[2009]

SOLUTION

The **refractive index** of a transparent medium is defined as the ratio of the speed of light in vacuum (or air) to the speed of light in that medium.

$$n = \frac{c}{v}$$

where c is the speed of light in vacuum and v is the speed of light in the medium. Since $\boldsymbol{v < c}$ for any real medium, n is always greater than 1 and has no unit.

Equivalently, by Snell's law at an interface from air into the medium, $\boldsymbol{n = \frac{\sin i}{\sin r}}$, where i is the angle of incidence and r the angle of refraction.

For a ray passing through a triangular prism, the angle of deviation δ depends on the angle of incidence i. As i is increased from a small value, the deviation δ first decreases, reaches a single minimum value δ_m called the **angle of minimum deviation**, and then increases again.

At minimum deviation the ray passes symmetrically through the prism, so the angle of incidence equals the angle of emergence ($\boldsymbol{i = e}$) and the refracted ray inside the prism is parallel to its base.

Graph description: plot the angle of incidence i on the x-axis and the angle of deviation δ on the y-axis. The curve is U-shaped (a smooth dip). It starts high at small i, falls to a flat lowest point δ_m at the symmetric position $\boldsymbol{i = e}$, then rises again as i increases further. For any deviation greater than δ_m there are two values of i (giving the same δ), but at $\boldsymbol{\delta = \delta_m}$ there is only one value of i.

Final answer. Refractive index $\boldsymbol{n = c/v}$; the $\boldsymbol{\delta}$ versus $\boldsymbol{i}$ graph is U-shaped with a single minimum $\boldsymbol{\delta_m}$ at the symmetric position $\boldsymbol{i = e}$.

← *Back to Q201*

Q202

Draw a ray diagram of a reflecting telescope. State the advantages of this telescope over a refracting telescope.

[2008]

SOLUTION

A reflecting telescope (Cassegrain type) uses a large concave parabolic mirror as the objective instead of a lens.

Ray diagram description: draw a wide curved concave parabolic mirror (the **primary mirror**) at the back of the tube, with its reflecting surface facing the incoming light. Show two or three parallel rays coming in from a distant object and striking the primary mirror. These rays reflect and converge toward the principal focus F of the primary mirror.

Before reaching F, the converging beam meets a small **convex secondary mirror** placed on the axis near the front of the tube. Mark this small mirror facing the primary. The secondary mirror reflects the beam back along the axis, sending it through a small hole drilled at the centre of the primary mirror.

Behind the hole, place the **eyepiece** (a convex lens). The eyepiece receives the converging beam and forms a final magnified image. In normal adjustment the final image is at infinity and is viewed by the relaxed eye. Label: primary mirror (concave, parabolic), secondary mirror (convex), focus F, eyepiece, eye.

Advantages over a refracting telescope:

1. No chromatic aberration, because a mirror reflects all wavelengths of light identically, whereas a lens refracts different colours by different amounts.

2. Spherical aberration is greatly reduced by using a parabolic mirror, which brings all parallel rays to a single sharp focus.

3. A large mirror can be supported firmly from behind, so very large apertures are possible. A large lens can only be held at its edges and sags under its own weight.

4. It is lighter, cheaper and easier to make, since only one surface of the mirror needs to be polished accurately, and it gives a brighter image (greater light-gathering power).

Final answer. The reflecting (Cassegrain) telescope uses a concave parabolic primary plus convex secondary mirror and eyepiece, with no chromatic aberration, low spherical aberration and a feasibly large aperture.

← Back to Q202

Q203

A ray of light passing through an equilateral triangular glass prism from air undergoes minimum deviation when angle of incidence is $\frac{3}{4}$ *of the angle of prism. Calculate the speed of light in the prism.*

[2008]

SOLUTION

For an equilateral glass prism the refracting angle is $A = 60°$. At minimum deviation the angle of incidence equals the angle of emergence.

The angle of incidence is given as three-quarters of the prism angle:

$$i = \frac{3}{4}\mathrm{A} = \frac{3}{4} \times 60° = 45°$$

At minimum deviation, $\delta_m = i + e - A = 2i - A$ (since $i = e$).

$$\delta_m = 2 \times 45° - 60° = 90° - 60° = 30°$$

Using the prism formula for refractive index:

$$n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$$

$$n = \sin(60° + 30°2)\sin(60°2) = \sin 45° \sin 30°$$

$$n = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414$$

The speed of light in the prism is found from $n = \frac{c}{v}$, so $v = \frac{c}{n}$.

$$v = \frac{3 \times 10^8}{\sqrt{2}} = \frac{3 \times 10^8}{1.414}$$

$$v \approx 2.12 \times 10^8 \ m\ s^{-1}$$

Hence the **speed of light in the prism is about $2.12 \times 10^8\ m\ s^{-1}$**.

COMMON MISTAKE

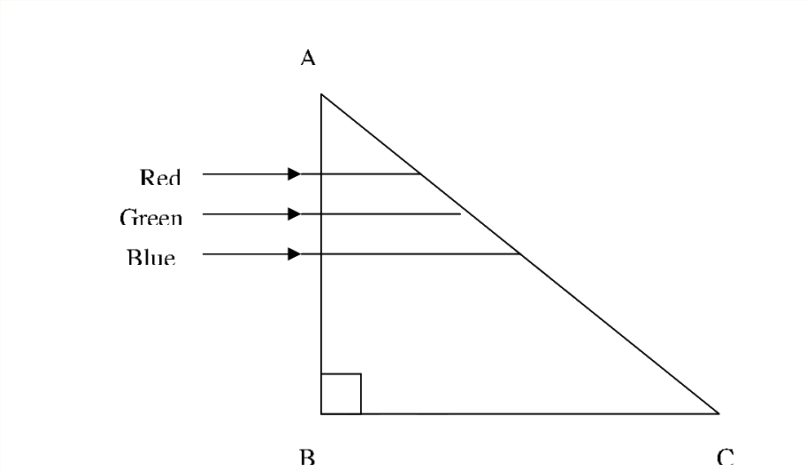
Watch the order: first get $\delta_m = 2i - A$*, then n from the prism formula, and only then* $v = c/n$.

Final answer. The speed of light in the prism is $v \approx 2.12 \times 10^8\ m\ s^{-1}$.

← *Back to Q203*

Q204

In the figure given below, light rays of blue, green, red wavelengths are incident on an isosceles right-angled prism. Explain with reason, which ray of light will be transmitted through the face AC. The refractive index of the prism for red, green, blue light are 1.39, 1.424, *and* 1.476 *respectively.*



[2008]

SOLUTION

The prism is an isosceles right-angled prism, so light striking the hypotenuse AC after entering through the vertical face does so at an angle of incidence of 45° inside the glass.

Total internal reflection at this hypotenuse face occurs only when the angle of incidence 45° is greater than the critical angle θ_c for that colour.

The critical angle is given by $\theta_c = \frac{1}{n}$.

The condition for total internal reflection is $45° > \theta_c$, which means $\sin 45° > \theta_c$, i.e.

$$\frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2} \approx 1.414$$

So a colour undergoes total internal reflection (and is NOT transmitted through AC) if its refractive index exceeds 1.414. A colour with $n < 1.414$ does not undergo TIR and is transmitted (refracted out) through AC.

Checking each colour:

Red: $n = 1.39 < 1.414$, so no total internal reflection. Red light is refracted and transmitted through face AC.

Green: $n = 1.424 > 1.414$, so total internal reflection occurs. Green is reflected, not transmitted.

Blue: $n = 1.476 > 1.414$, so total internal reflection occurs. Blue is reflected, not transmitted.

Therefore **only the red ray is transmitted through face AC**, because its refractive index is the only one for which the critical angle exceeds 45°.

Final answer. Only the red ray is transmitted through face AC, since for red $n = 1.39 < \sqrt{2}$ so it does not undergo total internal reflection.

← *Back to Q204*

Q205

A convex lens of refractive index 1.5 has a focal length of 18 cm in air. Calculate the change in its focal length when it is immersed in water of refractive index 4/3.

[2007]

SOLUTION

By the lens-maker's formula, $\frac{1}{f} = (n_{rel} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$, where n_{rel} is the refractive index of the lens material relative to the surrounding medium.

The radii of curvature R_1 and R_2 are fixed by the shape of the lens, so the bracket $\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ is the same in air and in water.

In air, $\mathbf{n_{rel}} = \frac{\mathbf{1.5}}{\mathbf{1}} = \mathbf{1.5}$, so

$$\frac{1}{f_{air}} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0.5\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

In water, $\mathbf{n_{rel}} = \frac{\mathbf{1.5}}{\mathbf{4/3}} = \frac{\mathbf{1.5 \times 3}}{\mathbf{4}} = \frac{\mathbf{9}}{\mathbf{8}}$, so

$$\frac{1}{f_{water}} = \left(\frac{9}{8} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{8}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Dividing the two relations, the common bracket cancels:

$$\frac{f_{water}}{f_{air}} = \frac{0.5}{1/8} = 0.5 \times 8 = 4$$

$$f_{water} = 4 \times f_{air} = 4 \times 18 = 72 \text{ cm}$$

The change in focal length is

$$\Delta f = f_{water} - f_{air} = 72 - 18 = 54 \text{ cm}$$

Hence the focal length **increases by 54 cm** (from 18 cm to 72 cm) when the lens is immersed in water, because the relative refractive index falls and the lens becomes weaker.

Final answer. In water $\mathbf{f_{water} = 72}$ **cm**, so the focal length increases by $\mathbf{\Delta f = 54}$ **cm**.

← *Back to Q205*

Q206

Define resolving power of a compound microscope. How does the resolving power of a compound microscope change when
(i) refractive index of the medium between the object and objective lens increases?
(ii) wavelength of the radiation used is increased?

[2007]

SOLUTION

The **resolving power of a compound microscope** is defined as the reciprocal of the least distance between two close points on the object that the microscope can just show as separate (distinct).

$$R.P. = \frac{1}{d_{min}} = \frac{2\,n\theta}{1.22\,\lambda}$$

where λ is the wavelength of light used, n is the refractive index of the medium between the object and the objective lens, and θ is the half-angle of the cone of light entering the objective. The quantity $n\theta$ is called the **numerical aperture** of the objective.

(i) When the refractive index n of the medium between the object and the objective is increased: since $\boldsymbol{R.P. \propto n}$, the resolving power increases.

This is why an oil-immersion objective is used; a drop of oil of high refractive index between the slide and the objective raises n and gives a higher resolving power.

(ii) When the wavelength λ of the radiation used is increased: since $\boldsymbol{R.P. \propto \frac{1}{\lambda}}$, the resolving power decreases.

A shorter wavelength (for example ultraviolet light) therefore gives a higher resolving power and a sharper, more detailed image.

Final answer. Resolving power = $\frac{2n\theta}{1.22\lambda}$; it increases when n increases and decreases when λ increases.

← *Back to Q206*

Q207

Draw a labelled ray diagram of a reflecting type telescope. Write its any one advantage over refracting type telescope.

[2006]

SOLUTION

A reflecting type telescope (Cassegrain) uses a large concave parabolic mirror as the objective.

Labelled ray diagram description: draw a wide concave parabolic mirror at the rear of the tube, the **primary (objective) mirror**, with its reflecting face towards the incoming light. Show two parallel rays from a distant object striking this mirror. After reflection they converge towards the focus of the primary mirror.

On the axis, near the front, place a small **convex secondary mirror** facing the primary. The converging beam is intercepted by this secondary mirror, which reflects it back along the axis, passing through a small hole at the centre of the primary mirror.

Behind the hole place the **eyepiece** (a convex lens) and the **eye**. Mark the light path with arrows: parallel rays from object → primary mirror (concave, parabolic) → secondary mirror (convex) → through hole → eyepiece → eye. In normal adjustment the final image is formed at infinity and viewed by the relaxed eye.

One advantage over a refracting telescope: a reflecting telescope is **free from chromatic aberration**, because a mirror reflects all wavelengths (colours) of light equally, whereas the lens of a refracting telescope bends different colours by different amounts and produces coloured fringes.

Final answer. The reflecting telescope uses a concave parabolic primary and convex secondary mirror with an eyepiece, and its key advantage is freedom from chromatic aberration.

← *Back to Q207*

Q208

The image of a candle is formed by a convex lens on a screen. The lower half of the lens is painted black to make it completely opaque. Draw the ray diagram to show the image formation. How will this image be different from the one obtained when the lens is not painted black?

[2005]

SOLUTION

Ray diagram description: draw a vertical convex lens with its principal axis horizontal. Place the candle (object) to the left, beyond the focal point, standing upright on the axis. Shade the lower half of the lens to show it is painted black (opaque); only the upper half transmits light. From the tip of the candle flame draw at least two rays that pass only through the upticked (unpainted) upper half of the lens. One ray parallel to the axis bends through the focus on the far side; a second ray through the optical centre goes straight. These two rays meet on the other side of the lens to locate the inverted image tip. Similarly rays from the base of the candle (which also use only the upper half) converge to form the base of the image on the screen.

The key point is that every point of the object sends rays to the whole lens, and each unblocked portion of the lens is able, by itself, to form a complete image of the object. So even though the lower half is opaque, the upper half still receives light from every point of the candle and forms a full, complete image of the entire candle.

Effect of painting the lower half black: the image is of the **same size, at the same position, and is complete (the full candle, not half)**, because the geometry of refraction is unchanged.

The only difference is in brightness. Now only the upper half of the lens lets light through, so only about half the original amount of light reaches the screen. Therefore the image becomes **less bright (dimmer)**, roughly half the intensity, compared with the image formed by the fully open lens.

CONCEPT

A blocked half-lens does not chop off half the image; it only reduces brightness, since every point still sends rays through the open half.

Final answer. A complete, same-size image forms at the same position, but it is dimmer (about half the brightness) because only half the lens transmits light.

← *Back to Q208*

Q209

A compound microscope with an objective of 1.0 cm focal length and an eye-piece of 2.0 cm focal length has a tube length of 20 cm. Calculate the magnifying power of the microscope, if the final image is formed at the near point of the eye.

Or

The magnifying power of an astronomical telescope in the normal adjustment position is 100. The distance between the objective and the eye-piece is 101 cm. Calculate the focal lengths of the objective and of the eye-piece.

[2004]

SOLUTION

Compound microscope: objective focal length $\mathbf{f_o = 1.0\ cm}$, eyepiece focal length $\mathbf{f_e = 2.0\ cm}$, tube length $\boldsymbol{L = 20\ cm}$, and the final image is at the near point $\boldsymbol{D = 25\ cm}$.

The magnifying power of a compound microscope (final image at the near point) is

$$M = m_o \times m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

The linear magnification of the objective is

$$m_o = \frac{L}{f_o} = \frac{20}{1.0} = 20$$

The angular magnification of the eyepiece (image at near point) is

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{2.0} = 1 + 12.5 = 13.5$$

Therefore the magnifying power is

$$M = 20 \times 13.5 = 270$$

Hence the **magnifying power of the microscope is 270**.

OR (Astronomical telescope): magnifying power in normal adjustment $\boldsymbol{m = 100}$ and the length $\boldsymbol{L = f_o + f_e = 101\ cm}$.

In normal adjustment, $\boldsymbol{m = \frac{f_o}{f_e} = 100}$, which gives $\mathbf{f_o = 100\ f_e}$.

Substituting in the length equation:

$$f_o + f_e = 101 \Rightarrow 100\, f_e + f_e = 101$$

$$101\, f_e = 101 \Rightarrow f_e = 1 \text{ cm}$$

$$f_o = 100 \times 1 = 100 \text{ cm}$$

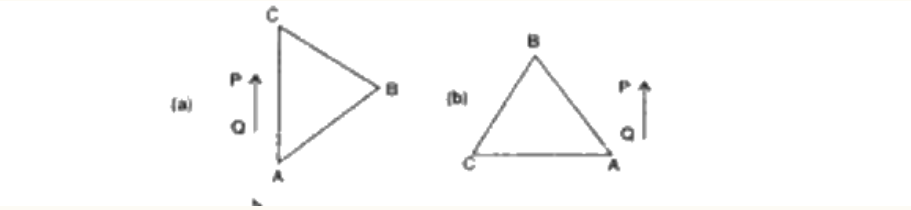
Hence the **objective has focal length 100 cm and the eyepiece 1 cm**.

Final answer. Microscope: $M = 270$. OR Telescope: $f_o = 100$ cm and $f_e = 1$ cm.

← *Back to Q209*

Q210

An object is placed in front of a right angled prism ABC in two positions (a) and (b) as shown. The prism is made of crown glass with critical angle of 41°. *Trace the path of two rays from P and Q, (i) in (a), normal to the hypotenuse and (ii) in (b), parallel to the hypotenuse.*



[2003]

SOLUTION

The prism is a right-angled (totally reflecting) prism with angles 90°, 45°, 45°, and the critical angle of the crown glass is $\theta_c = 41°$. Total internal reflection takes place whenever a ray strikes a glass-air face at an angle of incidence greater than 41°.

(a) Ray normal to the hypotenuse: the ray PQ enters perpendicular (normal) to the hypotenuse face, so it goes straight in without bending. Inside the prism it strikes one of the two perpendicular faces at an angle of incidence of 45°.

Since **45°> 41°**, total internal reflection occurs at this face. The ray is turned through 90° and travels to the second perpendicular face, where it again meets

the surface at 45°.

Again **45°> 41°**, so a second total internal reflection occurs. The ray is turned through another 90° and finally emerges back out through the hypotenuse, travelling parallel but opposite to the incident ray (turned through a total of 180°). This is the retro-reflecting action used to send light back the way it came.

(b) Ray parallel to the hypotenuse: this ray enters normally through one of the perpendicular faces, so it passes straight in without bending. Inside the prism it strikes the hypotenuse at an angle of incidence of 45°.

Since **45°> 41°**, total internal reflection occurs at the hypotenuse. The ray is turned through 90° and emerges normally through the other perpendicular face.

Thus the ray is **deviated through 90°**. This 90° deviation by a single total internal reflection is the action used in a periscope.

In both cases the bending is by total internal reflection (not silvering), because the angle of incidence inside the glass (45°) exceeds the critical angle (41°). Such prisms are used in binoculars and periscopes.

Final answer. In (a) the ray suffers two total internal reflections and returns through **180°**; in (b) it suffers one total internal reflection at the hypotenuse and is deviated by **90°**.

← *Back to Q210*

Q211

(a) *Draw the ray diagram to show the image formation by a refracting telescope and write the expression for angular magnification for the telescope in normal adjustment.*
(b) *Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope.*

[2026 • Set 55-1-1]

SOLUTION

(a) Ray diagram for a refracting telescope in normal adjustment: A long-focal-length convex lens O (the objective, focal length f_o, large aperture) is drawn on the left and a short-focal-length convex lens E (the eyepiece, focal length f_e) on the right, mounted on the same axis. The objective and eyepiece are separated so that the second focal point of the objective coincides with the first focal point of the eyepiece at a common point $\mathbf{F_o = F_e}$.

A parallel beam coming from a distant object enters the objective at a small angle α to the axis. The objective refracts this beam to form a real, inverted, diminished image I_1 (height h) at the common focal plane F_o.

Since I_1 lies at the focus of the eyepiece, the eyepiece sends out a parallel emergent beam, so the final image is virtual, inverted and at infinity, subtending angle β at the eye. Label the rays so that α is the angle the incoming rays make at the objective and β is the angle the emergent rays make at the eye.

The angular magnification (magnifying power) is the ratio of the angle subtended at the eye by the final image to that subtended by the object.

$$m = \frac{\beta}{\alpha}$$

From the geometry, $\alpha \approx \alpha = \frac{h}{f_o}$ and $\beta \approx \beta = \frac{h}{f_e}$.

$$m = \frac{\beta}{\alpha} = \frac{h/f_e}{h/f_o} = \frac{f_o}{f_e}$$

So the magnifying power in normal adjustment is $m = \frac{f_o}{f_e}$, which is **large when the objective has a long focal length and the eyepiece a short focal length**.

(b) Two reasons a reflecting telescope is preferred over a refracting telescope:

(i) A concave mirror has no **chromatic aberration**, because reflection bends all wavelengths by the same amount, whereas a lens refracts different colours by different amounts and so produces coloured fringes around the image.

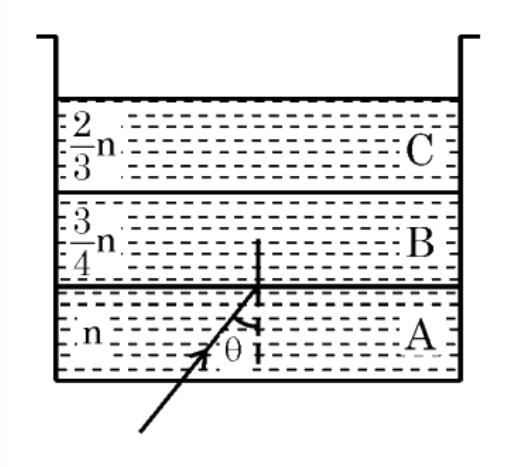
(ii) A large mirror can be supported firmly from behind over its whole back surface, so very **large apertures** are possible. This gives much greater light-gathering power and higher resolution, while a large lens sags under its own weight and can only be supported at its rim.

Final answer. In normal adjustment the magnifying power is $m = f_o/f_e$; reflecting telescopes are preferred because mirrors are free of chromatic aberration and can be made with very large apertures.

← Back to Q211

Q212

(a) *State the two conditions under which total internal reflection occurs.*
(b) *A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n,* $\frac{3}{2}$n *and* $\frac{2}{3}$n, *respectively. A laser beam is incident at the interface between A and B at an angle* θ *as shown in figure. Prove that the beam does not enter region C at all for* $\theta > \frac{2}{3}$.



[2026 • Set 55-1-1]

SOLUTION

(a) The two conditions for total internal reflection are:

(i) Light must travel from an optically **denser** medium to an optically **rarer** medium (higher to lower refractive index).

(ii) The angle of incidence in the denser medium must be **greater than the critical angle** θ_c, where $\theta_c = \frac{n_{rarer}}{n_{denser}}$.

(b) The three layers from top to bottom are A (index n), B (index $\frac{3}{2}$n) and C (index $\frac{2}{3}$n). The laser starts in A and strikes the A-B interface at angle θ to the normal; let it refract into B at angle θ_B, then strike the B-C interface at θ_B and refract into C at θ_C.

Applying Snell's law at the A-B interface:

$$n\theta = \frac{3}{2}\text{n}\,\theta_B$$

$$\theta_B = \frac{2}{3}\theta$$

Applying Snell's law at the B-C interface:

$$\frac{3}{2}\text{n}\,\theta_B = \frac{2}{3}\text{n}\,\theta_C$$

$$\theta_C = \frac{(3/2)n}{(2/3)n}\theta_B = \frac{9}{4}\theta_B$$

Substituting $\theta_B = \frac{2}{3}\theta$:

$$\theta_C = \frac{9}{4}\times\frac{2}{3}\theta = \frac{3}{2}\theta$$

The beam can enter C only if a real refracted ray exists there, i.e. only if $\theta_C \le 1$. If $\theta_C > 1$, no refraction is possible and total internal reflection takes place at the B-C interface, so the beam never enters C.

The condition $\theta_C > 1$ gives:

$$\frac{3}{2}\theta > 1$$

$$\theta > \frac{2}{3}$$

Hence, whenever $\theta > \frac{2}{3}$, total internal reflection occurs at the B-C boundary and **the beam does not enter region C at all**, as required.

Final answer. Since $\theta_C = \frac{3}{2}\theta$, the condition $\theta > \frac{2}{3}$ makes $\theta_C > 1$, so TIR occurs at the B-C interface and the beam never enters C.

← *Back to Q212*

Q213

Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case.

[2026 • Set 55-1-2]

SOLUTION

Ray diagram for a compound microscope with the final image at the least distance of distinct vision D: Draw two convex lenses on a common axis, the objective O (short focal length f_o) on the left and the eyepiece E (focal length f_e) on the right. Place a small object AB just beyond the objective's focus, so the object distance is slightly greater than f_o.

The objective forms a real, inverted, magnified image $A'B'$ inside the focal length of the eyepiece (between E and its focus F_e). The eyepiece then acts as a simple magnifier and produces a final image $A''B''$ that is virtual, inverted (with respect to the original object) and further magnified, located at the near point at distance D from the eye. Mark f_o at the objective, f_e at the eyepiece, the tube length, and the final image distance D.

The magnifying power is $\boldsymbol{m = m_o \times m_e}$, the product of the objective's linear magnification and the eyepiece's angular magnification.

For the objective, the linear magnification is

$$m_o = \frac{v_o}{u_o} \approx \frac{L}{f_o}$$

where L is the tube length (the distance between the objective's focal point and the eyepiece, with $L \gg f_o$), and $\boldsymbol{u_o \approx f_o}$ since the object is very close to the focus.

The eyepiece behaves as a simple microscope with its final image at D, so its angular magnification is

$$m_e = 1 + \frac{D}{f_e}$$

Multiplying the two contributions:

$$m = m_o \times m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

This is the **magnifying power of a compound microscope** when the final image is at the near point. It is made large by choosing a **small f_o, a small f_e and a long tube length L**.

Final answer. The magnifying power with the final image at the near point is $m = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$.

← *Back to Q213*

Q214

(a) Why does one prefer to view the image formed at infinity than that formed at near point in microscope/telescope?
(b) Consider lenses L_1, L_2 *and* L_3 *as specified in the following table. Which of them will you select as objective and eyepiece for constructing best possible (i) telescope (ii) compound microscope? Give reason for your answer.Lens*$L_1L_2L_3$*Power6 D3 D10 DAperture1 cm8 cm1 cm*

[2026 • Set 55-1-3]

SOLUTION

(a) When the final image is formed at infinity, the rays entering the eye are parallel, so the **eye does not have to accommodate** (the ciliary muscles stay relaxed). This is the most relaxed, least-tiring state for the eye, so it can be used comfortably for long observations. Forming the image at the near point gives slightly greater magnification, but the eye then has to keep accommodating, which causes strain. So for comfort one prefers the image at infinity (normal adjustment).

(b) The power and focal length are related by $P = \frac{1}{f}$, so a small focal length means a large power, and a large aperture means more light gathered.

$$f_1 = \frac{1}{6}\ m = 16.7\ \text{cm},\ f_2 = \frac{1}{3}\ m = 33.3\ \text{cm},\ f_3 = \frac{1}{10}\ m = 10\ \text{cm}$$

(i) Telescope: the objective should have a **long focal length and a large aperture** (high magnification and good light gathering), while the eyepiece should have a **short focal length**.

Objective = $\mathbf{L_2}$ (longest focal length 33.3 cm and the largest aperture, 8 cm).

Eyepiece = $\mathbf{L_3}$ (short focal length 10 cm); L_1 would also serve but L_3 gives a higher magnification f_o/f_e.

(ii) Compound microscope: **both lenses need short focal lengths**, and the objective should have the shortest focal length of all (since $m \propto 1/f_o$ and $\propto 1/f_e$).

Objective = $\mathbf{L_3}$ (shortest focal length 10 cm, giving the largest objective magnification).

Eyepiece = $\mathbf{L_1}$ (next shortest focal length 16.7 cm).

Thus for the telescope, objective L_2 and eyepiece L_3; for the compound microscope, objective L_3 and eyepiece L_1.

COMMON MISTAKE

For a telescope the objective wants a LONG focal length, but for a microscope the objective wants a SHORT one. Mixing these two up is the most common slip in this question.

Final answer. Image at infinity is preferred because the eye stays relaxed (no accommodation); telescope: objective $\mathbf{L_2}$, eyepiece $\mathbf{L_3}$; compound microscope: objective $\mathbf{L_3}$, eyepiece $\mathbf{L_1}$.

← Back to Q214

Q215

A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface.

[2026 • Set 55-3-1]

SOLUTION

Initially the ray inside the glass hits the horizontal glass-air surface at the critical angle, so the angle of refraction into air would be 90°. The critical angle for the glass-air pair satisfies

$$\theta_c^{ga} = \frac{1}{n_g} = \frac{1}{3/2} = \frac{2}{3}$$

When a thin horizontal water layer is placed on the slab, the ray now passes through three media in turn: glass → water → air. The ray hits the glass-water interface at the same angle θ_c^{ga} (the geometry inside the glass is unchanged), and we apply Snell's law there.

At the glass-water interface, with θ_w the refraction angle in water:

$$n_g\theta_c^{ga} = n_w\theta_w$$

$$\frac{3}{2}\times\frac{2}{3} = \frac{4}{3}\theta_w$$

$$1 = \frac{4}{3}\theta_w$$

$$\theta_w = \frac{3}{4}$$

The water layer has parallel faces, so the ray strikes the water-air surface at the same angle θ_w. Applying Snell's law at the water-air interface, with θ_a the emergence angle into air:

$$n_w\theta_w = (1)\theta_a$$

$$\frac{4}{3}\times\frac{3}{4}=\theta_a$$

$$\theta_a=1$$

$$\theta_a=90^\circ$$

So the ray emerges at 90° into air, i.e. it just **grazes along the water-air surface**. Physically, since the original ray was at the glass-air critical angle, the product $n\theta$ across the layers is preserved, so adding the intermediate water layer does not change the final emergence angle.

Final answer. The ray emerges at **90°** to the normal, i.e. it grazes along the water-air surface.

← Back to Q215

Q216

A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the:
(a) angle of minimum deviation for the prism, and
(b) refractive index of the material of the prism.

[2026 • Set 55-3-3]

SOLUTION

For an equilateral prism the refracting angle is $A = 60^\circ$. When a ray passes **symmetrically** through the prism, the angle of incidence equals the angle of emergence ($i = e$) and the prism is in the minimum-deviation condition. Here $i = 45^\circ$.

(a) At minimum deviation the deviation is related to the incidence angle by

$$\delta_m = 2i - A$$

$$\delta_m = 2(45^\circ) - 60^\circ$$

$$\delta_m = 90^\circ - 60^\circ = 30^\circ$$

So the angle of minimum deviation is $\delta_m = 30^\circ$.

(b) The refractive index of the prism material is given by the prism formula

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$n = \sin(60° + 30°2)\sin(60°2) = \sin 45°\sin 30°$$

$$n = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$n = \sqrt{2} \approx 1.414$$

Hence the angle of minimum deviation is $30°$ and the refractive index of the prism material is $\sqrt{2} \approx 1.414$.

Final answer. $\delta_m = 30°$ and the refractive index $n = \sqrt{2} \approx 1.414$.

← *Back to Q216*

Q217

(a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam?
(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.
(c) Explain the transmission of optical signal through an optical fibre by a diagram.

[2025 • Set 55-2-1]

SOLUTION

(a) When a parallel beam enters water obliquely, every ray bends **towards the normal** because water is denser than air. Since all the rays bend by the same amount towards the normal, the beam becomes more steeply inclined and the rays crowd closer together, so the **width of the beam decreases** inside the water.

(b) Bent-straw explanation with ray diagram: Draw the water surface as a horizontal line and the straw going down into the water at an angle, with its tip

T below the surface. Light coming from the submerged tip T travels up and strikes the water-air surface from below.

At the water-air surface the ray goes from denser water to rarer air, so it bends **away from the normal** (angle of refraction greater than angle of incidence).

The eye receives these refracted rays and, assuming light travels in straight lines, traces them back in straight lines. These backward extensions meet at a point T' which is **higher (closer to the surface)** than the real tip T.

So T' is the virtual image of the tip, raised towards the surface, and the submerged part of the straw appears lifted up. The result is that the straw seems to be **bent upward at the point where it meets the water surface**.

(c) Optical-fibre diagram: Draw a thin glass cylinder consisting of an inner **core** of refractive index n_1 surrounded by an outer **cladding** of slightly lower refractive index n_2 (so $\mathbf{n_1 > n_2}$). A light signal enters one end of the core at a small angle to the axis and strikes the core-cladding boundary.

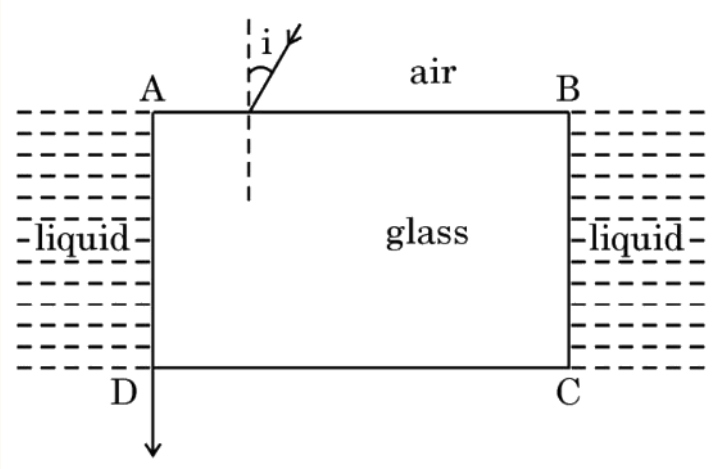
Because the core is denser than the cladding and the ray meets the boundary at an angle **greater than the critical angle**, it undergoes **total internal reflection**. The ray reflects repeatedly from wall to wall (a zig-zag path) and travels along the fibre with almost no loss, finally emerging at the far end. Thus the optical signal is guided along the fibre even when it is bent.

Final answer. The beam narrows in water; the straw appears bent up because light from the tip refracts away from the normal at the surface; and in an optical fibre the signal travels by repeated total internal reflection inside the denser core.

← *Back to Q217*

Q218

A rectangular glass slab ABCD *(refractive index* 1.5*) is surrounded by a transparent liquid (refractive index* 1.25*) as shown in the figure. A ray of light is incident on face* AB *at an angle i such that it is refracted out grazing the face* AD*. Find the value of angle i.*



[2025 • Set 55-5-1]

SOLUTION

The glass slab ABCD has refractive index $\mathbf{n_g = 1.5}$ and is surrounded by a liquid of refractive index $\mathbf{n_l = 1.25}$. The ray enters face AB at angle i, refracts to angle r inside the glass, travels to the adjacent face AD, and just **grazes out of AD**, which means it emerges along AD at 90° to the normal of AD. Grazing emergence means the ray strikes AD exactly at the critical angle θ_c for the glass-liquid pair.

The critical angle at the glass-liquid interface (AD):

$$\theta_c = \frac{n_l}{n_g} = \frac{1.25}{1.5} = \frac{5}{6}$$

The faces AB and AD are perpendicular, so the angle of refraction r at AB and the angle of incidence θ_c at AD are complementary:

$$r = 90° - \theta_c$$

$$\sin r = \sin(90° - \theta_c) = \theta_c$$

$$\theta_c = \sqrt{1 - \sin^2\theta_c} = \sqrt{1 - (\frac{5}{6})^2} = \sqrt{1 - \frac{25}{36}} = \sqrt{\frac{11}{36}} = \frac{\sqrt{11}}{6}$$

$$\sin r = \frac{\sqrt{11}}{6} \approx 0.553$$

Applying Snell's law at face AB (liquid to glass):

$$n_l \sin i = n_g \sin r$$

$$1.25 \sin i = 1.5 \times 0.553$$

$$1.25 \sin i = 0.829$$

$$\sin i = \frac{0.829}{1.25} = 0.663$$

$$i = \sin^{-1}(0.663) \approx 41.5^\circ$$

Hence the ray must be incident on face AB at an angle of about **41.5°** for it to graze out of face AD.

CONCEPT

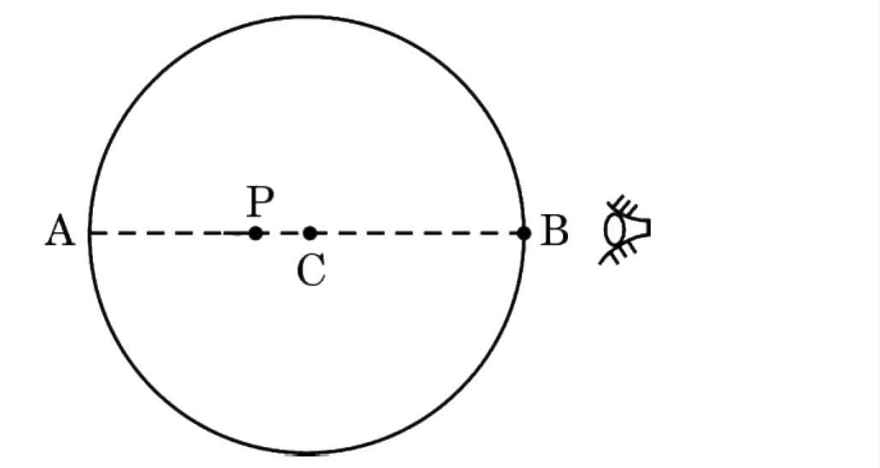
Grazing emergence from a face always means the ray hit that face exactly at the critical angle. Use the right-angle geometry to link that critical angle to the refraction angle at the entry face.

Final answer. The angle of incidence on face **AB** is $i \approx 41.5^\circ$.

← *Back to Q218*

Q219

*An air bubble is trapped at point P (***CP = 1.75 cm***) in a spherical glass ball (*$n = 1.5$*) of radius* 7 cm *as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram.*



[2025 • Set 55-5-2]

SOLUTION

The glass ball has refractive index $n = 1.5$ and radius $R = 7$ cm, with centre C. The air bubble at P acts as the object, with $CP = 1.75$ cm, and we view it from side B (the right). Light from P travels through the glass and refracts at the curved right-hand surface of the ball.

Taking the right surface as the refracting surface and measuring distances from its pole, with light travelling left to right:

Object distance (bubble is on the far side of the centre from B):

$$u = -(R + CP) = -(7 + 1.75) = -8.75 \text{ cm}$$

At this surface light goes from glass into air, so $n_1 = 1.5$ (glass, incident side) and $n_2 = 1$ (air). The centre of curvature lies to the left of the surface, so

$$R = -7 \text{ cm}$$

Applying the refraction-at-a-spherical-surface formula:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1}{v} - \frac{1.5}{-8.75} = \frac{1 - 1.5}{-7}$$

$$\frac{1}{v} + \frac{1.5}{8.75} = \frac{-0.5}{-7} = \frac{1}{14}$$

$$\frac{1.5}{8.75} = \frac{6}{35}$$

$$\frac{1}{v} = \frac{1}{14} - \frac{6}{35}$$

Taking the LCD as 70: $\frac{1}{14} = \frac{5}{70}$ and $\frac{6}{35} = \frac{12}{70}$, so

$$\frac{1}{v} = \frac{5 - 12}{70} = \frac{-7}{70} = -\frac{1}{10}$$

$$v = -10 \text{ cm}$$

The negative sign means the image is formed on the same side as the incoming light, i.e. **inside the glass ball**, 10 cm from the right surface (measured back into the glass, towards the bubble). The image is therefore **virtual** and erect, as seen by an observer on side B.

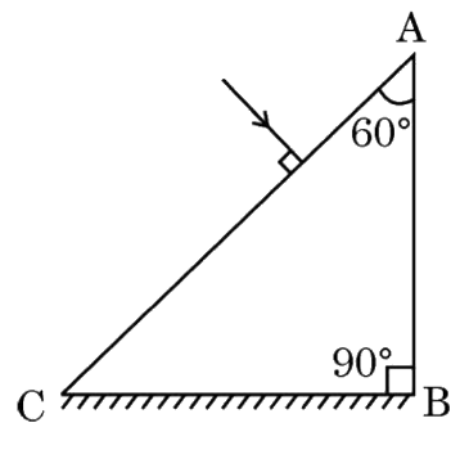
Ray diagram: Draw the spherical ball with centre C and the bubble P to the left of C. Two rays leave P, refract at the right curved surface (bending as they enter air), and diverge on emerging. Their backward extensions inside the glass meet at the image point I, 10 cm from the right surface, showing the virtual image on the same side as P.

Final answer. The image is virtual, formed **10 cm** from the viewing (B-side) surface inside the glass ball.

← *Back to Q219*

Q220

A right-angled prism ABC *(refractive index* $\sqrt{2}$*) is kept on a plane mirror as shown in the figure. A ray of light is incident normally on the face* AC.



(a) *Trace the path of the ray as it passes through the prism.*
(b) *Find the angle of deviation produced by the prism.*

[2025 • Set 55-5-3]

SOLUTION

The right-angled prism ABC has the right angle at B ($\angle B = 90°$), with $\angle A = 60°$ and $\angle C = 30°$. It rests on a plane mirror along its base BC, and a ray of light falls **normally on face AC**, with refractive index $\mu = \sqrt{2}$.

First find the critical angle for the glass-air interface:

$$\theta_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$$

$$\theta_c = 45^\circ$$

(a) Path of the ray:

Since the ray strikes face AC normally, it enters the prism **without bending** and travels straight towards the base BC.

From the geometry of the 30-60-90 prism, this ray meets the base BC at an angle of 30° to the normal of BC (equivalently 60° to the base surface), because the ray is perpendicular to AC and $\angle A = 60^\circ$.

At the base the angle of incidence is 30°, which is **less than the critical angle** 45°, so total internal reflection does **not** occur at the glass surface here. However, BC is silvered (it lies on the plane mirror), so the ray is **reflected by the mirror** at 30°, just like reflection from any plane mirror.

The reflected ray travels back up through the prism and emerges out of face AC, again meeting AC normally so it leaves without bending. So the diagram shows the ray entering normally through AC, going down to the mirror at BC, reflecting there, and coming back out through AC.

(b) Angle of deviation:

The only turning of the ray happens at the plane mirror BC. A plane mirror deviates a ray through $180^\circ - 2i$, where i is the angle of incidence on the mirror.

$$\text{Deviation} = 180^\circ - 2(30^\circ) = 120^\circ$$

Equivalently, the angle between the original incident ray and the final emergent ray is $\mathbf{2 \times 60^\circ = 120^\circ}$ (twice the glancing angle the ray makes with the mirror surface).

$$\text{Angle of deviation} = 120^\circ$$

Hence the ray is reflected by the base mirror and emerges from AC deviated by 120° from its original direction.

Final answer. The ray enters **AC** normally, strikes the base mirror **BC** at **30°** (no TIR, since $\mathbf{30^\circ < \theta_c = 45^\circ}$), is reflected by the mirror, and emerges from **AC**; the angle of deviation produced is **120°**.

← Back to Q220

Q221

Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula.

[2025 • Set 55-6-1]

SOLUTION

For a concave mirror to form a real, inverted and magnified image, the object must be placed between the focus F and the centre of curvature C.

Ray diagram: Draw a concave mirror with pole P, principal focus F and centre of curvature C marked on the principal axis. Place an erect object AB (with B on the axis and A at the top) between F and C. From A, draw two rays.

Ray 1: a ray AD parallel to the principal axis strikes the mirror at D and reflects through the focus F.

Ray 2: a ray AE passing through the centre of curvature C strikes the mirror normally and retraces its own path.

These two reflected rays intersect at A' beyond C. Drop a perpendicular $A'B'$ on the axis. Then $A'B'$ is the real, inverted and magnified image.

Derivation of the mirror formula: In ABP and $A'B'P$, the incident and reflected rays make equal angles with the principal axis (laws of reflection), so the triangles are similar.

$$\frac{A'B'}{AB} = \frac{PB'}{PB} \quad ...(1)$$

Similarly, in DPF and $A'B'F$ (where **DP ≈ AB** for a small aperture), they are similar, giving

$$\frac{A'B'}{AB} = \frac{B'F}{PF} \quad ...(2)$$

From (1) and (2),

$$\frac{\text{PB'}}{\text{PB}}=\frac{B'F}{\text{PF}}=\frac{\text{PB'}-\text{PF}}{\text{PF}}$$

Applying the sign convention (distances measured from the pole P, taken positive along incident light), $\mathbf{PB=-}u$, $\mathbf{PB'=-}v$, $\mathbf{PF=-}f$.

$$\frac{-v}{-u}=\frac{-v-(-f)}{-f}$$

$$\frac{v}{u}=\frac{v-f}{f}$$

$$\text{vf=uv-uf}$$

Dividing throughout by uvf,

$$\frac{1}{u}=\frac{1}{f}-\frac{1}{v}$$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

This is the required **mirror formula**, where u is the object distance, v the image distance and f the focal length.

Final answer. The image forms beyond C (real, inverted, magnified) and the mirror formula is $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$.

← Back to Q221

Q222

A ray of light is incident at an angle i on a parallel sided glass slab of thickness 'd' and gets refracted into the slab at angle r. Draw a ray diagram to show its path as it emerges out of the slab. Hence, obtain an expression for the lateral shift of the ray. Under what condition will the shift be minimum?

[2025 • Set 55-6-2]

SOLUTION

Ray diagram: Draw a parallel-sided glass slab PQRS of thickness d and refractive index n. An incident ray AB strikes the top face at point B at angle of

incidence i. It bends towards the normal and refracts into the slab along BC at angle r. At the bottom face it bends away from the normal and emerges as ray CD, parallel to AB but laterally displaced. The perpendicular distance between the produced incident ray and the emergent ray is the lateral shift s.

Since the two faces are parallel, the emergent ray CD is parallel to the incident ray AB; the slab only shifts the ray sideways without changing its direction.

Derivation of the lateral shift: Let **BC=** be the path length inside the slab. Drop a perpendicular from C to the extended incident ray, meeting it at N; the lateral shift is $\boldsymbol{s}$**=CN**.

In the right triangle formed, the angle between BC and the extended incident direction is $(i-r)$, so

$$s=\sin(i-r) \quad ...(1)$$

From the geometry of the slab (thickness d measured along the normal),

$$\cos r=\frac{d}{} \Rightarrow =\frac{d}{\cos r} \quad ...(2)$$

Substituting (2) in (1),

$$s=\frac{d\sin(i-r)}{\cos r}$$

Condition for minimum shift: The shift s is zero (minimum) when $\boldsymbol{\sin(i-r)=0}$, i.e. when $\boldsymbol{i=r}$. This happens only at **normal incidence** $\boldsymbol{(i=0)}$, when the ray passes straight through the slab undeviated and undisplaced.

Final answer. The lateral shift is $\boldsymbol{s=\frac{d\sin(i-r)}{\cos r}}$, and it is minimum (zero) at normal incidence $\boldsymbol{i=0}$.

← *Back to Q222*

Q223

(a) A concave mirror has radius of curvature 20 cm. *Calculate the distance of an object from the mirror so as to form an image of magnification* −2. *Also find the*

location of the image.
(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object? Justify your answer.

[2025 • Set 55-6-3]

SOLUTION

(a) For the concave mirror, radius of curvature R=**20 cm**, so the focal length is

$$f=\frac{R}{2}=\frac{20}{2}=10\text{ cm}$$

Using the sign convention, f=**−10 cm** (focus is in front of a concave mirror).

The magnification is m=**−2** (real, inverted, magnified image), and

$$m=-\frac{v}{u}=-2 \Rightarrow v=2u$$

Substituting in the mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$,

$$\frac{1}{2u}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1+2}{2u}=\frac{1}{f}$$

$$\frac{3}{2u}=\frac{1}{f}$$

$$u=\frac{3f}{2}=\frac{3(-10)}{2}=-15\text{ cm}$$

$$v=2u=2(-15)=-30\text{ cm}$$

The negative signs show that both object and image lie in front of the mirror (real image). So the **object is 15 cm from the mirror** and the **image forms 30 cm in front of the mirror**.

(b) Yes, the mirror will still form the image. The image is formed by reflection from the silvered (annular) part of the mirror, and even after the central coating is removed, the outer ring still reflects light. Rays from each point of the object continue to be reflected by the remaining surface and converge to form the image at the same position with the same size. Only the **brightness of**

the image decreases, because fewer rays now reach the mirror and contribute to the image.

Final answer. Object is at 15 cm and image at 30 cm in front of the mirror; the image still forms (only dimmer) if the central coating is removed.

← *Back to Q223*

Q224

Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium.

[2023 • Set 55-2-1]

SOLUTION

Critical angle: For a given pair of media, the critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90°, i.e. the refracted ray just grazes along the surface separating the two media.

Total internal reflection (TIR): When light travels from a denser medium to a rarer medium and the angle of incidence in the denser medium exceeds the critical angle, no refraction takes place and the whole of the light is reflected back into the denser medium. This phenomenon is called total internal reflection.

Two conditions for TIR are: (i) light must travel from a denser to a rarer medium, and (ii) the angle of incidence must be greater than the critical angle.

Relation between critical angle and refractive index: Let light pass from a denser medium of refractive index n_{denser} to a rarer medium of refractive index n_{rarer}. Applying Snell's law at the interface, with angle of incidence $=\theta_c$ and angle of refraction $=90°$,

$$n_{denser}\theta_c = n_{rarer} \sin 90°$$

$$n_{denser}\theta_c = n_{rarer}$$

$$\theta_c = \frac{n_{rarer}}{n_{denser}}$$

If the rarer medium is air ($\mathbf{n_{rarer}=1}$) and the denser medium has refractive index n, then

$$\theta_c = \frac{1}{n}$$

$$\boldsymbol{n = \frac{1}{\theta_c}}$$

Final answer. The required relation is $\theta_c = \frac{n_{rarer}}{n_{denser}}$, or $n = \frac{1}{\theta_c}$ when the rarer medium is air.

← *Back to Q224*

Q225

A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle.

[2023 • Set 55-2-2]

SOLUTION

Ray diagram: Draw a prism ABC with refracting angle A at the apex. A ray strikes the first refracting face AB at angle of incidence i and refracts into the prism at angle r_1. It then travels to the second face AC, where it meets the normal at angle r_2. The ray is said to just suffer total internal reflection at this second face, so r_2 equals the critical angle θ_c.

Geometry of the prism: The normals drawn at the two refracting faces meet inside the prism, and from the quadrilateral formed it follows that

$$r_1 + r_2 = A$$

Since the ray just suffers total internal reflection at the second face, the angle of incidence there equals the critical angle,

$$r_2 = \theta_c$$

$$\Rightarrow r_1 = A - r_2 = A - \theta_c$$

Applying Snell's law at the first face AB (air to glass of refractive index μ),

$$\sin i = \mu \sin r_1$$

$$\sin i = \mu \sin(A - \theta_c)$$

where the critical angle is given by $\theta_c = \frac{1}{\mu}$. This is the required relation between the angle of incidence i, the angle of prism A and the critical angle θ_c.

Final answer. The required relation is $\boldsymbol{\sin i = \mu \sin(A - \theta_c)}$, with $\theta_c = \frac{1}{\mu}$.

← *Back to Q225*

Q226

A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δ_m.

[2023 • Set 55-2-3]

SOLUTION

A monochromatic ray of light enters a prism of refracting angle A and suffers deviation. As the angle of incidence is increased, the deviation first decreases to a minimum value δ_m and then increases. At minimum deviation the ray passes symmetrically through the prism.

Ray diagram: Draw prism ABC with apex angle A. The ray enters face AB at angle i_1, refracts to r_1, travels parallel to the base BC, meets face AC at r_2, and emerges at i_2. At minimum deviation the path is symmetric.

At minimum deviation, the angles satisfy

$$i_1 = i_2 = i \; r_1 = r_2 = r$$

From the prism geometry,

$$r_1 + r_2 = A$$

$$2r=A \Rightarrow r=\frac{A}{2}$$

The total deviation is $\delta=(i_1+i_2)-A$. At minimum deviation,

$$\delta_m=2i-A$$

$$i=\frac{A+\delta_m}{2}$$

Applying Snell's law at the first refracting face,

$$n=\frac{\sin i}{\sin r}$$

Substituting $i=\frac{A+\delta_m}{2}$ and $r=\frac{A}{2}$,

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

This is the **prism formula** giving the refractive index in terms of the angle of prism A and the angle of minimum deviation δ_m.

Final answer. The refractive index is $n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$.

← Back to Q226

Q227

(i) Trace the path of a ray of light PQ *which is incident at an angle i on one face of a glass prism of angle A. It then emerges out from the other face at an angle e. Use the ray diagram to prove that the angle through which the ray is deviated is given by* $\angle\,\delta = \angle\,i + \angle\,e - \angle\,A$.

(ii) What will be the minimum value of δ *if the ray passes symmetrically through the prism?*

[2022 • Set 55-1-1]

SOLUTION

(i) Ray diagram: Draw prism ABC with refracting angle A at the apex. The incident ray PQ strikes the first face AB at angle of incidence i and refracts into the prism at angle r_1. The refracted ray QR travels inside and meets the second face AC, where it refracts at angle r_2 inside and emerges as ray RS at angle of emergence e. The incident ray PQ produced forward and the emergent ray RS produced backward meet at point T; the angle between them is the angle of deviation δ.

At the first face, the ray deviates by $(i-r_1)$, and at the second face it deviates by $(e-r_2)$. The total deviation is the sum of these,

$$\delta=(i-r_1)+(e-r_2)$$

$$\delta=(i+e)-(r_1+r_2) \ldots(1)$$

In the quadrilateral formed by the two normals and the two faces, the normals meet at an angle, and from the geometry of the triangle QRN inside the prism,

$$r_1+r_2=A \ldots(2)$$

Substituting (2) in (1),

$$\delta=i+e-A$$

Hence proved.

(ii) Minimum value of δ (symmetric pass): When the ray passes symmetrically through the prism, $\boldsymbol{i=e}$ and $\mathbf{r_1=r_2}$. The deviation is minimum, $\boldsymbol{\delta=\delta_m}$, and

$$\delta_m=2i-A$$

$$i=\frac{A+\delta_m}{2}$$

Combined with $\boldsymbol{r=\frac{A}{2}}$ and Snell's law, this gives the refractive index

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

from which the minimum deviation δ_m for a given n and A is obtained.

Final answer. $\delta = i + e - A$ is proved, and for a symmetric pass δ is minimum with $\delta_m = 2i - A$.

← Back to Q227

Q228

A point source in air is kept 24 cm *in front of a concave spherical glass surface* ($\mu_g = 1.5$) *and radius of curvature* 60 cm. *Find the nature of the image formed and its distance from the point source.*

[2022 • Set 55-1-1]

SOLUTION

Refraction takes place at a single spherical surface, with light going from air into glass. Here the rarer medium is air ($n_1 = 1$) and the denser medium is glass ($n_2 = 1.5$).

Applying the sign convention (distances measured from the pole, positive along incident light): the object is in front, so $u = -24$ cm. The surface is concave towards the incident light, so its centre of curvature lies on the same side as the object, giving $R = -60$ cm.

The refraction formula at a single spherical surface is

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Substituting the values,

$$\frac{1.5}{v} - \frac{1}{-24} = \frac{1.5 - 1}{-60}$$

$$\frac{1.5}{v} + \frac{1}{24} = \frac{0.5}{-60} = -\frac{1}{120}$$

$$\frac{1.5}{v} = -\frac{1}{120} - \frac{1}{24}$$

$$\frac{1.5}{v} = -\frac{1}{120} - \frac{5}{120} = -\frac{6}{120} = -\frac{1}{20}$$

$$v = -1.520 = -30 \text{ cm}$$

The negative sign shows the image is formed on the same side as the object (in front of the surface), so the image is **virtual and erect**, at 30 cm in front of the surface.

Distance of the image from the point source: the source is at 24 cm and the image at 30 cm, both on the same side of the surface, so

$$\text{distance} = |v| - |u| = 30 - 24 = 6 \text{ cm}$$

The image lies 6 cm beyond the source (farther from the surface).

Final answer. The image is virtual and erect, 30 cm in front of the surface, i.e. 6 cm beyond the point source.

← *Back to Q228*

Q229

Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres.

[2022 • Set 55-1-1]

SOLUTION

An optical fibre works on the principle of **total internal reflection** of light.

Structure and diagram: An optical fibre consists of a thin cylindrical glass (or plastic) core of refractive index n_1 surrounded by a cladding of slightly lower refractive index n_2 **($n_1 > n_2$)**. Draw a long fibre showing the inner core and outer cladding. A ray of light enters one end of the core and strikes the core-cladding boundary. Show the ray bouncing repeatedly in a zigzag path, each bounce being a total internal reflection at the core-cladding interface, until it comes out at the far end.

Working: Light entering the fibre within the acceptance angle strikes the core-cladding interface at an angle greater than the critical angle for that pair of media. Since the light travels from the denser core to the rarer cladding and the angle of incidence exceeds the critical angle, total internal reflection occurs again and again. The light is thus guided along the fibre with almost no loss of intensity, even when the fibre is bent.

The critical angle at the core-cladding boundary is given by

$$\theta_c = \frac{n_2}{n_1}$$

which is less than 1 since $\mathbf{n_2 1}$.

Two uses of optical fibres: (1) In high-speed, long-distance telecommunication for transmitting telephone, internet and television signals. (2) In medicine, as an endoscope, to examine internal organs of the body during diagnosis and surgery.

Final answer. Optical fibres work on total internal reflection; they are used in telecommunication and in medical endoscopy.

← *Back to Q229*

Q230

A thin equiconvex lens of radius of curvature R made of material of refractive index μ_1 *is kept coaxially, in contact with an equiconcave lens of the same radius of curvature and refractive index* μ_2 $(\boldsymbol{\mu_2 > \mu_1})$. *Find:*
(i) the ratio of their powers, and
(ii) the power of the combination and its nature.

[2022 • Set 55-1-1]

SOLUTION

Setup: An equiconvex lens of refractive index μ_1 (both surfaces of radius R) is placed in contact with an equiconcave lens of refractive index μ_2 having the same radius R, with $\boldsymbol{\mu_2 > \mu_1}$.

Power of the equiconvex lens: For an equiconvex lens the first surface is convex $(\mathbf{R_1 = +R})$ and the second is concave towards the object $(\mathbf{R_2 = -R})$. Using the lens maker's formula,

$$P_1 = (\mu_1 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$P_1 = (\mu_1 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (\mu_1 - 1)\cdot\frac{2}{R}$$

Power of the equiconcave lens: For an equiconcave lens $\mathbf{R_1 = -R}$ and $\mathbf{R_2 = +R}$,

$$P_2 = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{R}\right) = -(\mu_2 - 1)\cdot\frac{2}{R}$$

(i) Ratio of powers:

$$\frac{P_1}{P_2} = \frac{(\mu_1 - 1)\cdot\frac{2}{R}}{-(\mu_2 - 1)\cdot\frac{2}{R}}$$

$$\frac{P_1}{P_2} = -\frac{(\mu_1 - 1)}{(\mu_2 - 1)}$$

(ii) Power of the combination: For thin lenses in contact, the powers add,

$$P = P_1 + P_2$$

$$P = (\mu_1 - 1)\cdot\frac{2}{R} - (\mu_2 - 1)\cdot\frac{2}{R}$$

$$P = \frac{2}{R}[(\mu_1 - 1) - (\mu_2 - 1)]$$

$$P = \frac{2(\mu_1 - \mu_2)}{R}$$

Since $\mu_2 > \mu_1$, the quantity $(\mu_1 - \mu_2)$ is negative, so P is negative. Hence the combination behaves as a **diverging (concave) lens**.

Final answer. $\frac{P_1}{P_2} = -\frac{\mu_1 - 1}{\mu_2 - 1}$ and the combination has power $P = \frac{2(\mu_1 - \mu_2)}{R}$, which is negative, so it is diverging.

← *Back to Q230*

Q231

Find the two possible positions of an object kept in front of a lens of +5.0 *D, so that the image formed in both cases is four times magnified.*

[2022 • Set 55-1-2]

SOLUTION

The power of the lens is $P = +5.0\ D$, so its focal length is $f = \frac{1}{P} = \frac{1}{5}\ m = 0.20\ m = +20\ \text{cm}$. The positive value tells us it is a converging (convex) lens.

The magnitude of magnification is $|m| = 4$. A convex lens can give a four-times magnified image in two ways: a real, inverted image and a virtual, erect image. We treat both cases separately.

Using the magnification relation $m = \frac{v}{u}$ together with the thin lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.

Case 1: Real image (inverted, $m = -4$), so $v = -4u$.

$$\frac{1}{-4u} - \frac{1}{u} = \frac{1}{20}$$

$$\frac{1}{-4u} - \frac{4}{4u} = \frac{1}{20}$$

$$\frac{-1-4}{4u} = \frac{1}{20} \Rightarrow \frac{-5}{4u} = \frac{1}{20}$$

$$4u = -5 \times 20 = -100 \Rightarrow u = -25\ \text{cm}$$

Case 2: Virtual image (erect, $m = +4$), so $v = +4u$.

$$\frac{1}{4u} - \frac{1}{u} = \frac{1}{20}$$

$$\frac{1-4}{4u} = \frac{1}{20} \Rightarrow \frac{-3}{4u} = \frac{1}{20}$$

$$4u = -3 \times 20 = -60 \Rightarrow u = -15\ \text{cm}$$

Hence the object must be placed at a distance of 25 cm (for the real, four-times image) or 15 cm (for the virtual, four-times image) in front of the lens. **The two positions are 25 cm and 15 cm.**

TIP

For magnification problems with a fixed |m|, always check both signs of m. A convex lens gives a real magnified image (object beyond f) and a virtual magnified image (object inside f), so two positions normally exist.

Final answer. The object must be placed at 25 cm (real image) or 15 cm (virtual image) in front of the lens.

← Back to Q231

Q232

For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water?

[2022 • Set 55-1-3]

SOLUTION

For a prism, the refractive index is related to the angle of the prism A and the angle of minimum deviation δ_m by the prism formula:

$$n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$$

Here it is given that $\boldsymbol{\delta_m = A}$ and $\boldsymbol{n = \sqrt{3}}$. Substituting $\boldsymbol{\delta_m = A}$:

$$\sqrt{3} = \frac{\sin(\frac{A + A}{2})}{\sin(\frac{A}{2})} = \frac{\sin A}{\sin(\frac{A}{2})}$$

Using the identity $\boldsymbol{\sin A = 2\sin(\frac{A}{2})\cos(\frac{A}{2})}$:

$$\sqrt{3} = \frac{2\sin(\frac{A}{2})\cos(\frac{A}{2})}{\sin(\frac{A}{2})} = 2\cos(\frac{A}{2})$$

$$\cos(\frac{A}{2}) = \frac{\sqrt{3}}{2} \Rightarrow \frac{A}{2} = 30° \Rightarrow A = 60°$$

So the **angle of the prism is $A = 60°$**.

Effect of immersing the prism in water: The minimum deviation depends on the refractive index of the prism material relative to the surrounding medium.

In water the relative refractive index becomes

$$n_{rel} = \frac{n_{glass}}{n_{water}} = \frac{\sqrt{3}}{1.33} \approx 1.30$$

which is smaller than $\sqrt{3} \approx 1.73$. Since a smaller refractive index in the prism formula gives a smaller value of δ_m, **the angle of minimum deviation δ decreases when the prism is immersed in water.**

Final answer. The angle of the prism is A = 60 degrees, and the deviation delta decreases when the prism is immersed in water.

← *Back to Q232*

Q233

(i) State the conditions for total internal reflection to take place.
(ii) A tank is filled with a transparent liquid to height 'H'. A coin suspended by a thread in the liquid is gradually lowered till it touches the bottom. The apparent depth is determined corresponding to different positions of the coin.
(a) Plot a graph showing variation of the apparent depth with the real depth of the coin.
(b) What is the physical significance of the slope of the graph?

[2022 • Set 55-2-1]

SOLUTION

(i) Conditions for total internal reflection (TIR):

1. Light must travel from an optically denser medium to an optically rarer medium (for example, from glass or liquid into air).

2. The angle of incidence in the denser medium must be greater than the critical angle for that pair of media.

(ii) When an object lies inside a liquid of refractive index n and is viewed from above (nearly normally), the apparent depth and real depth are related by:

$$n = \frac{\text{real depth}}{\text{apparent depth}} = \frac{d_{real}}{d_{app}}$$

Rearranging gives:

$$d_{app} = \frac{1}{n} d_{real}$$

This is of the form $y = mx$, a straight line passing through the origin with d_{app} on the y-axis and d_{real} on the x-axis.

(a) Graph: Plot apparent depth d_{app} (vertical axis) against real depth d_{real} (horizontal axis). The graph is a straight line starting from the origin (0,0), rising to the right with a constant positive slope equal to $\frac{1}{n}$. Since $n > 1$, the slope is less than 1, so the line lies below the 45° line $d_{app} = d_{real}$. Label the line with slope $\frac{1}{n}$.

(b) Physical significance of the slope: The slope of the line equals $\frac{1}{n}$, the reciprocal of the refractive index of the liquid. Therefore the **refractive index of the liquid can be found from the slope as** $n = \frac{1}{\text{slope}}$.

Final answer. TIR needs denser-to-rarer travel with incidence above the critical angle; the apparent-depth vs real-depth graph is a straight line through the origin of slope 1/n, so n = 1/slope.

← *Back to Q233*

Q234

Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power.

[2022 • Set 55-2-1]

SOLUTION

Ray diagram (in words): Draw the principal axis horizontally. On the left place the objective lens O, a large convex lens of long focal length f_o; on the right place the eyepiece lens E, a small convex lens of short focal length f_e. Both share the same principal axis. From a distant object, two parallel rays (parallel

to each other but inclined slightly to the axis at angle α) enter the objective. The objective converges them to form a real, inverted, diminished image I_1 of height h' at its focal point F_o. In normal adjustment, the eyepiece is positioned so that this image I_1 lies exactly at the focus F_e of the eyepiece as well (the two foci coincide). The eyepiece therefore sends out a parallel beam, and the final image is formed at infinity, viewed by a fully relaxed eye. Mark α as the angle the incoming parallel rays make with the axis at the objective, and β as the angle the emergent parallel rays make with the axis at the eyepiece. The tube length is $L = f_o + f_e$.

Magnifying power: The magnifying power m is the ratio of the angle β subtended at the eye by the final image to the angle α subtended at the unaided eye by the object:

$$m = \frac{\beta}{\alpha}$$

From the intermediate image I_1 of height h' at the common focal plane, the angle subtended at the objective is

$$\alpha \approx \alpha = \frac{h'}{f_o}$$

and the angle subtended at the eyepiece is

$$\beta \approx \beta = \frac{h'}{f_e}$$

Therefore

$$m = \frac{\beta}{\alpha} = \frac{h'/f_e}{h'/f_o} = \frac{f_o}{f_e}$$

Hence the **magnifying power of an astronomical telescope in normal adjustment is** $m = \frac{f_o}{f_e}$, the ratio of the focal length of the objective to that of the eyepiece.

Final answer. The magnifying power in normal adjustment is m = f_o / f_e.

← *Back to Q234*

Q235

*A converging lens made of glass (**μ = 1.5**) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length*
(a) in air, and
(b) when it is immersed in a liquid of refractive index 1.25.

[2022 • Set 55-2-1]

SOLUTION

We use the lens maker's formula. For a converging (biconvex) lens, with light incident on the first face of radius R_1 and emerging from the second face of radius R_2, sign convention gives $\mathbf{R_1 = +10\ cm}$ and $\mathbf{R_2 = -20\ cm}$.

(a) In air, the medium outside has refractive index 1, so

$$\frac{1}{f_{air}} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_{air}} = (1.5 - 1)\left(\frac{1}{10} - \frac{1}{-20}\right)$$

$$\frac{1}{f_{air}} = (0.5)\left(\frac{1}{10} + \frac{1}{20}\right) = (0.5)\left(\frac{2+1}{20}\right) = (0.5)\left(\frac{3}{20}\right)$$

$$\frac{1}{f_{air}} = \frac{3}{40}\ \text{cm}^{-1} \Rightarrow f_{air} = \frac{40}{3} \approx 13.3\ \text{cm}$$

(b) When immersed in a liquid of refractive index $\mathbf{n_l = 1.25}$, we replace $(\mu - 1)$ by $\left(\frac{\mu}{n_l} - 1\right)$:

$$\frac{1}{f_{liq}} = \left(\frac{\mu}{n_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_{liq}} = \left(\frac{1.5}{1.25} - 1\right)\left(\frac{3}{20}\right)$$

$$\frac{1}{f_{liq}} = (1.2 - 1)\left(\frac{3}{20}\right) = (0.2)\left(\frac{3}{20}\right) = \frac{0.6}{20} = \frac{3}{100}\ \text{cm}^{-1}$$

$$f_{liq} = \frac{100}{3} \approx 33.3\ \text{cm}$$

The focal length increases from about 13.3 cm in air to about 33.3 cm in the liquid, so the lens becomes weaker (less converging) because the liquid reduces the relative refractive index. $\mathbf{f_{air} \approx 13.3\ cm}$ **and** $\mathbf{f_{liq} \approx 33.3\ cm.}$

Final answer. The focal length is about 13.3 cm in air and about 33.3 cm in the liquid of refractive index 1.25.

← *Back to Q235*

Q236

(a) Is the speed of light in glass independent of the colour of light? Give reason.
(b) A small bulb is placed at the bottom of a tank containing water to a depth of 70 *cm. Find the area of the surface of water through which light from the bulb can emerge out. Given refractive index of water is* $\frac{4}{3}$.

[2022 • Set 55-2-2]

SOLUTION

(a) No, the speed of light in glass is not independent of the colour of light. Glass has a different refractive index for each colour (it is a dispersive medium); the refractive index is greater for violet light than for red light. Since the speed in the medium is $v = \frac{c}{n}$, a larger n means a smaller speed. Hence violet light travels slower than red light in glass, and **the speed depends on the colour.**

(b) Light from the bulb can emerge through the surface only within a circular patch directly above the bulb. Rays that strike the water surface at the critical angle θ_c form the boundary of this circle; rays striking at angles greater than θ_c undergo total internal reflection and cannot escape.

The critical angle for water-to-air is given by:

$$\theta_c = \frac{1}{n} = \frac{1}{4/3} = \frac{3}{4}$$

Then $\theta_c = \sqrt{1 - \sin^2\theta_c} = \sqrt{1 - \frac{9}{16}} = \sqrt{\frac{7}{16}} = \frac{\sqrt{7}}{4}$, so

$$\theta_c = \frac{\theta_c}{\theta_c} = \frac{3/4}{\sqrt{7}/4} = \frac{3}{\sqrt{7}}$$

The radius of the circular patch at depth $H = 70$ cm is:

$$r = H\theta_c = 70 \times \frac{3}{\sqrt{7}} = \frac{210}{\sqrt{7}} \approx 79.4 \text{ cm}$$

The area of the surface through which light emerges is:

$$A = \pi r^2 = \pi \left(\frac{210}{\sqrt{7}}\right)^2 = \pi \times \frac{44100}{7} = \pi \times 6300$$

$$A \approx 3.14 \times 6300 \approx 1.98 \times 10^4 \text{ cm}^2 \approx 1.98 \text{ m}^2$$

Hence **light emerges through a circular area of about 1.98 m^2 (radius ≈ 79.4 cm).**

Final answer. Speed in glass depends on colour; light emerges through a circular area of about 1.98 m^2 (radius ~79.4 cm).

← *Back to Q236*

Q237

The refractive index of an equilateral triangular prism kept in air is $\sqrt{2}$. Calculate:
(a) the angle of minimum deviation.
(b) the critical angle for the prism.

[2022 • Set 55-2-2]

SOLUTION

For an equilateral triangular prism, the angle of the prism is $A = 60°$, and the refractive index is $n = \sqrt{2}$.

(a) Angle of minimum deviation: At minimum deviation the prism formula is

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Substituting $A = 60°$ and $n = \sqrt{2}$:

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ + \delta_m}{2}\right)}{\sin 30^\circ}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ + \delta_m}{2}\right)}{0.5} \Rightarrow \sin\left(\frac{60^\circ + \delta_m}{2}\right) = \sqrt{2} \times 0.5 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\frac{60^\circ + \delta_m}{2} = 45^\circ \Rightarrow 60^\circ + \delta_m = 90^\circ \Rightarrow \delta_m = 30^\circ$$

So the **angle of minimum deviation is $\delta_m = 30^\circ$**.

(b) Critical angle: The critical angle θ_c for the prism material with respect to air is given by

$$\theta_c = \frac{1}{n} = \frac{1}{\sqrt{2}}$$

$$\theta_c = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

So the **critical angle for the prism is $\theta_c = 45^\circ$.**

Final answer. The angle of minimum deviation is 30 degrees and the critical angle is 45 degrees.

← *Back to Q237*

Q238

An equiconvex lens forms a two times enlarged real image when an object is kept 16 *cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept* 16 *cm in front of one of these lenses, then find the nature and position of the image formed.*

[2022 • Set 55-2-3]

SOLUTION

First, find the focal length of the original equiconvex lens. The object is at $u = -16$ **cm** and the image is real and two times enlarged, so $m = -2$ and $v = -m\,u$

$\cdot(-1)$; using $m = \frac{v}{u} = -2$ gives $v = -2u = -2(-16) = +32$ **cm** (real image on the opposite side).

Apply the thin lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{32} - \frac{1}{-16} = \frac{1}{f}$$

$$\frac{1}{32} + \frac{1}{16} = \frac{1}{32} + \frac{2}{32} = \frac{3}{32} = \frac{1}{f}$$

$$f = \frac{32}{3} \approx 10.67 \text{ cm}$$

For an equiconvex lens both faces have equal radius R, so the lens maker's formula gives

$$\frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (n-1)\frac{2}{R}$$

When this lens is cut into two identical plano-convex lenses, each piece has one curved face of radius R and one flat face ($R = \infty$):

$$\frac{1}{f'} = (n-1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \frac{(n-1)}{R}$$

Comparing, $\frac{1}{f'} = \frac{1}{2}\cdot\frac{1}{f}$, so each plano-convex lens has

$$f' = 2f = 2 \times \frac{32}{3} = \frac{64}{3} \approx 21.3 \text{ cm}$$

Now place the object again at $u = -16$ **cm** in front of one plano-convex lens. Since the object distance (16 cm) is now less than the focal length (21.3 cm), the object lies inside the focus and a virtual image is expected.

$$\frac{1}{v} = \frac{1}{f'} + \frac{1}{u} = \frac{3}{64} + \frac{1}{-16}$$

$$\frac{1}{v} = \frac{3}{64} - \frac{4}{64} = -\frac{1}{64}$$

$$v = -64 \text{ cm}$$

The negative sign shows the image is on the same side as the object (virtual). Hence **the image is virtual and erect, formed 64 cm in front of the plano-convex lens (on the object side).**

COMMON MISTAKE

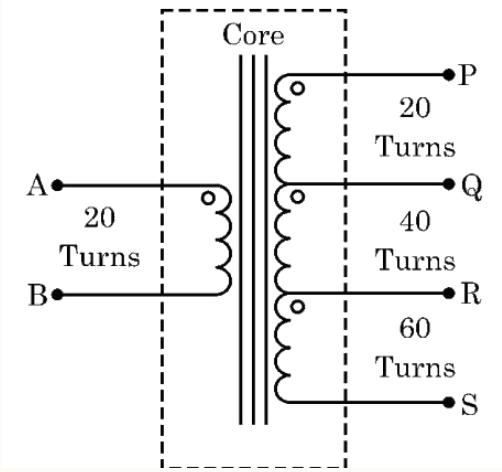
Cutting an equiconvex lens along the principal axis (into two plano-convex halves) doubles the focal length of each half, because each piece keeps one curved surface but loses the second. Do not confuse this with cutting perpendicular to the axis, which leaves f unchanged.

Final answer. The image is virtual and erect, formed 64 cm in front of the plano-convex lens on the same side as the object.

← *Back to Q238*

Q239

A ray is incident on a prism of material of refractive index $\sqrt{2}$ *at point M such that it grazes along* NC *after emerging from the prism, as shown in the figure.*



Find:

(a) the critical angle for the prism.

(b) the angle of refraction at face AB.

[2022 • Set 55-2-3]

SOLUTION

The refractive index of the prism material is $n = \sqrt{2}$, and the prism is equilateral, so $A = 60°$.

(a) Critical angle: For the prism material in air,

$$\theta_c = \frac{1}{n} = \frac{1}{\sqrt{2}}$$

$$\theta_c = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45°$$

So the **critical angle is $\theta_c = 45°$**.

(b) Angle of refraction at face AB: The ray enters the prism at point M on face AB, is refracted into the prism, and then strikes the second face. The condition that the ray grazes along NC after the prism means that at the second face the ray meets the surface exactly at the critical angle, so the angle of refraction at the second face is $r_2 = \theta_c = 45°$ (the grazing emergence corresponds to incidence at the critical angle on that face).

For a prism, the two angles of refraction inside the prism are related to the prism angle by:

$$r_1 + r_2 = A$$

Substituting $A = 60°$ and $r_2 = 45°$:

$$r_1 = A - r_2 = 60° - 45° = 15°$$

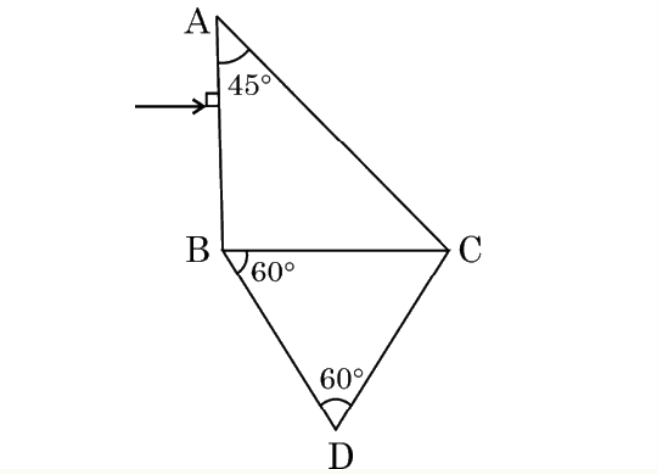
Hence the **angle of refraction at face AB is $r_1 = 15°$.**

Final answer. The critical angle is 45 degrees, and the angle of refraction at face AB is 15 degrees.

← *Back to Q239*

Q240

(a) Write two necessary conditions for total internal reflection.
(b) Two prisms ABC and DBC are arranged as shown in figure.



The critical angles for the two prisms with respect to air are 41.1° *and* 45° *respectively. Trace the path of the ray through the combination.*

[2022 • Set 55-3-1]

SOLUTION

(a) Two necessary conditions for total internal reflection:

1. Light must travel from an optically denser medium towards an optically rarer medium.

2. The angle of incidence in the denser medium must be greater than the critical angle for that pair of media.

(b) Tracing the ray through the two-prism combination: The critical angle of prism ABC is $\theta_c^{ABC} = 41.1°$ and that of prism DBC is $\theta_c^{DBC} = 45°$. In the arrangement shown, the ray strikes the common face BC at an angle of incidence of 45° (the 45° face of the right-angled prisms).

At the face BC inside prism ABC, the angle of incidence is 45°. Compare it with the critical angle of ABC:

$$45° > 41.1° = \theta_c^{ABC}$$

Since the incidence angle exceeds the critical angle of ABC, the ray undergoes total internal reflection inside prism ABC and is turned through 90°.

The reflected ray then travels on and meets a face of prism DBC at an angle of incidence of 45°. Compare with the critical angle of DBC:

$$45° = 45° = \theta_c^{DBC}$$

Since the incidence angle is not greater than the critical angle of DBC (it just equals it), total internal reflection does not occur in DBC; the ray refracts at the surface and grazes out (emerges along the face) into air.

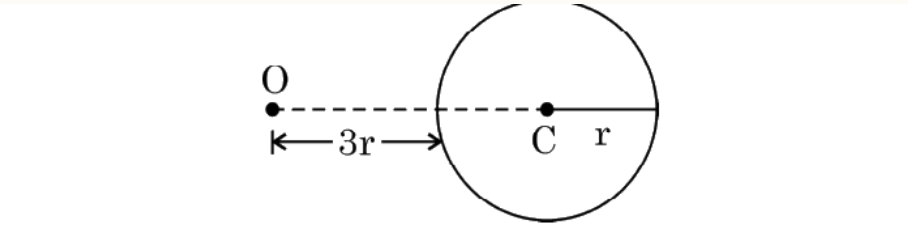
Hence the path is: the ray enters the combination, undergoes **total internal reflection inside prism ABC** (because **45°> 41.1°**), and then **refracts out of prism DBC** without total internal reflection (because 45° is not greater than its critical angle 45°).

Final answer. TIR needs denser-to-rarer travel with incidence above the critical angle; the ray totally internally reflects in prism ABC and then refracts out of prism DBC.

← Back to Q240

Q241

(a) An object is placed in front of a converging lens. Obtain the conditions under which the magnification produced by the lens is (i) negative and (ii) positive.
(b) A point object is placed at O in front of a glass sphere as shown in figure.



Show the formation of image by the sphere.

[2022 • Set 55-3-1]

SOLUTION

Part (a): The linear magnification of a thin lens is given by $m=\frac{v}{u}=\frac{h'}{h}$, where u and v are object and image distances measured with the sign convention.

(i) Negative magnification: For a real object placed at **u<0** in front of a converging lens, a real image forms on the opposite side, so **v>0**.

Then $m=\frac{v}{u}=\frac{(+)}{(-)}<0$, which means the image is real and inverted.

This happens when the object is placed beyond the focus, i.e. $|u|>f$.

(ii) Positive magnification: When the object is placed between the focus and the lens, i.e. $|u|$, **the image is virtual and forms on the same side as the object, so** $v<0$.

Then $m=\frac{v}{u}=\frac{(-)}{(-)}>0$, which means the image is virtual and erect.

So magnification is negative when $|u|>f$ (real, inverted image) and positive when $|u|$ **(virtual, erect image).**

Part (b): The image is formed by applying the refraction-at-a-spherical-surface formula twice, once at each surface of the sphere, in sequence.

The governing relation at a single spherical refracting surface is $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$.

At the first surface (light going from air into glass): take $n_1=1$ (air), $n_2=1.5$ (glass), $R=+R$ (centre of curvature on the far side), and u as the distance of O from the first surface.

Solving this gives the position of the intermediate image I_1 formed by the first surface inside the glass.

At the second surface (light going from glass back into air): now $n_1=1.5$, $n_2=1$, the radius is $-R$, and the object for this surface is the intermediate image I_1, with its distance measured from the second surface.

Solving again gives the position of the final image I.

Diagram description: Draw a circle for the glass sphere. Mark O to the left on the principal axis (the line through both poles and the centre C). Rays from O strike the left (first) surface, bend toward the axis on entering the denser glass, travel inside and converge toward I_1, then strike the right (second) surface where they bend again on leaving into air, finally meeting at the real image I on the right of the sphere. Label the first surface, second surface, centre C, O, I_1 and final image I.

Thus the sphere forms the final image by **two successive refractions**, the first surface giving I_1 and the second surface using I_1 as its object to give the final

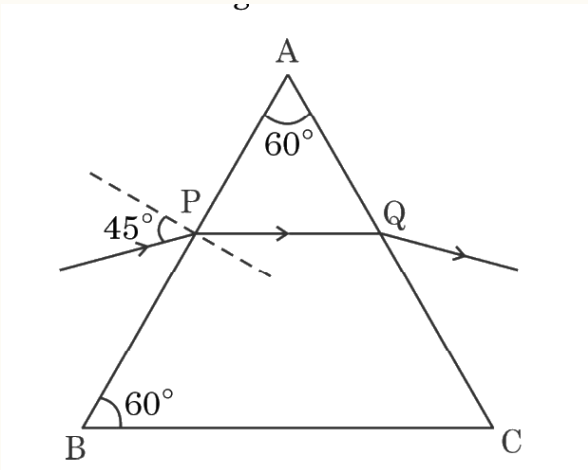
image I.

Final answer. Magnification is negative (real, inverted) for |u|>f and positive (virtual, erect) for |u|<f; the sphere forms the image by applying the refraction formula at both surfaces in sequence.

← *Back to Q241*

Q242

A ray of light is incident on a prism at an angle of 45° *and passes symmetrically as shown in the figure. Calculate:*
(a) the angle of minimum deviation,
(b) the refractive index of the material of the prism, and
(c) the angle of refraction at the point P.



[2022 • Set 55-4-1]

SOLUTION

When a ray passes symmetrically through a prism, the angle of incidence equals the angle of emergence, $\mathbf{i_1=i_2}$, and this is exactly the condition for **minimum deviation**. Here $\mathbf{i_1=45°}$ and the prism is equilateral, so $\boldsymbol{A=60°}$.

(a) At minimum deviation the deviation is given by $\boldsymbol{\delta_m=i_1+i_2-A}$.

Since $\mathbf{i_1=i_2=45°}$, $\boldsymbol{\delta_m=2i_1-A}$.

$$\delta_m=2(45°)-60°$$

$\delta_m = 90° - 60° = 30°$.

So the **angle of minimum deviation is 30°**.

(b) The refractive index of the prism material is found from the prism formula

$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}.$$

$$n = \frac{\sin\left(\frac{60°+30°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \frac{\sin 45°}{\sin 30°}.$$

$$n = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

So the **refractive index $n = \sqrt{2} \approx 1.414$**.

(c) At minimum deviation the ray inside the prism runs parallel to the base, and the two refraction angles are equal, with $r_1 + r_2 = A$.

Since $r_1 = r_2$, each equals $\frac{A}{2} = \frac{60°}{2} = 30°$.

So the **angle of refraction at P is 30°**.

> **TIP**
>
> Spot the symmetric passage first: $i_1 = i_2$ instantly tells you the prism is at minimum deviation, which unlocks both $\delta_m = 2i - A$ and n directly.

Final answer. Angle of minimum deviation = 30 °, refractive index = √2 ≈ 1.414, and angle of refraction at P = 30 °.

← *Back to Q242*

Q243

(i) Draw a labelled ray diagram showing the formation of the image at infinity by an astronomical telescope.
(ii) A telescope consists of an objective of focal length 150 cm and an eyepiece of focal length 6.0 cm. If the final image is formed at infinity, then calculate:

(I) the length of the tube in this adjustment, and
(II) the magnification produced.

[2022 • Set 55-4-1]

SOLUTION

Part (i): Ray diagram for an astronomical telescope with the final image at infinity (normal adjustment).

Draw a long horizontal principal axis. On the left place a large convex lens, the objective O of focal length f_o; on the right place a smaller convex lens, the eyepiece E of focal length f_e. The two lenses are separated by the tube length $\boldsymbol{L=f_o+f_e}$, so the common focus point between them lies at the focal point of both.

A parallel beam from a distant object arrives at O inclined at a small angle α to the axis. The objective brings this beam to a real, inverted, diminished image I_1 at its focus F_o, which is also at the focus F_e of the eyepiece. The eyepiece then sends out a parallel beam (because I_1 is at its focus), inclined at the larger angle β to the axis. Mark the rays leaving E as parallel, so the **final image is at infinity**, viewed by a relaxed eye. Label f_o, f_e, I_1, the angles α at the objective and β at the eyepiece.

Part (ii)(I): In normal adjustment the length of the tube equals the sum of the two focal lengths.

$$L=f_o+f_e$$

$$L=150\text{ cm}+6.0\text{ cm}$$

$\boldsymbol{L=156\text{ cm}}$.

So the **length of the tube is 156 cm**.

Part (ii)(II): The magnifying power for the final image at infinity is $\boldsymbol{m=\frac{f_o}{f_e}}$.

$$m=\frac{150\text{ cm}}{6.0\text{ cm}}$$

$\boldsymbol{m=25}$.

So the **magnification produced is 25** (the image is inverted relative to the object).

Final answer. Tube length = 156 cm and magnification = 25.

← Back to Q243

Q244

(i) Draw a labelled ray diagram showing the formation of the image at least distance of distinct vision by a compound microscope.
(ii) A small object is placed at a distance of 3.0 cm from a magnifier of focal length 4.0 cm. Find:
(I) the position of the image formed, and
(II) the linear magnification produced.

[2022 • Set 55-4-1]

SOLUTION

Part (i): Ray diagram for a compound microscope with the final image at the least distance of distinct vision (D=**25 cm**).

Draw a horizontal principal axis with two convex lenses: the objective O (short focal length f_o) on the left and the eyepiece E (focal length f_e) on the right. Place a tiny object AB just beyond the focus F_o of the objective. Two rays from the top of the object pass through the objective and form a real, inverted, magnified intermediate image I_1 between the eyepiece and its focus F_e.

This intermediate image I_1 acts as the object for the eyepiece, which works as a simple magnifier and forms a final virtual, inverted (relative to the original object), magnified image I at the near point, a distance D from the eye. Label the object AB, f_o, f_e, I_1, the final image I, the tube length L between the objective image and eyepiece, and the distance D.

The total magnification is $M=\frac{L}{f_o}\left(1+\frac{D}{f_e}\right)$.

Part (ii): Here the magnifier (a single convex lens) is used. Given u=**−3.0 cm**, f=**+4.0 cm**.

Using the thin lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

$$\frac{1}{v}-\frac{1}{-3}=\frac{1}{4}$$

$$\frac{1}{v}=\frac{1}{4}-\frac{1}{3}$$

$$\frac{1}{v}=\frac{3-4}{12}=-\frac{1}{12}$$

$v=-12$ cm.

(I) So the **image is formed at 12 cm on the same side as the object** (the negative sign shows it is virtual).

(II) The linear magnification is $m=\frac{v}{u}$.

$m=\frac{-12}{-3}=+4$.

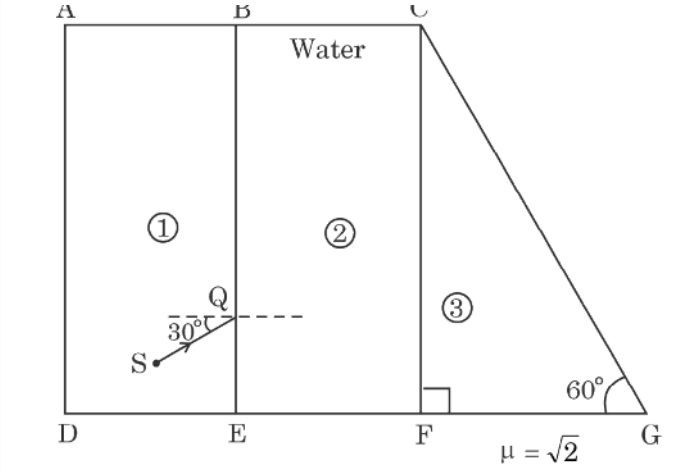
So the **magnification is +4**, meaning the image is virtual, erect and four times the size of the object.

Final answer. Image is virtual, formed 12 cm from the lens on the object side, with magnification +4.

← *Back to Q244*

Q245

The figure shows a water column BCFE surrounded by two media (1) and (2) of the same refractive index $\sqrt{2}$. *A ray of light from a point source S is incident on surface BE at an angle of* 30°.
(a) Trace the path of the ray through media (1) and (2) as it emerges out of face CG.
(b) Find the angle of emergence at face CG.



[2022 • Set 55-4-2]

SOLUTION

The water column BCFE (refractive index $\mathbf{n_w = \frac{4}{3}}$) is sandwiched between media (1) and (2), both of the same refractive index $\boldsymbol{n = \sqrt{2}}$. A ray from S strikes face BE at 30°.

Part (a): Tracing the path.

At face BE the light goes from medium (1) into water. Applying Snell's law $\mathbf{n \sin 30° = n_w \sin r}$.

$$\mathbf{\sin r = \frac{n \sin 30°}{n_w} = \frac{\sqrt{2} \times 0.5}{4/3} = \frac{0.707}{1.333} = 0.530.}$$

So $\boldsymbol{r \approx 32°}$, and the ray bends toward the normal on entering the denser water.

The ray then travels across the water column and meets face CG, where it passes from water into medium (2). Since BE and CG are the parallel side faces of the column, the angle of incidence inside the water at CG equals $\boldsymbol{r \approx 32°}$.

At CG, going from water back into a medium of index $\sqrt{2}$, the ray bends away from the normal and emerges into medium (2).

Diagram description: Draw the vertical water column with parallel faces BE (left) and CG (right). The ray from S hits BE at 30°, refracts toward the normal to about 32° inside the water, crosses to CG, and emerges out of CG bending away from the normal, finally running parallel to the original incident ray.

Part (b): The angle of emergence at CG.

Apply Snell's law at CG: $\mathbf{n_w \sin r = n \sin e}$, where $\boldsymbol{n = \sqrt{2}}$.

$$\frac{4}{3}\sin 32^\circ=\sqrt{2}\sin e$$

Combining both surfaces, the water term cancels because the entry and exit media have the same index $\sqrt{2}$: $\sqrt{2}\sin 30^\circ=\sqrt{2}\sin e$.

$$\sin e=\sin 30^\circ$$

$e=30^\circ$.

So the **angle of emergence at CG is 30°**, and the emergent ray is parallel to the incident ray because the first and last media have equal refractive index.

CONCEPT

When the first and last media share the same refractive index, the slab refractions cancel: the exit angle always equals the entry angle, no matter what is sandwiched between.

Final answer. The angle of emergence at face CG is 30 °, with the emergent ray parallel to the incident ray.

← *Back to Q245*

Q246

A diver looking up through water $(\mu = \frac{5}{3})$ *sees the outside world contained in a circular area on the surface of water. If the diver's eyes are* $\sqrt{7}$ *m below the surface of water, then calculate the area of the circle.*

[2022 • Set 55-4-3]

SOLUTION

Because of total internal reflection, a diver looking up can see the whole outside world only within a cone whose half-angle is the critical angle θ_c. Light from beyond this cone is totally internally reflected and never reaches the eye.

The critical angle is given by $\theta_c=\frac{1}{\mu}$.

$\theta_c = \frac{1}{5/3} = \frac{3}{5}$.

From this, $\theta_c = \sqrt{1-(\frac{3}{5})^2} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$.

So $\theta_c = \frac{\theta_c}{\theta_c} = \frac{3/5}{4/5} = \frac{3}{4}$.

The visible patch on the surface is a circle of radius r, where the depth is $h = \sqrt{7}$ m and $r = h\theta_c$.

$r = \sqrt{7} \times \frac{3}{4} = \frac{3\sqrt{7}}{4}$ m.

The area of the circle is $A = \pi r^2$.

$$A = \pi(\frac{3\sqrt{7}}{4})^2 = \pi \times \frac{9 \times 7}{16}$$

$$A = \frac{63\pi}{16} \text{ m}^2$$

$A \approx 12.4 \text{ m}^2$.

So the **area of the circle through which the diver sees the outside world is** $\frac{63\pi}{16} \approx 12.4 \text{ m}^2$.

Final answer. The area of the circle is $63\pi/16 \approx 12.4$ m^2.

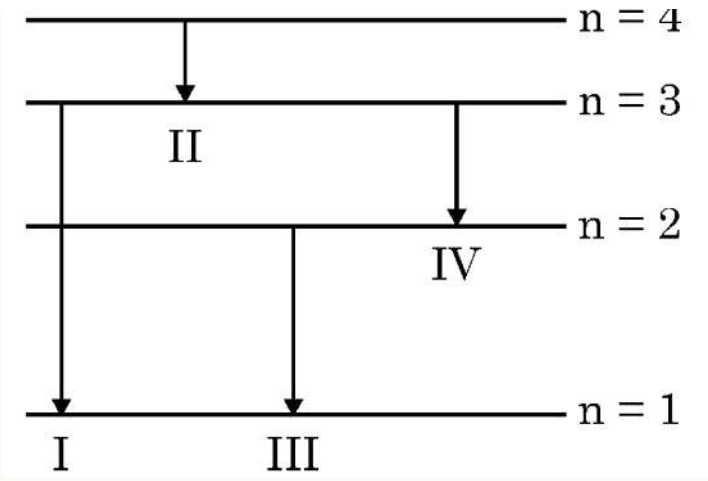
← *Back to Q246*

Q247

A ray of light passes through a prism of refractive index $\sqrt{2}$ as shown in the figure. Find:

(i) The angle of incidence ($\angle r_2$) at face AC.

(ii) The angle of minimum deviation for this prism.



[2022 • Set 55-5-1]

SOLUTION

The prism has refractive index $n=\sqrt{2}$. The ray enters normally at face AB, so the angle of refraction at the first face is $r_1=0$. The prism (refracting) angle is $A=45°$.

Part (i): The angle of incidence at the second face AC.

For any prism the two internal angles satisfy $r_1+r_2=A$.

$$0+r_2=45°$$

$r_2=45°$.

So the **angle of incidence at face AC is $r_2=45°$**.

We check this against the critical angle: $\theta_c=\frac{1}{n}=\frac{1}{\sqrt{2}}$, giving $\theta_c=45°$. Since $r_2=\theta_c=45°$, the ray strikes AC exactly at the critical angle and grazes along the surface.

Part (ii): The angle of minimum deviation. Here the prism angle is $A=45°$ (not 60°), and we use the prism formula $n=\frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$.

$\sin(\frac{A+\delta_m}{2})=n\sin\frac{A}{2}=\sqrt{2}\sin 22.5°$.

$\sin(\frac{45°+\delta_m}{2})=\sqrt{2}\times 0.3827=0.5412$.

$$\frac{45°+\delta_m}{2}=\sin^{-1}(0.5412)=32.77°$$

$$45^\circ + \delta_m = 65.54^\circ$$

$\delta_m = 20.54^\circ \approx 20.5^\circ$.

So the **angle of minimum deviation is approximately 20.5°**.

Final answer. The angle of incidence at face AC is $r_2 = 45^\circ$ (equal to the critical angle), and the angle of minimum deviation is approximately 20.5 °.

← *Back to Q247*

Q248

With the help of a ray diagram explain the working of a reflecting telescope. Mention two advantages of a reflecting telescope over a refracting telescope.

[2022 • Set 55-5-1]

SOLUTION

A reflecting telescope uses a large concave mirror as its objective instead of a lens. The common design is the Cassegrain telescope.

Working: Light from a distant object enters the tube as a parallel beam and falls on a large concave parabolic primary mirror at the back. This mirror reflects and converges the beam toward a small convex secondary mirror placed near the front of the tube, on the axis. The secondary mirror reflects the converging beam back down the tube and through a small hole at the centre of the primary mirror. Just behind this hole sits the eyepiece, which magnifies the real image and lets the eye view a greatly enlarged final image. In normal adjustment the final image is at infinity for a relaxed eye.

Ray diagram description: Draw the tube as a horizontal cylinder. At the back (right) place a large concave parabolic mirror (primary), labelled objective. Parallel rays from the distant object enter from the left, reflect off the primary, and converge toward a small convex secondary mirror near the front (left). The secondary reflects them back, sending the beam through a central hole in the primary to a small eyepiece behind it, where the eye views the image. Label: primary concave mirror, secondary convex mirror, hole, eyepiece, eye.

The angular magnification is $m = \frac{f_o}{f_e}$, where f_o is the focal length of the primary mirror and f_e that of the eyepiece.

Two advantages of a reflecting telescope over a refracting telescope:

1. **No chromatic aberration**, because a mirror reflects all wavelengths of light through the same angle, unlike a lens which refracts different colours differently.

2. **Larger apertures and brighter, sharper images** are possible, because a heavy mirror can be supported firmly from behind over its whole back surface, whereas a large lens can only be held at its rim and sags under its own weight. Mirrors are also free from spherical aberration when made parabolic.

Final answer. A reflecting telescope uses a concave primary mirror with a convex secondary mirror feeding the eyepiece; its main advantages are no chromatic aberration and the ability to use large, bright apertures.

← *Back to Q248*

Q249

(i) Define SI unit of power of a lens.
(ii) A plano convex lens is made of glass of refractive index 1.5. *The radius of curvature of the convex surface is* 25 cm. *Calculate the focal length of the lens.*
(iii) If an object is placed 50 cm *in front of the lens, find the nature and position of the image formed.*

[2022 • Set 55-5-1]

SOLUTION

Part (i): The SI unit of power of a lens is the **dioptre (D)**. One dioptre is the power of a lens whose focal length is one metre. In general $P = \frac{1}{f}$, where f is the focal length in metres, so $\mathbf{1\ D = 1\ m^{-1}}$.

Part (ii): For a plano-convex lens, one surface is flat ($\mathbf{R_2 = \infty}$) and the other is convex with radius $\mathbf{R_1 = +25\ cm}$, glass index $\mathbf{n = 1.5}$.

Using the lens maker's formula $\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$.

$$\frac{1}{f}=(1.5-1)(\frac{1}{25}-\frac{1}{\infty})$$

$$\frac{1}{f}=0.5\times\frac{1}{25}=\frac{1}{50}\ \text{cm}^{-1}$$

$f=50$ cm.

So the **focal length is 50 cm** (a converging lens).

Part (iii): The object is placed 50 cm in front, so $u=-50$ cm and $f=+50$ cm.

Using the thin lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{50}+\frac{1}{-50}$$

$$\frac{1}{v}=\frac{1}{50}-\frac{1}{50}=0$$

$v\to\infty$.

Since the object sits exactly at the focal point, the refracted rays emerge parallel and the **image is formed at infinity**.

Final answer. Power is measured in dioptre **(D)**; the lens focal length is 50 cm, and with the object at 50 cm (the focus) the image forms at infinity.

← Back to Q249

Q250

With the help of a ray diagram, show how a compound microscope forms a magnified image of a tiny object, at least distance of distinct vision. Hence derive an expression for the magnification produced by it.

[2022 • Set 55-5-3]

SOLUTION

Ray diagram (final image at least distance of distinct vision D):

Draw a horizontal principal axis with two convex lenses, the objective O (small focal length f_o) on the left and the eyepiece E (focal length f_e) on the right, separated by the tube length L. Place a tiny object AB just beyond the objective's focus F_o. Rays from B pass through the objective and form a real, inverted, magnified intermediate image $A'B'$ (I_1) lying between the eyepiece and its focus F_e. This image acts as the object for the eyepiece, which behaves as a simple magnifier and produces a virtual, magnified final image $A''B''$ (I) at the near point, a distance D from the eye. Label AB, f_o, f_e, $A'B'$, the final image $A''B''$, the tube length L and the distance D.

Derivation of magnification: The total magnification is the product of the objective and eyepiece magnifications, $M = m_o \times m_e$.

The objective produces linear magnification $m_o = \frac{h'}{h} = \frac{v_o}{u_o}$. Since the object is placed close to the focus and the image is formed near the eyepiece, this is approximately $m_o = \frac{L}{f_o}$, where L is the tube length (the separation of the image $A'B'$ from the objective).

The eyepiece acts as a simple microscope with the final image at the near point D, for which the angular magnification is $m_e = 1 + \frac{D}{f_e}$.

Multiplying the two gives $M = m_o \times m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$.

So the **magnification of the compound microscope is** $M = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$.

For high magnification both f_o and f_e should be small and the tube length L large.

Final answer. The compound microscope produces magnification $M = (L/f_o)(1 + D/f_e)$ at the least distance of distinct vision.

← *Back to Q250*

Q251

What is the difference in the construction of an astronomical telescope and a compound microscope? The focal lengths of the objective and eyepiece of a compound

microscope are 1.25 *cm and* 5.0 *cm, respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of* 30 *when the final image is formed at the near point.*

[2020 • Set 55-2-1]

SOLUTION

Difference in construction. In an astronomical telescope the objective lens has a large focal length and a large aperture, because it must gather light from very distant objects and form a bright real image; the eyepiece has a comparatively short focal length. In a compound microscope both the objective and the eyepiece have very short focal lengths, and the focal length of the objective is even smaller than that of the eyepiece, because the object is placed very close to (just beyond) the focus of the objective.

Numerical part. For a compound microscope, $\mathbf{f_o = 1.25\ cm}$, $\mathbf{f_e = 5.0\ cm}$, and the final image is at the near point $\boldsymbol{D = 25\ cm}$.

The angular magnification of the eyepiece (image at near point) is

$$m_e = 1 + \frac{D}{f_e}$$

$$m_e = 1 + \frac{25}{5.0} = 6$$

The total magnification is $\boldsymbol{M = m_o \times m_e}$, so the objective magnification is

$$m_o = \frac{M}{m_e} = \frac{30}{6} = 5$$

For the objective, $\mathbf{m_o = \frac{v_o}{u_o}}$. Taking magnitudes, $\mathbf{v_o = 5\,|u_o|}$.

Applying the lens formula to the objective, $\mathbf{\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}}$. Using magnitudes with u_o negative and v_o positive,

$$\frac{1}{5\,|u_o|} + \frac{1}{|u_o|} = \frac{1}{1.25}$$

$$\frac{1+5}{5\,|u_o|} = 0.8$$

$$\frac{6}{5\,|u_o|} = 0.8$$

$$|u_o| = \frac{6}{5 \times 0.8} = \frac{6}{4} = 1.5 \text{ cm}$$

So the object must be placed at **1.5 cm** in front of the objective lens.

Final answer. The object should be placed 1.5 cm in front of the objective lens.

← *Back to Q251*

Q252

(a) Using the necessary ray diagram, derive the mirror formula for a concave mirror.
(b) In the magnified image of a measuring scale (with equidistant markings) lying along the principal axis of a concave mirror, the markings are not equidistant. Explain.

[2020 • Set 55-3-1]

SOLUTION

(a) Mirror formula for a concave mirror. Consider a concave mirror of pole P, centre of curvature C and principal focus F. An object AB is placed beyond C on the principal axis, with B on the axis. A ray from A parallel to the principal axis reflects through F; a second ray from A passing through C retraces its path. These two reflected rays meet at A', forming a real, inverted image $A'B'$ between C and F, with B' on the axis.

In the right-angled triangles $A'B'F$ and the small triangle at the pole formed by the parallel ray hitting near P (taking the aperture small), the triangles $A'B'F$ and MPF are similar (where M is the point where the parallel ray strikes the mirror, **MP ≈ AB**). Hence

$$\frac{A'B'}{AB} = \frac{B'F}{PF}$$

Also triangles $A'B'C$ and ABC are similar, giving

$$\frac{A'B'}{AB} = \frac{B'C}{BC}$$

Therefore $\frac{B'F}{PF} = \frac{B'C}{BC}$.

Using $B'F = PB' - PF$, $B'C = PC - PB'$, $BC = PB - PC$, and applying the sign convention with $PB = -u$, $PB' = -v$, $PF = -f$, $PC = -R = -2f$,

$$\frac{(-v) - (-f)}{-f} = \frac{(-2f) - (-v)}{(-u) - (-2f)}$$

On simplifying this proportion we obtain

$$\frac{1}{v} + \frac{1}{u} = \frac{2}{R} = \frac{1}{f}$$

which is the required **mirror formula**.

(b) Non-uniform image of a scale. The linear (transverse) magnification of a mirror is $m = -\frac{v}{u}$, but along the axis the longitudinal magnification is

$$m_{long} = -\frac{dv}{du} = -\frac{v^2}{u^2} = -m^2$$

Since v changes with u, the value of m (and hence m^2) is different for markings at different distances. The markings nearer the mirror have a different magnification from those farther away, so equal object spacings are imaged as **unequal spacings**. That is why the markings of the magnified scale are not equidistant.

Final answer. The mirror formula is $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$; the image markings are unequal because longitudinal magnification $-m^2$ varies along the axis.

← *Back to Q252*

Q253

Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm*, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is* 50 cm *from the mirror, find the distance of Q from the mirror.*

[2020 • Set 55-4-1]

SOLUTION

For a concave mirror the magnitude of magnification for a real image is $|m| = \frac{v}{|u|}$, and image size = $|m| \times$ object size.

Object P: $u = -50$ cm, $f = -20$ cm. Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$,

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-50}$$

$$\frac{1}{v} = -\frac{1}{20} + \frac{1}{50} = \frac{-5+2}{100} = -\frac{3}{100}$$

$$v = -\frac{100}{3} \approx -33.3 \text{ cm}$$

$$|m_P| = \frac{|v|}{|u|} = \frac{100/3}{50} = \frac{2}{3}$$

Both images have equal size. Let object sizes be h_P and h_Q with $\mathbf{h_P = 3\,h_Q}$. Equal image sizes means

$$|m_P|\, h_P = |m_Q|\, h_Q$$

$$|m_Q| = |m_P| \times \frac{h_P}{h_Q} = \frac{2}{3} \times 3 = 2$$

For Q (real image), $|\mathbf{m_Q}| = \frac{\mathbf{v_Q}}{|\mathbf{u_Q}|} = \mathbf{2}$, so $\mathbf{v_Q = 2\,|u_Q|}$ (a real image is in front of the mirror, so $\mathbf{v_Q = -2|u_Q|}$, $\mathbf{u_Q = -|u_Q|}$).

Applying the mirror formula with magnitudes,

$$\frac{1}{2\,|u_Q|} + \frac{1}{|u_Q|} = \frac{1}{20}$$

$$\frac{1+2}{2\,|u_Q|} = \frac{1}{20}$$

$$\frac{3}{2\,|u_Q|} = \frac{1}{20}$$

$$|u_Q| = \frac{3 \times 20}{2} = 30 \text{ cm}$$

So object Q is at a distance of **30 cm** from the mirror.

Final answer. Object Q is placed 30 cm from the concave mirror.

← Back to Q253

Q254

An object is placed in front of a concave mirror of focal length 12 cm. *There are two possible positions of the object for which the image formed is three times the size of the object.*
(a) Draw the ray diagram for each case, and
(b) Find the distance between the two positions of the object.

[2020 • Set 55-4-2]

SOLUTION

A concave mirror of focal length $f = -12$ **cm** produces an image three times the object size in two ways: a real, inverted image (case 1) and a virtual, erect image (case 2).

(a) Ray diagrams.

Case 1 (real image, $m = -3$): The object stands between the focus F and the centre of curvature C, with its foot on the axis. A ray parallel to the axis reflects through F; a ray through C retraces back. They meet beyond C to form a magnified, inverted, real image on the same side as the object.

Case 2 (virtual image, $m = +3$): The object stands between the pole P and the focus F. A ray parallel to the axis reflects through F; a ray directed towards C retraces. The reflected rays diverge, and their backward extensions meet behind the mirror to form a magnified, erect, virtual image.

(b) Distance between the two object positions.

Case 1, real image: $m = -\frac{v}{u} = -3 \Rightarrow v = 3u$.

Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ with $v = 3u$,

$$\frac{1}{3u} + \frac{1}{u} = \frac{1}{-12}$$

$$\frac{1+3}{3u} = -\frac{1}{12}$$

$$\frac{4}{3u} = -\frac{1}{12}$$

$$u = -\frac{4 \times 12}{3} = -16 \text{ cm}$$

Case 2, virtual image: $m = -\frac{v}{u} = +3 \Rightarrow v = -3u$.

$$\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$$

$$\frac{-1+3}{3u} = -\frac{1}{12}$$

$$\frac{2}{3u} = -\frac{1}{12}$$

$$u = -\frac{2 \times 12}{3} = -8 \text{ cm}$$

Distance between the two positions of the object

$$= |-16 - (-8)| = 8 \text{ cm}$$

The two object positions are **8 cm** apart.

Final answer. The two object positions are 8 cm apart (at 16 cm and 8 cm from the mirror).

← *Back to Q254*

Q255

A concave mirror forms a real image of an object kept at a distance 9 cm *from it. If the object is taken away from the mirror by* 6 cm, *the image size reduces to* $\frac{4}{9}$ *th of its previous size. Find the focal length of the mirror.*

[2020 • Set 55-4-3]

SOLUTION

For a mirror the magnification can be written as $m = \dfrac{f}{f-u}$, and for a real image the image size scales as $|m|$.

Initially the object is at $\mathbf{u_1 = -9\ cm}$. After moving it 6 cm farther, $\mathbf{u_2 = -15\ cm}$. The new image size is $\dfrac{4}{9}$ of the old one, so

$$\frac{m_2}{m_1} = \frac{4}{9}$$

Using $m = \dfrac{f}{f-u}$,

$$\frac{m_2}{m_1} = \frac{\dfrac{f}{f-u_2}}{\dfrac{f}{f-u_1}} = \frac{f-u_1}{f-u_2}$$

Substituting $\mathbf{u_1 = -9}$, $\mathbf{u_2 = -15}$,

$$\frac{f-(-9)}{f-(-15)} = \frac{f+9}{f+15} = \frac{4}{9}$$

Cross-multiplying,

$$9(f+9) = 4(f+15)$$

$$9f + 81 = 4f + 60$$

$$5f = 60 - 81 = -21$$

$$f = -4.2\ \text{cm}$$

The negative sign confirms a concave mirror. The **focal length is 4.2 cm**.

COMMON MISTAKE

tip

Final answer. The focal length of the concave mirror is 4.2 cm.

← *Back to Q255*

Q256

An optical instrument uses a lens of power 100 *D for objective lens and* 50 *D for its eyepiece. When the tube length is kept at* 25 *cm, the final image is formed at infinity.*
(a) Identify the optical instrument.
(b) Calculate the magnification produced by the instrument.

[2020 • Set 55-5-1]

SOLUTION

(a) Identifying the instrument. The focal lengths from the given powers are

$$f_o = \frac{1}{P_o} = \frac{1}{100}\ m = 0.01\ m = 1\ \text{cm}$$

$$f_e = \frac{1}{P_e} = \frac{1}{50}\ m = 0.02\ m = 2\ \text{cm}$$

Both focal lengths are very small and the object is placed close to the objective, which is the construction of a **compound microscope**.

(b) Magnification. For a compound microscope with the final image at infinity (normal adjustment), the eyepiece angular magnification is $\frac{D}{f_e}$ and the objective magnification is $\frac{L}{f_o}$, where L is the tube length and $\boldsymbol{D = 25\ \text{cm}}$.

$$M = \frac{L}{f_o} \times \frac{D}{f_e}$$

$$M = \frac{25}{1} \times \frac{25}{2}$$

$$M = 25 \times 12.5$$

$$M = 312.5$$

The instrument is a compound microscope with a magnification of about **312.5**.

Final answer. The instrument is a compound microscope; its magnification is about 312.5.

← *Back to Q256*

Q257

An optical instrument uses an objective lens of power 100 *D and an eyepiece of power* 40 *D. The final image is formed at infinity when the tube length of the instrument is kept at* 20 *cm.*
(a) Identify the optical instrument.
(b) Calculate the angular magnification produced by the instrument.

[2020 • Set 55-5-3]

SOLUTION

(a) Identifying the instrument. Converting the powers to focal lengths,

$$f_o = \frac{1}{P_o} = \frac{1}{100}\ m = 0.01\ m = 1\ \text{cm}$$

$$f_e = \frac{1}{P_e} = \frac{1}{40}\ m = 0.025\ m = 2.5\ \text{cm}$$

Both focal lengths are short, with a small objective focal length and the object placed near it, which is the construction of a **compound microscope**.

(b) Angular magnification. With the final image at infinity (normal adjustment), tube length $L = 20$ **cm** and $D = 25$ **cm**,

$$M = \frac{L}{f_o} \times \frac{D}{f_e}$$

$$M = \frac{20}{1} \times \frac{25}{2.5}$$

$$M = 20 \times 10$$

$$M = 200$$

The instrument is a compound microscope with an angular magnification of **200**.

Final answer. The instrument is a compound microscope; its angular magnification is 200.

← *Back to Q257*

Q258

Draw a labelled ray diagram of an astronomical telescope in the near point adjustment position.
A giant refracting telescope at an observatory has an objective lens of focal length 15 *m and an eyepiece of focal length* 1.0 cm. *If this telescope is used to view the Moon, find the diameter of the image of the Moon formed by the objective lens. The diameter of the Moon is* 3.48×10^6 *m, and the radius of lunar orbit is* 3.8×10^8 *m.*

[2019 • Set 55-1-1]

SOLUTION

Ray diagram (near point adjustment). Draw the objective lens O (large focal length f_o) and the eyepiece E (small focal length f_e) on a common axis. Parallel rays from a distant object enter the objective at a small angle α to the axis and form a real, inverted image $A'B'$ in the focal plane of the objective. This image lies just inside the focus of the eyepiece. The eyepiece, acting as a magnifier, forms the final magnified, virtual, inverted image at the near point, a distance $\boldsymbol{D = 25\ cm}$ from the eyepiece. Mark the angle α subtended at the objective and the larger angle β subtended by the final image at the eye.

Numerical part. The Moon subtends an angle at the objective equal to

$$\alpha = \frac{\text{diameter of Moon}}{\text{radius of lunar orbit}}$$

$$\alpha = \frac{3.48 \times 10^6}{3.8 \times 10^8}\ \text{rad}$$

$$\alpha = 9.16 \times 10^{-3}\ \text{rad}$$

The objective forms the image in its focal plane, so the diameter d of the Moon's image is

$$d = f_o \times \alpha$$

$$d = 15 \times 9.16 \times 10^{-3}\ m$$

$$d = 0.137\ m = 13.7\ \text{cm}$$

The diameter of the image of the Moon formed by the objective is about **13.7 cm**.

Final answer. The image of the Moon formed by the objective is about 0.137 m (13.7 cm) in diameter.

← *Back to Q258*

Q259

A triangular prism of refracting angle 60° *is made of a transparent material of refractive index* $2/\sqrt{3}$. *A ray of light is incident normally on the face* KL *as shown in the figure. Trace the path of the ray as it passes through the prism and calculate the angle of emergence and angle of deviation.*

[2019 • Set 55-2-1]

SOLUTION

Given an equilateral prism with refracting angle $A = 60°$ and refractive index $n = \frac{2}{\sqrt{3}}$.

Path of the ray. The ray strikes face KL normally, so it enters without bending (angle of refraction at the first face $r_1 = 0$). It travels straight to the second refracting face. Since the angle of the prism is 60°, the geometry gives the angle of incidence inside the prism at the second face as

$$r_2 = A - r_1 = 60° - 0° = 60°$$

Critical angle. The critical angle for the material is

$$\sin\theta_c = \frac{1}{n} = \frac{\sqrt{3}}{2}$$

$$\theta_c = 60°$$

Since the angle of incidence at the second face equals the critical angle ($r_2 = \theta_c = 60°$), the ray is refracted exactly along the second face, so it just grazes the surface. The **angle of emergence is 90°** (grazing emergence).

Angle of deviation. The total deviation is the sum of the deviations at the two faces. At the first face there is no deviation ($\mathbf{r_1 = 0}$, normal incidence). At the second face the ray bends from the normal direction (60° inside) to 90° at emergence, so

$$\delta = (i_1 - r_1) + (e - r_2)$$

$$\delta = (0 - 0) + (90° - 60°)$$

$$\delta = 30°$$

The ray emerges grazing the second face with **angle of emergence 90° and angle of deviation 30°**.

CONCEPT

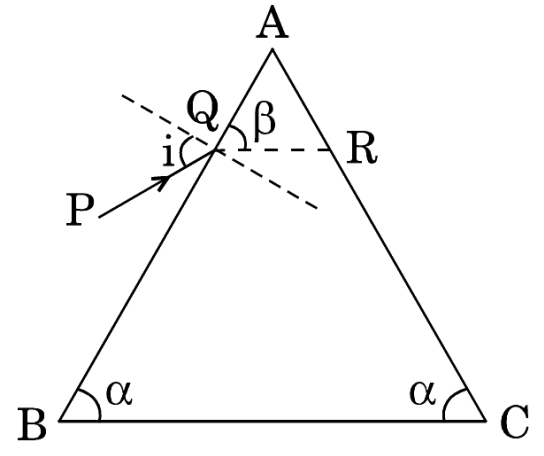
concept

Final answer. The angle of emergence is 90 degrees (grazing emergence) and the angle of deviation is 30 degrees.

← *Back to Q259*

Q260

A ray of light incident on the face AB *of an isosceles triangular prism makes an angle of incidence (i) and deviates by angle* β *as shown in the figure. Show that in the position of minimum deviation* $\mathbf{2\beta = \alpha}$. *Also find out the condition when the refracted ray* QR *suffers total internal reflection.*



[2019 • Set 55-2-2]

SOLUTION

Setting up the prism geometry. Let the refracting (apex) angle of the isosceles prism be α, the angle of incidence on face AB be i, the angle of emergence be e, and the internal angles of refraction at the two faces be r_1 and r_2. In the figure β is the deviation the ray suffers at the first face, that is $\beta = i - r_1$.

The standard prism relations are

$$r_1 + r_2 = \alpha$$

$$\delta = (i - r_1) + (e - r_2) = i + e - \alpha$$

Condition at minimum deviation. At minimum deviation the ray passes symmetrically through the prism, so

$$i = e \text{ and } r_1 = r_2 = r$$

From $r_1 + r_2 = \alpha$ this gives

$$2r = \alpha \Rightarrow r = \frac{\alpha}{2}$$

By symmetry the total deviation is shared equally between the two faces, so the single face deviation β equals half the total minimum deviation,

$$\delta_m = 2\beta$$

Inside the prism the refracted ray QR runs parallel to the base. The geometry of the symmetric path makes the angle that QR turns through at each face equal to the refraction angle there, so the per face deviation equals r,

$$\beta = r = \frac{\alpha}{2}$$

Multiplying both sides by 2,

$$2\beta = \alpha$$

which is the required result, $\mathbf{2\beta = \alpha}$.

Condition for total internal reflection of QR. The ray QR strikes the second face at the internal angle $\mathbf{r_2 = \alpha - r_1}$. Total internal reflection occurs when this angle exceeds the critical angle θ_c, where

$$\theta_c = \frac{1}{n}$$

Hence the condition is

$$r_2 > \theta_c$$

$$\alpha - r_1 > \theta_c$$

So QR suffers total internal reflection when $\mathbf{(\alpha - r_1) > \sin^{-1}(\frac{1}{n})}$.

Final answer. At minimum deviation $\mathbf{2\beta = \alpha}$; the ray **QR** undergoes total internal reflection when $\mathbf{(\alpha - r_1) > \theta_c}$, where $\mathbf{\theta_c = \frac{1}{n}}$.

← *Back to Q260*

Q261

(a) When a convex lens of focal length 30 cm *is in contact with a concave lens of focal length* 20 cm*, find out if the system is converging or diverging.*
(b) Obtain the expression for the angle of incidence of a ray of light which is incident on the face of a prism of refracting angle A so that it suffers total internal reflection at the other face. (Given the refractive index of the glass of the prism is μ.*)*

[2019 • Set 55-2-3]

SOLUTION

(*a*) For two thin lenses in contact, the equivalent focal length F obeys the combination rule.

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

Using sign convention, the converging (convex) lens has $\mathbf{f_1=+30\ cm}$ and the diverging (concave) lens has $\mathbf{f_2=-20\ cm}$.

$$\frac{1}{F}=\frac{1}{30}+\frac{1}{(-20)}=\frac{1}{30}-\frac{1}{20}$$

$$\frac{1}{F}=\frac{2-3}{60}=-\frac{1}{60}\ \text{cm}^{-1}$$

$$F=-60\ \text{cm}$$

Since F is negative, the combination behaves as a **diverging** system of focal length 60 cm.

(*b*) Let the ray strike the first face at angle of incidence i and refract at angle r_1. At the second face the refraction angle is r_2. For a prism of refracting angle A, the two internal angles add up to A.

$$r_1+r_2=A$$

For the ray to just suffer total internal reflection at the second face, the angle there must equal the critical angle θ_c.

$$r_2=\theta_c$$

$$\Rightarrow r_1=A-\theta_c$$

Applying Snell's law at the first face (air to glass),

$$\sin i=\mu \sin r_1=\mu \sin(A-\theta_c)$$

The critical angle is fixed by the refractive index through $\theta_c=\frac{1}{\mu}$, so $\theta_c=\sin^{-1}(\frac{1}{\mu})$.

Hence the required **angle of incidence** is

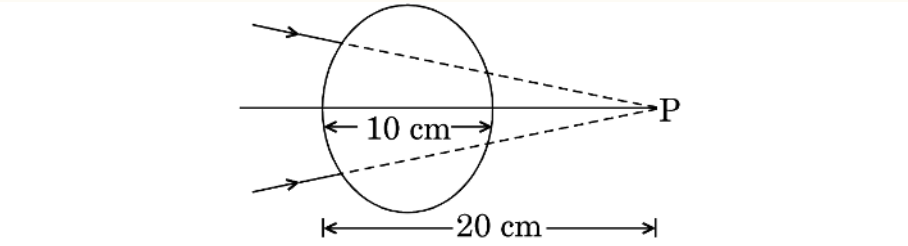
$$i=\sin^{-1}[\mu \sin(A-\sin^{-1}\frac{1}{\mu})]$$

Final answer. (a) The system is diverging with F = −60 cm. (b) $i = \sin^{-1}[\mu \sin(A - \theta_c)]$, where $\sin \theta_c = 1/\mu$.

← *Back to Q261*

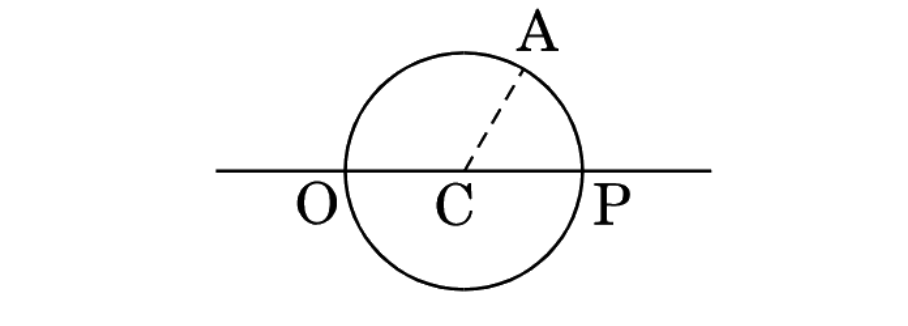
Q262

A converging beam of light travelling in air converges at a point P as shown in the figure. When a glass sphere of refractive index 1.5 *is introduced in between the path of the beam, calculate the new position of the image. Also draw the ray diagram for the image formed.*



OR

A point 'O' marked on the surface of a glass sphere of diameter 20 cm *is viewed through glass from the position directly opposite to the point O. If the refractive index of the glass is* 1.5, *find the position of the image formed. Also, draw the ray diagram for the formation of the image.*



[2019 • Set 55-3-1]

SOLUTION

This is the OR alternative, which is fully numerical and is the part that can be marked exactly. A point *O* marked on the surface of a glass sphere of diameter 20 cm is viewed from the diametrically opposite side, through the glass.

Light from O travels inside the glass and refracts only at the far (curved) surface where it leaves into air, so refraction occurs at that single spherical surface.

The refraction at a spherical surface is governed by

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

Here light goes from glass to air, so $\mathbf{n_1=1.5}$ (object side, glass) and $\mathbf{n_2=1}$ (image side, air).

The object O lies on the surface diametrically opposite, at the full diameter from the refracting surface, so $\boldsymbol{u=-20\ cm}$.

The refracting surface is concave towards the incoming light, so its centre of curvature lies on the object side, giving $\boldsymbol{R=-10\ cm}$.

Substituting these values,

$$\frac{1}{v}-\frac{1.5}{-20}=\frac{1-1.5}{-10}$$

$$\frac{1}{v}+\frac{1.5}{20}=\frac{-0.5}{-10}=0.05$$

$$\frac{1}{v}+0.075=0.05$$

$$\frac{1}{v}=0.05-0.075=-0.025\ \text{cm}^{-1}$$

$$v=-40\ \text{cm}$$

The negative sign shows the image is on the same side as the object, so it is **virtual** and lies 40 cm from the refracting surface, i.e. 20 cm behind the sphere on the far side.

Ray diagram: draw the sphere with centre C. Mark O on the left surface. Two rays from O travel through the glass to the right (far) surface; on emerging into air they bend away from the normal (glass to rarer medium). Produce the emergent rays backward as dashed lines; they meet 40 cm from the right surface, locating the virtual image I. Label O, C, the far surface, the emergent rays and the dashed back-projections meeting at I.

COMMON MISTAKE

When light leaves a denser medium, n_1 is the glass index and n_2 the air index in $n_2/v - n_1/u = (n_2-n_1)/R$; mixing these up is the commonest sign error here.

Final answer. The image is virtual, formed 40 cm from the refracting surface (v = −40 cm), i.e. 20 cm behind the sphere.

← Back to Q262

Q263

Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

OR

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

[2019 • Set 55-4-1]

SOLUTION

Refracting (astronomical) telescope in normal adjustment: it uses two convex lenses, an objective of large focal length f_o and an eyepiece of small focal length f_e.

Ray diagram: parallel rays from a distant object enter the objective O and converge to form a real, inverted, diminished image $A'B'$ at its focus. This image lies at the common focal plane, which is also the focus of the eyepiece E. The eyepiece then sends out a parallel beam to the relaxed (normal) eye, so the final image is at infinity. Mark f_o from objective to the intermediate image and f_e from intermediate image to eyepiece. Show the angle α subtended by the object at the objective and β subtended by the final image at the eye.

The angular (magnifying) power in normal adjustment is the ratio of these angles.

$$m=\frac{\beta}{\alpha}=\frac{f_o}{f_e}$$

For large **resolution**, the objective aperture (diameter D) must be large, because the limit of resolution is given by

$$\Delta\theta=\frac{1.22\,\lambda}{D}$$

A larger D gives a smaller $\Delta\theta$, hence finer resolving power and also greater light-gathering power so faint objects are seen.

Limitations: making the objective lens very large is difficult because a big lens sags under its own weight, is hard to support (held only at its rim), absorbs some light, and suffers more from chromatic and spherical aberration. These practical limits on lens size are why large research telescopes are built as **reflectors** rather than refractors.

OR (a) Graph of deviation δ versus angle of incidence i: a U-shaped curve. As i increases, δ first decreases, reaches a single lowest value δ_m (the angle of minimum deviation) when $\boldsymbol{i=e}$ (the ray passes symmetrically), and then increases again. Mark δ_m at the bottom of the curve and the symmetric value of i on the horizontal axis.

OR (b) At minimum deviation the path is symmetric, so $\boldsymbol{i=e}$ and $\mathbf{r_1=r_2=}\boldsymbol{r}$. Since $\mathbf{r_1+r_2=}\boldsymbol{A}$,

$$r=\frac{A}{2}$$

The total deviation is $\boldsymbol{\delta=i+e-A}$; at minimum deviation $\boldsymbol{\delta=\delta_m}$ and $\boldsymbol{i=e}$, giving

$$\delta_m=2i-A\Rightarrow i=\frac{A+\delta_m}{2}$$

Applying Snell's law at the first face,

$$n=\frac{\sin i}{\sin r}$$

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Final answer. Angular magnification m = f_o/f_e; large resolution needs a large objective aperture, limited by lens sag, absorption and aberration.

← *Back to Q263*

Q264

A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

OR

A convex lens of focal length 20 cm and a concave lens of focal length 15 cm are kept 30 cm apart with their principal axes coincident. When an object is placed 30 cm in front of the convex lens, calculate the position of the final image formed by the combination. Would this result change if the object were placed 30 cm in front of the concave lens? Give reason.

[2019 • Set 55-5-1]

SOLUTION

A fixed object-to-screen distance with a convex lens giving a sharp image at two positions is the displacement (Bessel) method. Let the object-screen separation be L and the separation between the two lens positions be d.

For a fixed $L>4f$ there are two lens positions giving a real image; the focal length is

$$f=\frac{L^2-d^2}{4L}$$

Here $L=90$ **cm** and $d=20$ **cm**.

$$f=\frac{(90)^2-(20)^2}{4\times 90}$$

$$f \approx \frac{8100-400}{360} = \frac{7700}{360}$$

$$f \approx 21.4 \text{ cm}$$

Hence the **focal length of the lens is about 21.4 cm**.

OR: Convex lens $\mathbf{f_1 = +20\ cm}$, concave lens $\mathbf{f_2 = -15\ cm}$, separated by 30 cm; object 30 cm in front of the convex lens.

Refraction at the convex lens, with $\mathbf{u_1 = -30\ cm}$:

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{20}$$

$$\frac{1}{v_1} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$$

$$v_1 = +60 \text{ cm}$$

This image would form 60 cm to the right of the convex lens, but the concave lens is only 30 cm to the right. So it acts as a virtual object for the concave lens at a distance $\mathbf{60 - 30 = 30\ cm}$ behind it, giving $\mathbf{u_2 = +30\ cm}$.

Refraction at the concave lens, $\mathbf{f_2 = -15\ cm}$:

$$\frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f_2}$$

$$\frac{1}{v_2} - \frac{1}{30} = \frac{1}{-15}$$

$$\frac{1}{v_2} = -\frac{1}{15} + \frac{1}{30} = \frac{-2+1}{30} = -\frac{1}{30}$$

$$v_2 = -30 \text{ cm}$$

The final image is **virtual, 30 cm to the left of the concave lens** (i.e. 0 cm from the convex lens, since the lenses are 30 cm apart).

Yes, the result would change if the object were placed 30 cm in front of the concave lens instead. The combined system is not symmetric: the lenses have different focal lengths and are separated, so light entering from the concave

side meets the elements in a different order and at different distances. The intermediate image positions, and therefore the final image, are different.

TIP

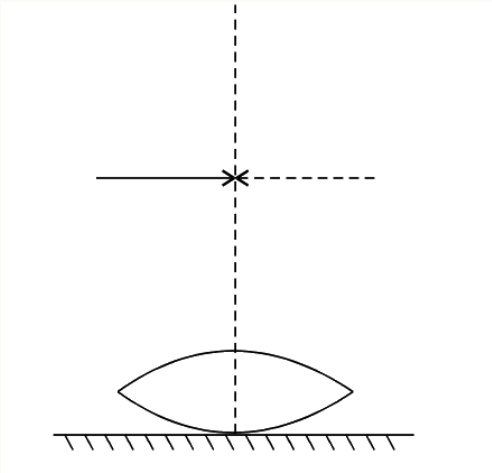
In two-lens problems, always carry the image of the first lens forward as the object of the second, adjusting u by the separation and watching for a virtual object (positive u).

Final answer. f ≈ 21.4 cm. (OR: final image is virtual, 30 cm to the left of the concave lens; yes, the result changes if the object faces the concave lens first.)

← *Back to Q264*

Q265

A symmetric biconvex lens of radius of curvature R and made of glass of refractive index 1.5*, is placed on a layer of liquid placed on top of a plane mirror as shown in the figure. An optical needle with its tip on the principal axis of the lens is moved along the axis until its real, inverted image coincides with the needle itself. The distance of the needle from the lens is measured to be x. On removing the liquid layer and repeating the experiment, the distance is found to be y. Obtain the expression for the refractive index of the liquid in terms of x and y.*



[2018]

SOLUTION

When the symmetric biconvex lens rests on the liquid film over the plane mirror and the needle's real inverted image coincides with the needle, the needle is at the focus of the whole arrangement, so the measured distance equals the equivalent focal length.

With the liquid present, the system is the glass biconvex lens in contact with a plano-concave liquid lens (the liquid fills the gap between the curved lower face of the glass lens and the flat mirror). The plane mirror simply reflects. The equivalent focal length x obeys the contact-lens rule.

$$\frac{1}{x}=\frac{1}{\mathrm{f}_g}+\frac{1}{\mathrm{f}_l}$$

When the liquid is removed, only the glass lens acts, and the measured distance is y.

$$\frac{1}{y}=\frac{1}{\mathrm{f}_g}$$

So $\mathbf{f_g=y}$. Subtracting gives the liquid-lens focal length.

$$\frac{1}{\mathrm{f}_l}=\frac{1}{x}-\frac{1}{y}=\frac{y-x}{\mathrm{xy}}$$

The liquid lens is plano-concave: its flat face (on the mirror) has $\boldsymbol{R=\infty}$, and its upper face matches the lower face of the biconvex lens, of radius of curvature $-R$ as seen by the liquid. By the lens maker's formula for the liquid of refractive index n_l,

$$\frac{1}{\mathrm{f}_l}=(\mathrm{n}_l-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)=(\mathrm{n}_l-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)$$

$$\frac{1}{\mathrm{f}_l}=-\frac{(\mathrm{n}_l-1)}{R}$$

For the glass biconvex lens, with $\mathbf{R_1=+R}$, $\mathbf{R_2=-R}$ and $\mathbf{n_g=1.5}$,

$$\frac{1}{\mathrm{f}_g}=(1.5-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=0.5\times\frac{2}{R}=\frac{1}{R}$$

Since $\mathbf{f_g=y}$, this gives $\boldsymbol{R=y}$.

Now equate the two expressions for $\frac{1}{\mathrm{f}_l}$:

$$-\frac{(n_l-1)}{R}=\frac{y-x}{xy}$$

Replacing $R=y$ and solving for n_l:

$$n_l-1=-\frac{R(y-x)}{xy}=-\frac{y(y-x)}{xy}=-\frac{y-x}{x}=\frac{x-y}{x}$$

$$n_l=1+\frac{x-y}{x}=\frac{2x-y}{x}$$

Hence the **refractive index of the liquid** is

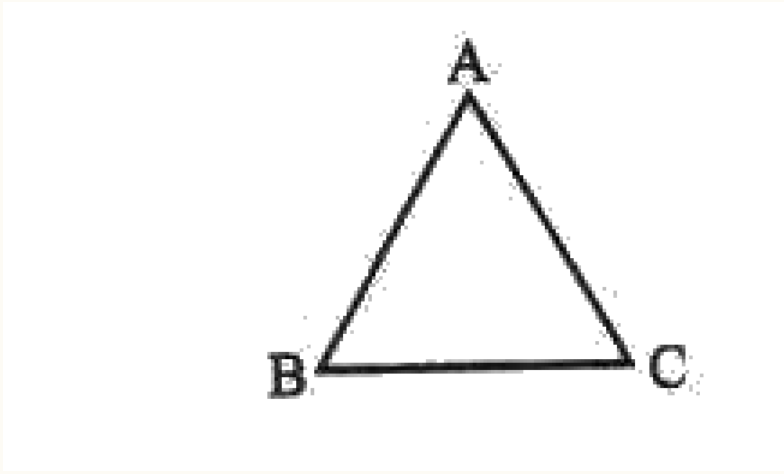
$$n_l=\frac{2x-y}{x}$$

Final answer. The refractive index of the liquid is n_l = (2x − y)/x.

← *Back to Q265*

Q266

(i) A ray of light incident on face AB of an equilateral glass prism, shows minimum deviation of 30°. Calculate the speed of light through the prism.



(ii) Find the angle of incidence at face AB so that the emergent ray grazes along the face AC.

[2017]

SOLUTION

(*i*) For an equilateral prism the refracting angle is $A=60°$, and the angle of minimum deviation is $\delta_m=30°$. The refractive index follows from the prism formula.

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$n=\sin(60°+30°2)\sin(60°2)=\sin 45°\sin 30°$$

$$n=\frac{1/\sqrt{2}}{1/2}=\frac{2}{\sqrt{2}}=\sqrt{2}\approx 1.414$$

The speed of light in the prism is

$$v=\frac{c}{n}=\frac{3\times 10^8}{\sqrt{2}}$$

$$v\approx 2.12\times 10^8\ m/s$$

Hence the **speed of light through the prism is about $2.12\times 10^8\ m/s$**.

(ii) For the emergent ray to graze along face AC, it must just undergo total internal reflection there, so the angle inside at the second face equals the critical angle.

$$\theta_c=\frac{1}{n}=\frac{1}{\sqrt{2}}\Rightarrow \theta_c=45°$$

So $r_2=45°$. Using $r_1+r_2=A$,

$$r_1=A-r_2=60°-45°=15°$$

Applying Snell's law at face AB,

$$\sin i=n\sin r_1=\sqrt{2}\sin 15°$$

$$\sin i=1.414\times 0.2588\approx 0.366$$

$$i=\sin^{-1}(0.366)\approx 21.5°$$

Hence the required **angle of incidence at face AB is about 21.5°**.

Final answer. (i) v ≈ 2.12 × 10^8 m/s. (ii) i ≈ 21.5 °.

← Back to Q266

Q267

(a) Monochromatic light of wavelength 589 nm *is incident from air on a water surface. If* μ *for water is* 1.33*, find the wavelength, frequency and speed of the refracted light.*
(b) A double convex lens is made of a glass of refractive index 1.55*, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is* 20 cm.

[2017]

SOLUTION

(*a*) When light passes from air into water, its frequency stays the same (it is fixed by the source), while its speed and wavelength reduce by the factor μ.

The frequency is found from the air values, where $\lambda_{air}=589\ \text{nm}=589\times 10^{-9}\ m$.

$$\nu=\frac{c}{\lambda_{air}}=\frac{3\times 10^{8}}{589\times 10^{-9}}$$

$$\nu\approx 5.09\times 10^{14}\ \text{Hz}$$

This frequency is unchanged on entering water.

The speed of light in water is

$$v=\frac{c}{\mu}=\frac{3\times 10^{8}}{1.33}$$

$$v\approx 2.26\times 10^{8}\ m/s$$

The wavelength in water is

$$\lambda_{w}=\frac{\lambda_{air}}{\mu}=\frac{589}{1.33}$$

$$\lambda_{w}\approx 443\ \text{nm}$$

So in water: **frequency $\nu\approx 5.09\times 10^{14}$ Hz, speed $v\approx 2.26\times 10^{8}\ m/s$, wavelength $\lambda_{w}\approx 443$ nm.**

(*b*) For a double convex lens with both faces of the same radius of curvature, $\mathbf{R_1}=+\boldsymbol{R}$ and $\mathbf{R_2}=-\boldsymbol{R}$. The lens maker's formula gives

$$\frac{1}{f}=(n-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$$

$$\frac{1}{f}=(n-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=(n-1)\,\frac{2}{R}$$

Solving for R,

$$R=2(n-1)f$$

Substituting $\boldsymbol{n}=\mathbf{1.55}$ and $\boldsymbol{f}=\mathbf{20\ cm}$,

$$R=2\times(1.55-1)\times 20=2\times 0.55\times 20$$

$$R=22\ \mathrm{cm}$$

Hence each face needs a **radius of curvature of 22 cm**.

CONCEPT

Frequency never changes when light crosses into a new medium; only speed and wavelength scale by 1/μ. Use the air wavelength to get ν, then divide for the rest.

Final answer. (a) ν ≈ 5.09 × 10^{14} Hz, v ≈ 2.26 × 10^{8} m/s, λ_w ≈ 443 nm. (b) R = 22 cm.

← *Back to Q267*

Q268

(a) Draw a ray diagram depicting the formation of the image by an astronomical telescope in normal adjustment.
(b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? Give reason.

Lenses	Power (D)	Aperture (cm)
L_1	3	8
L_2	6	1
L_3	10	1

[2017]

SOLUTION

(*a*) Astronomical (refracting) telescope in normal adjustment: it uses a convex objective of large focal length f_o and a convex eyepiece of small focal length f_e.

Ray diagram: parallel rays from the distant object fall on the objective *O*, which converges them to a real, inverted, diminished image *A'B'* at its focus. This intermediate image lies at the common focal plane, which is also the focus of the eyepiece *E*. The eyepiece then refracts these rays into a parallel beam, so the final image is virtual and at infinity for the relaxed eye. Mark f_o between the objective and the intermediate image, f_e between the intermediate image and the eyepiece, the angle α subtended by the object at the objective and the larger angle β subtended by the final image at the eye.

The magnifying power in this adjustment is

$$m=\frac{f_o}{f_e}$$

(*b*) For an astronomical telescope, the objective should have a long focal length and a large aperture (to give high magnification and good light gathering and resolution), while the eyepiece should have a short focal length.

Among the three lenses, the one with the largest focal length and largest aperture (lowest power, biggest diameter) is chosen as the **objective**, and the one with the smallest focal length (highest power, small aperture) is chosen as the **eyepiece**.

So the lens with the longest focal length and widest aperture is used as the objective, and the lens with the shortest focal length is used as the eyepiece. This maximizes $m=\frac{f_o}{f_e}$ while keeping a bright, well-resolved image.

Final answer. Use the long-focal-length, large-aperture lens as the objective and the short-focal-length lens as the eyepiece; m = f_o/f_e.

← *Back to Q268*

Q269

(a) Draw a ray diagram showing the formation of image by a reflecting telescope.
(b) Write two advantages of a reflecting telescope over a refracting telescope.

[2017]

SOLUTION

(*a*) Reflecting telescope (Cassegrain type) ray diagram: parallel rays from a distant object fall on a large concave parabolic primary mirror placed at the back of the tube. The primary reflects and converges the beam towards its focus. Before the beam converges, a small convex secondary mirror near the open front intercepts it and reflects it back through a small hole at the centre of the primary mirror. The rays then form a real, inverted image just behind the primary, which is viewed and magnified by the eyepiece *E* placed behind the hole. Label the concave primary mirror, the convex secondary mirror, the hole in the primary, the eyepiece *E* and the final image.

The magnifying power is

$$m=\frac{f_o}{f_e}$$

where f_o is the focal length of the concave primary mirror.

(*b*) Two advantages of a reflecting telescope over a refracting telescope:

First, there is **no chromatic aberration**, because a mirror reflects all wavelengths through the same angle, whereas a lens refracts different colours by different amounts and so produces coloured fringes.

Second, a much **larger aperture** is achievable, because a mirror can be supported firmly from behind over its whole back surface, while a large lens can be held only at its rim and sags and distorts under its own weight; the

larger mirror also gives greater light-gathering power and higher resolving power.

Final answer. Reflecting telescope: free of chromatic aberration and allows a larger, better-supported aperture (more light and resolution).

← *Back to Q269*

Q270

(a) Draw a ray diagram for the formation of image by a compound microscope.
(b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct a compound microscope?
(c) Define resolving power of a microscope and write one factor on which it depends.

[2017]

SOLUTION

(*a*) Compound microscope ray diagram: the object AB is placed just beyond the focus of the short-focal-length objective O (between f_o and $2f_o$). The objective forms a real, inverted and magnified image $A'B'$ on the far side. This intermediate image $A'B'$ falls within the focal length of the eyepiece E, which then acts as a simple magnifier and produces a virtual, further magnified and (relative to the object) inverted final image $A''B''$ at the least distance of distinct vision D. Label the object AB, the objective O with its focus, the real intermediate image $A'B'$, the eyepiece E, and the final image $A''B''$ at distance D from the eye.

The total magnification is the product of the two stages,

$$m = m_o \times m_e = \frac{v_o}{u_o}\left(1+\frac{D}{f_e}\right)$$

(*b*) A compound microscope needs high magnification, and since $m \propto \frac{1}{f_o f_e}$, both the objective and the eyepiece should have **short focal lengths (high power)**. So from the given lenses, choose the two with the smallest focal lengths: the one with the shortest focal length as the **objective**, and the next shortest as the **eyepiece**.

(*c*) The **resolving power** of a microscope is the reciprocal of the smallest distance between two close points on the object that can still be seen as separate (just resolved). It is given by

$$R.P.=\frac{1}{d_{min}}=\frac{2\,n\theta}{1.22\,\lambda}$$

It depends on the **numerical aperture** $n\theta$ of the objective (and inversely on the wavelength λ of light used); using a medium of higher refractive index, such as oil immersion, raises the numerical aperture and so increases the resolving power.

Final answer. Use the two shortest-focal-length lenses (shortest as objective, next as eyepiece); resolving power = $2n\sin\theta / 1.22\lambda$, depending on numerical aperture (and λ).

← *Back to Q270*

Q271

(i) What is total internal reflection? Under what conditions does it occur?
(ii) Find a relation between critical angle and refractive index.
(iii) Name one phenomenon which is based on total internal reflection.

[2016]

SOLUTION

(i) Total internal reflection (TIR) is the phenomenon in which a ray of light travelling from an optically denser medium to an optically rarer medium is completely reflected back into the denser medium at the interface, with no part of it being refracted into the rarer medium.

When light goes from a denser to a rarer medium, the refracted ray bends away from the normal. As the angle of incidence increases, the angle of refraction increases faster, and at a particular angle called the critical angle θ_c, the refracted ray just grazes the interface (angle of refraction **=90°**). For any angle of incidence greater than θ_c, refraction is no longer possible and the light is totally reflected back.

The two necessary conditions for TIR are: (1) Light must travel from an optically denser medium to an optically rarer medium. (2) The angle of incidence in the denser medium must be greater than the critical angle, i.e. $i > \theta_c$.

(ii) Applying Snell's law at the interface when the angle of incidence equals the critical angle θ_c, the refracted ray grazes along the surface so the angle of refraction is 90°. Taking the denser medium as 1 and rarer as 2:

$$n_1\theta_c = n_2 \sin 90^\circ$$

$$n_1\theta_c = n_2$$

$$\theta_c = \frac{n_2}{n_1}$$

If the rarer medium is air ($\mathbf{n_2 = 1}$) and the denser medium is glass of refractive index n, then

$$\theta_c = \frac{1}{n} \Rightarrow n = \frac{1}{\theta_c}$$

This is the required relation between the **critical angle** and the **refractive index**.

(iii) One phenomenon based on total internal reflection is the working of an **optical fibre** (used in communication). The sparkling brilliance of cut diamonds and the formation of a mirage are also based on TIR.

Final answer. TIR is complete reflection from denser to rarer medium for i greater than the critical angle, with sin theta_c = n2/n1 (= 1/n for glass-air); optical fibres are based on it.

← *Back to Q271*

Q272

(a) Draw a schematic diagram of a reflecting telescope.
(b) State the advantages of reflecting telescope over refracting telescope.
OR

(i) Draw a schematic ray diagram of a compound microscope when image is formed at distance of distinct vision.
(ii) Write the expression for resolving power of a compound microscope. How can the resolving power of a microscope be increased?

[2016]

SOLUTION

(a) Schematic diagram of a reflecting telescope (Cassegrain type): Draw a wide tube. At the closed back end place a large concave parabolic mirror, the objective (primary) mirror. Parallel rays from a distant star enter the open front and travel to this primary mirror, which converges them. Before they meet, the converging beam strikes a small convex secondary mirror mounted on the axis near the front, facing the primary. This secondary reflects the beam back along the axis and sends it through a small central hole in the primary mirror. Just behind that hole, place a convex lens eyepiece through which the observer's eye views the final magnified image. Label: parallel rays from star, concave parabolic primary mirror, small convex secondary mirror, hole in primary, eyepiece, eye.

(b) Advantages of a reflecting telescope over a refracting telescope:

1. A mirror is free from chromatic aberration because reflection does not depend on wavelength, whereas a lens objective splits colours and produces chromatic aberration.

2. By using a parabolic mirror, spherical aberration is also greatly reduced.

3. A large mirror can be supported firmly from behind over its whole back surface, so very large apertures are possible; a large lens can be supported only at its edges and sags under its own weight, limiting its size.

4. A larger aperture gives higher resolving power and a brighter image; reflecting telescopes are also lighter and cheaper to build for the same aperture.

OR

(i) Ray diagram of a compound microscope (final image at the least distance of distinct vision, D**=25 cm**): The tiny object AB is placed just beyond the focus f_o of the objective. The objective (short focal length) forms a real, inverted, magnified intermediate image $A'B'$ inside the tube, positioned just within the

focus f_e of the eyepiece. The eyepiece then acts as a simple magnifier and forms a final image $A''B''$ that is virtual, inverted (relative to the object) and highly magnified, situated at the near point a distance D from the eye. Label: object AB beyond f_o, objective O, intermediate image $A'B'$, eyepiece E, final virtual image $A''B''$ at distance D.

(ii) The resolving power of a compound microscope is

$$\text{RP}=\frac{1}{d_{\min}}=\frac{2n\theta}{1.22\,\lambda}$$

where λ is the wavelength of light used, n is the refractive index of the medium between the object and objective, and $n\theta$ is the numerical aperture.

The resolving power can be **increased** by (a) using light of **shorter wavelength** λ, and (b) increasing the **numerical aperture** $n\theta$, for example by using an oil-immersion objective (oil has $\boldsymbol{n>1}$).

Final answer. Reflecting telescopes avoid chromatic aberration and allow larger apertures; microscope RP = 2n sin theta / (1.22 lambda), raised by smaller lambda or larger numerical aperture.

← *Back to Q272*

Q273

(i) A screen is placed at a distance of 100 cm *from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by* 20 cm. *Calculate the focal length of the lens used.*
(ii) A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal length. What is the focal length of the combination?

[2016]

SOLUTION

(i) This is the displacement (Bessel's) method of finding focal length. When an object and screen are kept a fixed distance D apart (with $\boldsymbol{D>4f}$), there are two positions of a convex lens between them that form a sharp image on the screen. If d is the distance between these two lens positions, the focal length is given by

$$f=\frac{D^2-d^2}{4D}$$

Here the object-to-screen distance D=**100 cm** and the separation between the two lens positions d=**20 cm**.

Substituting:

$$f=\frac{(100)^2-(20)^2}{4\times 100}$$

$$f=\frac{10000-400}{400}$$

$$f=\frac{9600}{400}$$

$$f=24 \text{ cm}$$

So the **focal length of the lens is 24 cm**.

(ii) For two thin lenses kept coaxially in contact, the equivalent focal length F is given by

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

The converging lens has focal length $\mathbf{f_1}=+f$ and the diverging lens has equal magnitude focal length $\mathbf{f_2}=-f$. Therefore

$$\frac{1}{F}=\frac{1}{f}+\frac{1}{-f}=\frac{1}{f}-\frac{1}{f}=0$$

$$\Rightarrow F\to\infty$$

So the **focal length of the combination is infinite**; the combination has zero power and behaves like a plane glass plate, neither converging nor diverging the light.

CONCEPT

In Bessel's method, the two sharp-image positions are interchangeable: object and image distances simply swap between the two locations of the lens.

Final answer. (i) f = 24 cm; (ii) the combination has infinite focal length (zero power), acting like a plain glass plate.

← *Back to Q273*

Q274

(i) For a glass prism ($\mu=\sqrt{3}$) the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.
(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu=\sqrt{3}$.

[2016]

SOLUTION

(i) For a prism, the refractive index is related to the angle of the prism A and the angle of minimum deviation δ_m by the prism formula

$$\mu=\frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$$

Here it is given that the angle of minimum deviation equals the angle of the prism, so $\delta_m=A$, and $\mu=\sqrt{3}$. Substituting $\delta_m=A$:

$$\mu=\frac{\sin(\frac{A+A}{2})}{\sin(\frac{A}{2})}=\frac{\sin A}{\sin(\frac{A}{2})}$$

Using $\sin A=2\sin(\frac{A}{2})\cos(\frac{A}{2})$:

$$\mu=\frac{2\sin(\frac{A}{2})\cos(\frac{A}{2})}{\sin(\frac{A}{2})}=2\cos(\frac{A}{2})$$

$$\sqrt{3}=2\cos(\frac{A}{2})$$

$$\cos\left(\frac{A}{2}\right)=\frac{\sqrt{3}}{2}$$

$$\frac{A}{2}=30^\circ$$

$$A=60^\circ$$

So the **angle of the prism is 60°**.

(ii) Ray diagram for a right-angled isosceles prism (45°–45°–90°) with $\mu=\sqrt{3}$: Draw the prism with the right angle at the top and the two equal faces vertical and horizontal, the hypotenuse being the slanted face. A ray strikes one of the equal faces normally (along the normal), so it enters without bending and travels straight to the hypotenuse, where it meets the surface at an angle of incidence of 45°.

The critical angle for this glass is

$$\theta_c=\frac{1}{\mu}=\frac{1}{\sqrt{3}}=0.577$$

$$\theta_c=\sin^{-1}(0.577)\approx 33.6^\circ$$

Since the angle of incidence on the hypotenuse (45°) is greater than the critical angle (33.6°), total internal reflection occurs at the hypotenuse. The ray turns through 90° and emerges normally through the second equal face. So in the diagram show the ray bending sharply by 90° at the hypotenuse and leaving perpendicular to the other equal face.

Final answer. (i) The angle of the prism is 60 degrees; (ii) the ray undergoes total internal reflection at the hypotenuse (45 deg greater than theta_c = 33.6 deg) and turns through 90 degrees.

← *Back to Q274*

Q275

(i) Name the phenomenon on which the working of an optical fibre is based.
(ii) What are the necessary conditions for this phenomenon to occur?

(iii) Draw a labelled diagram of an optical fibre and show how light propagates through the optical fibre using this phenomenon.

[2016]

SOLUTION

(i) The working of an optical fibre is based on the phenomenon of **total internal reflection (TIR)**.

(ii) The necessary conditions for total internal reflection to occur are:

1. Light must travel from an optically denser medium to an optically rarer medium (here, from the core to the cladding).

2. The angle of incidence at the boundary must be greater than the critical angle θ_c, where

$$\theta_c = \frac{n_{rarer}}{n_{denser}} = \frac{n_2}{n_1}$$

with n_1 the refractive index of the core (denser) and n_2 that of the cladding (rarer), $\mathbf{n_1 > n_2}$.

(iii) Labelled diagram of an optical fibre: Draw a long cylindrical fibre with two layers. The inner cylinder is the core, made of glass of higher refractive index $\mathbf{n_1 \approx 1.5}$. Surrounding it is a thin outer layer, the cladding, of slightly lower refractive index n_2 ($\mathbf{n_2 1}$). A protective outer jacket may also be shown.

Now show light propagation: a ray of light enters one end of the core and strikes the core-cladding boundary at an angle of incidence greater than the critical angle. Because it is going from the denser core to the rarer cladding at $\boldsymbol{i > \theta_c}$, it undergoes total internal reflection back into the core. It strikes the opposite wall, again at $\boldsymbol{i > \theta_c}$, and is again totally internally reflected. In this way the ray zig-zags down the entire length of the fibre, repeatedly reflecting off the walls, and emerges at the far end with almost no loss of energy. Label: core (n_1), cladding (n_2), incident ray, totally internally reflected zig-zag ray path, emergent ray.

Final answer. Optical fibres work on total internal reflection (core denser than cladding, incidence greater than critical angle), so light zig-zags along the core to the far end.

← Back to Q275

Q276

(a) Give two reasons to explain why reflecting telescopes are preferred over refracting type.
(b) Use mirror equation to show that convex mirror always produces a virtual image independent of the location of the object.

[2015]

SOLUTION

(a) Two reasons why reflecting telescopes are preferred over refracting telescopes:

1. A reflecting telescope uses a mirror as its objective, and reflection is independent of wavelength, so the image is free from chromatic aberration. A refracting telescope uses a lens objective, which disperses different colours to different foci and so suffers from chromatic aberration.

2. A large mirror can be supported firmly over its whole back surface, so very large apertures are possible. This gives higher resolving power and a brighter image. A large lens objective can be held only at its rim and sags under its own weight, limiting the aperture (it also absorbs some light).

(b) For a convex mirror, the focal length is positive, $f>0$. Take a real object placed anywhere in front of the mirror, so its object distance is negative, $u<0$.

The mirror equation is

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

Rearranging for the image distance:

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

For a convex mirror f is positive, so $\frac{1}{f}$ is positive. For a real object u is negative, so $-\frac{1}{u}$ is also positive. Therefore

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}>0$$

$$\Rightarrow v>0$$

A positive image distance means the image is formed behind the mirror, that is, it is a **virtual image**. Since this result holds for every negative value of u, the convex mirror **always produces a virtual image whatever the position of the object**. (The image is also erect and diminished.)

Final answer. Reflecting telescopes avoid chromatic aberration and allow larger apertures; for a convex mirror f greater than 0 and u less than 0 give 1/v greater than 0, so v is always positive (virtual image) for any object position.

← *Back to Q276*

Q277

(i) A giant refracting telescope has an objective lens of focal length 15 m. If an eye piece of focal length 1.0 cm is used, what is the angular magnification of the telescope? (ii) If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.48×10^6 *m and the radius of lunar orbit is* 3.8×10^8 *m.*

[2015]

SOLUTION

(i) For a telescope in normal adjustment, the angular (magnifying) power is the ratio of the focal length of the objective to the focal length of the eyepiece:

$$m=\frac{f_o}{f_e}$$

Here $\mathbf{f_o=15\ m}$ and $\mathbf{f_e=1.0\ cm=0.01\ m}$. Substituting:

$$m=\frac{15}{0.01}$$

$$m=1500$$

So the **angular magnification of the telescope is 1500**.

(ii) The objective forms the image of the moon in its focal plane. The angular size of the moon as seen from the Earth is the ratio of its diameter to its distance:

$$\theta=\frac{\text{diameter of moon}}{\text{radius of lunar orbit}}=\frac{3.48\times 10^{6}}{3.8\times 10^{8}}$$

$$\theta=9.16\times 10^{-3}\ \text{rad}$$

The image is formed at the focal plane of the objective, so its diameter d subtends the same angle θ at the objective:

$$d=f_o\,\theta$$

$$d=15\times 9.16\times 10^{-3}$$

$$d=0.1374\ m$$

$$d\approx 13.7\ \text{cm}$$

So the **diameter of the image of the moon formed by the objective is about 13.7 cm** (0.137 m).

Final answer. (i) Angular magnification m = 1500; (ii) diameter of the moon's image = 0.137 m (about 13.7 cm).

← *Back to Q277*

Q278

(a) Draw a labelled ray diagram showing the formation of a final image by a compound microscope at least distance of distinct vision.
(b) The total magnification produced by a compound microscope is 20. The magnification produced by the eye piece is 5. The microscope is focussed on a certain object. The distance between the objective and eyepiece is observed to be 14 cm. If least distance of distinct vision is 20 cm, calculate the focal length of the objective and the eye piece.

[2014]

SOLUTION

(a) Ray diagram of a compound microscope (final image at the least distance of distinct vision $\boldsymbol{D=25\ \text{cm}}$): The small object AB is placed just beyond the focus of the objective (u_o slightly greater than f_o). The objective, of short focal length f_o, forms a real, inverted, magnified intermediate image $A'B'$ inside the tube. This intermediate image lies just within the focal length of the eyepiece. The eyepiece, acting as a simple magnifier, then forms the final image $A''B''$, which is virtual, magnified and inverted with respect to the object, located at the least distance of distinct vision D from the eye. Label: object AB, objective O with focus f_o, intermediate image $A'B'$, eyepiece E with focus f_e, final virtual image $A''B''$ at distance $\boldsymbol{D=25\ \text{cm}}$.

(b) Given: total magnification $\boldsymbol{M=20}$, eyepiece magnification $\mathbf{M_e=5}$, tube length $\boldsymbol{L=14\ \text{cm}}$, $\boldsymbol{D=20\ \text{cm}}$.

The objective magnification is

$$m_o=\frac{M}{M_e}=\frac{20}{5}=4$$

For the eyepiece (image at near point), $\mathbf{M_e=1+\frac{D}{f_e}}$:

$$5=1+\frac{20}{f_e}$$

$$\frac{20}{f_e}=4$$

$$f_e=5\ \text{cm}$$

The object distance for the eyepiece is found from the lens equation; for the eyepiece forming the image at D, $\mathbf{u_e=-\frac{f_e\,D}{D+f_e}}$:

$$u_e=-\frac{5\times 20}{20+5}=-\frac{100}{25}=-4\ \text{cm}$$

The intermediate image (formed by the objective) lies at the object point of the eyepiece, so the image distance of the objective is

$$v_o=L-|u_e|=14-4=10\ \text{cm}$$

From the objective magnification $\mathbf{m_o=\frac{v_o}{|u_o|}=4}$:

$$|u_o| = \frac{v_o}{4} = \frac{10}{4} = 2.5 \text{ cm}$$

Now apply the lens equation to the objective, $\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$, with $v_o = +10$, $u_o = -2.5$:

$$\frac{1}{f_o} = \frac{1}{10} + \frac{1}{2.5} = \frac{1}{10} + \frac{4}{10} = \frac{5}{10} = \frac{1}{2}$$

$$f_o = 2 \text{ cm}$$

So the **focal length of the objective is 2 cm and that of the eyepiece is 5 cm**.

Final answer. Focal length of the objective = 2 cm and focal length of the eyepiece = 5 cm.

← *Back to Q278*

Q279

(a) A mobile phone lies along the principal axis of a concave mirror. Show, with the help of a suitable diagram, the formation of its image. Explain why magnification is not uniform.
(b) Suppose the lower half of the concave mirror's reflecting surface is covered with an opaque material. What effect this will have on the image of the object ? Explain.

[2014]

SOLUTION

(a) Ray diagram and explanation: Place the mobile phone along the principal axis of the concave mirror, with one end nearer the mirror and the other end farther away (the whole phone lying between the mirror and beyond the centre of curvature, say). To locate the image of each end, use two standard rays from each end point: a ray parallel to the axis that reflects through the focus F, and a ray through the centre of curvature C that retraces its path. These rays form the image of the near end and the image of the far end at different points, so the image of the phone is also a strip lying along the axis.

The magnification is not uniform because different points of the phone are at different object distances u from the mirror, and the magnification

$$m=-\frac{v}{u}$$

depends on u. The end of the phone closer to the mirror has a smaller $|u|$ and is magnified differently from the far end, which has a larger $|u|$. Since each point along the length is at a different u, the longitudinal magnification varies continuously along the phone. Hence the image is **distorted (non-uniformly magnified)**, appearing stretched or compressed rather than a faithful scaled copy.

(b) If the lower half of the concave mirror is covered with opaque material, the image is still formed at the **same position and with the same size**. This is because every point of the object sends rays to the whole mirror, and the rays reflected from the upper (uncovered) half are by themselves enough to form the complete image at the correct location. Covering half the mirror only removes half of the reflected rays, so the number of rays reaching the image is reduced. Therefore the image becomes **less bright (its intensity is reduced to about half)**, but its position, size and completeness are unchanged.

COMMON MISTAKE

A common mistake is to say covering the lower half cuts off the lower half of the image. It does not: the whole image still forms, only dimmer.

Final answer. (a) Magnification varies because each point is at a different u and m = -v/u, so the image is distorted; (b) the image stays at the same position and size but becomes about half as bright.

← *Back to Q279*

Q280

A convex lens of focal length 20 cm is placed coaxially with a convex mirror of radius of curvature 20 cm. The two are kept 15 cm apart. A point object is placed 40 cm in

front of the convex lens. Find the position of the image formed by this combination. Draw the ray diagram showing the image formation.

[2014]

SOLUTION

Refraction at the convex lens first. Given $\mathbf{f_L = +20\ cm}$ and the object placed 40 cm in front of the lens, so $\boldsymbol{u = -40\ cm}$. Using the lens equation

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_L}$$

$$\frac{1}{v_1} = \frac{1}{f_L} + \frac{1}{u} = \frac{1}{20} + \frac{1}{-40} = \frac{2-1}{40} = \frac{1}{40}$$

$$v_1 = +40\ \text{cm}$$

So the lens alone would form a real image 40 cm to its right.

Now this image acts as the object for the convex mirror. The mirror is 15 cm to the right of the lens, so the would-be image lies

$$40 - 15 = 25\ \text{cm}$$

behind the mirror. Since the rays are intercepted by the mirror before they actually meet, this is a **virtual object** for the mirror, at a distance of 25 cm behind it.

For the convex mirror, the radius of curvature is $\mathbf{R_M = 20\ cm}$, so

$$f_M = \frac{R_M}{2} = +10\ \text{cm}$$

Taking the virtual object behind the mirror as $\mathbf{u_M = +25\ cm}$ and $\mathbf{f_M = +10\ cm}$, apply the mirror equation

$$\frac{1}{v_M} + \frac{1}{u_M} = \frac{1}{f_M}$$

$$\frac{1}{v_M} = \frac{1}{f_M} - \frac{1}{u_M} = \frac{1}{10} - \frac{1}{25} = \frac{5-2}{50} = \frac{3}{50}$$

$$v_M = \frac{50}{3} \approx 16.7\ \text{cm}$$

The positive sign shows the image is formed 16.7 cm behind the mirror (on its far side). Measured from the lens it lies at **15+16.7=31.7 cm** on the far side of the mirror.

Ray diagram: Draw the convex lens on the left, the convex mirror 15 cm to its right, and the point object 40 cm in front of the lens on the principal axis. Show the rays from the object refracting through the lens and converging toward a point 40 cm behind the lens, but being intercepted by the convex mirror. The mirror reflects them so that they appear to diverge from a point $\frac{50}{3}$ cm behind the mirror, marking the final virtual image.

So the **final image is formed about 16.7 cm behind the convex mirror** (i.e. 31.7 cm from the lens), and it is virtual.

CONCEPT

When a converging beam meets a mirror before focusing, treat the would-be image as a virtual object and take its distance as positive (it lies behind the mirror).

Final answer. The final image forms about 16.7 cm (50/3 cm) behind the convex mirror, i.e. roughly 31.7 cm from the lens on its far side, and is virtual.

← *Back to Q280*

Q281

Three rays of light - red (*R*)*, green* (*G*) *and blue* (*B*) *- are incident on the face* AB *of a right-angled prism* ABC*. The refractive indices of the material of the prism for red, green and blue wavelengths are* 1.39*,* 1.44 *and* 1.47 *respectively. Trace the path of the rays through the prism.*
How will the situation change if these rays were incident normally on one of the faces of an equilateral prism?

[2013]

SOLUTION

Total internal reflection (TIR) occurs at a surface only when light travels from a denser to a rarer medium and the angle of incidence inside the prism exceeds the critical angle θ_c, where $\theta_c = \frac{1}{n}$.

In the right-angled prism ABC, angle $A = 90°$ and the other two angles are 45° each. A ray entering normally (or near-normally) through face AB strikes the hypotenuse AC at an angle of incidence of 45°.

For TIR at the hypotenuse we need $45° > \theta_c$, i.e. $\sin 45° > \theta_c = \frac{1}{n}$.

$$\frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2} = 1.414.$$

So any colour with refractive index greater than 1.414 is totally internally reflected at the hypotenuse.

For red, $n_R = 1.39 < 1.414$, so the critical angle for red is larger than 45°. Red light does NOT undergo TIR; it is partly refracted and passes out through the hypotenuse face AC (with some weak reflection).

For green, $n_G = 1.44 > 1.414$, so 45° exceeds its critical angle and green light undergoes total internal reflection at AC, then emerges through the face BC after a 90° deflection.

For blue, $n_B = 1.47 > 1.414$, so blue light also undergoes total internal reflection at AC and emerges through face BC.

Ray-diagram description: Draw the right-angled prism with the right angle at A, hypotenuse AC sloping down to the right, and the three rays R, G, B entering horizontally through the vertical face AB. At the hypotenuse, the green and blue rays bend through 90° (reflected downward) and leave through the horizontal base BC; the red ray continues through and exits the hypotenuse AC. Label θ_c at AC.

Now consider an equilateral prism (each angle 60°) with the rays incident normally on one face. Entering normally, the rays pass straight in without bending and strike the second face at an angle of incidence of 60°.

The critical angle for the prism material (taking $n \approx 1.4$ to 1.47) is about $\theta_c = \sin^{-1}\left(\frac{1}{n}\right) \approx 42°$ to $45°$, which is less than 60°.

Since $60° > \theta_c$ for ALL three colours (red, green and blue), every ray now undergoes total internal reflection at the second face and emerges through the third face. **In the equilateral prism all three colours behave identically and all are totally internally reflected**, unlike the right-angled prism where only green and blue showed TIR.

> **TIP**
>
> The cut-off here is n greater than 1.414, so check each refractive index against root 2 rather than memorising which colour reflects.

Final answer. In the right-angled prism, green and blue undergo TIR (emerge through BC) while red refracts out through the hypotenuse; in the equilateral prism with normal incidence all three colours undergo TIR since 60 degrees exceeds the critical angle for each.

← *Back to Q281*

Q282

(i) Draw a neat labelled ray diagram of an astronomical telescope in normal adjustment. Explain briefly its working.
(ii) An astronomical telescope uses two lenses of powers 10 *D and* 1 *D. What is its magnifying power in normal adjustment?*
OR
(i) Draw a neat labelled ray diagram of a compound microscope. Explain briefly its working.
(ii) Why must both the objective and the eye-piece of a compound microscope have short focal lengths?

[2012]

SOLUTION

(i) Astronomical telescope in normal adjustment.

Ray-diagram description: Draw two convex lenses on a common axis. The objective O on the left has a long focal length f_o and a large aperture; the eyepiece E on the right has a short focal length f_e. A parallel beam from a

distant object arrives at O inclined at a small angle α to the axis. It is brought to a focus, forming a real, inverted, diminished image $A'B'$ in the common focal plane of the two lenses (the point $\mathbf{F_o = F_e}$). The eyepiece is positioned so this image lies at its focus, so the rays leaving E emerge parallel, making a larger angle β with the axis. The final image is virtual, inverted and at infinity. Mark f_o between O and the common focus, f_e between that focus and E, and the angles α (at O) and β (at E).

Working: The objective collects parallel rays from the distant object and forms a real inverted image $A'B'$ at its focus. This image acts as the object for the eyepiece, which is used as a simple magnifier. In normal adjustment the image $A'B'$ is at the focus of the eyepiece, so the final image forms at infinity and the eye is fully relaxed.

Length of telescope in normal adjustment: $\boldsymbol{L = f_o + f_e}$.

Magnifying power: $\boldsymbol{m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}}$.

(ii) Powers given are 10 D and 1 D, so the focal lengths are $\boldsymbol{f = \frac{1}{P}}$.

$\boldsymbol{f = \frac{1}{10} = 0.10\ m = 10\ \text{cm}}$ and $\boldsymbol{f = \frac{1}{1} = 1\ m = 100\ \text{cm}}$.

The objective must have the longer focal length and the eyepiece the shorter, so $\mathbf{f_o = 100\ cm}$ and $\mathbf{f_e = 10\ cm}$.

$\boldsymbol{m = \frac{f_o}{f_e} = \frac{100}{10} = 10}$.

The magnifying power in normal adjustment is $m = 10$.

OR (i) Compound microscope.

Ray-diagram description: Two convex lenses of short focal length on a common axis. The object AB is placed just beyond the focus of the objective (u_o slightly greater than f_o). The objective forms a real, inverted, magnified image $A'B'$ inside the tube, between the eyepiece and its focus. This image acts as the object for the eyepiece, which (used as a magnifier) forms a final image $A''B''$ that is virtual, inverted (relative to the object) and highly magnified, viewed at the least distance of distinct vision $\boldsymbol{D = 25\ \text{cm}}$. Label f_o, f_e, the tube length L, image $A'B'$ and final image $A''B''$.

Working: The objective forms a real, magnified intermediate image $A'B'$ of the close object. The eyepiece then magnifies this image further like a simple microscope, giving a large overall magnification $M = m_o \times m_e \approx \frac{L}{f_o} \times \frac{D}{f_e}$.

(ii) Both focal lengths are short because the total magnification is $M \approx \frac{L}{f_o} \times \frac{D}{f_e}$. A small f_o makes the objective magnification $\frac{L}{f_o}$ large, and a small f_e makes the eyepiece angular magnification $\frac{D}{f_e}$ large. **Hence keeping both focal lengths short gives a high overall magnifying power.**

Final answer. The magnifying power of the astronomical telescope in normal adjustment is m = 10.

← *Back to Q282*

Q283

An illuminated object and a screen are placed 90 cm *apart. Determine the focal length and nature of the lens required to produce a clear image on the screen, twice the size of the object.*

[2012]

SOLUTION

A clear image on a screen must be a real image, and a real, magnified image of a real object is produced only by a convex (converging) lens, so the required lens is convex.

The image is twice the size of the object, so the magnitude of the linear magnification is $|m|=2$, giving $|v|=2|u|$.

The object and screen are 90 cm apart, and for a real image the object distance and image distance lie on opposite sides of the lens, so $|u|+|v|=90$ **cm**.

Substituting $|v|=2|u|$: $|u|+2|u|=90$.

$3|u|=90 \Rightarrow |u|=30$ **cm** and $|v|=60$ **cm**.

Using the sign convention (object on the left), $u=-30$ **cm** and $v=+60$ **cm**.

Lens formula: $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

$\frac{1}{f}=\frac{1}{60}-\frac{1}{(-30)}=\frac{1}{60}+\frac{1}{30}$.

$\frac{1}{f}=\frac{1+2}{60}=\frac{3}{60}=\frac{1}{20}$.

$f=+20$ cm.

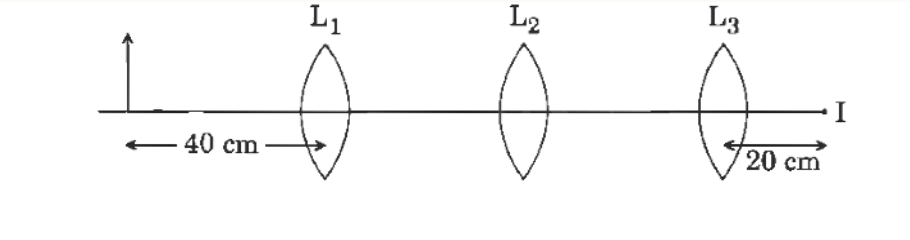
The positive sign confirms the lens is **convex (converging) with focal length $f=20$ cm**.

Final answer. A convex (converging) lens of focal length 20 cm is required.

← *Back to Q283*

Q284

You are given three lenses L_1, L_2 *and* L_3 *each of focal length* 20 cm. *An object is kept at* 40 cm *in front of* L_1, *as shown. The final real image is formed at the focus 'I' of* L_3. *Find the separations between* L_1, L_2 *and* L_3.



[2012]

SOLUTION

Each lens has $f=20$ cm and the object is placed 40 cm in front of L_1, so $u_1=-40$ cm.

For L_1: $\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f}$.

$\frac{1}{v_1}=\frac{1}{20}+\frac{1}{(-40)}=\frac{1}{20}-\frac{1}{40}=\frac{2-1}{40}=\frac{1}{40}$.

$v_1=+40$ cm, so L_1 forms a real image 40 cm to its right.

The final image is at the focus I of L_3. An image is formed exactly at the focus of L_3 only when the rays entering L_3 are parallel to the axis, that is, when the object for L_3 is effectively at infinity.

Parallel rays leave L_2 only when the object for L_2 lies at its focus, i.e. the rays from L_1 must converge towards a point 20 cm in front of L_2.

The image of L_1 is at 40 cm from L_1. Let the separation $\mathbf{L_1L_2}=d$. This image lies at distance $(40-d)$ beyond L_2 and acts as a virtual object for L_2, but for parallel emergent rays it must fall on the focus of L_2.

So the distance of the L_1-image from L_2 must equal $f=20$ **cm**: $40-d=20$.

$d=20$ **cm**, hence the separation $\mathbf{L_1L_2=20}$ **cm**.

Since the rays between L_2 and L_3 are parallel to the axis, they converge to the focus of L_3 regardless of how far L_3 is placed. **The separation $\mathbf{L_2L_3}$ can be any value (the parallel beam is focused at the focus of $\mathbf{L_3}$ whatever the gap), while $\mathbf{L_1L_2=20}$ cm.**

CONCEPT

When a final image lands exactly on a lens focus, the rays reaching that lens must be parallel, which fixes the previous lens at its own focus.

Final answer. The separation L1L2 = 20 cm, and L2L3 can be any distance since the beam between L2 and L3 is parallel.

← *Back to Q284*

Q285

Define power of a lens. Write its units. Deduce the relation $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$ *for two thin lenses kept in contact coaxially.*

[2012]

SOLUTION

Power of a lens is its ability to converge or diverge a beam of light, defined as the reciprocal of the focal length in metres: $P=\frac{1}{f}$.

The SI unit of power is the dioptre (D), where $\mathbf{1\ D = 1\ m^{-1}}$ is the power of a lens of focal length one metre. Power is positive for a convex lens and negative for a concave lens.

Consider two thin lenses of focal lengths f_1 and f_2 placed in contact coaxially. Let an object O be at distance u from the combination.

The first lens L_1 alone forms an image at distance v_1: $\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$ ...(i)

This image acts as the object for the second lens L_2. Since the lenses are in contact, the object distance for L_2 is v_1, and L_2 forms the final image at distance v: $\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$ ...(ii)

Adding equations (i) and (ii): $\frac{1}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_1} + \frac{1}{f_2}$.

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$ ...(iii)

If the combination behaves as a single equivalent lens of focal length f, then $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ ...(iv)

Comparing (iii) and (iv): $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$.

Hence $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$, which in terms of power is $P = P_1 + P_2$.

Final answer. Power P = 1/f (unit dioptre, D); for two thin lenses in contact 1/f = 1/f1 + 1/f2.

← *Back to Q285*

Q286

A convex lens made up of glass of refractive index 1.5 *is dipped, in turn, in (i) a medium of refractive index* 1.65*, (ii) a medium of refractive index* 1.33.
(a) Will it behave as a converging or a diverging lens in the two cases?
(b) How will its focal length change in the two media?

[2011]

SOLUTION

When a lens of refractive index n_l is immersed in a medium of refractive index n_m, the lens-maker's formula becomes $\frac{1}{f}=(\frac{n_l}{n_m}-1)(\frac{1}{R_1}-\frac{1}{R_2})$.

For a convex lens in air the bracket $(\frac{1}{R_1}-\frac{1}{R_2})$ is positive, so the sign and size of f depend on the factor $(\frac{n_l}{n_m}-1)$. Here $\mathbf{n_l=1.5}$.

(a) and (b), case (i): medium of refractive index $\mathbf{n_m=1.65}$.

$\frac{n_l}{n_m}=\frac{1.5}{1.65}\approx 0.91<1$, so $(\frac{n_l}{n_m}-1)$ is negative.

Hence f becomes negative. The lens (denser-medium convex shape) now behaves as a **diverging (concave-acting) lens**, because the surrounding medium is optically denser than the glass.

Case (ii): medium of refractive index $\mathbf{n_m=1.33}$ (water).

$\frac{n_l}{n_m}=\frac{1.5}{1.33}\approx 1.13>1$, so $(\frac{n_l}{n_m}-1)$ is positive but much smaller than the air value $(1.5-1)=0.5$.

Hence f stays positive, so the lens remains a **converging lens**, but since the factor is smaller, $\frac{1}{f}$ is smaller and the **focal length increases** (the lens becomes much weaker, with f roughly four times its air value).

Quantitatively, $\frac{f_{water}}{f_{air}}=\frac{n_l-1}{\frac{n_l}{n_m}-1}=\frac{0.5}{0.128}\approx 4$.

Summary: (i) in the denser medium ($n_m=1.65$) it acts as a diverging lens with negative focal length; (ii) in water ($n_m=1.33$) it remains converging but with a focal length about four times longer than in air.

CONCEPT

A lens converges only when the surrounding medium is rarer than the lens material; if the medium is denser, a convex lens turns diverging.

Final answer. In the medium of refractive index 1.65 the lens diverges (f negative); in water it still converges but with a much longer focal length (about 4 times that in air).

← Back to Q286

Q287

Use the mirror equation to show that
(a) an object placed between f and 2f of a concave mirror produces a real image beyond 2f.
(b) a convex mirror always produces a virtual image independent of the location of the object.
(c) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image.

[2011]

SOLUTION

The mirror formula is $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, which gives $\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$. The sign convention is: distances measured against the incident light are negative, so for a concave mirror f and $R=2f$ are negative, and a real object has u negative.

(a) Concave mirror, object between f and 2f.

Let $f=-|f|$ and the object lie between f and 2f, so $|f|<|u|<2|f|$ with u negative, i.e. $u=-|u|$.

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{|f|}+\frac{1}{|u|}.$$

Since $|u|<2|f|$, we have $\frac{1}{|u|}>\frac{1}{2|f|}$, so $\frac{1}{v}=\frac{1}{|u|}-\frac{1}{|f|}>\frac{1}{2|f|}-\frac{1}{|f|}=-\frac{1}{2|f|}$.

Also since $|u|>|f|$, $\frac{1}{|u|}<\frac{1}{|f|}$, so $\frac{1}{v}<0$, meaning v is negative (image is real, on the same side).

Combining, $-\frac{1}{2|f|}<\frac{1}{v}<0$, which means $|v|>2|f|$. **The image is real and lies beyond 2f.**

(b) Convex mirror, any object position.

For a convex mirror f is positive, $f=+|f|$, and for a real object $u=-|u|$.

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{|f|}+\frac{1}{|u|}.$$

Both terms are positive, so $\frac{1}{v}>0$, giving $v>0$ for every value of u. A positive v means the image is formed behind the mirror. **Hence a convex mirror always forms a virtual image, whatever the object position.**

(c) Concave mirror, object between pole and focus.

Now $|u|<|f|$ with $u=-|u|$ and $f=-|f|$.

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{|f|}+\frac{1}{|u|}=\frac{1}{|u|}-\frac{1}{|f|}.$$

Since $|u|<|f|$, $\frac{1}{|u|}>\frac{1}{|f|}$, so $\frac{1}{v}>0$, i.e. $v>0$. The image is behind the mirror and therefore virtual.

The magnification is $m=-\frac{v}{u}$. With $v>0$ and $u<0$, m is positive, so the image is erect.

Because $\frac{1}{v}=\frac{1}{|u|}-\frac{1}{|f|}$ is small (close to zero), $|v|$ is large and $|v|>|u|$, so $|m|=\frac{|v|}{|u|}>1$. **The image is virtual, erect and enlarged.**

Final answer. Using the mirror equation: (a) the image is real and beyond 2f; (b) a convex mirror always gives a virtual image; (c) the image is virtual, erect and enlarged.

← *Back to Q287*

Q288

A compound microscope uses an objective lens of focal length 4 cm *and eyepiece lens of focal length* 10 cm. *An object is placed at* 6 cm *from the objective lens. Calculate the magnifying power of the compound microscope. Also calculate the length of the microscope.*

[2011]

SOLUTION

Given the objective focal length $f_o = 4$ cm, eyepiece focal length $f_e = 10$ cm and object distance from the objective $u_o = -6$ cm (object on the left). Take the least distance of distinct vision $D = 25$ cm and the final image at infinity (normal adjustment).

First find the image formed by the objective using the lens formula $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$.

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{4} + \frac{1}{(-6)} = \frac{1}{4} - \frac{1}{6}.$$

$$\frac{1}{v_o} = \frac{3-2}{12} = \frac{1}{12} \Rightarrow v_o = 12 \text{ cm}.$$

Magnification of the objective: $m_o = \frac{v_o}{u_o} = \frac{12}{-6} = -2$.

Magnification of the eyepiece (final image at infinity): $m_e = \frac{D}{f_e} = \frac{25}{10} = 2.5$.

Total magnifying power: $M = m_o \times m_e = (-2) \times 2.5 = -5$.

The negative sign shows the final image is inverted, so the magnifying power has magnitude $|M| = 5$.

Length of the microscope is the distance between the two lenses. With the final image at infinity, the intermediate image lies at the focus of the eyepiece, so $L = v_o + f_e$.

$L = 12 + 10 = 22$ cm.

The magnifying power of the microscope is $M = -5$ (magnitude 5, inverted) and its length is $L = 22$ cm.

Final answer. The magnifying power is 5 (image inverted) and the length of the microscope is 22 cm.

← *Back to Q288*

Q289

A giant refracting telescope at an observatory has an objective lens of focal length 15 *m. If an eyepiece lens of focal length* 1.0 cm *is used, find the angular magnification of*

the telescope.
If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.42×10^{6} *m and the radius of the lunar orbit is* 3.8×10^{8} *m.*

[2011]

SOLUTION

Given the objective focal length $\mathbf{f_o=15\ m}$ and the eyepiece focal length $\mathbf{f_e=1.0\ cm=0.01\ m}$.

The angular magnification of a telescope in normal adjustment is $m=\frac{f_o}{f_e}$.

$$m=\frac{15}{0.01}=1500.$$

The angular magnification is $m=1500$.

For the diameter of the moon's image, first find the angle subtended by the moon at the objective. The moon of diameter $\mathbf{D_m=3.42\times 10^6\ m}$ is at distance $\mathbf{r=3.8\times 10^8\ m}$, so its angular size is $\theta=\frac{D_m}{r}$.

$$\theta=\frac{3.42\times 10^{6}}{3.8\times 10^{8}}=9.0\times 10^{-3}\ \text{rad}.$$

The objective forms the moon's image in its focal plane. The image subtends the same angle θ at the objective, so the image diameter is $d=f_o\,\theta$.

$$d=15\times 9.0\times 10^{-3}=0.135\ m.$$

Converting, $d=0.135\ m=13.5$ cm.

The angular magnification is 1500 and the diameter of the moon's image formed by the objective is 13.5 cm.

CONCEPT

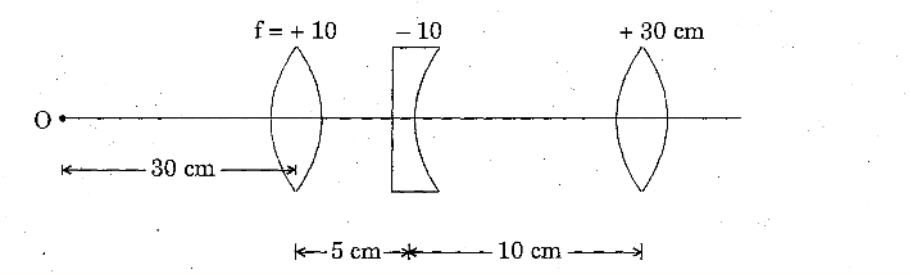
The image at the objective's focal plane subtends the same angle theta as the distant object, so its size is simply f_o times theta.

Final answer. Angular magnification = 1500 and the diameter of the moon's image is 0.135 m (13.5 cm).

← Back to Q289

Q290

Find the position of the image formed of the object 'O' by the lens combination given in the figure.



[2011 • Set 55-2-1]

SOLUTION

The combination has lens L_1 of focal length $\mathbf{f_1=+10\ cm}$, then L_2 of focal length $\mathbf{f_2=+30\ cm}$ placed 5 cm to its right; the object *O* is 30 cm in front of L_1. We treat the lenses one after another, using the image from each as the object for the next.

For L_1: object distance $\mathbf{u_1=-30\ cm}$, $\mathbf{f_1=+10\ cm}$.

$$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1} \Rightarrow \frac{1}{v_1}=\frac{1}{10}+\frac{1}{(-30)}=\frac{1}{10}-\frac{1}{30}.$$

$$\frac{1}{v_1}=\frac{3-1}{30}=\frac{2}{30}=\frac{1}{15} \Rightarrow v_1=+15\ cm.$$

So L_1 would form its image 15 cm to its right, which is $\mathbf{15-5=10\ cm}$ to the right of L_2. This acts as the object for L_2, on the far (right) side, so it is a virtual object with $\mathbf{u_2=+10\ cm}$.

For L_2: $\mathbf{u_2=+10\ cm}$, $\mathbf{f_2=+30\ cm}$.

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2} \Rightarrow \frac{1}{v_2}=\frac{1}{30}+\frac{1}{10}=\frac{1+3}{30}=\frac{4}{30}=\frac{2}{15}.$$

$$v_2=\frac{15}{2}=+7.5\ cm.$$

The final image is formed 7.5 cm to the right of L_2, so it is a real image.

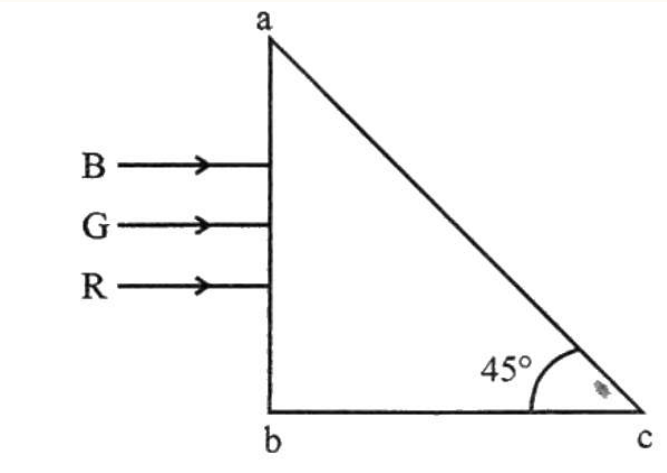
The final image is real and forms 7.5 cm to the right of the second lens L_2.

Final answer. The final image is real and forms 7.5 cm to the right of the second lens L2.

← *Back to Q290*

Q291

Three light rays red (R), green (G) and blue (B) are incident on a right angled prism abc *at face* ab. *The refractive indices of the material of the prism for red, green and blue wavelengths are* 1.39, 1.44 *and* 1.47 *respectively. Out of the three which colour ray will emerge out of face* ac*? Justify your answer. Trace the path of these rays after passing through face* ab.



[2009]

SOLUTION

The prism is a right-angled isosceles prism with angles 45°– 45°– 90°. The three rays strike face ab normally, pass straight in without bending, and then hit the hypotenuse face ac at an angle of incidence of 45°.

At face ac a ray will emerge (refract out) only if it does NOT undergo total internal reflection. Total internal reflection occurs when the angle of incidence inside the prism exceeds the critical angle i_c, where

$$\sin i_c = \frac{1}{n}$$

Here the angle of incidence at ac is 45°, so a ray escapes only if $\mathbf{45° < i_c}$, i.e. only if $\mathbf{\sin 45° < \sin i_c}$.

$$\sin 45° < \frac{1}{n} \Rightarrow \frac{1}{\sqrt{2}} < \frac{1}{n} \Rightarrow n < \sqrt{2} \approx 1.414$$

Now check each colour against this condition $n < 1.414$:

Red: $n_R = 1.39 < 1.414$, so $i_c > 45°$. The condition for escape is satisfied, so red refracts out through face ac.

Green: $n_G = 1.44 > 1.414$, so $i_c < 45°$. The 45° incidence exceeds the critical angle, so green suffers total internal reflection at ac.

Blue: $n_B = 1.47 > 1.414$, so again $i_c < 45°$ and blue also suffers total internal reflection at ac.

Therefore only the **red ray emerges out of face ac**, while green and blue are totally internally reflected.

Path tracing: All three rays enter normally at ab and travel undeviated to the hypotenuse ac. At ac, the red ray bends away from the normal and emerges out of ac. The green and blue rays are turned through 90° by total internal reflection at ac and travel down to face bc, where they strike normally and pass straight out of the prism through face bc.

TIP

Set the escape condition as n < 1/sin(angle of incidence). At 45 degrees this gives the famous n < 1.414 threshold for a right-angled prism.

Final answer. Only the red ray (n = 1.39 < 1.414) emerges from face ac; green and blue undergo total internal reflection at ac and exit through face bc.

← Back to Q291

Q292

Draw a plot showing the variation of power of a lens with the wavelength of the incident light.
A diverging lens of refractive index 1.5 *and of focal length* 20 cm *in air has the same*

radii of curvature for both sides. If it is immersed in a liquid of refractive index 1.7*, calculate the focal length of the lens in the liquid.*

[2008]

SOLUTION

Plot of power versus wavelength: The power of a lens is $P = \frac{1}{f} = (n-1)(\frac{1}{R_1} - \frac{1}{R_2})$, so $P \propto (n-1)$. Since the refractive index n of the lens material decreases as the wavelength λ of light increases (normal dispersion: n is larger for violet, smaller for red), the power P also decreases as λ increases.

The graph is a curve starting high at short wavelengths (violet end) and falling steadily as wavelength increases towards the red end. Plot P on the vertical axis and λ on the horizontal axis: the curve slopes downward from left (high P, small λ) to right (low P, large λ), concave upward.

Numerical part. For the lens in air, refractive index of glass $\mathbf{n_g = 1.5}$ and focal length $\mathbf{f_a = -20}$ cm (diverging lens, so negative). Both radii are equal in magnitude; for a biconcave lens $\mathbf{R_1 = -R}$ and $\mathbf{R_2 = +R}$. Using the lens-maker's formula in air:

$$\frac{1}{f_a} = (n_g - 1)(\frac{1}{R_1} - \frac{1}{R_2}) = (1.5-1)(\frac{1}{-R} - \frac{1}{R}) = 0.5(-\frac{2}{R}) = -\frac{1}{R}$$

When the lens is immersed in the liquid of refractive index $\mathbf{n_l = 1.7}$, the relative refractive index becomes $\frac{\mathbf{n_g}}{\mathbf{n_l}} = \frac{\mathbf{1.5}}{\mathbf{1.7}}$, so:

$$\frac{1}{f_l} = (\frac{n_g}{n_l} - 1)(\frac{1}{R_1} - \frac{1}{R_2}) = (\frac{1.5}{1.7} - 1)(-\frac{2}{R})$$

Dividing the two relations to eliminate the geometry term:

$$\frac{f_l}{f_a} = \frac{(n_g - 1)}{(\frac{n_g}{n_l} - 1)} = \frac{1.5-1}{\frac{1.5}{1.7} - 1} = \frac{0.5}{0.8824 - 1} = \frac{0.5}{-0.1176}$$

$$\frac{f_l}{f_a} = -4.25$$

$$f_l = -4.25 \times f_a = -4.25 \times (-20) = +85 \text{ cm}$$

The focal length in the liquid is $\mathbf{f_l = +85\ cm}$. The positive sign means the diverging lens in air now behaves as a converging lens in the liquid, because

the liquid is optically denser than the lens material ($\mathbf{n_l > n_g}$), which reverses the lens action.

Final answer. Focal length in the liquid is +85 cm; the lens now acts as a converging lens.

← *Back to Q292*

Q293

Define the term 'resolving power' of an astronomical telescope. How does it get affected on
(i) increasing the aperture of the objective lens?
(ii) increasing the wavelength of the light used?
Justify your answer in each case.

[2007]

SOLUTION

Resolving power of an astronomical telescope is defined as the reciprocal of the smallest angular separation dθ between two distant point objects (such as two stars) that the telescope can just distinguish as separate. By the Rayleigh criterion, this limit of resolution is:

$$d\theta = \frac{1.22\,\lambda}{D}$$

where λ is the wavelength of light used and D is the diameter (aperture) of the objective lens. Therefore the resolving power is:

$$\text{Resolving power} = \frac{1}{d\theta} = \frac{D}{1.22\,\lambda}$$

(i) Effect of increasing the aperture D of the objective: Since the resolving power $\propto \boldsymbol{D}$, increasing the aperture of the objective lens increases the resolving power. A larger aperture produces a smaller diffraction (Airy) disc for each star, so two close stars can be resolved more easily.

(ii) Effect of increasing the wavelength λ: Since the resolving power $\propto \frac{1}{\lambda}$, increasing the wavelength of the light used decreases the resolving power. A longer wavelength produces a larger diffraction disc, so the two images overlap more and become harder to separate.

Final answer. Resolving power = D/(1.22 lambda); it increases when the objective aperture increases and decreases when the wavelength increases.

← *Back to Q293*

Q294

A double convex lens made of a glass of refractive index 1.6 has its both surfaces of equal radii of curvature of 30 cm each. An object of height 5 cm is placed at a distance of 12.5 cm from the lens. Calculate the size of the image formed.

[2007]

SOLUTION

First find the focal length of the lens using the lens-maker's formula. For a double-convex (biconvex) lens with equal radii, $\mathbf{R_1 = +30}$ cm and $\mathbf{R_2 = -30}$ cm, with $\boldsymbol{n = 1.6}$:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.6-1)\left(\frac{1}{30} - \frac{1}{-30}\right)$$

$$\frac{1}{f} = 0.6 \times \frac{2}{30} = 0.6 \times \frac{1}{15} = \frac{0.6}{15} = \frac{1}{25}$$

$$f = +25 \text{ cm}$$

Now apply the thin-lens formula. The object is placed at $\boldsymbol{u = -12.5}$ cm (real object, to the left), and $\boldsymbol{f = +25}$ cm:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{25} + \frac{1}{-12.5} = \frac{1}{25} - \frac{2}{25} = -\frac{1}{25}$$

$$v = -25 \text{ cm}$$

The image is virtual and on the same side as the object. The magnification is:

$$m = \frac{v}{u} = \frac{-25}{-12.5} = +2$$

The positive magnification confirms the image is virtual and erect. The size of the image is:

$$h' = m \times h = 2 \times 5 = 10 \text{ cm}$$

The image is **10 cm tall**, virtual, erect and magnified two times.

Final answer. The image is 10 cm tall, virtual, erect and twice magnified.

← *Back to Q294*

Q295

A convex lens made up of glass of refractive index 1.5 *is dipped, in turn, in:*
(i) medium A of refractive index 1.65
(ii) medium B of refractive index 1.33
Explain, giving reasons, whether it will behave as a converging lens or a diverging lens in each of these two media.

[2006]

SOLUTION

The behaviour of a lens placed in a surrounding medium is governed by the lens-maker's formula written with the medium:

$$\frac{1}{f} = \left(\frac{n_g}{n_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

The geometry term $\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ is fixed (positive for a convex lens). So the sign of the focal length, and hence whether the lens converges or diverges, depends entirely on the sign of the factor $\left(\frac{n_g}{n_m} - 1\right)$. Here $\mathbf{n_g = 1.5}$.

(i) In medium A, $\mathbf{n_A = 1.65}$:

$$\frac{n_g}{n_A} = \frac{1.5}{1.65} = 0.91 < 1 \Rightarrow \left(\frac{n_g}{n_A} - 1\right) < 0$$

The factor is negative, so $\frac{1}{f}$ becomes negative and f is negative. The convex lens therefore behaves as a **diverging lens in medium A**. This is because the medium is optically denser than the glass ($\mathbf{n_A > n_g}$), which reverses the usual bending of the rays.

(ii) In medium B, $\mathbf{n_B = 1.33}$:

$$\frac{n_g}{n_B} = \frac{1.5}{1.33} = 1.13 > 1 \Rightarrow \left(\frac{n_g}{n_B} - 1\right) > 0$$

The factor is positive, so $\frac{1}{f}$ stays positive and f is positive. The convex lens still behaves as a **converging lens in medium B**, because the glass is still denser than the medium ($\mathbf{n_g > n_B}$). However, since $\left(\frac{n_g}{n_B} - 1\right)$ is smaller than in air, its focal length increases (it becomes a weaker converging lens).

CONCEPT

The single deciding factor is whether the lens material is denser (converges as usual) or rarer (action reverses) than the surrounding medium.

Final answer. In medium A (n = 1.65) the convex lens behaves as a diverging lens; in medium B (n = 1.33) it remains a converging lens but weaker.

← *Back to Q295*

Q296

A figure divided into squares, each of size 1 mm^2*, is being viewed at a distance of* 9 cm *through a magnifying lens of focal length* 10 cm*, held close to the eye.*
(a) *Draw a ray diagram showing the formation of the image.*
(b) *What is the magnification produced by the lens? How much is the area of each square in the virtual image?*
(c) *What is the angular magnification of the lens?*

[2005]

SOLUTION

Given a magnifying lens (simple microscope) of focal length $f = +10$ cm, with the object (the squared figure) at distance $u = -9$ cm, held close to the eye, with the least distance of distinct vision $D = 25$ cm.

(a) Ray diagram: The object lies between the lens and its focus (since **9 cm < 10 cm**). Draw the convex lens with its principal axis. Place the object as a small upright arrow within the focal length on the left. From the top of the object draw one ray parallel to the axis, which after refraction passes through the focus F on the right. Draw a second ray through the optical centre, which goes straight undeviated. These two refracted rays diverge on the right side, so extend them backwards (dashed) to the left; they meet to form a virtual, erect and magnified image, larger and farther than the object on the same side as the object.

(b) Magnification and area. Using the lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} + \frac{1}{-9} = \frac{9 - 10}{90} = -\frac{1}{90}$$

$$v = -90 \text{ cm}$$

The linear (transverse) magnification is:

$$m = \frac{v}{u} = \frac{-90}{-9} = 10$$

So each square of side 1 mm becomes a square of side $10 \times 1 = 10$ mm in the image. The area of each square in the virtual image is:

$$A' = (10 \text{ mm}) \times (10 \text{ mm}) = 100 \text{ mm}^2 = 1 \text{ cm}^2$$

(c) Angular magnification. The angular magnification of a simple magnifier (when the eye is held close to the lens) is the ratio of the least distance of distinct vision to the object distance:

$$m_\theta = \frac{D}{|u|} = \frac{25}{9} \approx 2.8$$

So the **linear magnification is 10**, the **image area of each square is 100 mm^2**, and the **angular magnification is about 2.8**.

> **COMMON MISTAKE**
>
> *Linear magnification (v/u) and angular magnification (D/u) are different quantities; the area scales as the square of the linear magnification, not the angular one.*

Final answer. Linear magnification = 10, image area of each square = 100 mm^2, angular magnification = 25/9 about 2.8.

← Back to Q296

Q297

An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

[2026 • Set 55-4-1]

SOLUTION

An astronomical telescope is used to view distant objects (stars, planets) in normal adjustment, with the final image formed at infinity.

Ray diagram description: Draw two thin convex lenses on a common principal axis. The left lens is the objective (large focal length f_o, large aperture); the right lens is the eyepiece (small focal length f_e, small aperture). Parallel rays come from a distant object and enter the objective at a small angle α to the axis. The objective converges them to form a real, inverted, diminished image $A'B'$ at its focus F_o. This intermediate image lies at the common focal point, since in normal adjustment the focus of the eyepiece coincides with the focus of the objective. The eyepiece then treats $A'B'$ as its object and, acting as a magnifier, produces a highly magnified virtual image at infinity. The rays finally emerge parallel, making a larger angle β with the axis as they enter the eye.

Working: The objective collects the parallel rays from the distant object and forms a real inverted image at its focal plane. The eyepiece is positioned so that this image lies at (or within) its focal length, so the eyepiece magnifies it like a simple microscope. The final image seen by the eye is virtual, inverted relative to the object, and greatly magnified in angular size.

Magnifying power in normal adjustment is the ratio of the angle β subtended at the eye by the final image to the angle α subtended by the object at the unaided eye:

$$m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$$

The length of the telescope tube (distance between the two lenses) in normal adjustment is:

$$L = f_o + f_e$$

To obtain high magnification the objective should have a **large focal length f_o** and the eyepiece a **small focal length f_e**.

Final answer. Magnifying power m = f_o / f_e and tube length L = f_o + f_e in normal adjustment.

← *Back to Q297*

Q298

A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Considering image formation by a single spherical surface successively at the two surfaces of a lens, lens maker's formula is obtained. It is useful to design lenses of desired focal length using surfaces of suitable radii of curvature. This formula helps us obtain a relation between u, v and f for a lens. Lenses form images of objects and they are used in a number of optical devices, for example microscopes and telescopes.

[2023 • Set 55-5-1]

SOLUTION

Consider a thin lens of refractive index n placed in air, made of two spherical surfaces of radii R_1 and R_2. An object is placed at O at distance u on the principal axis. We use the formula for refraction at a single spherical surface, $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$, twice, once at each surface.

At the first surface (radius R_1): light goes from air ($n_1 = 1$) into glass ($n_2 = n$). If the image formed by this surface alone is at distance v_1:

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1} \quad \ldots(i)$$

At the second surface (radius R_2): the image I_1 from the first surface acts as the (virtual) object. Light now goes from glass (n) into air (1). Since the lens is thin, the object distance for the second surface is still v_1:

$$\frac{1}{v} - \frac{n}{v_1} = \frac{1-n}{R_2} = -\frac{n-1}{R_2} \quad \ldots(ii)$$

Adding equations (i) and (ii), the $\frac{n}{v_1}$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = (n-1)\frac{1}{R_1} - (n-1)\frac{1}{R_2}$$

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \ldots(iii)$$

For an object at infinity ($u = \infty$), the image is formed at the focus, so $v = f$. Substituting in (iii):

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

This is the **lens-maker's formula**. Comparing with (iii), we also obtain the thin-lens relation $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. Lens designers use this formula to choose the radii R_1 and R_2 and the material index n needed to produce a lens of a desired focal length f.

Final answer. The lens-maker's formula 1/f = (n - 1)(1/R1 - 1/R2) follows by adding the refraction relations at the two surfaces.

← *Back to Q298*

Q299

Mrs. Rashmi Singh broke her reading glasses. When she went to the shopkeeper to order new spects, he suggested that she should get spectacles with plastic lenses instead of glass lenses. On getting the new spectacles, she found that the new ones were thicker than the earlier ones. She asked this question to the shopkeeper but he could not offer satisfactory explanation for this. At home, Mrs. Singh raised the same question to her daughter Anuja who explained why plastic lenses were thicker.
(a) Write two qualities displayed each by Anuja and her mother.
(b) How do you explain this fact using lens maker's formula?

[2017]

SOLUTION

(a) Two qualities (values) displayed by each:

Anuja: scientific awareness and knowledge of optics; helpful and patient nature, willingness to explain a concept to her mother.

Mrs. Rashmi Singh: a curious and inquisitive mind, eager to know the reason behind an observation; a rational outlook, not blindly accepting things but seeking a proper explanation.

(b) Explanation using the lens-maker's formula. For a lens of refractive index n with equal radii of curvature R, the power is given by:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a given pair of spectacles, both lenses must have the same focal length f (the same corrective power), so the product $(n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ must be the same for both.

Plastic has a smaller refractive index ($\boldsymbol{n \approx 1.49}$) than glass ($\boldsymbol{n \approx 1.52}$). Since $(n-1)$ is smaller for plastic, the curvature term $\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ must be correspondingly larger to keep $\frac{1}{f}$ the same. A larger curvature term means smaller radii of curvature, i.e. the surfaces must be more sharply curved (more bulging).

Therefore, to give the same power, the **plastic lens must be more strongly curved and hence thicker** than the glass lens. This is the explanation Anuja

gave.

Final answer. Plastic has lower refractive index, so to give the same focal power its surfaces must be more curved (smaller R), making the lens thicker.

← *Back to Q299*

Q300

(a) *Using the relation for refraction at a curved spherical surface, derive the expression for lens maker's formula.*
(b) *Three lenses* L_1, L_2 *and* L_3, *each of focal length 40 cm, are placed coaxially. The distance between* L_1 *and* L_2 *and between* L_2 *and* L_3 *are 120 cm and 20 cm respectively. An object is kept at a distance of 80 cm to the left of lens* L_1. *Find the distance of the final image formed from the object.*
OR
(a) *Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. Using this diagram, derive the mirror formula.*
(b) *A concave mirror produces a two times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.*

[2026 • Set 55-1-1]

SOLUTION

(a) Lens-maker's formula. Consider a thin lens of refractive index n in air, with surfaces of radii R_1 and R_2. Apply the single-surface refraction formula $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ at each surface.

At surface 1, light passes from air ($n_1 = 1$) into glass ($n_2 = n$), forming an image at v_1:

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1} \quad ...(1)$$

At surface 2, this image acts as object; light passes from glass (n) into air (1), and since the lens is thin the object distance is v_1:

$$\frac{1}{v} - \frac{n}{v_1} = -\frac{n-1}{R_2} \quad ...(2)$$

Adding (1) and (2):

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For an object at infinity, $v = f$, which gives the lens-maker's formula:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

(b) Three-lens system. Each lens has $f = 40$ cm. The object is 80 cm to the left of L_1, so $u_1 = -80$ cm. Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ at L_1:

$$\frac{1}{v_1} = \frac{1}{40} + \frac{1}{-80} = \frac{2-1}{80} = \frac{1}{80}$$

$$v_1 = +80 \text{ cm}$$

So image I_1 forms 80 cm to the right of L_1. Since L_2 is 120 cm from L_1, this image is $120 - 80 = 40$ cm to the LEFT of L_2. It is a real object for L_2, so $u_2 = -40$ cm:

$$\frac{1}{v_2} = \frac{1}{40} + \frac{1}{-40} = 0$$

$$v_2 = \infty$$

So L_2 sends out a parallel beam (the object was at its focus). These parallel rays now fall on L_3. A parallel beam is focused at the focal point of L_3:

$v_3 = f = +40$ **cm** from L_3

The final image forms 40 cm to the right of L_3. The total distance of the final image from the object is the sum of all spacings:

$$\text{Distance} = 80 + 120 + 20 + 40 = 260 \text{ cm}$$

The final image is formed **260 cm from the object**.

OR (Concave mirror alternative).

(a) Ray diagram and mirror formula. Draw a concave mirror with pole *P*, centre of curvature *C* and focus *F* (with *F* midway between *P* and *C*) on the principal axis. Place an upright object AB between *F* and *C*. From the top *A*, draw one ray

parallel to the axis; after reflection it passes through F. Draw a second ray through C; it strikes the mirror normally and retraces its path. The two reflected rays meet beyond C to form a real, inverted and magnified image $A'B'$.

Derivation: Let the object distance be u, image distance v and radius of curvature $R = 2f$. Using the geometry of the two reflected rays and similar triangles formed by the object, image, pole P and centre C, and applying the sign convention (distances measured from the pole, against the incident direction taken negative), one obtains the mirror formula:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

(b) Magnified virtual image. A virtual image by a concave mirror is erect, so the magnification is positive: $m = +2$. The object is 10 cm in front of the mirror, so $u = -10$ cm. Using $m = -\frac{v}{u}$:

$$+2 = -\frac{v}{-10} = \frac{v}{10} \Rightarrow v = +20 \text{ cm}$$

The positive v confirms a virtual image behind the mirror. Apply the mirror formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{20} + \frac{1}{-10} = \frac{1-2}{20} = -\frac{1}{20}$$

$$f = -20 \text{ cm}$$

The **focal length of the concave mirror is 20 cm** (the negative sign confirms it is concave).

TIP

When a lens forms its image exactly at the focus of the next lens, that lens outputs a parallel beam; the following lens then focuses it at its own focal plane.

Final answer. Lens system: final image is 260 cm from the object. OR concave mirror: focal length = 20 cm (f = −20 cm).

← *Back to Q300*

Q301

(a) Draw a labelled ray diagram of a refracting telescope when it forms image of a distant object at infinity. Derive expression for its magnifying power.

[2026 • Set 55-2-1]

SOLUTION

An astronomical refracting telescope in normal adjustment uses two coaxial thin convex lenses: an objective O of large focal length f_o and large aperture, and an eyepiece E of small focal length f_e, mounted so their principal axes coincide.

Ray diagram (described): Two parallel rays from a distant object arrive at the objective O inclined at a small angle α to the principal axis. They converge to form a real, inverted, diminished intermediate image $A'B'$ of height h' at the focus of the objective. This focal plane is made to coincide with the focus of the eyepiece E. Rays from $A'B'$ then pass through E and emerge parallel, so the final magnified virtual image is formed at infinity for the relaxed (normal-adjustment) eye. Label O and E on the axis, $A'B'$ between them at the common focus, mark f_o from O to $A'B'$ and f_e from $A'B'$ to E, the angle α at O and the angle β that the final emergent rays make at the eye.

The distance between the two lenses (tube length) is the sum of their focal lengths.

$$L = f_o + f_e$$

The magnifying power is the ratio of the angle β subtended at the eye by the final image to the angle α subtended at the unaided eye by the object.

$$m = \frac{\beta}{\alpha}$$

From the intermediate image $\boldsymbol{A'B' = h'}$ at the common focus, the angle subtended at the eye (image at infinity) is

$$\beta \approx \beta = \frac{h'}{f_e}$$

The angle subtended at the objective (and so at the unaided eye) by the distant object equals

$$\alpha \approx \alpha = \frac{h'}{f_o}$$

Dividing, the height h' cancels.

$$m = \frac{\beta}{\alpha} = \frac{h'/f_e}{h'/f_o}$$

$$m = \frac{f_o}{f_e}$$

Thus a long-focal-length objective and a short-focal-length eyepiece give high **magnifying power**. The image is inverted, which is unimportant for astronomical viewing.

TIP

In normal adjustment the final image is at infinity, so $\boldsymbol{m = f_o/f_e}$. If the image is at the near point D instead, the eyepiece contributes an extra factor and $\boldsymbol{m = \frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)}$.

Final answer. The magnifying power in normal adjustment is $\boldsymbol{m = f_o/f_e}$, with tube length $\boldsymbol{L = f_o + f_e}$.

← *Back to Q301*

Q302

(b) (i) In a telescope the objective has much larger aperture than the eyepiece. Why? (ii) Write two advantages of reflecting telescope over refracting telescope.

[2026 • Set 55-2-1]

SOLUTION

(b)(i) The objective is given a much larger aperture than the eyepiece for two linked reasons.

First, light-gathering power. The objective collects light from faint distant objects, and the amount of light gathered is proportional to the area of the aperture.

$$\text{Light gathered} \propto D^2$$

A larger diameter D therefore gathers far more light and forms a much **brighter image**, letting us see faint stars.

Second, resolving power. The smallest angular separation that can be resolved is

$$\Delta\theta = \frac{1.22\,\lambda}{D}$$

so the resolving power $\frac{1}{\Delta\theta} \propto \frac{D}{\lambda}$ increases with aperture. A large objective therefore resolves finer detail of the object. The eyepiece only has to magnify the small intermediate image formed near the axis, so it can be made small with a short focal length for high magnification.

(b)(ii) Two advantages of a reflecting telescope over a refracting telescope:

1. No chromatic aberration. A reflecting telescope uses a mirror as objective, and reflection is the same for all wavelengths, so all colours focus at the same point. A refracting telescope uses a lens objective, which disperses different colours to slightly different foci, producing **chromatic aberration** (coloured fringes).

2. Large aperture with light, stable mounting. A mirror can be supported firmly across its whole back surface, so very large diameters are possible (giving high light-gathering and resolving power). A large lens can be supported only at its edges, so it sags under its own weight, distorting the image, and a thick lens also absorbs light. Reflectors are therefore cheaper to build large and give brighter, sharper images.

Final answer. A large objective aperture maximises light-gathering and resolving power; reflectors beat refractors by having no chromatic aberration and by allowing large, well-supported mirrors.

← *Back to Q302*

Q303

(i) A point object is kept in front of a convex spherical surface of radius of curvature R. Draw the ray diagram to show the formation of image and derive the relation between the object and image distance (u and v) in terms of refractive index n of the medium and R.

[2026 • Set 55-3-1]

SOLUTION

Consider a point object P lying on the principal axis in medium 1 of refractive index n_1, in front of a convex spherical refracting surface of radius of curvature R and centre of curvature C. The light refracts into medium 2 of refractive index n_2 (with $\mathbf{n_2 > n_1}$) and forms a real image I on the axis.

Ray diagram (described): Draw the spherical surface with pole O on the axis and centre C to the right. One ray PO travels along the axis undeviated. A second paraxial ray from P strikes the surface at a point A at small height h above the axis; the normal at A is the line AC (a radius). The ray bends towards the normal (entering a denser medium) and meets the axis at the image I. Mark the angle of incidence i (between PA and AC), the angle of refraction r (between AC and AI), and the three small angles the lines PA, CA, IA make with the axis, namely α at P, γ at C, and β at I. Drop a perpendicular AM of height h to the axis.

For paraxial rays all angles are small, so Snell's law $\mathbf{n_1 \sin i = n_2 \sin r}$ becomes

$$n_1\, i = n_2\, r$$

From the exterior-angle property of triangles, the angle of incidence is an exterior angle of triangle PAC and the angle of refraction is an exterior angle of triangle IAC.

$$i = \alpha + \gamma$$

$$r = \gamma - \beta$$

For small angles, expressing each angle through the perpendicular height $\boldsymbol{h \approx}$ **AM** and the distances $\mathbf{PO = -}\boldsymbol{u}$, $\mathbf{IO = }\boldsymbol{v}$, $\mathbf{CO = }\boldsymbol{R}$ (using the sign convention, distances measured against the incident light are negative):

$$\alpha \approx \frac{h}{-u},\ \beta \approx \frac{h}{v},\ \gamma \approx \frac{h}{R}$$

Substituting i and r into $\mathbf{n_1}\, i = \mathbf{n_2}\, r$:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

$$n_1\left(\frac{h}{-u} + \frac{h}{R}\right) = n_2\left(\frac{h}{R} - \frac{h}{v}\right)$$

Dividing throughout by h and collecting terms:

$$\frac{-n_1}{u} + \frac{n_1}{R} = \frac{n_2}{R} - \frac{n_2}{v}$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2}{R} - \frac{n_1}{R}$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

This is the **refraction-at-a-single-spherical-surface** relation. Applying it in turn to both surfaces of a thin lens gives the lens-maker's formula.

Final answer. The required relation is $\boldsymbol{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$.

← *Back to Q303*

Q304

(ii) A convex lens of focal length of 20 cm *is used to form the image of an object placed* 30 cm *away from the lens. Find the position and nature of the image formed.*

[2026 • Set 55-3-1]

SOLUTION

Given a thin convex (converging) lens of focal length $f = +20$ **cm** with the object placed 30 cm in front of it. Using the Cartesian sign convention, distances measured against the incident light are negative, so the object distance is

$$u = -30 \text{ cm}, f = +20 \text{ cm}$$

Apply the thin-lens equation.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-30}$$

$$\frac{1}{v} = \frac{3-2}{60} = \frac{1}{60}$$

$$v = +60 \text{ cm}$$

The positive sign of v shows the image is formed 60 cm on the far side of the lens (opposite to the object), so it is a **real** image that can be caught on a screen.

Now find the magnification.

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$

The negative sign shows the image is **inverted**, and $|m| = 2$ shows it is magnified to twice the object's size.

Hence the image is formed 60 cm from the lens on the side opposite the object; it is real, inverted, and twice as large as the object.

Final answer. The image is real, inverted and magnified two times, formed **60 cm** from the lens on the side opposite the object.

← *Back to Q304*

Q305

(i) Two thin converging lenses of focal length f_1 *and* f_2 *are placed coaxially in contact. Derive expression for the focal length of the combination.*

[2026 • Set 55-3-1]

SOLUTION

Let two thin converging lenses L_1 and L_2 of focal lengths f_1 and f_2 be placed coaxially in contact with each other. Let a point object O be placed on the common principal axis at distance u.

Ray diagram (described): Draw the two thin lenses touching, sharing one optic axis. The object O is to the left at distance u. The first lens alone would form an image I_1 at distance v_1. This image I_1 then acts as the object for the second lens, which forms the final image I at distance v. Since the lenses are in contact (separation zero), the object distance for the second lens is the same v_1.

For the first lens, taking I_1 as its image:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

The image I_1 acts as the object for the second lens, which forms the final image I at distance v. Because the lenses are in contact, the object distance for the second lens is v_1:

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

Adding the two equations, the term $\frac{1}{v_1}$ cancels:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

If the combination is replaced by a single equivalent lens of focal length F giving the same image for the same object, then by definition

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$$

Comparing the two boxed relations:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Since power $P = 1/f$, this is equivalent to $P = P_1 + P_2$. For N thin lenses in contact this generalises to $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \cdots + \frac{1}{f_N}$, i.e. **powers simply add**.

Final answer. The combined focal length is given by $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$, equivalently $P = P_1 + P_2$.

← Back to Q305

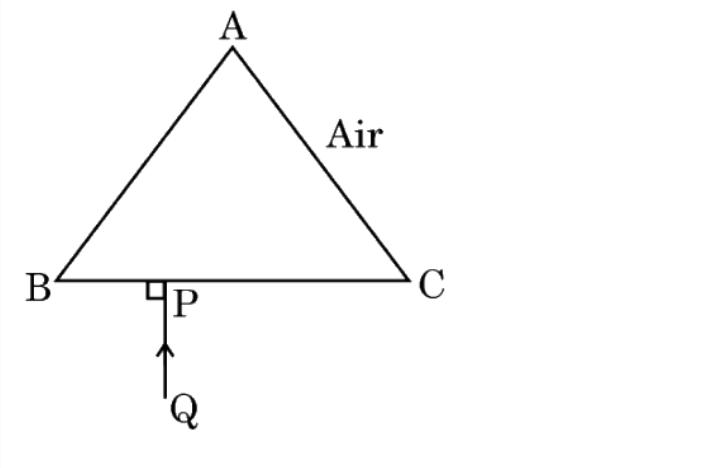
Q306

(a)

(i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum deviation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation.



OR

(b)

(i) What is the difference between a ray and a wavefront?

(ii) A plane wave is incident on a reflecting surface. Using Huygens principle, show how it is reflected from the surface. Hence, verify the law of reflection.

(iii) Depict refraction of a plane wave by a convex lens.

[2026 • Set 55-4-1]

SOLUTION

(a)(i) The refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in that medium.

$$n = \frac{c}{v}$$

Here c is the speed of light in vacuum and v is its speed in the medium. A larger n means light travels slower in that medium (an optically denser medium).

(a)(ii) Prism formula. Let a ray pass through a prism of refracting angle A, with angles of incidence and emergence i and e, internal refraction angles r_1 and r_2, and deviation δ. In general

$$\delta = i + e - A,\ r_1 + r_2 = A$$

At minimum deviation $\boldsymbol{\delta = \delta_m}$ the ray passes symmetrically through the prism, so $\boldsymbol{i = e}$ and $\boldsymbol{r_1 = r_2 = r}$.

$$r_1 + r_2 = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$$

$$\delta_m = i + i - A \Rightarrow i = \frac{A + \delta_m}{2}$$

Applying Snell's law at the first face, with the prism material of index n in air:

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

(a)(iii) Ray QP incident normally on face BC, $\boldsymbol{n = 1.5}$. First find the critical angle.

$$\theta_c = \frac{1}{n} = \frac{1}{1.5} = 0.667$$

$$\theta_c = \sin^{-1}(0.667) \approx 41.8^\circ$$

Ray path (described): Since QP is incident normally on BC, it enters the prism straight through without bending. Inside the prism it travels to the hypotenuse face AC of the right-angled prism, striking it at 45° (the geometry of the 45°-45°-90° prism). Because

$$45^\circ > \theta_c\ (\approx 41.8^\circ)$$

the angle of incidence inside the denser medium exceeds the critical angle, so **total internal reflection** occurs at AC. The ray is turned through 90° and then strikes face AB normally, emerging without bending. Mark on the diagram: ray QP entering BC at 90°, hitting AC at 45°, reflecting through 90°, and leaving AB along the new normal direction. Such totally reflecting prisms are used in periscopes and binoculars.

CONCEPT

Total internal reflection needs two conditions together: light must go from denser to rarer medium, and the angle of incidence must exceed the critical angle $\theta_c = \sin^{-1}(1/n)$.

Final answer. $n = c/v$; the prism formula is $n = \dfrac{\sin[(A+\delta_m)/2]}{\sin(A/2)}$; and since the internal angle **45°** exceeds the critical angle **≈ 41.8°**, the ray undergoes total internal reflection at AC and turns through **90°**.

← *Back to Q306*

Q307

(b) (i) With the help of a ray diagram, describe the construction and working of a compound microscope.
(ii) (I) The real image of an object placed between f and 2f from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there? Explain.
(II) Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances? Explain.

[2026 • Set 55-5-1]

SOLUTION

(b)(i) Compound microscope. It has two converging lenses on a common axis: an objective of short focal length f_o and small aperture, and an eyepiece of focal length f_e, separated by the tube length L.

Ray diagram (described): Place a small object AB just beyond the objective's focus F_o (between f_o and $2f_o$). The objective forms a real, inverted, magnified intermediate image $A'B'$ on the far side, inside the focal length of the eyepiece. The eyepiece, used as a simple magnifier, then forms a virtual, erect (relative to $A'B'$), still-more-magnified final image $A''B''$ at the near point (least distance of distinct vision $\boldsymbol{D = 25\ \text{cm}}$). Label objective O, eyepiece E, foci F_o and F_e, the object AB, intermediate image $A'B'$, final image $A''B''$, and the tube length L between the lenses.

Working: the objective produces the first magnification, then the eyepiece magnifies that intermediate image again, giving a large total magnification.

$$m = m_o \times m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

(b)(ii)(I) Object between f and 2f, screen removed. Yes, the **image is still there**. The lens makes the refracted rays actually converge and cross at a definite point in space, so a real image exists there independently of any screen. The screen does not create the image; it merely intercepts the converging light and scatters it so our eye can see it. If we place a screen at that location the image appears; remove it and the image still exists at the same point in space (it can also be seen by looking directly into the converging beam).

(b)(ii)(II) Plane and convex mirrors producing real images. Yes, both can form a **real image** when the object is a virtual object, i.e. when a converging beam of light is directed at the mirror. For a plane mirror, if a converging beam heading toward a point behind the mirror is intercepted, the reflected rays actually meet in front of the mirror, forming a real image. Likewise a convex mirror receiving a converging beam (virtual object) can reflect the rays to a real focus in front of it. The general rule is that a real image forms whenever the reflected or refracted rays actually converge and meet at a point.

Final answer. The compound microscope gives $\boldsymbol{m = \frac{L}{f_o}\left(1+\frac{D}{f_e}\right)}$; the real image persists without a screen, and both plane and convex mirrors give real images for a virtual (converging-beam) object.

← *Back to Q307*

Q308

(i) A thin pencil of length (f/4) is placed coinciding with the principal axis of a mirror of focal length f. The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror.
(ii) A ray of light is incident on a refracting face AB of a prism ABC at an angle of 45°. *The ray emerges from face AC and the angle of deviation is* 15°. *The angle of prism is* 30°. *Show that the emergent ray is normal to the face AC from which it emerges out. Find the refraction index of the material of the prism.*

[2025 • Set 55-2-1]

SOLUTION

(i) A thin pencil of length $\frac{f}{4}$ lies along the principal axis of a concave mirror of focal length f, and its real, enlarged image just touches the pencil. The phrase just touches means one end of the pencil coincides with its own image. A point coincides with its image only at the centre of curvature, where $\boldsymbol{u = v = 2f}$.

So the farther end of the pencil sits at the centre of curvature, and since the image must be enlarged the pencil extends from the centre of curvature towards the mirror. Taking magnitudes with the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (object distances measured from the pole), the two ends are

$$u_{far} = 2f \text{ (touching end, at centre of curvature)}$$

$$u_{near} = 2f - \frac{f}{4} = \frac{7f}{4}$$

The far end at $\boldsymbol{u = 2f}$ images onto itself, giving the touch point at 2f. Now find the image of the near end at $\mathbf{u_{near}} = \frac{\mathbf{7f}}{\mathbf{4}}$.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u_{near}} = \frac{1}{f} - \frac{4}{7f} = \frac{7-4}{7f} = \frac{3}{7f}$$

$$v = \frac{7f}{3}$$

The image therefore runs from 2f (image of the far end) to $\frac{7f}{3}$ (image of the near end), so the image length is

$$L_{img} = \left|\frac{7f}{3} - 2f\right| = \left|\frac{7f - 6f}{3}\right| = \frac{f}{3}$$

Since $\frac{f}{3} > \frac{f}{4}$, the image is indeed enlarged, as required. The longitudinal magnification is the ratio of image length to object length.

$$m = \frac{L_{img}}{L_{obj}} = \frac{f/3}{f/4} = \frac{4}{3}$$

The magnification produced by the mirror is $\frac{4}{3}$.

(ii) Prism with $i = 45°$, $A = 30°$, $\delta = 15°$. Using the deviation relation:

$$\delta = i + e - A$$

$$e = A + \delta - i = 30° + 15° - 45° = 0°$$

Since the emergence angle is $0°$, the emergent ray leaves face AC along its normal, i.e. perpendicular to AC.

With $e = 0°$, the refraction angle at the second face is $r_2 = 0°$. From $r_1 + r_2 = A$:

$$r_1 = A - r_2 = 30° - 0° = 30°$$

Apply Snell's law at the first face AB:

$$\sin i = n \sin r_1$$

$$\sin 45° = n \sin 30°$$

$$n = \sin 45° \sin 30° = 1/\sqrt{2}\,1/2 = 2\sqrt{2} = \sqrt{2}$$

The refractive index of the prism is $n = \sqrt{2} \approx 1.414$.

TIP

When the angle of emergence comes out as $0°$, it means the ray hits the second face along its normal, so $r_2 = 0$ and $r_1 = A$. That is the quickest route to proving the emergent ray is perpendicular to that face.

Final answer. (i) The longitudinal magnification produced is $\frac{4}{3}$; (ii) since $e = 0°$ the emergent ray is normal to AC, and the refractive index is $n = \sqrt{2} \approx 1.41$.

← Back to Q308

Q309

(a)

(i) An object is placed 30 cm *from a thin convex lens of focal length* 10 cm. *The lens forms a sharp image on a screen. If a thin concave lens is placed in contact with the convex lens, the sharp image on the screen is formed when the screen is moved by* 45 cm *from its initial position. Calculate the focal length of the concave lens.*

(ii) Calculate the angle of minimum deviation of an equilateral prism. The refractive index of the prism is $\sqrt{3}$. *Calculate the angle of incidence for this case of minimum deviation also.*

OR

(b)

(i) A physics teacher wants to demonstrate interference with the help of double slit experiment using a laser beam of 633 nm *wavelength. Since the hall is large enough, interference pattern is formed on the wall* 5.0 *m from the slits. For clear and comfortable view by all the students they want the fringe width* 5 mm.

(I) Find the slit separation for obtaining the desired interference pattern.

(II) How far will the first minimum be from the central maximum ?

(ii) A parallel beam of light of wavelength 650 nm *passes through a slit of width* 0.6 mm. *The diffraction pattern is obtained on a screen kept* 60 cm *away from the slit. Find the distance between first order minima on both sides of the central maximum.*

[2025 • Set 55-4-1]

SOLUTION

(a)(i) Convex lens first. Convex lens $\mathbf{f_1 = +10\ cm}$, object at $\boldsymbol{u = -30\ cm}$.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{-30} = \frac{3-1}{30} = \frac{2}{30}$$

$$v = 15\ \text{cm}$$

So the sharp image initially forms 15 cm from the lens. When the concave lens is added in contact, the screen must be moved 45 cm farther (the image becomes less convergent, so it forms farther away).

$$v' = 15 + 45 = 60 \text{ cm}$$

For the combination with the same object $u = -30$ cm and new image $v' = 60$ cm, the equivalent focal length F is:

$$\frac{1}{F} = \frac{1}{v'} - \frac{1}{u} = \frac{1}{60} - \frac{1}{-30} = \frac{1}{60} + \frac{2}{60} = \frac{3}{60} = \frac{1}{20}$$

$$F = 20 \text{ cm}$$

Now use the lenses-in-contact relation with $f_1 = 10$ cm and the unknown concave lens f_2:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f_2} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{20} - \frac{1}{10} = \frac{1-2}{20} = -\frac{1}{20}$$

$$f_2 = -20 \text{ cm}$$

The negative sign confirms it is a **concave (diverging) lens of focal length 20 cm**.

(a)(ii) Equilateral prism, $n = \sqrt{3}$, so $A = 60°$ and $A/2 = 30°$. Use the minimum-deviation prism formula:

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sqrt{3} = \frac{\sin\left(\frac{60° + \delta_m}{2}\right)}{\sin 30° = 2 \sin\left(\frac{60° + \delta_m}{2}\right)}$$

$$\sin\left(\frac{60° + \delta_m}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\frac{60° + \delta_m}{2} = 60°$$

$$60°+ \delta_m = 120° \Rightarrow \delta_m = 60°$$

The angle of incidence in this minimum-deviation case is

$$i = \frac{A + \delta_m}{2} = 60°+ 60°2 = 60°$$

$$\delta_m = 60°,\ i = 60°$$

Final answer. The concave lens has focal length **−20 cm**; for the equilateral prism the minimum deviation is $\boldsymbol{\delta_m = 60°}$ at an angle of incidence $\boldsymbol{i = 60°}$.

← *Back to Q309*

Q310

(a) (i) *Draw a ray diagram of a reflecting telescope (Cassegrain) and explain the formation of image. State two important advantages that a reflecting telescope has over a refracting telescope.*
(ii) *In a refracting telescope, the focal length of the objective is* 50 *times the focal length of the eyepiece. When the final image is formed at infinity, the length of the tube is* 102 cm. *Find the focal lengths of the two lenses.*
OR
(b) (i) *Write any two advantages of a compound microscope over a simple microscope. Draw a ray diagram for the image formation at the near point by a compound microscope and explain it.*
(ii) *A thin planoconcave lens with its curved face of radius of curvature R is made of glass of refractive index* n_1. *It is placed coaxially in contact with a thin equiconvex lens of same radius of curvature of refractive index* n_2. *Obtain the power of the combination lens.*

[2025 • Set 55-5-1]

SOLUTION

(a)(i) Cassegrain reflecting telescope. It uses a large concave parabolic primary mirror at the rear of the tube and a small convex secondary mirror near the open front, with the eyepiece behind a central hole in the primary.

Ray diagram (described): Parallel rays from a distant object enter the open front of the tube and fall on the large concave primary mirror at the back. The primary reflects them forward, converging toward its focus. Before they converge, they strike the small convex secondary mirror placed on the axis near the front. The secondary reflects the converging beam back down the tube and through a small hole at the centre of the primary mirror. Behind this hole the eyepiece *E* receives the light and forms the final magnified image (at infinity for the relaxed eye). Label the parabolic primary (concave), the secondary (convex), the central hole, and the eyepiece.

Working: the primary mirror gathers light and forms a real intermediate image; the secondary mirror lengthens the effective focal length and relays the image through the hole to the eyepiece, which is used as a magnifier.

Two advantages over a refracting telescope:

1. No chromatic aberration, because reflection is identical for all wavelengths (a lens objective disperses colours).

2. Large apertures are possible, since a mirror can be supported across its whole back without sagging, giving greater light-gathering and resolving power.

(a)(ii) Numerical. Given $\mathbf{f_o = 50\ f_e}$ and, in normal adjustment, tube length $\mathbf{L = f_o + f_e = 102\ cm}$.

$$f_o + f_e = 102$$

$$50\ f_e + f_e = 102$$

$$51\ f_e = 102$$

$$f_e = 2\ \text{cm}$$

$$f_o = 50 \times 2 = 100\ \text{cm}$$

$$f_o = 100\ \text{cm},\ f_e = 2\ \text{cm}$$

(b)(i) Two advantages of a compound microscope over a simple microscope:

1. It gives a much higher magnification (of the order of hundreds), because magnification occurs in two stages, whereas a simple microscope gives only about ten times.

2. The two-stage magnification (objective then eyepiece) allows fine control and far greater resolving detail than a single lens.

Ray diagram for image at the near point (described): A small object AB is placed just beyond the objective focus F_o. The objective forms a real, inverted, magnified intermediate image $A'B'$ inside the focal length of the eyepiece. The eyepiece then acts as a magnifier and forms a virtual, magnified final image $A''B''$ at the near point $\boldsymbol{D = 25\ \text{cm}}$. Label objective O, eyepiece E, foci, object AB, intermediate image $A'B'$ and final image $A''B''$.

(b)(ii) Plano-concave (index n_1, curved radius R) in contact with equiconvex lens (index n_2, each radius R). Use the lens-maker's formula $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.

For the plano-concave lens, one face is plane ($R_1 = \infty$) and the curved concave face has radius $-R$:

$$\frac{1}{f_1} = (n_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\cdot(-1)\right) = -\frac{(n_1 - 1)}{R}$$

For the equiconvex lens, the two faces have radii $+R$ and $-R$:

$$\frac{1}{f_2} = (n_2 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = \frac{2(n_2 - 1)}{R}$$

The power of the combination is the sum of the powers.

$$P = \frac{1}{f_1} + \frac{1}{f_2} = -\frac{(n_1 - 1)}{R} + \frac{2(n_2 - 1)}{R}$$

$$P = \frac{-(n_1 - 1) + 2(n_2 - 1)}{R} = \frac{-n_1 + 1 + 2n_2 - 2}{R}$$

$$P = \frac{2n_2 - n_1 - 1}{R}$$

Final answer. (a) For the telescope, $\boldsymbol{f_o = 100\ \text{cm}}$ and $\boldsymbol{f_e = 2\ \text{cm}}$; (b) the combination power is $\boldsymbol{P = \frac{2n_2 - n_1 - 1}{R}}$.

← *Back to Q310*

Q311

(a) (i) Draw a ray diagram to show the image formation by a compound microscope. Obtain the expression for the total magnification of the microscope when the final image is formed at infinity.
(ii) In a compound microscope, an object is placed at a distance of 1·5 cm *from the objective of focal length* 1·25 cm. *The eyepiece has a focal length of* 5 cm. *The final image is formed at infinity. Calculate the distance between the objective and the eyepiece.*
OR
(b) (i) Using Huygens' principle, explain the refraction of a plane wavefront, propagating in air, at a plane interface between air and glass. Hence verify Snell's law.
(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

[2025 • Set 55-6-1]

SOLUTION

(a)(i) Ray diagram for a compound microscope with the final image at infinity. Draw the principal axis horizontally. Place the objective lens O on the left and the eyepiece lens E on the right, both converging lenses on the same axis. A small object AB is kept just beyond the focus of the objective (between f_o and $2f_o$), with B on the axis and A above it.

The objective forms a real, inverted, magnified intermediate image $A'B'$ on the far side of O. This image is made to fall exactly at the first focus F_e of the eyepiece. The eyepiece then acts as a simple microscope and the rays leaving it become parallel, so the final image $A''B''$ is formed at infinity and is virtual, inverted (with respect to the object) and highly magnified. Mark the tube length L as the distance between the second focus of the objective and the first focus of the eyepiece.

The magnification of the objective is the ratio of the size of the intermediate image to the object,

$$m_o = \frac{h'}{h} = \frac{L}{f_o}$$

where L is the tube length and f_o the focal length of the objective.

When the final image is at infinity, the eyepiece is used in the relaxed-eye (normal) setting, so its angular magnification is

$$m_e = \frac{D}{f_e}$$

where D**=25 cm** is the least distance of distinct vision and f_e the focal length of the eyepiece.

The total magnification is the product of the two,

$$M = m_o \times m_e$$

$$M = \frac{L}{f_o} \times \frac{D}{f_e}$$

This is the required expression for the **total magnification** of a compound microscope with the final image at infinity.

(a)(ii) For the objective, using the sign convention u_o**=−1.5 cm** and f_o**=+1.25 cm**.

$$\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$$

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{1.25} - \frac{1}{1.5}$$

$$\frac{1}{v_o} = \frac{1.5-1.25}{1.251.5} = \frac{0.25}{1.875} = \frac{1}{7.5}$$

$$v_o = 7.5 \text{ cm}$$

Since the final image is at infinity, the intermediate image must lie at the focus of the eyepiece, so the object distance for the eyepiece equals f_e**=5 cm**.

The distance between the objective and the eyepiece is

$$L = v_o + f_e = 7.5 + 5 = 12.5 \text{ cm}$$

Hence the **distance between the objective and the eyepiece is 12.5 cm**.

OR

(b)(i) Refraction of a plane wavefront by Huygens' principle. Let XY be the plane interface separating air (medium 1, speed v_1) from glass (medium 2, speed v_2, with $v_2 1$). A plane wavefront AB travelling in air strikes the surface so that the

incident rays make an angle of incidence i with the normal; AB touches the surface at A first.

While the wavelet from B travels in air to reach the surface at C, the wavelet from A enters glass. By Huygens' principle every point on AB is a source of secondary wavelets.

In the time t that B reaches C, the distance covered in air is

$$BC=v_1\,t$$

and in the same time the secondary wavelet from A advances in glass by

$$AD=v_2\,t$$

The new refracted wavefront is CD, and the refracted rays make the angle of refraction r with the normal.

In right triangle ABC, the angle $\angle BAC=i$, so

$$\sin i=\frac{BC}{AC}=\frac{v_1\,t}{AC}$$

In right triangle ADC, the angle $\angle ACD=r$, so

$$\sin r=\frac{AD}{AC}=\frac{v_2\,t}{AC}$$

Dividing the two equations,

$$\frac{\sin i}{\sin r}=\frac{v_1}{v_2}=\text{constant}=n_{21}$$

Since $v_1=\frac{c}{n_1}$ and $v_2=\frac{c}{n_2}$, we get $\frac{v_1}{v_2}=\frac{n_2}{n_1}$, hence

$$n_1\sin i=n_2\sin r$$

which is **Snell's law**, verified using Huygens' principle.

(b)(ii) For a convex mirror the focal length is positive, $f>0$. A real object placed in front of the mirror has $u<0$.

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

With $f>0$ the term $\frac{1}{f}$ is positive, and with $u<0$ the term $-\frac{1}{u}$ is also positive.

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{|u|}>0$$

Therefore $v>0$ for every position of a real object. A positive v means the image lies behind the mirror, so it is always **virtual** (and, being behind the mirror, erect). Hence a convex mirror always forms a virtual image of an object kept in front of it.

TIP

In a compound microscope the tube length L is the distance between the second focus of the objective and the first focus of the eyepiece, not the full lens-to-lens distance, but for image at infinity the lens separation equals v_o+f_e.

Final answer. Distance between objective and eyepiece **=12.5 cm**; Snell's law $n_1 \sin i = n_2 \sin r$ verified; convex mirror gives $v>0$ so image is always virtual.

← *Back to Q311*

Q312

(a) (i) *Explain with the help of a labelled ray diagram the formation of final image by an astronomical telescope at infinity. Write the expression for its magnifying power.*
(ii) *The total magnification produced by a compound microscope is* 20. *The magnification produced by the eyepiece is* 5. *When the microscope is focussed on a certain object, the distance between the objective and eyepiece is observed to be* 14 cm. *Calculate the focal lengths of the objective and the eyepiece. (Given that the least distance of distinct vision* **= 25 cm**)
OR
(b) (i) *Two coherent light waves, each of intensity* I_0 *superpose each other and produce interference pattern on a screen. Obtain the expression for the resultant intensity at a point where the phase difference between the waves is* ϕ. *Write its maximum and minimum possible values.*

(ii) *In a single slit diffraction experiment, the aperture of the slit is* 3 mm *and the separation between the slit and the screen is* 1.5 *m. A monochromatic light of wavelength* 600 nm *is normally incident on the slit. Calculate the distance of (I) first order minimum, and (II) second order maximum, from the centre of the screen.*

[2025 • Set 55-7-1]

SOLUTION

(a)(i) Ray diagram for an astronomical telescope in normal adjustment (final image at infinity). Draw the principal axis. Place the objective O (large aperture, long focal length f_o) on the left and the eyepiece E (small focal length f_e) on the right. Parallel rays from a distant object arrive at O making an angle α with the axis.

The objective converges these rays to form a real, inverted, diminished image $A'B'$ at its focus F_o. The separation of the lenses is adjusted so that this image also lies at the first focus F_e of the eyepiece (so F_o and F_e coincide). The eyepiece then sends out parallel rays, and the final image is formed at infinity, subtending an angle β at the eye. Mark α at the objective and β at the eyepiece; the length of the telescope tube is $\boldsymbol{L=f_o+f_e}$.

The magnifying power is the ratio of the angle subtended at the eye by the final image to the angle subtended by the object at the objective,

$$m=\frac{\beta}{\alpha}$$

From the intermediate image of height h at the common focus, $\boldsymbol{\alpha=\frac{h}{f_o}}$ and $\boldsymbol{\beta=\frac{h}{f_e}}$. For small angles,

$$m=\frac{\beta}{\alpha}\approx\frac{\beta}{\alpha}=\frac{h/f_e}{h/f_o}$$

$$m=\frac{f_o}{f_e}$$

This is the **magnifying power** of an astronomical telescope in normal adjustment.

(a)(ii) Total magnification $\boldsymbol{M=20}$ and eyepiece magnification $\boldsymbol{m_e=5}$, so the objective magnification is

$$m_o=\frac{M}{m_e}=\frac{20}{5}=4$$

For the eyepiece, with the image at the near point D=**25 cm**,

$$m_e=1+\frac{D}{f_e}$$

$$5=1+\frac{25}{f_e}\Rightarrow\frac{25}{f_e}=4$$

$$f_e=\frac{25}{4}=6.25\text{ cm}$$

The object distance for the eyepiece (its real image is the final virtual image at −25 cm) is found from the lens formula,

$$\frac{1}{v_e}-\frac{1}{u_e}=\frac{1}{f_e}\Rightarrow\frac{1}{-25}-\frac{1}{u_e}=\frac{1}{6.25}$$

$$\frac{1}{u_e}=\frac{1}{-25}-\frac{1}{6.25}=\frac{-1-4}{25}=\frac{-5}{25}$$

$$u_e=-5\text{ cm}$$

So the intermediate image lies 5 cm in front of the eyepiece. Since the objective-eyepiece distance is 14 cm,

$$v_o=14-5=9\text{ cm}$$

The objective magnification gives $\mathbf{m_o=\frac{v_o}{|u_o|}=4}$, so

$$|u_o|=\frac{v_o}{4}=\frac{9}{4}=2.25\text{ cm},\ u_o=-2.25\text{ cm}$$

Now applying the lens formula to the objective,

$$\frac{1}{f_o}=\frac{1}{v_o}-\frac{1}{u_o}=\frac{1}{9}-\frac{1}{-2.25}=\frac{1}{9}+\frac{1}{2.25}$$

$$\frac{1}{f_o}=\frac{1}{9}+\frac{4}{9}=\frac{5}{9}$$

$$f_o=\frac{9}{5}=1.8\text{ cm}$$

Hence $\mathbf{f_o}$**=1.8 cm and** $\mathbf{f_e}$**=6.25 cm**.

OR

(b)(i) Two coherent waves of equal intensity I_0 superpose. Let their displacements be $\mathbf{y_1 = a\omega\, t}$ and $\mathbf{y_2 = a\cos(\omega\, t + \phi)}$, where ϕ is the constant phase difference and a the amplitude of each ($\mathbf{I_0 \propto a^2}$).

By the principle of superposition the resultant displacement is

$$y = y_1 + y_2 = a\omega\, t + a\cos(\omega\, t + \phi)$$

$$y = 2a\cos\frac{\phi}{2}\cos\left(\omega\, t + \frac{\phi}{2}\right)$$

The resultant amplitude is $\mathbf{A = 2a\cos\frac{\phi}{2}}$, so the resultant intensity ($\mathbf{I \propto A^2}$) is

$$I = 4a^2\cos^2\frac{\phi}{2} = 4I_0\cos^2\frac{\phi}{2}$$

Equivalently $\mathbf{I = 2I_0(1 + \phi)}$.

The maximum intensity occurs when $\mathbf{\cos^2\frac{\phi}{2} = 1}$, i.e. $\mathbf{\phi = 0, 2\pi, 4\pi,}$

$$I_{max} = 4I_0$$

The minimum intensity occurs when $\mathbf{\cos^2\frac{\phi}{2} = 0}$, i.e. $\mathbf{\phi = \pi, 3\pi,}$

$$I_{min} = 0$$

So the **resultant intensity is $\mathbf{I = 4I_0\cos^2(\phi/2)}$, with maximum $\mathbf{4I_0}$ and minimum 0**.

(b)(ii) Given slit width $\mathbf{a = 3\ mm = 3 \times 10^{-3}\ m}$, slit-to-screen distance $\mathbf{D = 1.5\ m}$, and wavelength $\mathbf{\lambda = 600\ nm = 6 \times 10^{-7}\ m}$.

(I) The first order minimum satisfies $\mathbf{a\theta = n\lambda}$ with $\mathbf{n = 1}$, and for small angles $\mathbf{\theta \approx \frac{x}{D}}$, so

$$x_1 = \frac{n\lambda\, D}{a} = \frac{1 \times 6 \times 10^{-7} \times 1.5}{3 \times 10^{-3}}$$

$$x_1 = \frac{9 \times 10^{-7}}{3 \times 10^{-3}} = 3 \times 10^{-4}\ m = 0.3\ mm$$

(II) The second order (secondary) maximum lies at $a\theta=(n+\frac{1}{2})\lambda$ with $n=2$, i.e. at $\frac{5\lambda}{2}$,

$$x_2=\frac{(2+12)\lambda D}{a}=\frac{2.5610^{-7}1.5}{310^{-3}}$$

$$x_2=\frac{2.2510^{-6}}{310^{-3}}=7.510^{-4}\,m=0.75\text{ mm}$$

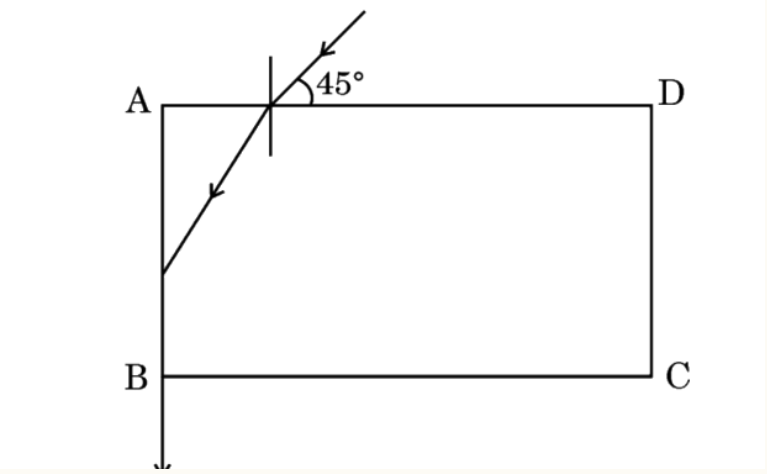
Hence the **first minimum is at 0.3 mm and the second maximum at 0.75 mm** from the centre.

Final answer. $f_o=1.8$ **cm**, $f_e=6.25$ **cm**; OR $I=4I_0\cos^2(\phi/2)$ with $I_{max}=4I_0$, $I_{min}=0$, first minimum at **0.3 mm** and second maximum at **0.75 mm**.

← *Back to Q312*

Q313

(i) Trace the path of a ray of light showing refraction through a triangular prism and hence obtain an expression for angle of deviation (δ) in terms of A, i and e, where symbols have their usual meanings. Draw a graph showing the variation of angle of deviation with the angle of incidence.
(ii) In the figure, a ray of light is incident on a transparent liquid contained in a thin glass box at an angle of 45° *with its one face. The emergent ray passes along the face* AB. *Find the refractive index of the liquid.*



[2024 • Set 55-3-1]

SOLUTION

(i) Refraction through a triangular prism. Draw a triangular prism with apex angle A at the top and refracting faces AB and AC. A ray PQ strikes face AB at Q with angle of incidence i, bends towards the normal as it enters the denser glass making angle of refraction r_1, travels as QR inside the prism, hits face AC at R with internal angle of incidence r_2, and finally emerges as RS bending away from the normal with angle of emergence e. Produce the incident ray PQ forward and the emergent ray SR backward to meet at T; the angle between them is the angle of deviation δ.

At surface AB the deviation is $(i-r_1)$ and at surface AC it is $(e-r_2)$. The total deviation is their sum,

$$\delta=(i-r_1)+(e-r_2)$$

$$\delta=(i+e)-(r_1+r_2)$$

In the quadrilateral AQOR (where O is where the two normals meet inside), the normals at Q and R make the angle A supplementary to $\angle$ QOR, giving $\angle$ **QOR=180°−A**. In triangle QOR,

$$r_1+r_2+\angle \text{ QOR}=180^\circ$$

$$r_1+r_2=180^\circ-\angle \text{ QOR}=A$$

Substituting $\mathbf{r_1+r_2=A}$,

$$\delta=i+e-A$$

This is the required expression for the **angle of deviation**.

Graph of δ versus i: the curve is U-shaped (a wide parabola-like dip). As i increases, δ first decreases, reaches a single minimum value δ_m at one particular angle of incidence (where $\boldsymbol{i=e}$ and $\mathbf{r_1=r_2=A/2}$), and then increases again. For any deviation other than δ_m there are two angles of incidence giving the same δ.

(ii) Light is incident at 45° on a face of the glass box holding the liquid, and the emergent ray passes along face AB. The ray grazes along AB, which means it strikes the liquid-glass (effectively liquid-to-rarer) boundary at the critical angle C, so total internal reflection is just on the verge.

From the geometry of the rectangular box, the angle of refraction at the first face and the angle of incidence at the second face are complementary. Refraction at the first face gives

$$45°=n\sin r$$

where r is the angle of refraction in the liquid, and at the second face the ray meets the boundary at the critical angle, with $r+C=90°$, so $C=90°-r$ and the emergent ray grazes along AB.

At the second (liquid to air) face, the critical-angle condition is

$$\sin C=\frac{1}{n}$$

Since $C=90°-r$, $\sin C=\cos r$, so $\cos r=\frac{1}{n}$. From the first face, $\sin r=45°n=1n2$.

Using $\sin^2 r+\cos^2 r=1$,

$$(\frac{1}{n2})^2+(\frac{1}{n})^2=1$$

$$\frac{1}{2n^2}+\frac{1}{n^2}=1$$

$$\frac{1+2}{2n^2}=1\Rightarrow\frac{3}{2n^2}=1$$

$$n^2=\frac{3}{2}\Rightarrow n=\sqrt{\frac{3}{2}}=1.22$$

Hence the **refractive index of the liquid is $\sqrt{3/2}\,1.22$**.

Final answer. $\delta=i+e-A$; the refractive index of the liquid is $n=\sqrt{3/2}\,1.22$.

← *Back to Q313*

Q314

(i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.
(ii) Why are multi-component lenses used for both the objective and the eyepiece in

optical instruments?
(iii) The magnification of a small object produced by a compound microscope is 200. *The focal length of the eyepiece is* 2 cm *and the final image is formed at infinity. Find the magnification produced by the objective.*

[2024 • Set 55-4-1]

SOLUTION

(i) Ray diagram and mirror equation for a convex mirror. Draw a convex mirror with pole P, focus F and centre of curvature C both behind the mirror on the principal axis. Place an erect object AB in front of the mirror, with B on the axis. Draw two rays from the top point A. The first ray travels parallel to the axis and, on reflection, diverges as if coming from the focus F behind the mirror. The second ray is directed towards C and retraces its path after reflection. The reflected rays diverge; on producing them backwards they meet behind the mirror at A', giving a virtual, erect, diminished image $A'B'$ between P and F.

For the derivation, consider a ray hitting the mirror at M. Drop a perpendicular MN from M to the axis. For paraxial rays N is very close to P. Triangles formed by the object, image and the focus/pole are similar. Using the property that the angle of incidence equals the angle of reflection, triangles ABP and $A'B'P$ are similar,

$$\frac{A'B'}{\mathrm{AB}}=\frac{\mathrm{PB'}}{\mathrm{PB}}$$

Triangles $A'B'F$ and MPF (with **MP≈ AB**) are similar,

$$\frac{A'B'}{\mathrm{AB}}=\frac{\mathrm{FB'}}{\mathrm{PF}}$$

Equating,

$$\frac{\mathrm{PB'}}{\mathrm{PB}}=\frac{\mathrm{FB'}}{\mathrm{PF}}=\frac{\mathrm{PF}-\mathrm{PB'}}{\mathrm{PF}}$$

Applying the sign convention (distances measured from P; for a convex mirror **PB=−u, PB'=+v, PF=+f**),

$$\frac{v}{-u}=\frac{f-v}{f}$$

$$\mathrm{vf}=-u(f-v)=-\mathrm{uf}+\mathrm{uv}$$

Dividing throughout by uvf,

$$\frac{1}{u}=-\frac{1}{v}+\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

This is the **mirror equation**, with $f=\frac{R}{2}$.

(ii) Multi-component (combination) lenses are used for the objective and the eyepiece because a single lens suffers from defects. Combining lenses of different materials and powers helps to minimise **chromatic aberration** (the different focusing of different colours) and **spherical aberration** (the blurring caused by marginal and paraxial rays focusing at different points). They also give a brighter, sharper image over a wider field of view. So multi-component lenses produce a clearer, distortion-free image.

(iii) For a compound microscope with the final image at infinity, the total magnification is

$$M=\mathrm{m}_o\times \mathrm{m}_e=\mathrm{m}_o\times\frac{D}{\mathrm{f}_e}$$

where m_o is the objective magnification, D**=25 cm** and $\mathbf{f}_e$**=2 cm**.

$$\mathrm{m}_e=\frac{D}{\mathrm{f}_e}=\frac{25}{2}=12.5$$

$$M=\mathrm{m}_o\times \mathrm{m}_e\Rightarrow \mathrm{m}_o=\frac{M}{\mathrm{m}_e}=\frac{200}{12.5}$$

$$\mathrm{m}_o=16$$

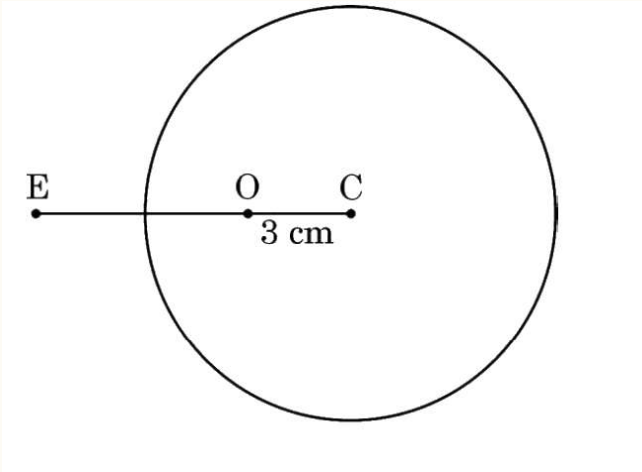
Hence the **magnification produced by the objective is 16**.

Final answer. Mirror equation $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$; multi-component lenses reduce chromatic and spherical aberration; objective magnification **=16.**

← *Back to Q314*

Q315

(i) A spherical surface of radius of curvature R separates two media of refractive indices n_1 *and* n_2*. A point object is placed in front of the surface at distance u in medium of refractive index* n_1 *and its image is formed by the surface at distance v, in the medium of refractive index* n_2*. Derive a relation between u and v.*
(ii) A solid glass sphere of radius 6.0 *cm has a small air bubble trapped at a distance* 3.0 *cm from its centre C as shown in the figure. The refractive index of the material of the sphere is* 1.5*. Find the apparent position of this bubble when seen through the surface of the sphere from an outside point E in air.*



[2023 • Set 55-1-1]

SOLUTION

(i) Refraction at a single spherical surface of radius R separating media of refractive indices n_1 and n_2. Draw the principal axis through the centre of curvature C and the pole P. A point object O lies in medium n_1 at distance u. A ray from O along the axis passes undeviated, and a second ray strikes the surface at a point M at a small height, bending according to Snell's law to form the image I in medium n_2 at distance v. Let the ray OM make angle α with the axis, the refracted ray MI make angle β, and MC make angle γ; the angles of incidence and refraction at M are i and r.

For paraxial rays the angles are small, so Snell's law $\mathbf{n_1 \sin i = n_2 \sin r}$ becomes

$$n_1\, i = n_2\, r$$

From the exterior-angle property of triangles, $\boldsymbol{i = \alpha + \gamma}$ and $\boldsymbol{\gamma = \beta + r}$, so $\boldsymbol{r = \gamma - \beta}$. Substituting,

$$n_1(\alpha+\gamma)=n_2(\gamma-\beta)$$

For small angles with **MN≈ *h*** the height of M above the axis, $\alpha\approx\frac{h}{PO}$, $\beta\approx\frac{h}{PI}$, $\gamma\approx\frac{h}{PC}$. Substituting and cancelling h,

$$n_1\left(\frac{1}{PO}+\frac{1}{PC}\right)=n_2\left(\frac{1}{PC}-\frac{1}{PI}\right)$$

Applying the sign convention **PO=−*u***, **PI=+*v***, **PC=+*R***,

$$n_1\left(\frac{1}{-u}+\frac{1}{R}\right)=n_2\left(\frac{1}{R}-\frac{1}{v}\right)$$

$$-\frac{n_1}{u}+\frac{n_1}{R}=\frac{n_2}{R}-\frac{n_2}{v}$$

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

This is the required **relation between *u* and *v*** for refraction at a spherical surface.

(ii) The bubble is at 3.0 cm from the centre C, on the near side towards the observer E. The observer views it through the nearer spherical surface, so light travels from the bubble (in glass) to the eye (in air). The distance of the bubble from the refracting surface (the near pole) is the radius minus the off-centre distance, **6.0−3.0=3.0 cm**.

Using the sign convention with light going from glass to air, take $\mathbf{n_1=1.5}$ (glass), $\mathbf{n_2=1}$ (air), object distance ***u*=−3.0 cm**, and the surface is concave towards the incoming light so ***R*=−6.0 cm**.

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

$$\frac{1}{v}-\frac{1.5}{-3.0}=\frac{1-1.5}{-6.0}$$

$$\frac{1}{v}+0.5=\frac{-0.5}{-6.0}=\frac{1}{12}$$

$$\frac{1}{v}=\frac{1}{12}-\frac{1}{2}=\frac{1-6}{12}=-\frac{5}{12}$$

$$v = -\frac{12}{5} = -2.4 \text{ cm}$$

The negative sign shows the image is virtual and on the same side as the bubble. So the bubble appears at 2.4 cm from the near surface (i.e. **6.0−2.4=3.6 cm** from the centre), shifted towards the observer.

Hence the **apparent position of the bubble is 2.4 cm below the surface (3.6 cm from the centre), on the same side as the observer**.

COMMON MISTAKE

When light leaves the glass sphere into air, n_1 is the glass index (1.5) and n_2 is air (1); a common slip is to keep n_1=1. Always set n_1 = medium where the object sits.

Final answer. $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$; the bubble appears **2.4 cm** from the near surface (3.6 cm from the centre).

← *Back to Q315*

Q316

(i) Draw a labelled ray diagram showing the image formation by a refracting telescope. Define its magnifying power. Write two limitations of a refracting telescope over a reflecting telescope.
(ii) The focal lengths of the objective and the eye-piece of a compound microscope are 1.0 cm *and* 2.5 cm *respectively. Find the tube length of the microscope for obtaining a magnification of* 300.

[2023 • Set 55-2-1]

SOLUTION

(i) Ray diagram and magnifying power of a refracting (astronomical) telescope. Draw the principal axis with the objective O (large focal length f_o) on the left and the eyepiece E (small focal length f_e) on the right. Parallel rays from a distant object reach O at a small angle α to the axis. The objective forms a real, inverted, diminished image $A'B'$ at its focus. In normal adjustment this image lies at the focus of the eyepiece, so the eyepiece sends out parallel rays and the

final, magnified, inverted image is seen at infinity, subtending angle β at the eye.

The magnifying power is defined as the ratio of the angle subtended at the eye by the final image to the angle subtended by the object at the unaided eye (objective),

$$M=\frac{\beta}{\alpha}$$

With the intermediate image of height h at the common focus, $\alpha=\frac{h}{f_o}$ and $\beta=\frac{h}{f_e}$, so for small angles

$$M=\frac{f_o}{f_e}$$

This is the **magnifying power** in normal adjustment.

Two limitations of a refracting telescope over a reflecting telescope:

1. A refracting telescope uses a large objective lens which suffers from **chromatic aberration** (it splits white light into colours), whereas a mirror in a reflecting telescope is free from it.

2. A large lens is heavy, sags under its own weight and is very difficult and costly to grind and support (it can only be held at the edges), while a mirror can be supported all over its back and is easier to make large.

(ii) For a compound microscope with the relaxed eye (final image at infinity), the magnitude of the magnification is

$$M=\frac{L}{f_o}\times\frac{D}{f_e}$$

Here f_o**=1.0 cm**, f_e**=2.5 cm**, D**=25 cm**, and M**=300**.

First the eyepiece magnification,

$$m_e=\frac{D}{f_e}=\frac{25}{2.5}=10$$

Then the objective magnification,

$$m_o=\frac{M}{m_e}=\frac{300}{10}=30$$

Since $\mathbf{m_o = \frac{L}{f_o}}$,

$$L = m_o \times f_o = 30 \text{ cm}$$

Hence the **tube length of the microscope is 30 cm**.

Final answer. Magnifying power $M = f_o/f_e$; the tube length of the microscope is **30 cm**.

← *Back to Q316*

Q317

(a) (i) Draw a ray diagram showing the formation of a real image of an object placed at a distance 'u' in front of a concave mirror of radius of curvature 'R'. Hence, obtain the relation for the image distance 'v' in terms of u and R.
(ii) A 1.8 m tall person stands in front of a convex lens of focal length 1 m, at a distance of 5 m. Find the position and height of the image formed.
OR
(b) (i) Draw a ray diagram showing refraction of a ray of light through a triangular glass prism. Hence, obtain the relation for the refractive index (μ) in terms of angle of prism (A) and angle of minimum deviation (δ_m).
(ii) The radii of curvature of the two surfaces of a concave lens are 20 cm each. Find the refractive index of the material of the lens if its power is −5.0 D.

[2023 • Set 55-3-1]

SOLUTION

(a)(i) Concave mirror forming a real image and the mirror relation. Draw a concave mirror with pole P, focus F, and centre of curvature C on the principal axis (**PC=*R***). Place an object AB beyond C, with B on the axis. From the top point A draw a ray parallel to the axis which reflects through F, and a second ray through C which retraces its path. The reflected rays meet in front of the mirror at A', forming a real, inverted, diminished image $A'B'$ between F and C.

From similar triangles ABC and $A'B'C$,

$$\frac{A'B'}{AB} = \frac{CB'}{CB}$$

and from similar triangles ABP and $A'B'P$,

$$\frac{A'B'}{AB}=\frac{PB'}{PB}$$

Equating, $\frac{CB'}{CB}=\frac{PB'}{PB}$. Writing distances about P with $\mathbf{PB=-u}$, $\mathbf{PB'=-v}$, $\mathbf{PC=-R}$,

$$\frac{PC-PB'}{PB-PC}=\frac{PB'}{PB}$$

$$\frac{(-R)-(-v)}{(-u)-(-R)}=\frac{-v}{-u}$$

$$\frac{v-R}{R-u}=\frac{v}{u}$$

Cross-multiplying, $\boldsymbol{u(v-R)=v(R-u)}$, so $\mathbf{uv-uR=vR-uv}$, giving $\mathbf{2uv=R(u+v)}$. Dividing by uvR,

$$\frac{2}{R}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{2}{R}-\frac{1}{u}$$

This gives the **image distance** v in terms of u and R (and equals $\frac{1}{f}$ with $\boldsymbol{f=R/2}$).

(a)(ii) Convex lens with $\boldsymbol{f=1\ m}$, object height $\boldsymbol{h=1.8\ m}$, $\boldsymbol{u=-5\ m}$.

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{1}+\frac{1}{-5}=1-0.2=0.8$$

$$v=\frac{1}{0.8}=1.25\ m$$

The image is real, formed 1.25 m on the other side of the lens. The magnification is

$$m=\frac{v}{u}=\frac{1.25}{-5}=-0.25$$

$$h'=m\times h=-0.25 1.8=-0.45\ m$$

The negative sign means the image is inverted, of height 0.45 m.

So the **image is real, inverted, 1.25 *m* from the lens and 0.45 *m* tall**.

OR

(b)(i) Refraction through a prism and the prism formula at minimum deviation. Draw a triangular prism of apex angle A with faces AB and AC. A ray strikes AB at angle of incidence i, refracts to r_1, travels through the prism, hits AC at internal angle r_2, and emerges at angle e. The angle between the produced incident ray and the emergent ray is the deviation δ. As before, $\boldsymbol{\delta = i + e - A}$ and $\boldsymbol{r_1 + r_2 = A}$.

At minimum deviation $\boldsymbol{\delta = \delta_m}$ the ray passes symmetrically, so $\boldsymbol{i = e}$ and $\boldsymbol{r_1 = r_2 = r}$. Then

$$r_1 + r_2 = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$$

$$\delta_m = i + e - A = 2i - A \Rightarrow i = \frac{A + \delta_m}{2}$$

Applying Snell's law at the first face,

$$\mu = \frac{\sin i}{\sin r}$$

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

This is the **prism formula** for the refractive index in terms of A and δ_m.

(b)(ii) Concave lens with both radii 20 cm and power $\boldsymbol{P = -5.0\ D}$, so $\boldsymbol{f = \frac{1}{P} = -0.2\ m = -20\ cm}$. For a biconcave lens $\boldsymbol{R_1 = -20\ cm}$ and $\boldsymbol{R_2 = +20\ cm}$.

Lens maker's formula,

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{-20} = (n - 1)\left(\frac{1}{-20} - \frac{1}{20}\right)$$

$$-\frac{1}{20} = (n - 1)\left(-\frac{2}{20}\right) = -\frac{(n - 1)}{10}$$

$$\frac{1}{20}=\frac{n-1}{10}\Rightarrow n-1=\frac{10}{20}=0.5$$

$$n=1.5$$

Hence the **refractive index of the lens material is 1.5**.

Final answer. $\frac{1}{v}=\frac{2}{R}-\frac{1}{u}$; image real, inverted, **1.25 *m*** away, **0.45 *m*** tall. OR $\mu=\frac{\sin[(A+\delta_m)/2]}{\sin(A/2)}$ and $n=1.5$.

← *Back to Q317*

Q318

(a) (i) Draw a ray diagram to show how the final image is formed at infinity in an astronomical refracting telescope. Obtain an expression for its magnifying power.
(ii) Two thin lenses L_1 *and* L_2, L_1 *being a convex lens of focal length 24 cm and* L_2 *a concave lens of focal length 18 cm are placed coaxially at a separation of 45 cm. A 1 cm tall object is placed in front of the lens* L_1 *at a distance of 36 cm. Find the location and height of the image formed by the combination.*

OR

(b) (i) Explain the working principle of an optical fibre with the help of a diagram. Mention one use of a light pipe.
(ii) A ray of light is incident at an angle of 60° *on one face of a prism with the prism angle* $A = 60°$. *The ray passes symmetrically through the prism. Find the angle of minimum deviation* (δ_m) *and refractive index of the material of the prism. If the prism is immersed in water, how will* δ_m *be affected? Justify your answer.*

[2023 • Set 55-4-1]

SOLUTION

(a)(i) Astronomical refracting telescope with the final image at infinity (normal adjustment). Draw the principal axis with the objective O (long focal length f_o) on the left and the eyepiece E (short focal length f_e) on the right. Parallel rays from a distant object reach O inclined at angle α to the axis. The objective forms a real, inverted, diminished image $A'B'$ at its focus. The lenses are separated so that this image lies at the focus of the eyepiece as well, hence the eyepiece

sends out parallel rays and the final image is formed at infinity, subtending angle β at the eye.

The magnifying power is the ratio of the angle subtended at the eye by the final image to the angle subtended by the object at the objective,

$$M=\frac{\beta}{\alpha}$$

With the common image of height h, $\alpha=\frac{h}{f_o}$ and $\beta=\frac{h}{f_e}$, so for small angles

$$M=\frac{\beta}{\alpha}=\frac{f_o}{f_e}$$

This is the **magnifying power** of the telescope, and the tube length is $L=f_o+f_e$.

(a)(ii) Refraction at L_1 (convex, $f_1=+24$ **cm**, $u_1=-36$ **cm**),

$$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1}$$

$$\frac{1}{v_1}=\frac{1}{24}+\frac{1}{-36}=\frac{3-2}{72}=\frac{1}{72}$$

$v_1=72$ **cm** (to the right of L_1)

This image would form 72 cm from L_1, but L_2 is only 45 cm away, so for L_2 it acts as a virtual object at

$$u_2=72-45=+27 \text{ cm}$$

Refraction at L_2 (concave, $f_2=-18$ **cm**),

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$$

$$\frac{1}{v_2}=\frac{1}{-18}+\frac{1}{27}=\frac{-3+2}{54}=-\frac{1}{54}$$

$$v_2=-54 \text{ cm}$$

The final image is virtual, 54 cm to the left of L_2. The total magnification is

$$m=m_1\times m_2=\frac{v_1}{u_1}\times\frac{v_2}{u_2}=\frac{72}{-36}\times\frac{-54}{27}$$

$$m=(-2)\times(-2)=4$$

$$h' = m \times h = 4 \times 1 = 4 \text{ cm}$$

Hence the **final image is virtual, 54 cm to the left of L_2, erect and 4 cm tall**.

OR

(b)(i) Working principle of an optical fibre. An optical fibre is a thin, long thread of glass or quartz consisting of a central core of higher refractive index surrounded by a cladding of slightly lower refractive index. Light entering one end strikes the core-cladding boundary at an angle greater than the critical angle, so it undergoes **total internal reflection** repeatedly and travels along the fibre without leaking out, even around bends, finally emerging at the other end with very little loss. Diagram: a long fibre with a core and cladding, showing a zig-zag ray reflecting back and forth from the core-cladding boundary along the length of the fibre. One use of a light pipe: it is used in medical instruments such as an endoscope to view internal organs, and to transmit signals in optical communication.

(b)(ii) The ray passes symmetrically, so this is minimum deviation with $A = 60°$ and $i = e = 60°$.

$$\delta_m = i + e - A = 60° + 60° - 60° = 60°$$

The refractive index from the prism formula,

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \sin(60° + 60°2)\sin(60°2)$$

$$\mu = 60°30° = 3/21/2 = 31.732$$

When the prism is immersed in water, the effective (relative) refractive index of glass with respect to water is smaller than its value in air,

$$\mu_{rel} = \frac{\mu_{glass}}{\mu_{water}} < \mu_{glass}$$

Since a smaller relative refractive index bends the ray less, the angle of minimum deviation **decreases**. So δ_m is reduced when the prism is placed in water.

Hence the **minimum deviation is 60°, $\mu = 31.732$, and δ_m decreases in water**.

Final answer. Final image virtual, **54 cm** left of L_2, height **4 cm**. OR δ_m=60°, **μ=31.732**, and δ_m decreases in water.

← *Back to Q318*

Q319

(a) (i) Draw a ray diagram to show the working of a compound microscope. Obtain the expression for the total magnification for the final image to be formed at the near point.
(ii) In a compound microscope an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eye-piece has a focal length of 5 cm and the final image is formed at the near point, find the magnifying power of the microscope.
OR
(b) (i) Draw a ray diagram for the formation of image of an object by an astronomical telescope, in normal adjustment. Obtain the expression for its magnifying power.
(ii) The magnifying power of an astronomical telescope in normal adjustment is 2.9 and the objective and the eyepiece are separated by a distance of 150 cm. Find the focal lengths of the two lenses.

[2023 • Set 55-5-1]

SOLUTION

(a)(i) Ray diagram for a compound microscope with the final image at the near point. Draw the objective O on the left and the eyepiece E on the right on a common axis. A small object AB is placed just beyond the focus of the objective. The objective forms a real, inverted, magnified intermediate image $A'B'$. This image lies just inside the first focus of the eyepiece (between E and F_e), so the eyepiece, acting as a simple magnifier, forms a virtual, magnified, further-inverted final image $A''B''$ at the near point D**=25 cm** from the eye on the same side as the object.

The objective magnification is

$$m_o=\frac{L}{f_o}\ (\text{or } m_o=\frac{v_o}{u_o})$$

The eyepiece, with the final image at the near point, gives

$$m_e = 1 + \frac{D}{f_e}$$

The total magnification is the product,

$$M = m_o \times m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

This is the required **total magnification** for the final image at the near point.

(a)(ii) For the objective, $\mathbf{u_o = -1.5\ cm}$, $\mathbf{f_o = 1.25\ cm}$.

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{1.25} - \frac{1}{1.5}$$

$$\frac{1}{v_o} = \frac{1.5 - 1.25}{1.25 1.5} = \frac{0.25}{1.875} = \frac{1}{7.5}$$

$$v_o = 7.5\ \text{cm}$$

The objective magnification,

$$m_o = \frac{v_o}{|u_o|} = \frac{7.5}{1.5} = 5$$

The eyepiece magnification with $\mathbf{f_e = 5\ cm}$, $\mathbf{D = 25\ cm}$,

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 1 + 5 = 6$$

The magnifying power,

$$M = m_o \times m_e = 5 6 = 30$$

Hence the **magnifying power of the microscope is 30**.

OR

(b)(i) Ray diagram for an astronomical telescope in normal adjustment. Draw the objective O (long f_o) and eyepiece E (short f_e) on a common axis. Parallel rays from a distant object enter O at angle α to the axis and form a real, inverted, diminished image $A'B'$ at the common focus of the two lenses. The eyepiece then renders the rays parallel, so the final inverted image is at infinity, subtending angle β at the eye.

The magnifying power,

$$M=\frac{\beta}{\alpha}=\frac{h/f_e}{h/f_o}=\frac{f_o}{f_e}$$

This is the **magnifying power** in normal adjustment.

(b)(ii) Given $M=\frac{f_o}{f_e}=2.9$ and the lens separation in normal adjustment $L=f_o+f_e=150$ cm.

From the first relation, $f_o=2.9\,f_e$. Substituting,

$$2.9f_e+f_e=150$$

$$3.9f_e=150\Rightarrow f_e=\frac{150}{3.9}=38.46\text{ cm}$$

$$f_o=2.938.46=111.54\text{ cm}$$

Hence the **focal lengths are f_o111.5 cm and f_e38.5 cm**.

COMMON MISTAKE

For the near-point setting use $m_e=1+D/f_e$; for image at infinity use $m_e=D/f_e$. Mixing the two is a frequent exam slip.

Final answer. Magnifying power of the microscope =30. OR f_o111.5 cm and f_e38.5 cm.

← Back to Q319

Q320

A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image.

***I.** In a compound microscope the images formed by the objective and the eye-piece are respectively*

(A) virtual, real (B) real, virtual (C) virtual, virtual (D) real, real

II. *The magnification due to a compound microscope does not depend upon*

(A) the aperture of the objective and the eye-piece (B) the focal length of the objective and the eye-piece (C) the length of the tube (D) the colour of the light used

III. *Which of the following is not correct in the context of a compound microscope?*

(A) Both the lenses are of short focal lengths. (B) The magnifying power increases by decreasing the focal lengths of the two lenses. (C) The distance between the two lenses is more than $(f_o + f_e)$*. (D) The microscope can be used as a telescope by interchanging the two lenses.*

IV. *A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be*

(A) 2 (B) 10 (C) 30 (D) 200

V. *The focal lengths of objective and eye-piece of a compound microscope are* 1.2 cm *and* 3.0 cm *respectively. The object is placed at a distance of* 1.25 cm *from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be*

(A) 100 (B) 150 (C) 200 (D) 250

[2022 • Set 55-3-1]

SOLUTION

I. In a compound microscope the objective forms a real, inverted and magnified intermediate image of the object. This image then acts as the object for the eyepiece, which behaves like a simple magnifier and forms a virtual, magnified final image. So the objective image is real and the eyepiece image is virtual. (A), (C) and (D) wrongly label one or both, so they are ruled out. The correct option is (B) real, virtual.

II. The magnification of a compound microscope is $M = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$. It depends on the tube length L and the focal lengths f_o and f_e, so options (B) and (C) do affect it. The colour of light changes f slightly through dispersion, so (D) has a minor effect. The **aperture** of the lenses only controls brightness and resolving power, not magnification, so it is the quantity magnification does not depend upon. The correct option is (A).

III. Check each statement. (A) Both lenses do have short focal lengths, which is true. (B) Magnifying power $\propto \frac{1}{f_o f_e}$, so decreasing the focal lengths increases it, true. (C) The lens separation in a microscope is indeed greater than $(f_o + f_e)$, true.

(D) A microscope cannot simply be turned into a telescope by interchanging the two lenses, because a telescope needs a long-focus objective and a short-focus eyepiece, the opposite arrangement. So statement (D) is not correct, and the answer is (D).

IV. The total magnification of a compound microscope is the product of the objective and eyepiece powers,

$$M = m_o \times m_e = 10 \cdot 20 = 200$$

So the magnification is 200, option (D). The smaller values 2, 10, 30 come from wrongly adding or mis-multiplying, so (A), (B), (C) are ruled out.

V. For the objective, $\mathbf{u_o = -1.25\ cm}$, $\mathbf{f_o = 1.2\ cm}$.

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{1.2} - \frac{1}{1.25}$$

$$\frac{1}{v_o} = \frac{1.25 - 1.2}{1.2 \cdot 1.25} = \frac{0.05}{1.5} = \frac{1}{30}$$

$$v_o = 30\ \text{cm}$$

$$m_o = \frac{v_o}{|u_o|} = \frac{30}{1.25} = 24$$

With the final image at infinity, $\mathbf{m_e = \frac{D}{f_e} = \frac{25}{3.0} = 8.33.}$

$$M = m_o \times m_e = 24 \cdot 8.33 \approx 200$$

So the magnifying power is about 200, option (C).

Final answer. I **(B)** real, virtual; II **(A)** aperture; III **(D)**; IV **(D)** 200; V **(C)** 200.

← *Back to Q320*

Q321

Two transparent media of refractive indices n_1 and n_2 are separated by a spherical transparent surface. The rays of light incident on the surface get refracted into the medium on the other side. The laws of refraction are valid at each point of the

spherical surface. A lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. The focal length of a lens is determined by the radii of curvature (R_1 and R_2) of its two surfaces and the refractive index (n) of the medium of the lens with respect to the surrounding medium. Depending on R_1 and R_2, a lens behaves as a diverging or a converging lens. The ability of a lens to diverge or converge a beam of light incident on it defines its power.

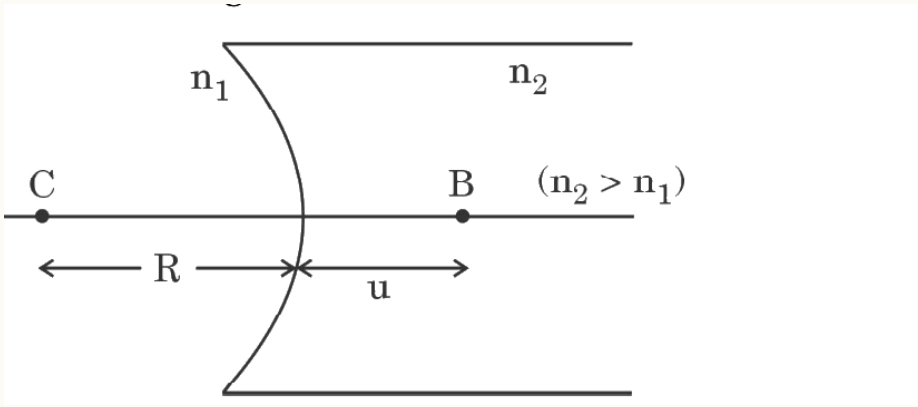
(a) An object is placed at the point B as shown in the figure. The object distance (u) and the image distance (v) are related as:

(i) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

(ii) $\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$

(iii) $\frac{n_2}{u} - \frac{n_1}{v} = \frac{n_2 - n_1}{R}$

(iv) $\frac{n_1}{u} - \frac{n_2}{v} = \frac{n_1 - n_2}{R}$



(b) A point object is placed in air at a distance R in front of a convex spherical refracting surface of radius of curvature R. If the medium on the other side of the surface is glass, then the image is:

(i) real and formed in glass.

(ii) real and formed in air.

(iii) virtual and formed in glass.

(iv) virtual and formed in air.

(c) An object is kept at 2F in front of an equiconvex lens. The image formed is:

(i) real and of the size of the object.

(ii) virtual and of the size of the object.

(iii) real and enlarged.

(iv) virtual and diminished.

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal

length 20 cm are placed coaxially in contact. The power of the combination is:

(i) $-5\ D$

(ii) $+5\ D$

(iii) $+15\ D$

(iv) $-15\ D$

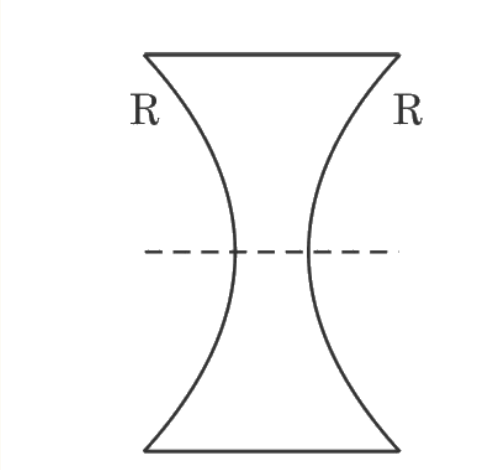
(e) An equiconcave lens of focal length f is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be:

(i) $\frac{f}{2}$

(ii) $\frac{f}{4}$

(iii) f

(iv) 2f



[2022 • Set 55-4-1]

SOLUTION

This is a five-part case-study question. Each part is answered separately below.

(a) For refraction at a single spherical surface separating a medium of index n_1 (object side) from n_2 (image side), with light travelling from medium 1 into medium 2, the relation between object distance u, image distance v and radius of curvature R is obtained by applying Snell's law in the paraxial approximation. The standard sign-convention form is

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

This matches option **(i)**.

(b) Here light goes from air into glass through a convex surface, so $\mathbf{n_1 = 1}$ (air), $\mathbf{n_2 = 1.5}$ (glass), object distance $\boldsymbol{u = -R}$ and radius $+R$ (convex toward the object).

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{(-R)} = \frac{1.5 - 1}{R}$$

$$\frac{1.5}{v} + \frac{1}{R} = \frac{0.5}{R}$$

$$\frac{1.5}{v} = \frac{0.5}{R} - \frac{1}{R} = -\frac{0.5}{R}$$

$$v = -3R$$

The negative sign means the image is on the same side as the object, i.e. in air, and it is virtual. So the answer is **(iv) virtual and formed in air**.

(c) For a thin lens, when an object is placed at 2F (i.e. $\boldsymbol{u = -2f}$), the lens equation $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ gives $\boldsymbol{v = +2f}$. The image forms at 2F on the other side, is real and inverted, and has magnification $\boldsymbol{m = \frac{v}{u} = -1}$, so it is the same size as the object. Hence **(i) real and of the size of the object**.

(d) For lenses in contact, powers add. Converging lens: $\mathbf{f_1 = +10\ cm = +0.10\ m}$, so $\mathbf{P_1 = +10\ D}$. Diverging lens: $\mathbf{f_2 = -20\ cm = -0.20\ m}$, so $\mathbf{P_2 = -5\ D}$.

$$P = P_1 + P_2 = 10 + (-5) = +5\ D$$

So the combination has power **(ii) +5 *D***.

(e) The equiconcave lens is cut along the dotted line, which runs along the principal axis (horizontal, through the diameter). Each half still keeps the SAME two curved surfaces with the same radii R_1 and R_2 as before. Since the lens maker's formula $\frac{1}{f} = (n-1)(\frac{1}{R_1} - \frac{1}{R_2})$ depends only on n, R_1, R_2 (not on the height of the lens), the focal length is unchanged. Hence each part has focal length **(iii) *f***.

COMMON MISTAKE

Cutting a lens along the principal axis (a horizontal cut through the diameter) leaves R1 and R2 unchanged, so f stays the same. A vertical cut perpendicular to the axis would change f instead.

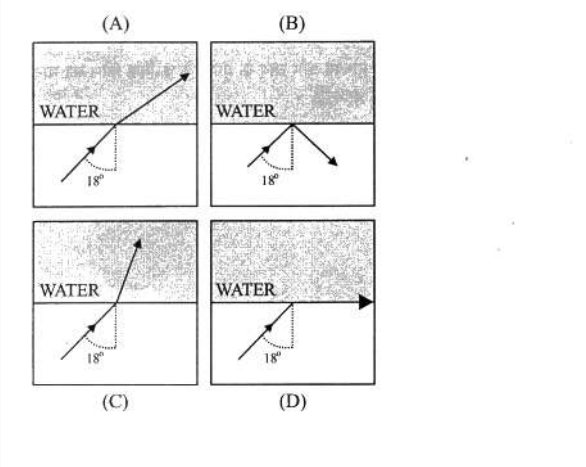
Final answer. (a) (i); (b) (iv); (c) (i); (d) (ii) +5 D; (e) (iii) f.

← Back to Q321

Q322

A ray of light travels from a denser to a rarer medium. After refraction, it bends away from the normal. When we keep increasing the angle of incidence, the angle of refraction also increases till the refracted ray grazes along the interface of two media. The angle of incidence for which it happens is called critical angle. If the angle of incidence is increased further the ray will not emerge and it will be reflected back in the denser medium. This phenomenon is called total internal reflection of light.

(i) A ray of light travels from a medium into water at an angle of incidence of 18°. *The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is* 20°. *Which one of the following figures best represents the correct path of the ray of light?*



(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, *to a depth d. The area of the surface of water through which light from the source can emerge, is:*

(a) $\frac{\pi d^2}{2(\mu^2-1)}$ *\quad (b)* $\frac{\pi d^2}{(\mu^2-1)}$ *\quad (c)* $\frac{\pi d^2}{\sqrt{\mu^2-1}}$ *\quad (d)* $\frac{2\pi d^2}{(\mu^2-1)}$

(iii) For which of the following media, with respect to air, the value of critical angle is maximum?

(a) Crown glass \quad (b) Flint glass \quad (c) Water \quad (d) Diamond

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 *and* 1.0 *respectively is:*

(a) 0° *\quad (b)* 30° *\quad (c)* 45° *\quad (d)* 60°

(v) The critical angle of pair of a medium and air is 30°*. The speed of light in the medium is:*

(a) 1×10^8 *m* s^{-1} *\quad (b)* 1.5×10^8 *m* s^{-1} *\quad (c)* 2.2×10^8 *m* s^{-1} *\quad (d)* 2.8×10^8 *m* s^{-1}

[2022 • Set 55-5-1]

SOLUTION

This case study has five parts based on total internal reflection (TIR). Each is answered below.

(i) The ray travels from a denser medium (index greater than water) into water at an angle of incidence 18°, and the critical angle for this interface is 20°. Since **18°< 20°**, the angle of incidence is less than the critical angle, so total internal reflection does NOT occur and the ray refracts into the water. Because light goes from a denser to a rarer medium, it bends away from the normal, so the angle of refraction is greater than 18°. Hence the correct figure is the one showing the ray crossing into water and bending away from the normal (with only a weak partially reflected ray).

(ii) A point source at the bottom emits light in all directions; only rays striking the surface within the critical angle θ_c can emerge, while rays beyond θ_c undergo TIR. Light therefore escapes through a circle of radius *r*, where

$$\theta_c = \frac{1}{\mu}$$

$$\theta_c = \frac{\theta_c}{\theta_c} = \frac{1/\mu}{\sqrt{1-1/\mu^2}} = \frac{1}{\sqrt{\mu^2-1}}$$

$$r = d\theta_c = \frac{d}{\sqrt{\mu^2-1}}$$

$$\text{Area} = \pi r^2 = \frac{\pi d^2}{\mu^2 - 1}$$

So the area is option **(b)** $\mathbf{\frac{\pi d^2}{(\mu^2 - 1)}}$.

(iii) The critical angle satisfies $\theta_c = \frac{1}{\mu}$, so the smaller the refractive index, the larger the critical angle. The refractive indices are: crown glass $\mu \approx 1.52$, flint glass $\mu \approx 1.62$, water $\mu \approx 1.33$, diamond $\mu \approx 2.42$. Water has the smallest refractive index, so it gives the largest critical angle ($\theta_c \approx \sin^{-1}(1/1.33) \approx 48.8°$, compared with about 41° for crown glass). Hence the value of the critical angle is maximum for **(c) Water**.

(iv) For media A (denser, $n_A = 2.0$) and B (rarer, $n_B = 1.0$), TIR occurs going from A to B.

$$\theta_c = \frac{n_B}{n_A} = \frac{1.0}{2.0} = \frac{1}{2}$$

$$\theta_c = \sin^{-1}\left(\frac{1}{2}\right) = 30°$$

So the critical angle is option **(b) 30°**.

(v) The critical angle with air is 30°, so

$$\theta_c = \frac{1}{n} \Rightarrow \sin 30° = \frac{1}{n}$$

$$\frac{1}{2} = \frac{1}{n} \Rightarrow n = 2$$

$$v = \frac{c}{n} = \frac{3\times 10^8}{2} = 1.5\times 10^8\ m\ s^{-1}$$

So the speed of light in the medium is option **(b) $\mathbf{1.5\times 10^8\ m\ s^{-1}}$**.

Final answer. (i) ray refracts into water bending away from normal; (ii) (b); (iii) (c) Water; (iv) (b) 30 degrees; (v) (b) 1.5x10^8 m/s.

← *Back to Q322*

Q323

(i) Draw a ray diagram of an astronomical refracting telescope in normal adjustment. Obtain an expression for its magnifying power. How can we increase the magnifying power of the telescope?
(ii) A beam of light converges at a point P. A lens is placed in the path of the beam at a distance of 25 cm *from P. The final image is formed at infinity. Calculate the power of the lens.*

[2021]

SOLUTION

(i) Astronomical refracting telescope in normal adjustment.

A refracting astronomical telescope uses two convex lenses mounted coaxially in a tube: an objective O of large focal length f_o and large aperture, and an eyepiece E of small focal length f_e. In normal adjustment the final image is at infinity so the eye views it in a relaxed state.

Ray diagram (description): Draw a long horizontal principal axis. At the left place the objective lens O (tall, large aperture), and to its right the smaller eyepiece E. Parallel rays arrive from a distant object at a small angle α to the axis. After passing through O they converge to form a real, inverted, diminished image $A'B'$ at the back focal plane of O, a distance f_o from O. This image plane is arranged to coincide with the front focal plane of E, so E is a distance f_e beyond $A'B'$. The eyepiece then sends out a parallel beam (emerging at angle β to the axis) that enters the eye, forming the final image at infinity. Mark f_o between O and $A'B'$, f_e between $A'B'$ and E, and the tube length $\boldsymbol{L = f_o + f_e}$.

The magnifying power m is the ratio of the angle β subtended at the eye by the final image to the angle α subtended at the unaided eye by the object.

$$m = \frac{\beta}{\alpha}$$

With the intermediate image $A'B'$ of height h, the angle at the objective is $\boldsymbol{\alpha = \frac{h}{f_o}}$ and the angle at the eyepiece is $\boldsymbol{\beta = \frac{h}{f_e}}$ (small angles, $\theta \approx \theta$).

$$m = \frac{\beta}{\alpha} = \frac{h/f_e}{h/f_o} = \frac{f_o}{f_e}$$

$$m = \frac{f_o}{f_e}$$

To increase the magnifying power we use an objective of larger focal length f_o and an eyepiece of smaller focal length f_e.

(ii) Power of the lens.

A converging beam is heading toward the point P. Since the beam is already converging, P lies on the far (transmission) side of the lens, so P acts as a virtual object: $u = +25$ cm. The final image is at infinity, so $v = \infty$.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{\infty} - \frac{1}{25} = \frac{1}{f}$$

$$\frac{1}{f} = -\frac{1}{25}\ \text{cm}^{-1}$$

$$f = -25\ \text{cm} = -0.25\ m$$

The focal length is negative, so the lens is a diverging (concave) lens. Its power is

$$P = \frac{1}{f(\text{in } m)} = \frac{1}{-0.25} = -4\ D$$

$P = -4\ D$ (diverging lens).

This makes physical sense: a beam that is already converging can only be made parallel (image pushed to infinity) by a diverging lens, which reduces the convergence.

> **TIP**
>
> When a beam is already converging toward P and the lens sends the image to infinity, P is a virtual object sitting at the focus, so |f| equals the lens-to-P distance.

Final answer. (i) m = f_o/f_e, increase by larger f_o and smaller f_e; (ii) f = −25 cm so P = −4 D (diverging lens).

← Back to Q323

Q324

(i) A coin is placed inside a denser medium. Why does it appear to be raised? Obtain an expression for the height through which the object appears to be raised in terms of refractive index of the medium and real depth.
(ii) A compound microscope consists of an objective lens of focal length 2 cm *and an eyepiece of focal length* 6.25 cm *separated by a distance of* 15 cm. *How far from the objective should an object be placed in order to obtain the final image at the least distance of distinct vision (*25 cm*)? Calculate the magnifying power of the microscope.*

[2021]

SOLUTION

(i) Why a submerged coin appears raised, and the shift formula.

A coin lying in a denser medium (say water) appears raised because light rays from it bend AWAY from the normal as they pass from the denser medium into the rarer medium (air) at the surface. To the observer looking down, these emergent rays appear to come from a point higher up than the real position, so the coin seems closer to the surface. The image is virtual and at a smaller (apparent) depth than the real depth.

Ray diagram (description): Draw a horizontal water surface. Place the real coin O at depth h (real depth) below the surface on the normal. From O draw one ray straight up along the normal (undeviated) and a second ray hitting the surface at a small angle, refracting away from the normal toward the eye. Extend the refracted ray backward (dashed); it meets the normal at a point I above O. The depth of I below the surface is the apparent depth.

Let the real depth be h and apparent depth be h'. Light goes from the denser medium (refractive index n) to air (index 1). Applying refraction at the surface for near-normal (paraxial) viewing, Snell's law gives

$$n = \frac{\text{real depth}}{\text{apparent depth}} = \frac{h}{h'}$$

$$h' = \frac{h}{n}$$

The height through which the coin appears raised (the apparent shift) is

$$\Delta = h - h' = h - \frac{h}{n}$$

$$\Delta = h(1 - \frac{1}{n})$$

(ii) Compound microscope.

Given: objective $\mathbf{f_o = 2\ cm}$, eyepiece $\mathbf{f_e = 6.25\ cm}$, tube length (lens separation) $\boldsymbol{L}$ $\mathbf{= 15\ cm}$, final image at the least distance of distinct vision $\boldsymbol{D}$ $\mathbf{= 25\ cm}$.

First find where the intermediate image must fall by working back from the eyepiece. The eyepiece forms the final image at $\mathbf{v_e = -25\ cm}$ (virtual, on the same side as the object for the eyepiece).

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\frac{1}{-25} - \frac{1}{u_e} = \frac{1}{6.25}$$

$$\frac{1}{u_e} = \frac{1}{-25} - \frac{1}{6.25} = -0.04 - 0.16 = -0.20$$

$$u_e = -5\ \text{cm}$$

So the intermediate (real) image lies 5 cm in front of the eyepiece. The objective therefore forms its image at

$$v_o = L - |u_e| = 15 - 5 = 10\ \text{cm}$$

Now apply the lens formula to the objective to find the object distance u_o:

$$\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$$

$$\frac{1}{10} - \frac{1}{u_o} = \frac{1}{2}$$

$$\frac{1}{u_o} = \frac{1}{10} - \frac{1}{2} = \frac{1-5}{10} = -\frac{4}{10} = -\frac{2}{5}$$

$$u_o = -2.5 \text{ cm}$$

So the object must be placed 2.5 cm in front of the objective lens.

Magnifying power for the final image at the near point:

$$M = \frac{v_o}{|u_o|}\left(1 + \frac{D}{f_e}\right)$$

$$M = \frac{10}{2.5}\left(1 + \frac{25}{6.25}\right)$$

$$M = 4 \times (1 + 4) = 4 \times 5 = 20$$

$$M = 20$$

Final answer. (i) apparent shift $\Delta = h(1 - 1/n)$; (ii) object at 2.5 cm from objective, magnifying power M = 20.

← *Back to Q324*

Q325

Draw the ray diagram of an astronomical telescope when the final image is formed at infinity. Write the expression for the resolving power of the telescope.

[2020 • Set 55-1-1]

SOLUTION

Astronomical telescope with final image at infinity (normal adjustment).

An astronomical refracting telescope uses two convex lenses mounted coaxially: an objective O of large focal length f_o and large aperture, and an eyepiece E of small focal length f_e.

Ray diagram (description): Draw a long horizontal principal axis. Place the large objective lens O at the left and the smaller eyepiece E to the right. A parallel beam from a distant object arrives inclined at a small angle α to the axis. After the objective the rays converge to a real, inverted, diminished image $A'B'$ formed at the focal plane of O, a distance f_o behind it. The eyepiece is positioned so that this image lies at its front focal point (a distance f_e in front of

E). The eyepiece then renders the rays parallel, emerging at an angle β to the axis, so the final image is formed at infinity and is viewed by a relaxed eye. Mark f_o from O to $A'B'$, f_e from $A'B'$ to E, and the tube length $L = f_o + f_e$. Also mark α (angle subtended by object at objective) and β (angle subtended by final image at eye).

The angular magnification in this adjustment is $m = \frac{f_o}{f_e}$.

Resolving power of the telescope.

The resolving power of a telescope is its ability to form separate (distinguishable) images of two distant objects that are close together. By Rayleigh's criterion, the smallest angular separation that can just be resolved is

$$\theta_{\text{min}} = \frac{1.22\,\lambda}{D}$$

where λ is the wavelength of light used and D is the diameter (aperture) of the objective lens. The resolving power is the reciprocal of this minimum angle:

$$\text{Resolving Power} = \frac{1}{\theta_{\text{min}}} = \frac{D}{1.22\,\lambda}$$

$$R.P. = \frac{D}{1.22\,\lambda}$$

Thus a larger objective aperture D gives higher resolving power.

Final answer. Ray diagram drawn (normal adjustment); resolving power = D/(1.22 λ), where D is the objective aperture.

← *Back to Q325*

Q326

An object is placed in front of a concave mirror. It is observed that a virtual image is formed. Draw the ray diagram to show the image formation and hence derive the mirror equation $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$.

[2020 • Set 55-1-1]

SOLUTION

Setup and ray diagram for a virtual image in a concave mirror.

A concave mirror forms a virtual image when the object is placed between the pole P and the focus F. The image is virtual, erect and magnified, and lies behind the mirror.

Ray diagram (description): Draw the concave mirror with pole P, principal focus F and centre of curvature C on the principal axis (C is farther than F). Place an upright object AB between P and F, with B on the axis and tip A above it. From the tip A draw two rays:

(1) A ray parallel to the principal axis strikes the mirror at a point M near the pole and reflects so as to pass through F.

(2) A ray directed toward the centre of curvature C strikes the mirror normally and retraces its path.

The reflected rays diverge in front of the mirror, so produce them backward (dashed) behind the mirror; they meet at A'. Drop a perpendicular $A'B'$ to the axis. $A'B'$ is the virtual, erect, magnified image behind the mirror.

Derivation of the mirror equation.

Consider the parallel ray from A hitting the mirror at M (close to P in the paraxial limit, so **MP ≈ AB**) and reflecting through F. Two pairs of similar triangles are used, working first with unsigned lengths.

Triangles $A'B'F$ and MPF are similar (both right-angled and sharing the angle at F), and **MP = AB**:

$$\frac{A'B'}{\mathrm{AB}} = \frac{\mathrm{FB'}}{\mathrm{FP}}$$

Triangles $A'B'P$ and ABP are similar (incident and reflected rays at P make equal angles with the axis):

$$\frac{A'B'}{\mathrm{AB}} = \frac{\mathrm{PB'}}{\mathrm{PB}}$$

Equating the two ratios:

$$\frac{\mathrm{FB'}}{\mathrm{FP}} = \frac{\mathrm{PB'}}{\mathrm{PB}}$$

The image B' is behind the mirror and the focus F is in front of it, so **FB' = FP + PB'**. Substituting:

$$\frac{\text{FP} + \text{PB}'}{\text{FP}} = \frac{\text{PB}'}{\text{PB}}$$

$$1 + \frac{\text{PB}'}{\text{FP}} = \frac{\text{PB}'}{\text{PB}}$$

Dividing through by PB':

$$\frac{1}{\text{PB}'} + \frac{1}{\text{FP}} = \frac{1}{\text{PB}}$$

Now apply the sign convention, with all distances measured from the pole P and the incident-light direction taken as positive. For this case the object distance is negative, the focal length of the concave mirror is negative, and the virtual image behind the mirror has a positive distance:

$$\text{PB} = -u,\ \text{PB}' = v,\ \text{FP} = -f$$

$$\frac{1}{v} + \frac{1}{(-f)} = \frac{1}{(-u)}$$

$$\frac{1}{v} - \frac{1}{f} = -\frac{1}{u}$$

Rearranging:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

This is the mirror equation, valid for all spherical mirrors when the sign convention is applied consistently.

Final answer. Ray diagram drawn (virtual image behind mirror); mirror equation 1/f = 1/v + 1/u derived correctly from similar triangles with the sign convention.

← *Back to Q326*

Q327

An astronomical telescope has an objective lens of focal length 20 *m and eyepiece of focal length* 1 cm.
(i) Find the angular magnification of the telescope.
(ii) If this telescope is used to view the Moon, find the diameter of the image formed by the objective lens. Given the diameter of the Moon is $3{\cdot}5 \times 10^6$ *m and radius of lunar orbit is* $3{\cdot}8 \times 10^8$ *m.*

[2020 • Set 55-1-1]

SOLUTION

Given: objective focal length $\mathbf{f_o = 20\ m}$, eyepiece focal length $\mathbf{f_e = 1\ cm = 0.01\ m}$.

(i) Angular magnification.

For a telescope in normal adjustment the angular magnification is

$$m = \frac{f_o}{f_e}$$

$$m = \frac{20\ m}{0.01\ m} = 2000$$

$$m = 2000$$

(ii) Diameter of the Moon's image formed by the objective.

The Moon subtends an angle θ at the objective equal to (diameter of Moon) divided by (distance to Moon):

$$\theta = \frac{\text{diameter of Moon}}{\text{radius of lunar orbit}} = \frac{3.5\times 10^6\ m}{3.8\times 10^8\ m}$$

$$\theta = 9.21\times 10^{-3}\ \text{rad}$$

The objective forms the image at its focal plane, so the image diameter d is the focal length times the angle subtended:

$$d = f_o\,\theta$$

$$d = 20\ m \times 9.21\times 10^{-3}\ \text{rad}$$

$$d = 0.184\ m = 18.4\ \text{cm}$$

$$d \approx 18.4 \text{ cm}$$

Final answer. (i) m = 2000; (ii) image diameter d ≈ 0.184 m = 18.4 cm.

← *Back to Q327*

Q328

An object is placed 30 cm *in front of a plano-convex lens with its spherical surface of radius of curvature* 20 cm. *If the refractive index of the material of the lens is* 1·5, *find the position and nature of the image formed.*

[2020 • Set 55-1-1]

SOLUTION

Given: object distance from a plano-convex lens, $u = -30$ **cm**; radius of the curved (convex) surface $R = 20$ **cm**; the plane surface has $R = \infty$; refractive index $n = 1.5$.

First find the focal length using the lens maker's formula. Take the curved surface facing the object with $R_1 = +20$ **cm** and the plane surface with $R_2 = \infty$.

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = (1.5-1)\left(\frac{1}{20} - \frac{1}{\infty}\right)$$

$$\frac{1}{f} = 0.5 \times \frac{1}{20} = \frac{1}{40}$$

$$f = +40 \text{ cm}$$

Now apply the thin-lens formula to locate the image.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{(-30)} = \frac{1}{40}$$

$$\frac{1}{v} + \frac{1}{30} = \frac{1}{40}$$

$$\frac{1}{v} = \frac{1}{40} - \frac{1}{30} = \frac{3-4}{120} = -\frac{1}{120}$$

$$v = -120 \text{ cm}$$

The negative value of v means the image is formed on the SAME side as the object (in front of the lens), so it is virtual and erect.

Magnification:

$$m = \frac{v}{u} = \frac{-120}{-30} = +4$$

The positive magnification of +4 confirms an erect, virtual image, magnified four times.

Virtual, erect image at 120 cm in front of the lens, magnified 4×.

(The object is within the focal length, **30 cm < 40 cm**, so a magnified virtual image is expected, as for a magnifying glass.)

Final answer. Virtual, erect image at v = −120 cm (120 cm in front of the lens), magnification m = +4 (magnified 4×).

← *Back to Q328*

Q329

(a) Derive lens maker's formula for a biconvex lens.
(b) A point object is placed at a distance of 12 *cm on the principal axis of a convex lens of focal length* 10 *cm. A convex mirror is placed coaxially on the other side of the lens at a distance of* 10 *cm. If the final image coincides with the object, sketch the ray diagram and find the focal length of the convex mirror.*
OR
(a) What is a wavefront? How does it propagate? Using Huygens' principle, explain reflection of a plane wavefront from a surface and verify the laws of reflection.
(b) A parallel beam of light of wavelength 500 *nm falls on a narrow slit and the resulting diffraction pattern is obtained on a screen* 1 *m away. If the first minimum is*

formed at a distance of 2.5 mm from the centre of the screen, find the (i) width of the slit, and (ii) distance of first secondary maximum from the centre of the screen.

[2020 • Set 55-2-1]

SOLUTION

(a) Lens maker's formula for a biconvex lens.

A thin lens is treated as two successive refractions at its two spherical surfaces. Let the lens material have index n_2, the surrounding medium index n_1, and surfaces of radii R_1 and R_2.

Refraction at the first surface (radius R_1): an object at distance u forms an image at distance v_1. Using the single-surface refraction relation,

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \quad ...(1)$$

Refraction at the second surface (radius R_2): the image I_1 acts as the object. Now light goes from the lens medium n_2 back into the medium n_1, so

$$\frac{n_1}{v} - \frac{n_2}{v_1} = \frac{n_1 - n_2}{R_2} \quad ...(2)$$

Add equations (1) and (2). The $\frac{n_2}{v_1}$ terms cancel:

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Divide through by n_1:

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For an object at infinity, the image forms at the focus ($v = f$), so the left side equals $\frac{1}{f}$. Writing $n = \frac{n_2}{n_1}$ (refractive index of lens relative to surroundings):

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

This is the lens maker's formula. For a biconvex lens $R_1 > 0$ and $R_2 < 0$, giving a positive (converging) f.

(b) Lens and convex mirror combination.

Given: convex lens $f = +10$ **cm**, object at 12 cm in front, convex mirror placed 10 cm behind the lens. The final image coincides with the object.

First, the lens forms an image of the object. With $u = -12$ **cm**, $f = +10$ **cm**:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{(-12)} = \frac{1}{10} - \frac{1}{12} = \frac{6-5}{60} = \frac{1}{60}$$

$$v = +60 \text{ cm}$$

So the lens alone would form a real image 60 cm to its right. The mirror is only 10 cm to the right of the lens, so this image lies **60 − 10 = 50 cm** behind the mirror, acting as a virtual object for the mirror.

Ray diagram (description): Draw the principal axis. Place the object 12 cm left of the convex lens. Rays refract through the lens and converge toward a point 60 cm to the right. Place the convex mirror 10 cm right of the lens; the converging rays strike the mirror before reaching that point. For the final image to coincide with the object, the rays must retrace their path, which happens only if they strike the mirror NORMALLY, i.e. they are heading straight toward the mirror's centre of curvature C.

For the rays to retrace, the point toward which they converge (50 cm behind the mirror) must be the centre of curvature C of the convex mirror. Hence

$$R = 50 \text{ cm}$$

$$f_m = \frac{R}{2} = \frac{50}{2} = 25 \text{ cm}$$

$$f_m = 25 \text{ cm (convex mirror)}$$

OR

(a) Wavefront and reflection by Huygens' principle.

A wavefront is the locus of all points of a medium that vibrate in the same phase. The direction of propagation is always perpendicular to the wavefront (along the rays). A wavefront propagates because, by Huygens' principle, every point on a wavefront acts as a fresh source of secondary spherical wavelets that

spread out with the wave speed v; the new wavefront after time t is the forward tangential envelope (forward common tangent) of all these secondary wavelets.

Reflection of a plane wavefront (description and proof): Let a plane wavefront AB be incident on a plane reflecting surface XY at angle of incidence i. Point A touches the surface first while B is still a distance away. While the disturbance travels from B to the surface at C (covering $\mathbf{BC = v\,t}$), a secondary wavelet from A expands into the same medium to radius $\mathbf{AE = v\,t}$. Drawing the tangent CE from C to this wavelet gives the reflected wavefront. Let the reflection angle be r.

In right triangles ABC and CEA (common hypotenuse AC):

$$\mathrm{BC} = \mathrm{AE} = v\,t$$

$$\angle\,\mathrm{ABC} = \angle\,\mathrm{CEA} = 90^\circ$$

$\mathbf{AC} = \mathbf{AC}$ (common)

So the two triangles are congruent (RHS), giving $\mathbf{\angle\,BAC = \angle\,ECA}$, i.e. the angle of incidence equals the angle of reflection:

$$i = r$$

Also the incident ray, reflected ray and normal all lie in the same plane (the plane of the diagram). These are the two laws of reflection, verified.

(b) Single-slit diffraction.

Given: $\mathbf{\lambda = 500\ nm = 500\times 10^{-9}\ m}$, screen distance $\mathbf{D = 1\ m}$, first minimum at $\mathbf{y_1 = 2.5\ mm = 2.5\times 10^{-3}\ m}$.

(i) The first minimum of a single slit of width a satisfies $\mathbf{a\theta = \lambda}$, and for small angles $\theta \approx \theta \approx \frac{y_1}{D}$:

$$a\,\frac{y_1}{D} = \lambda$$

$$a = \frac{\lambda\,D}{y_1} = \frac{(500\times 10^{-9})(1)}{2.5\times 10^{-3}}$$

$$a = \frac{5\times 10^{-7}}{2.5\times 10^{-3}} = 2\times 10^{-4}\ m = 0.2\ \mathrm{mm}$$

(ii) The first secondary maximum lies at $a\theta = \frac{3\lambda}{2}$, i.e. at $y = \frac{3}{2}y_1$ (since the first minimum is at y_1):

$$y = \frac{3}{2} \times 2.5 \text{ mm} = 3.75 \text{ mm}$$

$$a = 0.2 \text{ mm}, y = 3.75 \text{ mm}$$

CONCEPT

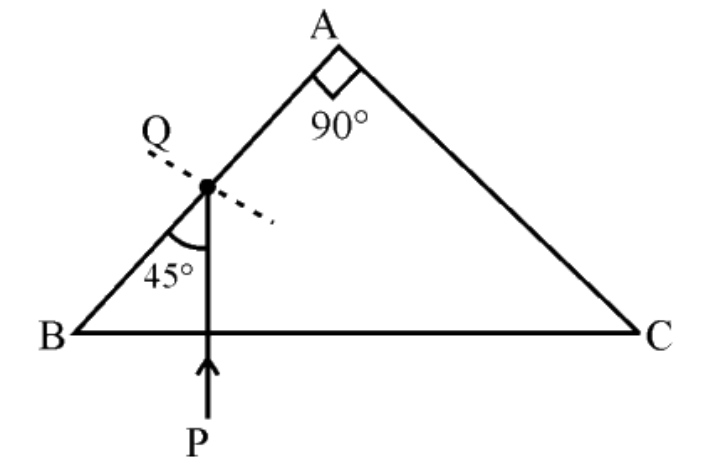
For the lens-mirror coincidence, the rays must hit the mirror normally and retrace, so the lens image must land exactly at the mirror's centre of curvature C, giving R directly.

Final answer. (a) $1/f = (n-1)(1/R_1 - 1/R_2)$; (b) convex mirror $f_m = 25$ cm. OR: laws of reflection verified by Huygens; slit width a = 0.2 mm, first secondary maximum at 3.75 mm.

← *Back to Q329*

Q330

(a) Draw the ray diagram showing refraction of ray of light through a glass prism. Derive the expression for the refractive index μ *of the material of prism in terms of the angle of prism A and angle of minimum deviation* δ_m.
(b) A ray of light PQ *enters an isosceles right angled prism* ABC *of refractive index* 1.5 *as shown in figure.*



(i) Trace the path of the ray through the prism.
(ii) What will be the effect on the path of the ray if the refractive index of the prism is 1.4*?*

OR

(a) Two thin lenses are placed coaxially in contact. Obtain the expression for the focal length of this combination in terms of the focal lengths of the two lenses.
(b) A converging lens of refractive index 1.5 *has a power of* 10 *D. When it is completely immersed in a liquid, it behaves as a diverging lens of focal length* 50 cm. *Find the refractive index of the liquid.*

[2020 • Set 55-4-1]

SOLUTION

(a) Refraction through a prism and the prism formula.

Ray diagram (description): Draw a triangular prism ABC with the refracting angle A at the apex. A ray PQ strikes face AB at Q with angle of incidence i, bends toward the normal, travels inside as QR making angle of refraction r_1, then strikes the second face AC at R with internal angle r_2, and emerges as RS bending away from the normal with angle of emergence e. The emergent ray is deviated from the original direction by the angle of deviation δ. Mark the normals at Q and R, the angles i, r_1, r_2, e, and the deviation δ between the incident-ray-produced and emergent ray.

At each surface light bends. The total deviation is

$$\delta = (i - r_1) + (e - r_2)$$

From the quadrilateral AQOR (with normals meeting at O), the angle A and the angle QOR are supplementary; combined with triangle QOR this gives

$$r_1 + r_2 = A$$

Substituting:

$$\delta = i + e - (r_1 + r_2) = i + e - A$$

At minimum deviation $\boldsymbol{\delta = \delta_m}$, the ray passes symmetrically through the prism, so $\boldsymbol{i = e}$ and $\boldsymbol{r_1 = r_2 = r}$. Then

$$r_1 + r_2 = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$$

$$\delta_m = i + i - A = 2i - A \Rightarrow i = \frac{A + \delta_m}{2}$$

Applying Snell's law at the first face ($\mu = \frac{\sin i}{\sin r}$):

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$$

$$\mu = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$$

(b) Ray through an isosceles right-angled prism, $n = 1.5$.

The prism ABC is right-angled at A with the two equal angles of 45°. The ray PQ enters normally through one of the perpendicular faces and then strikes the hypotenuse at 45°.

Find the critical angle for the glass-air interface:

$$\theta_c = \frac{1}{n} = \frac{1}{1.5} = 0.667$$

$$\theta_c = \sin^{-1}(0.667) = 41.8°$$

(i) The ray meets the hypotenuse at 45°. Since **$45° > \theta_c = 41.8°$**, total internal reflection occurs at the hypotenuse. The ray is reflected through 90° and emerges normally through the other perpendicular face. (Path: enters normally, undergoes TIR at the hypotenuse, exits at right angles to the entry direction.)

(ii) If the refractive index is 1.4:

$$\theta_c = \frac{1}{1.4} = 0.714 \Rightarrow \theta_c = 45.6°$$

Now the angle of incidence on the hypotenuse (45°) is LESS than the new critical angle (45.6°), so TIR no longer occurs. The ray is refracted and partly transmitted out through the hypotenuse (bending away from the normal), with only partial reflection.

OR

(a) Focal length of two thin lenses in contact.

Two thin lenses of focal lengths f_1 and f_2 are placed in contact coaxially. For an object at distance u, the first lens forms an image at v_1:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

This image acts as the object for the second lens (lenses in contact, so separation is zero), forming the final image at v:

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

Adding the two equations, the $\frac{1}{v_1}$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

If F is the focal length of the equivalent single lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$, so

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

In terms of power, $P = P_1 + P_2$.

(b) Refractive index of the liquid.

The lens has $n_g = 1.5$ and power $P = 10\ D$, so its focal length in air is

$$f_a = \frac{1}{P} = \frac{1}{10} = 0.1\ m = +10\ \text{cm}$$

In the liquid it behaves as a diverging lens of focal length $f_l = -50\ \text{cm}$.

Using the lens maker's formula in air and in liquid, the ratio is

$$\frac{f_l}{f_a} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_l} - 1\right)}$$

$$\frac{-50}{10} = \frac{1.5 - 1}{\frac{1.5}{n_l} - 1}$$

$$-5 = \frac{0.5}{\frac{1.5}{n_l} - 1}$$

$$\frac{1.5}{n_l} - 1 = \frac{0.5}{-5} = -0.1$$

$$\frac{1.5}{n_l} = 1 - 0.1 = 0.9$$

$$n_l = \frac{1.5}{0.9} = \frac{5}{3} = 1.67$$

$$n_l \approx 1.67$$

Since the liquid is optically denser than the glass ($\mathbf{n_l > n_g}$), the converging lens indeed acts as a diverging lens in it.

Final answer. (a) μ = sin((A+δ_m)/2)/sin(A/2); (b) n=1.5: TIR at hypotenuse, 90 ° turn; n=1.4: no TIR, ray emerges. OR: 1/F = 1/f_1 + 1/f_2; liquid index n_l ≈ 1.67.

← *Back to Q330*

Q331

(a) Define the term 'focal length of a mirror'. With the help of a ray diagram, obtain the relation between its focal length and radius of curvature.
(b) Calculate the angle of emergence (e) of the ray of light incident normally on the face AC *of a glass prism* ABC *of refractive index* $\sqrt{3}$. *How will the angle of emergence change qualitatively, if the ray of light emerges from the prism into a liquid of refractive index* 1.3 *instead of air ?*
OR
(a) Define the term 'resolving power of a telescope'. How will the resolving power be effected with the increase in
(i) Wavelength of light used.
(ii) Diameter of the objective lens.
Justify your answers.
(b) A screen is placed 80 *cm from an object. The image of the object on the screen is*

formed by a convex lens placed between them at two different locations separated by a distance 20 *cm. Determine the focal length of the lens.*

[2020 • Set 55-5-1]

SOLUTION

Part (a): The focal length of a mirror is the distance between the pole of the mirror and its principal focus, measured along the principal axis. It is the point where a beam of light parallel to the principal axis converges (concave mirror) or appears to diverge from (convex mirror) after reflection.

Ray diagram: Draw a concave mirror with pole P, centre of curvature C and principal focus F on the principal axis. A ray AB travelling parallel to the principal axis strikes the mirror at point B. The line CB joining the centre of curvature to B is the normal at B (a radius is always normal to the spherical surface). The reflected ray passes through the focus F. Mark the angle of incidence and angle of reflection on either side of the normal CB.

Let the angle of incidence be θ. Since AB ∥ PC, the angle $\angle$ **BCF = θ** (alternate angles). By the law of reflection, the angle of reflection $\angle$ BFC also equals θ.

Hence triangle BCF is isosceles, so **FB = FC**.

For a mirror of small aperture, B lies very close to P, so **FB ≈ FP =** f.

Therefore **FC = FP =** f.

Now **PC = PF + FC**, that is $R = f + f$.

$\Rightarrow R = 2f$, which gives $f = \frac{R}{2}$.

Thus the **focal length of a spherical mirror is half its radius of curvature**.

Part (b): The ray is incident normally on face AC, so it enters without bending and the angle of refraction at AC is $r_1 = 0°$. The prism here has the angle at A equal to 30° (the ray strikes the second face AB from inside).

From prism geometry, $r_1 + r_2 = A$, so $r_2 = A - r_1 = 30° - 0° = 30°$.

Applying Snell's law at the second face AB (glass to air), with $n = \sqrt{3}$:

$$n \sin r_2 = 1 \times \sin e$$

$$\sin e = \sqrt{3} \times \sin 30°$$

$$\sin e = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow e = 60^\circ$$

When the ray emerges into a liquid of refractive index 1.3 instead of air, Snell's law at AB becomes $n \sin r_2 = 1.3 \sin e'$:

$$\sin e' = \sqrt{3} \times \sin 30^\circ 1.3 = 0.8661.3 = 0.666$$

$$\Rightarrow e' \approx 41.8^\circ$$

Since the surrounding medium is now optically denser than air, the ray bends less on emerging, so the **angle of emergence decreases** (from 60° to about 42°).

TIP

Whenever a ray is incident normally on one face of a prism, that face contributes zero refraction, so the angle of refraction at the second face is simply equal to the angle of the prism (r2 = A).

Final answer. Part (a): f = R/2. Part (b): the angle of emergence in air is e = 60 degrees, and it decreases (to about 42 degrees) when the ray emerges into the liquid of refractive index 1.3.

← *Back to Q331*

Q332

Under what conditions is the phenomenon of total internal reflection of light observed? Obtain the relation between the critical angle of incidence and the refractive index of the medium.

[2019 • Set 55-1-1]

SOLUTION

Total internal reflection (TIR) is the phenomenon in which a ray of light travelling in a denser medium is completely reflected back into the same medium when it strikes the boundary with a rarer medium at an angle greater than the critical angle.

Conditions for total internal reflection:

(i) Light must travel from an optically denser medium towards an optically rarer medium, that is $\mathbf{n_1 > n_2}$ where n_1 is the refractive index of the denser medium.

(ii) The angle of incidence in the denser medium must be greater than the critical angle, that is $\boldsymbol{i > \theta_c}$.

Description of the diagram: Draw a horizontal boundary separating a denser medium (below) from a rarer medium (above). Show three rays starting from a point on the boundary. For a small angle of incidence, the ray refracts and bends away from the normal into the rarer medium. As the angle of incidence increases to the critical angle θ_c, the refracted ray grazes along the boundary, making an angle of refraction of 90°. For an angle of incidence greater than θ_c, no refraction occurs and the ray is totally reflected back into the denser medium, obeying the law of reflection.

Relation between critical angle and refractive index: The critical angle θ_c is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90°.

Applying Snell's law at the boundary for this special case (incidence angle = $\boldsymbol{\theta_c}$, refraction angle = **90**°):

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

Since $\mathbf{\sin 90^\circ = 1}$:

$$n_1 \sin \theta_c = n_2$$

$$\Rightarrow \sin \theta_c = \frac{n_2}{n_1} = \frac{n_{\text{rarer}}}{n_{\text{denser}}}$$

If the denser medium has refractive index n and the rarer medium is air ($\mathbf{n_2 = 1}$), then

$$\sin \theta_c = \frac{1}{n}$$

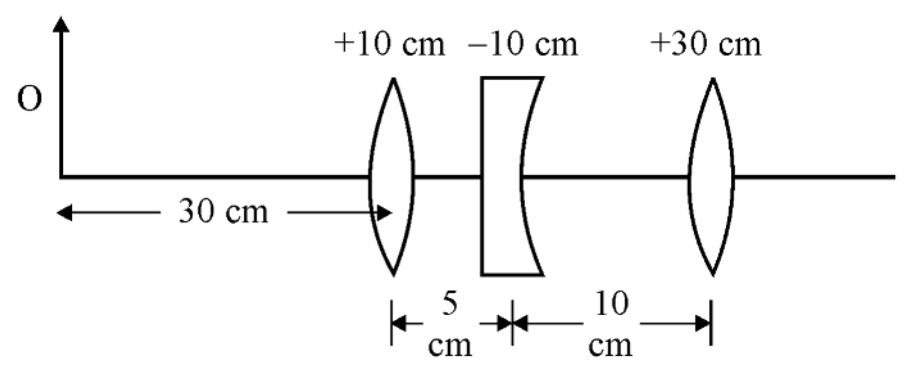
This is the required relation between the **critical angle and the refractive index of the medium**.

Final answer. TIR occurs when light goes from denser to rarer medium at an angle greater than the critical angle, and sin(theta_c) = n_rarer / n_denser, which becomes sin(theta_c) = 1/n for a medium-to-air boundary.

← *Back to Q332*

Q333

Three lenses of focal lengths +10 cm, −10 cm *and* +30 cm *are arranged coaxially as in the figure given below. Find the position of the final image formed by the combination.*



[2019 • Set 55-1-1]

SOLUTION

This is the standard NCERT three-lens problem. The lenses have focal lengths $\mathbf{f_1}$ = **+10** cm, $\mathbf{f_2}$ = **−10** cm and $\mathbf{f_3}$ = **+30** cm, arranged coaxially with a separation of 5 cm between the first and second lenses and 10 cm between the second and third lenses. A parallel beam of light (object at infinity) is incident on the first lens. We use the thin-lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for each lens in turn, where the image from one lens becomes the object for the next.

Refraction at the first lens L_1 ($\mathbf{f_1}$ = **+10** cm): The incident beam is parallel, so the object is at infinity. The image is formed at the focus of L_1.

$\frac{1}{\mathbf{v_1}} - \frac{1}{\mathbf{u_1}} = \frac{1}{\mathbf{f_1}}$ with $\mathbf{u_1} = \infty$ gives $\mathbf{v_1} = \mathbf{f_1} = \mathbf{+10\ cm}$.

This image I_1 lies 10 cm to the right of L_1, which is **10 − 5 = 5** cm to the right of the second lens L_2.

Refraction at the second lens L_2 ($\mathbf{f_2 = -10}$ cm): The image I_1 would form on the far side of L_2, so it acts as a virtual object with $\mathbf{u_2 = +5\ cm}$.

$$\frac{1}{v_2} = \frac{1}{f_2} + \frac{1}{u_2} = \frac{1}{-10} + \frac{1}{5}$$

$$\frac{1}{v_2} = \frac{-1+2}{10} = \frac{1}{10}$$

$$\Rightarrow v_2 = +10\ \text{cm}$$

So I_2 is formed 10 cm to the right of L_2. Since L_3 is exactly 10 cm to the right of L_2, the rays are converging towards the position of the third lens itself.

Refraction at the third lens L_3 ($\mathbf{f_3 = +30}$ cm): The object for L_3 is at its own pole, $\mathbf{u_3 = 0}$, meaning the rays are effectively parallel as they emerge from L_2 towards L_3. Equivalently, the beam incident on L_3 behaves like a parallel beam, so it converges to the focus of L_3.

$$v_3 = f_3 = +30\ \text{cm}$$

Therefore the **final image is formed 30 cm to the right of the third lens**.

COMMON MISTAKE

When tracking images through a multi-lens system, always shift the reference point: subtract the inter-lens separation from the previous image distance to get the object distance for the next lens, and keep the sign convention strict.

Final answer. The final image is formed 30 cm to the right of the third lens.

← *Back to Q333*

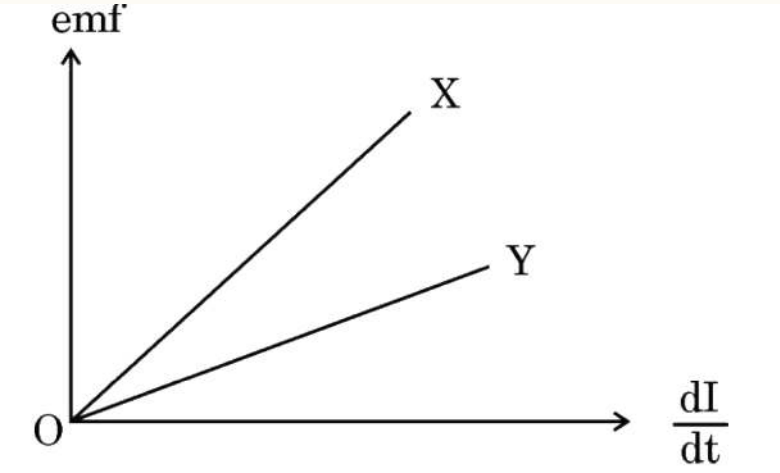
Q334

(a) Using the ray diagram for a system of two lenses of focal lengths f_1 and f_2 in contact with each other, show that the two lens system can be regarded as equivalent to a single lens of focal length f, where

$$\mathbf{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}.$$

Also write the relation for the equivalent power of the lens combination.

(b) Determine the position of the image formed by the lens combination given in the figure.



OR

(a) Explain, using a suitable diagram, how unpolarized light gets linearly polarized by scattering.

(b) Describe briefly the variation of the intensity of transmitted light when a polaroid sheet kept between two crossed polaroids is rotated. Draw the graph depicting the variation of intensity with the angle of rotation. How many maxima and minima would be observed when θ *varies from* 0 *to* π*?*

[2019 • Set 55-4-1]

SOLUTION

Part (a): Consider two thin lenses L_1 and L_2 of focal lengths f_1 and f_2 placed in contact with each other on a common principal axis.

Ray diagram description: Draw the common principal axis. Place a point object O on the axis at a distance u to the left of the lens pair. The first lens L_1 alone would form an image I_1 at distance v_1. The second lens L_2, in contact, then takes I_1 as its object and forms the final image I at distance v. Mark O, I_1 and I on the axis with the object and image distances labelled.

For the first lens L_1 forming image I_1:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

The image I_1 acts as the object for the second lens L_2 (the lenses are in contact, so the separation is zero). For L_2 forming the final image I:

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

Adding the two equations, the $\frac{1}{v_1}$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

If the combination is replaced by a single equivalent lens of focal length f, then $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. Comparing:

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

Since power $P = \frac{1}{f}$ (with f in metres), the equivalent power of the combination is

$$P = P_1 + P_2$$

Part (b): The figure shows the same three lenses $\mathbf{f_1 = +10}$ cm, $\mathbf{f_2 = -10}$ cm and $\mathbf{f_3 = +30}$ cm, separated by 5 cm and 10 cm respectively, with a parallel beam incident. Tracing image by image:

Lens 1: parallel beam, so $\mathbf{v_1 = f_1 = +10}$ cm (image 10 cm right of L_1).

Lens 2: object is virtual, $\mathbf{u_2 = 10 - 5 = +5}$ cm; $\frac{1}{v_2} = \frac{1}{-10} + \frac{1}{5} = \frac{1}{10}$, so $\mathbf{v_2 = +10}$ cm.

Lens 3: I_2 lies 10 cm right of L_2, which is exactly at L_3, so the beam reaching L_3 is parallel and converges to its focus: $\mathbf{v_3 = f_3 = +30}$ cm.

Thus the **final image is formed 30 cm to the right of the third lens**.

Final answer. Part (a): 1/f = 1/f1 + 1/f2 and P = P1 + P2; Part (b): the final image forms 30 cm to the right of the third lens.

← *Back to Q334*

Q335

(a) Draw a labelled ray diagram of compound microscope, when final image forms at the least distance of distinct vision.
(b) Why is its objective of short focal length and of short aperture, compared to its eyepiece? Explain.
(c) The focal length of the objective is 4 cm while that of eyepiece is 10 cm. The object is placed at a distance of 6 cm from the objective lens.
(i) Calculate the magnifying power of the compound microscope, if its final image is formed at the near point.
(ii) Also calculate length of the compound microscope.
OR
(a) With the help of a labelled ray diagram, explain the construction and working of a Cassegrain reflecting telescope.
(b) An amateur astronomer wishes to estimate roughly the size of the Sun using his crude telescope consisting of an objective lens of focal length 200 cm and an eyepiece of focal length 10 cm. By adjusting the distance of the eyepiece from the objective, he obtains an image of the Sun on a screen 40 cm behind the eyepiece. The diameter of the Sun's image is measured to be 6.0 cm. Estimate the Sun's size, given that the average Earth-Sun distance is 1.5×10^{11} *m.*

[2019 • Set 55-5-1]

SOLUTION

Part (a): Ray diagram of a compound microscope (final image at the least distance of distinct vision D).

Description: Draw two convex lenses on a common axis: the objective O (short focal length f_o) on the left and the eyepiece E (larger focal length f_e) on the right. Place a small object AB just beyond the focus F_o of the objective. The objective forms a real, inverted, magnified image $A'B'$ between the eyepiece and its focus F_e. The eyepiece then acts as a simple magnifier and forms a final image $A''B''$ that is virtual, inverted (relative to the object) and highly magnified, at the least distance of distinct vision $\boldsymbol{D = 25}$ cm from the eye. Label f_o, f_e, the intermediate image $A'B'$, the final image $A''B''$, and mark the distance D from the eyepiece to $A''B''$.

Part (b): In a compound microscope the magnifying power is $M = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)$, which is large when f_o is small (the object is placed close to the objective, giving a large linear magnification $\frac{v_o}{u_o}$). A short focal length of the objective therefore gives high magnification. The aperture of the objective is kept small so that only paraxial rays close to the axis are collected; this **reduces spherical and chromatic aberrations** and gives a sharp, clear image. The eyepiece has a larger focal length and aperture because it only needs to provide moderate angular magnification of an already enlarged image.

Part (c): Given $f_o = 4$ cm, $f_e = 10$ cm, object distance from objective $u_o = -6$ cm, near point $D = 25$ cm.

For the objective, using $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$:

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{4} + \frac{1}{-6}$$

$$\frac{1}{v_o} = \frac{3-2}{12} = \frac{1}{12}$$

$$\Rightarrow v_o = 12 \text{ cm}$$

Magnification of the objective: $m_o = \frac{v_o}{|u_o|} = \frac{12}{6} = 2$.

Magnification of the eyepiece (final image at near point): $m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{10} = 3.5$.

(i) Total magnifying power:

$$M = m_o \times m_e = 2 \times 3.5 = 7$$

(ii) Length of the microscope $L = v_o + |u_e|$, where $|u_e|$ is the object distance for the eyepiece. Using $\frac{1}{-D} - \frac{1}{u_e} = \frac{1}{f_e}$, the object distance is

$$|u_e| = \frac{f_e D}{f_e + D} = \frac{10 \times 25}{10 + 25} = \frac{250}{35} = 7.14 \text{ cm}$$

$$L = v_o + |u_e| = 12 + 7.14 = 19.14 \text{ cm}$$

Thus the **magnifying power is 7 and the length of the microscope is about 19.14 cm**.

Final answer. The magnifying power of the compound microscope is 7 and its length is about 19.14 cm.

← Back to Q335

Q336

(a) Draw a ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object.
(b) Obtain the mirror formula and write the expression for the linear magnification.
(c) Explain two advantages of a reflecting telescope over a refracting telescope.

[2018]

SOLUTION

Part (a): Ray diagram for a real, inverted, magnified image by a concave mirror.

Description: Draw a concave mirror with pole P, focus F and centre of curvature C on the principal axis. Place an object AB (an upright arrow) between F and C. From the top point B draw two rays: (i) a ray parallel to the principal axis that reflects through the focus F, and (ii) a ray passing through the centre of curvature C that retraces its path (since it hits the mirror normally). These two reflected rays intersect beyond C to form the image $A'B'$. The image $A'B'$ is real, inverted and magnified (larger than the object), located beyond C.

Part (b): Derivation of the mirror formula.

In the ray diagram, consider the ray BP reflected to B' and the parallel ray reflected through F. Using the geometry of the two right-angled triangles formed by the object, image and the axis, triangles $A'B'F$ and the triangle at the pole are similar, and triangles ABP and $A'B'P$ are similar.

From similar triangles ABP and $A'B'P$ (sharing the angle at P):

$$\frac{A'B'}{AB} = \frac{PA'}{PA}$$

From similar triangles $A'B'F$ and MPF (where M is the point where the parallel ray meets the mirror, with **MP ≈ AB**):

$$\frac{A'B'}{AB} = \frac{FA'}{FP} = \frac{PA' - PF}{PF}$$

Equating the two expressions:

$$\frac{PA'}{PA} = \frac{PA' - PF}{PF}$$

Applying the sign convention with **PA = −u**, **PA' = −v**, **PF = −f** and simplifying:

$$\frac{-v}{-u} = \frac{-v - (-f)}{-f}$$

This reduces to

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$$

The linear magnification is the ratio of image height to object height:

$$m = \frac{h'}{h} = -\frac{v}{u}$$

The negative sign shows that a real image is inverted.

Part (c): Two advantages of a reflecting telescope over a refracting telescope:

(i) A reflecting telescope uses a concave mirror as objective, which reflects all wavelengths through the same point, so it is **free from chromatic aberration**, unlike a lens which disperses different colours and causes coloured fringes.

(ii) A large mirror can be supported from behind along its whole back surface, so very large apertures are possible. This gives **greater light-gathering power and higher resolving power**, whereas a large lens can only be supported at its edges and tends to sag and distort under its own weight.

Final answer. The mirror formula 1/v + 1/u = 1/f = 2/R with m = -v/u, plus the reflecting telescope's freedom from chromatic aberration and its larger feasible aperture, are the key results.

← *Back to Q336*

Q337

(a) A point object is placed on the principal axis of a convex spherical surface of radius of curvature R, which separates the two media of refractive indices n_1 *and* n_2 ($\mathbf{n_2 > n_1}$). *Draw the ray diagram and deduce the relation between the object distance (u), image distance (v) and the radius of curvature (R) for refraction to take place at the convex spherical surface from rarer to denser medium.*

(b) A converging lens has a focal length of 20 cm *in air. It is made of a material of refractive index* 1·6. *If it is immersed in a liquid of refractive index* 1·3, *find its new focal length.*

OR

(a) Draw the ray diagram showing refraction of light through a glass prism and hence obtain the relation between the refractive index μ *of the prism, angle of prism and angle of minimum deviation.*

(b) Determine the value of the angle of incidence for a ray of light travelling from a medium of refractive index $\mathbf{\mu_1 = \sqrt{2}}$ *into the medium of refractive index* $\mathbf{\mu_2 = 1}$, *so that it just grazes along the surface of separation.*

[2017]

SOLUTION

Part (a): Refraction at a convex spherical surface (rarer to denser medium).

Diagram description: Draw a convex refracting surface separating a rarer medium of index n_1 (left) from a denser medium of index n_2 (right), with $\mathbf{n_2 > n_1}$. Mark the pole *P*, centre of curvature *C* and radius *R* on the principal axis. Place a point object *O* on the axis in the rarer medium. A ray OA strikes the surface at *A*; the normal CA (a radius) is drawn at *A*. The refracted ray bends towards the normal and meets the axis at the real image *I* in the denser medium. Mark the angles of incidence *i* and refraction *r*, and the angles α, β, γ that OA, IA and CA make with the axis.

For small (paraxial) angles, from the geometry of the triangles:

$$i = \alpha + \beta \text{ and } \beta = r + \gamma \Rightarrow r = \beta - \gamma$$

For small angles, Snell's law $\mathbf{n_1 \sin i = n_2 \sin r}$ becomes $\mathbf{n_1 i = n_2 r}$.

Substituting:

$$n_1(\alpha+\beta) = n_2(\beta-\gamma)$$

Using $\alpha = \frac{AN}{-u}$, $\beta = \frac{AN}{R}$, $\gamma = \frac{AN}{v}$ (where AN is the height of A above the axis) and applying the sign convention:

$$n_1\left(\frac{1}{-u}+\frac{1}{R}\right) = n_2\left(\frac{1}{R}-\frac{1}{v}\right)$$

Rearranging gives the required relation:

$$\frac{n_2}{v}-\frac{n_1}{u} = \frac{n_2-n_1}{R}$$

Part (b): A converging lens has focal length $f_a = 20$ cm in air, lens material $n_g = 1.6$, immersed in liquid $n_l = 1.3$. Using the lens maker's formula, the focal length in a surrounding medium of index n_l is related to the focal length in air by

$$\frac{f_l}{f_a} = \frac{(n_g-1)}{\left(\frac{n_g}{n_l}-1\right)}$$

Substituting the values:

$$\frac{f_l}{f_a} = \frac{1.6-1}{\frac{1.6}{1.3}-1} = \frac{0.6}{1.231-1} = \frac{0.6}{0.231}$$

$$\frac{f_l}{f_a} = 2.6$$

$$f_l = 2.6 \times f_a = 2.6 \times 20$$

$$\Rightarrow f_l = 52 \text{ cm}$$

Thus the **new focal length of the lens in the liquid is 52 cm** (it increases because the lens is now less converging in the denser surrounding medium).

Final answer. Part (a): n2/v - n1/u = (n2 - n1)/R; Part (b): the new focal length in the liquid is 52 cm.

← *Back to Q337*

Q338

(a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.
(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$ *th of the angle of prism. Calculate the speed of light in the prism.*

[2017]

SOLUTION

Part (a): Image formation by two thin convex lenses in contact and the power of the combination.

Ray diagram description: Draw two thin convex lenses L_1 and L_2 placed in contact on a common axis. A point object O is placed on the axis at distance u to the left. The first lens L_1 alone forms a real image I_1 at distance v_1. This image I_1 serves as the object for L_2, which forms the final image I at distance v. Mark O, I_1 and I on the axis.

For the first lens L_1:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

The image I_1 acts as the object for L_2 (separation zero, since the lenses are in contact):

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

Adding the two equations, the $\frac{1}{v_1}$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

For a single equivalent lens of focal length F, $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$, so

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Since power $P = \frac{1}{f}$ (with f in metres), the power of the combination is

$$P = P_1 + P_2$$

Part (b): Equilateral prism, so $A = 60°$. The deviation is minimum when the angle of incidence is $\frac{3}{4}$ of the prism angle:

$$i = \frac{3}{4} \times 60° = 45°$$

At minimum deviation, $i = e$, so

$$\delta_m = 2i - A = 2 \times 45° - 60° = 30°$$

The refractive index of the prism is

$$n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})} = \sin(60° + 30°2)\sin(60°2)$$

$$n = \sin 45° \sin 30° = 1/\sqrt{2}\,1/2 = \sqrt{2}$$

The speed of light in the prism is

$$v = \frac{c}{n} = \frac{3 \times 10^8}{\sqrt{2}}$$

$$\Rightarrow v \approx 2.12 \times 10^8\ m\ s^{-1}$$

Thus the **speed of light in the prism is about $2.12 \times 10^8\ m\ s^{-1}$**.

Final answer. Part (a): P = P1 + P2; Part (b): the speed of light in the prism is about 2.12 x 10^8 m/s.

← *Back to Q338*

Q339

(i) Derive the mathematical relation between refractive indices n_1 *and* n_2 *of two media and radius of curvature R for refraction at a convex spherical surface. Consider*

the object to be a point since lying on the principal axis in rarer medium of refractive index n_1 *and a real image formed in the denser medium of refractive index* n_2. *Hence, derive lens maker's formula.*

(ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 *and radius of curvature* 20 cm. *The distance of light source from the glass surface is* 100 cm. *At what position is the image formed?*

OR

(a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.

(b) You are given three lenses of power 0.5 *D,* 4 *D and* 10 *D to design a telescope.*

(i) Which lenses should be used as objective and eyepiece? Justify your answer.

(ii) Why is the aperture of the objective preferred to be large?

[2016]

SOLUTION

Part (i): Refraction at a single convex spherical surface and the lens maker's formula.

Diagram description: Draw a convex refracting surface of radius R separating a rarer medium n_1 (left) from a denser medium n_2 (right), with the pole P and centre of curvature C on the axis. A point object O in the rarer medium sends a paraxial ray to point A on the surface; the normal is the radius CA. The refracted ray meets the axis at the real image I in the denser medium. Mark angles i, r and the small angles α, β, γ.

For paraxial rays, $i = \alpha + \beta$ and $r = \beta - \gamma$. Snell's law for small angles gives $n_1 i = n_2 r$:

$$n_1(\alpha + \beta) = n_2(\beta - \gamma)$$

Writing the small angles as $\alpha = \frac{h}{-u}$, $\beta = \frac{h}{R}$, $\gamma = \frac{h}{v}$ and applying the sign convention:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Lens maker's formula: A thin lens has two refracting surfaces of radii R_1 and R_2. Apply the above relation at each surface, treating the image from the first surface as the object for the second.

At the first surface (rarer to denser, radius R_1), with image distance v_1:

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$$

At the second surface (denser to rarer, radius R_2), the image I_1 acts as object:

$$\frac{n_1}{v} - \frac{n_2}{v_1} = \frac{n_1 - n_2}{R_2}$$

Adding these two equations, the $\frac{n_2}{v_1}$ terms cancel:

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Dividing by n_1 and writing $\boldsymbol{n = \frac{n_2}{n_1}}$ (refractive index of lens relative to surroundings), and using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for an object at infinity:

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Part (ii): Light from a point source in air falls on a convex glass surface, $\mathbf{n_1 = 1}$, $\mathbf{n_2 = 1.5}$, $\boldsymbol{R = +20}$ cm, object distance $\boldsymbol{u = -100}$ cm.

Using $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$:

$$\frac{1.5}{v} - \frac{1}{-100} = \frac{1.5 - 1}{20}$$

$$\frac{1.5}{v} + \frac{1}{100} = \frac{0.5}{20} = \frac{1}{40}$$

$$\frac{1.5}{v} = \frac{1}{40} - \frac{1}{100} = \frac{5 - 2}{200} = \frac{3}{200}$$

$$v = \frac{1.5 \times 200}{3} = \frac{300}{3} = 100 \text{ cm}$$

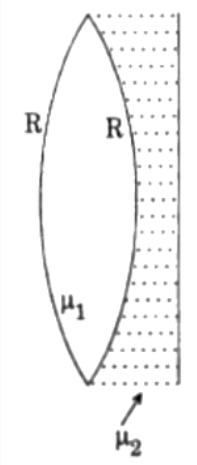
Thus the **image is formed 100 cm from the surface, on the same side as the refracted ray (inside the glass)**.

Final answer. Part (i): n2/v - n1/u = (n2 - n1)/R leading to the lens maker's formula; Part (ii): the image forms 100 cm beyond the surface.

← Back to Q339

Q340

A biconvex lens with its two faces of equal radius of curvature R is made of a transparent medium of refractive index μ_1. It is kept in contact with a medium of refractive index μ_2 as shown in the figure.
(a) Find the equivalent focal length of the combination.
(b) Obtain the condition when this combination acts as a diverging lens.
(c) Draw the ray diagram for the case $\mu_2 > (\mu_1 + 1)/2$, when the object is kept far away from the lens. Point out the nature of the image formed by the system.



[2015]

SOLUTION

A biconvex lens of material μ_1 with both faces of radius R is in contact with a second medium of refractive index μ_2 on one of its curved faces. We treat the system as a combination of two thin lenses in contact: the original biconvex glass lens, plus a plano-concave lens formed by the second medium μ_2 filling the space against the curved face.

Part (a): Equivalent focal length.

For the biconvex lens of material μ_1 in air, with $\mathbf{R_1} = +R$ and $\mathbf{R_2} = -R$, the lens maker's formula gives

$$\frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (\mu_1 - 1)\frac{2}{R}$$

The second medium μ_2 in contact with the curved face acts as a plano-concave lens (curved face of radius R against the glass, flat outer face). Its focal length is

$$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right) = -\frac{(\mu_2 - 1)}{R}$$

For lenses in contact, $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$:

$$\frac{1}{f} = \frac{2(\mu_1 - 1)}{R} - \frac{(\mu_2 - 1)}{R}$$

$$\frac{1}{f} = \frac{2\mu_1 - 2 - \mu_2 + 1}{R} = \frac{2\mu_1 - \mu_2 - 1}{R}$$

$$f = \frac{R}{2\mu_1 - \mu_2 - 1}$$

Part (b): Condition for the combination to act as a diverging lens.

The combination is diverging when its focal length is negative, that is when the denominator is negative:

$$2\mu_1 - \mu_2 - 1 < 0$$

$$\Rightarrow \mu_2 > 2\mu_1 - 1$$

Part (c): Ray diagram for $\mu_2 > \frac{\mu_1 + 1}{2}$, distant object.

The condition $\mu_2 > \frac{\mu_1 + 1}{2}$ is weaker than the diverging condition $\mu_2 > 2\mu_1 - 1$ for the usual values, so here the denominator $2\mu_1 - \mu_2 - 1$ is still positive and the combination is **converging** (a real lens of positive focal length).

Ray diagram description: Draw the combination on the axis. Since the object is very far away, the incident rays are parallel to the principal axis. The converging combination brings these parallel rays to a focus on the far side, at the focal point F of the system. The image formed is **real, inverted and diminished**, located at the focal plane of the combination.

Thus the system acts as a converging lens and forms a real, inverted, point-sized image of the distant object at its focus.

CONCEPT

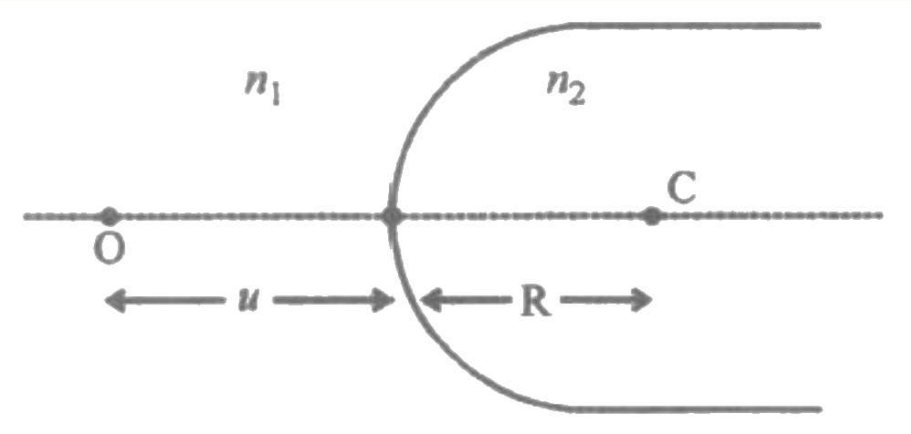
When a second medium sits against one curved face of a lens, model that medium as an extra thin lens (here plano-concave) and just add the powers; do not try to apply the lens maker's formula to the whole thing at once.

Final answer. f = R / (2μ1 - μ2 - 1); it diverges when μ2 > 2μ1 - 1; and for μ2 > (μ1+1)/2 it still converges, forming a real, inverted image at its focus for a distant object.

← *Back to Q340*

Q341

(a) A point object 'O' is kept in a medium of refractive index n_1 *in front of a convex spherical surface of radius of curvature R which separates the second medium of refractive index* n_2 *from the first one, as shown in the figure.*
Draw the ray diagram showing the image formation and deduce the relationship between the object distance and the image distance in terms of n_1, n_2 *and R.*
(b) When the image formed above acts as a virtual object for a concave spherical surface separating the medium n_2 *from* n_1 ($\mathbf{n_2 > n_1}$)*, draw this ray diagram and write the similar (similar to (a)) relation. Hence obtain the expression for the lens maker's formula.*



[2015]

SOLUTION

(a) When light refracts at a single spherical surface, the relation is found using Snell's law in the paraxial (small-angle) form together with the geometry of the

figure.

Ray diagram: Draw the principal axis horizontally. Mark the pole P of the convex spherical surface, its centre of curvature C on the right at distance R, and the point object O on the axis in medium 1 (refractive index n_1). A ray OA strikes the surface at point A a small height above the axis. The normal at A is the line AC. The ray bends towards the normal (since $\mathbf{n_2 > n_1}$) and, together with a second ray OP going straight along the axis, meets at the real image I in medium 2 (refractive index n_2). Label the angle of incidence i (between OA and AC) and angle of refraction r (between AC and AI). Mark $\boldsymbol{\alpha = \angle AOP}$, $\boldsymbol{\beta = \angle AIP}$ and $\boldsymbol{\gamma = \angle ACP}$.

By Snell's law at A,

$$n_1 \sin i = n_2 \sin r$$

For paraxial rays the angles are small, so $\boldsymbol{\sin i \approx i}$, $\boldsymbol{\sin r \approx r}$, giving

$$n_1\, i = n_2\, r$$

From the exterior-angle property of a triangle, i is the exterior angle of triangle OAC and γ is the exterior angle of triangle IAC:

$$i = \alpha + \gamma \qquad \gamma = r + \beta \Rightarrow r = \gamma - \beta$$

For small angles, $\theta\theta$, and with $\mathbf{AP \approx h}$ the height of A,

$$\alpha = \frac{h}{PO},\ \beta = \frac{h}{PI},\ \gamma = \frac{h}{PC}$$

Substituting in $\mathbf{n_1\, i = n_2\, r}$:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

$$n_1\left(\frac{h}{PO} + \frac{h}{PC}\right) = n_2\left(\frac{h}{PC} - \frac{h}{PI}\right)$$

Cancelling h:

$$\frac{n_1}{PO} + \frac{n_2}{PI} = \frac{n_2 - n_1}{PC}$$

Applying the sign convention ($\mathbf{PO = -u}$, $\mathbf{PI = +v}$, $\mathbf{PC = +R}$):

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

This is the required relation for **refraction at a single spherical surface**.

(b) Now treat a thin lens as two such surfaces. Light first refracts at surface 1 (radius R_1, going from n_1 to n_2) forming image I_1 at distance v_1; this I_1 acts as the (virtual) object for surface 2 (radius R_2, going from n_2 back to n_1).

Ray diagram: Draw the biconvex (or biconcave) thin lens. The first surface alone would form I_1; before the rays reach I_1 they strike the second (concave-to-the-light) surface, so I_1 serves as a virtual object, and the second refraction forms the final image I.

At the first surface:

$$\frac{n_2}{v_1}-\frac{n_1}{u}=\frac{n_2-n_1}{R_1} \quad ...(1)$$

At the second surface (object I_1 in medium n_2, image in medium n_1):

$$\frac{n_1}{v}-\frac{n_2}{v_1}=\frac{n_1-n_2}{R_2}=-\frac{n_2-n_1}{R_2} \quad ...(2)$$

Adding (1) and (2), the $\frac{n_2}{v_1}$ terms cancel:

$$\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Dividing by n_1:

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For a lens in air, $\mathbf{n_1=1}$ and $\mathbf{n_2=n}$, and for an object at infinity $v=f$. Hence the **lens maker's formula**:

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Final answer. Refraction at a single surface gives $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$, and adding the two surface relations gives the lens maker's formula $\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.

← Back to Q341

Q342

(a) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain expression for total magnification when the image is formed at infinity.
(b) Distinguish between myopia and hypermetropia. Show diagrammatically how these defects can be corrected.
OR
(a) State Huygen's principle. Using this principle draw a diagram to show how a plane wave front incident at the interface of the two media gets refracted when it propagates from a rarer to a denser medium. Hence verify Snell's law of refraction.
(b) When monochromatic light travels from a rarer to a denser medium, explain the following, giving reasons:
(i) Is the frequency of reflected and refracted light same as the frequency of incident light?
(ii) Does the decrease in speed imply a reduction in the energy carried by light wave?

[2013]

SOLUTION

(a) A compound microscope uses two convex lenses of short focal length, the objective (f_o) and the eyepiece (f_e), with $\mathbf{f_o e}$.

Ray diagram: Place the object AB just beyond the focus F_o of the objective. The objective forms a real, inverted, magnified image $A'B'$ on the far side. This image lies within the focal length of the eyepiece, which acts as a simple magnifier and produces a final virtual, inverted, highly magnified image $A''B''$. Label: object AB, objective O, intermediate image $A'B'$, eyepiece E, tube length L (distance between the two foci F_o and F_e' side), final image $A''B''$ at infinity, and the eye behind the eyepiece.

The magnifying power is the product of the linear magnification of the objective and the angular magnification of the eyepiece:

$$M = m_o \times m_e$$

For the objective the intermediate image forms near the focus, so

$$m_o = \frac{v_o}{u_o} \approx \frac{L}{f_o}$$

When the final image is at infinity, the eyepiece is a simple microscope in normal adjustment:

$$m_e = \frac{D}{f_e}$$

Therefore the total magnification when the image is formed at infinity:

$$M = \frac{L}{f_o} \cdot \frac{D}{f_e}$$

where L is the tube length and D=**25 cm** is the least distance of distinct vision.

(b) **Myopia (short-sightedness):** A person can see near objects clearly but distant objects appear blurred. The image of a distant object forms in front of the retina because the eyeball is too long or the lens is too converging. The far point is at a finite distance instead of infinity. Correction: use a concave (diverging) lens of suitable focal length so that parallel rays from a distant object appear to come from the eye's far point. Diagram: parallel rays diverged by a concave lens, then focused exactly on the retina.

Hypermetropia (long-sightedness): A person can see distant objects clearly but near objects appear blurred. The image of a near object forms behind the retina because the eyeball is too short or the lens is too weakly converging. The near point is farther than 25 cm. Correction: use a convex (converging) lens so that rays from an object at 25 cm appear to come from the eye's actual near point. Diagram: rays from a near object converged a little by a convex lens, then focused on the retina.

(OR)

(a) **Huygens' principle:** Every point on a given wavefront acts as a fresh source of secondary wavelets which spread out in all directions with the speed of light in that medium. The new wavefront at a later instant is the forward common tangent (envelope) to all these secondary wavelets.

Refraction from rarer to denser medium (verification of Snell's law): Let a plane wavefront AB in the rarer medium (speed v_1) strike the plane interface XY at angle of incidence i, with A touching first. In the time t the wavelet from B travels **BC**=$v_1 t$ in the rarer medium to reach C on the interface, the wavelet

from A travels $\mathbf{AE = v_2\, t}$ into the denser medium (speed $\mathbf{v_2}$**1**). The refracted wavefront is CE.

Diagram: incident wavefront AB, interface XY, refracted wavefront CE, normal at the point of incidence; angle of incidence i between incident ray and normal equals $\angle$ BAC, angle of refraction r equals $\angle$ ACE.

In triangle ABC, $\sin i = \dfrac{BC}{AC} = \dfrac{v_1\, t}{AC}$

In triangle ACE, $\sin r = \dfrac{AE}{AC} = \dfrac{v_2\, t}{AC}$

Dividing:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = n_{21} = \text{constant}$$

This is **Snell's law** of refraction, verified from Huygens' principle.

(b) (i) When light goes from a rarer to a denser medium, the **frequency of the reflected and refracted light stays the same as that of the incident light**. Frequency is decided by the source; it does not change on reflection or refraction. The speed and the wavelength change, but their ratio (the frequency $\nu = v/\lambda$) is unchanged because the wave cannot be created or destroyed at the boundary, so the number of wave crests arriving per second equals the number leaving.

(ii) No. A decrease in speed does **not** mean a reduction in the energy carried by the wave. The energy of a light wave depends on the number of photons and the frequency ($E = h\nu$ per photon), not on the speed. Since the frequency is unchanged, each photon's energy is unchanged; the wave simply travels slower in the denser medium.

Final answer. Total magnification of the microscope (image at infinity) is $M = \dfrac{L}{f_o} \cdot \dfrac{D}{f_e}$; in the OR part Snell's law $\dfrac{\sin i}{\sin r} = \dfrac{v_1}{v_2}$ follows from Huygens' principle, the frequency stays the same, and the energy is unchanged.

← *Back to Q342*

Q343

Define magnifying power of a telescope. Write its expression.
A small telescope has an objective lens of focal length 150 cm *and an eye piece of focal length* 5 cm. *If this telescope is used to view a* 100 *m high tower* 3 km *away, find the height of the final image when it is formed* 25 cm *away from the eye piece.*
OR
How is the working of a telescope different from that of a microscope?
The focal lengths of the objective and eyepiece of a microscope are 1.25 cm *and* 5 cm *respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of* 30 *in normal adjustment.*

[2012]

SOLUTION

The **magnifying power of a telescope** is the ratio of the angle subtended at the eye by the final image to the angle subtended at the unaided eye by the object.

$$m=\frac{\beta}{\alpha}$$

In normal adjustment (final image at infinity) this equals $m=\frac{f_o}{f_e}$, where f_o and f_e are the focal lengths of the objective and eyepiece.

Numerical: The tower is the object. Its angular size at the objective is

$$\alpha=\frac{\text{height of tower}}{\text{distance of tower}}=\frac{100\ m}{3000\ m}=\frac{1}{30}\ \text{rad}$$

The objective ($\mathbf{f_o=150\ cm}$) forms a real image of the tower in its focal plane. The height of this image is

$$h_o=\alpha\ f_o=\frac{1}{30}\times 150\ \text{cm}=5\ \text{cm}$$

This image acts as the object for the eyepiece ($\mathbf{f_e=5\ cm}$), and the final image is formed at $\boldsymbol{D}$**=25 cm** from the eyepiece. The eyepiece's linear magnification is

$$m_e=1+\frac{D}{f_e}=1+\frac{25}{5}=1+5=6$$

Hence the height of the final image is

$$h = m_e \times h_o = 6 \times 5 \text{ cm} = 30 \text{ cm}$$

So the **final image is 30 cm high**.

(OR)

Difference between a telescope and a microscope: A telescope is used to see distant, large objects (stars, towers) and increases their angular size; its objective has a large focal length and large aperture, and the object lies far beyond the objective so the intermediate image is small. A microscope is used to see very small, near objects and increases their linear size; both its lenses have very short focal lengths and the object is placed just outside the objective's focus, giving a large intermediate image.

Numerical: For a compound microscope in normal adjustment (final image at infinity), the magnifying power is

$$M = \frac{v_o}{|u_o|} \times \frac{D}{f_e}$$

The eyepiece part is

$$\frac{D}{f_e} = \frac{25}{5} = 5$$

Given total $M = 30$, the objective magnification must be

$$|m_o| = \frac{M}{D/f_e} = \frac{30}{5} = 6$$

So $\frac{v_o}{|u_o|} = 6 \Rightarrow v_o = 6\,|u_o| = -6u_o$ (with u_o negative).

Using the lens formula for the objective $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$ with $f_o = 1.25$ cm and $v_o = -6u_o$:

$$\frac{1}{-6u_o} - \frac{1}{u_o} = \frac{1}{1.25}$$

$$\frac{-1-6}{6u_o} = \frac{1}{1.25}$$

$$\frac{-7}{6u_o} = \frac{1}{1.25}$$

$$u_o = \frac{-7 \times 1.25}{6} = \frac{-8.75}{6} = -1.46 \text{ cm}$$

So the **object must be placed about 1.46 cm in front of the objective** (just beyond its focal length of 1.25 cm).

COMMON MISTAKE

A common slip is to multiply α by the full magnifying power to get the image height. Instead, find the objective image height $\mathbf{h_o = \alpha f_o}$ first, then apply only the eyepiece magnification $(1+D/f_e)$ for a near-point image.

Final answer. The final image of the tower is 30 cm high; for the microscope the object must be placed about 1.46 cm in front of the objective.

← *Back to Q343*

Q344

(a) Draw a ray diagram to show refraction of a ray of monochromatic light passing through a glass prism.
Deduce the expression for the refractive index of glass in terms of angle of prism and angle of minimum deviation.
(b) Explain briefly how the phenomenon of total internal reflection is used in fibre optics.

[2011 • Set 55-1-1]

SOLUTION

(a) A prism deviates a ray of light because refraction occurs at its two faces.

Ray diagram: Draw a triangular prism with refracting angle A at the top. A monochromatic ray strikes the first face at angle of incidence i_1, bends towards the normal and refracts at angle r_1. Inside the prism the ray travels to the second face, meeting it at angle r_2, then bends away from the normal and emerges at angle i_2. Extend the incident and emergent rays to meet; the angle between them is the angle of deviation δ. Label the normals at both faces, the angles i_1, r_1, r_2, i_2, the prism angle A at the apex, and δ between the produced incident ray and the emergent ray.

In the quadrilateral formed by the two normals and the two faces, the normals meet the faces at right angles, so the angle at the apex plus the angle between the normals is 180°. In the triangle formed by the ray inside and the two normals,

$$r_1+r_2=A$$

The deviation at the first face is (i_1-r_1) and at the second face is (i_2-r_2), so the total deviation is

$$\delta=(i_1-r_1)+(i_2-r_2)=(i_1+i_2)-(r_1+r_2)$$

$$\delta=i_1+i_2-A$$

As i_1 is increased, δ first decreases, reaches a minimum value δ_m, then increases. At minimum deviation the ray passes symmetrically through the prism, so

$$i_1=i_2=ir_1=r_2=r$$

Then $\mathbf{r_1+r_2=A}$ gives

$$r=\frac{A}{2}$$

and $\boldsymbol{\delta_m=2i-A}$ gives

$$i=\frac{A+\delta_m}{2}$$

Applying Snell's law at the first face, $\boldsymbol{n=\frac{\sin i}{\sin r}}$:

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

This is the required expression for the refractive index of the prism material.

(b) **Total internal reflection in fibre optics:** An optical fibre is a thin, flexible thread with a central core of refractive index n_1 surrounded by a cladding of slightly lower refractive index n_2 ($\mathbf{n_1>n_2}$). Light entering one end of the fibre, within the acceptance cone, strikes the core-cladding boundary at an angle

greater than the critical angle. Since it travels from the denser core to the rarer cladding and exceeds the critical angle, it undergoes total internal reflection, with no loss of intensity. The ray suffers repeated total internal reflections all along the fibre and travels the full length even if the fibre is bent, emerging from the far end with very little energy loss. This is the basis of high-speed, long-distance optical communication and of medical endoscopes.

Final answer. The refractive index of the prism is $n=\frac{\sin((A+\delta_m)/2)}{\sin(A/2)}$, and optical fibres guide light by repeated total internal reflection at the core-cladding boundary.

← *Back to Q344*

Q345

(a) Obtain lens makers formula using the expression $\frac{n_2}{v}-\frac{n_1}{u}=\frac{(n_2-n_1)}{R}$.
Here the ray of light propagating from a rarer medium of refractive index (n_1) *to a denser medium of refractive index* (n_2) *is incident on the convex side of spherical refracting surface of radius of curvature R.*
(b) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the pole. Using this diagram, derive the magnification formula for the image formed.

[2011 • Set 55-1-1]

SOLUTION

(a) Start from the given single-surface relation and apply it to the two surfaces of a thin lens.

Consider a thin convex lens of material refractive index n in air, with surfaces of radii R_1 (first) and R_2 (second). Light from object O at distance u first refracts at surface 1 (air to glass, radius R_1) forming image I_1 at distance v_1.

Using $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ with $n_1=1$, $n_2=n$:

$$\frac{n}{v_1}-\frac{1}{u}=\frac{n-1}{R_1} \quad ...(i)$$

I_1 now acts as the object for surface 2 (glass to air, radius R_2), forming the final image I at distance v. Here $\mathbf{n_1}=n$ (glass side), $\mathbf{n_2}=1$ (air side):

$$\frac{1}{v}-\frac{n}{v_1}=\frac{1-n}{R_2} \quad ...(ii)$$

Adding (i) and (ii), the $\frac{n}{v_1}$ terms cancel:

$$\frac{1}{v}-\frac{1}{u}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For an object at infinity ($u\to\infty$), the image forms at the focus, so $v=f$:

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

This is the **lens maker's formula**.

(b) **Concave mirror, object between focus F and pole P:**

Ray diagram: Draw the concave mirror with pole P, focus F and centre of curvature C on the principal axis. Place the object AB (an upright arrow) between F and P. Draw two rays from the top B: (1) a ray parallel to the principal axis, which reflects and passes through F; (2) a ray directed towards C (or the ray through F region) that reflects back parallel/retraces. The reflected rays diverge in front of the mirror, so they do not meet in front; their backward extensions meet behind the mirror. The image $A'B'$ formed behind the mirror is virtual, erect and magnified. Label A, B, A', B', F, C, P and the principal axis.

Magnification derivation: The two right-angled triangles formed by the object and image with the pole are similar (the incident and reflected rays at the pole make equal angles with the axis). Comparing triangle ABP (object) and triangle $A'B'P$ (image):

$$\frac{A'B'}{AB}=\frac{PA'}{PA}$$

Here $\mathbf{A'B'}=h'$ (image height), $\mathbf{AB}=h$ (object height), PA' corresponds to the image distance and PA to the object distance. Applying the sign convention (distances measured from P, with $\mathbf{PA}=-u$ for the real object and $\mathbf{PA'}=+v$ for the virtual image behind the mirror):

$$\frac{h'}{h}=-\frac{v}{u}$$

Therefore the **magnification** of the mirror is

$$m=\frac{h'}{h}=-\frac{v}{u}$$

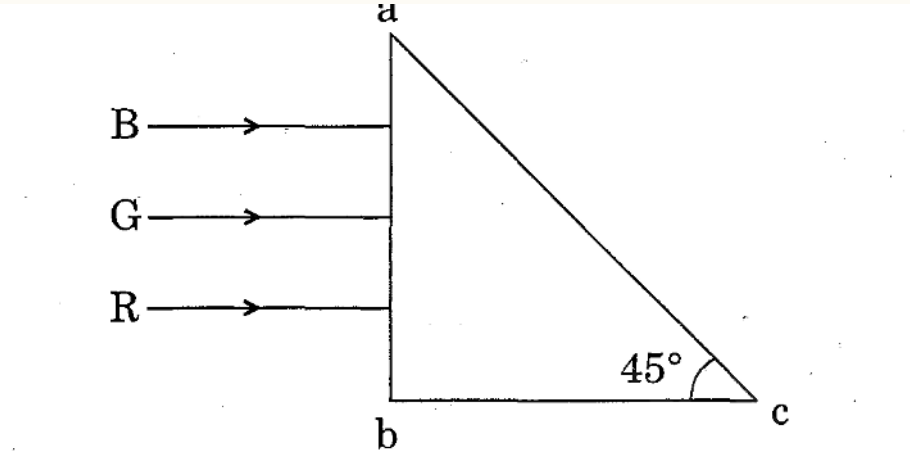
For this case the image is virtual and erect, so m comes out positive and greater than 1, confirming an enlarged upright image.

Final answer. The lens maker's formula is $\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$, and for the concave mirror the magnification is $m=\frac{h'}{h}=-\frac{v}{u}$.

← Back to Q345

Q346

(i) A ray of monochromatic light is incident on one of the faces of an equilateral triangular prism of refracting angle A. Trace the path of ray passing through the prism. Hence derive an expression for the refractive index of the material of the prism in terms of the angle of minimum deviation and its refracting angle.
(ii) Three light rays red (R), green (G) and blue (B) are incident on the right angled prism abc at face ab. The refractive indices of the material of the prism for red, green and blue wavelengths are respectively 1.39, 1.44 and 1.47. Trace the paths of these rays reasoning out the difference in their behaviour.



[2011 • Set 55-2-1]

SOLUTION

(i) When monochromatic light passes through an equilateral prism (refracting angle A), it refracts at both faces and is deviated.

Ray diagram: Draw the prism with apex angle A. The ray strikes the first face at angle of incidence i_1, refracts at r_1, crosses the prism to the second face where the internal angle is r_2, and emerges at angle i_2. Produce the incident and emergent rays to meet; the angle between them is the deviation δ. Mark the normals to both faces, the angles i_1, r_1, r_2, i_2, the apex angle A, and δ.

From the geometry (the quadrilateral with the two normals), the internal angles satisfy

$$r_1+r_2=A$$

The total deviation is the sum of the deviations at the two faces:

$$\delta=(i_1-r_1)+(i_2-r_2)=(i_1+i_2)-A$$

At minimum deviation the ray travels symmetrically, so

$$i_1=i_2=i,\ r_1=r_2=r=\frac{A}{2}$$

and

$$\delta_m=2i-A \Rightarrow i=\frac{A+\delta_m}{2}$$

By Snell's law at the first face:

$$n=\frac{\sin i}{\sin r}=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$n=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

(ii) The right-angled prism abc has a 45° angle, so a ray entering normally through face ab strikes the hypotenuse ac at an angle of incidence of 45°. Total internal reflection occurs there only if 45° exceeds the critical angle θ_c, where

$$\theta_c=\frac{1}{n}$$

The condition for total internal reflection at 45° is

$$\sin 45^\circ > \theta_c = \frac{1}{n} \Rightarrow n > \frac{1}{\sin 45^\circ = \sqrt{2} \approx 1.414}$$

Now compare each colour:

Red, $\mathbf{n_R = 1.39 < 1.414}$: the angle 45° is less than its critical angle, so red does **not** undergo total internal reflection. It is refracted and passes out through the hypotenuse face ac (mostly transmitted, slightly bent).

Green, $\mathbf{n_G = 1.44 > 1.414}$: 45° exceeds the critical angle, so green undergoes **total internal reflection** at ac and emerges through the third face bc.

Blue, $\mathbf{n_B = 1.47 > 1.414}$: similarly 45° exceeds its critical angle, so blue also undergoes **total internal reflection** at ac and emerges through face bc.

Thus red is transmitted through the hypotenuse, while green and blue are totally internally reflected and leave through the other face, because the critical angle decreases as the refractive index increases.

Final answer. The refractive index is $\boldsymbol{n = \frac{\sin((A+\delta_m)/2)}{\sin(A/2)}}$; red ($\boldsymbol{n = 1.39}$) is refracted out of the hypotenuse while green (**1.44**) and blue (**1.47**) undergo total internal reflection there.

← *Back to Q346*

Q347

Draw a ray diagram to show the working of a compound microscope. Deduce an expression for the total magnification when the final image is formed at the near point.
In a compound microscope, an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eye piece has a focal length of 5 cm and the final image is formed at the near point, estimate the magnifying power of the microscope.

[2010]

SOLUTION

A compound microscope uses two short-focal-length convex lenses to view tiny objects.

Ray diagram: Place the object AB just beyond the focus F_o of the objective O. The objective forms a real, inverted, magnified image $A'B'$ on the other side, inside the focal length of the eyepiece E. The eyepiece acts as a simple magnifier and forms the final virtual, inverted, magnified image $A''B''$ at the near point D. Label: AB (object), O (objective), $A'B'$ (intermediate real image), E (eyepiece), $A''B''$ (final image at distance D from the eye), object distance u_o and image distance v_o at the objective.

The total magnification is the product of the objective's linear magnification and the eyepiece's angular magnification:

$$M = m_o \times m_e$$

For the objective,

$$m_o = \frac{v_o}{u_o}$$

When the final image is at the near point, the eyepiece acts as a simple microscope and

$$m_e = 1 + \frac{D}{f_e}$$

Hence the total magnification at the near point:

$$M = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)$$

Numerical: Given $\mathbf{u_o = -1.5\ cm}$, $\mathbf{f_o = 1.25\ cm}$, $\mathbf{f_e = 5\ cm}$, $\mathbf{D = 25\ cm}$.

First find v_o using the lens formula $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$:

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{1.25} + \frac{1}{(-1.5)}$$

$$\frac{1}{v_o} = \frac{1}{1.25} - \frac{1}{1.5} = 0.8 - 0.667 = 0.1333\ \text{cm}^{-1}$$

$$v_o = 7.5\ \text{cm}$$

Objective magnification:

$$m_o = \frac{v_o}{u_o} = \frac{7.5}{-1.5} = -5$$

Eyepiece magnification:

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 1 + 5 = 6$$

Total magnifying power:

$$M = m_o \times m_e = (-5) \times 6 = -30$$

The negative sign shows the final image is inverted. So the **magnifying power is 30** (image inverted and magnified 30 times).

Final answer. The magnifying power of the compound microscope is $M = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right) = -30$, i.e. magnitude 30 (inverted).

← *Back to Q347*

Q348

Trace the rays of light showing the formation of an image due to a point object placed on the axis of a spherical surface separating the two media of refractive indices n_1 *and* n_2. *Establish the relation between the distances of the object, the image and the radius of curvature from the central point of the spherical surface.*
Hence derive the expression of the lens maker's formula.

[2009]

SOLUTION

Refraction at a single spherical surface separating media of refractive indices n_1 and n_2.

Ray diagram: Draw the principal axis. Mark the pole P of the convex spherical surface, its centre of curvature C at distance R, and the point object O on the axis in medium 1 (index n_1). A ray OA from the object strikes the surface at A (small height h above the axis); the normal at A is along AC. Since $n_2 > n_1$, the ray bends towards the normal and, together with the axial ray OP, meets at the real

image I in medium 2 (index n_2). Mark the angle of incidence i ($\angle$ OAC), angle of refraction r ($\angle$ CAI), and the angles $\alpha = \angle$ **AOP**, $\beta = \angle$ **AIP**, $\gamma = \angle$ **ACP**.

Snell's law at A:

$$n_1 \sin i = n_2 \sin r$$

For paraxial rays the angles are small, so $\sin i \approx i$, $\sin r \approx r$:

$$n_1 i = n_2 r$$

Using the exterior-angle theorem for triangles OAC and IAC:

$$i = \alpha + \gamma,\ \gamma = r + \beta \Rightarrow r = \gamma - \beta$$

For small angles, with A close to the pole so $PA \approx h$,

$$\alpha \approx \frac{h}{PO},\ \beta \approx \frac{h}{PI},\ \gamma \approx \frac{h}{PC}$$

Substituting into $n_1 i = n_2 r$:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

$$n_1\left(\frac{h}{PO} + \frac{h}{PC}\right) = n_2\left(\frac{h}{PC} - \frac{h}{PI}\right)$$

Cancelling h:

$$\frac{n_1}{PO} + \frac{n_2}{PI} = \frac{n_2 - n_1}{PC}$$

Applying the sign convention ($PO = -u$, $PI = +v$, $PC = +R$):

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

This is the required relation between object distance, image distance and radius of curvature.

Lens maker's formula: A thin lens of material index n in air has two surfaces, radii R_1 and R_2. Apply the above relation at each surface.

At surface 1 (air to glass, $n_1 = 1$, $n_2 = n$), forming intermediate image at v_1:

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1} \quad \ldots(1)$$

The intermediate image is the object for surface 2 (glass to air, $\mathbf{n_1}=n$, $\mathbf{n_2}=1$), forming final image at v:

$$\frac{1}{v}-\frac{n}{v_1}=\frac{1-n}{R_2} \quad ...(2)$$

Adding (1) and (2):

$$\frac{1}{v}-\frac{1}{u}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For an object at infinity, $v=f$:

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Final answer. The single-surface relation is $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$, which leads to the lens maker's formula $\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.

← *Back to Q348*

Q349

Draw the labelled ray diagram for the formation of image by a compound microscope. Derive the expression for the total magnification of a compound microscope. Explain why both the objective and the eyepiece of a compound microscope must have short focal lengths.

[2009]

SOLUTION

Ray diagram for a compound microscope: Draw the objective O and eyepiece E, both short-focal-length convex lenses, on a common axis. Place the object AB just beyond the objective's focus F_o. The objective forms a real, inverted, magnified intermediate image $A'B'$ between the lenses, lying just within the focal length of the eyepiece. The eyepiece, acting as a simple magnifier, forms the final virtual, inverted, magnified image $A''B''$ at the near point (or at infinity in normal adjustment). Label: AB (object), F_o (objective focus), O (objective), $A'B'$ (intermediate image), E (eyepiece), $A''B''$ (final image), and the eye.

Total magnification is the product of the objective's linear magnification and the eyepiece's angular magnification:

$$M = m_o \times m_e$$

For the objective, the intermediate image forms close to the focal plane, so

$$m_o = \frac{v_o}{u_o} \approx \frac{L}{f_o}$$

where L is the tube length (separation of objective and eyepiece foci).

For the eyepiece acting as a simple magnifier with the final image at the near point,

$$m_e = 1 + \frac{D}{f_e}$$

Therefore the total magnification at the near point:

$$M = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

When the final image is at infinity (normal adjustment), $\mathbf{m_e = \frac{D}{f_e}}$, giving

$$M = \frac{L}{f_o} \cdot \frac{D}{f_e}$$

Why both focal lengths must be short: The objective magnification is $\mathbf{m_o = \frac{L}{f_o}}$, so a **small $\mathbf{f_o}$ makes $\mathbf{m_o}$ large**, giving a big intermediate image. The eyepiece magnification is $\mathbf{m_e = 1 + \frac{D}{f_e}}$ (or D/f_e), so a **small $\mathbf{f_e}$ makes $\mathbf{m_e}$ large**. Since the total magnification is the product of these two, keeping both f_o and f_e small makes M very large (several hundred), which is exactly what a microscope needs to resolve very small objects. Large focal lengths would give magnifications close to 1 and the instrument would be useless.

Final answer. The total magnification is $\boldsymbol{M = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)}$, and short $\mathbf{f_o}$ and $\mathbf{f_e}$ are needed because $\boldsymbol{M}$ is inversely proportional to both, so small focal lengths give large magnification.

← Back to Q349

Q350

(a) (i) Draw a labelled ray diagram to show the formation of image in an astronomical telescope for a distant object.
(ii) Write three distinct advantages of a reflecting type telescope over a refracting type telescope.
(b) A convex lens of focal length 10 cm *is placed coaxially* 5 cm *away from a concave lens of focal length* 10 cm. *If an object is placed* 30 cm *in front of the convex lens, find the position of the final image formed by the combined system.*

[2009]

SOLUTION

(a)(i) Astronomical telescope, image of a distant object:

Ray diagram: Draw the objective O (large focal length f_o) and eyepiece E (short focal length f_e) on a common axis. Parallel rays from a distant object enter the objective at a small angle α to the axis and converge to form a real, inverted image $A'B'$ at the common focus of the two lenses (the objective focal plane). This image lies just within the focus of the eyepiece, which acts as a magnifier and forms the final image at infinity, subtending the larger angle β at the eye. Label the parallel incident rays, O, $A'B'$ (real inverted intermediate image), E, the angles α and β, and the final image at infinity. In normal adjustment the magnifying power is $m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$.

(ii) Three advantages of a reflecting (mirror objective) telescope over a refracting (lens objective) telescope:

1. A reflecting telescope is free from chromatic aberration, because a mirror does not split light into colours (no dispersion), whereas a lens objective produces chromatic aberration.

2. Using a parabolic mirror removes spherical aberration, giving a sharper image; large lenses suffer from spherical aberration.

3. A large mirror can be supported all along its back surface, so it does not sag under its own weight and gathers more light (greater resolving power and

brightness), while a large lens can be supported only at the edges and tends to deform; mirrors are also cheaper and easier to make large.

(b) Numerical: Convex lens $\mathbf{f_1 = +10\ cm}$, concave lens $\mathbf{f_2 = -10\ cm}$, separation 5 cm, object 30 cm in front of the convex lens.

For the convex lens, $\mathbf{u_1 = -30\ cm}$. Using $\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$:

$$\frac{1}{v_1} = \frac{1}{f_1} + \frac{1}{u_1} = \frac{1}{10} + \frac{1}{(-30)}$$

$$\frac{1}{v_1} = \frac{3-1}{30} = \frac{2}{30} = \frac{1}{15}$$

$$v_1 = +15\ \text{cm}$$

So the convex lens alone would form an image 15 cm to its right. This image acts as the object for the concave lens, which is 5 cm to the right of the convex lens. The image lies $\mathbf{15 - 5 = 10\ cm}$ beyond the concave lens, so it is a virtual object for it:

$$u_2 = +10\ \text{cm}$$

For the concave lens, $\frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f_2}$:

$$\frac{1}{v_2} = \frac{1}{f_2} + \frac{1}{u_2} = \frac{1}{(-10)} + \frac{1}{10} = 0$$

$$v_2 = \infty$$

The final image is formed at **infinity** (the emergent rays are parallel). Equivalently, the combined system sends the distant-image rays out parallel, so the final image is at infinity.

CONCEPT

When light passes through a lens combination, the image from the first lens becomes the object for the second; convert its position relative to the second lens (here 10 cm past it, giving a virtual object with $\mathbf{u_2 = +10}$ cm) before applying the lens formula again.

Final answer. The final image is formed at infinity (the emergent rays from the concave lens are parallel).

← Back to Q350

Q351

(a) With the help of a suitable ray diagram, derive the mirror formula for a concave mirror.
(b) The near point of a hypermetropic person is 50 cm *from the eye. What is the power of the lens required to enable the person to read clearly a book held at* 25 cm *from the eye?*

[2009]

SOLUTION

(a) The mirror formula connects the object distance u, the image distance v, and the focal length f of a spherical mirror.

Ray diagram: Take a concave mirror with pole P, centre of curvature C and principal focus F. Place an object AB (with A on the principal axis and B above it) beyond C. From B draw one ray parallel to the principal axis, which after reflection passes through F. Draw a second ray through C, which strikes the mirror normally and retraces its path. The two reflected rays intersect at B', giving the real, inverted image $A'B'$ between C and F, with A' on the axis.

In right-angled triangles $A'B'F$ and PEF (PE is the part of the mirror up to the axis), they are similar, so

$$\frac{A'B'}{\mathrm{PE}}=\frac{A'F}{\mathrm{PF}}$$

Since **PE = AB** (aperture small), this gives

$$\frac{A'B'}{\mathrm{AB}}=\frac{A'F}{\mathrm{PF}}=\frac{\mathrm{PA'-PF}}{\mathrm{PF}}$$

Also triangles ABP and $A'B'P$ are similar (the ray BP reflects to $B'P$, angles of incidence and reflection equal), so

$$\frac{A'B'}{AB}=\frac{PA'}{PA}$$

Equating the two ratios,

$$\frac{PA'-PF}{PF}=\frac{PA'}{PA}$$

Applying the sign convention (distances measured from P, opposite to incident light are negative): $\mathbf{PA=-}u$, $\mathbf{PA'=-}v$, $\mathbf{PF=-}f$.

$$\frac{-v-(-f)}{-f}=\frac{-v}{-u}$$

$$\frac{-v+f}{-f}=\frac{v}{u}$$

Cross-multiplying: $u(-v+f)=-fv$, i.e. $-uv+uf=-fv$.

Dividing throughout by uvf,

$$-\frac{1}{f}+\frac{1}{v}=-\frac{1}{u}$$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

This is the **mirror formula** $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$.

(b) A hypermetropic (long-sighted) eye has its near point pushed out to 50 cm. To read a book held at the normal near point of 25 cm, the spectacle lens must take the object at 25 cm and form a virtual image at the eye's own near point, 50 cm, so the eye can see it comfortably.

Here $u=-25$ **cm** (object on the same side as incident light) and $v=-50$ **cm** (virtual image, same side).

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{-50}-\frac{1}{-25}=\frac{1}{f}$$

$$-\frac{1}{50}+\frac{1}{25}=\frac{1}{f}$$

$$\frac{-1+2}{50}=\frac{1}{f}\Rightarrow\frac{1}{f}=\frac{1}{50}$$

f**=+50 cm=+0.5** m (a converging lens).

$$P=\frac{1}{f(\text{in } m)}=\frac{1}{0.5}=+2\ D$$

So the person needs a converging lens of power **+2** D.

Final answer. (a) Mirror formula derived: 1/v + 1/u = 1/f. (b) The required lens is a converging lens of power +2 D (f = +50 cm).

← *Back to Q351*

Q352

(a) For a ray of light traveling from a denser medium of refractive index n_1 *to a rarer medium of refractive index* n_2*, prove that* $\frac{n_2}{n_1} = \sin i_c$*, where* i_c *is the critical angle of incidence for the media.*
(b) Explain with the help of a diagram, how the above principle is used for transmission of video signals using optical fibres.

[2008]

SOLUTION

(a) When light travels from a denser medium (refractive index n_1) to a rarer medium (refractive index n_2, with $n_1 > n_2$), the refracted ray bends away from the normal. As the angle of incidence increases, the angle of refraction increases faster.

The critical angle i_c is that particular angle of incidence in the denser medium for which the angle of refraction in the rarer medium becomes 90°, so the refracted ray just grazes along the interface.

Diagram: Draw a horizontal boundary separating the lower denser medium (n_1) from the upper rarer medium (n_2). A ray in the denser medium meets the boundary at angle i_c to the normal; the refracted ray runs along the surface at 90° to the normal.

Applying Snell's law at the interface (denser to rarer):

$$n_1 \sin i_c = n_2 \sin 90^\circ$$

$$n_1 \sin i_c = n_2 \times 1$$

$$\sin i_c = \frac{n_2}{n_1}$$

$\frac{n_2}{n_1} = \sin i_c$ (proved).

For any angle of incidence greater than i_c, no refraction is possible and the light is completely reflected back into the denser medium. This is called **total internal reflection** (TIR).

(b) An optical fibre is a very thin, long glass thread. It has a central glass core of higher refractive index n_1 surrounded by a cladding (sheath) of lower refractive index n_2 ($\mathbf{n_1 > n_2}$).

Diagram: Draw a long cylindrical fibre with a darker inner core and a lighter outer cladding. A light ray enters one end of the core and is shown striking the core-cladding boundary at successive points, each time at an angle greater than the critical angle, so it zig-zags down the fibre by repeated total internal reflection and emerges at the far end.

A video signal is first converted into an electrical signal and then into a light (optical) signal by a transmitter. This light is fed into the core. Because each ray meets the core-cladding interface at an angle larger than the critical angle, it undergoes repeated **total internal reflection** and travels along the fibre with almost no loss of energy, even around bends, for very long distances.

At the receiving end the light signal is reconverted into an electrical signal and then into the video signal. Thus optical fibres carry video, audio and data signals over thousands of kilometres with negligible loss, forming the backbone of modern communication.

Final answer. (a) From Snell's law at the critical angle, $n_2/n_1 = \sin i_c$ is proved. (b) Optical fibres guide light/video signals by repeated total internal reflection in a high-index core.

← *Back to Q352*

Q353

Draw a labeled ray diagram of a compound microscope and write an expression for its magnifying power.
The focal lengths of the objective and eye-lens of a compound microscope are 2 cm, 6.25 cm *respectively. The distance between the lenses is* 15 cm. *(i) How far from the objective lens will the object be kept, so as to obtain the final image at the near point of the eye? (ii) Also calculate its magnifying power.*

[2008]

SOLUTION

A compound microscope uses two convex lenses: an objective of small focal length f_o and an eyepiece of slightly larger focal length f_e.

Ray diagram: Draw the principal axis. Place a small object AB just beyond the objective's focus F_o. The objective forms a real, inverted, magnified image $A'B'$ inside the focal length of the eyepiece. The eyepiece, acting as a simple magnifier, forms a final image $A''B''$ that is virtual, inverted (with respect to the object) and highly magnified, formed at the near point (distance D) on the same side as the object. Mark the objective, eyepiece, F_o, F_e, the intermediate image $A'B'$ and the final image $A''B''$.

Magnifying power (final image at the near point):

$$M = m_o \times m_e = \frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)$$

Given $\mathbf{f_o = 2\ cm}$, $\mathbf{f_e = 6.25\ cm}$, tube length $\mathbf{L = 15\ cm}$, $\mathbf{D = 25\ cm}$.

(i) First find where the eyepiece must take its object so the final image is at the near point. For the eyepiece, $\mathbf{v_e = -25\ cm}$:

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{6.25}$$

$$\frac{1}{u_e} = -\frac{1}{25} - \frac{4}{25} = -\frac{5}{25} = -\frac{1}{5}$$

$$u_e = -5\ \text{cm}$$

So the intermediate image $A'B'$ is 5 cm in front of the eyepiece. Since the lenses are 15 cm apart, the objective must form this image at

$$v_o = L - |u_e| = 15 - 5 = 10 \text{ cm}$$

For the objective:

$$\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$$

$$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2}$$

$$\frac{1}{u_o} = \frac{1-5}{10} = -\frac{4}{10} = -\frac{2}{5}$$

$$u_o = -2.5 \text{ cm}$$

The object must be kept **2.5 cm from the objective lens**.

(ii) Magnifying power:

$$m_o = \frac{v_o}{u_o} = \frac{10}{-2.5} = -4$$

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{6.25} = 1 + 4 = 5$$

$$M = m_o \times m_e = (-4) \times 5 = -20$$

The negative sign shows the final image is inverted. So the magnifying power has magnitude **20**.

> **TIP**
>
> When the final image is at the near point, always start at the eyepiece (v_e = -D) to get u_e, subtract from the tube length for v_o, then back-solve the objective for u_o.

Final answer. (i) The object is placed 2.5 cm from the objective. (ii) The magnifying power is 20 (image inverted).

← *Back to Q353*

Q354

Draw a labeled ray diagram of an astronomical telescope, in the normal adjustment position and write the expression for its magnifying power.
An astronomical telescope uses an objective lens of focal length 15 cm *and eye-lens of focal length* 1 cm. *What is the angular magnification of the telescope?*
If this telescope is used to view Moon, what is the diameter of the image of Moon formed by the objective lens? (Diameter of Moon = 3.5×10^6 *m and Radius of lunar orbit* = 3.8×10^8 *m)*

[2008]

SOLUTION

An astronomical telescope uses two convex lenses: an objective of large focal length f_o and wide aperture, and an eyepiece of small focal length f_e. In normal adjustment the final image is formed at infinity, so the eye views it in a relaxed state.

Ray diagram: Draw the principal axis. Parallel rays from a distant object enter the objective and converge to form a real, inverted image $A'B'$ at the objective's focus F_o. This focus coincides with the focus F_e of the eyepiece, so the eyepiece sends out a parallel beam (final image at infinity) which the eye receives. Mark the objective, eyepiece, the common focal point (F_o, F_e), the intermediate image $A'B'$, the angle α subtended by the object at the objective and the angle β subtended by the final image at the eye.

The angular magnification (magnifying power) in normal adjustment is

$$m=\frac{\beta}{\alpha}=\frac{f_o}{f_e}$$

Given $\mathbf{f_o}$**=15 cm** and $\mathbf{f_e}$**=1 cm**:

$$m=\frac{f_o}{f_e}=\frac{15}{1}=15$$

So the **angular magnification is 15**.

Now for the image of the Moon formed by the objective. The Moon subtends an angle at the objective equal to (its diameter)/(distance of orbit):

$$\theta = \frac{D_{moon}}{r} = \frac{3.5 \times 10^{6}}{3.8 \times 10^{8}}$$

$$\theta = 9.21 \times 10^{-3} \text{ rad}$$

The objective forms the image at its focal plane, so the image diameter is

$$d = f_o \, \theta$$

$$d = 15 \text{ cm} \times 9.21 \times 10^{-3}$$

$$d = 0.138 \text{ cm} \approx 1.4 \text{ mm}$$

So the diameter of the Moon's image formed by the objective is about **1.4 mm**.

Final answer. Angular magnification = 15; the diameter of the Moon's image formed by the objective is about 1.4 mm.

← *Back to Q354*

Q355

With the help of a ray diagram, show the formation of image of a point object by refraction of light at a spherical surface separating two media of refractive indices n_1 *and* n_2 ($\mathbf{n_2 > n_1}$) *respectively. Using this diagram, derive the relation:*

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Write the sign conventions used. What happens to the focal length of a convex lens when it is immersed in water?

[2004]

SOLUTION

Consider a convex spherical surface of radius of curvature R, with pole P and centre of curvature C, separating a rarer medium (n_1) on the left from a denser medium (n_2) on the right, with $\mathbf{n_2 > n_1}$.

Ray diagram: A point object O lies on the principal axis in medium 1. One ray OP travels along the axis undeviated. A second ray OA strikes the surface at point A (just above the axis), bends towards the normal CA (since it enters a denser medium), and the two refracted rays meet at I on the axis, forming the real image. Mark angles: i = angle of incidence at A, r = angle of refraction, $\alpha = \angle$ **AOP**, $\beta = \angle$ **AIP**, $\gamma = \angle$ **ACP**.

By Snell's law at A:

$$n_1 \sin i = n_2 \sin r$$

For paraxial rays the angles are small, so $\sin i \approx i$ and $\sin r \approx r$:

$$n_1 \, i = n_2 \, r$$

From the exterior-angle property of triangles, i is the exterior angle of triangle AOC and γ is the exterior angle of triangle AIC:

$$i = \alpha + \gamma$$

$$\gamma = r + \beta \Rightarrow r = \gamma - \beta$$

Substituting in $n_1 \, i = n_2 \, r$:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

For small angles, with $AP \approx h$ the height of A above the axis,

$$\alpha \approx \frac{h}{PO}, \; \beta \approx \frac{h}{PI}, \; \gamma \approx \frac{h}{PC}$$

$$n_1\left(\frac{h}{PO} + \frac{h}{PC}\right) = n_2\left(\frac{h}{PC} - \frac{h}{PI}\right)$$

Dividing by h:

$$\frac{n_1}{PO} + \frac{n_1}{PC} = \frac{n_2}{PC} - \frac{n_2}{PI}$$

Now apply the sign convention: $PO = -u$, $PI = +v$, $PC = +R$.

$$\frac{n_1}{-u} + \frac{n_1}{R} = \frac{n_2}{R} - \frac{n_2}{v}$$

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2}{R}-\frac{n_1}{R}$$

$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ (derived).

Sign conventions used: all distances are measured from the pole P. Distances measured in the direction of the incident light are positive; those against it are negative. Heights measured upward from the axis are positive, downward are negative. Hence the real object distance u is negative, the real image distance v is positive, and R is positive for this convex surface (centre C on the side of outgoing light).

Convex lens in water: The lens-maker relation gives $\frac{1}{f}=(\frac{n_g}{n_m}-1)(\frac{1}{R_1}-\frac{1}{R_2})$. In air $\frac{n_g}{n_m}=1.5$, but in water $\frac{n_g}{n_w}\approx\frac{1.5}{1.33}\approx 1.13$. So the factor $(\frac{n_g}{n_m}-1)$ falls from 0.5 to about 0.13, making $\frac{1}{f}$ smaller. Therefore the **focal length increases** (the lens becomes weaker and converges light less strongly) when a convex lens is immersed in water.

Final answer. The relation $n_2/v - n_1/u = (n_2 - n_1)/R$ is derived; immersing a convex lens in water increases its focal length (it becomes weaker).

← *Back to Q355*

Q356

Derive the relation between distance of object, distance of image and radius of curvature of a convex spherical surface, when refraction takes place from a rarer medium of refractive index n_1 *to a denser medium of refractive index* n_2 *and the image produced is real.*
State assumptions and convention of signs used.

[2003]

SOLUTION

Consider refraction at a convex spherical surface of radius R, pole P and centre of curvature C, separating a rarer medium (n_1) from a denser medium (n_2), with $n_2 > n_1$. We want the relation when a real image is formed.

Ray diagram: A point object O lies on the principal axis in the rarer medium. The axial ray OP passes straight through. A second ray OA meets the surface at A close to the axis, bends towards the normal CA on entering the denser medium, and meets the axis at I, forming a real image. Mark i (incidence), r (refraction), $\alpha = \angle$ **AOP**, $\beta = \angle$ **AIP**, $\gamma = \angle$ **ACP**.

Assumptions: (i) the aperture of the surface is small and the rays are paraxial, so all angles are small and $\theta \approx \theta\theta$; (ii) the object is a point on the principal axis; (iii) the refraction is at a single spherical surface.

Snell's law at A:

$$n_1 \sin i = n_2 \sin r$$

For small angles:

$$n_1\, i = n_2\, r$$

Using exterior-angle geometry:

$$i = \alpha + \gamma$$

$$\gamma = r + \beta \Rightarrow r = \gamma - \beta$$

Substituting:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

With A at height h above the axis,

$$\alpha \approx \frac{h}{PO}, \beta \approx \frac{h}{PI}, \gamma \approx \frac{h}{PC}$$

$$n_1\left(\frac{h}{PO} + \frac{h}{PC}\right) = n_2\left(\frac{h}{PC} - \frac{h}{PI}\right)$$

Dividing by h:

$$\frac{n_1}{PO} + \frac{n_1}{PC} = \frac{n_2}{PC} - \frac{n_2}{PI}$$

Apply the sign convention: **PO** $= -u$, **PI** $= +v$, **PC** $= +R$.

$$-\frac{n_1}{u} + \frac{n_1}{R} = \frac{n_2}{R} - \frac{n_2}{v}$$

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2}{R}-\frac{n_1}{R}$$

$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ (the required relation).

Sign conventions used: all distances are measured from the pole P; distances in the direction of incident light are positive and those opposite are negative; heights above the axis are positive. Hence u is negative (real object), v is positive (real image), and R is positive for the convex surface.

Final answer. For refraction from a rarer to a denser medium at a convex surface, $n_2/v - n_1/u = (n_2 - n_1)/R$ is derived using paraxial assumptions and the sign convention.

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Q357

Draw a ray diagram to show the formation of image of an object placed between the optical centre and focus of the convex lens. Write the characteristics of image formed. Using this diagram, derive the relation between object distance, image distance and focal length of the convex lens. Write the assumptions and convention of signs used.

[2003]

SOLUTION

When an object is placed between the optical centre O and the focus F of a convex lens, the lens acts as a simple magnifier.

Ray diagram: Draw a convex lens with principal axis, optical centre O, and foci F on both sides. Place the object AB (foot A on the axis) between O and the first focus. From the top point B draw two rays: (i) one parallel to the principal axis, which after refraction passes through the far focus F; (ii) one straight through the optical centre O undeviated. After refraction these two rays diverge, so they do not meet on the far side. Produce them backwards (on the object side); they meet at B', giving a virtual image $A'B'$ that is erect and magnified, on the same side as the object.

Characteristics of the image: virtual, erect, magnified, and formed on the same side as the object (beyond the object, farther from the lens).

Assumptions: the lens is thin, the rays are paraxial (small angles, close to the axis), and the light is monochromatic.

Sign conventions: all distances are measured from the optical centre O; distances in the direction of incident light are positive and those opposite are negative; heights above the principal axis are positive, below are negative.

Derivation of the lens formula: In the ray diagram, triangles formed by the object, image and the optical centre are similar. The ray through the optical centre gives similar triangles $A'B'O$ and ABO, so

$$\frac{A'B'}{\text{AB}}=\frac{\text{OA'}}{\text{OA}}$$

The ray parallel to the axis, refracting through F, gives similar triangles $A'B'F$ and OPF (where **OP=AB**, the height at the lens), so

$$\frac{A'B'}{\text{AB}}=\frac{A'F}{\text{OF}}=\frac{\text{OA'+OF}}{\text{OF}}$$

Equating the two expressions for $\frac{A'B'}{\text{AB}}$:

$$\frac{\text{OA'}}{\text{OA}}=\frac{\text{OA'+OF}}{\text{OF}}$$

Now apply the sign convention. For this virtual image with the object between O and F: **OA=$-u$**, **OA'=$-v$**, **OF=$-f$**.

$$\frac{-v}{-u}=\frac{-v+(-f)}{-f}$$

$$\frac{v}{u}=\frac{-v-f}{-f}=\frac{v+f}{f}$$

Cross-multiplying: **vf=$u(v+f)$=uv+uf**.

Dividing throughout by uvf:

$$\frac{1}{u}=\frac{1}{f}+\frac{1}{v}$$

Rearranging gives the **lens formula** $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

Final answer. The image is virtual, erect and magnified on the same side; the lens formula $1/v - 1/u = 1/f$ is derived from similar triangles and the sign convention.

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